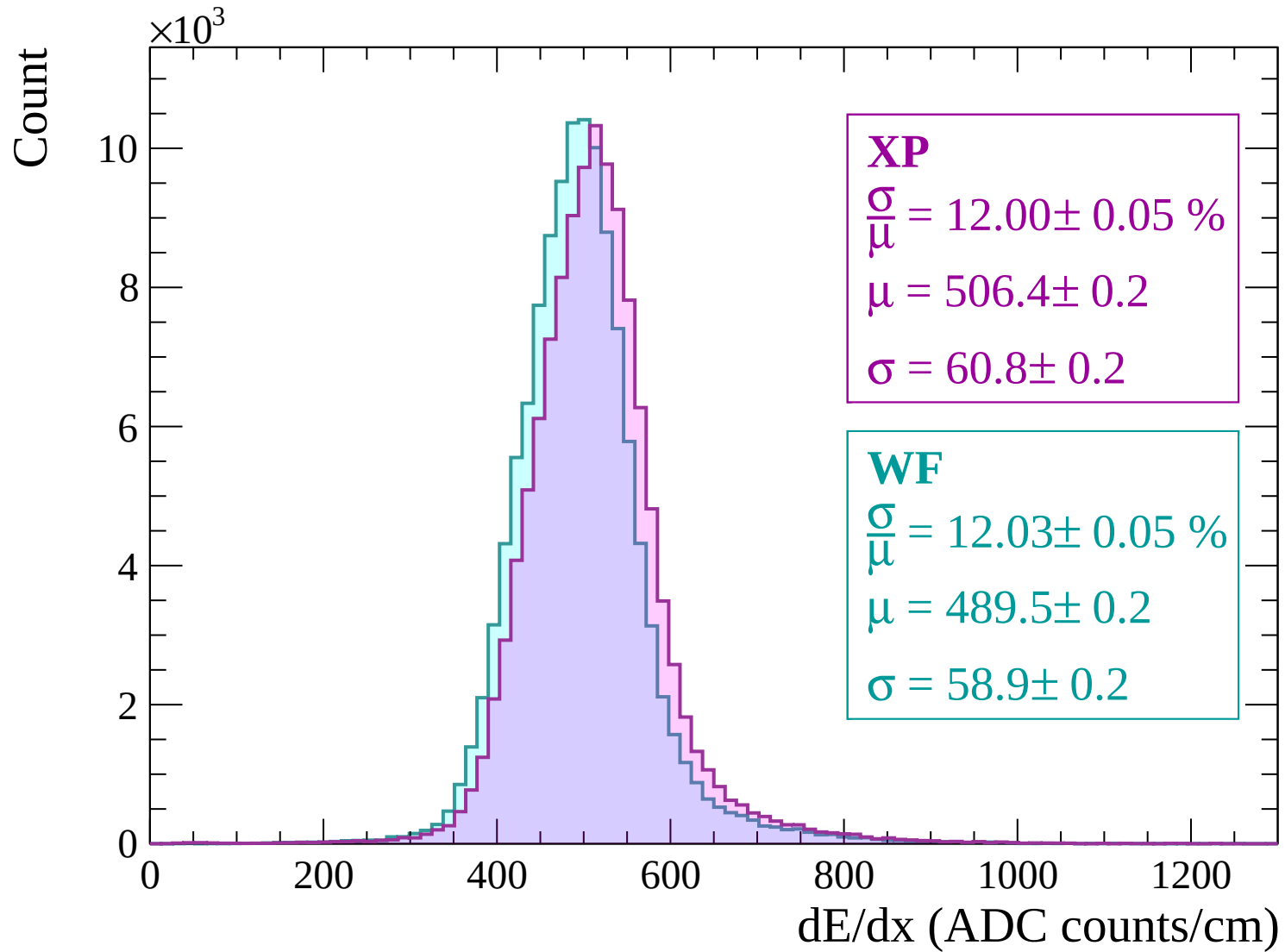
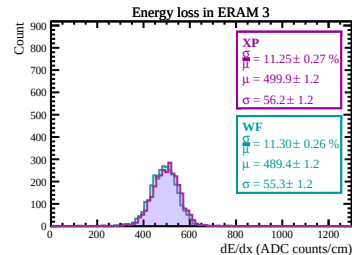
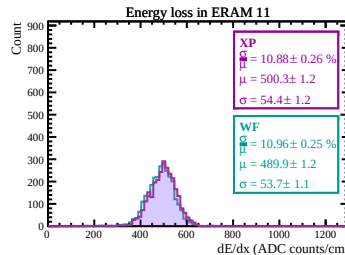
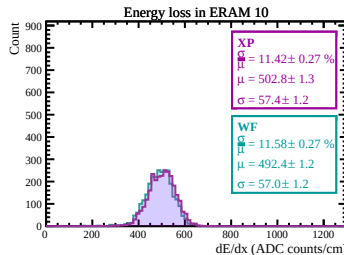
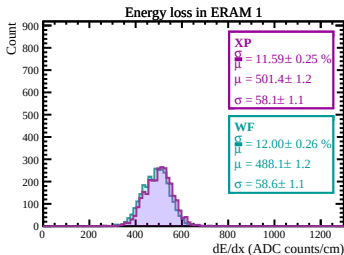
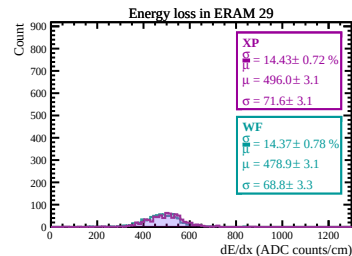
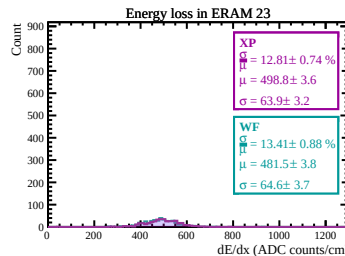
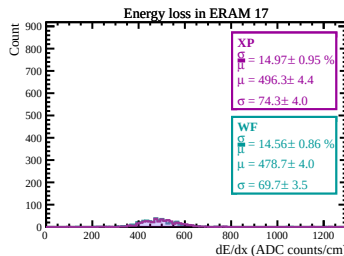
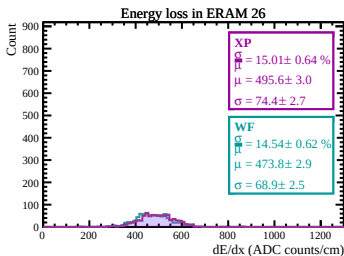
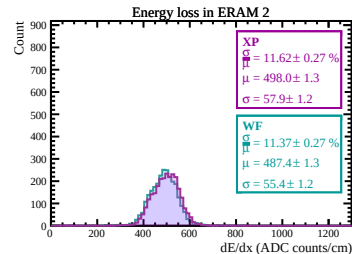
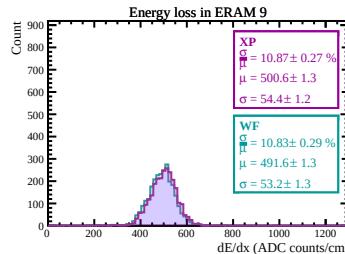
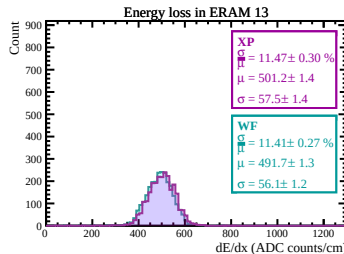
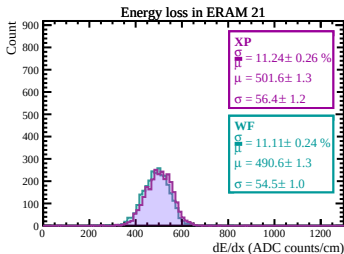
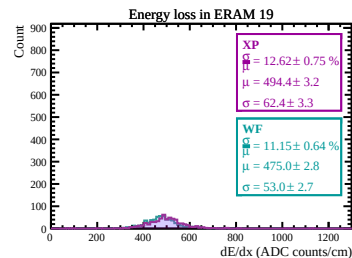
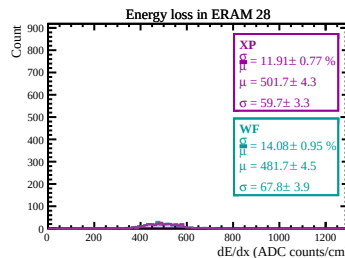
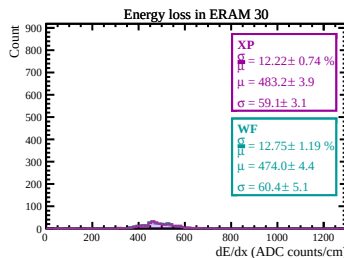
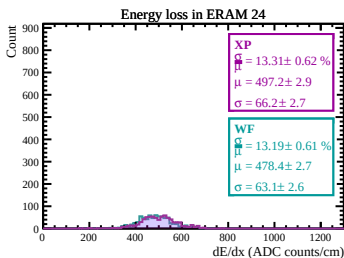
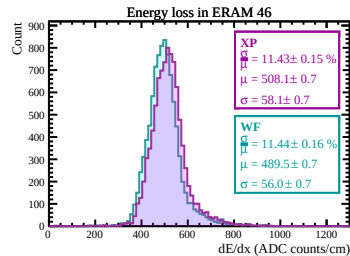
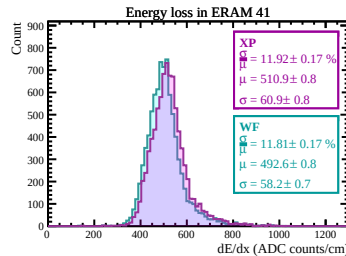
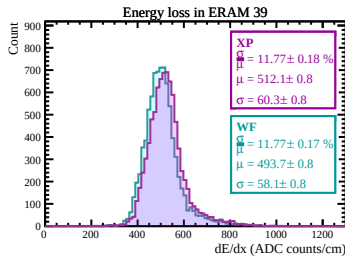
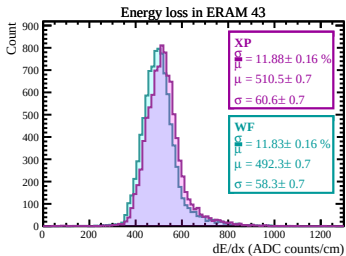
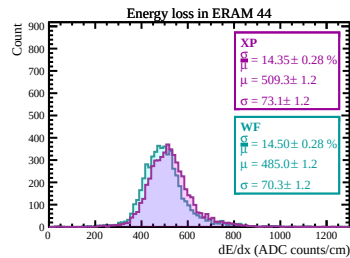
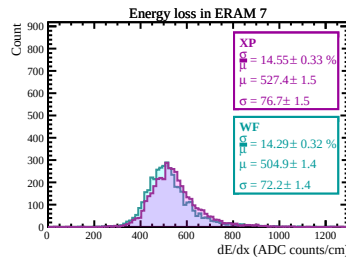
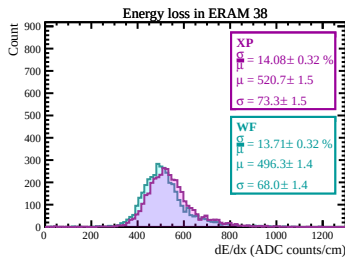
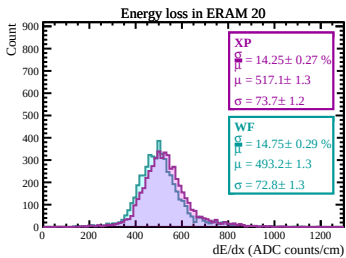
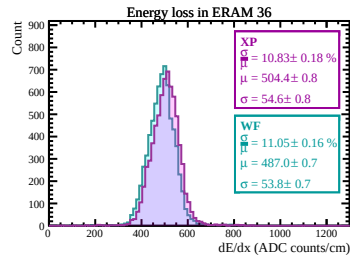
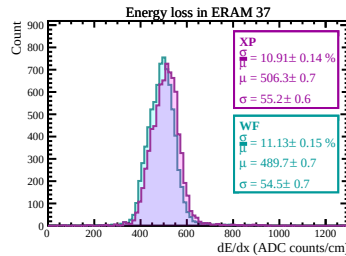
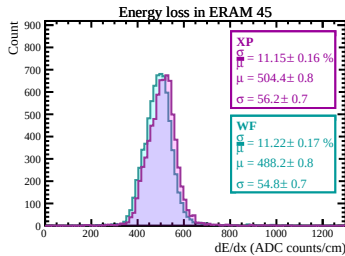
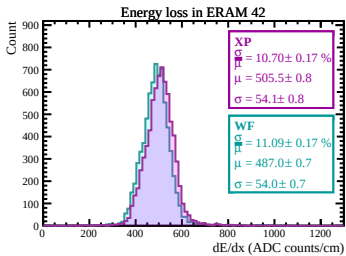
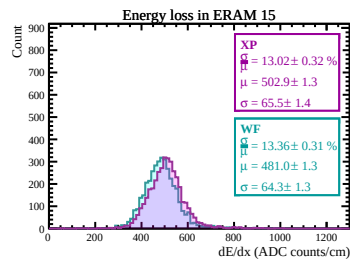
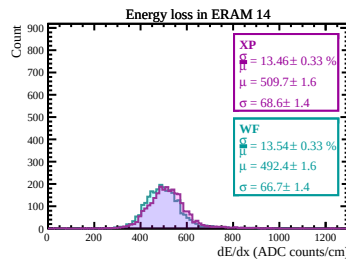
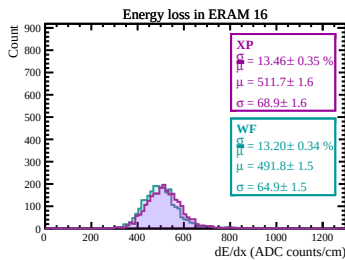
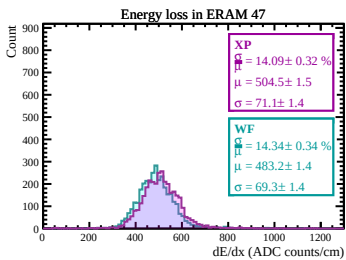
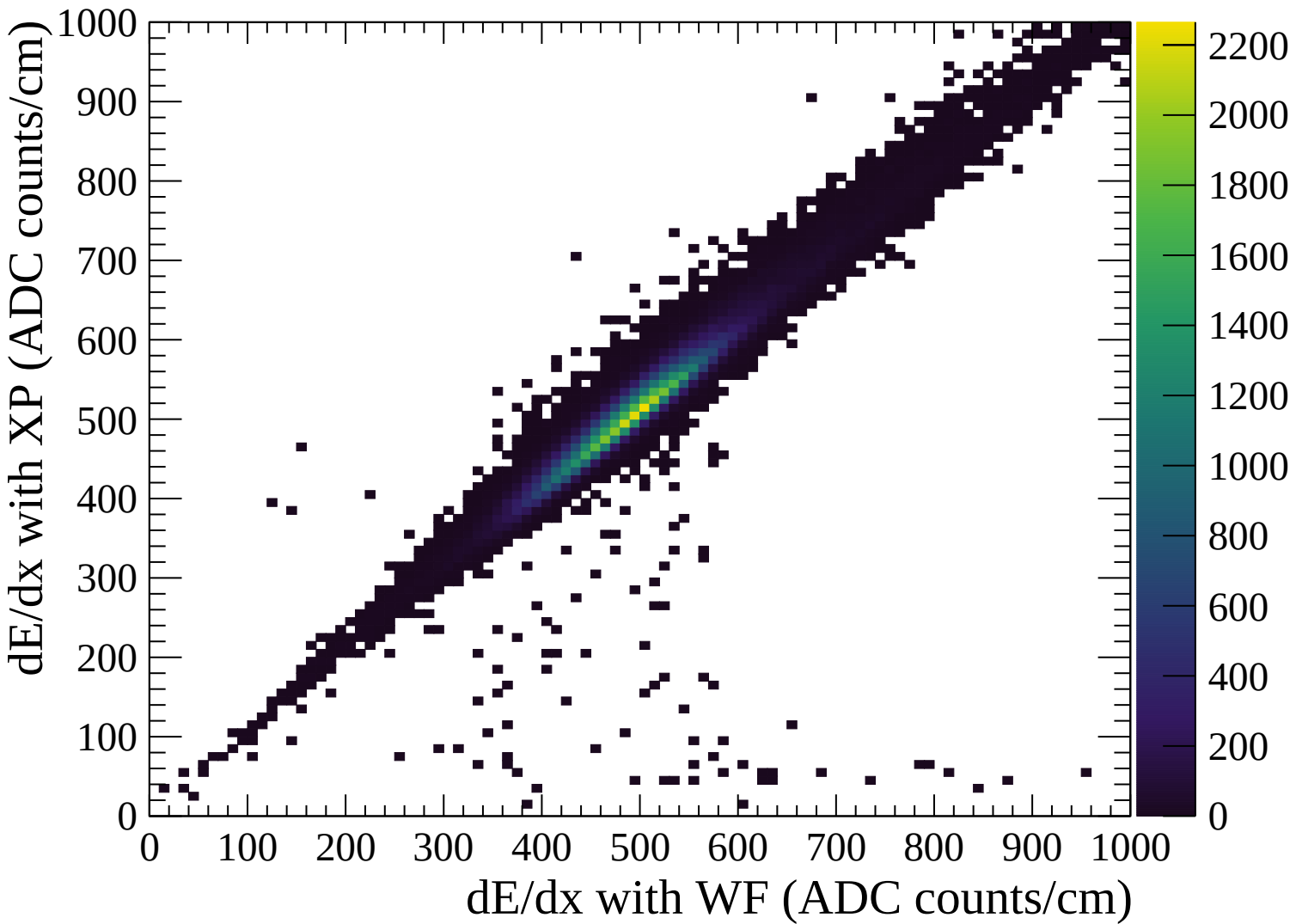
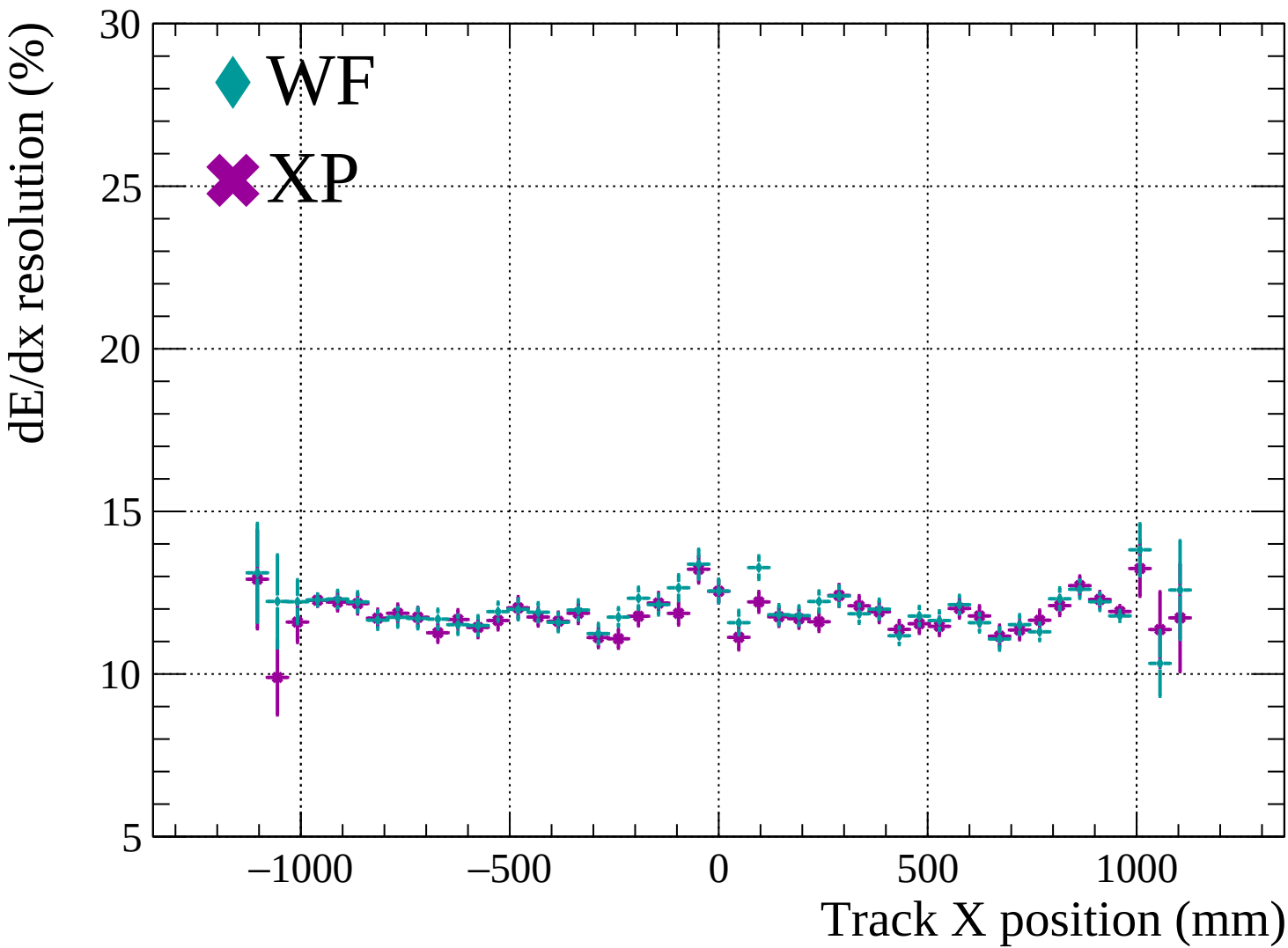


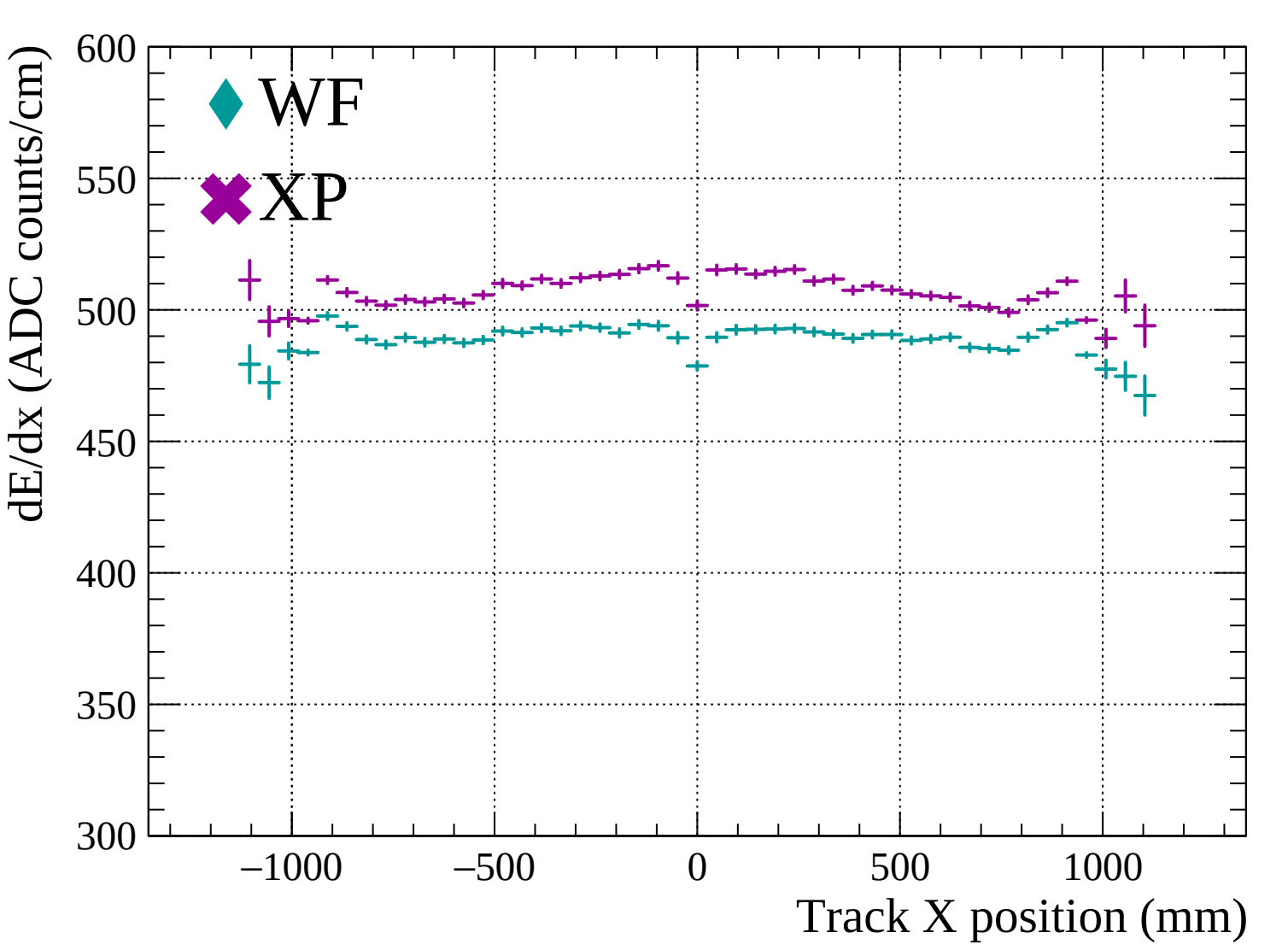


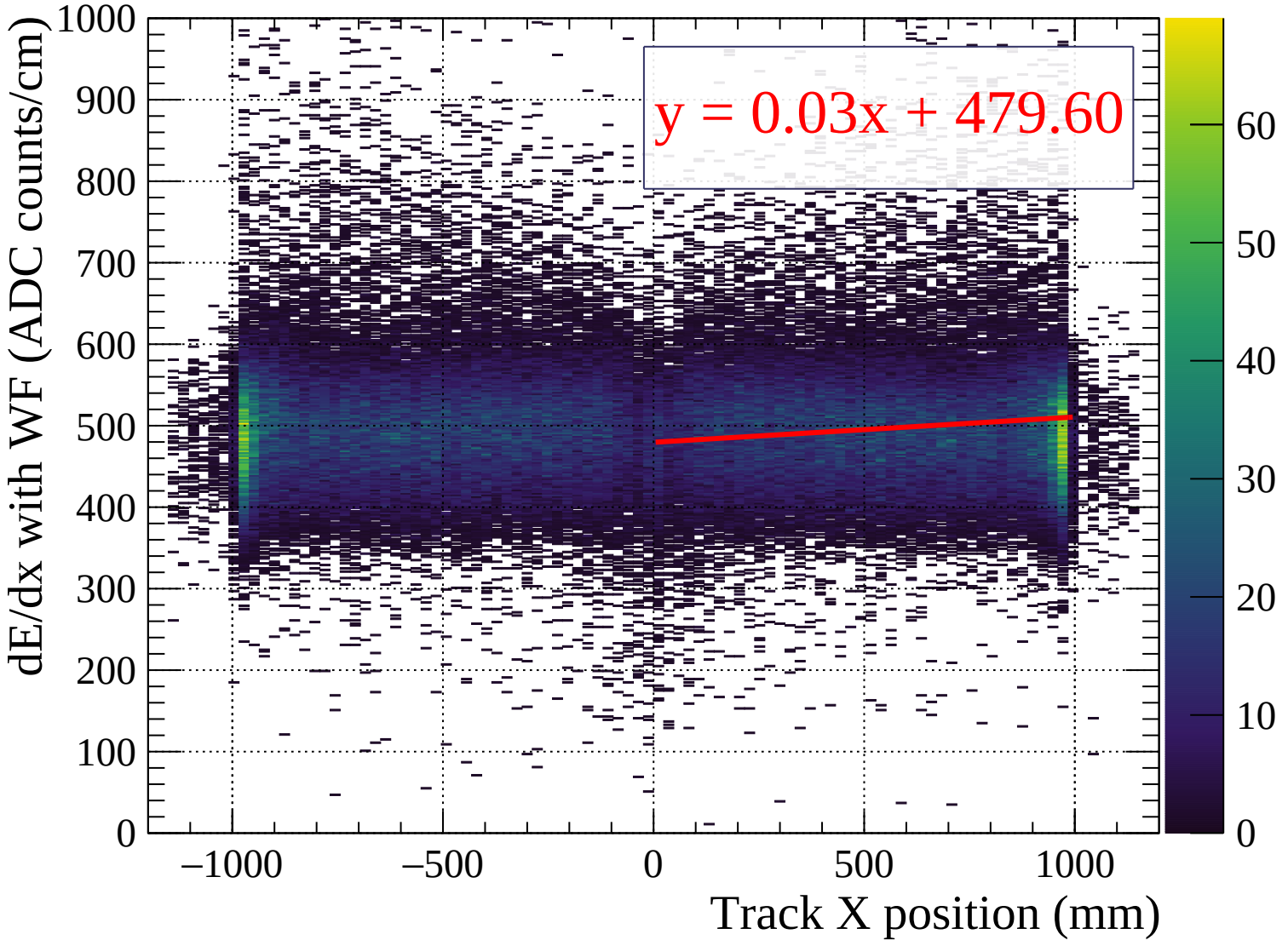


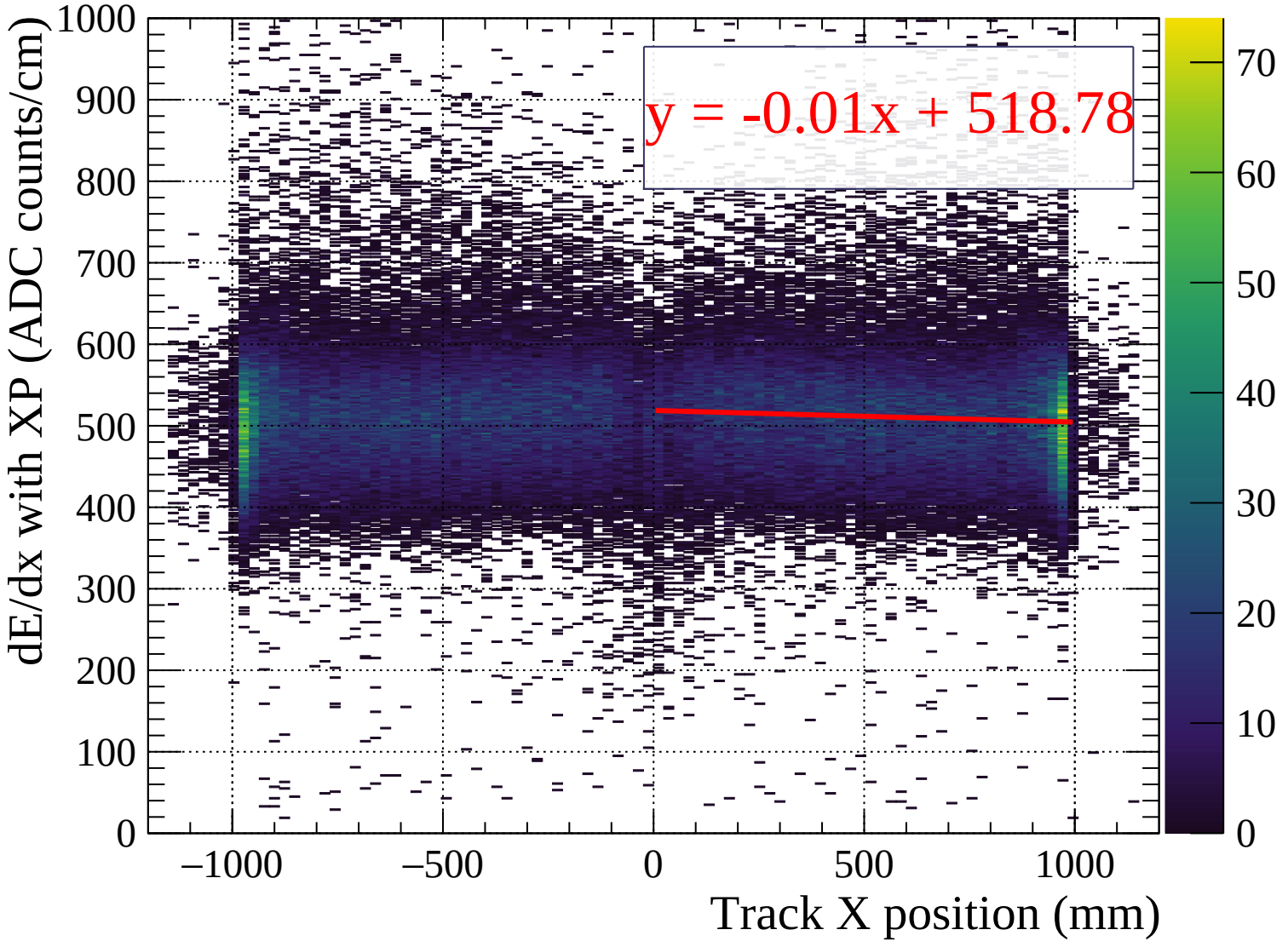


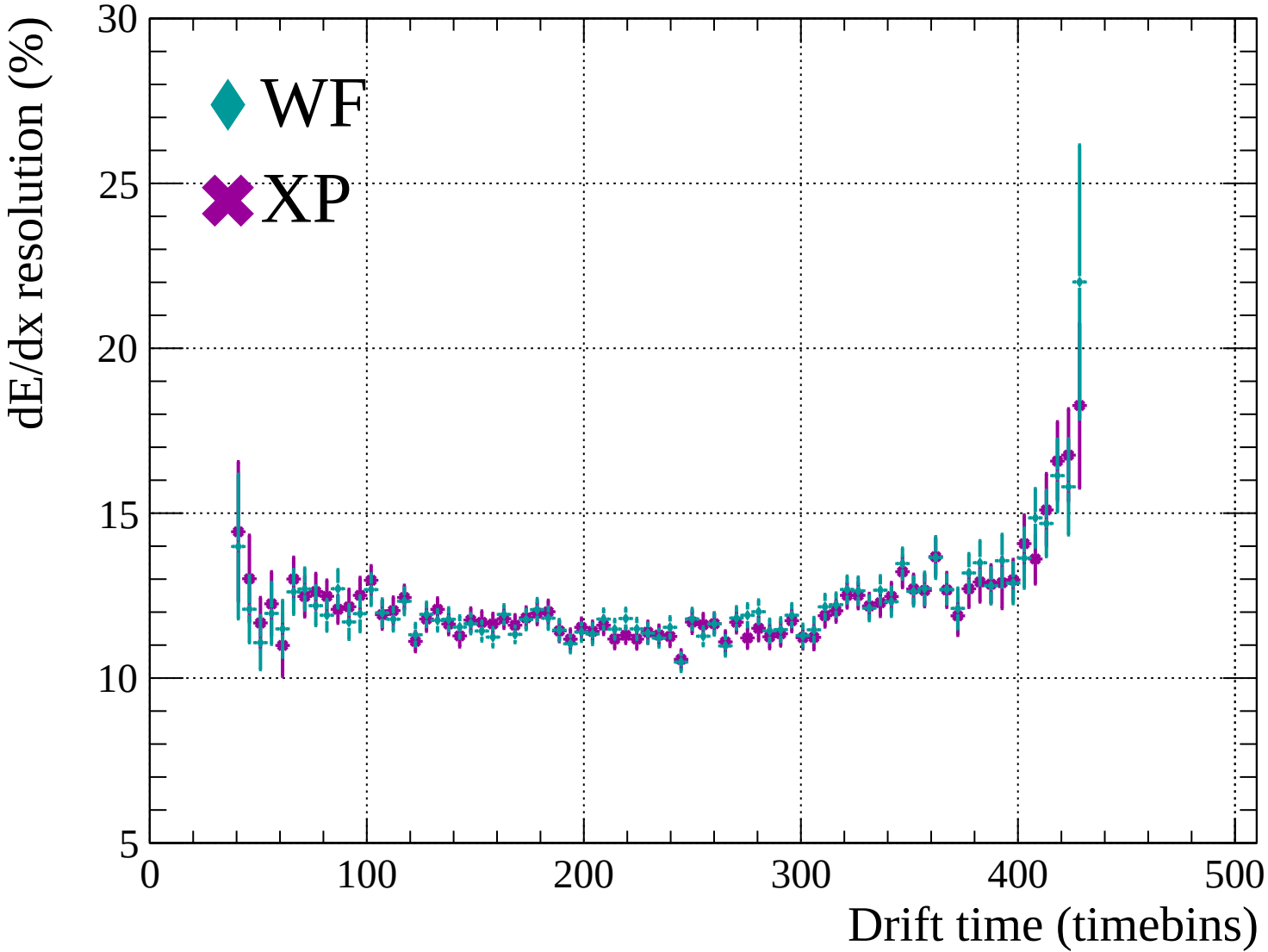


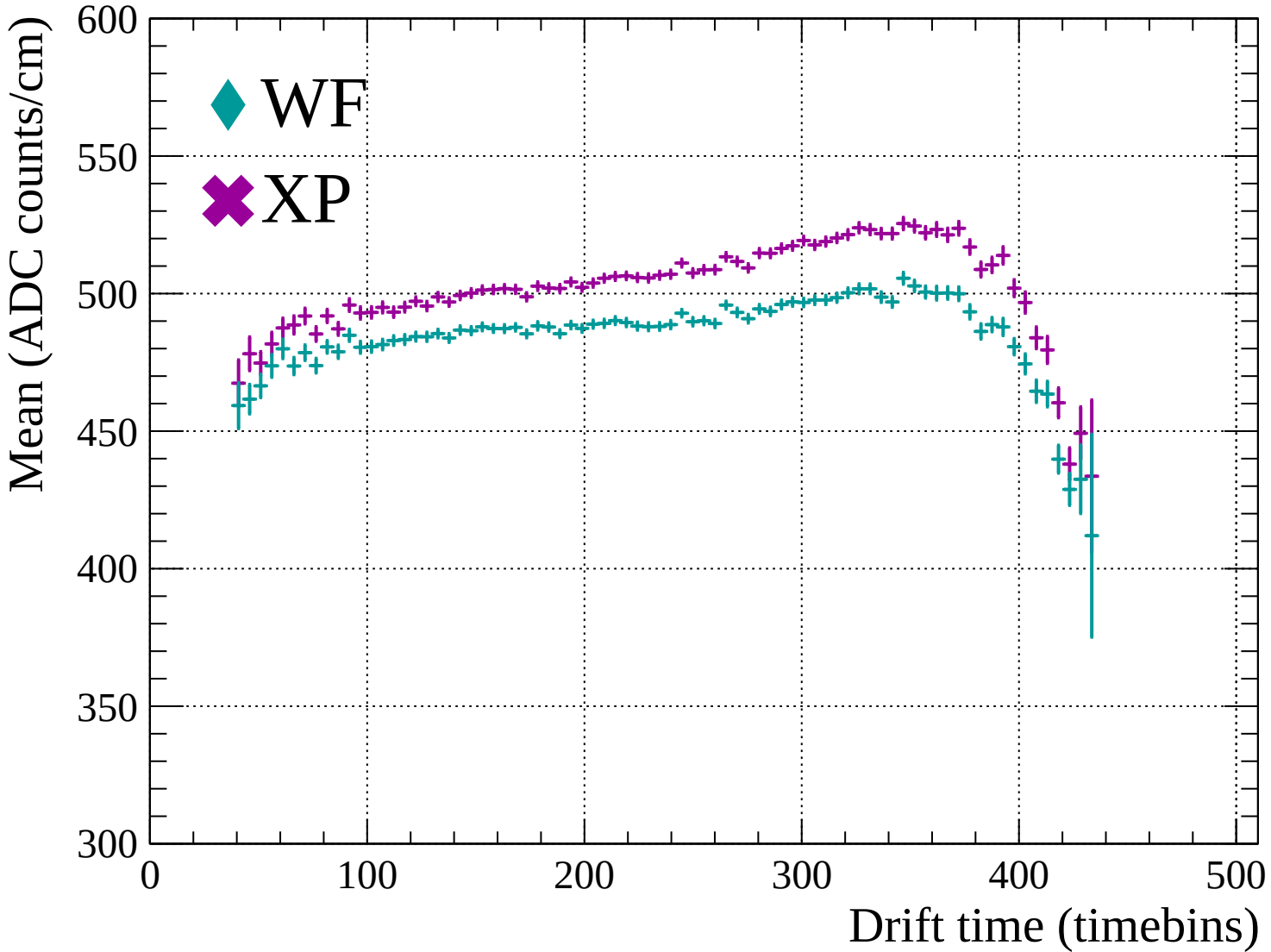


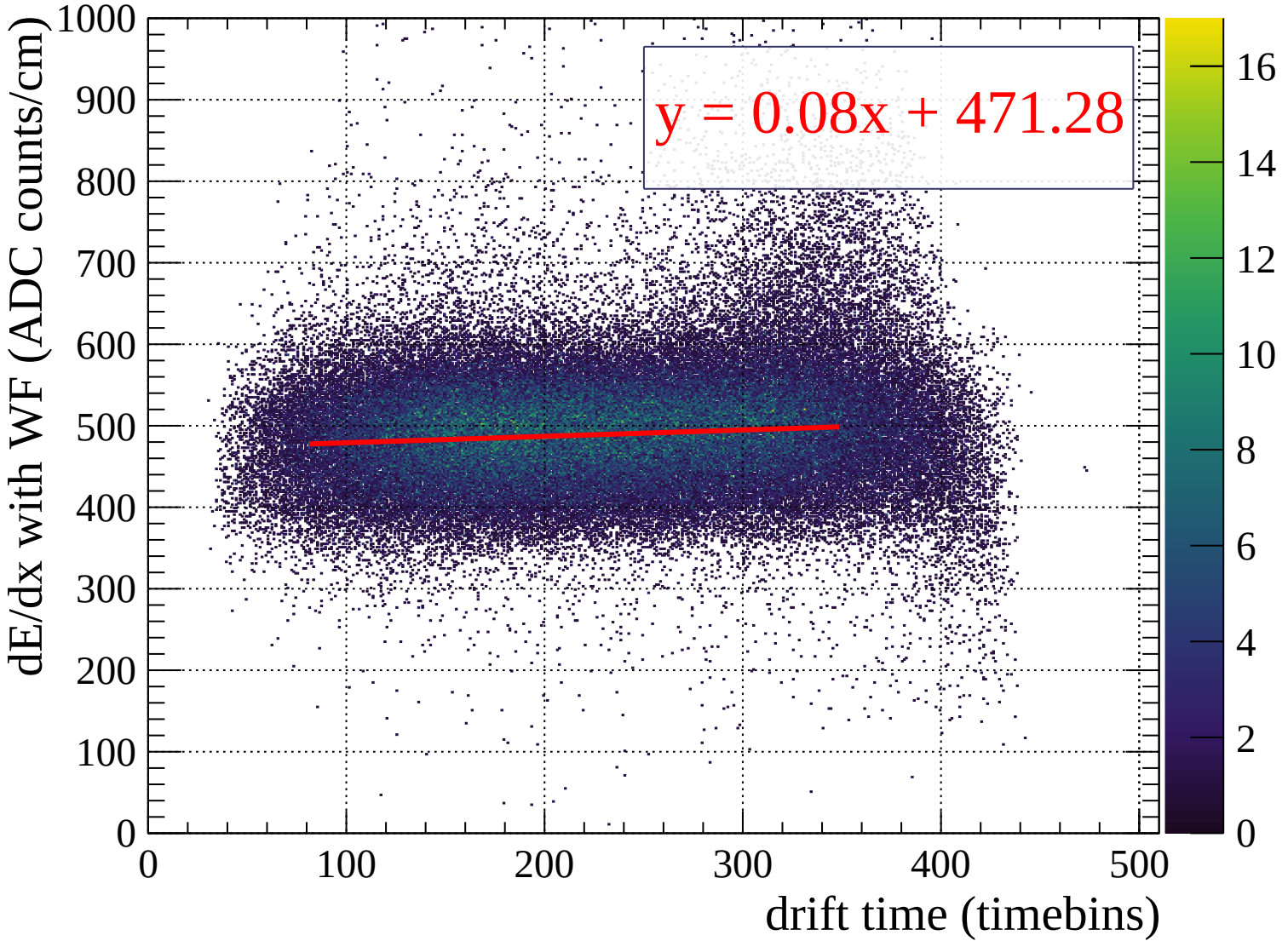


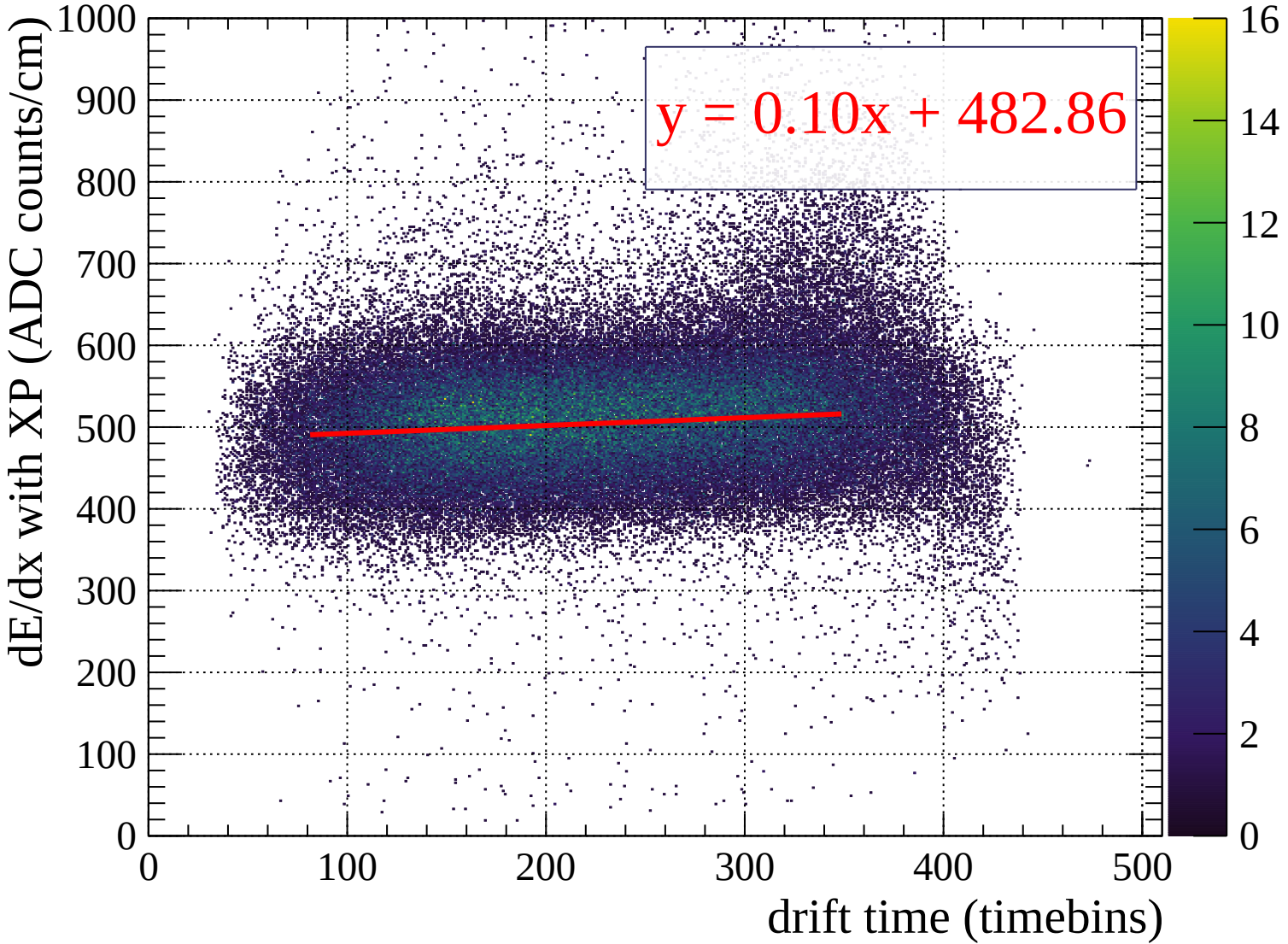


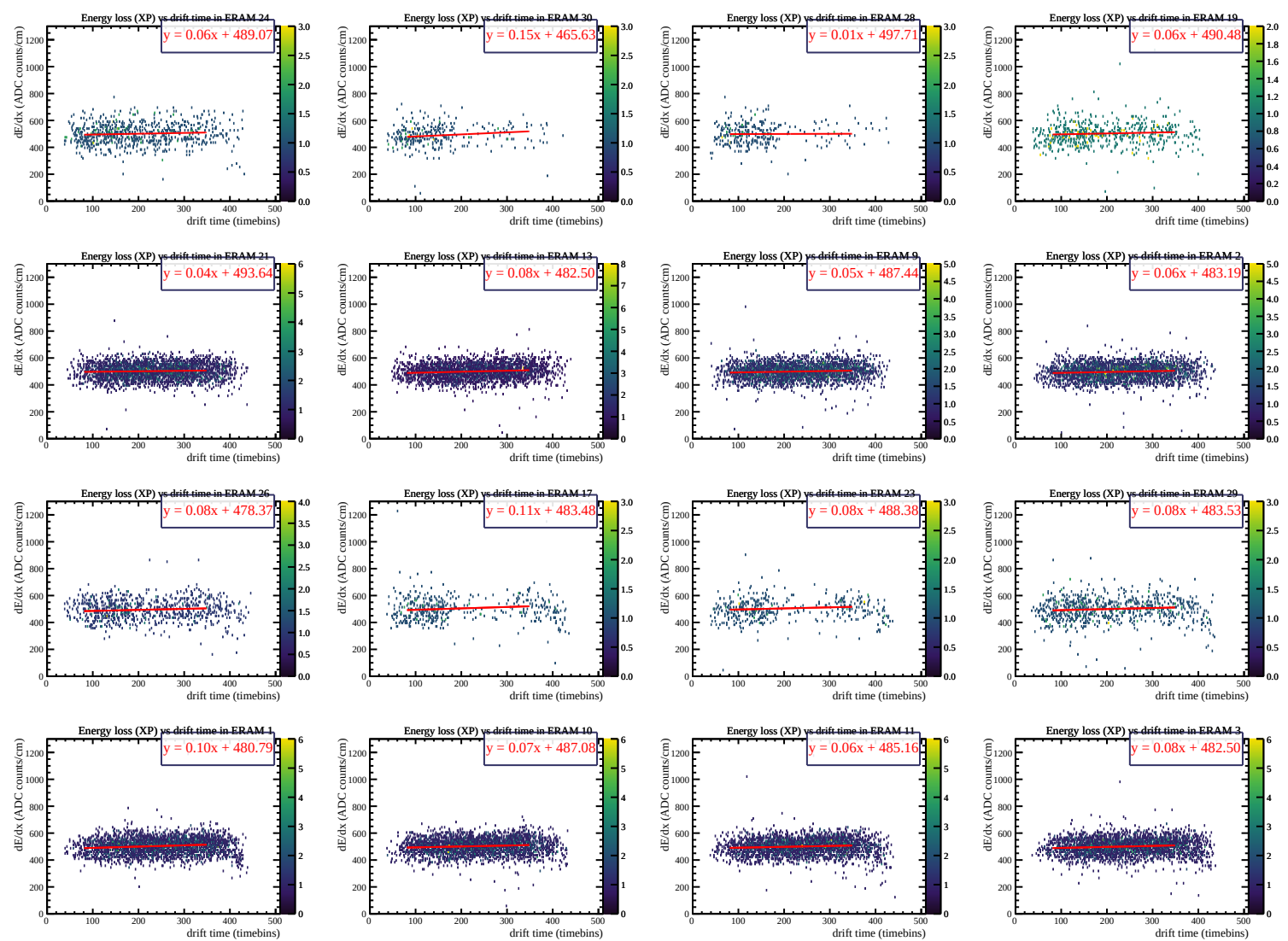


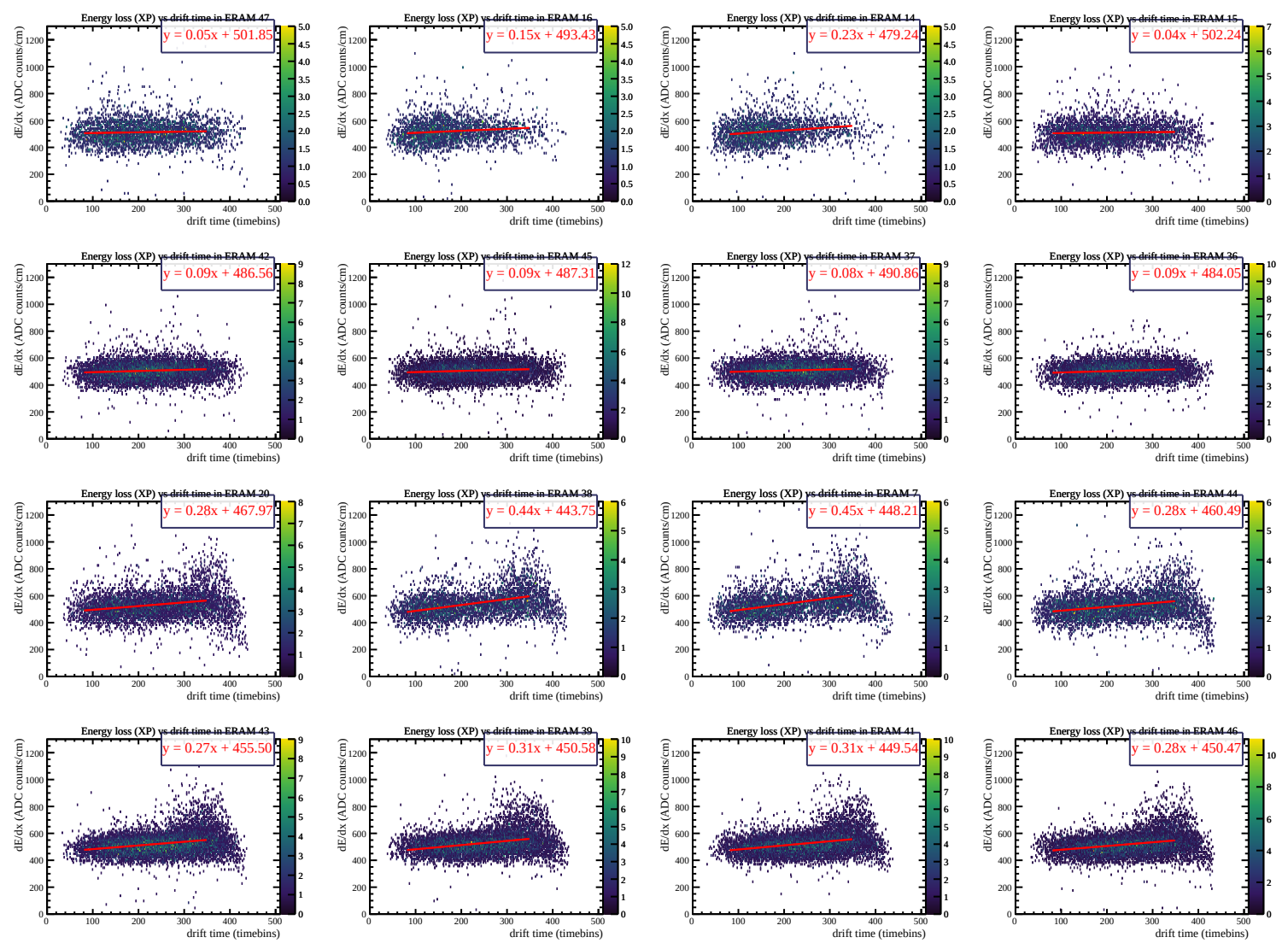


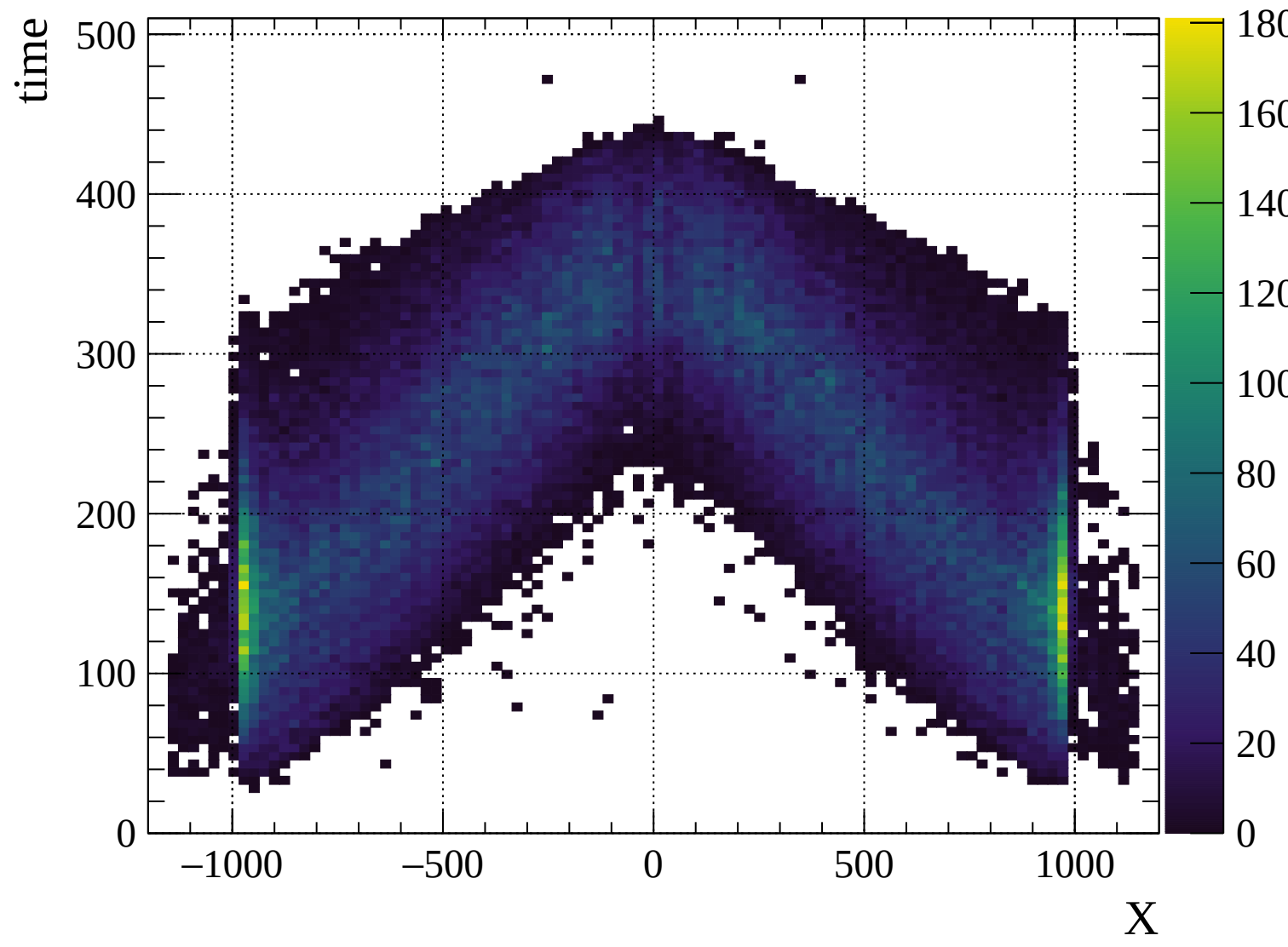


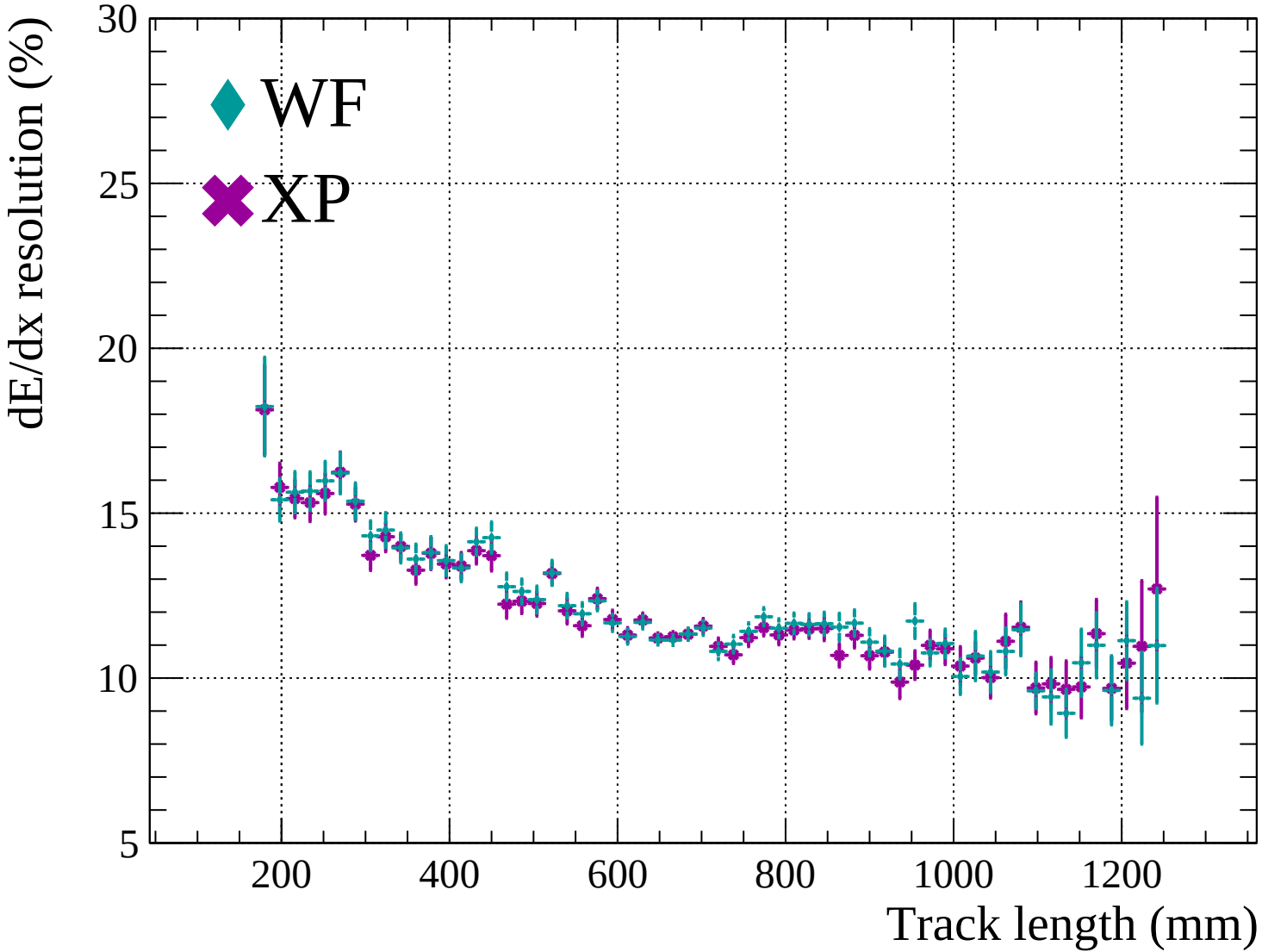


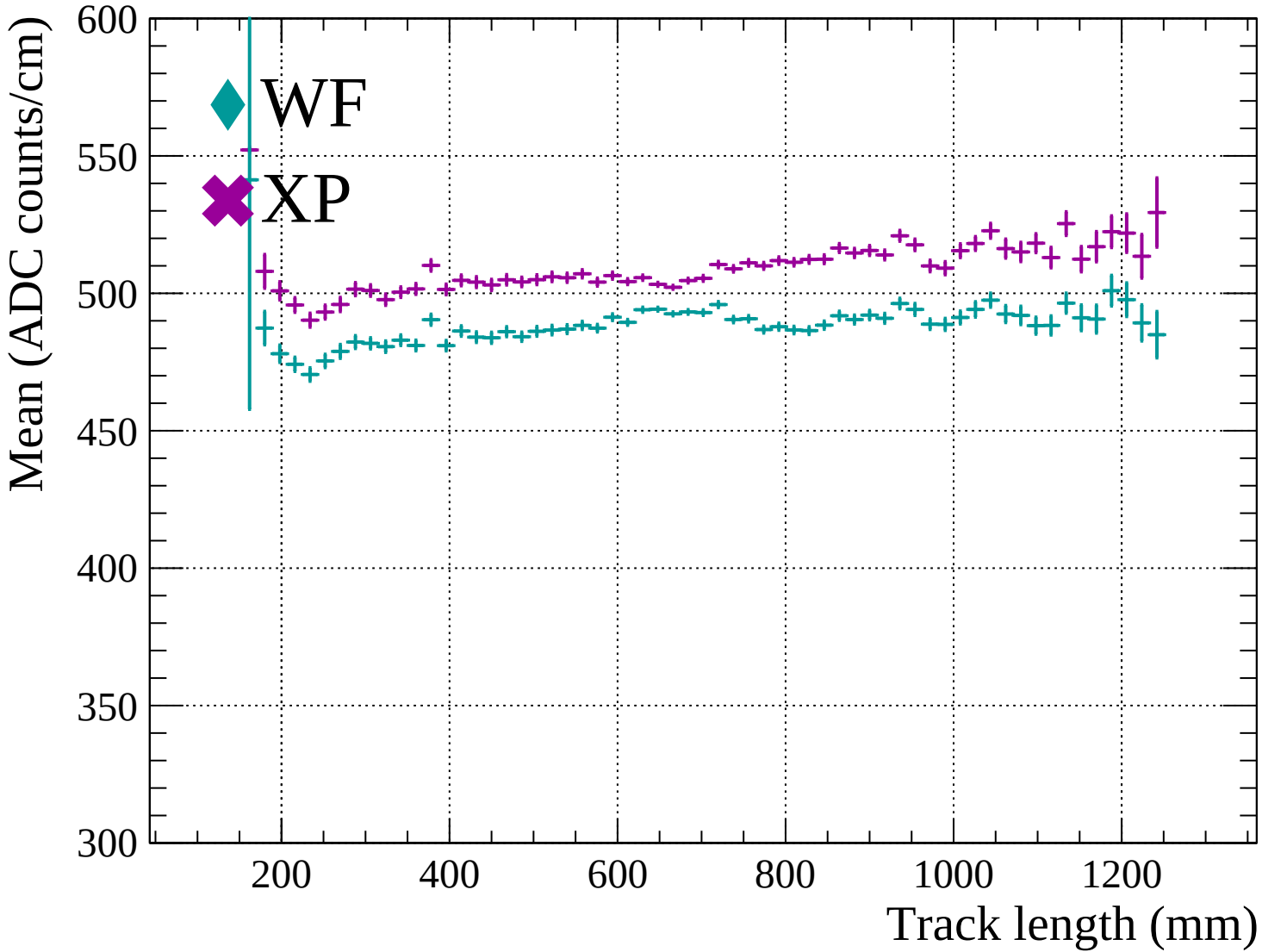


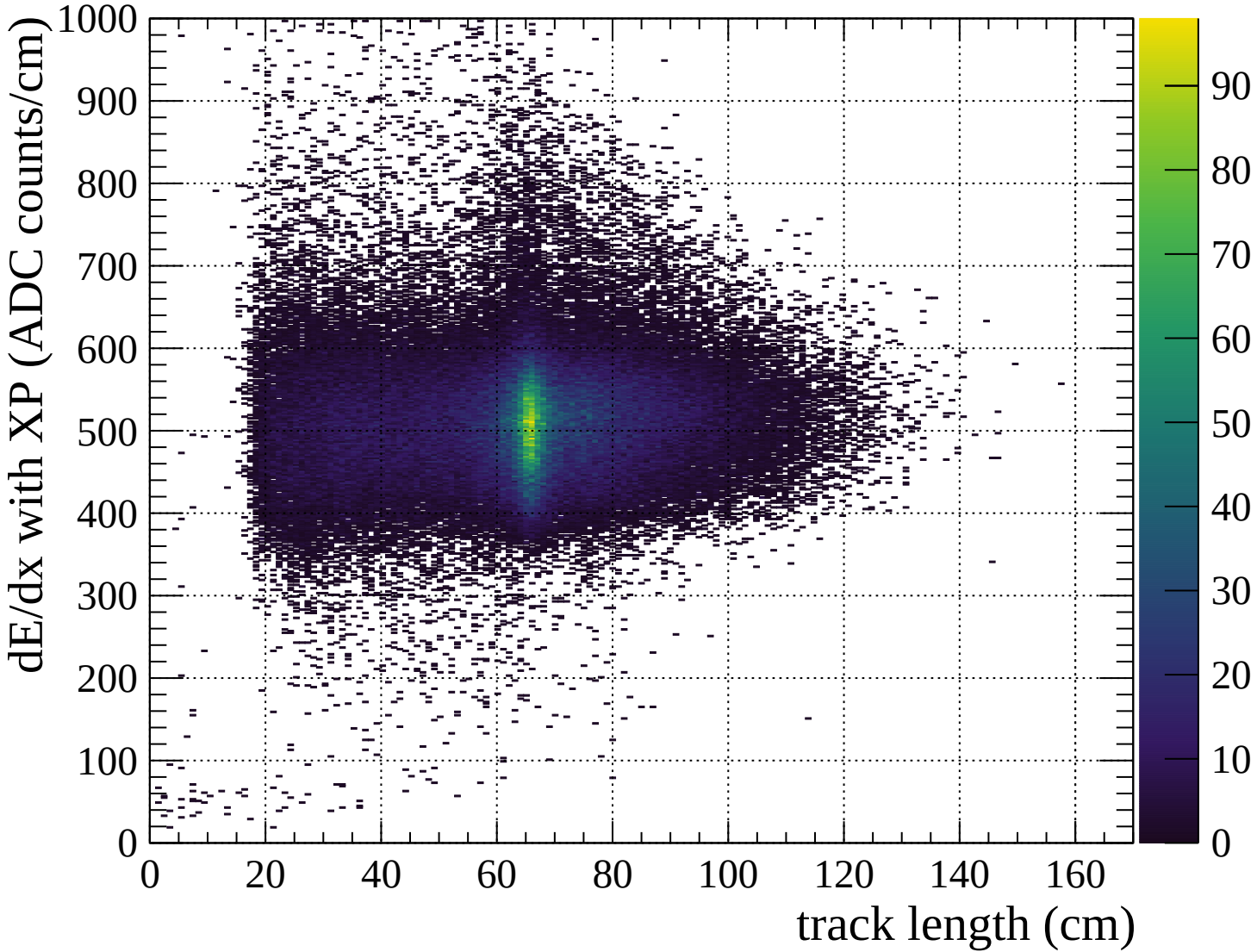


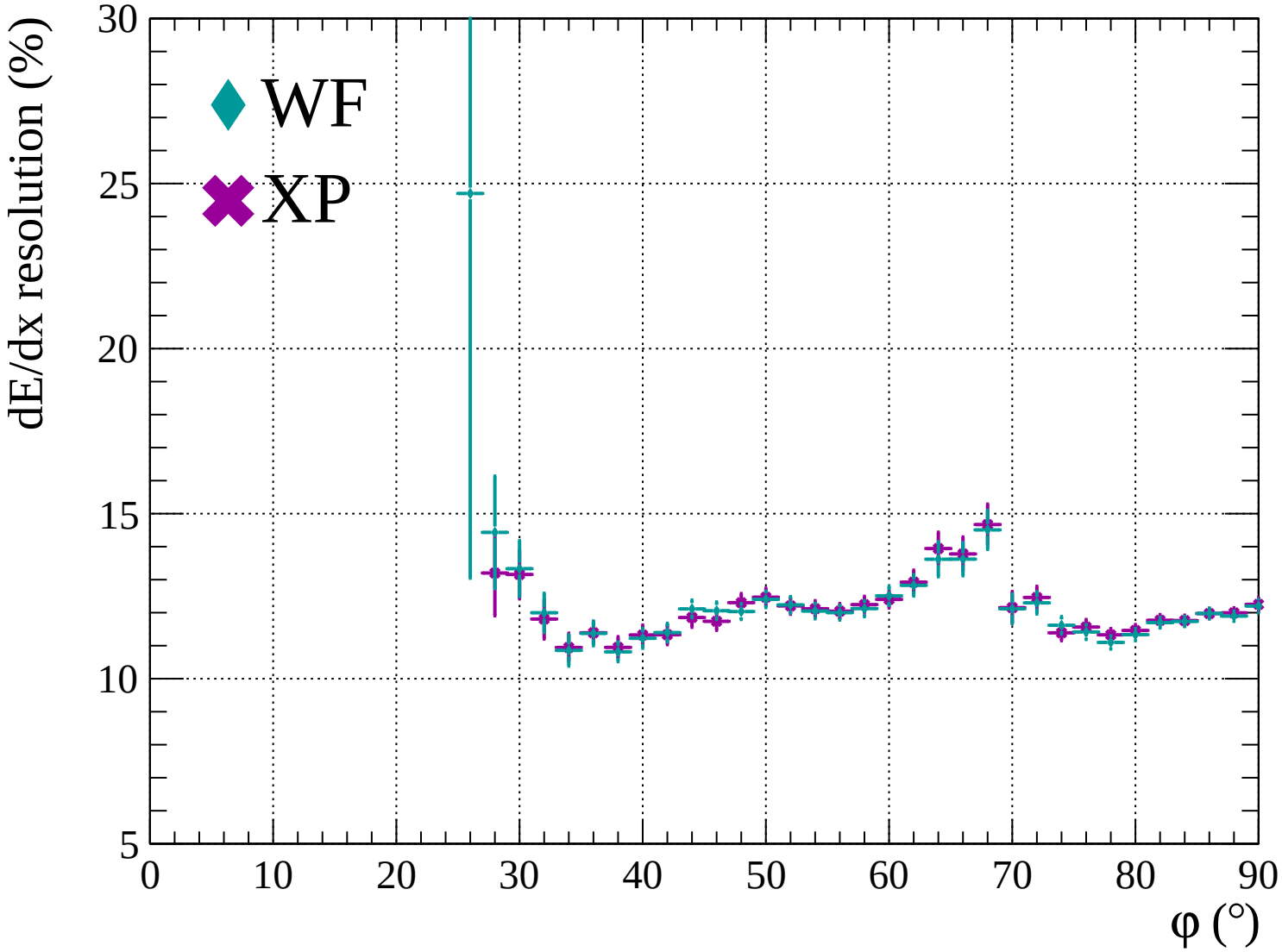


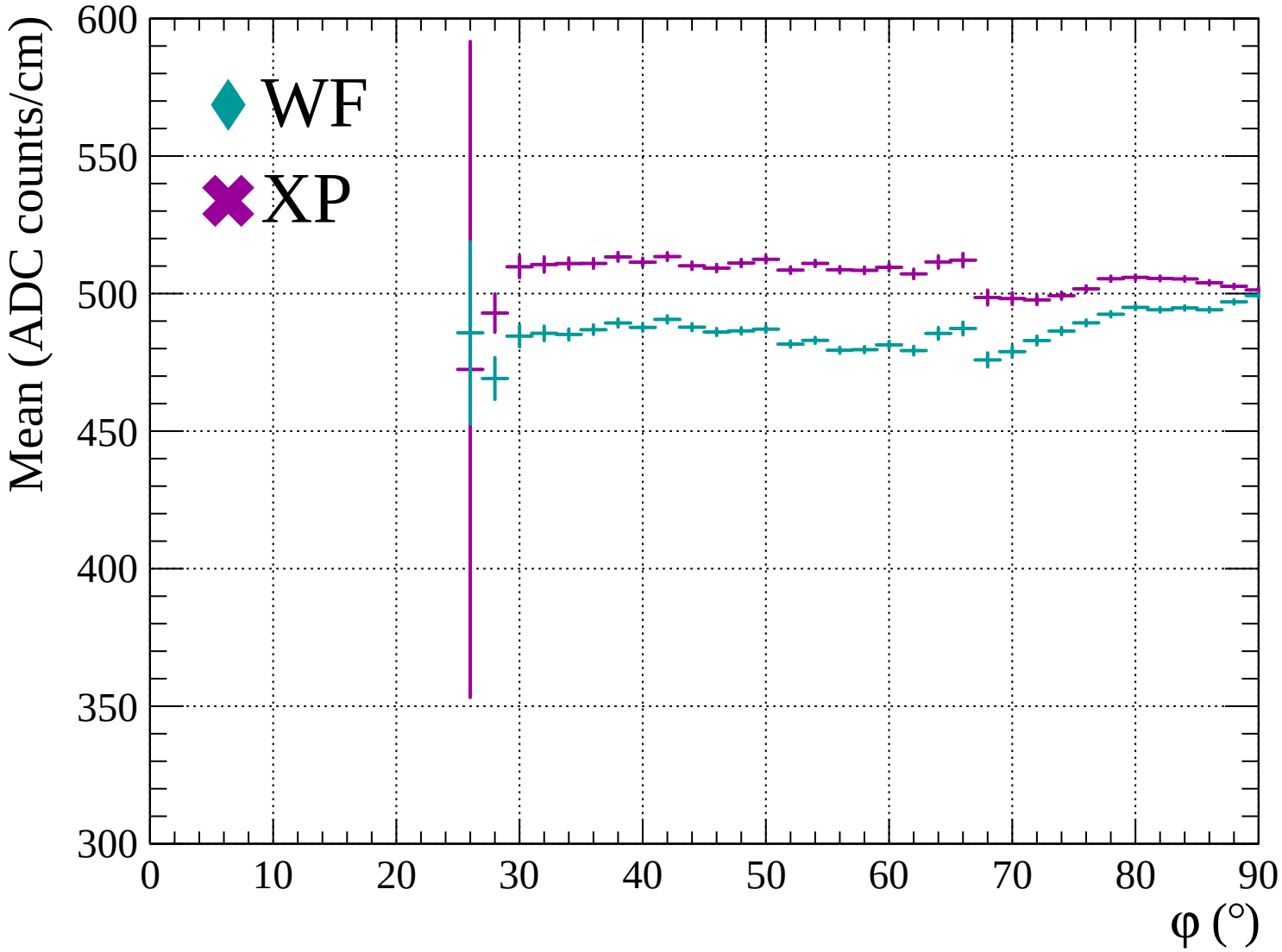


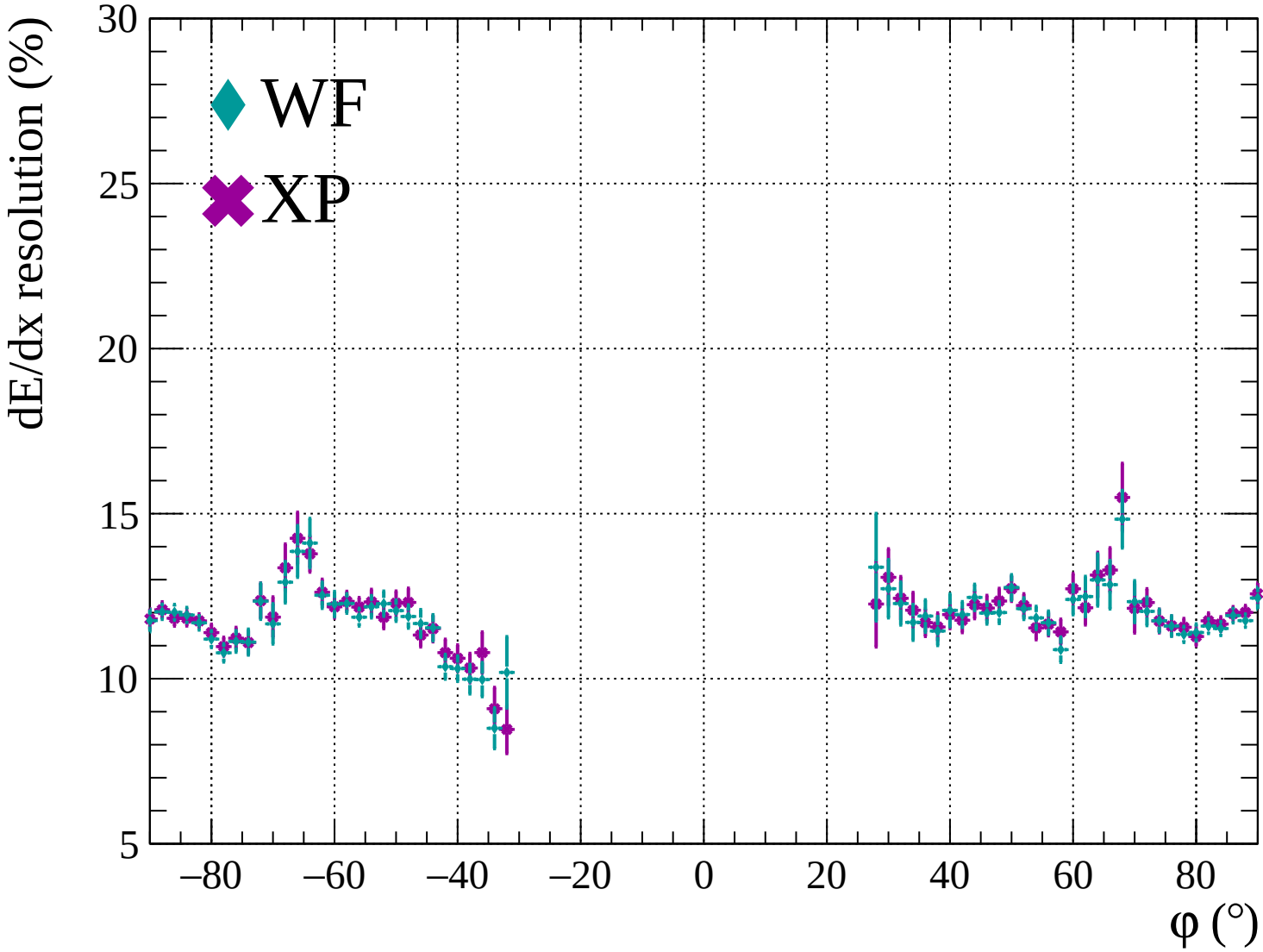


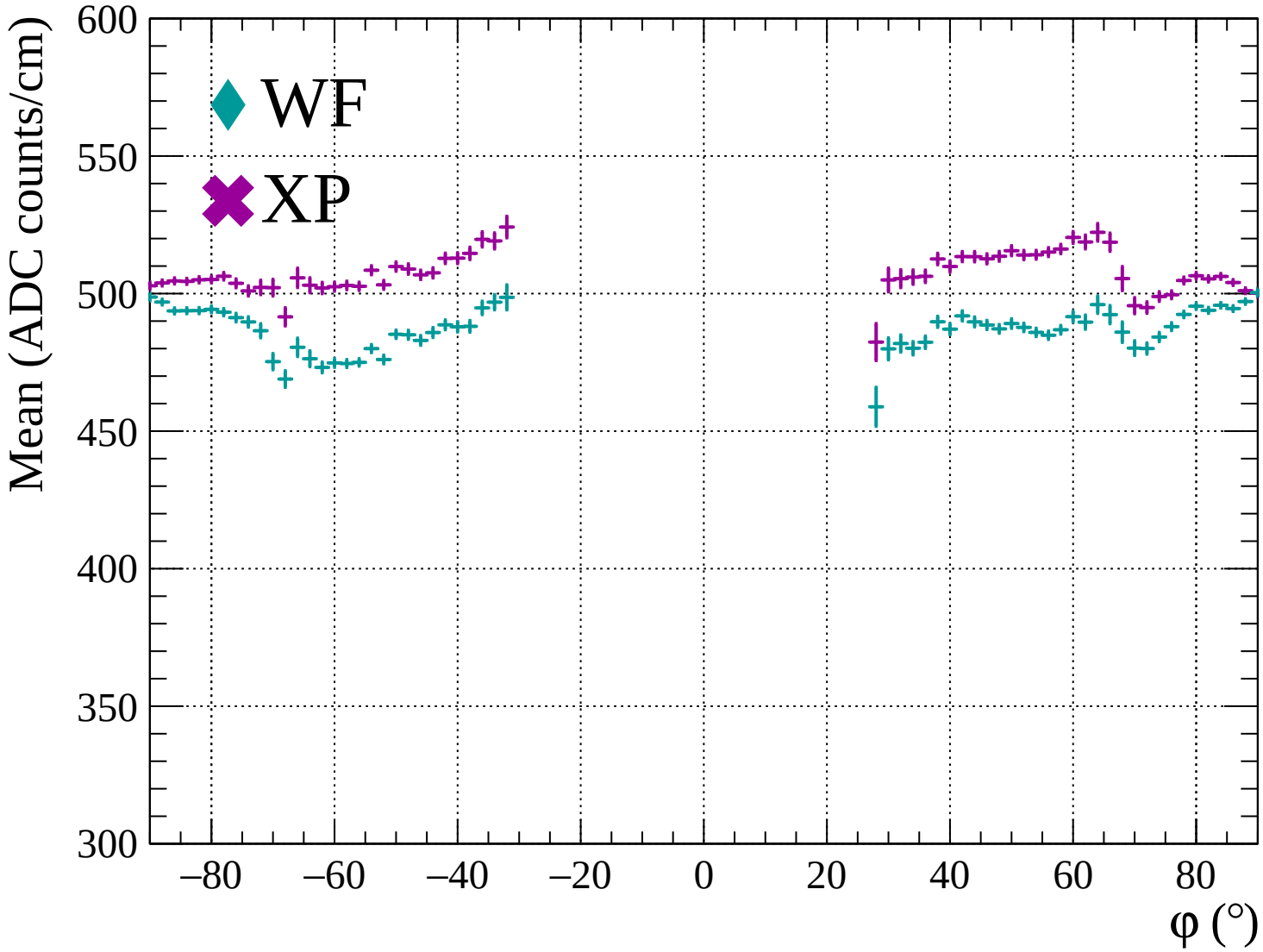


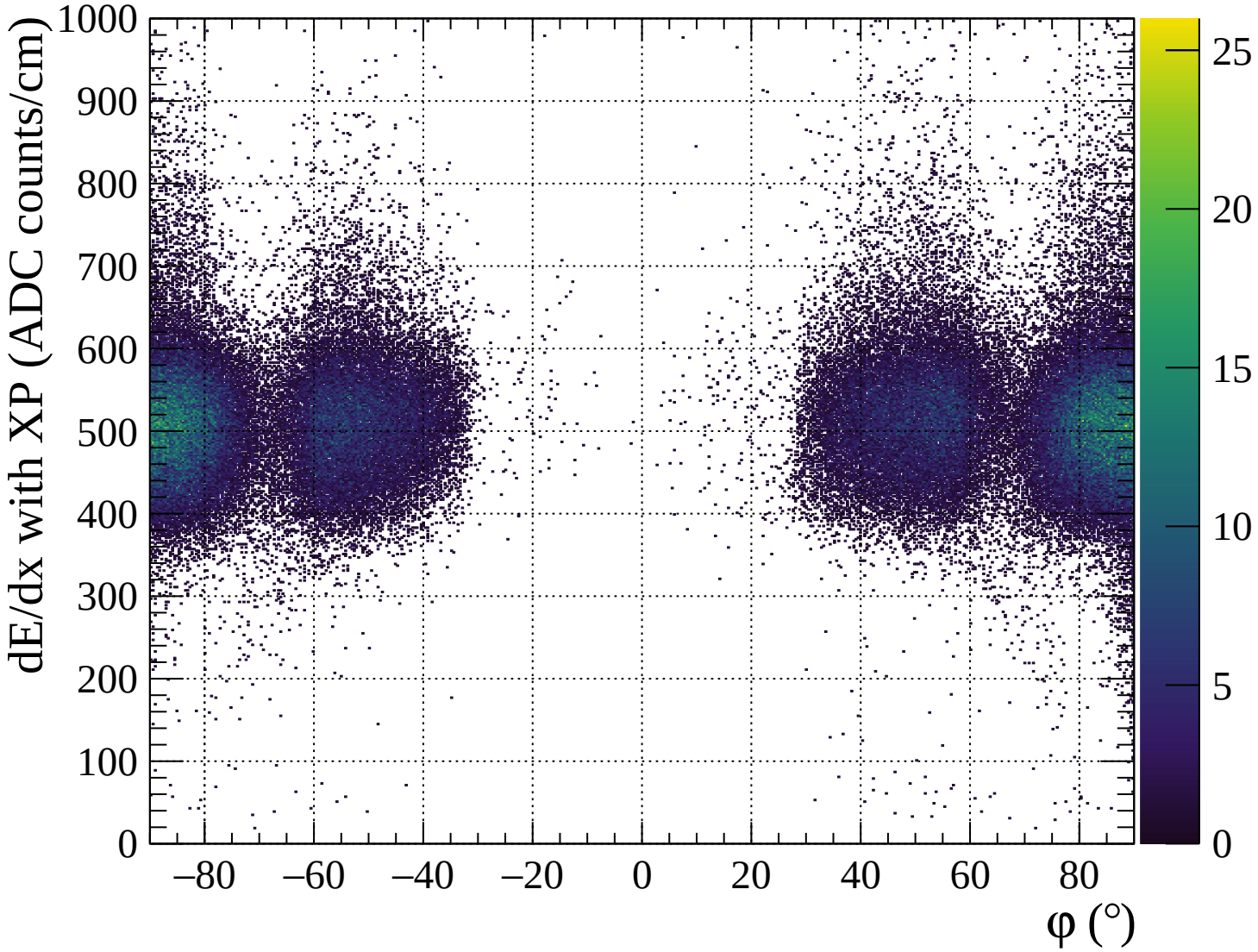


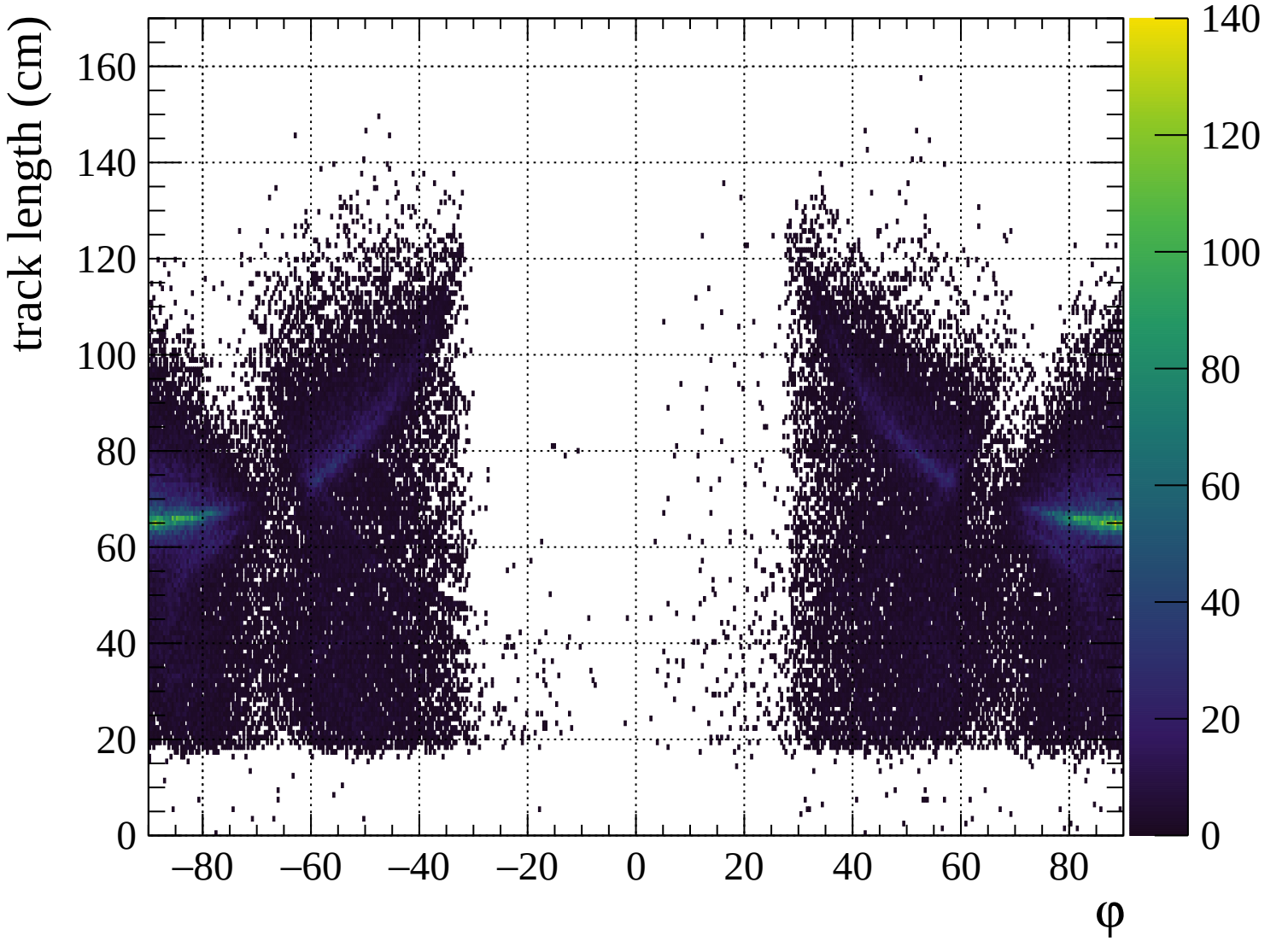


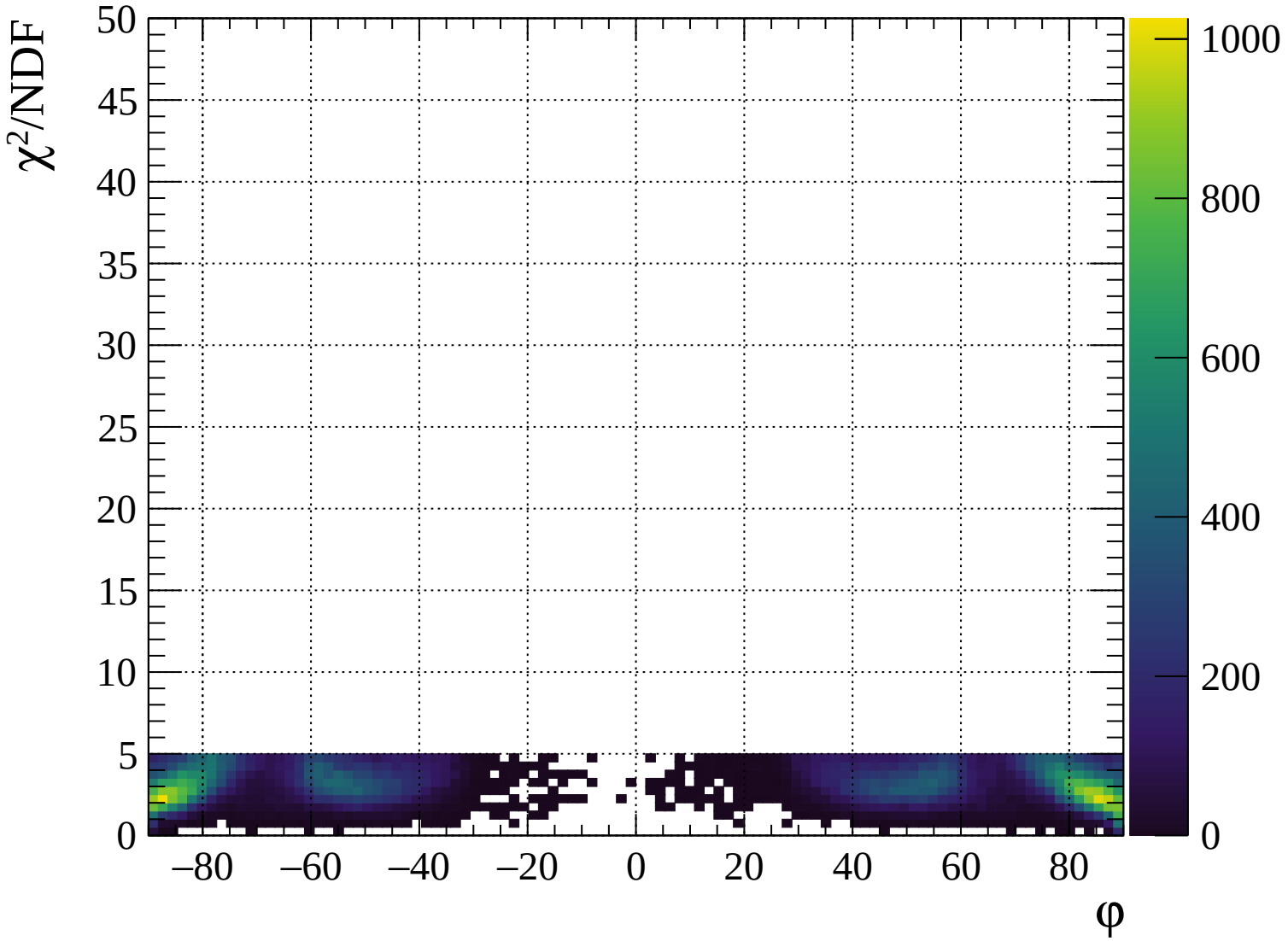


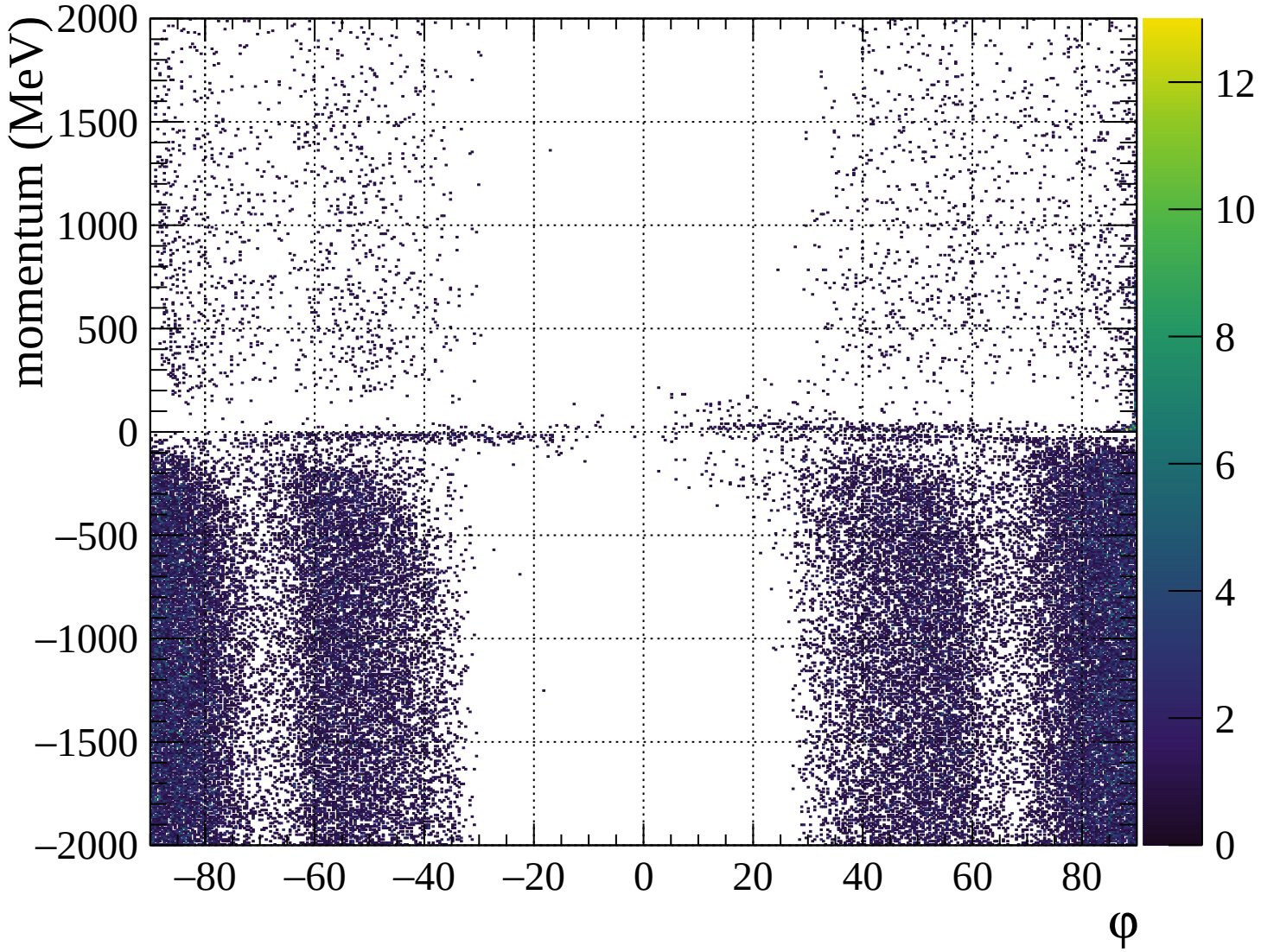


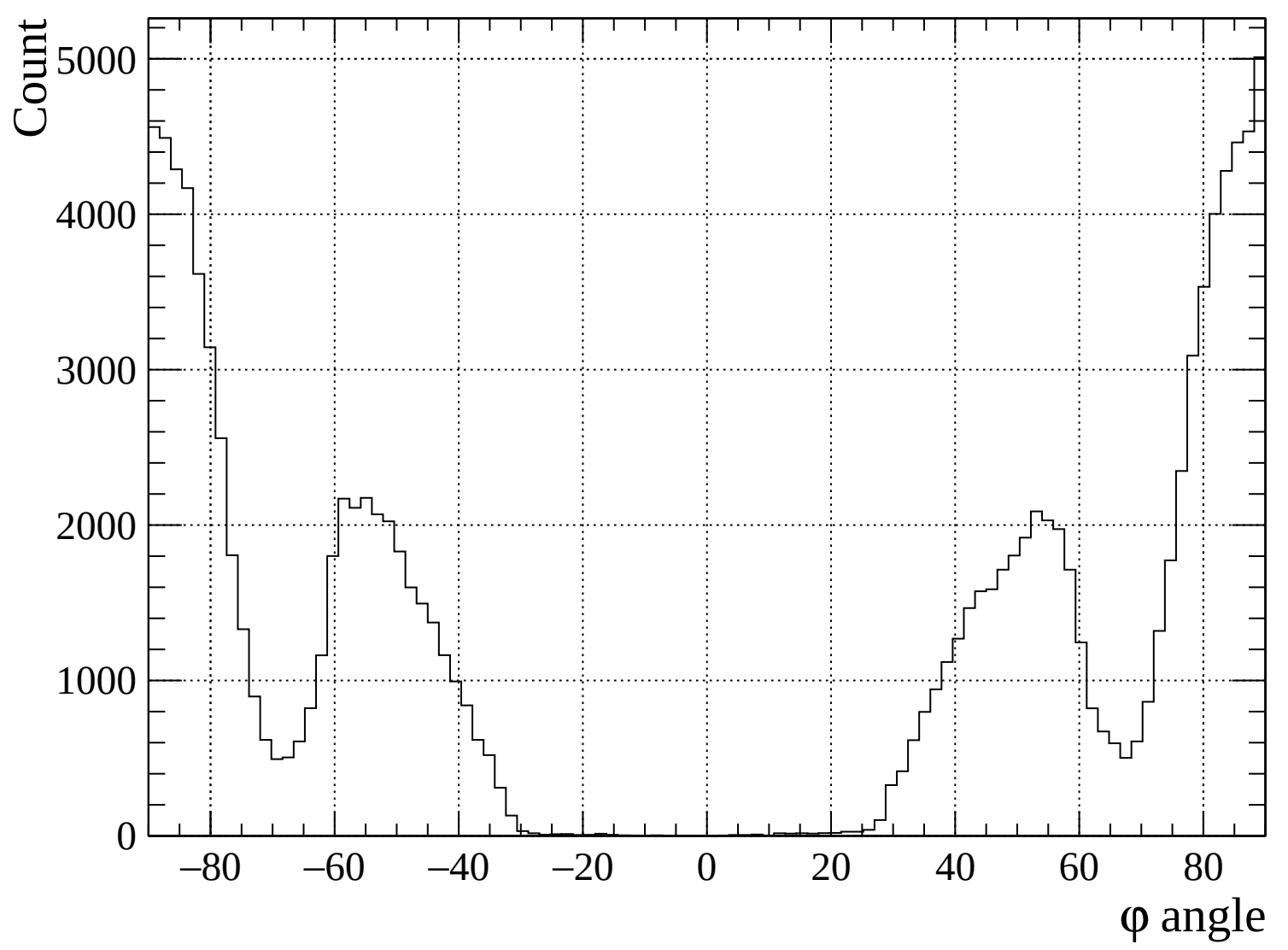


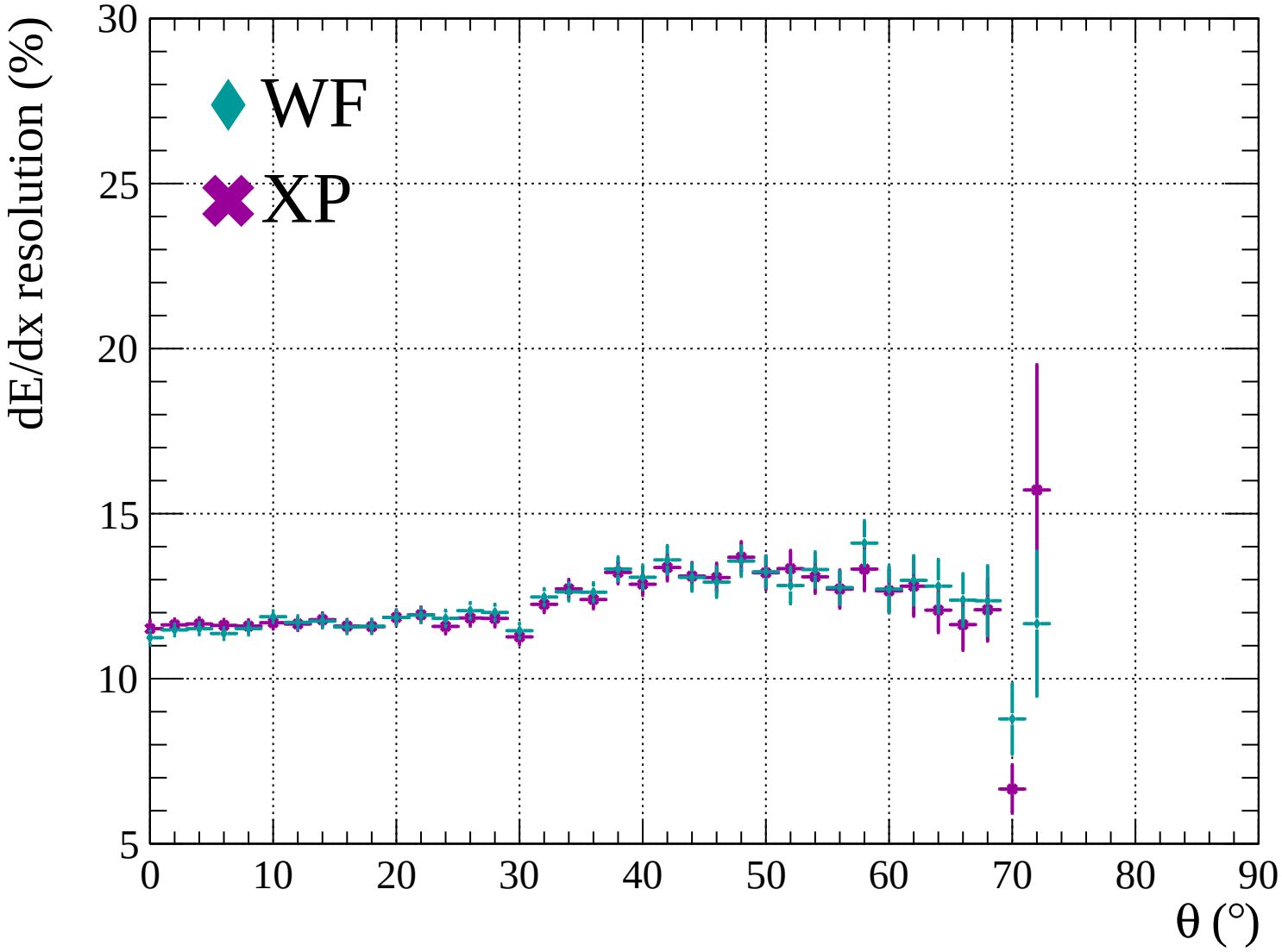


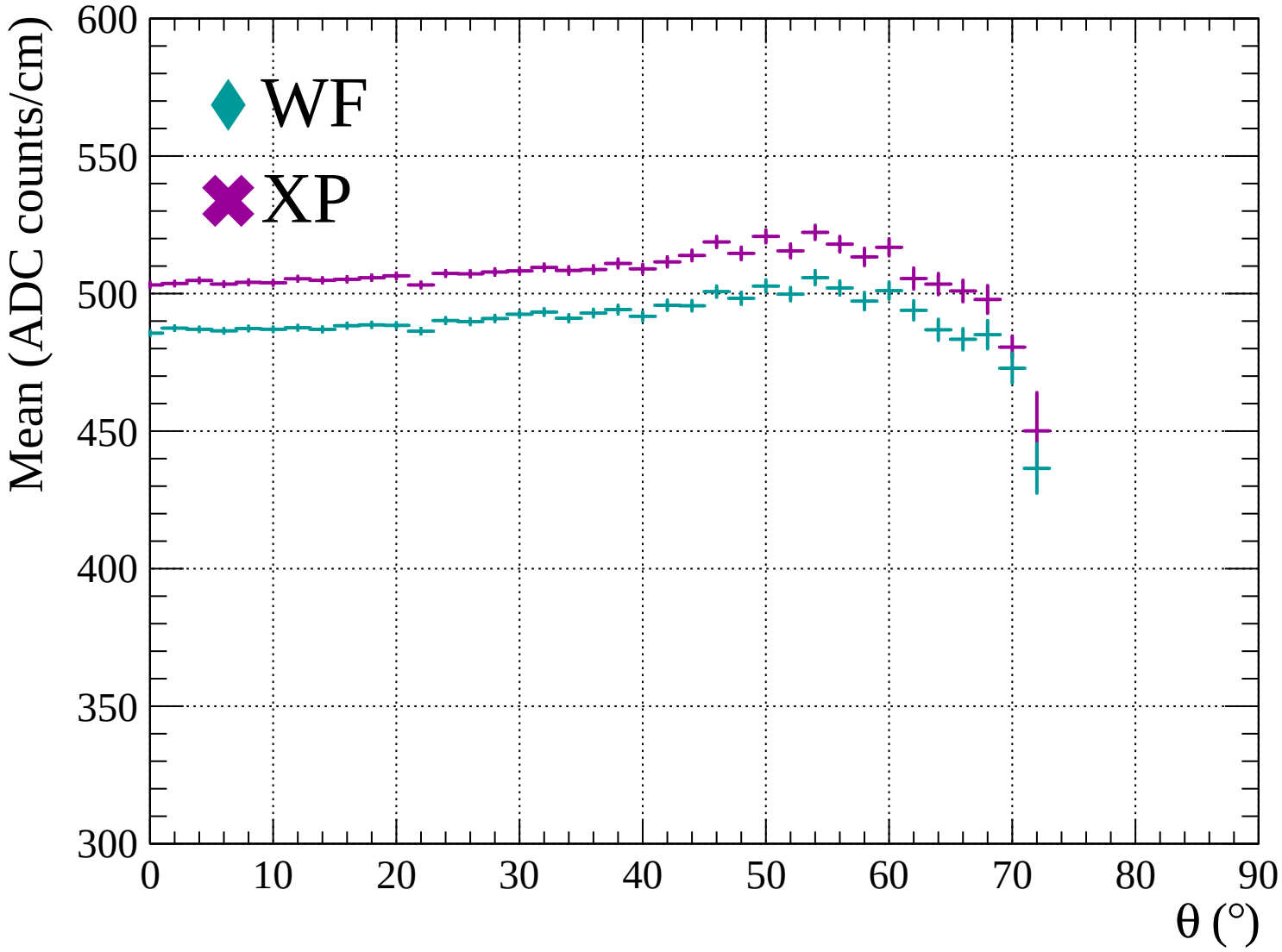


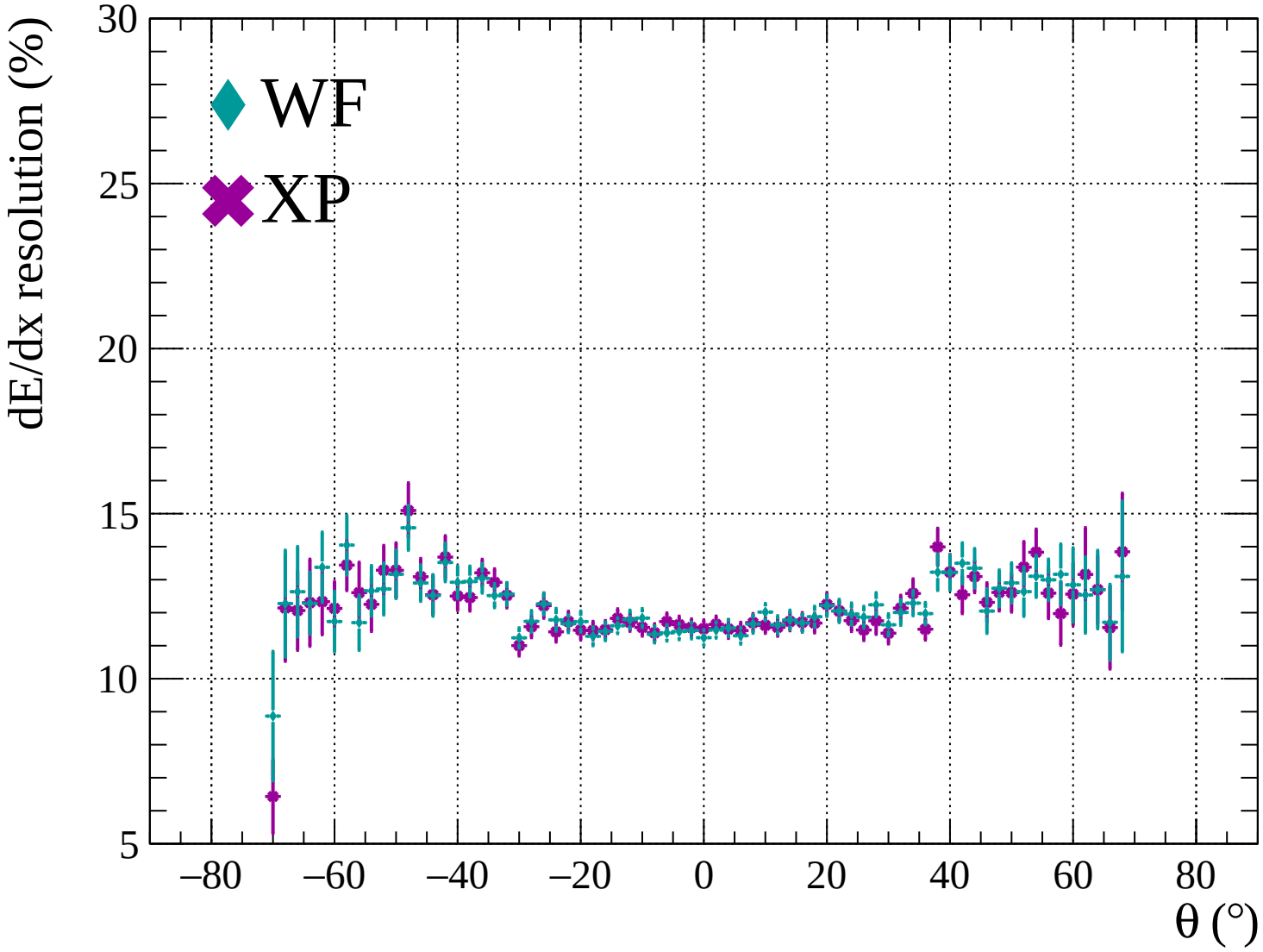


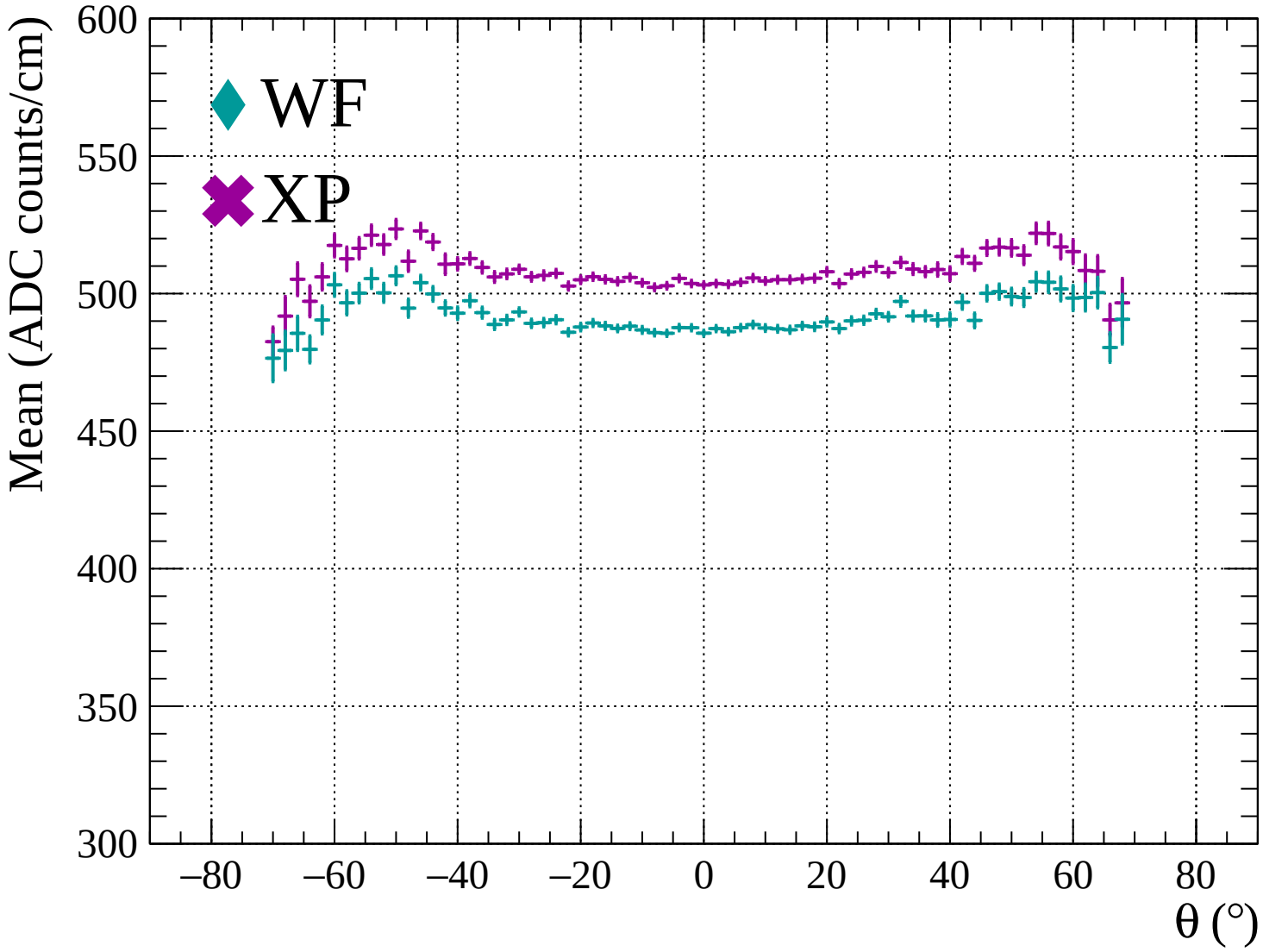


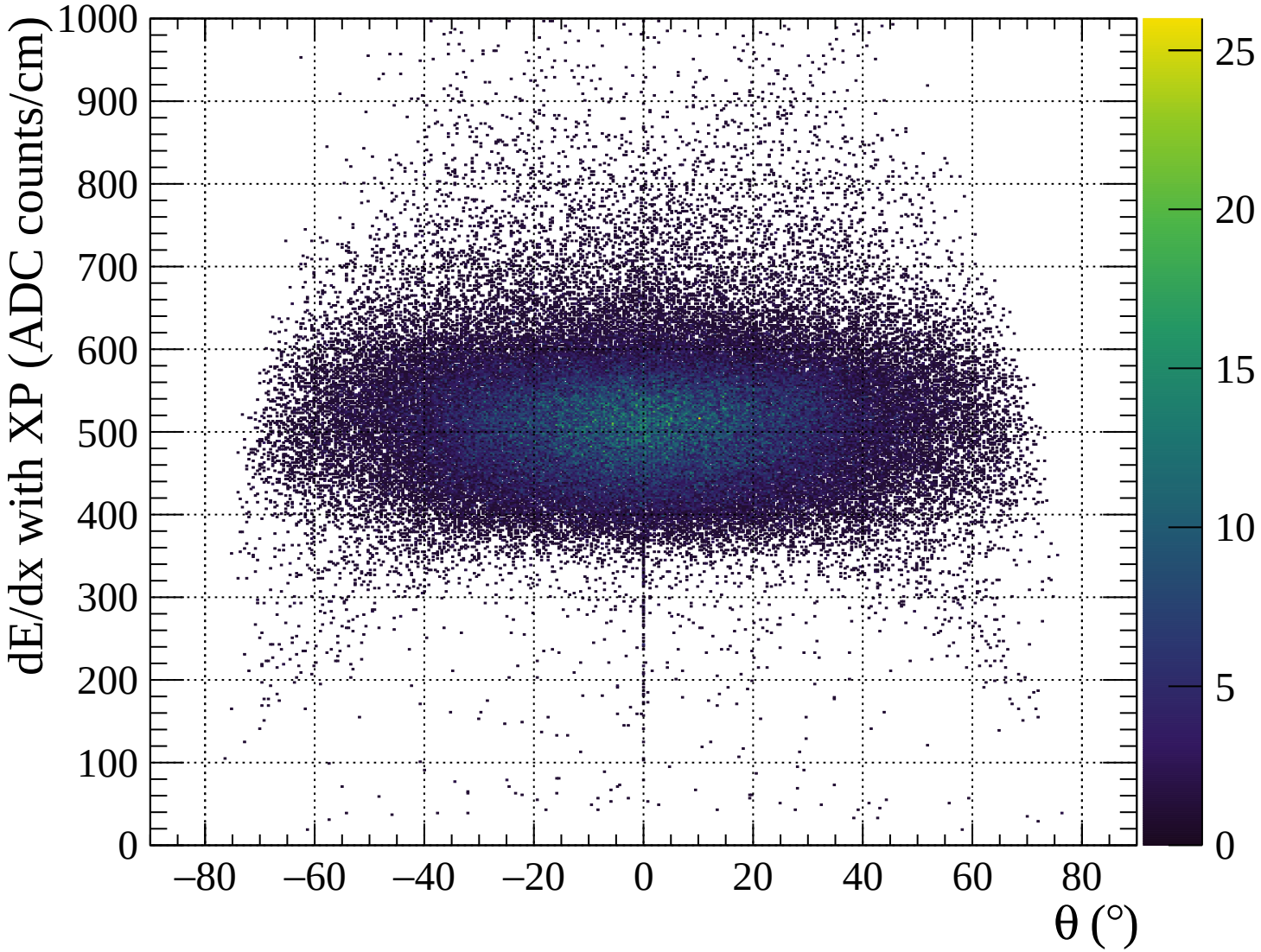


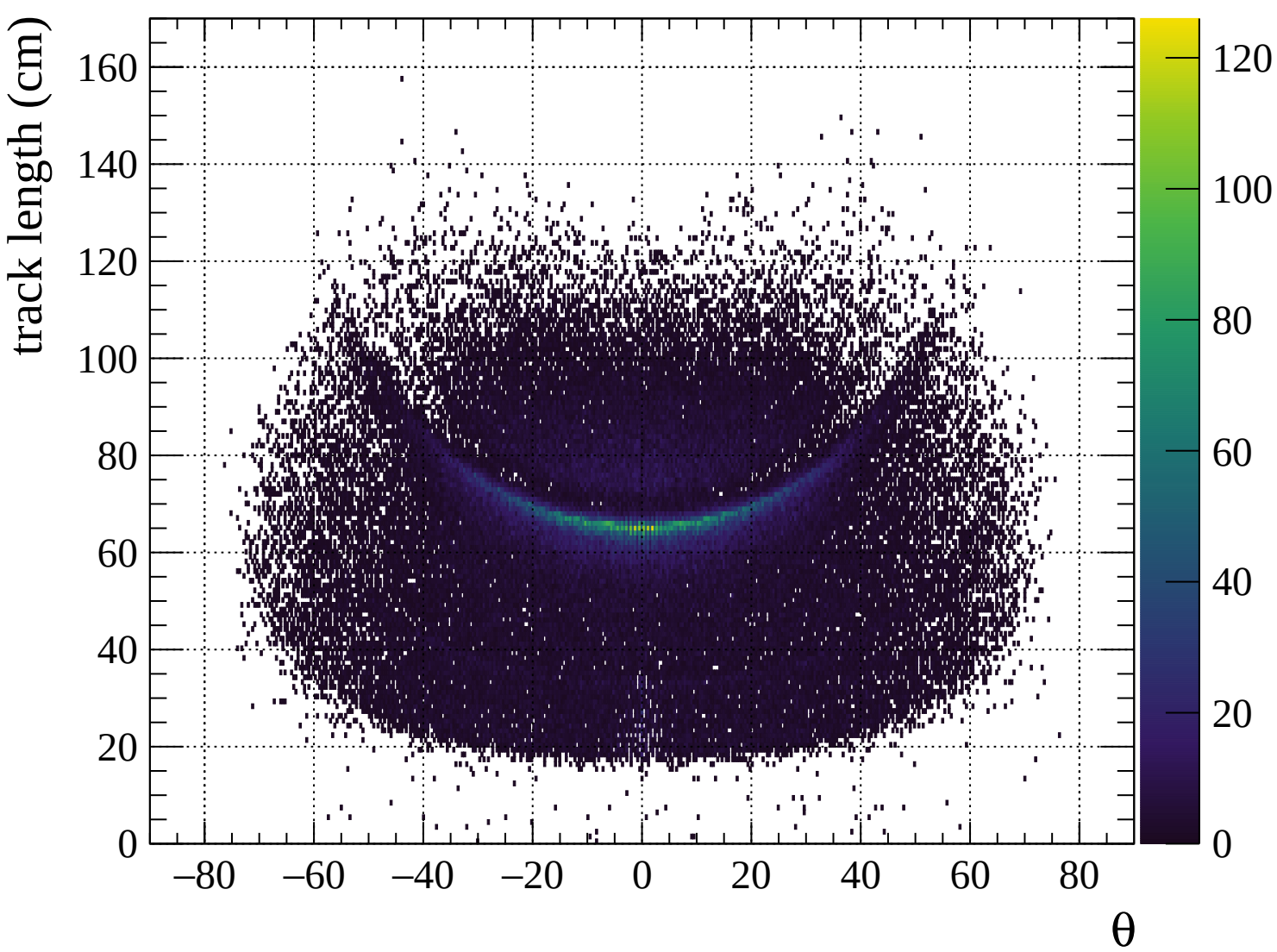


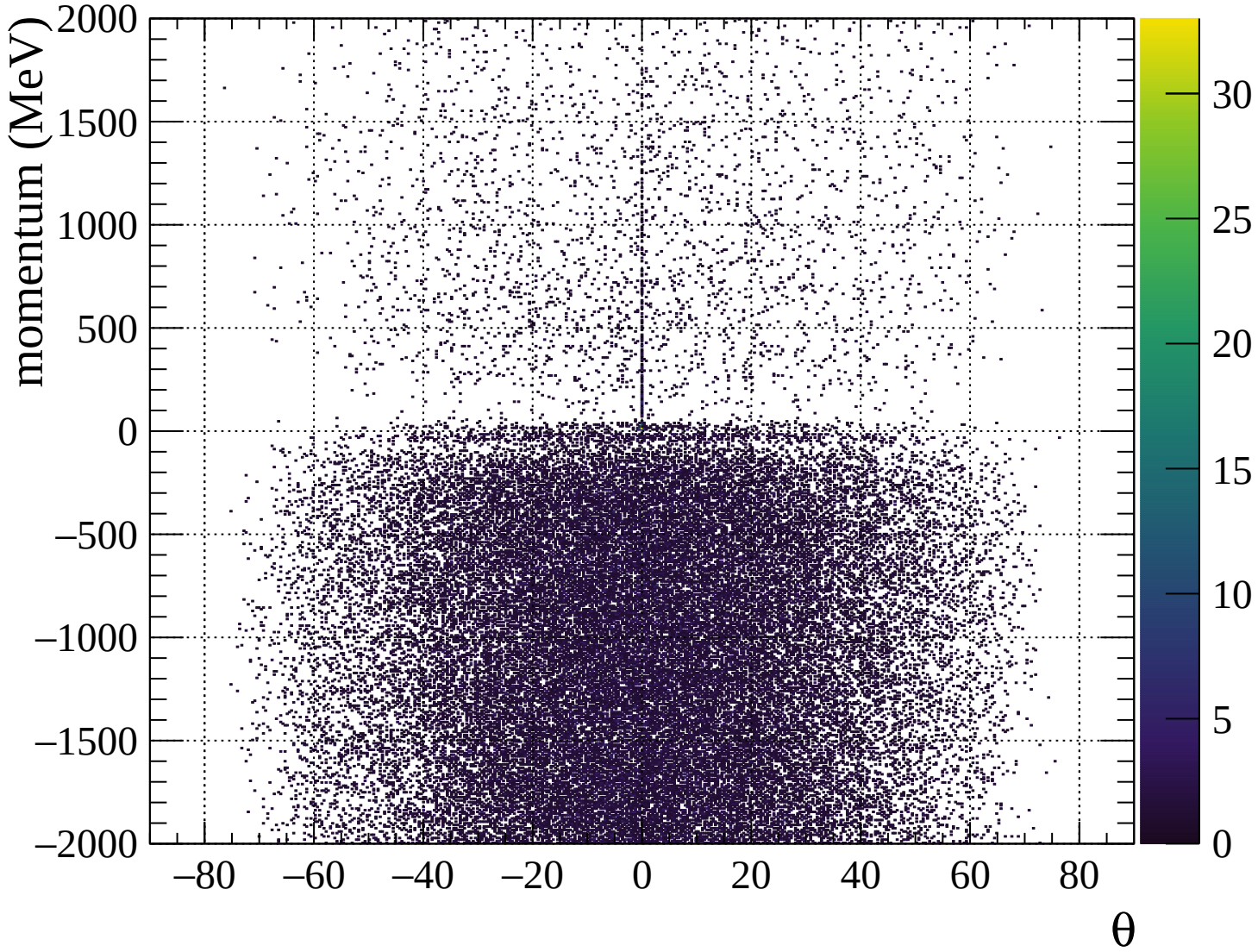


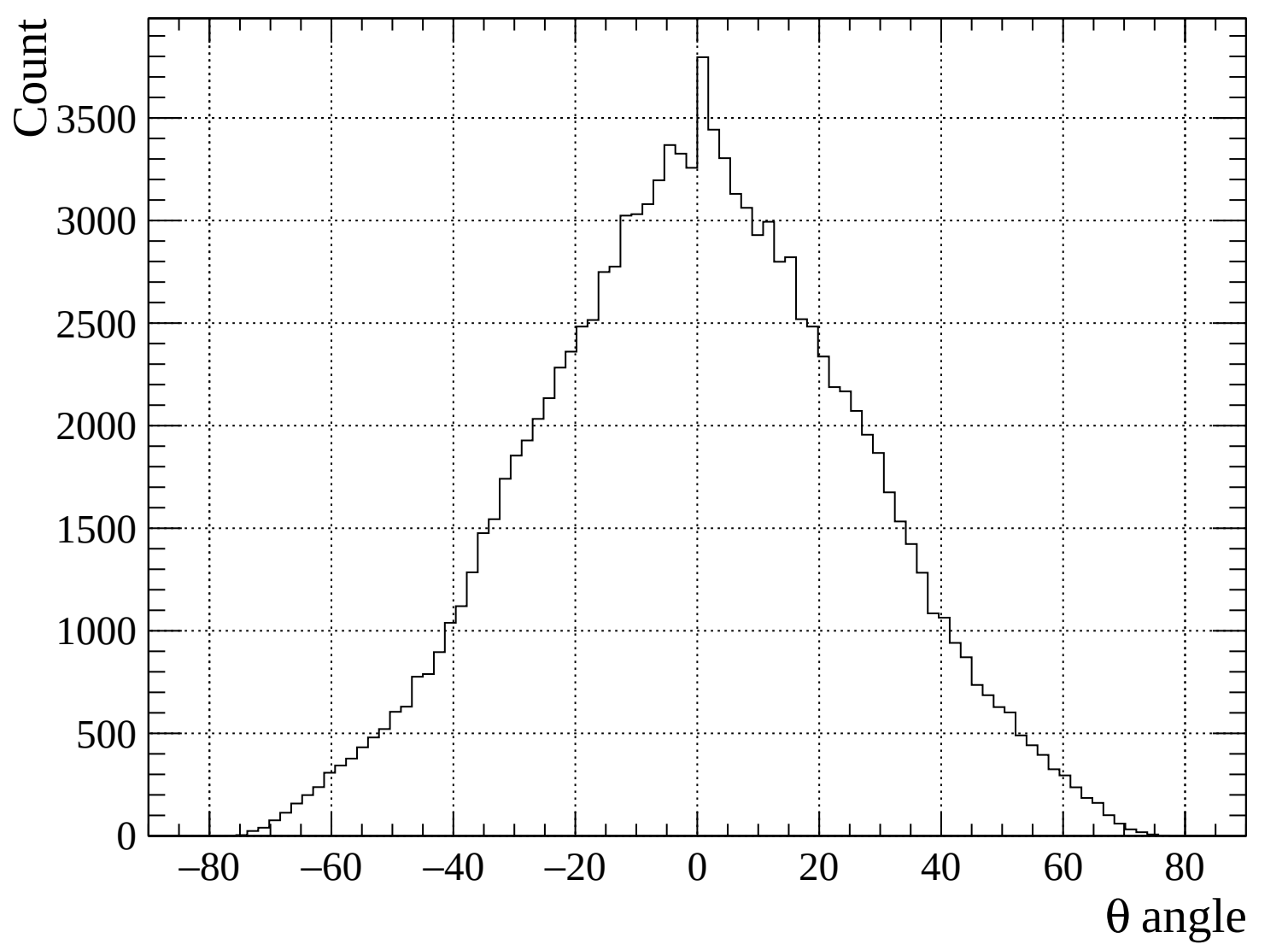


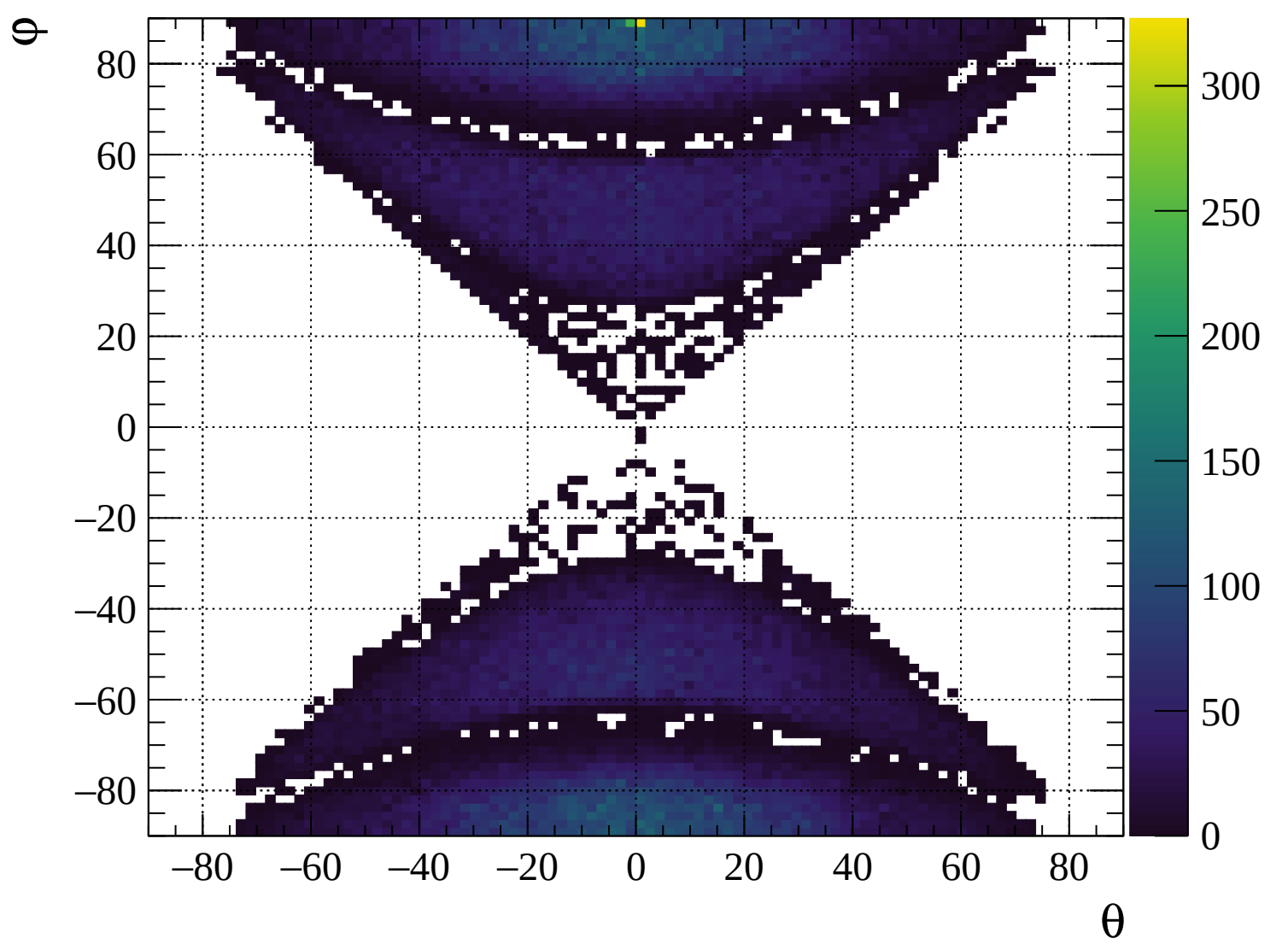


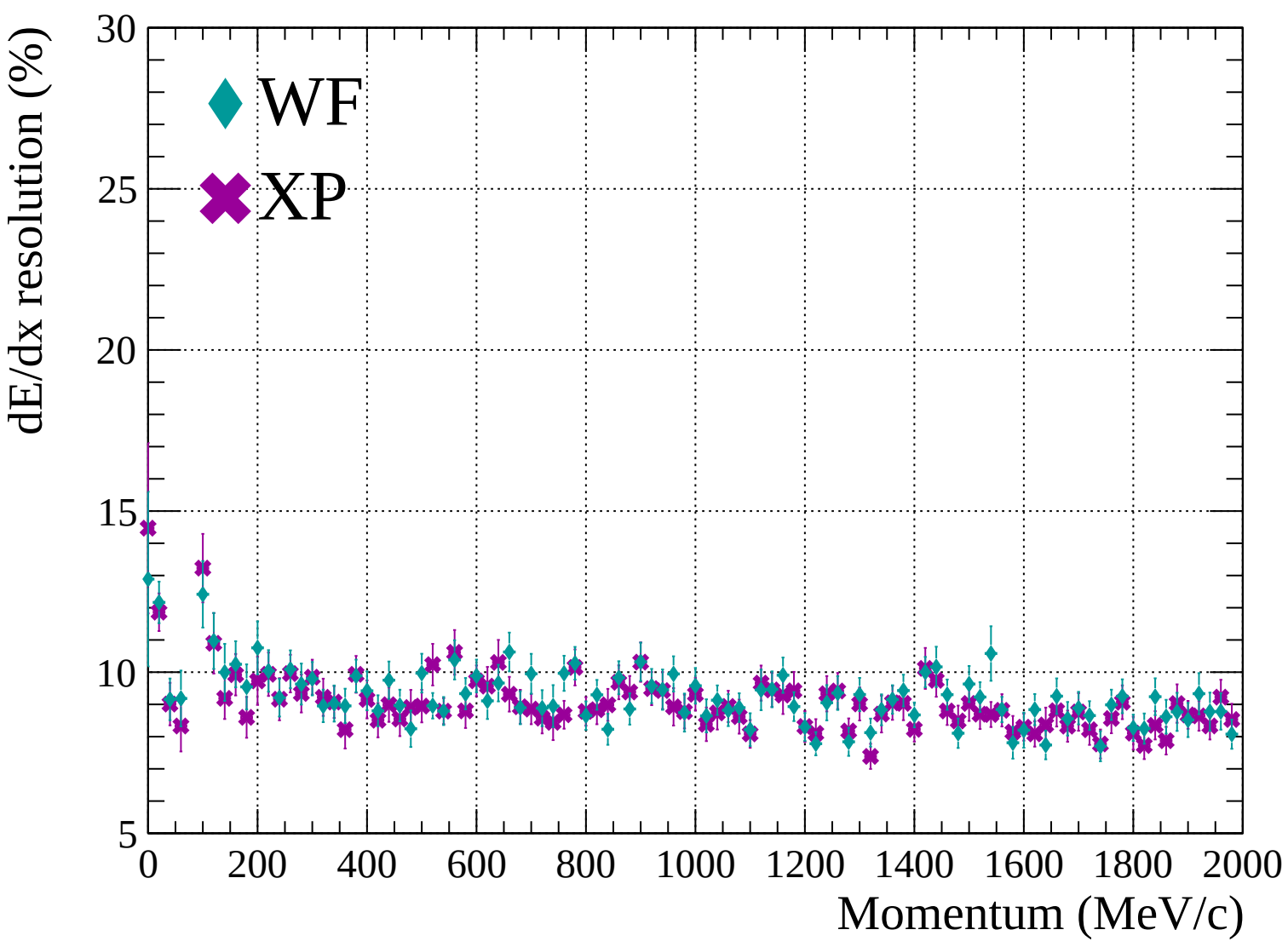


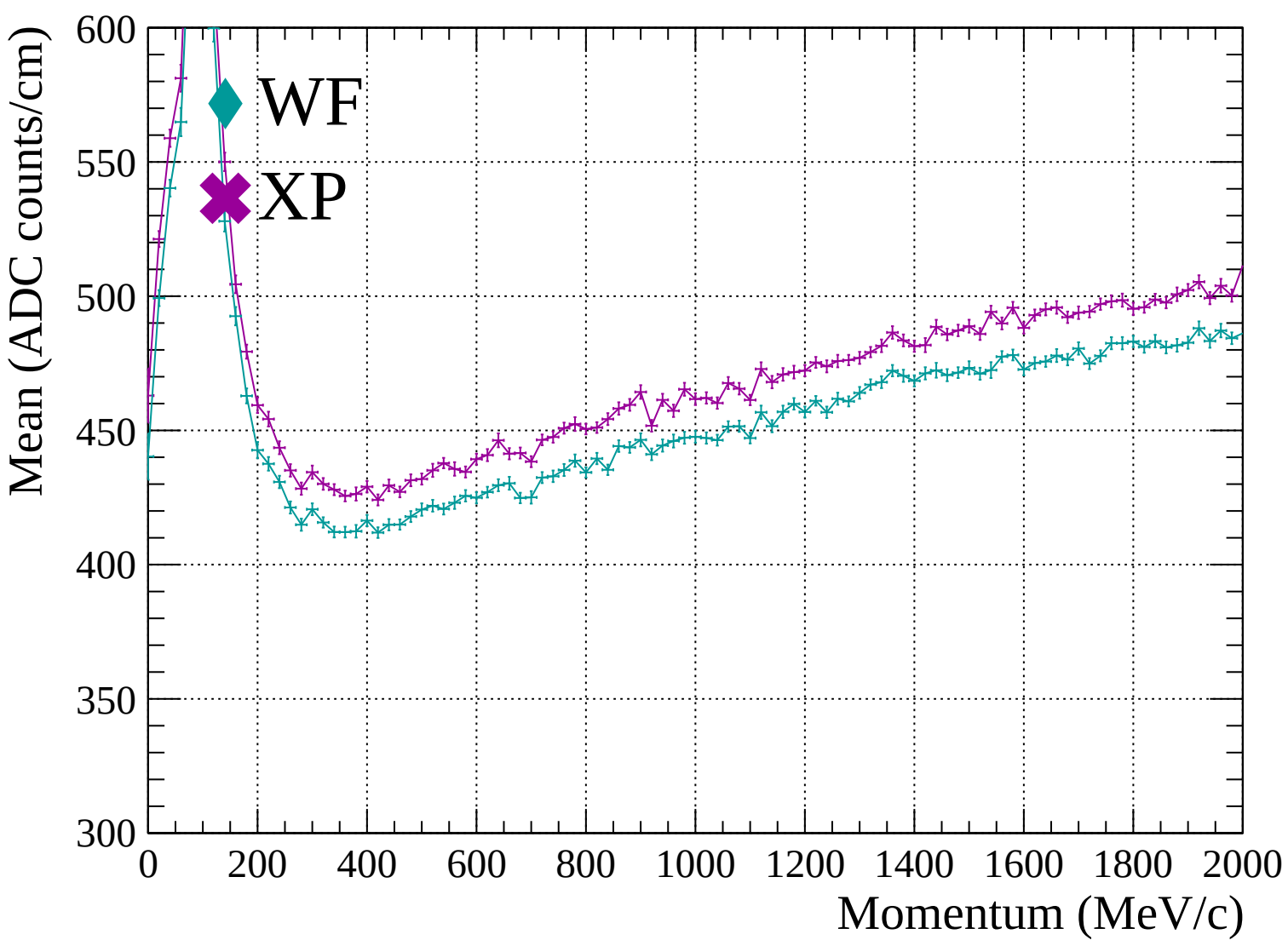


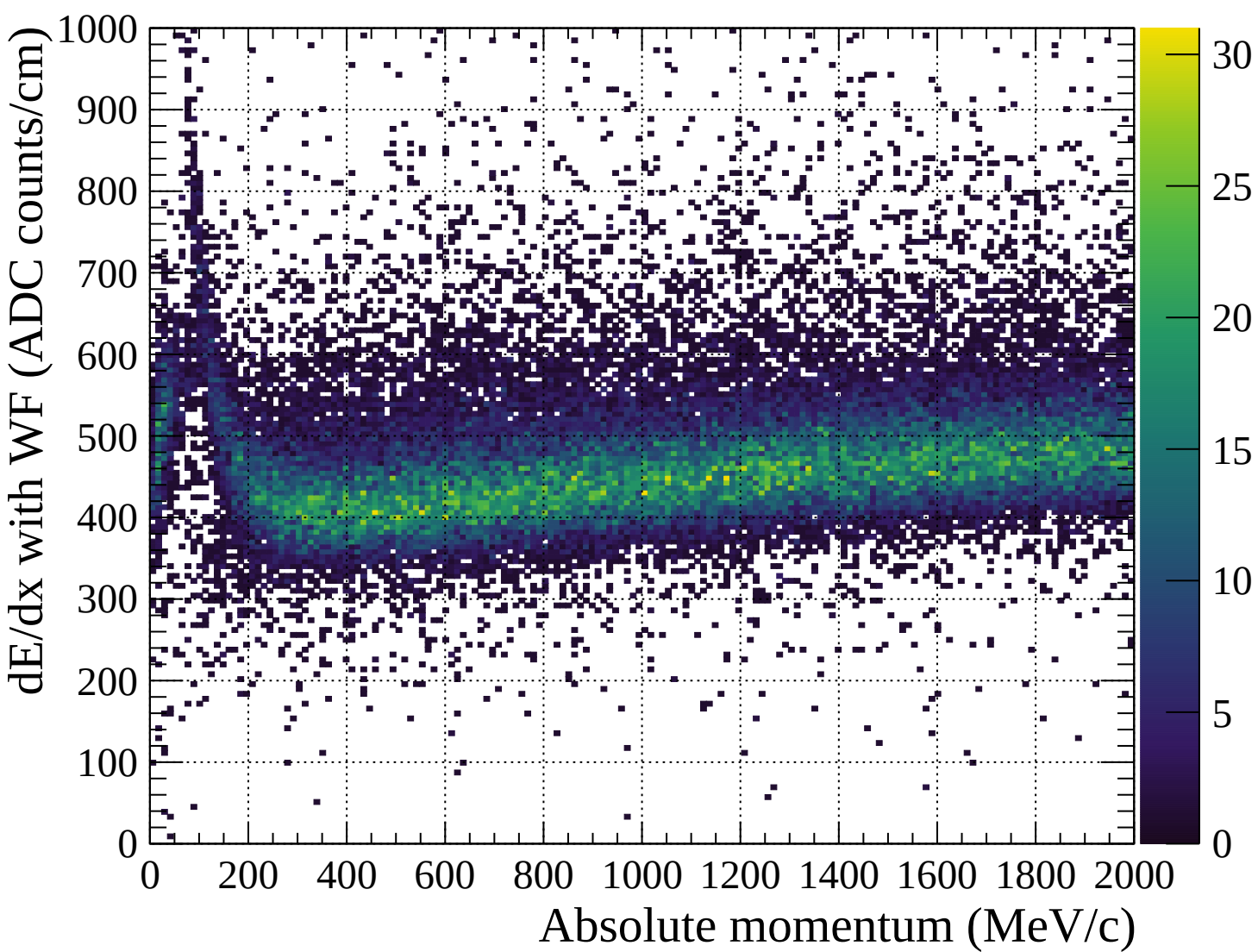


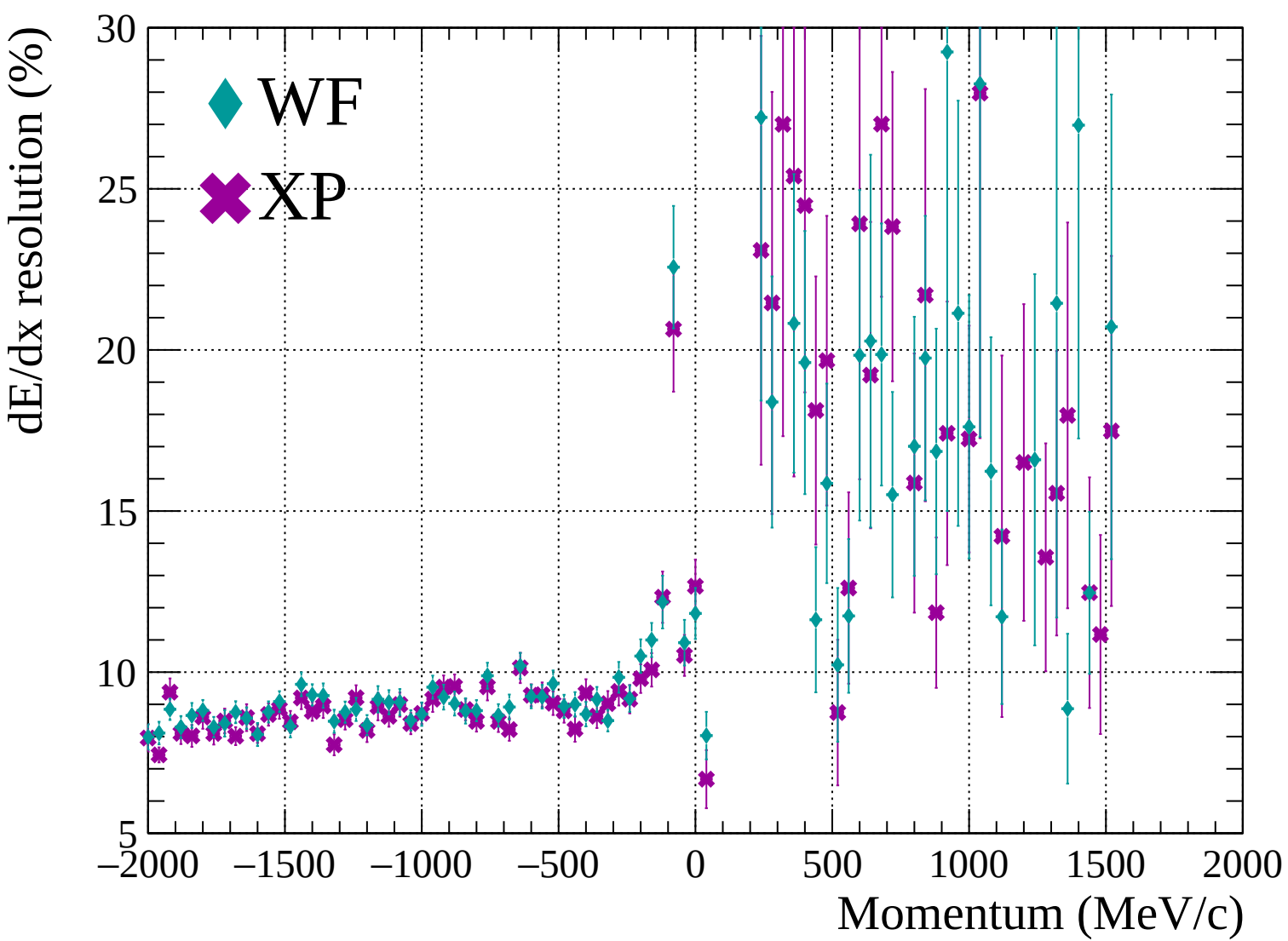


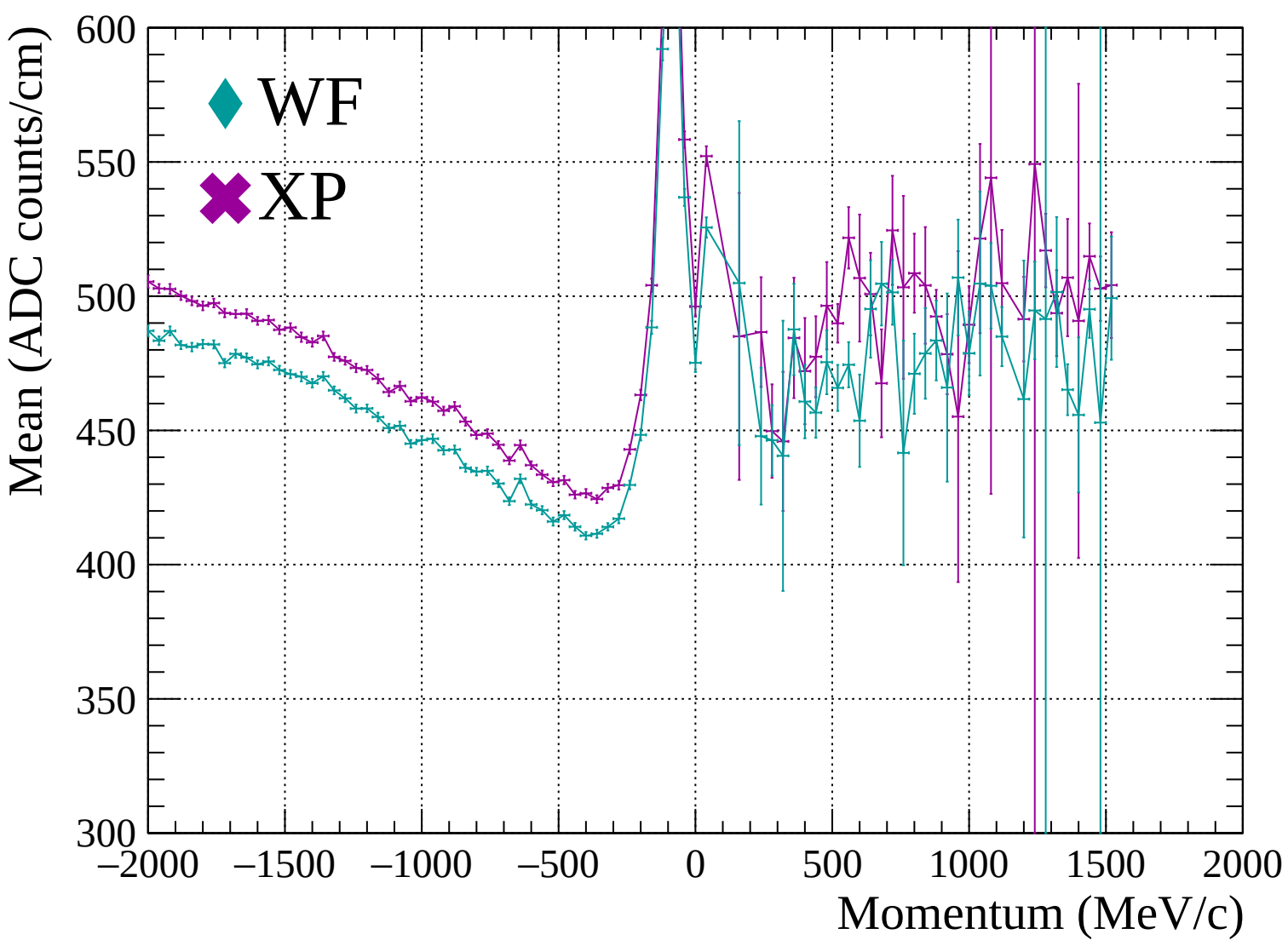


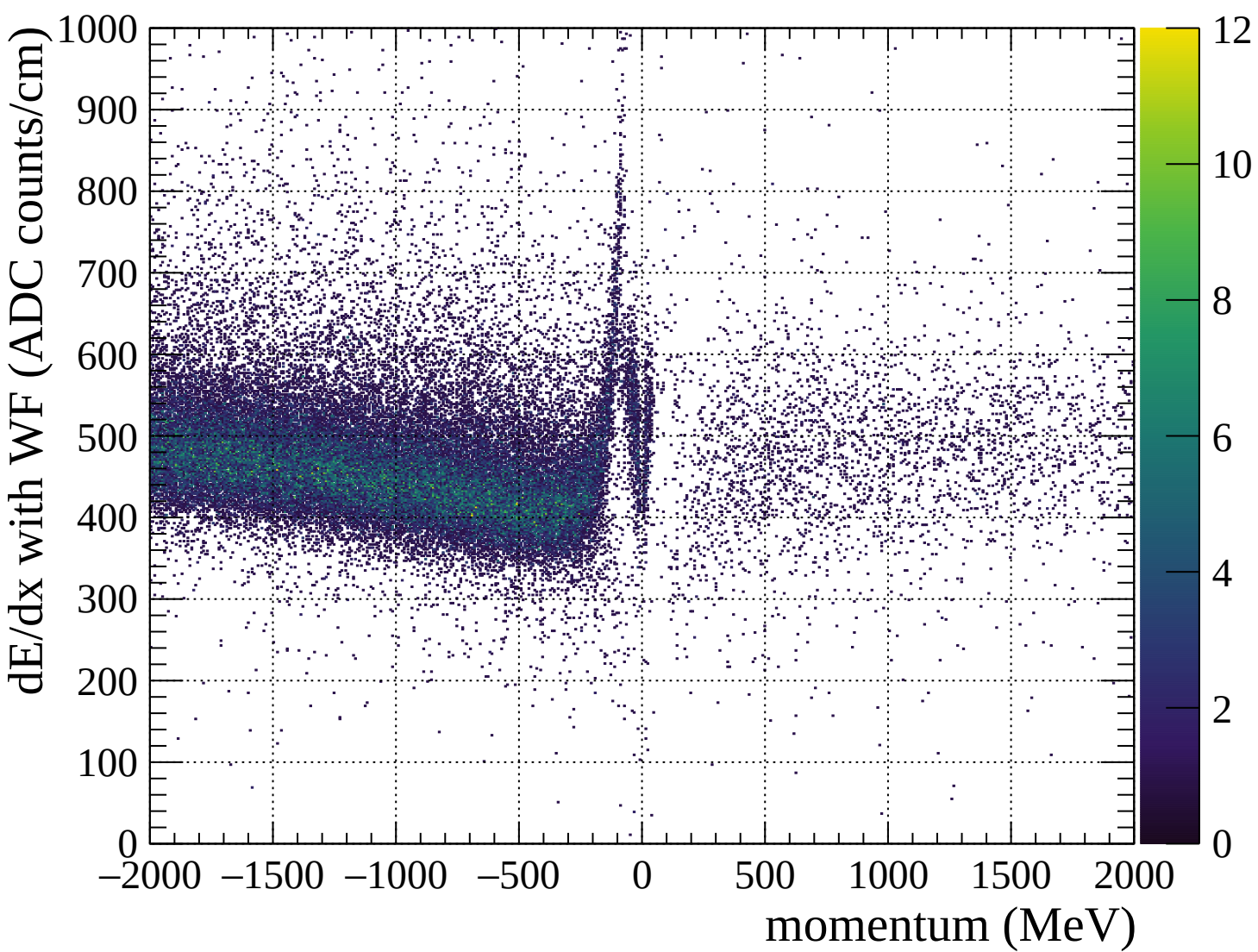


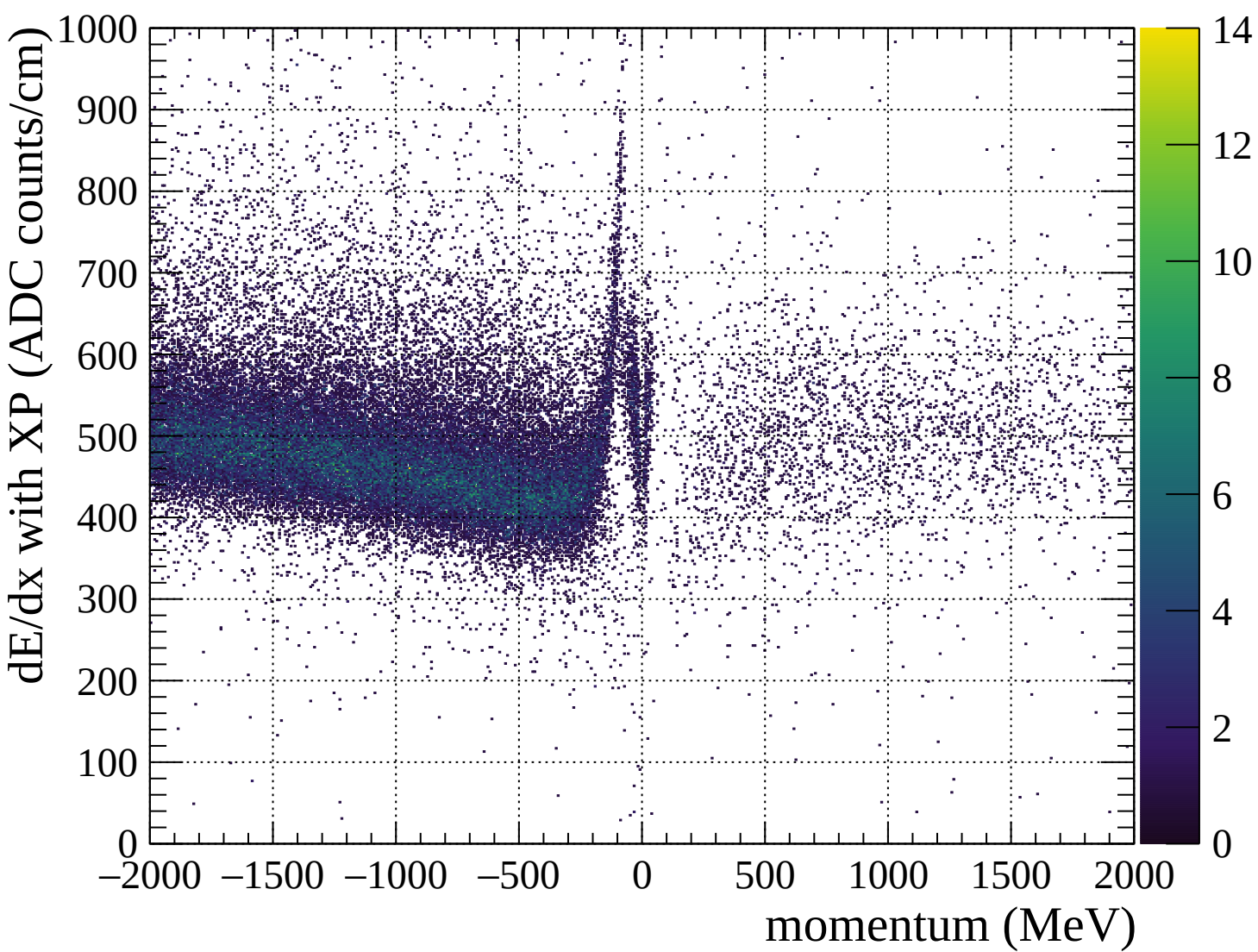


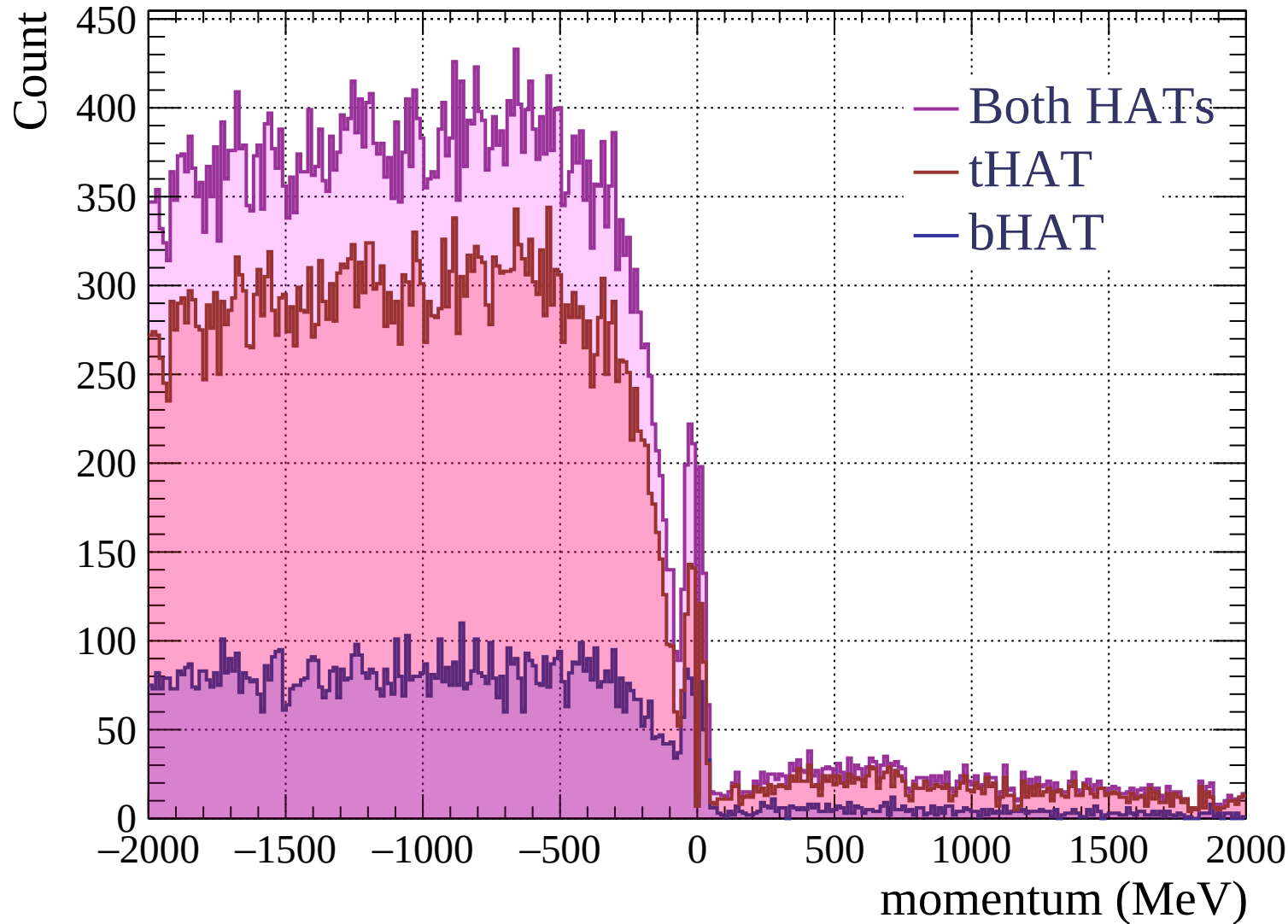


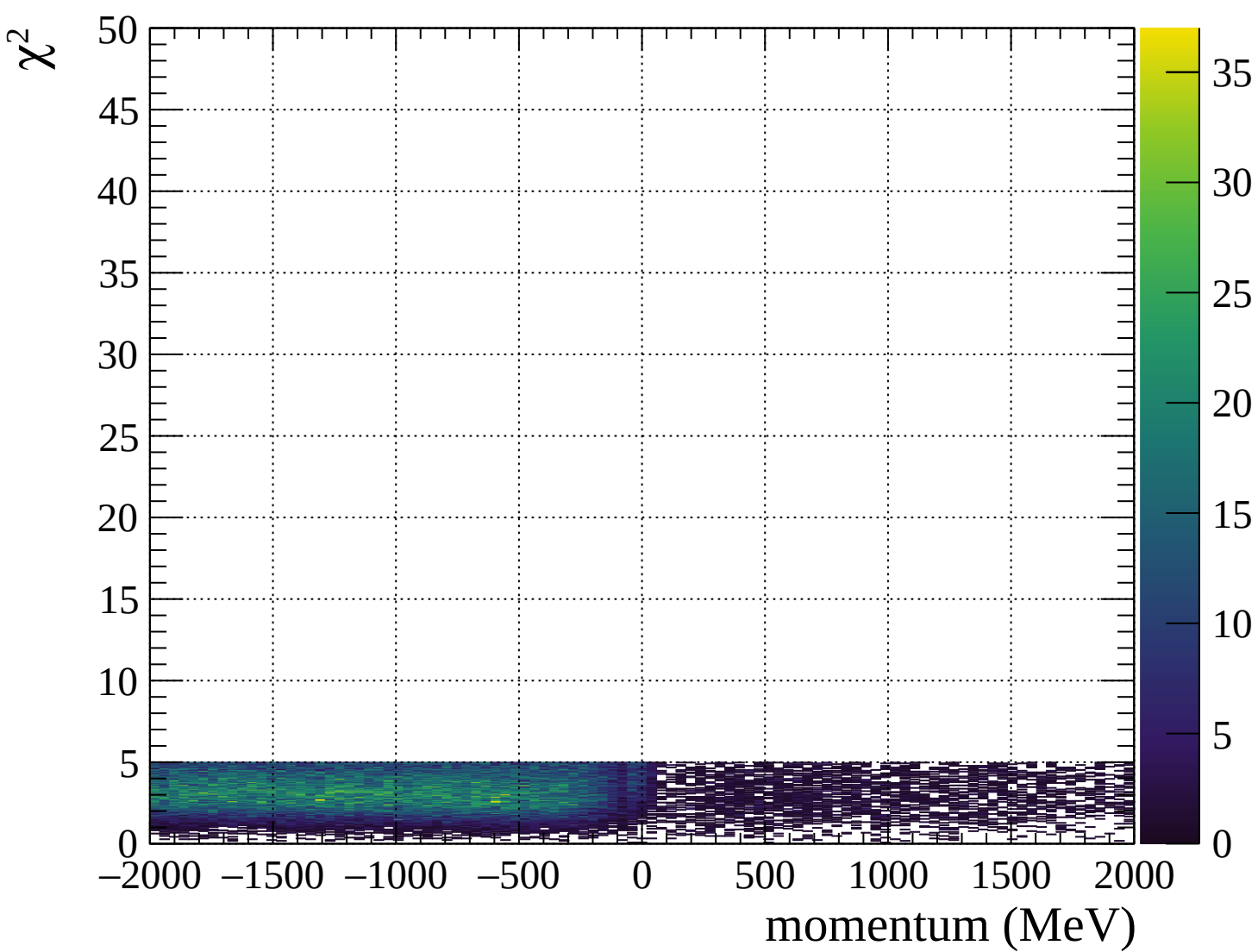




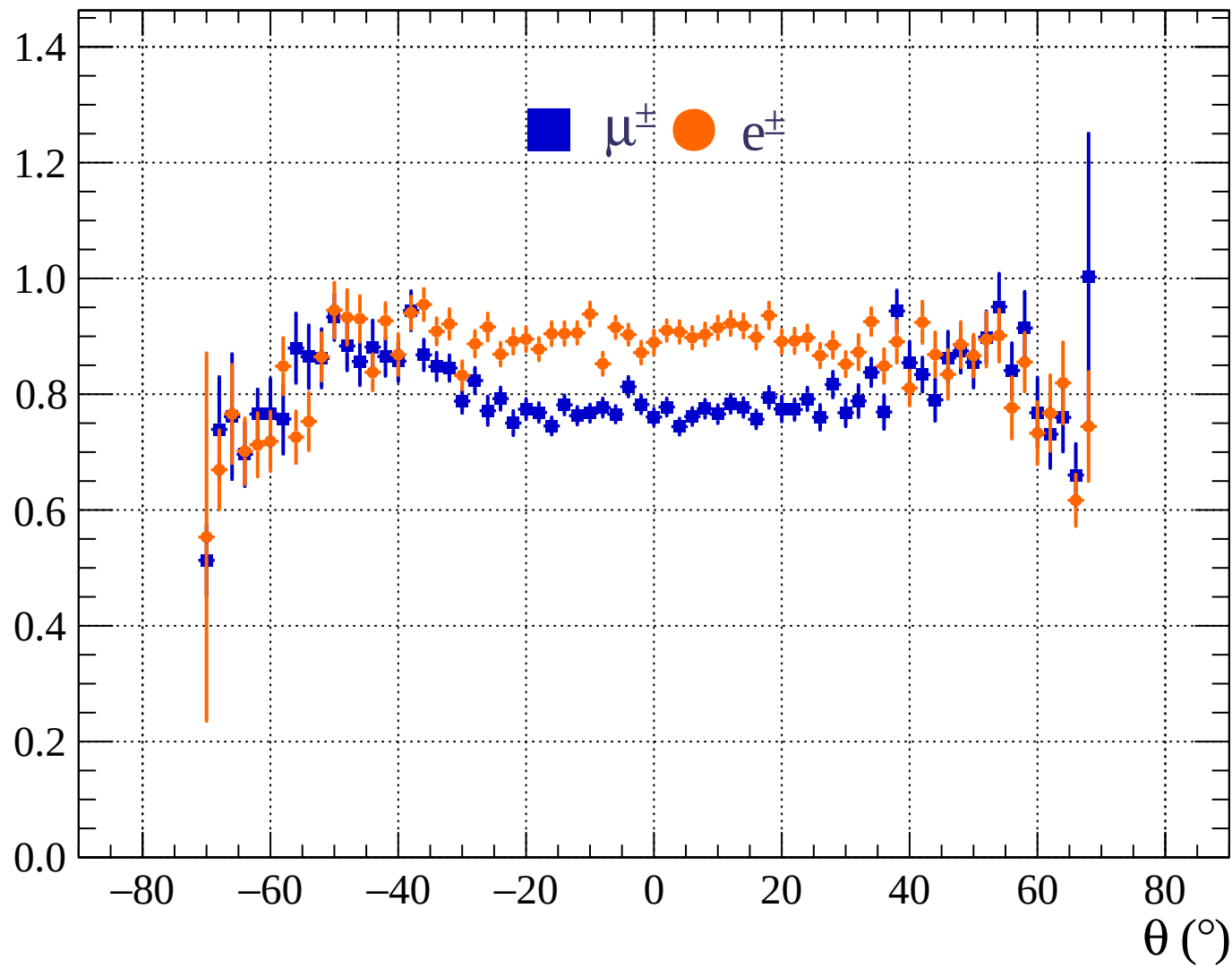


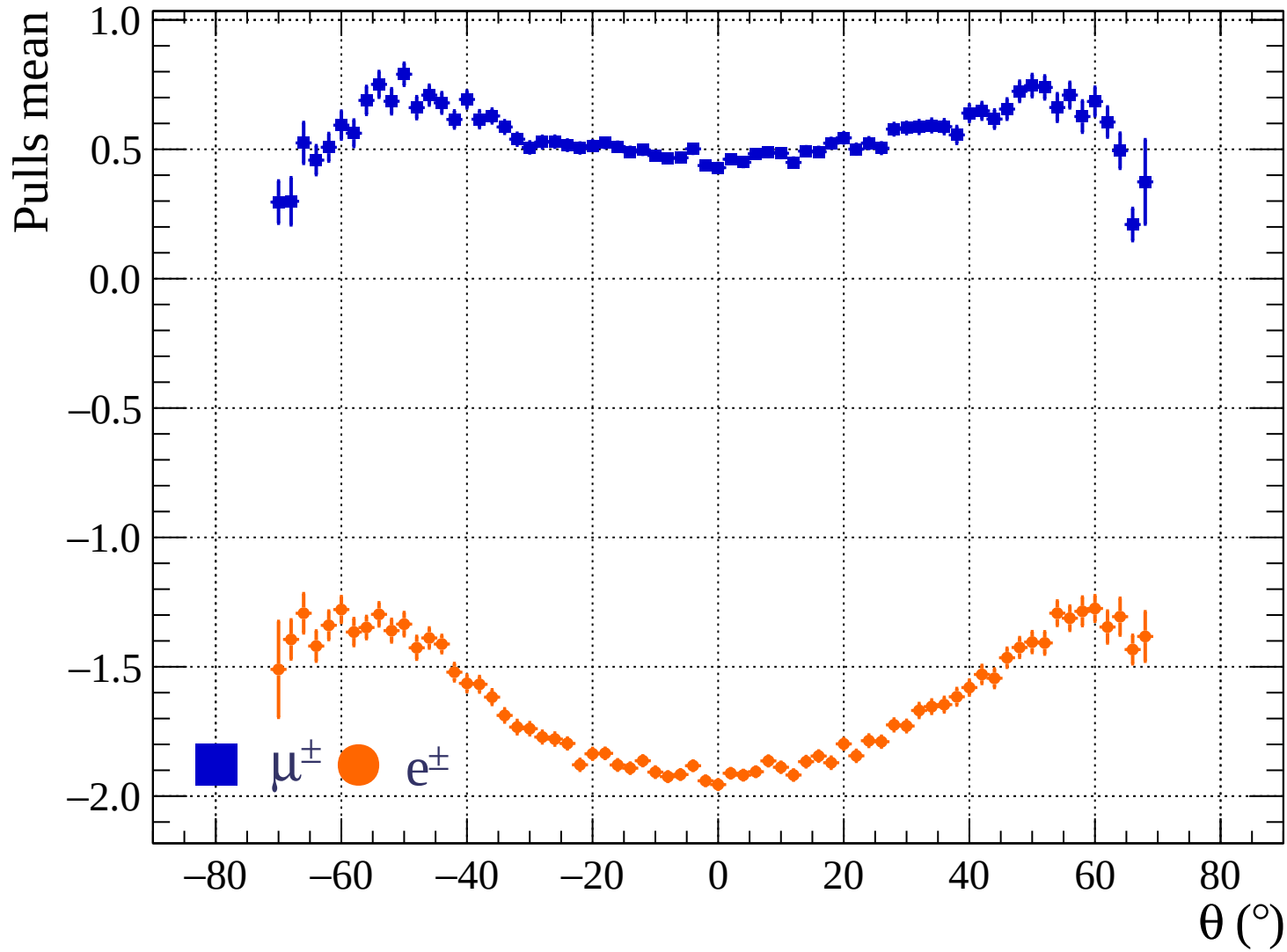




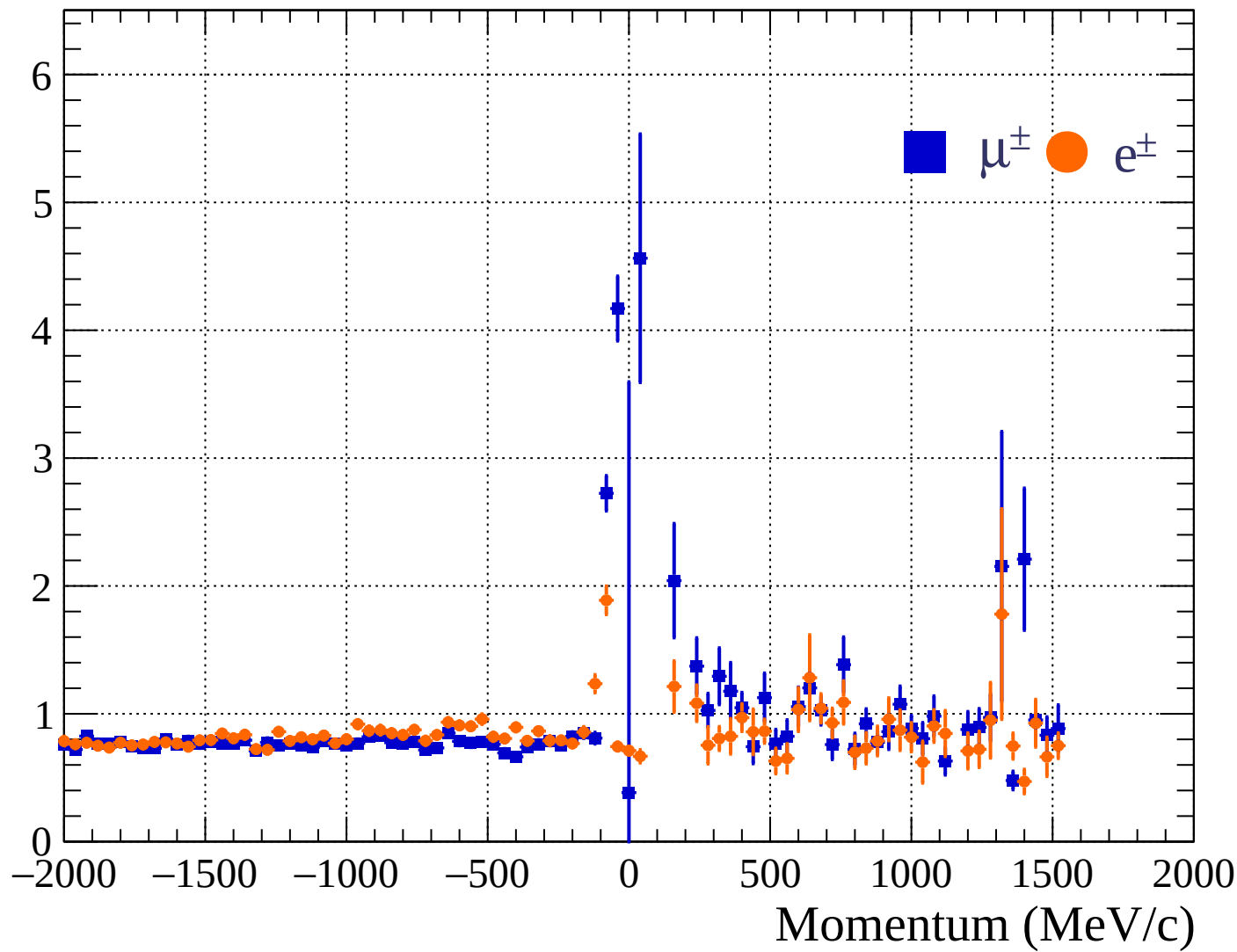


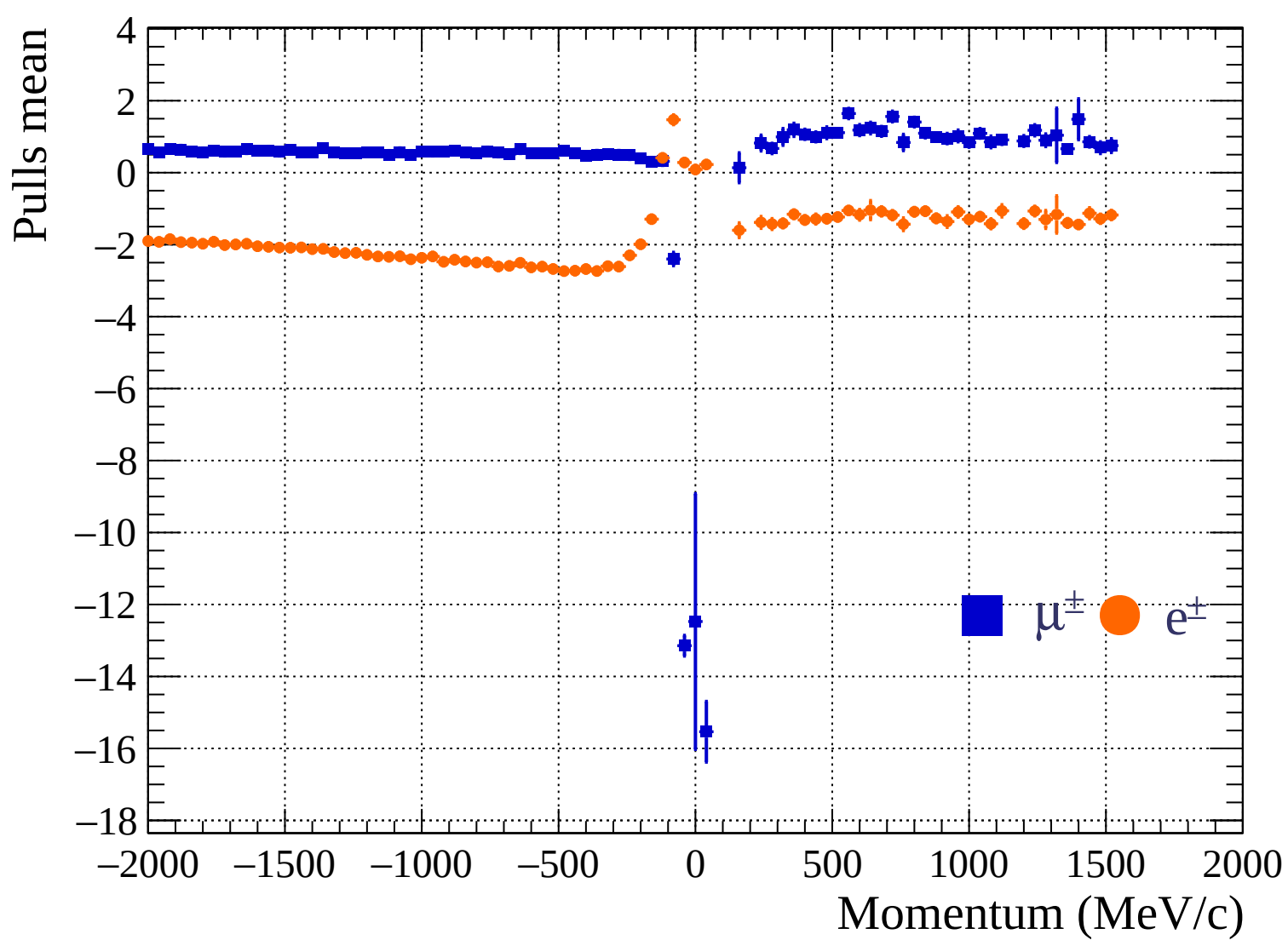
Pulls standard deviation

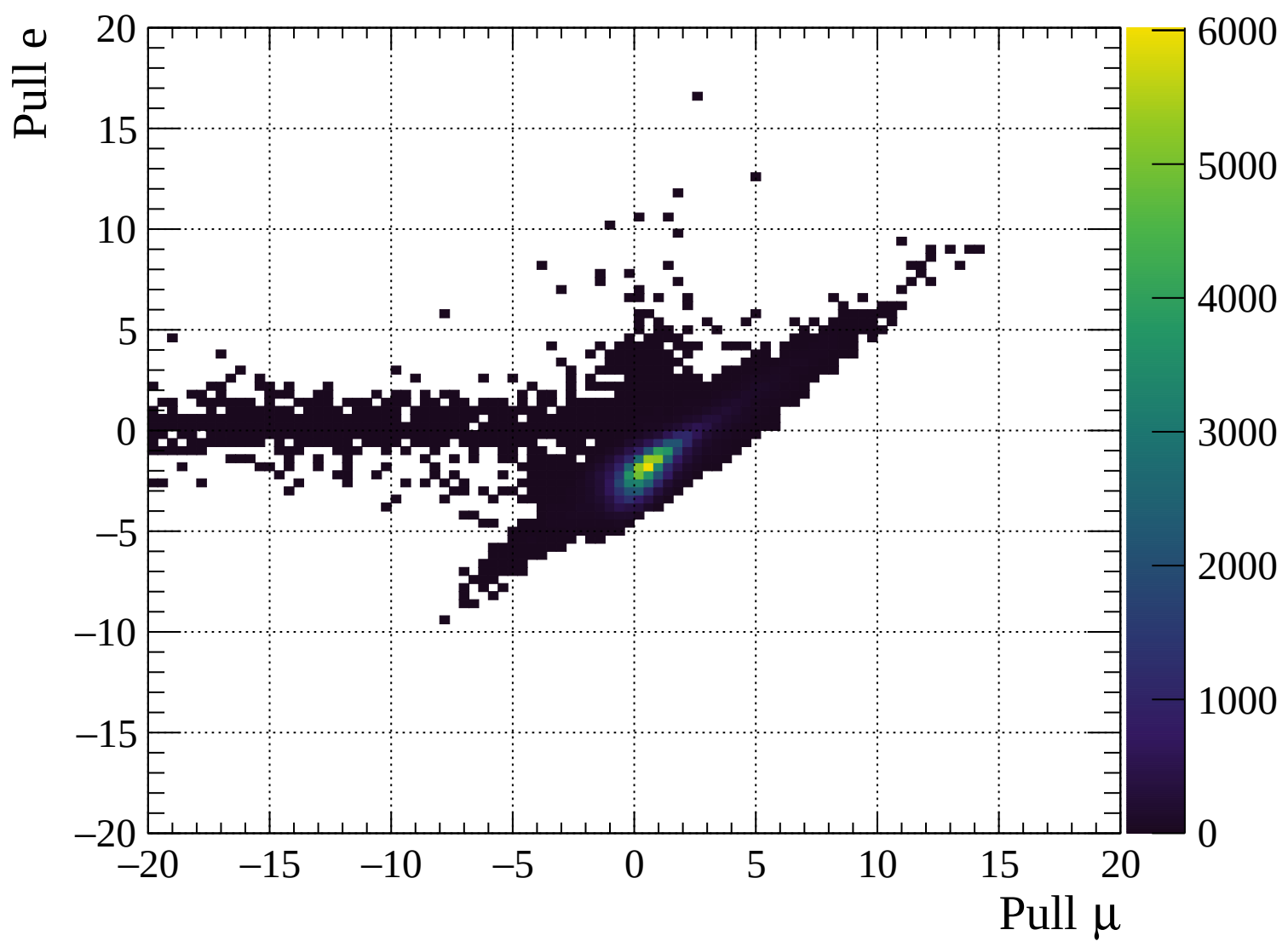


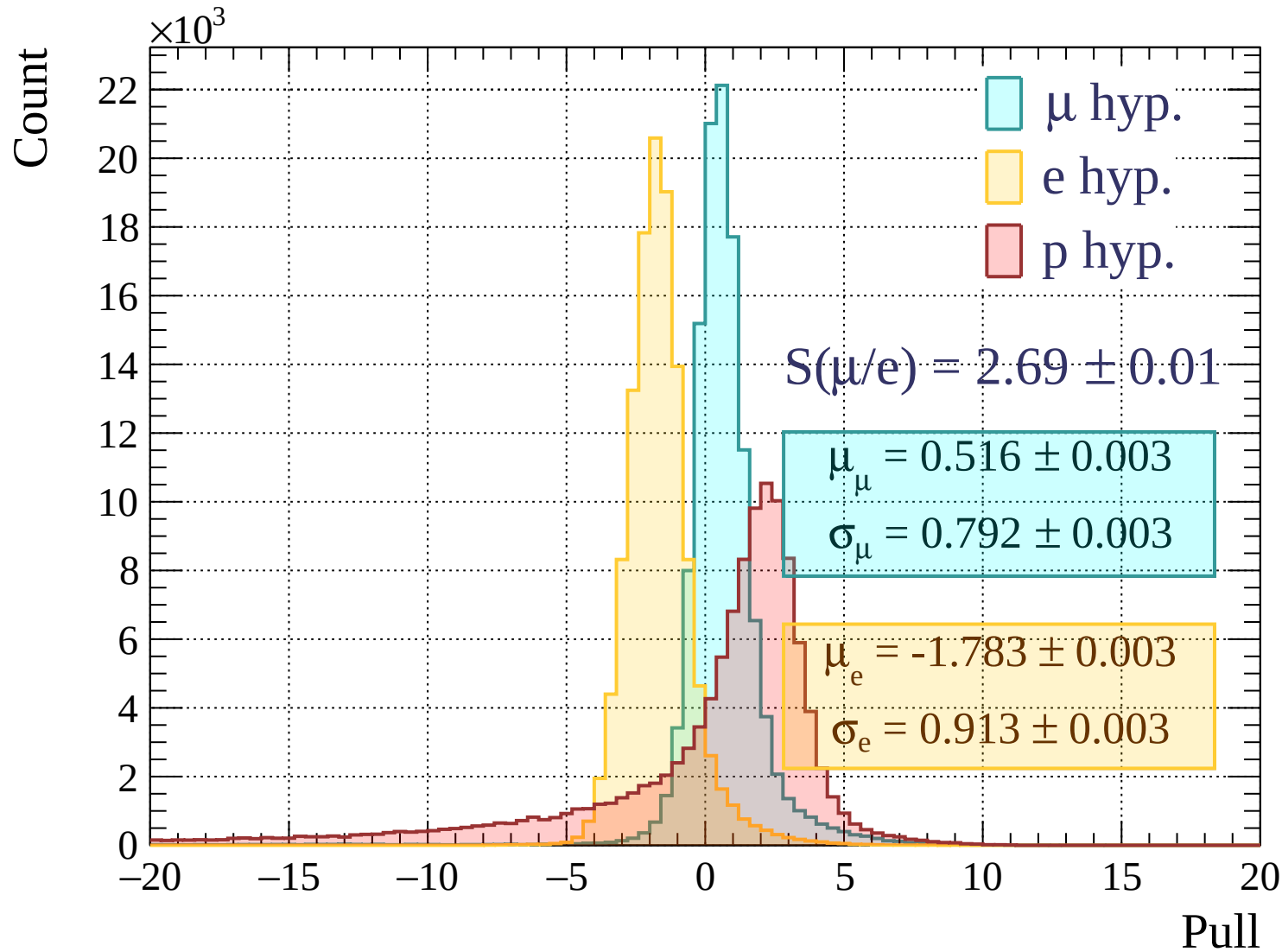


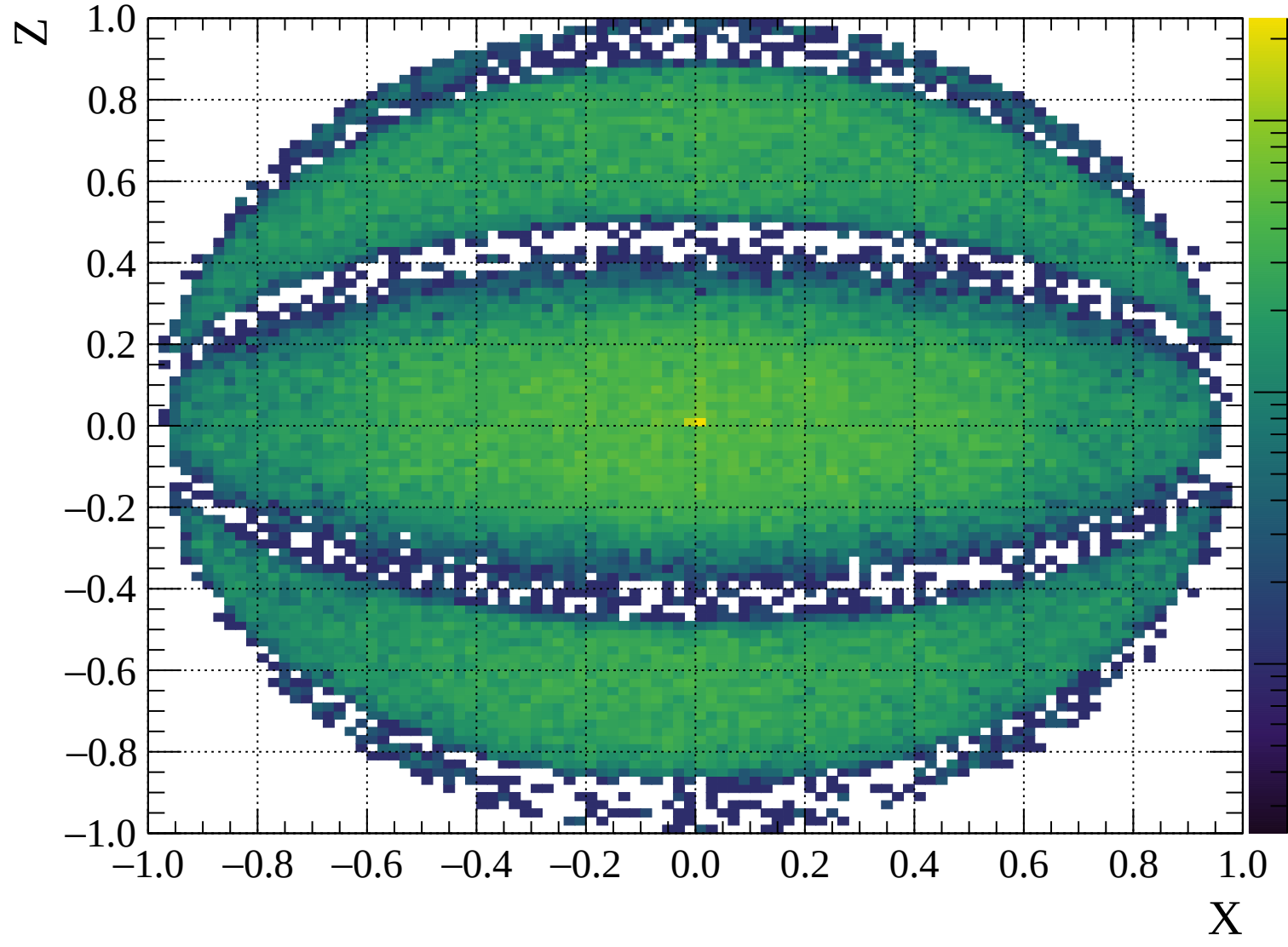
Pulls standard deviation

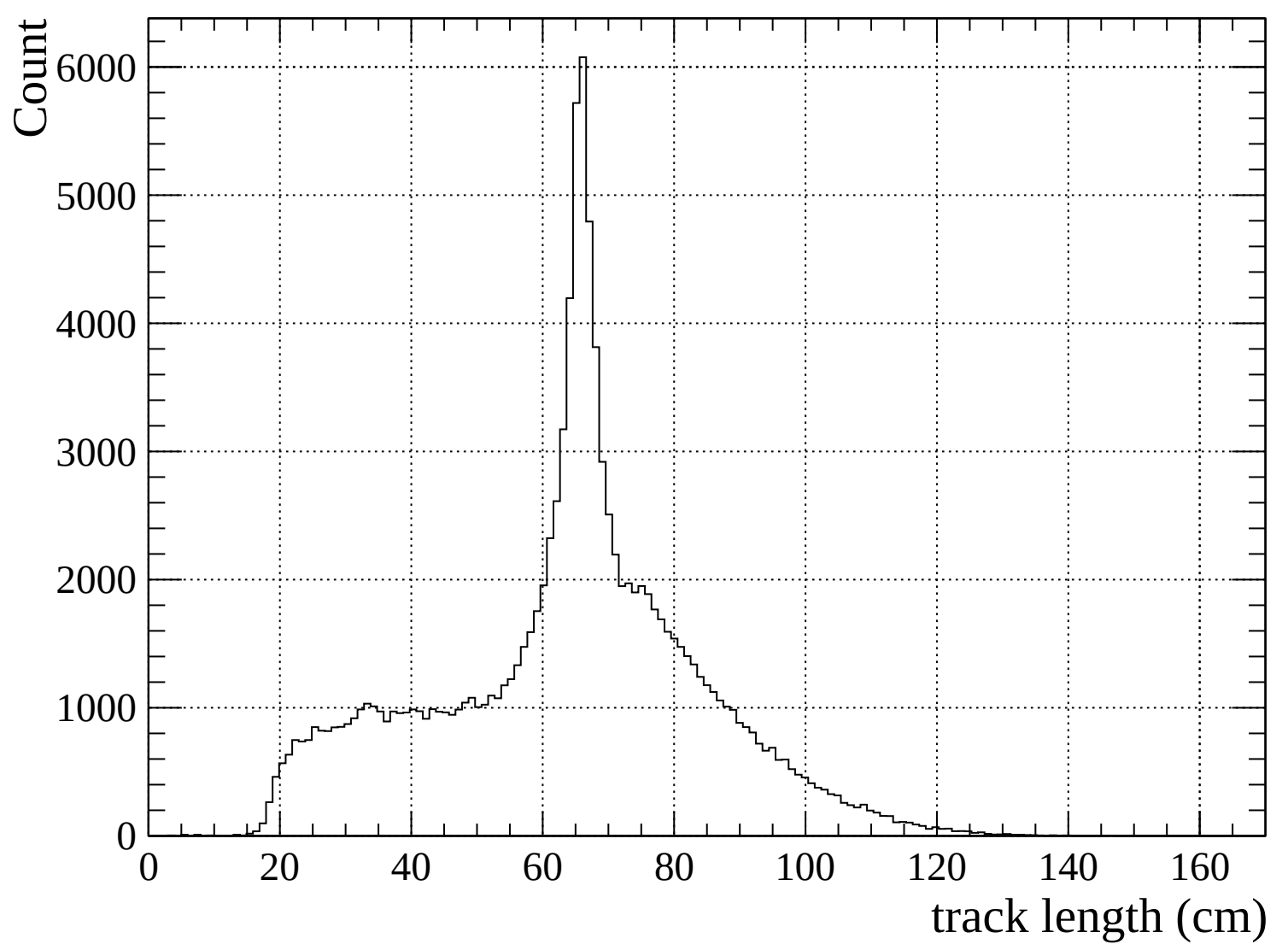


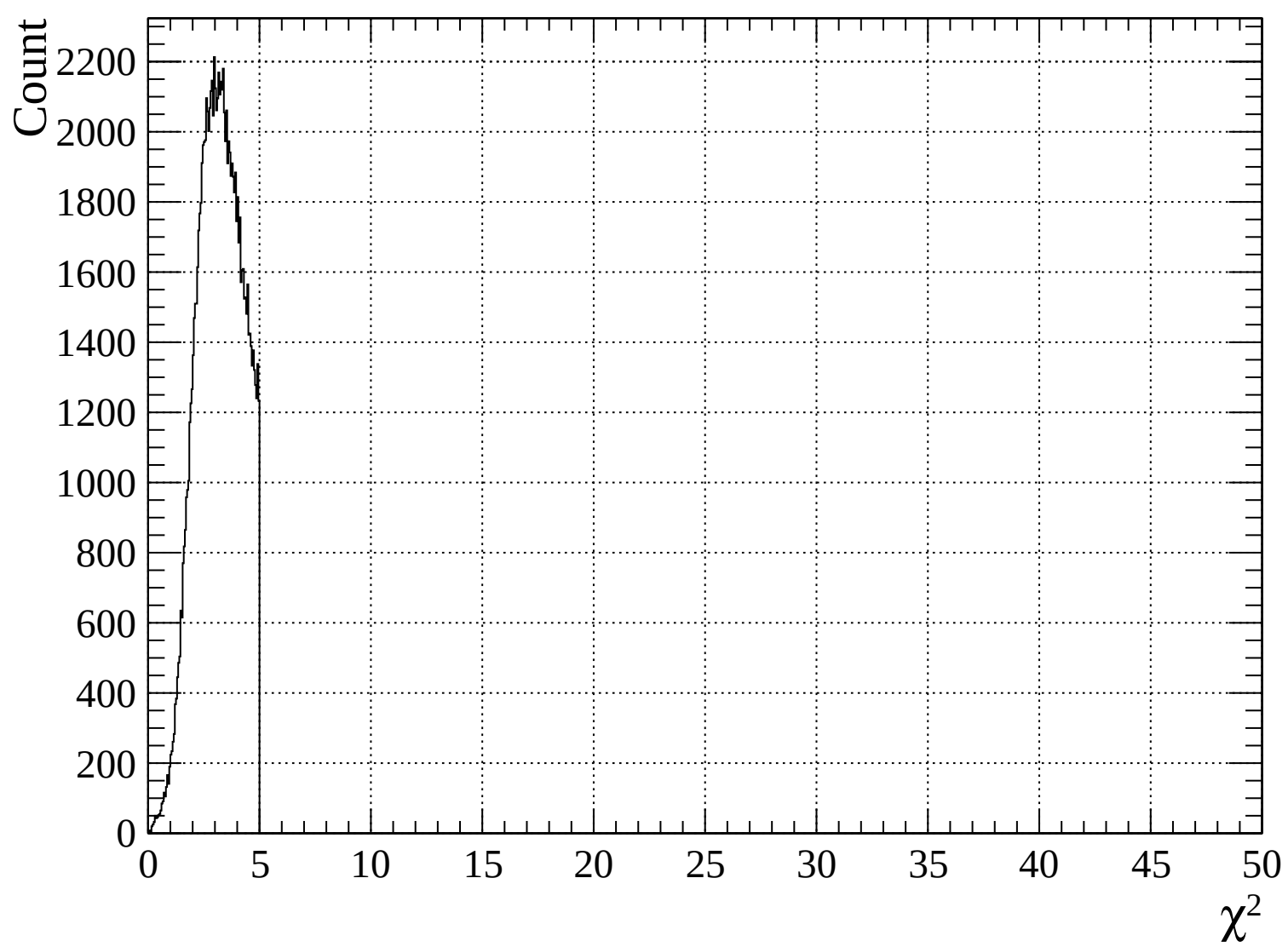


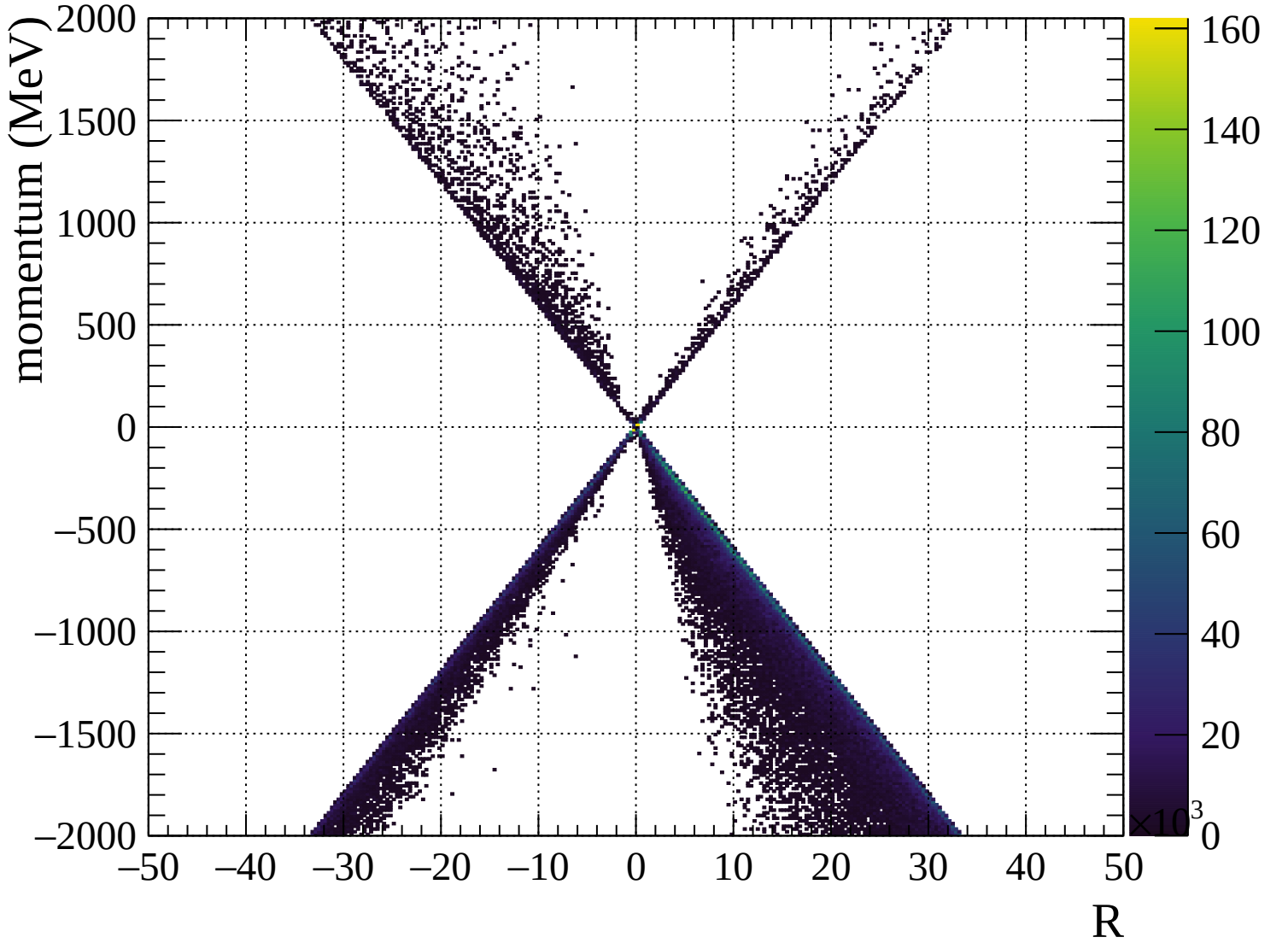


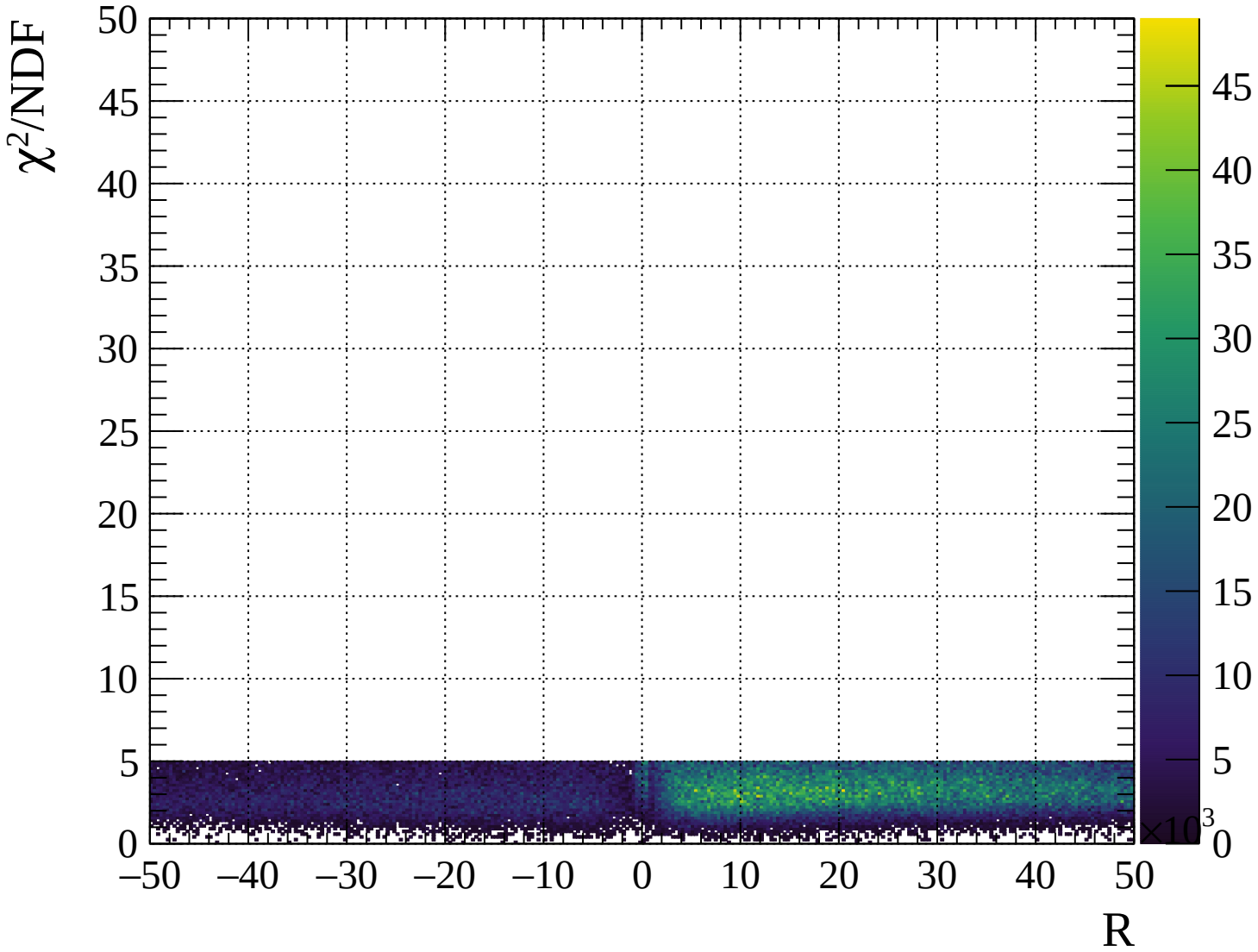






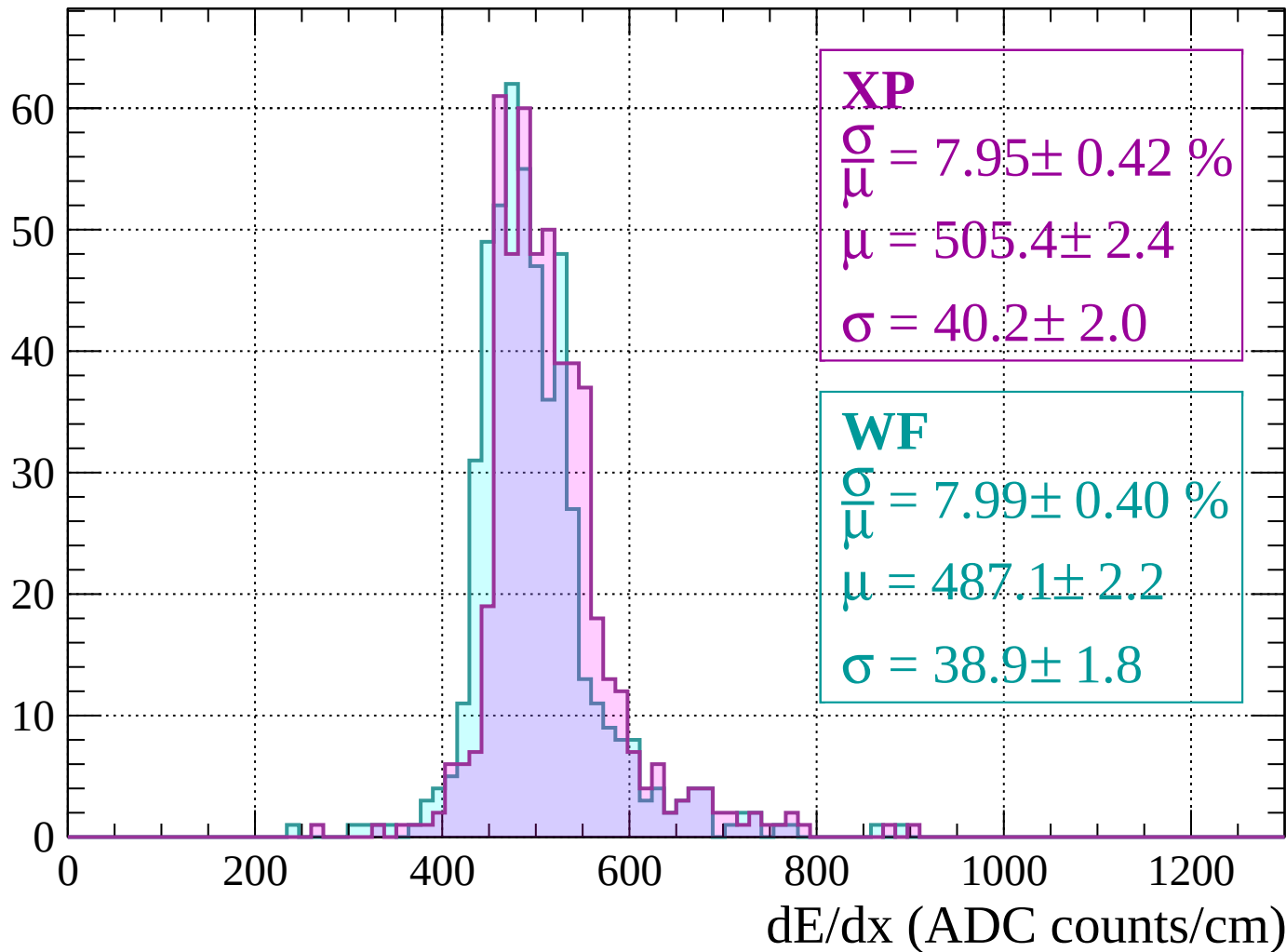






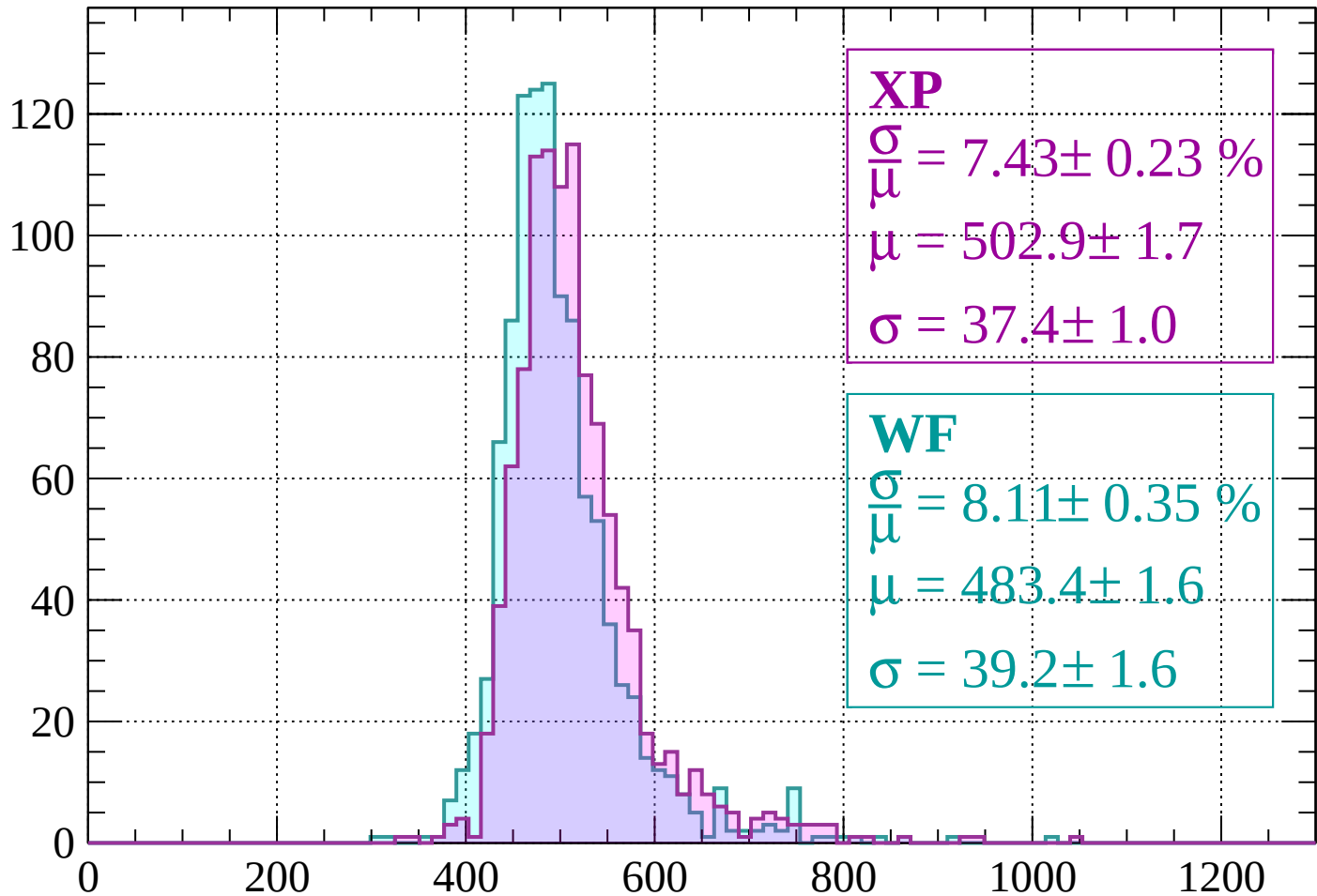
Energy loss | $-2000 < p < -1960$

Count



Energy loss | -1960 < p < -1920

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.43 \pm 0.23 \%$$

$$\mu = 502.9 \pm 1.7$$

$$\sigma = 37.4 \pm 1.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.11 \pm 0.35 \%$$

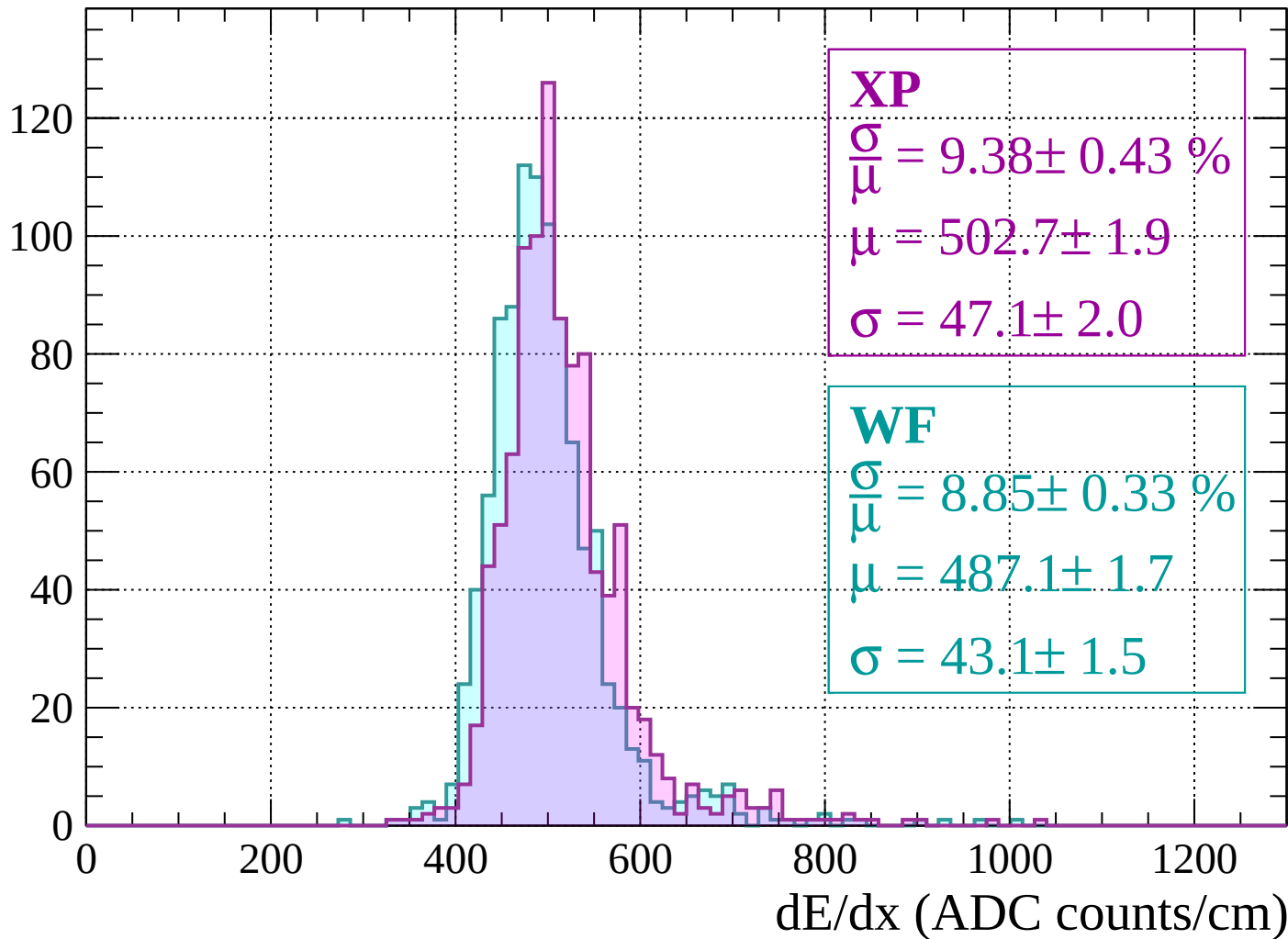
$$\mu = 483.4 \pm 1.6$$

$$\sigma = 39.2 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1920 < p < -1880$

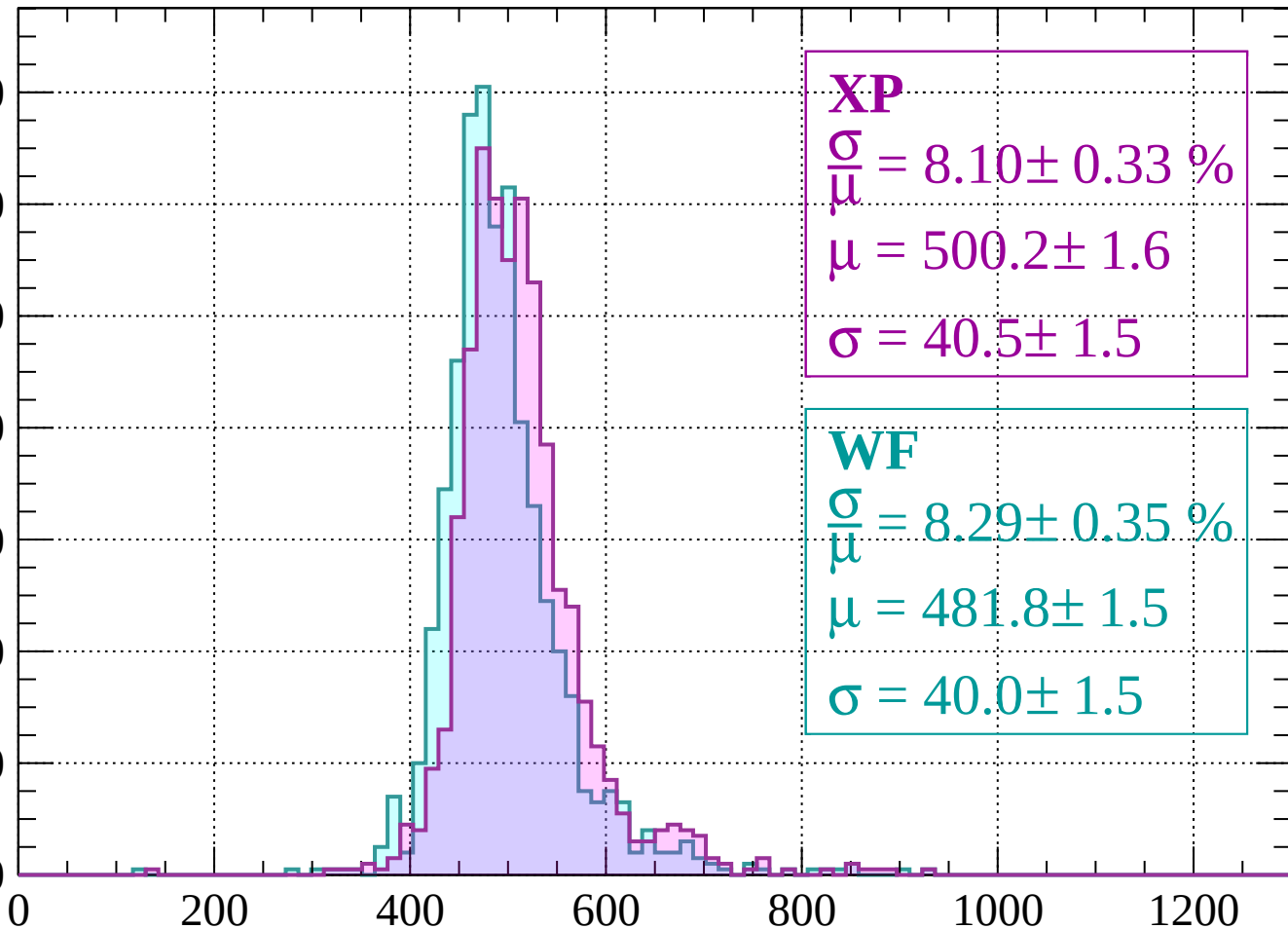
Count



Energy loss | $-1880 < p < -1840$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.10 \pm 0.33 \%$$

$$\mu = 500.2 \pm 1.6$$

$$\sigma = 40.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.35 \%$$

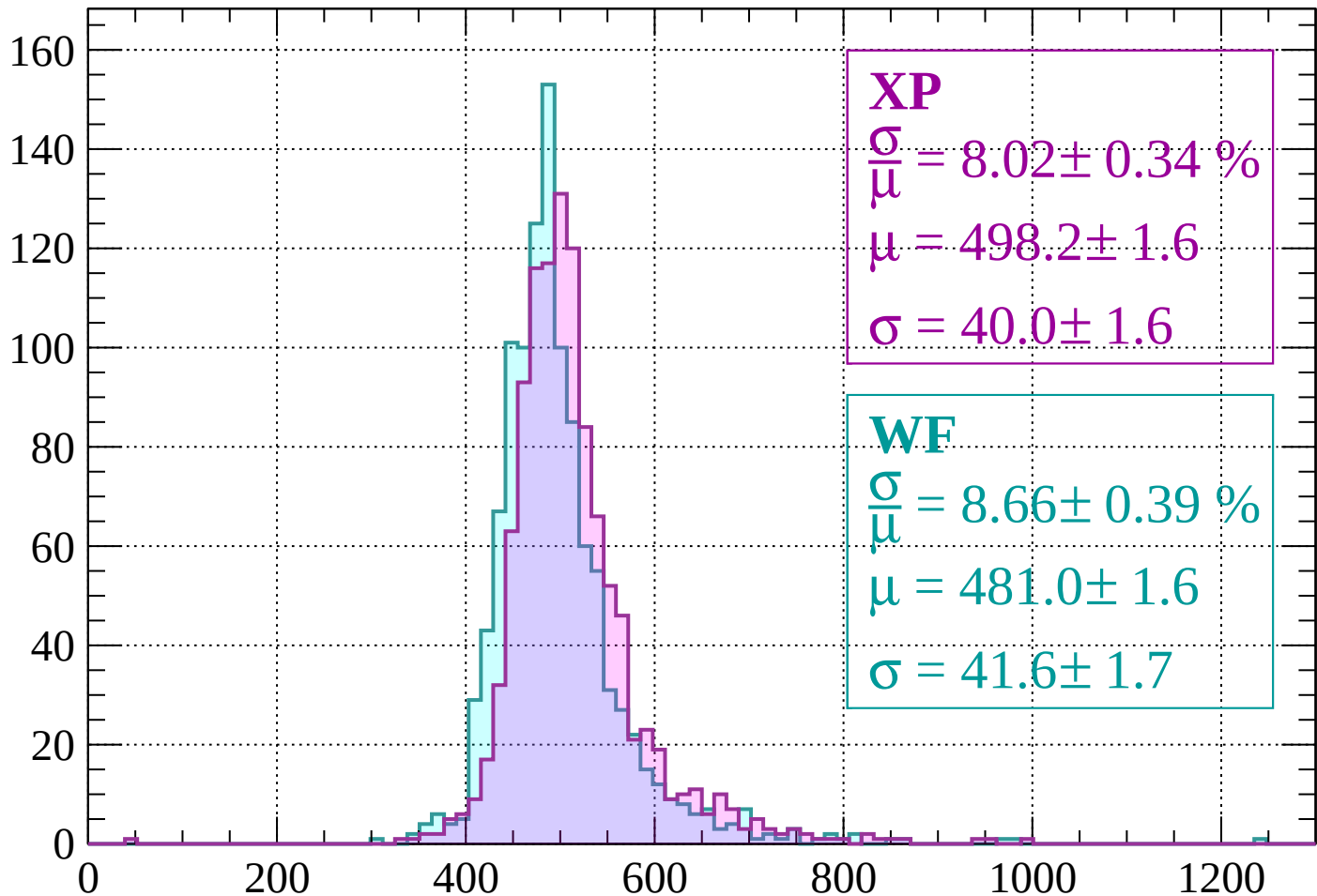
$$\mu = 481.8 \pm 1.5$$

$$\sigma = 40.0 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.34 \%$$

$$\mu = 498.2 \pm 1.6$$

$$\sigma = 40.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.39 \%$$

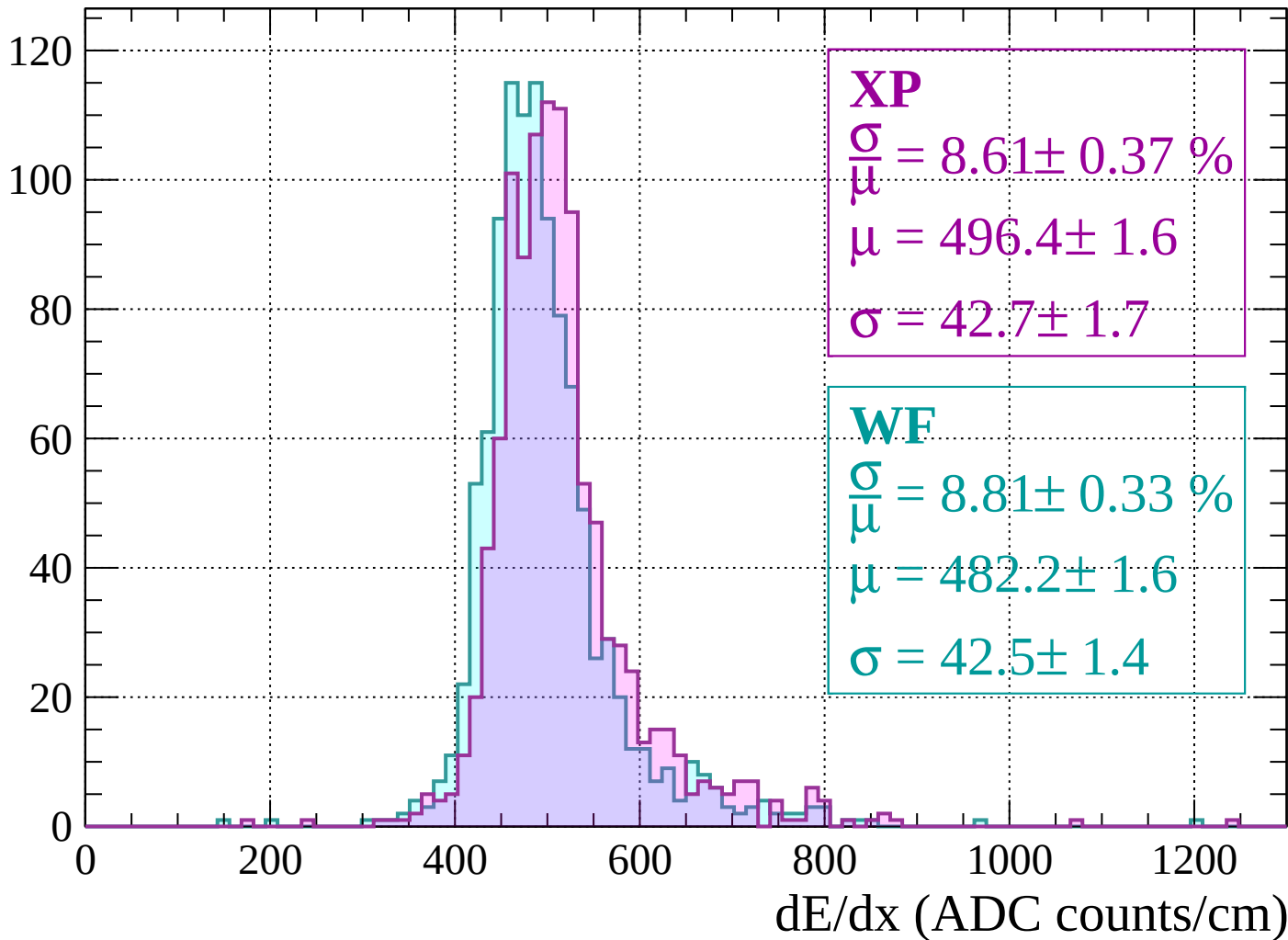
$$\mu = 481.0 \pm 1.6$$

$$\sigma = 41.6 \pm 1.7$$

dE/dx (ADC counts/cm)

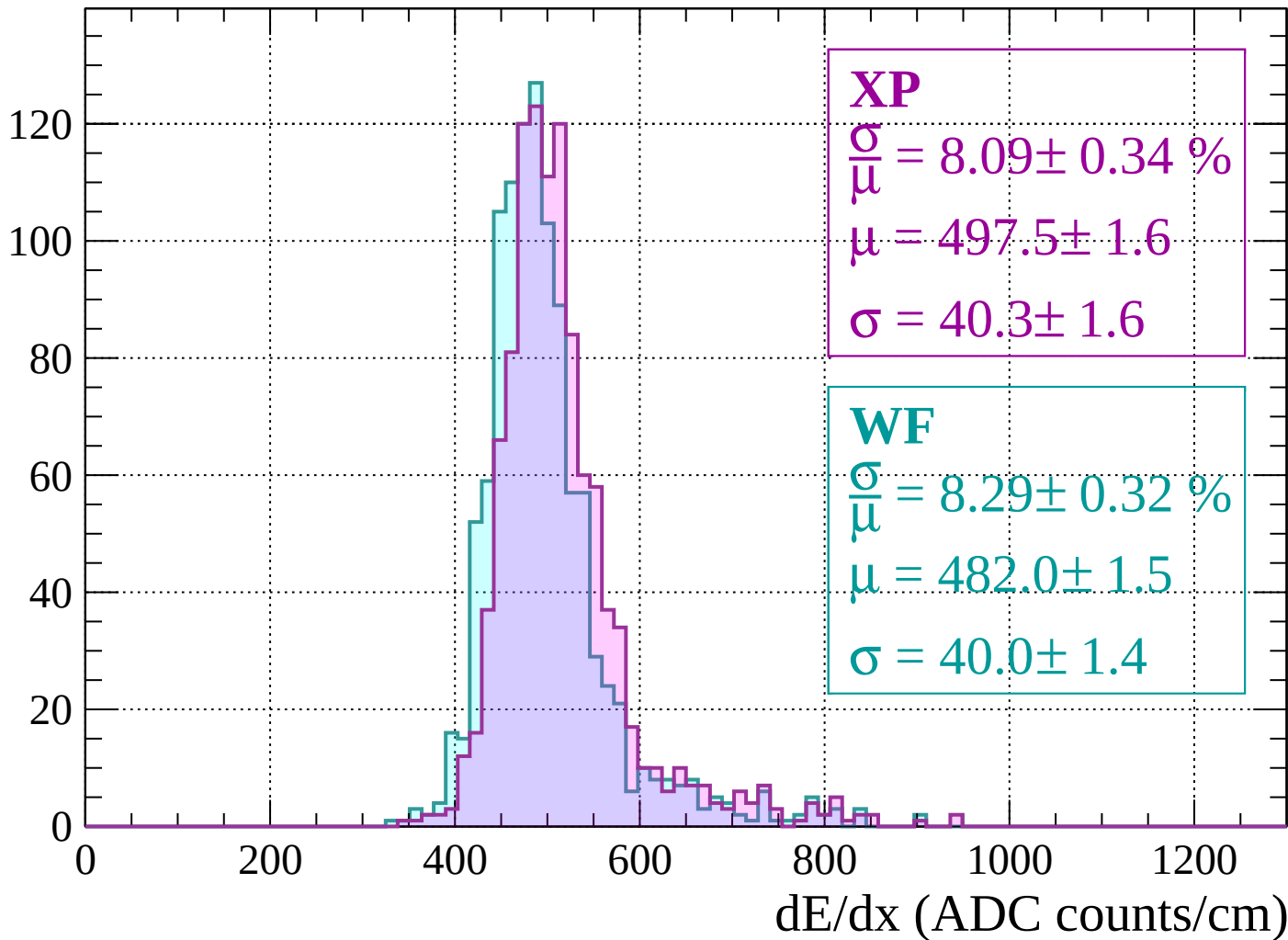
Energy loss | $-1800 < p < -1760$

Count



Energy loss | $-1760 < p < -1720$

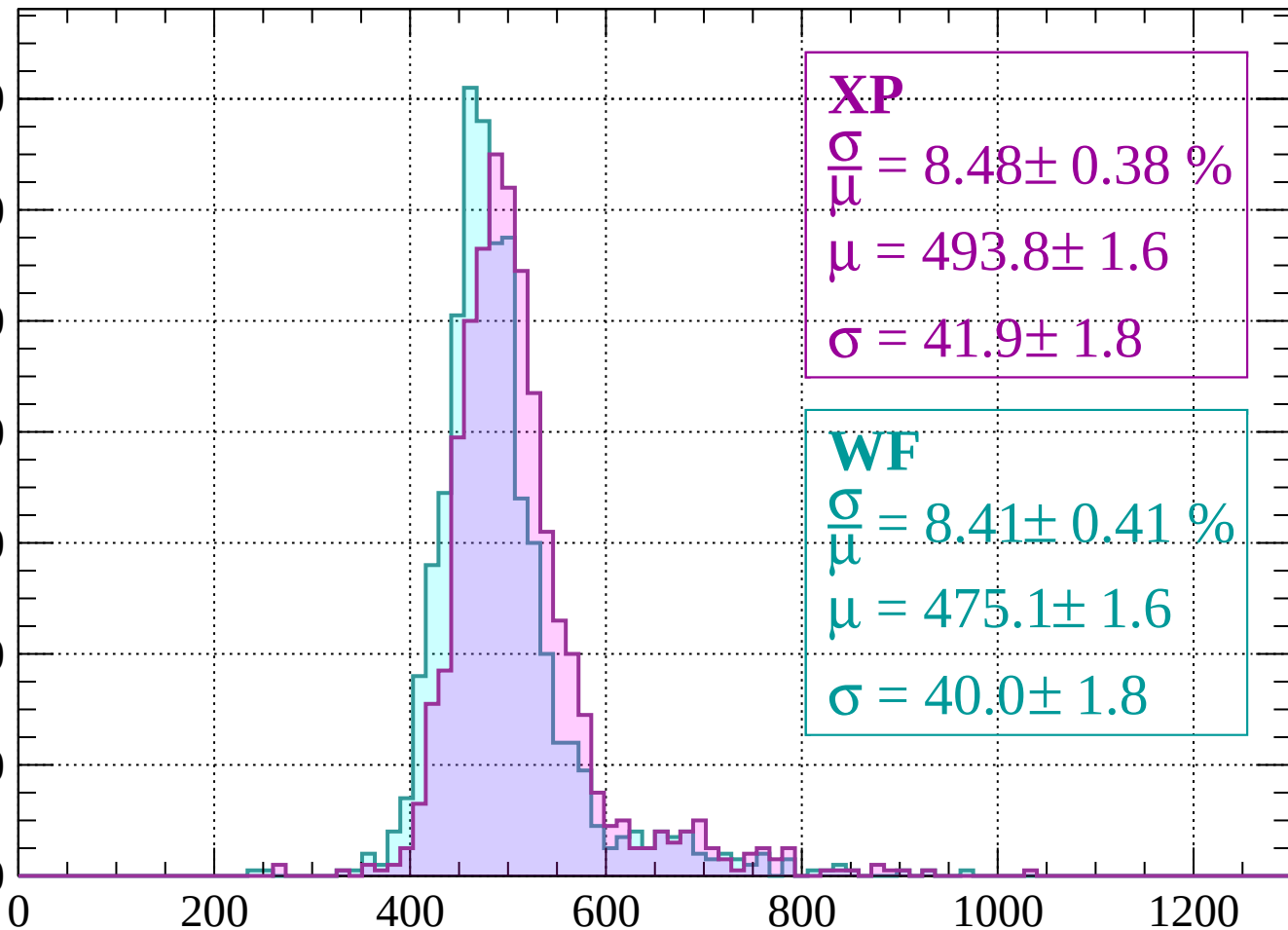
Count



Energy loss | $-1720 < p < -1680$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.48 \pm 0.38 \%$$

$$\mu = 493.8 \pm 1.6$$

$$\sigma = 41.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.41 \pm 0.41 \%$$

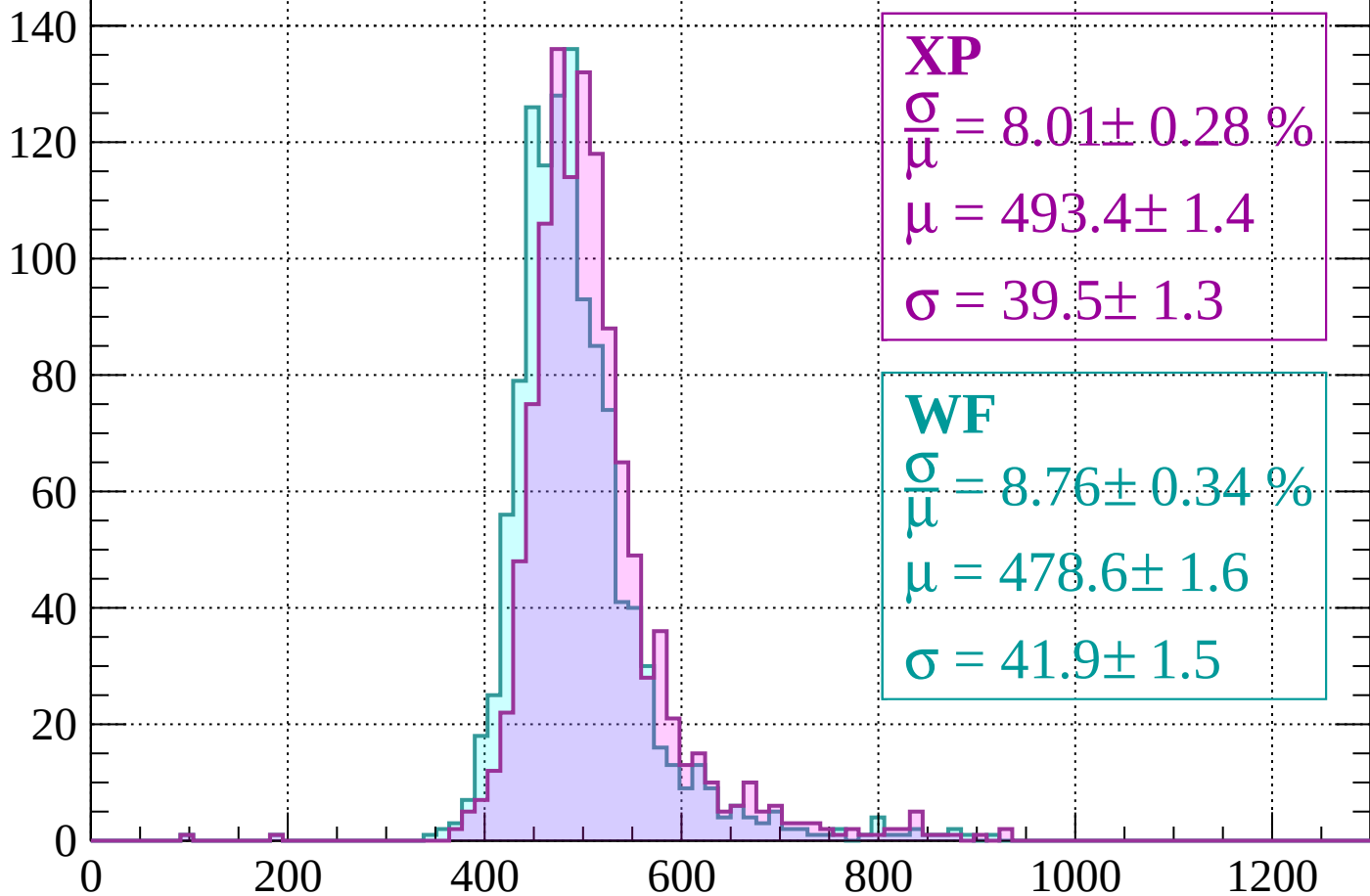
$$\mu = 475.1 \pm 1.6$$

$$\sigma = 40.0 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 8.01 \pm 0.28 \%$$

$$\mu = 493.4 \pm 1.4$$

$$\sigma = 39.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.76 \pm 0.34 \%$$

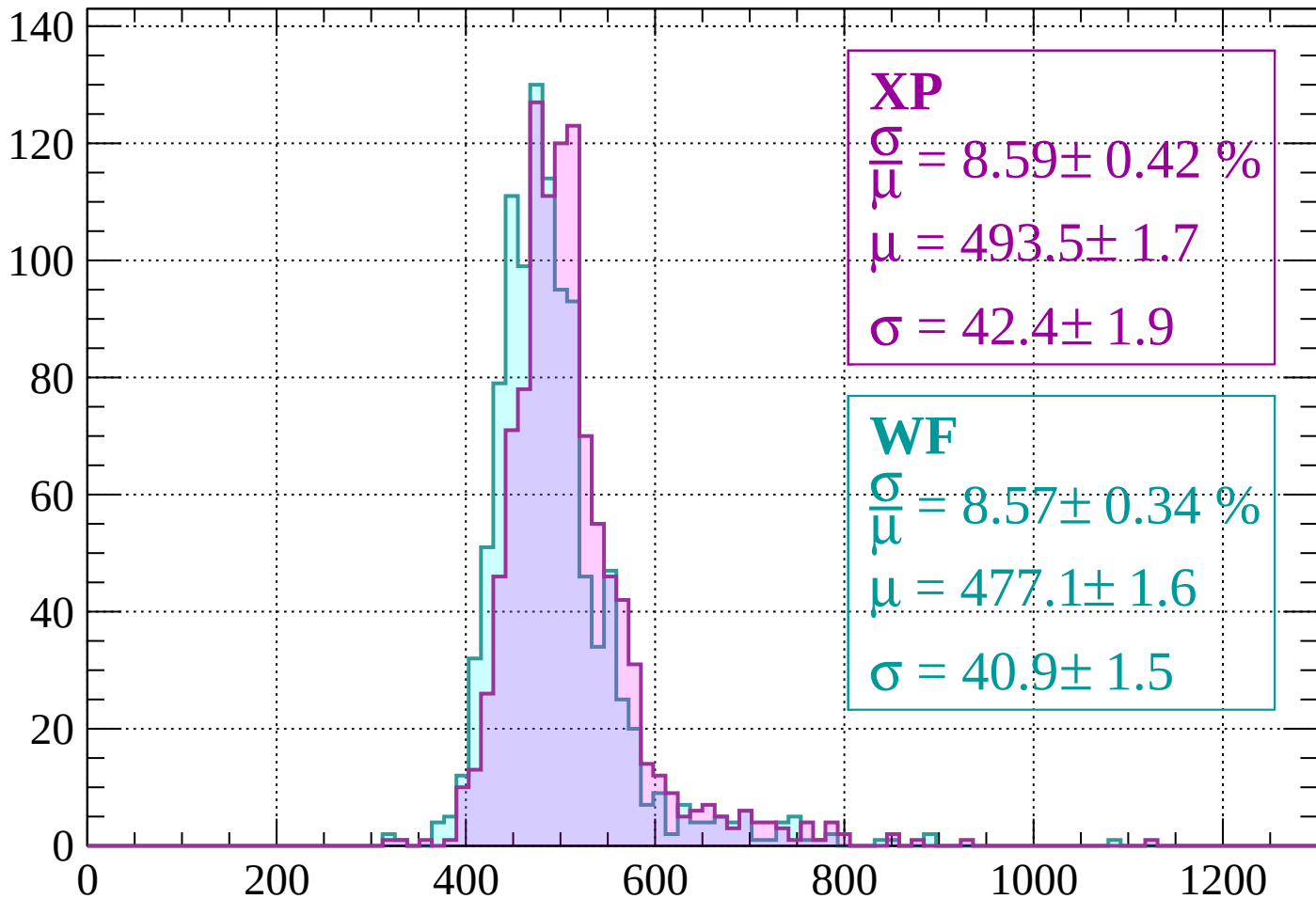
$$\mu = 478.6 \pm 1.6$$

$$\sigma = 41.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 8.59 \pm 0.42 \%$$

$$\mu = 493.5 \pm 1.7$$

$$\sigma = 42.4 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.57 \pm 0.34 \%$$

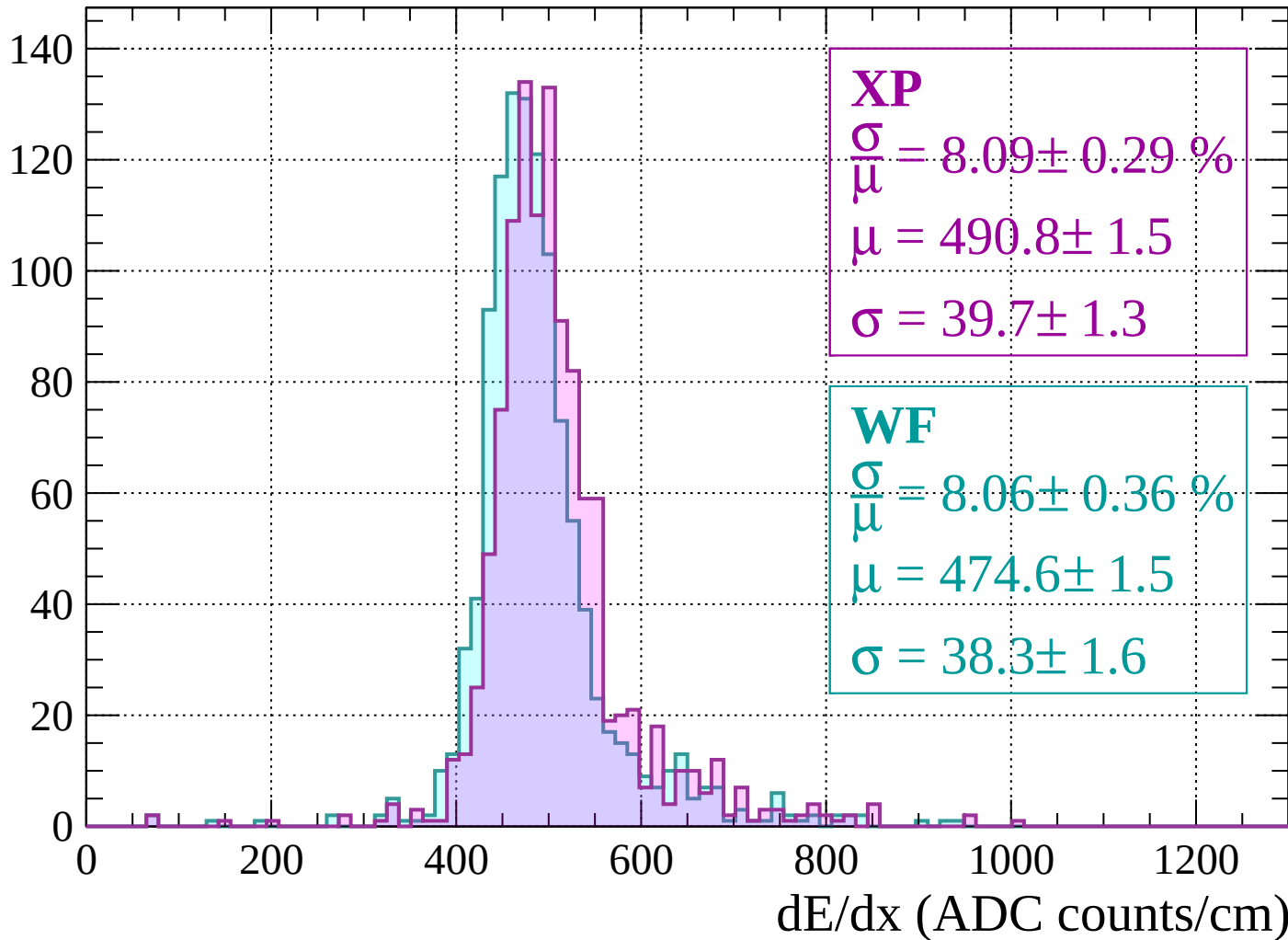
$$\mu = 477.1 \pm 1.6$$

$$\sigma = 40.9 \pm 1.5$$

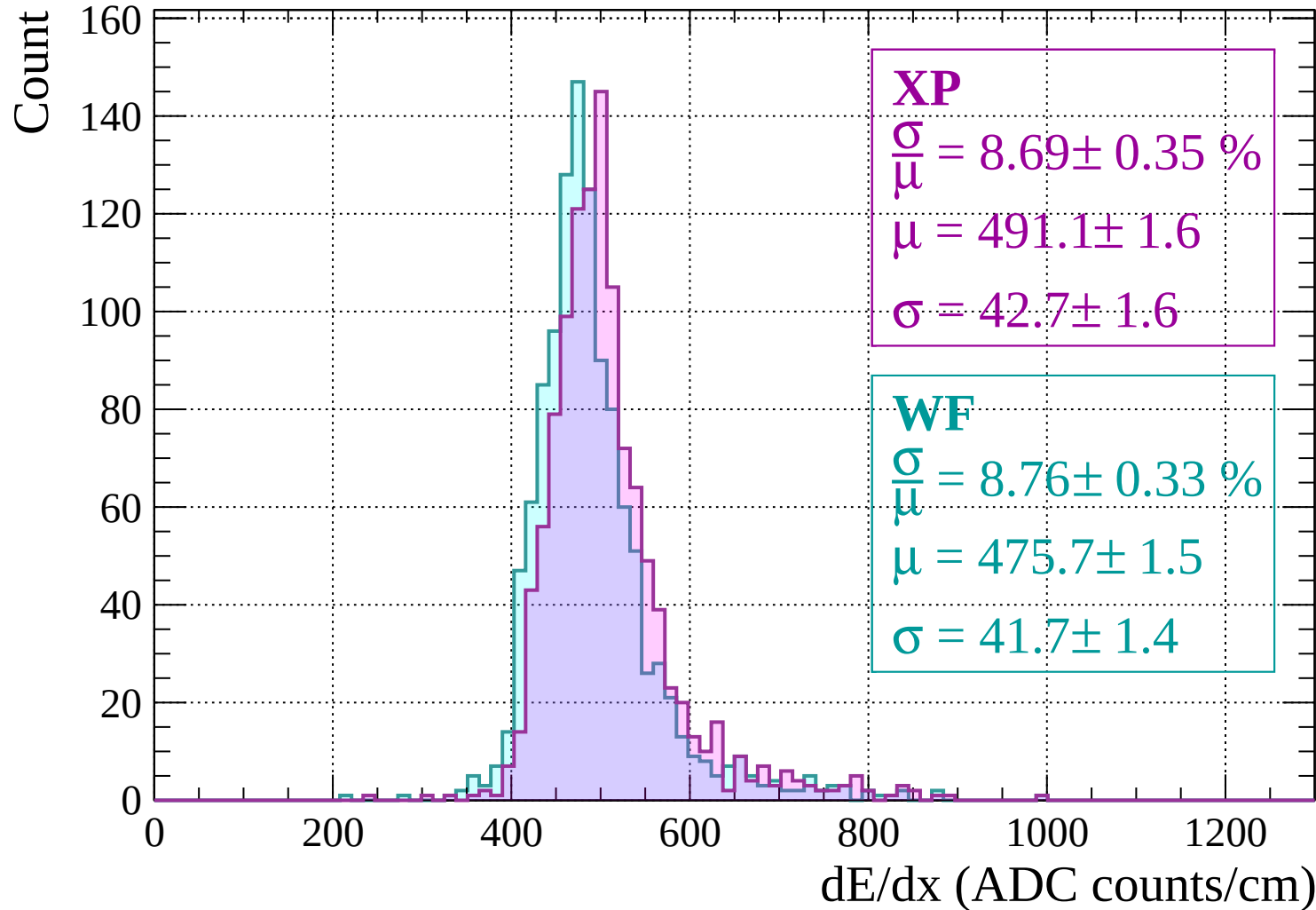
dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1560$

Count

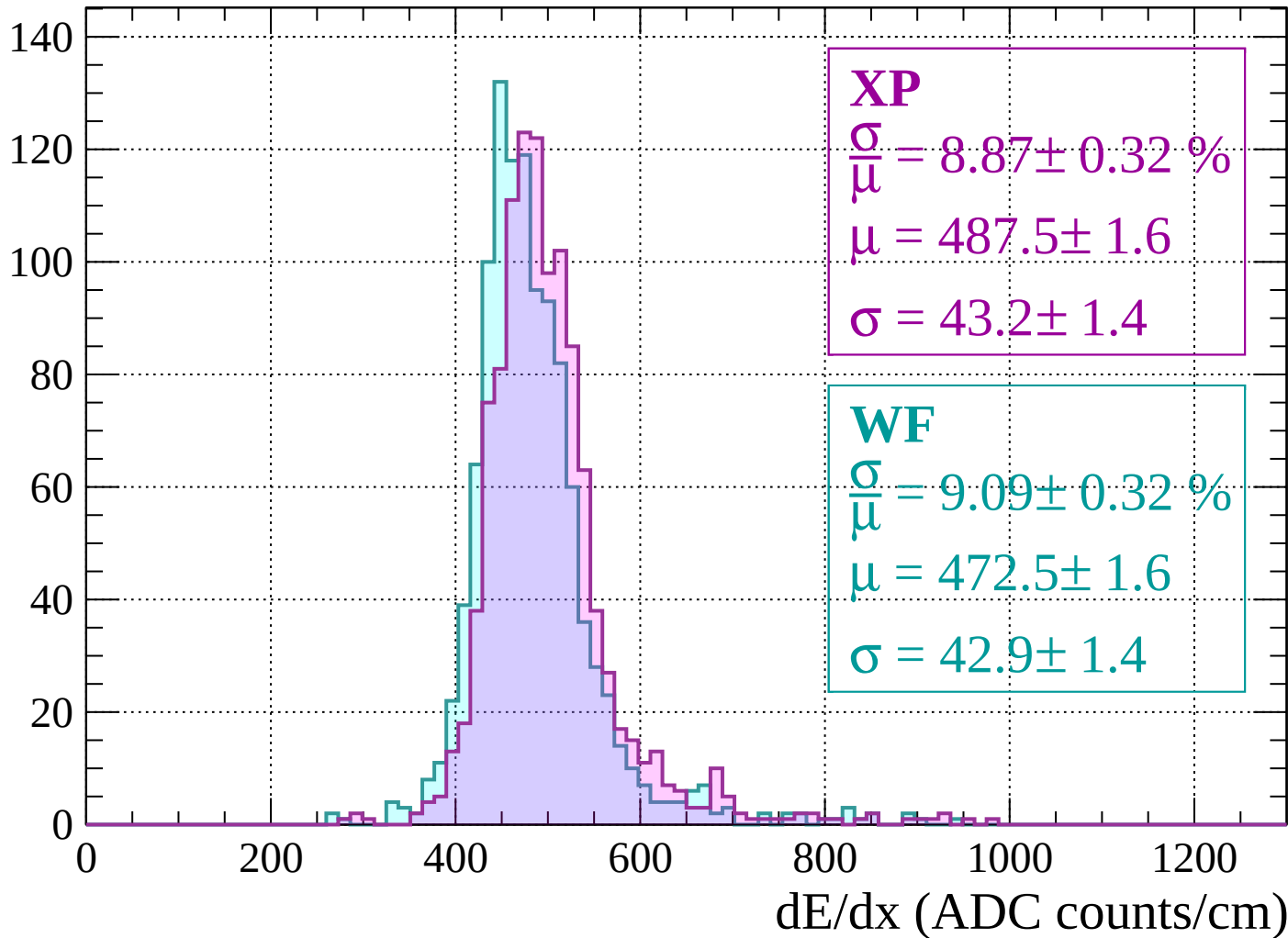


Energy loss | -1560 < p < -1520



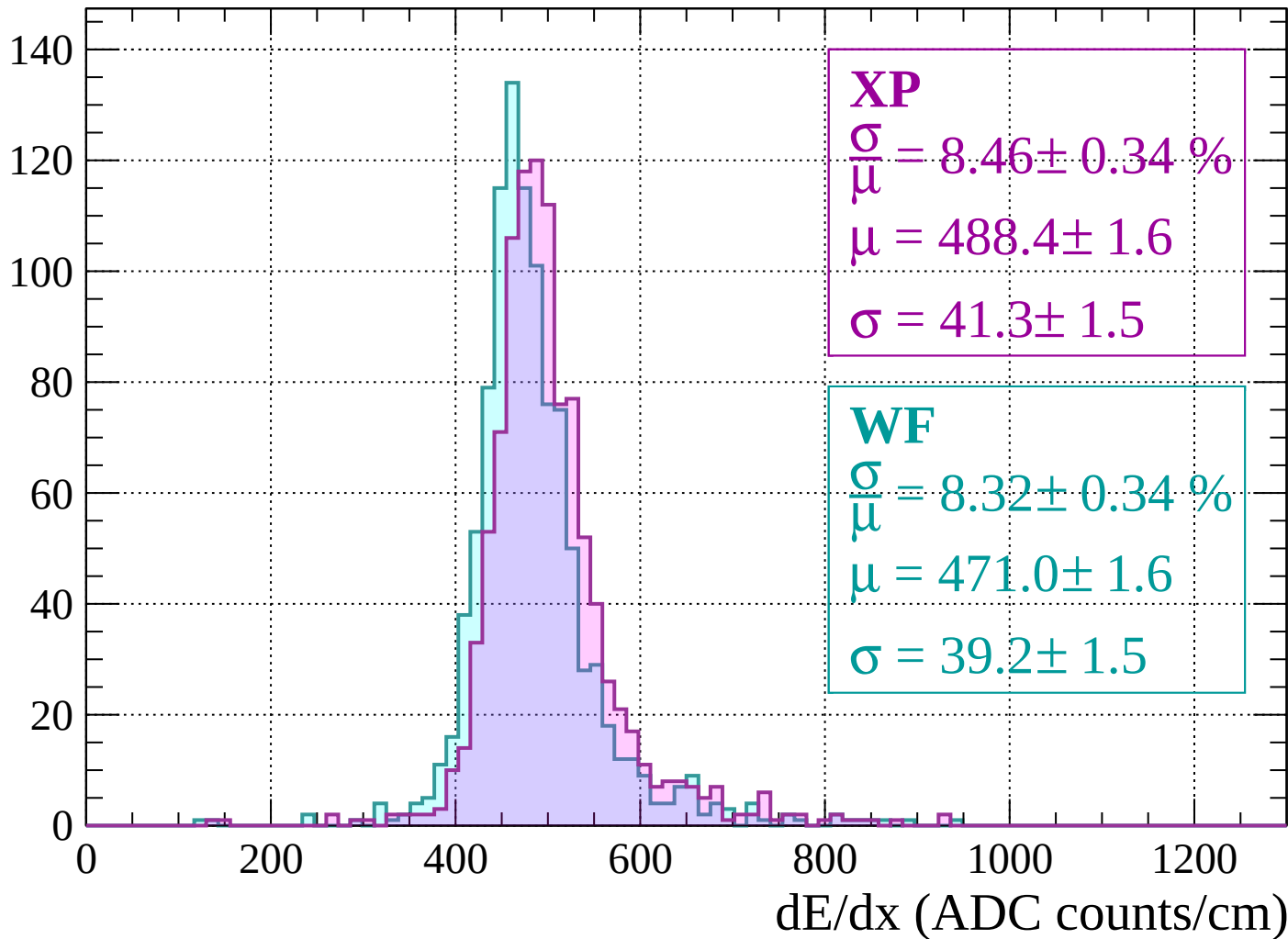
Energy loss | $-1520 < p < -1480$

Count



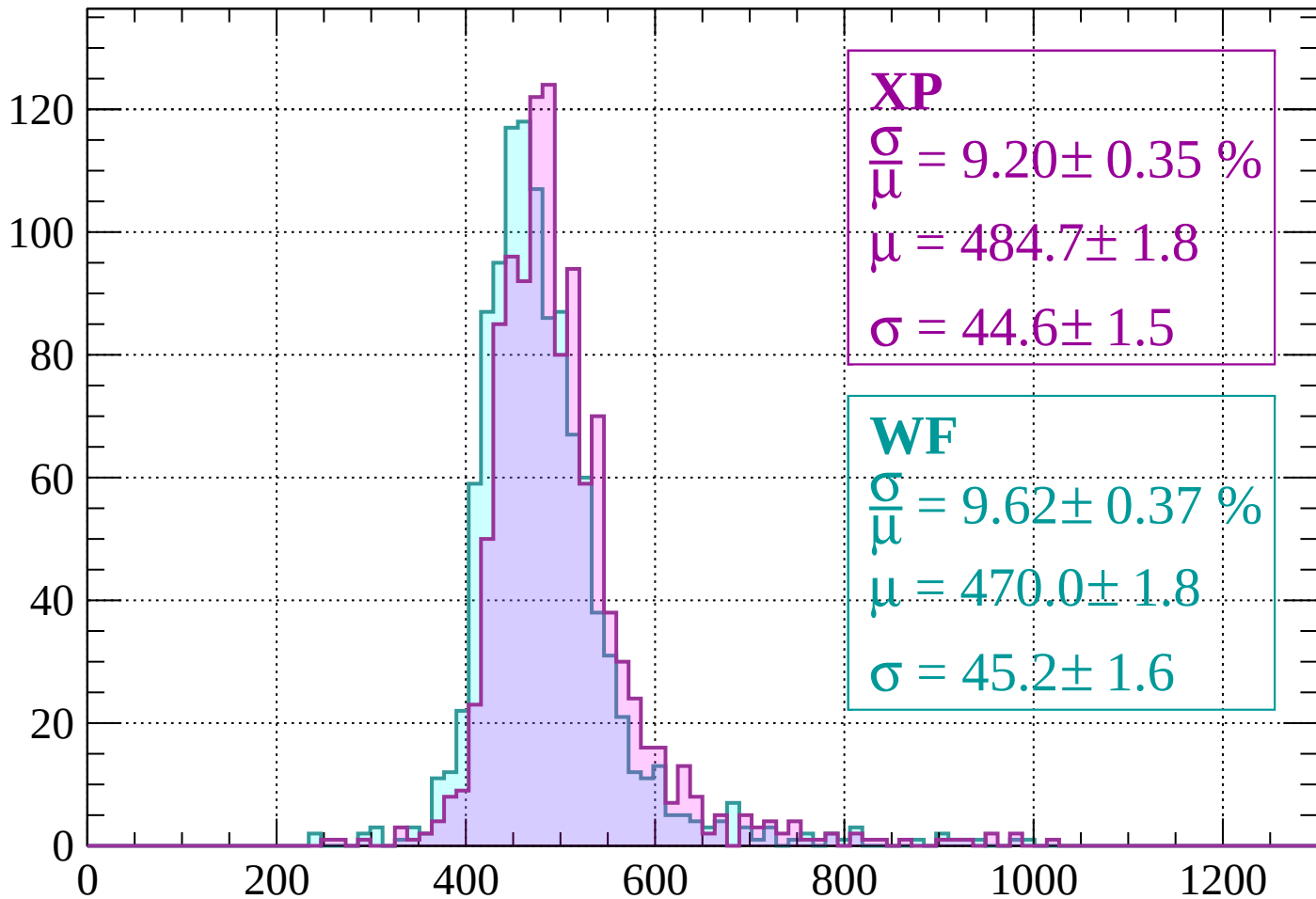
Energy loss | $-1480 < p < -1440$

Count



Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.35 \%$$

$$\mu = 484.7 \pm 1.8$$

$$\sigma = 44.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.62 \pm 0.37 \%$$

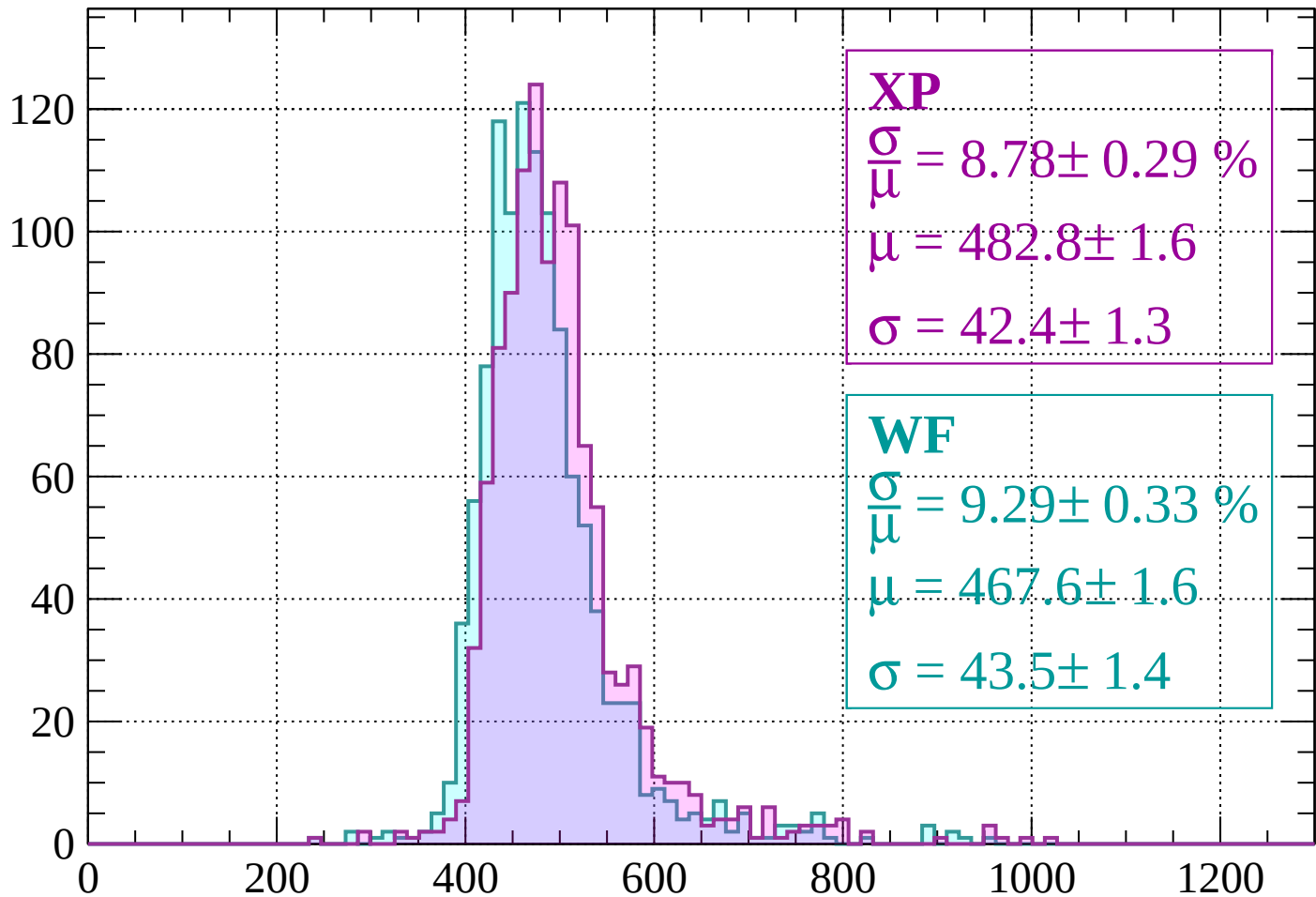
$$\mu = 470.0 \pm 1.8$$

$$\sigma = 45.2 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.29 \%$$

$$\mu = 482.8 \pm 1.6$$

$$\sigma = 42.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.29 \pm 0.33 \%$$

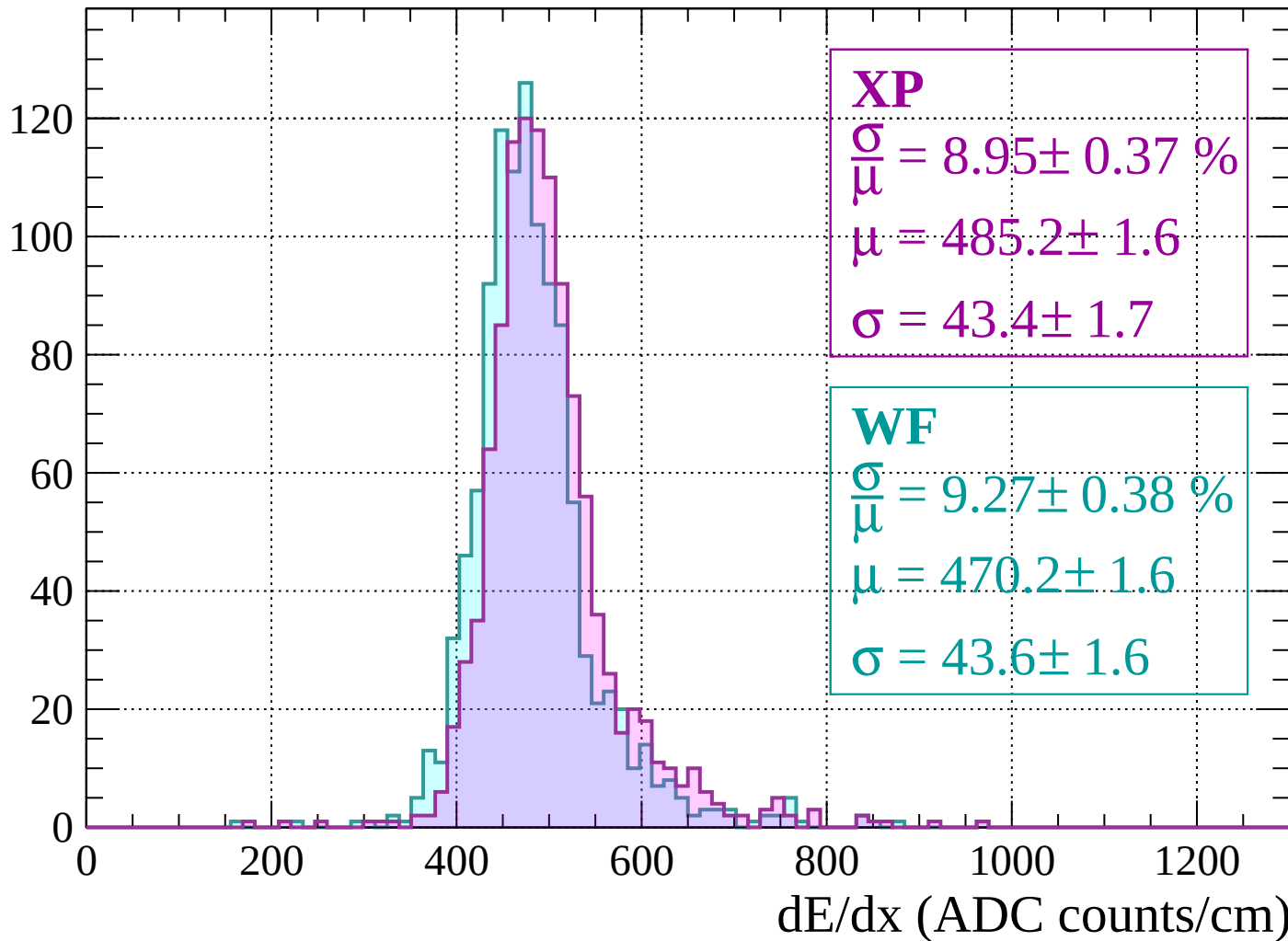
$$\mu = 467.6 \pm 1.6$$

$$\sigma = 43.5 \pm 1.4$$

dE/dx (ADC counts/cm)

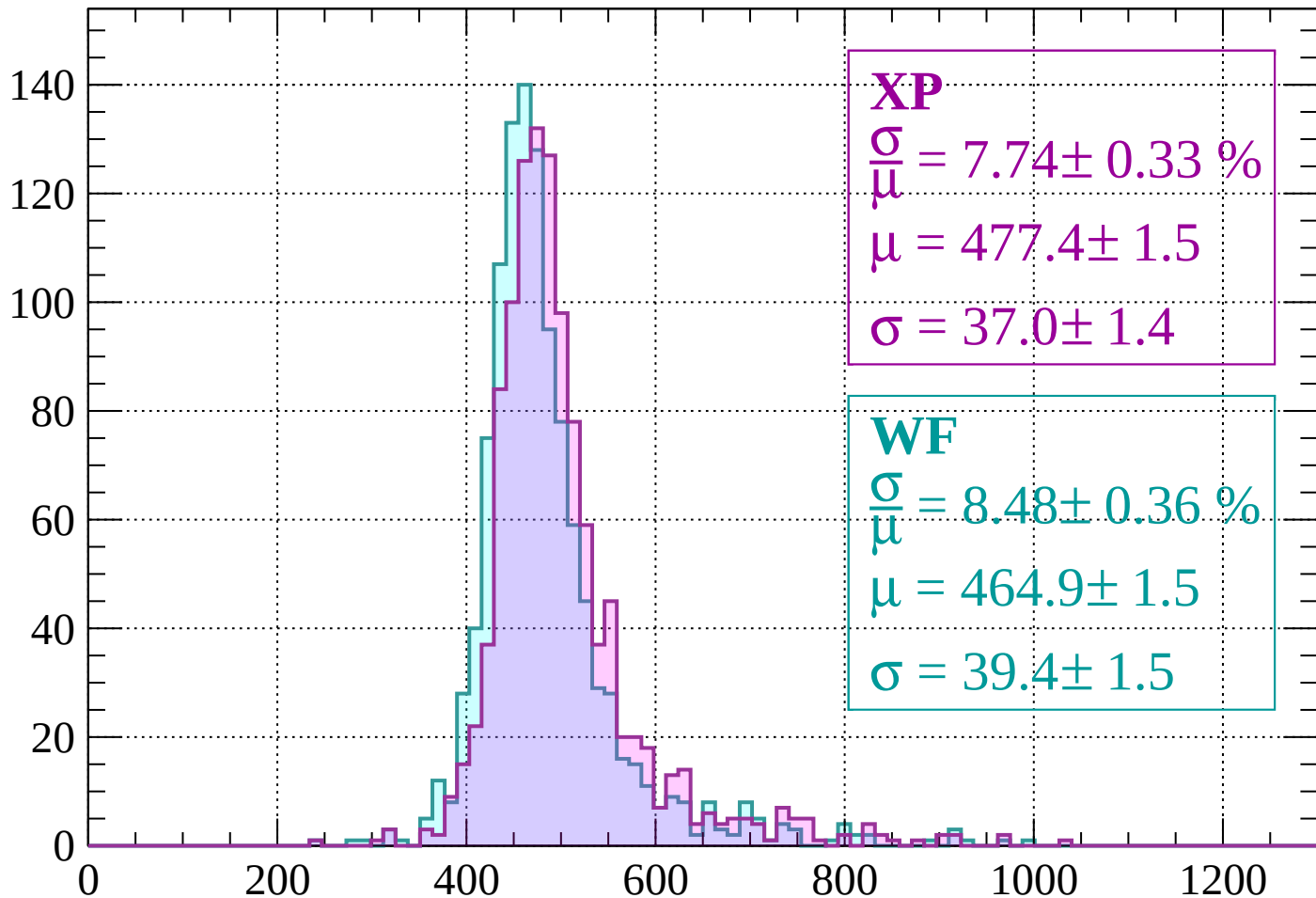
Energy loss | $-1360 < p < -1320$

Count



Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.74 \pm 0.33 \%$$

$$\mu = 477.4 \pm 1.5$$

$$\sigma = 37.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.48 \pm 0.36 \%$$

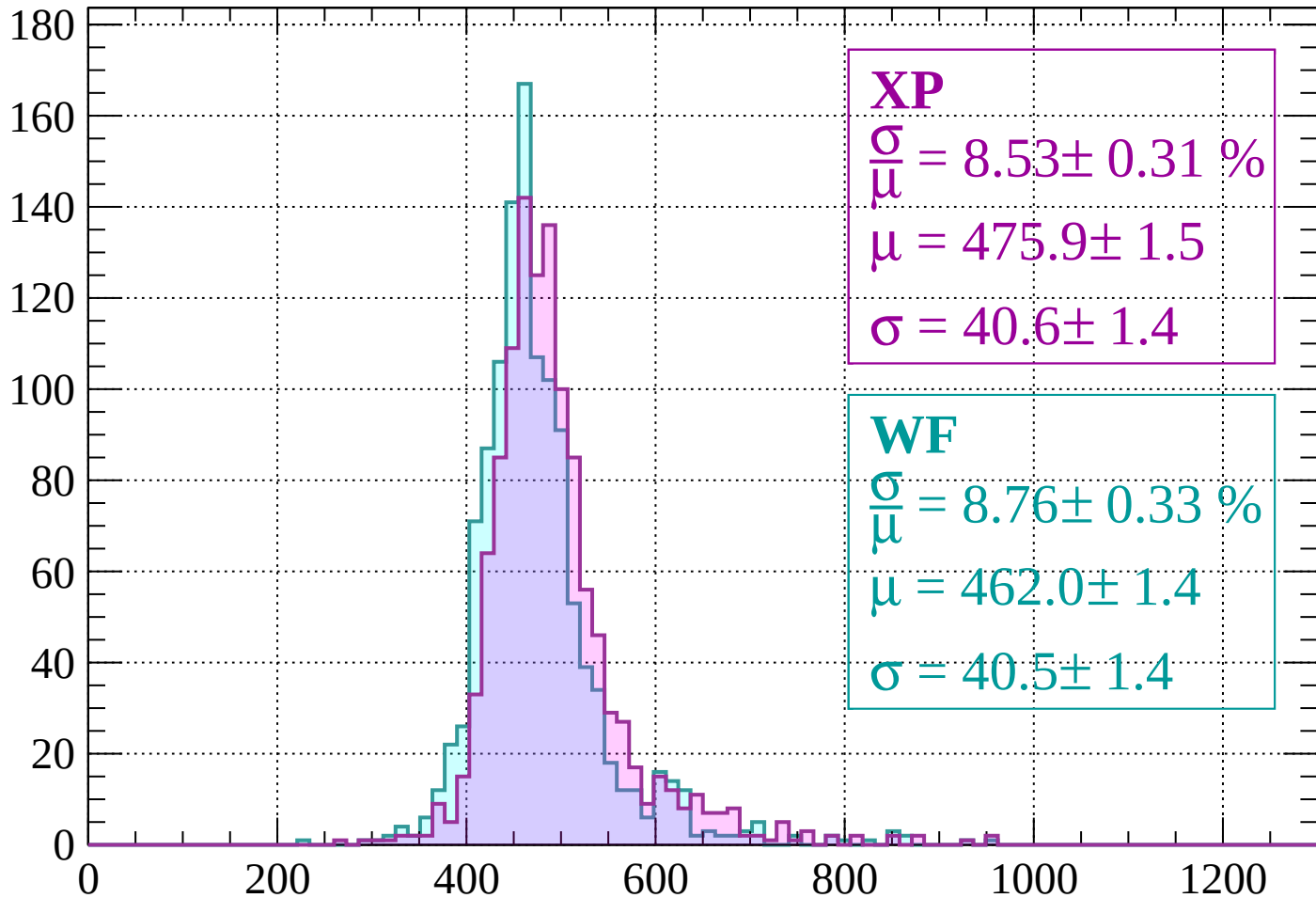
$$\mu = 464.9 \pm 1.5$$

$$\sigma = 39.4 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.53 \pm 0.31 \%$$

$$\mu = 475.9 \pm 1.5$$

$$\sigma = 40.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.76 \pm 0.33 \%$$

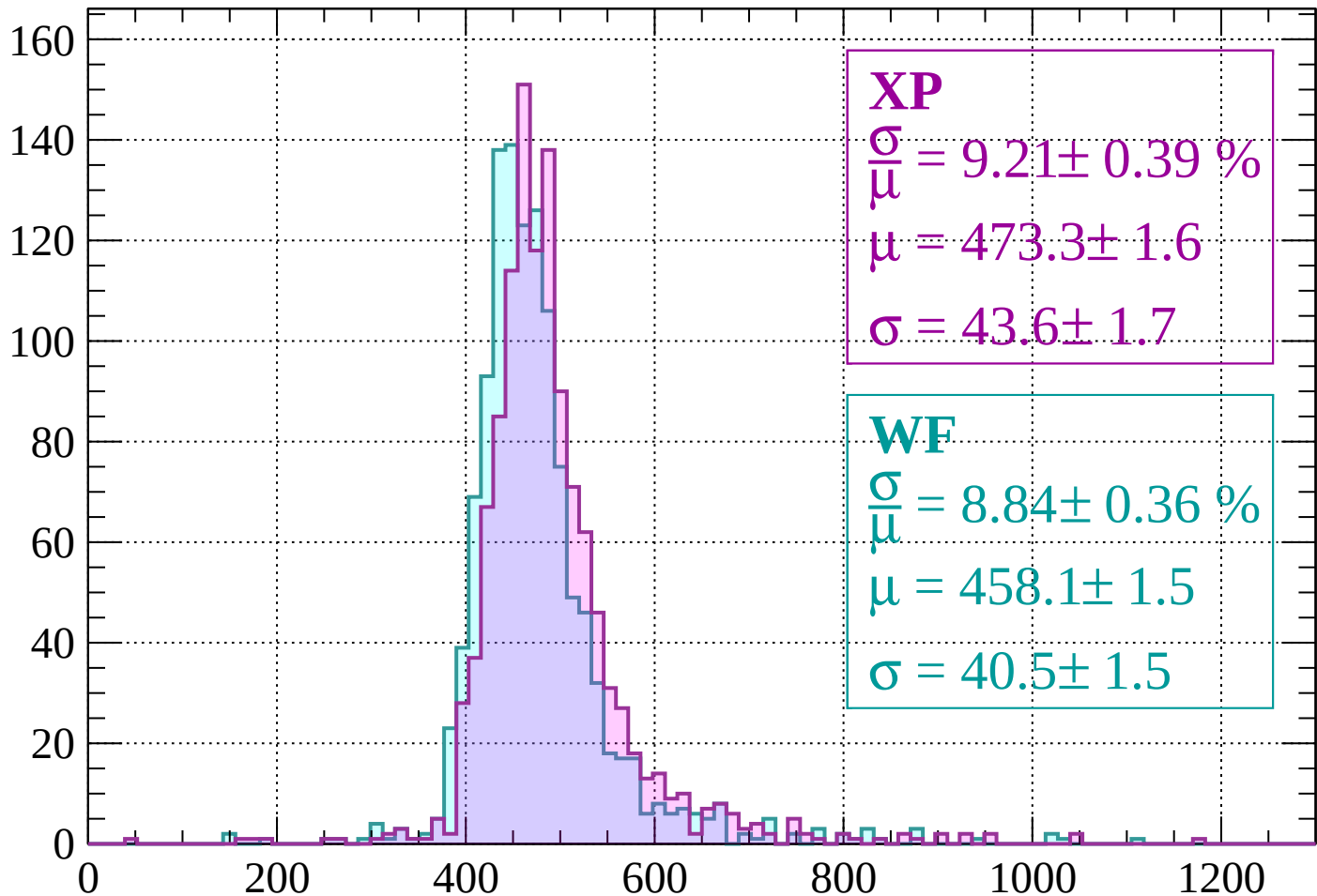
$$\mu = 462.0 \pm 1.4$$

$$\sigma = 40.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.39 \%$$

$$\mu = 473.3 \pm 1.6$$

$$\sigma = 43.6 \pm 1.7$$

WF

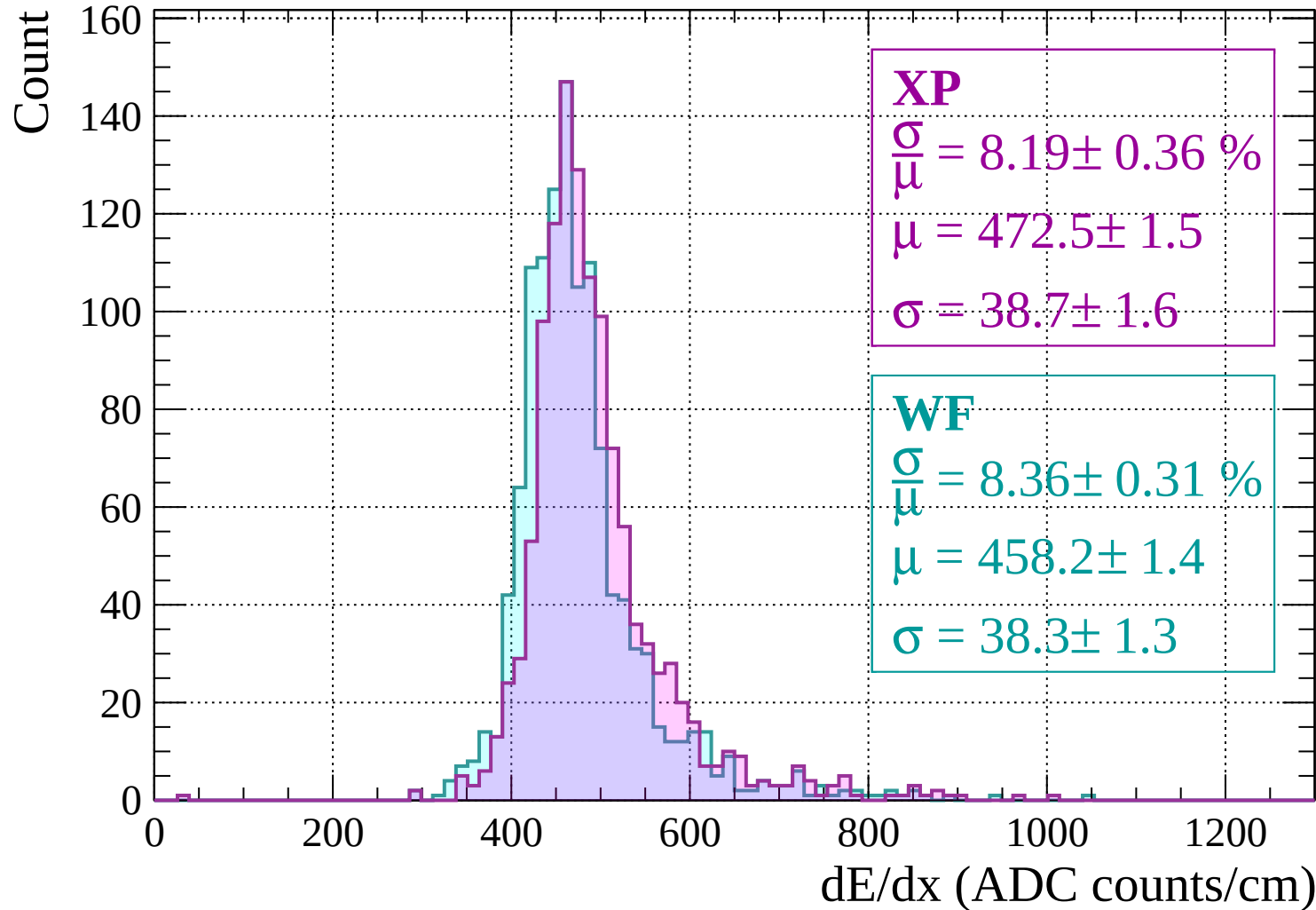
$$\frac{\sigma}{\mu} = 8.84 \pm 0.36 \%$$

$$\mu = 458.1 \pm 1.5$$

$$\sigma = 40.5 \pm 1.5$$

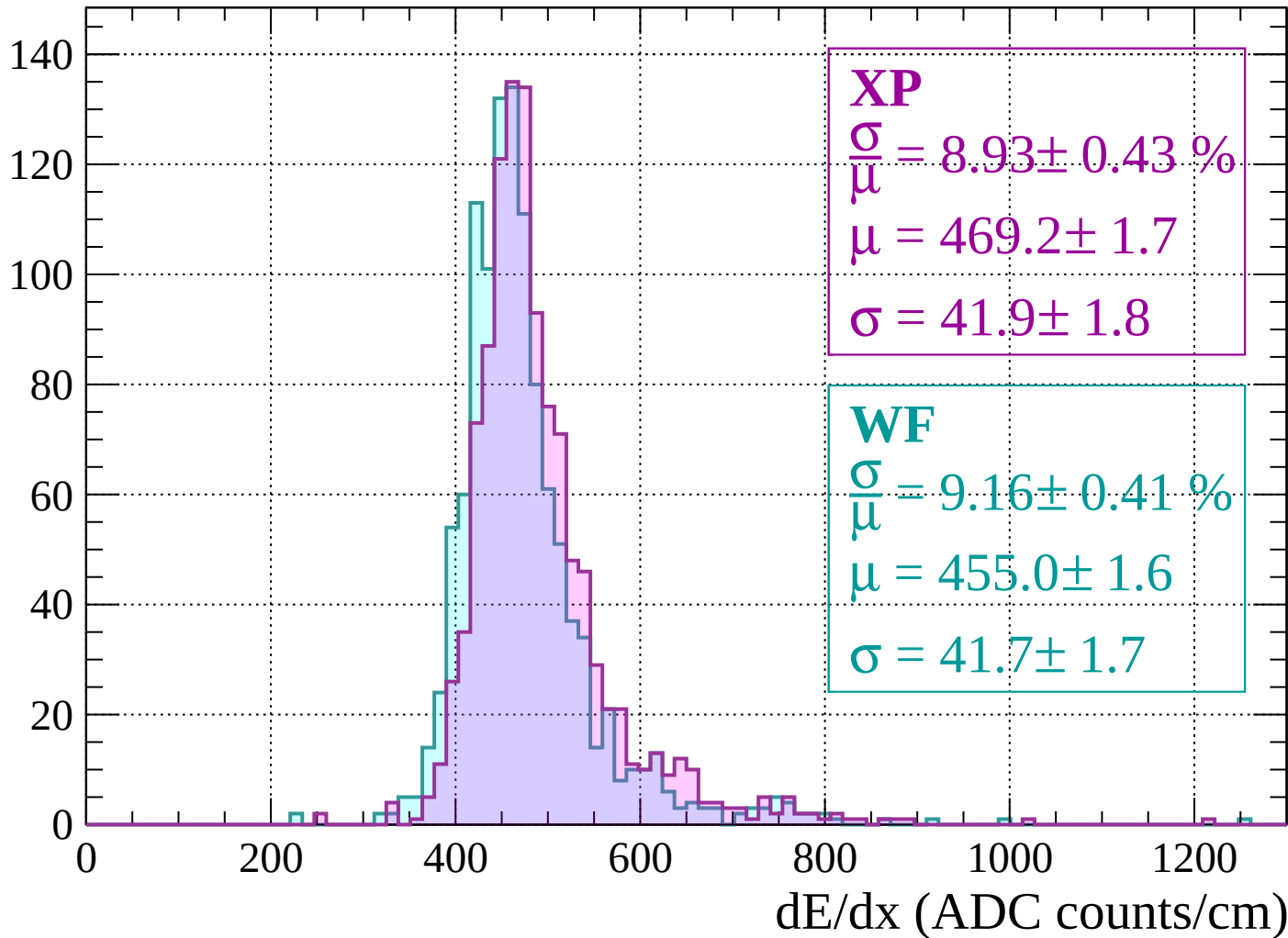
dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160

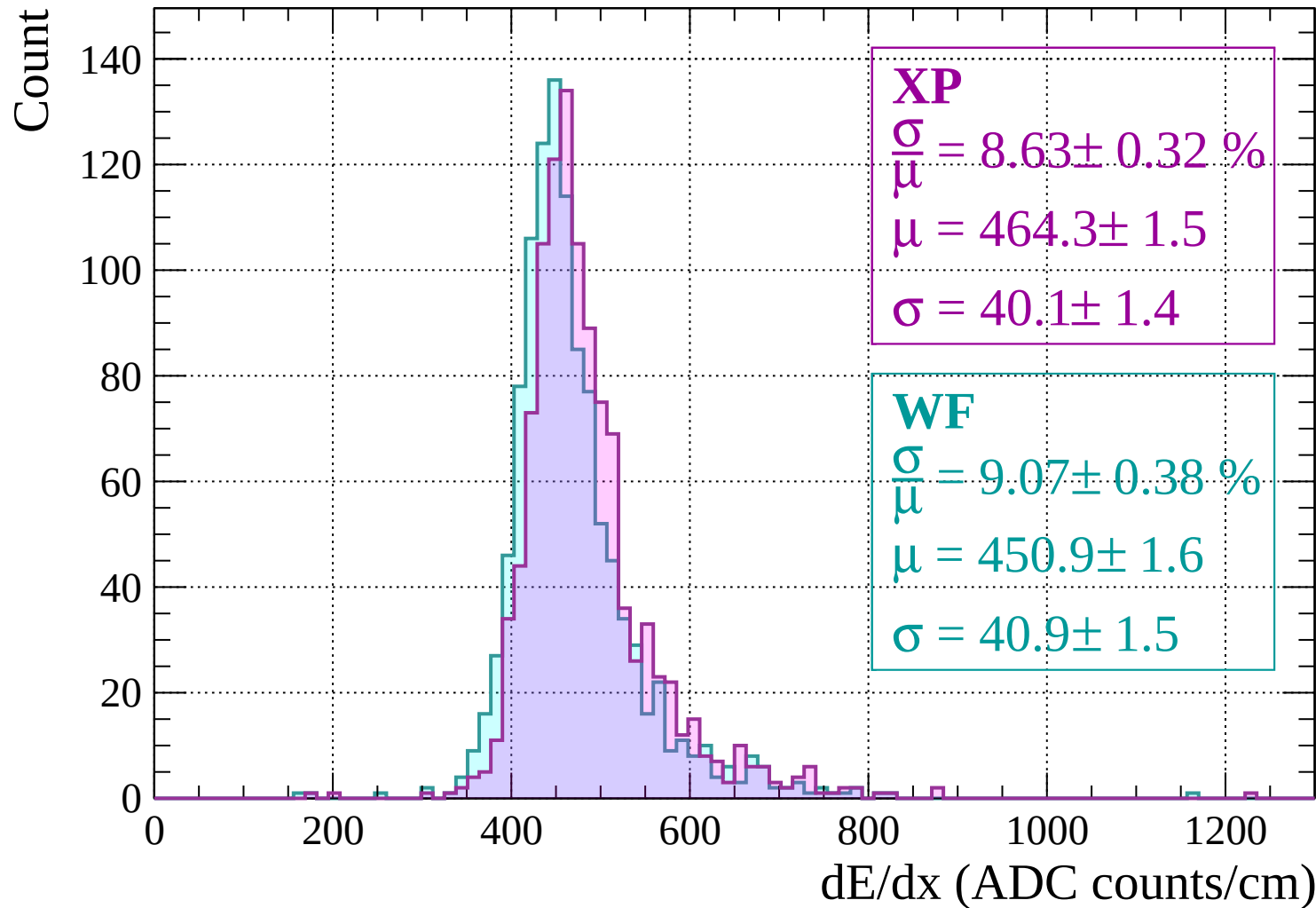


Energy loss | $-1160 < p < -1120$

Count

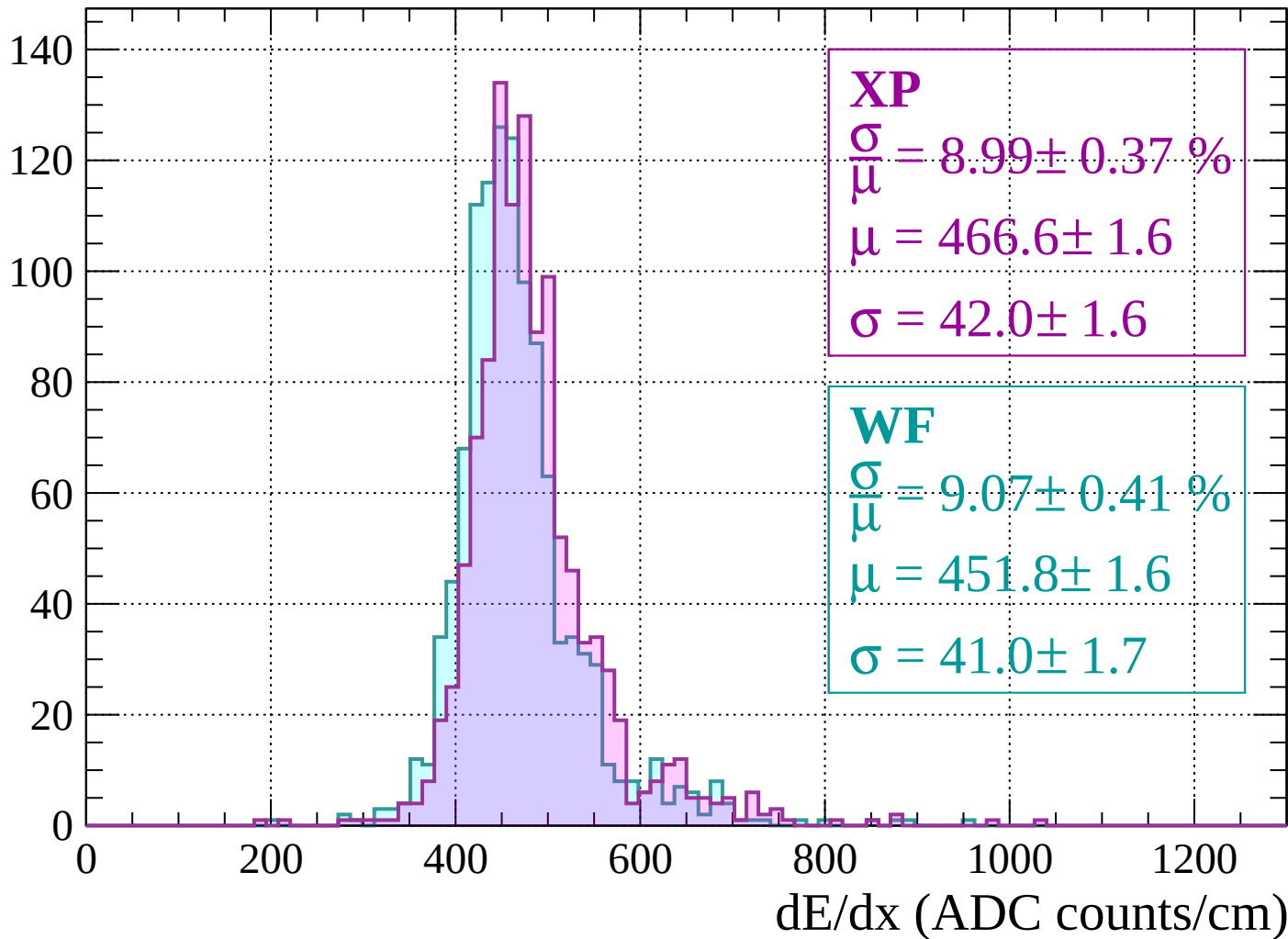


Energy loss | -1120 < p < -1080



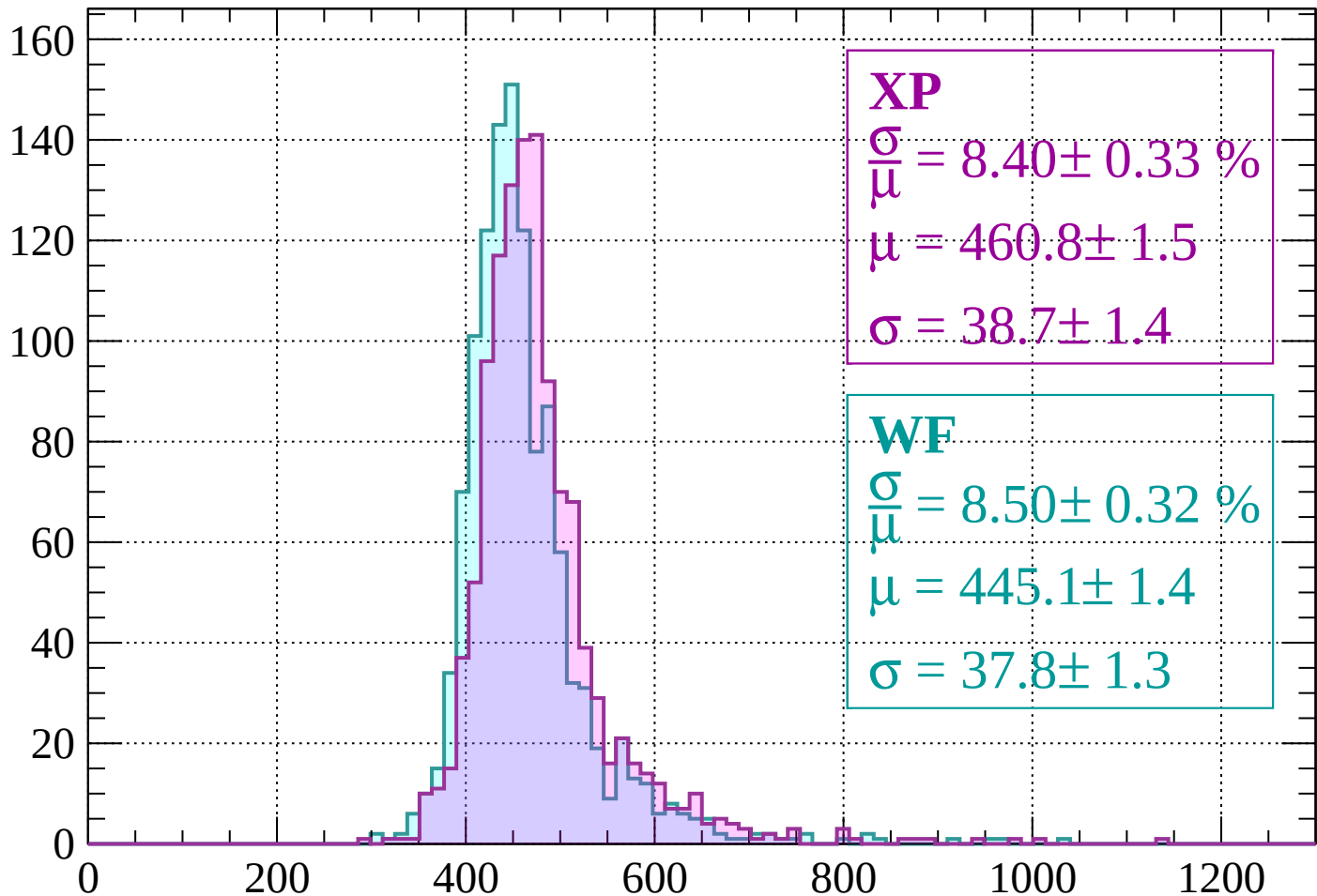
Energy loss | -1080 < p < -1040

Count



Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.33 \%$$

$$\mu = 460.8 \pm 1.5$$

$$\sigma = 38.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.32 \%$$

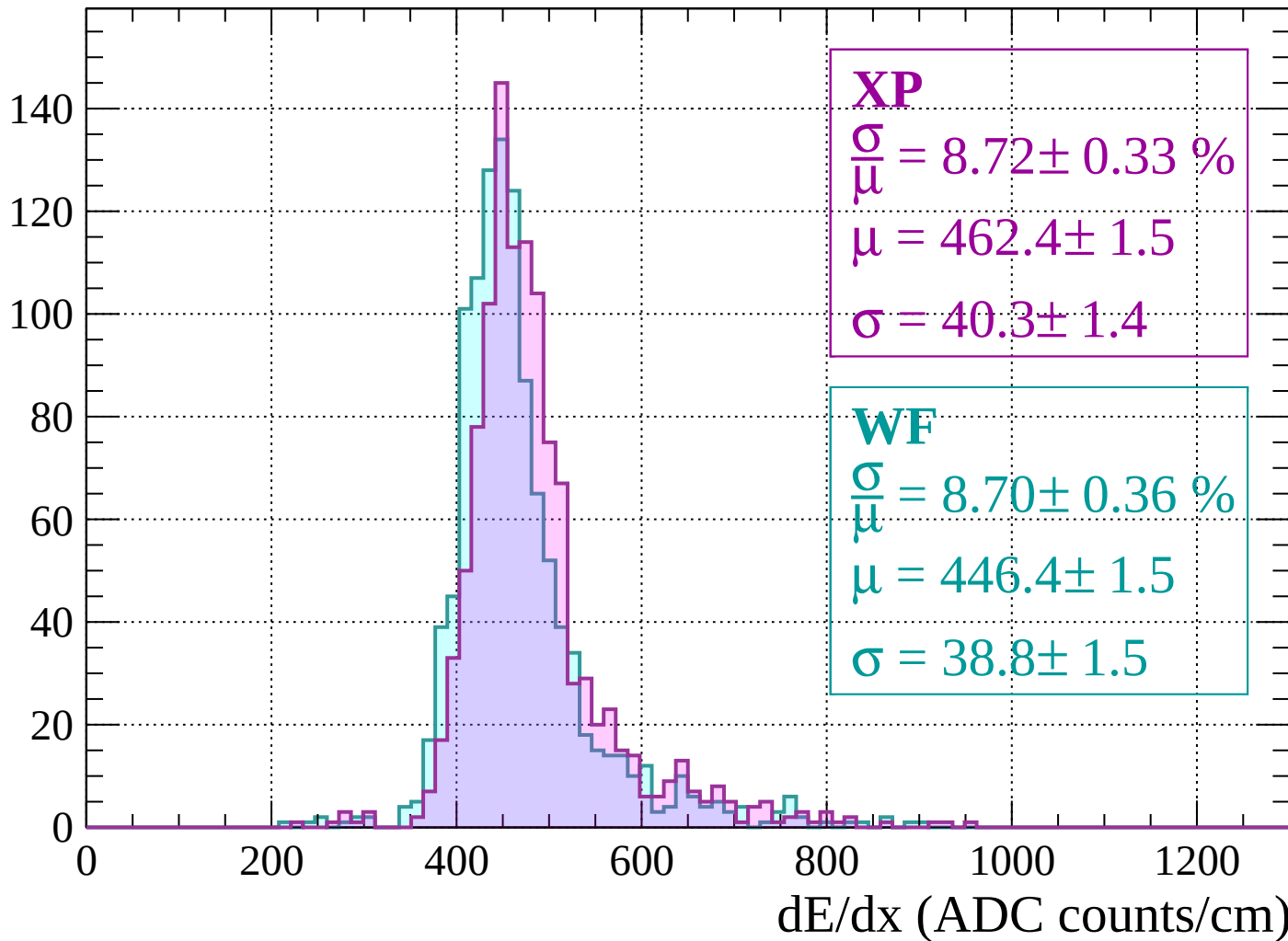
$$\mu = 445.1 \pm 1.4$$

$$\sigma = 37.8 \pm 1.3$$

dE/dx (ADC counts/cm)

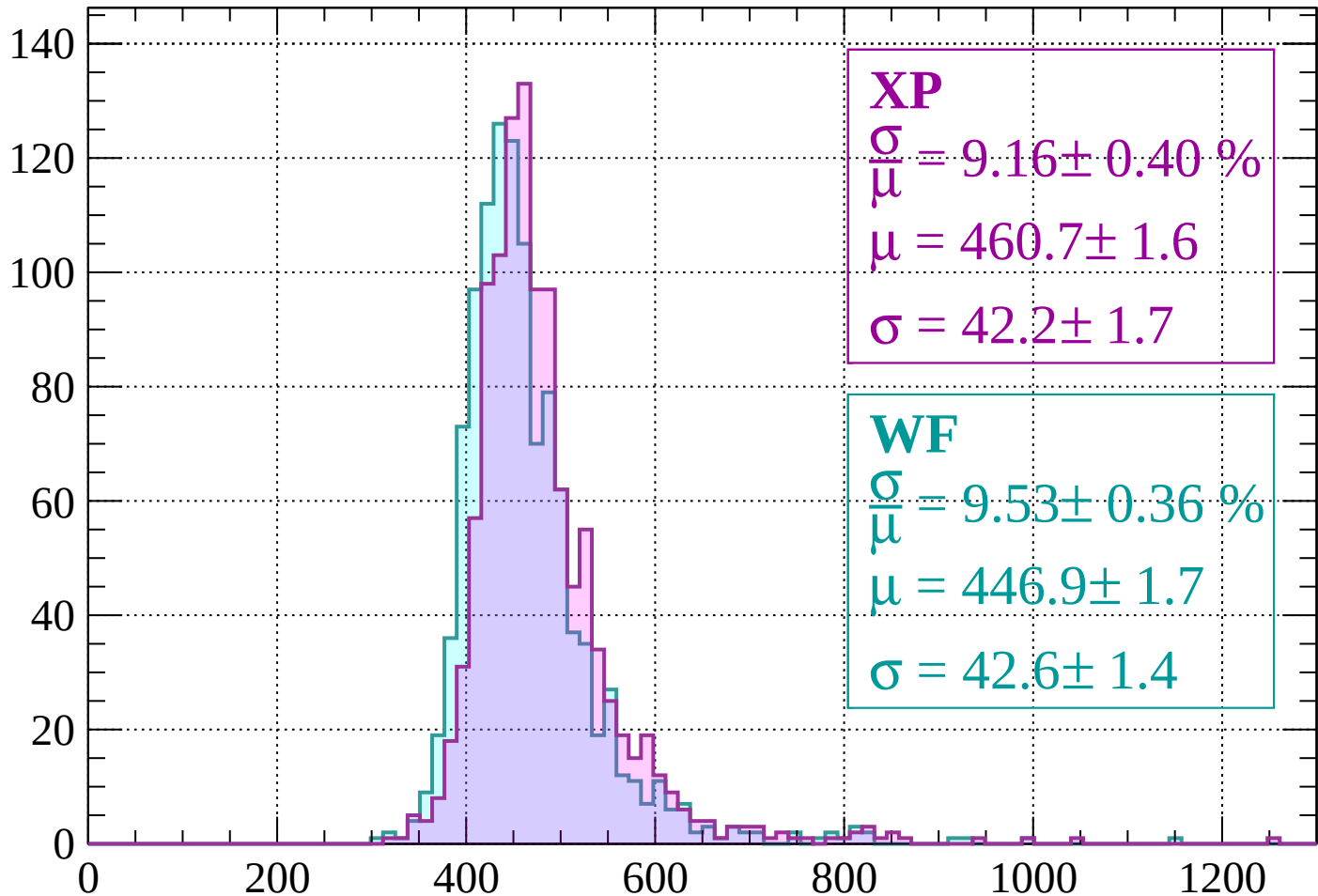
Energy loss | $-1000 < p < -960$

Count



Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.40 \%$$

$$\mu = 460.7 \pm 1.6$$

$$\sigma = 42.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.53 \pm 0.36 \%$$

$$\mu = 446.9 \pm 1.7$$

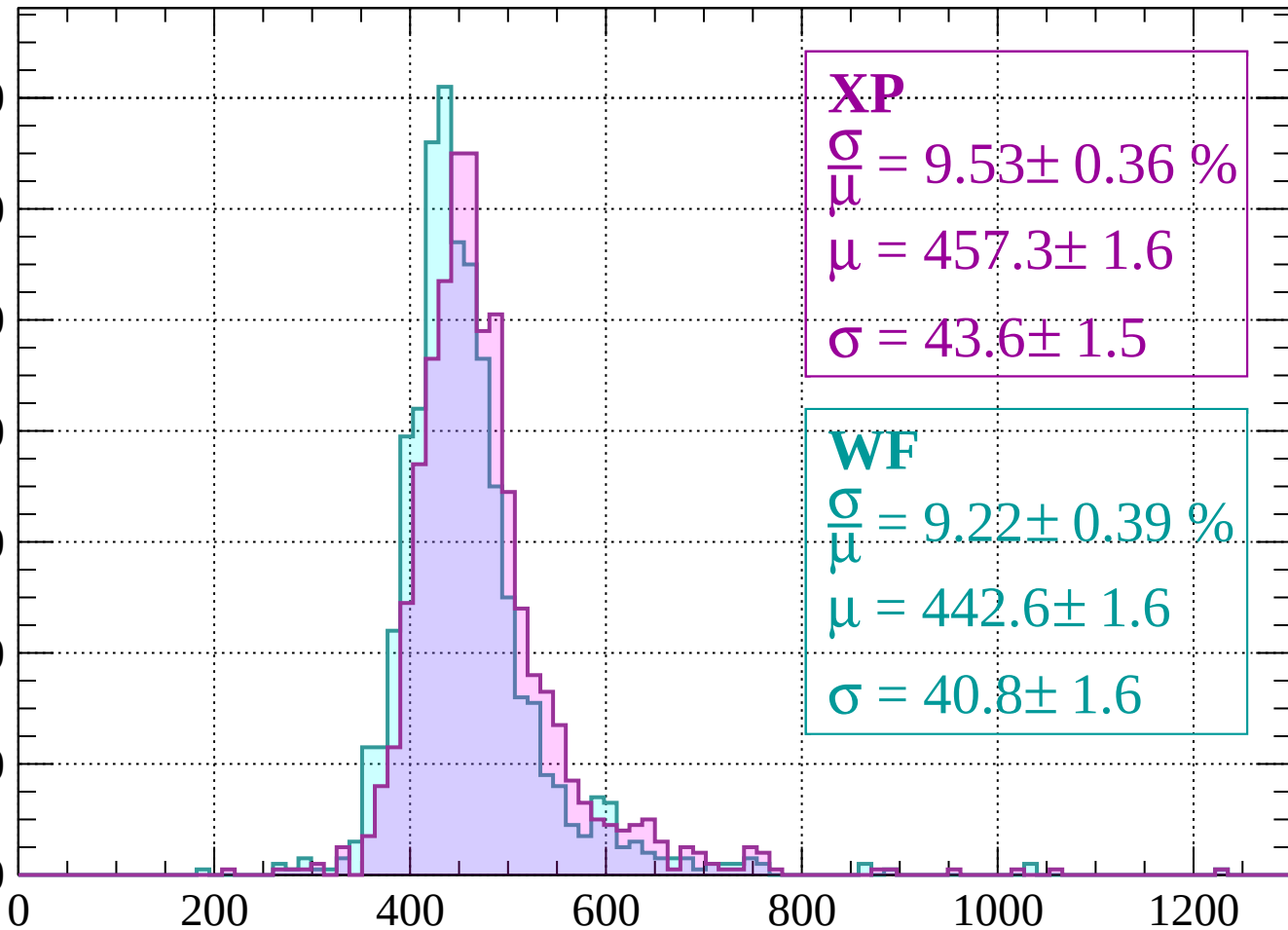
$$\sigma = 42.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.36 \%$$

$$\mu = 457.3 \pm 1.6$$

$$\sigma = 43.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.39 \%$$

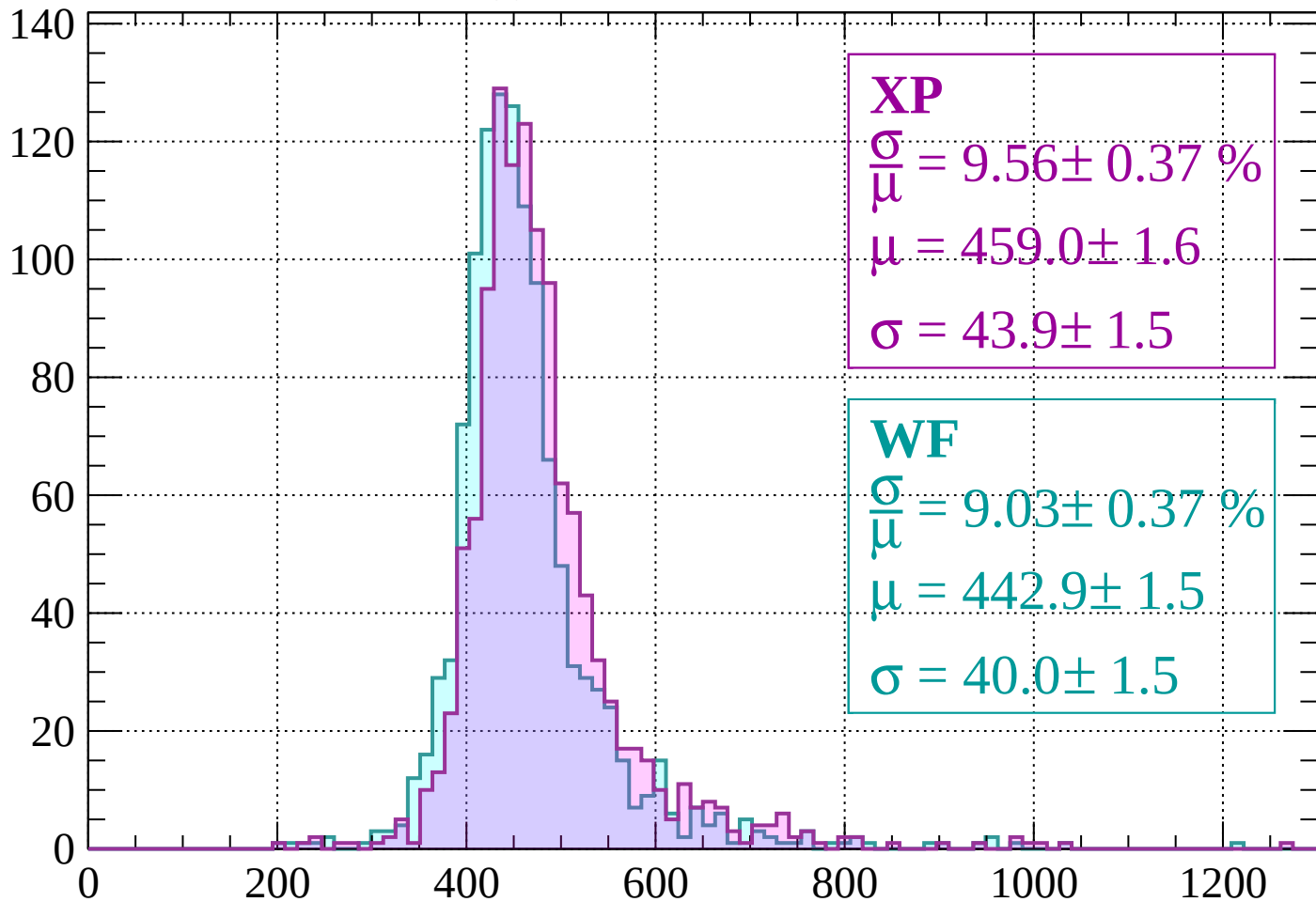
$$\mu = 442.6 \pm 1.6$$

$$\sigma = 40.8 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.56 \pm 0.37 \%$$

$$\mu = 459.0 \pm 1.6$$

$$\sigma = 43.9 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.03 \pm 0.37 \%$$

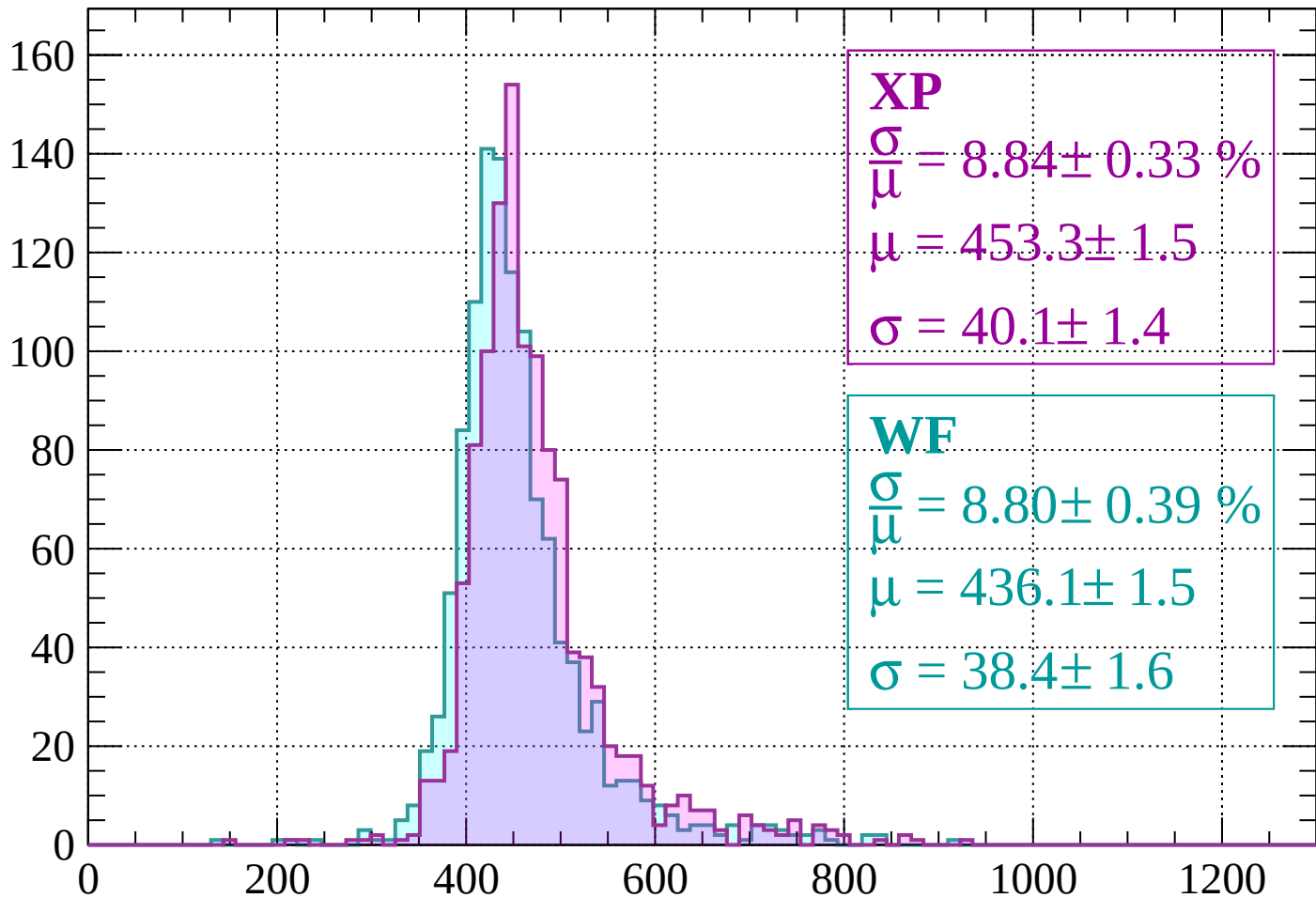
$$\mu = 442.9 \pm 1.5$$

$$\sigma = 40.0 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.84 \pm 0.33 \%$$

$$\mu = 453.3 \pm 1.5$$

$$\sigma = 40.1 \pm 1.4$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.80 \pm 0.39 \%$$

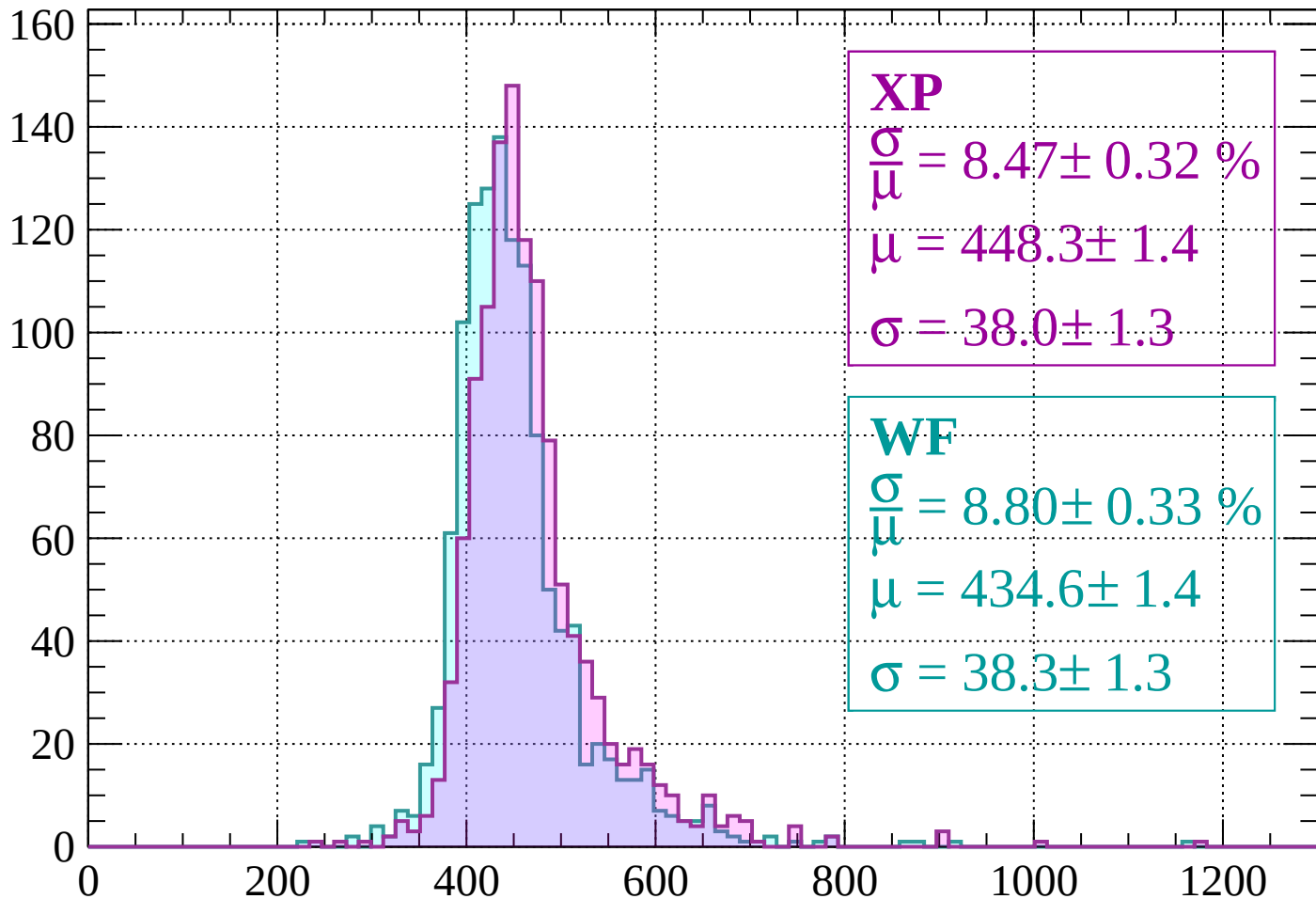
$$\mu = 436.1 \pm 1.5$$

$$\sigma = 38.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.32 \%$$

$$\mu = 448.3 \pm 1.4$$

$$\sigma = 38.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.80 \pm 0.33 \%$$

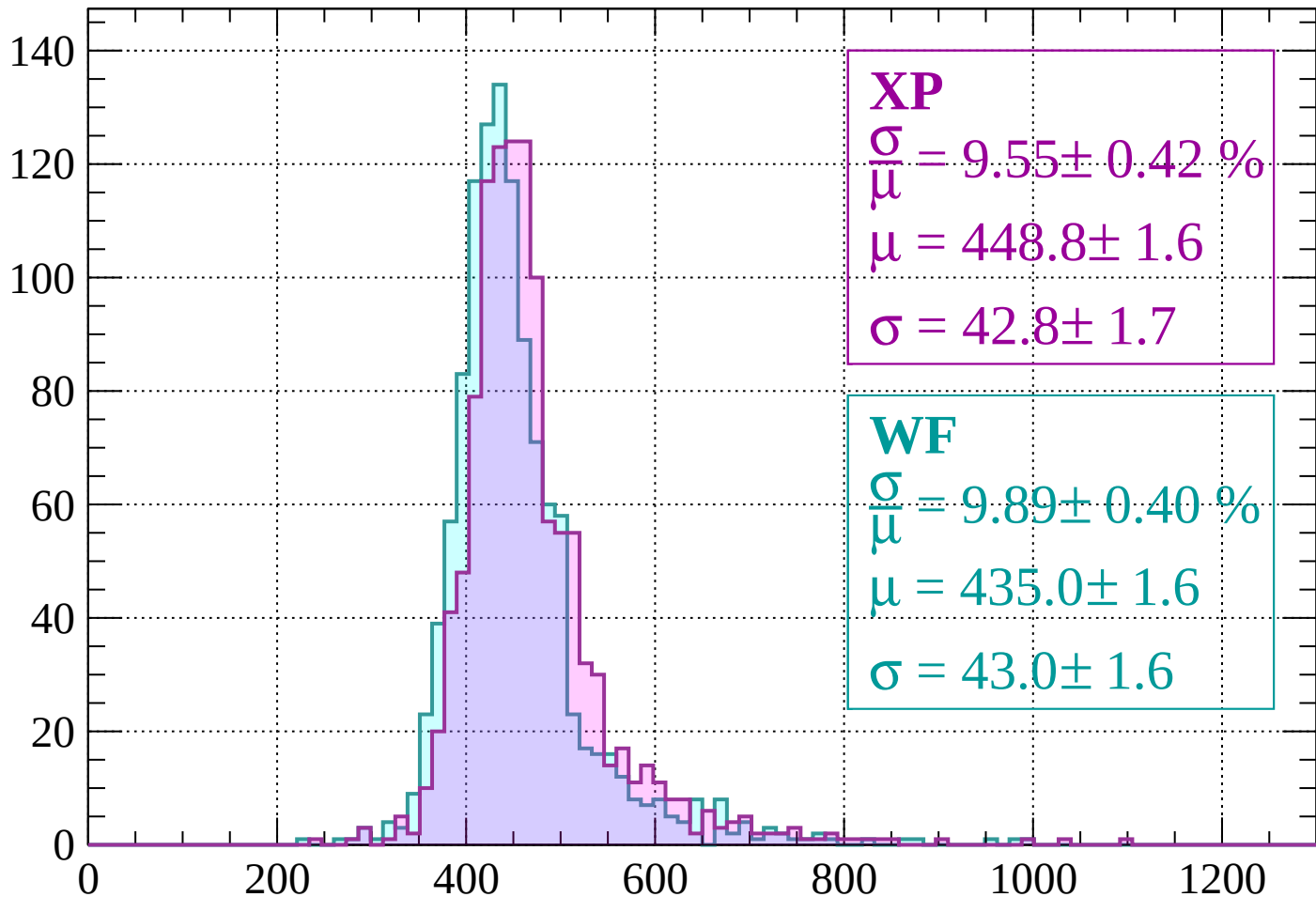
$$\mu = 434.6 \pm 1.4$$

$$\sigma = 38.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.42 \%$$

$$\mu = 448.8 \pm 1.6$$

$$\sigma = 42.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.40 \%$$

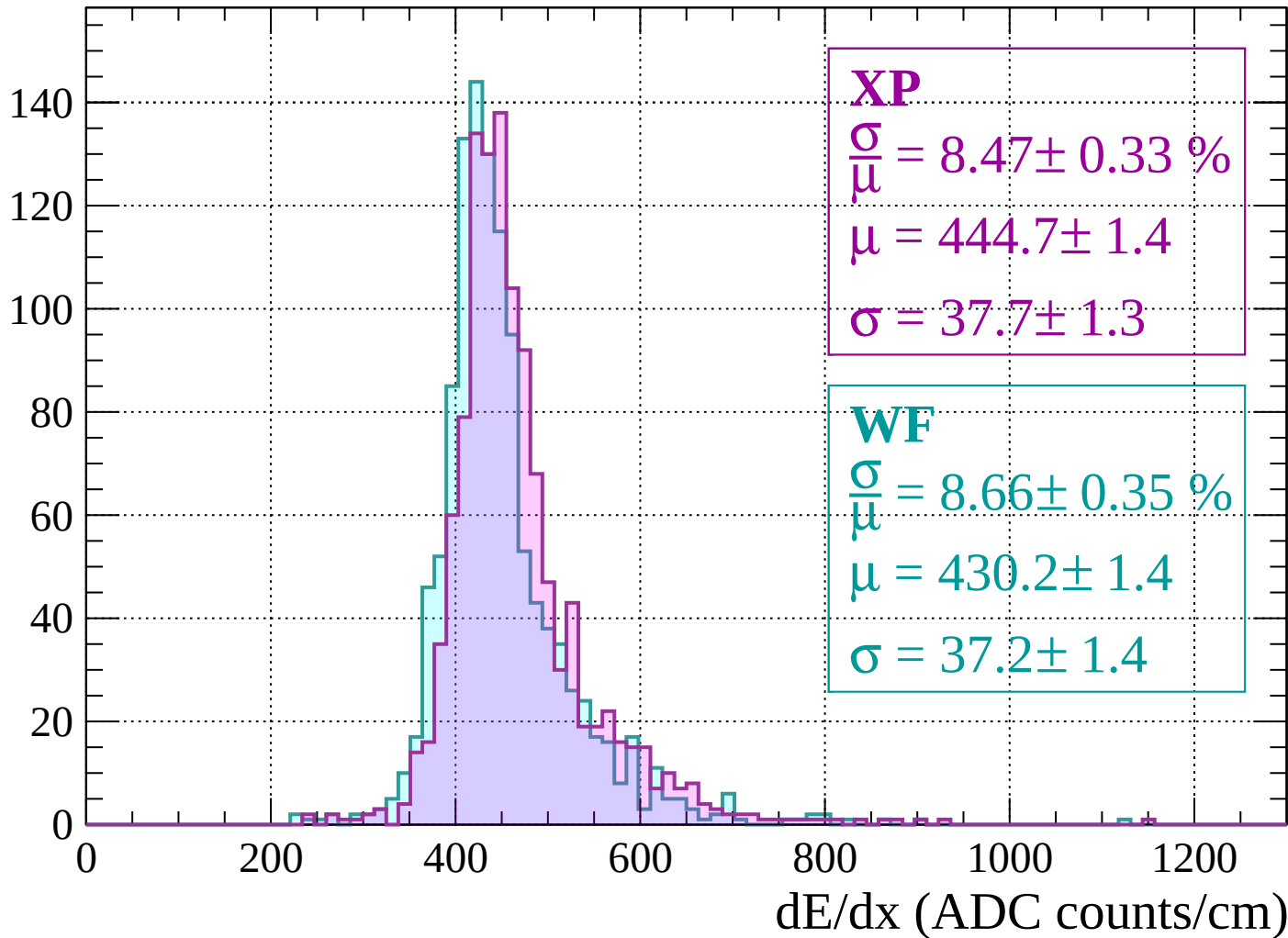
$$\mu = 435.0 \pm 1.6$$

$$\sigma = 43.0 \pm 1.6$$

dE/dx (ADC counts/cm)

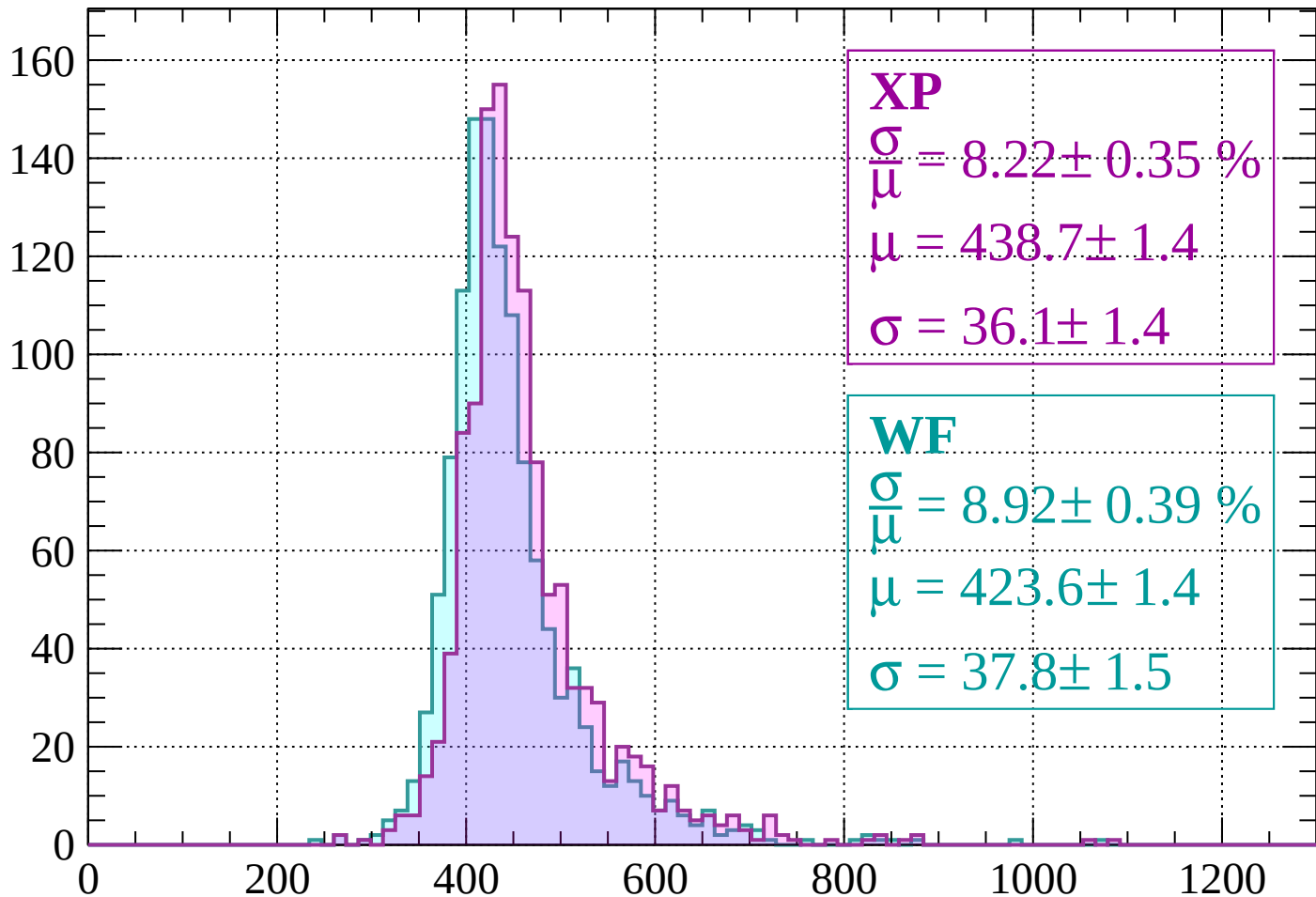
Energy loss | $-720 < p < -680$

Count



Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.35 \%$$

$$\mu = 438.7 \pm 1.4$$

$$\sigma = 36.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.92 \pm 0.39 \%$$

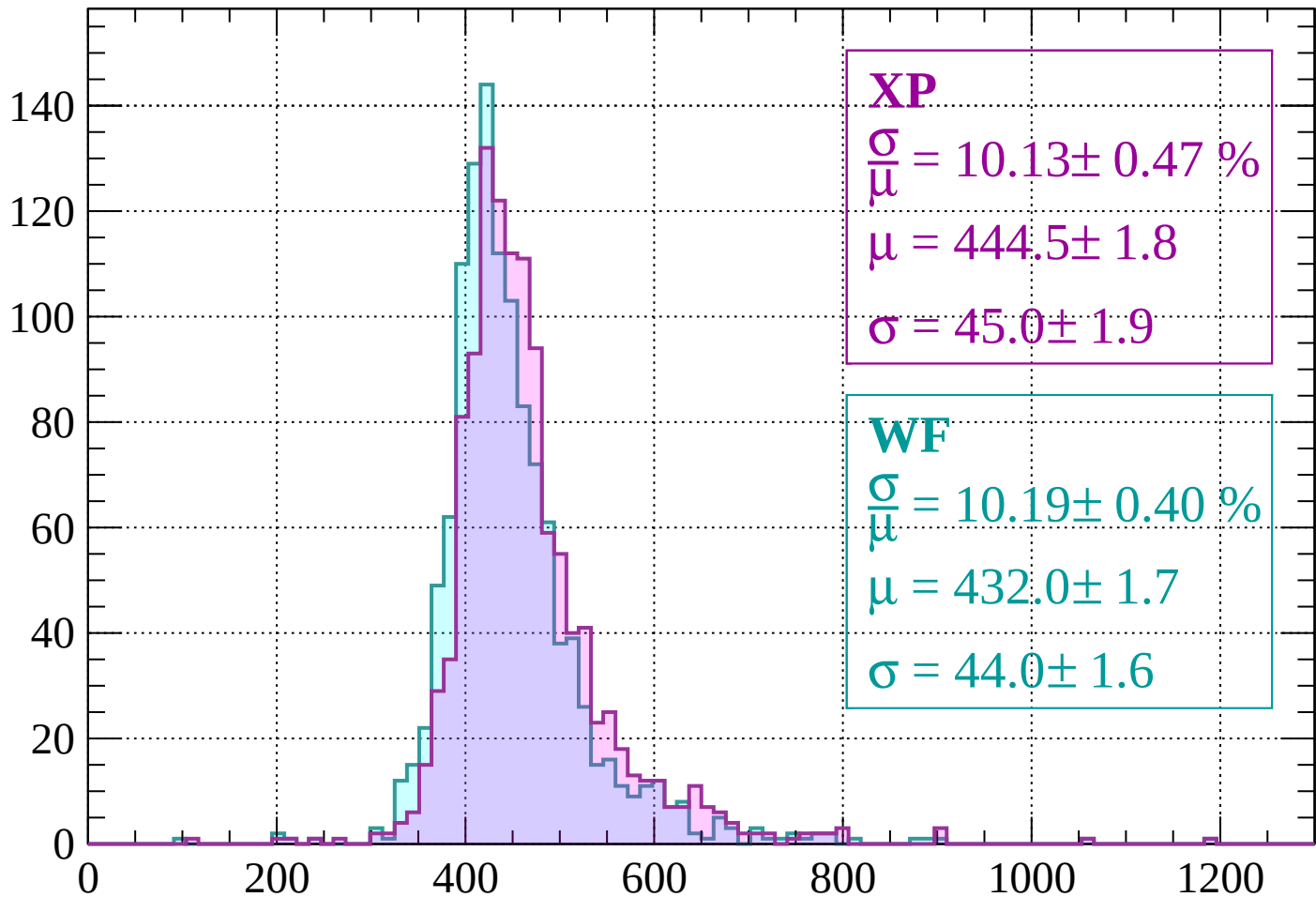
$$\mu = 423.6 \pm 1.4$$

$$\sigma = 37.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 10.13 \pm 0.47 \%$$

$$\mu = 444.5 \pm 1.8$$

$$\sigma = 45.0 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.19 \pm 0.40 \%$$

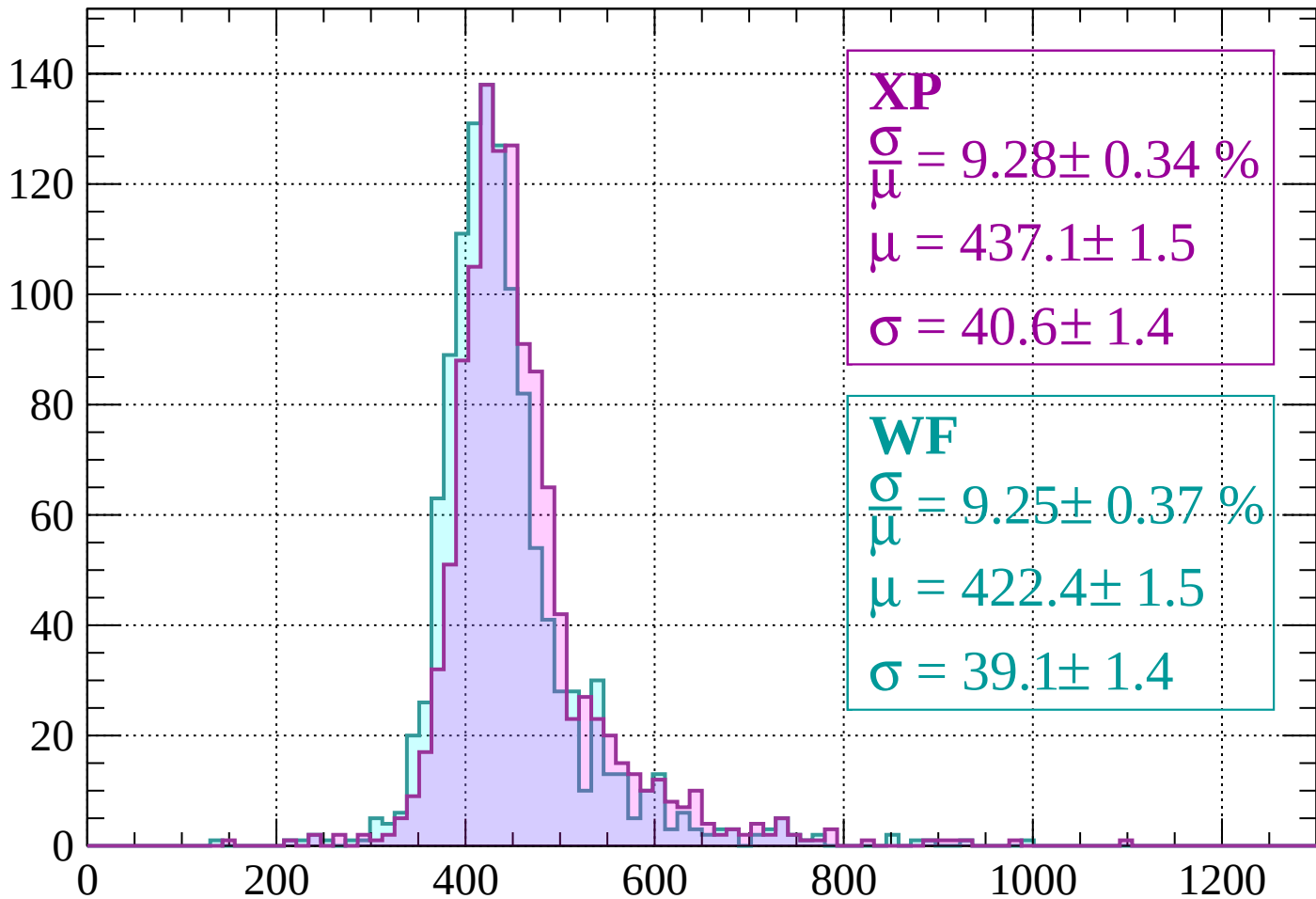
$$\mu = 432.0 \pm 1.7$$

$$\sigma = 44.0 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 9.28 \pm 0.34 \%$$

$$\mu = 437.1 \pm 1.5$$

$$\sigma = 40.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.25 \pm 0.37 \%$$

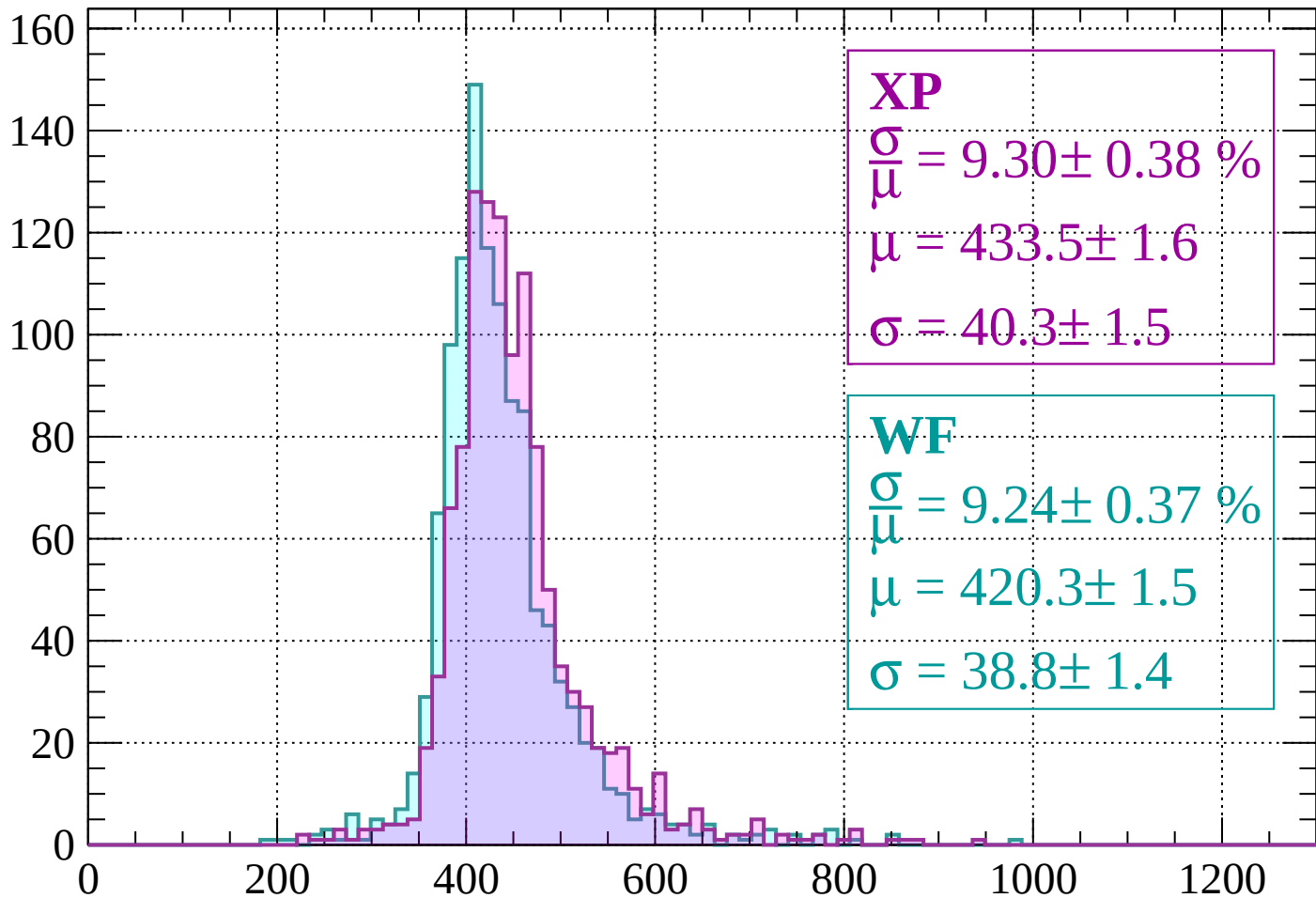
$$\mu = 422.4 \pm 1.5$$

$$\sigma = 39.1 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.38 \%$$

$$\mu = 433.5 \pm 1.6$$

$$\sigma = 40.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.37 \%$$

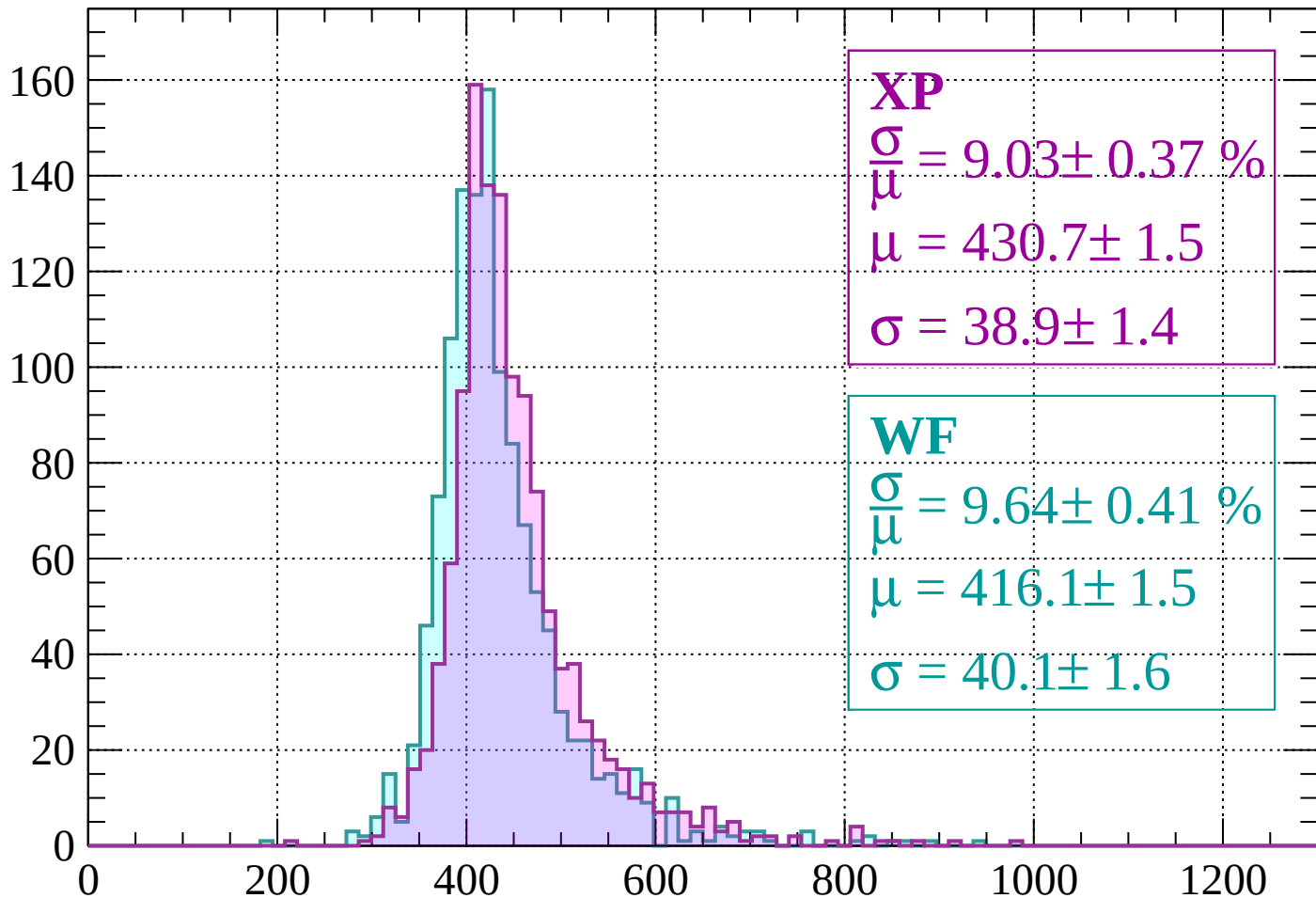
$$\mu = 420.3 \pm 1.5$$

$$\sigma = 38.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.37 \%$$

$$\mu = 430.7 \pm 1.5$$

$$\sigma = 38.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.41 \%$$

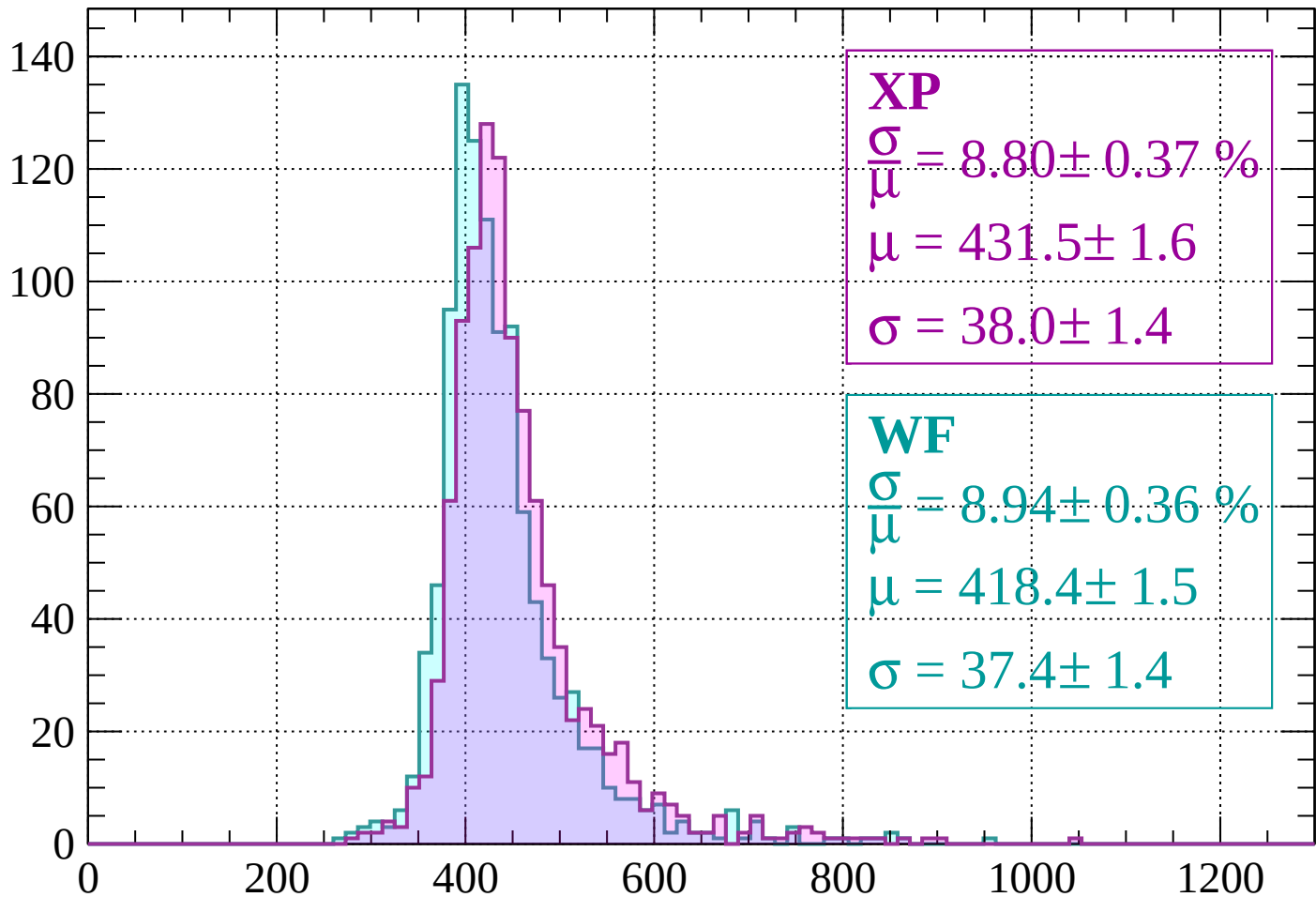
$$\mu = 416.1 \pm 1.5$$

$$\sigma = 40.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.80 \pm 0.37 \%$$

$$\mu = 431.5 \pm 1.6$$

$$\sigma = 38.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.94 \pm 0.36 \%$$

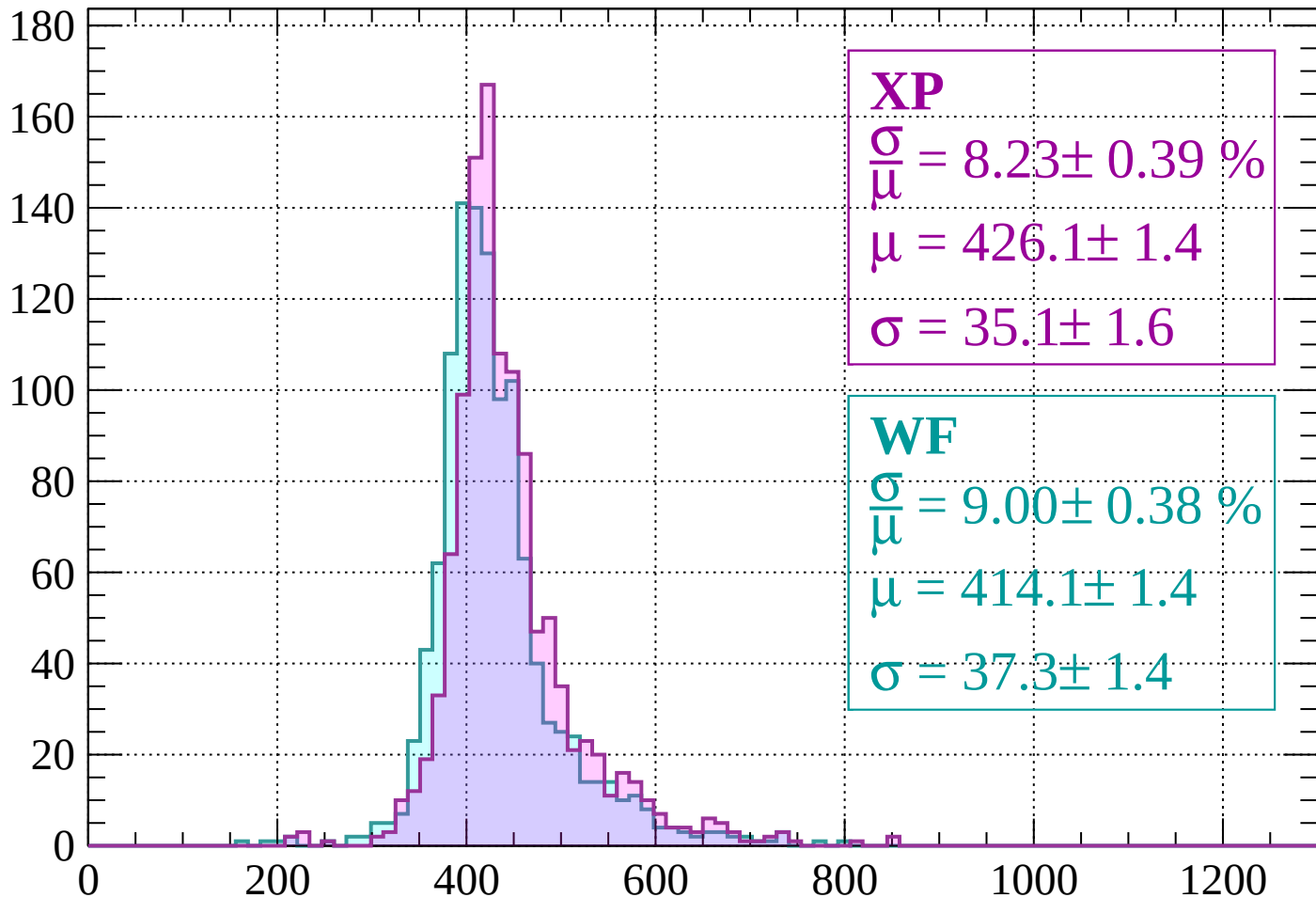
$$\mu = 418.4 \pm 1.5$$

$$\sigma = 37.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 8.23 \pm 0.39 \%$$

$$\mu = 426.1 \pm 1.4$$

$$\sigma = 35.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.00 \pm 0.38 \%$$

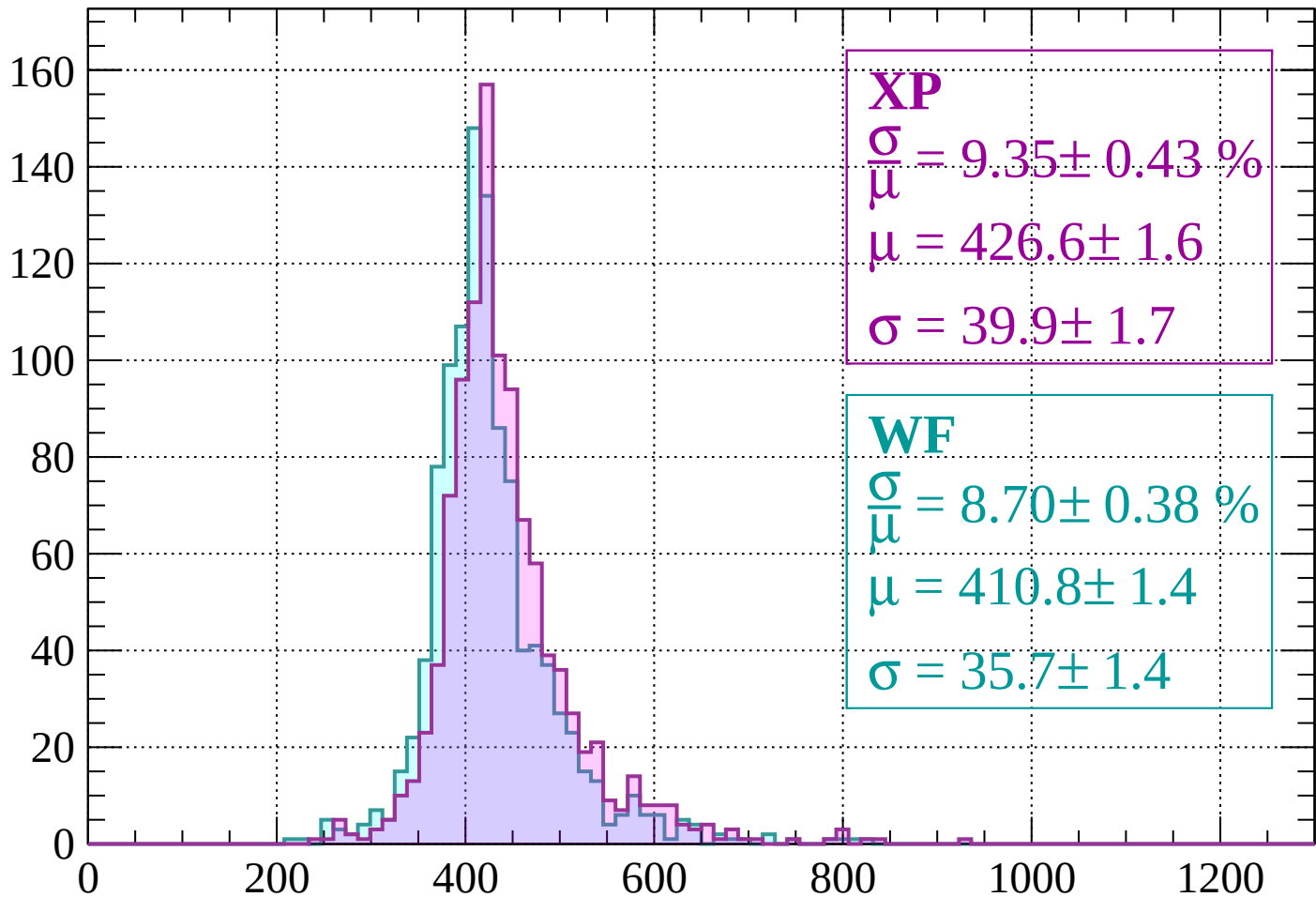
$$\mu = 414.1 \pm 1.4$$

$$\sigma = 37.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 9.35 \pm 0.43 \%$$

$$\mu = 426.6 \pm 1.6$$

$$\sigma = 39.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.38 \%$$

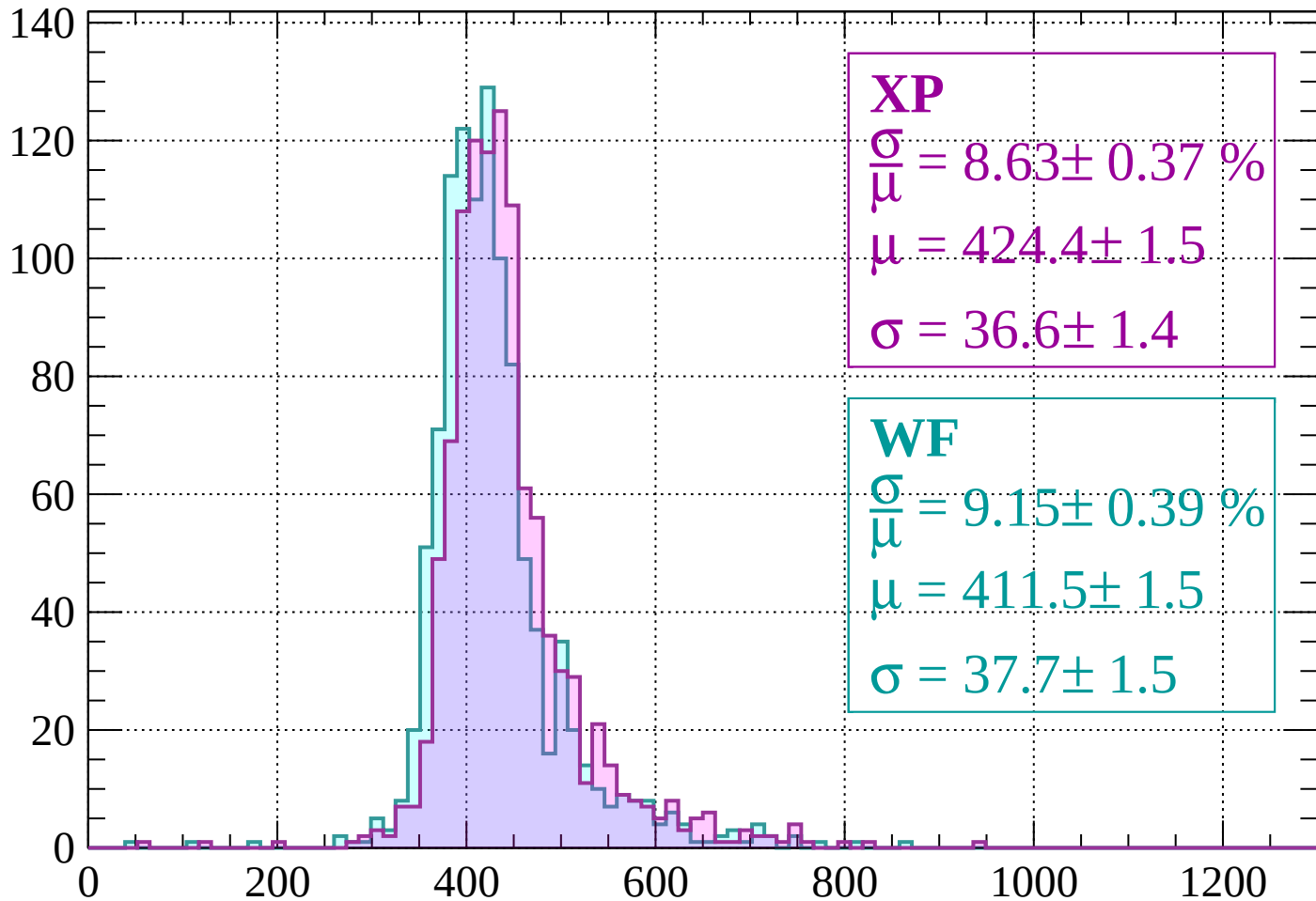
$$\mu = 410.8 \pm 1.4$$

$$\sigma = 35.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.63 \pm 0.37 \%$$

$$\mu = 424.4 \pm 1.5$$

$$\sigma = 36.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.39 \%$$

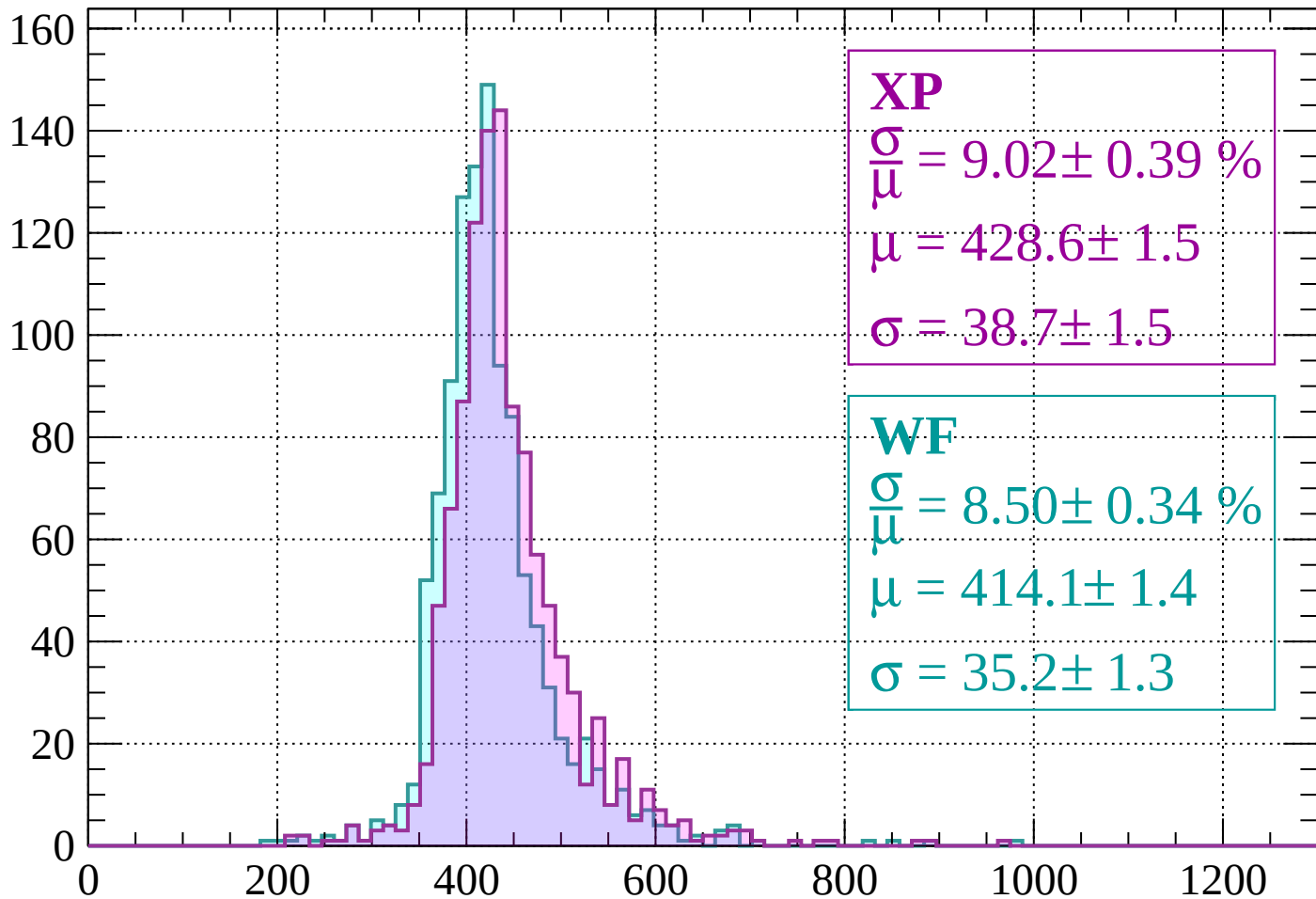
$$\mu = 411.5 \pm 1.5$$

$$\sigma = 37.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.39 \%$$

$$\mu = 428.6 \pm 1.5$$

$$\sigma = 38.7 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.34 \%$$

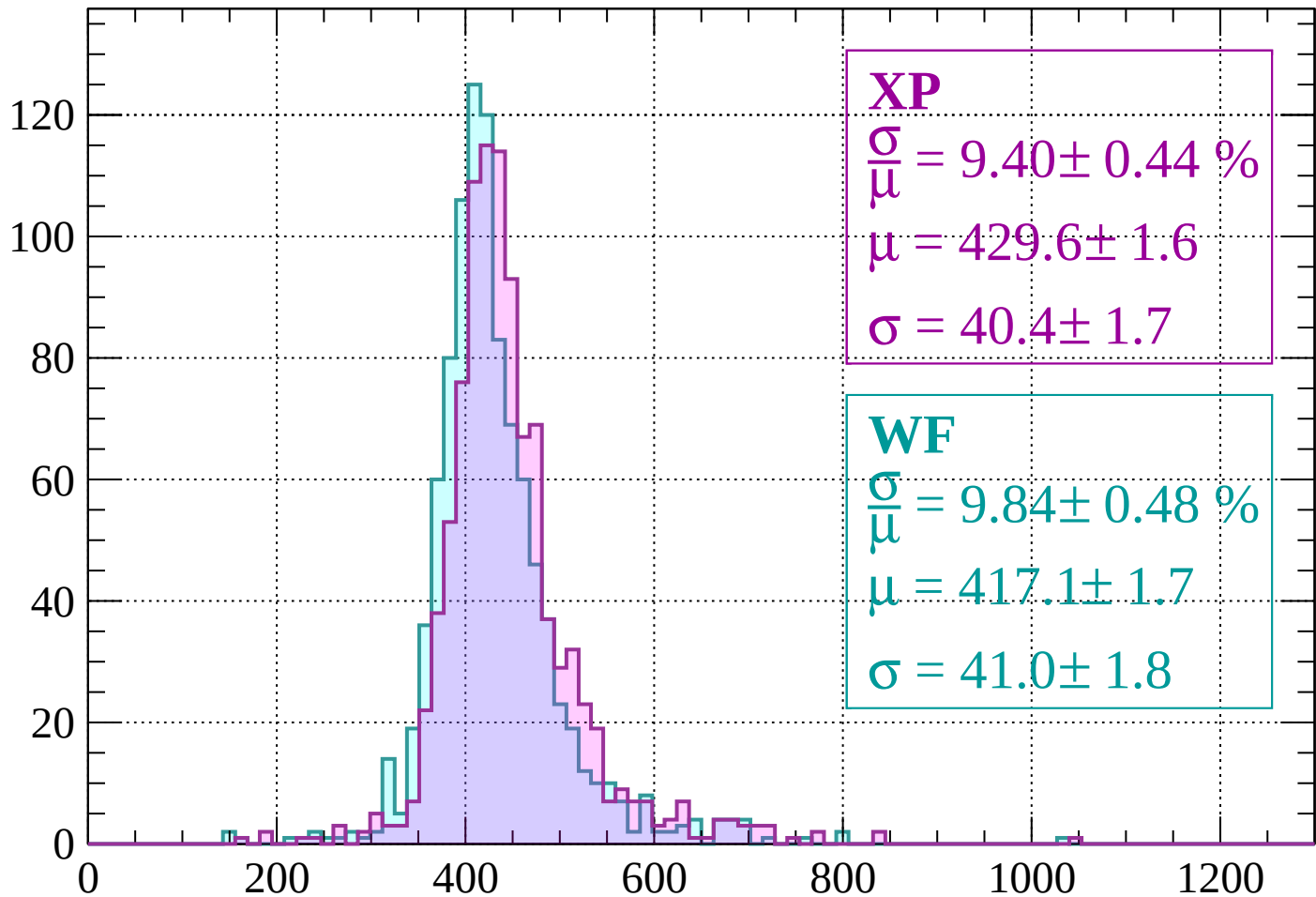
$$\mu = 414.1 \pm 1.4$$

$$\sigma = 35.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.44 \%$$

$$\mu = 429.6 \pm 1.6$$

$$\sigma = 40.4 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.84 \pm 0.48 \%$$

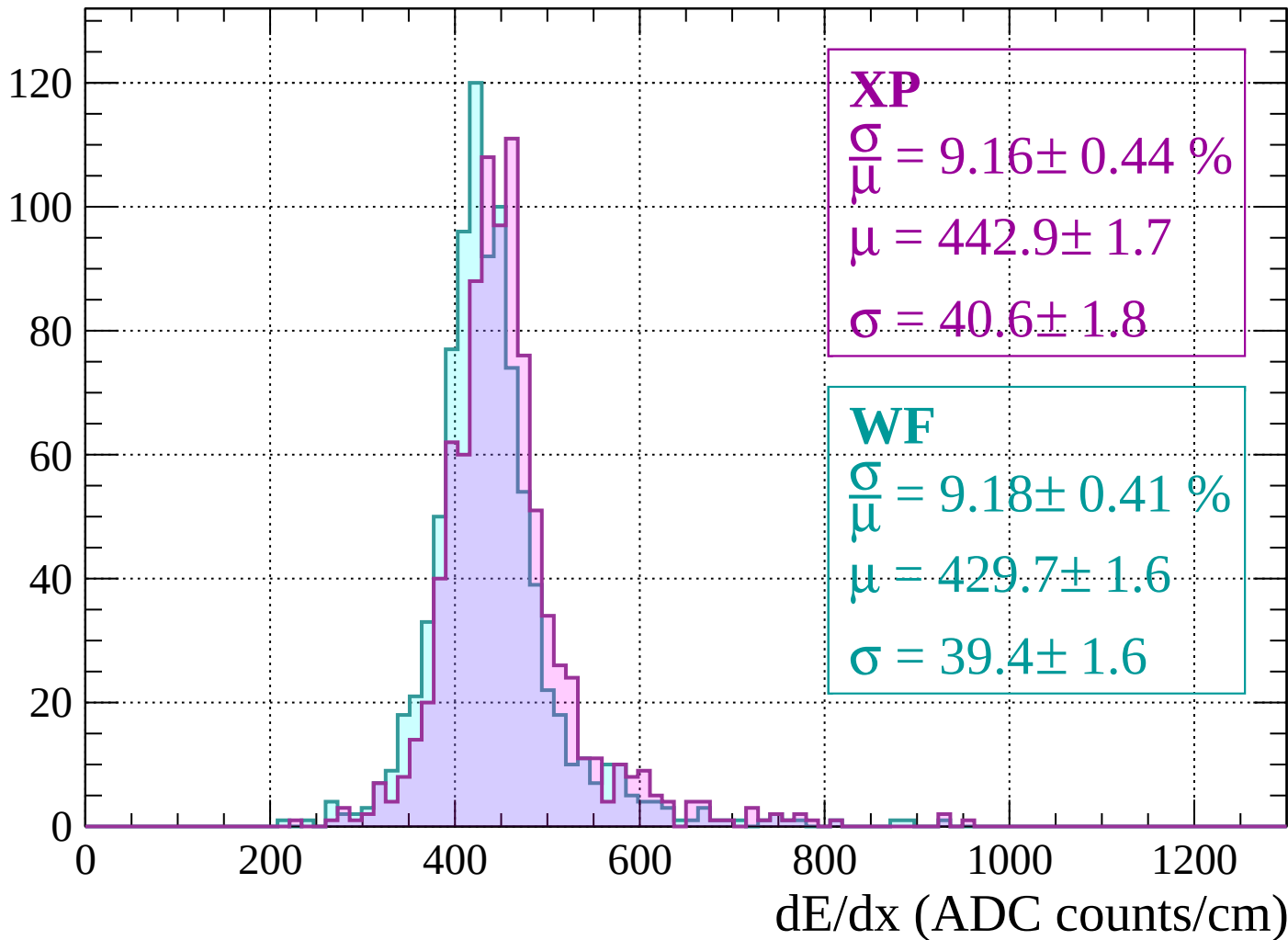
$$\mu = 417.1 \pm 1.7$$

$$\sigma = 41.0 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



Energy loss | $-200 < p < -160$

Count

XP

$$\frac{\sigma}{\mu} = 9.80 \pm 0.45 \%$$

$$\mu = 463.2 \pm 2.0$$

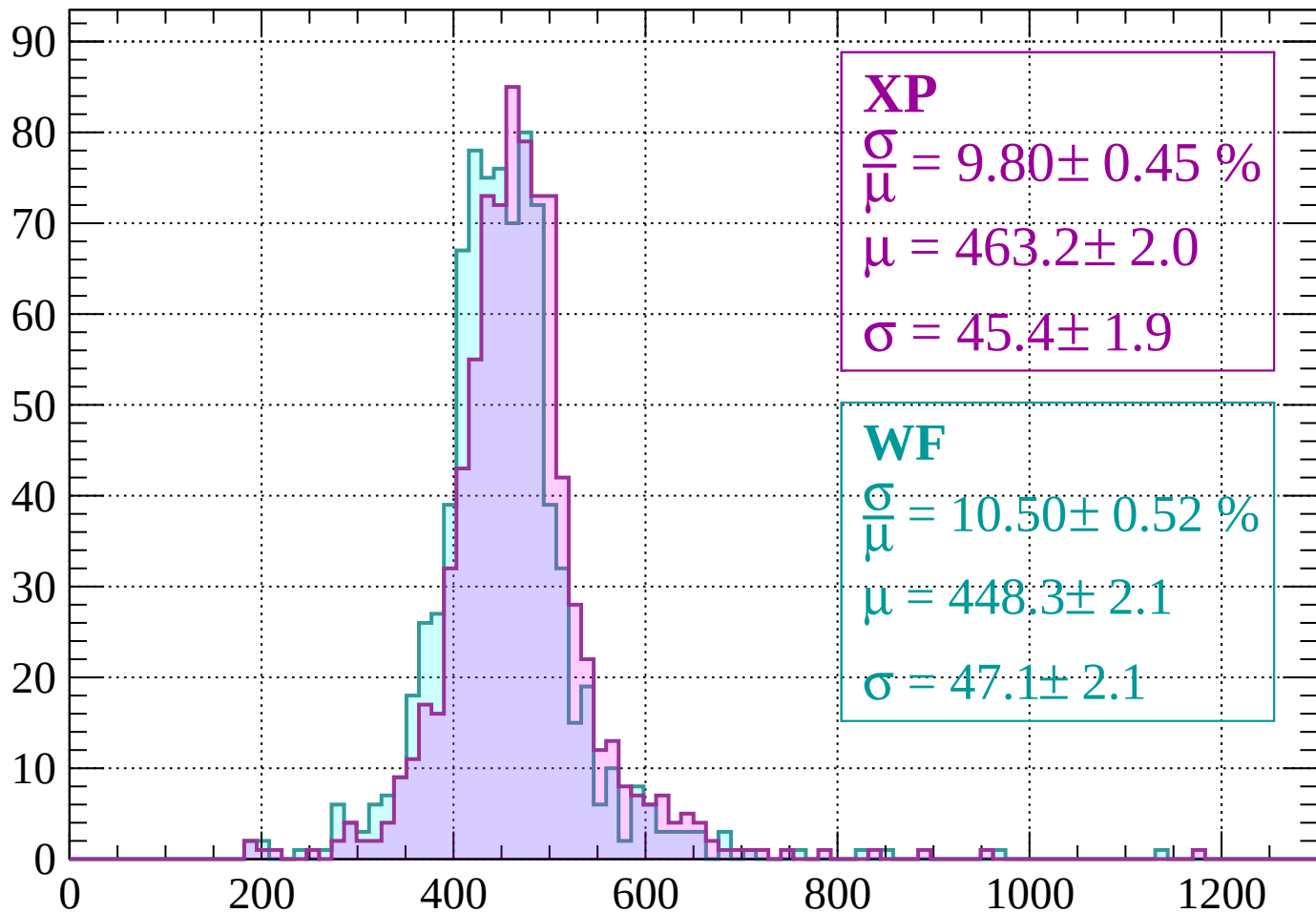
$$\sigma = 45.4 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 10.50 \pm 0.52 \%$$

$$\mu = 448.3 \pm 2.1$$

$$\sigma = 47.1 \pm 2.1$$



dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count

XP

$$\frac{\sigma}{\mu} = 10.07 \pm 0.52 \%$$

$$\mu = 504.1 \pm 2.5$$

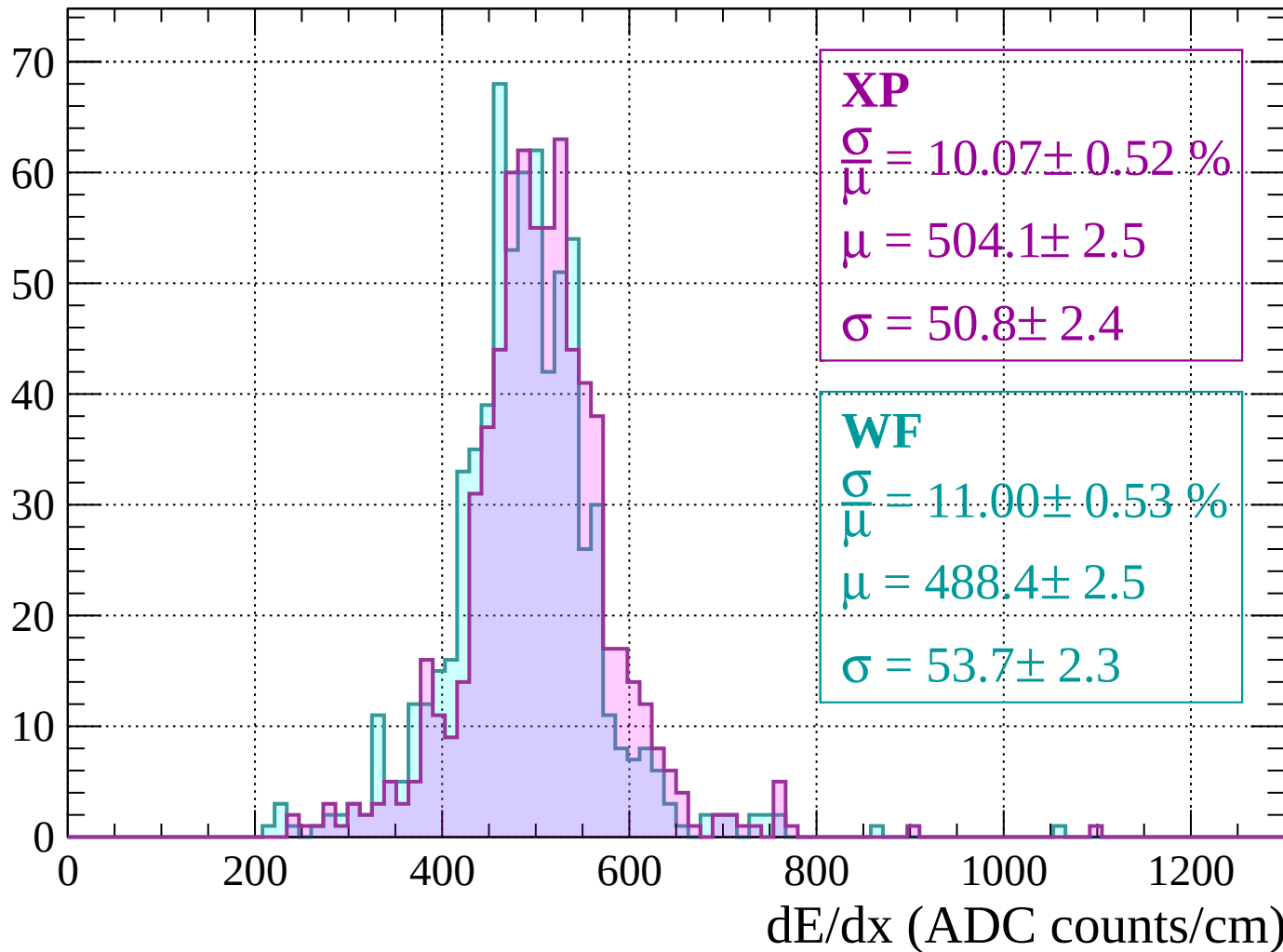
$$\sigma = 50.8 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 11.00 \pm 0.53 \%$$

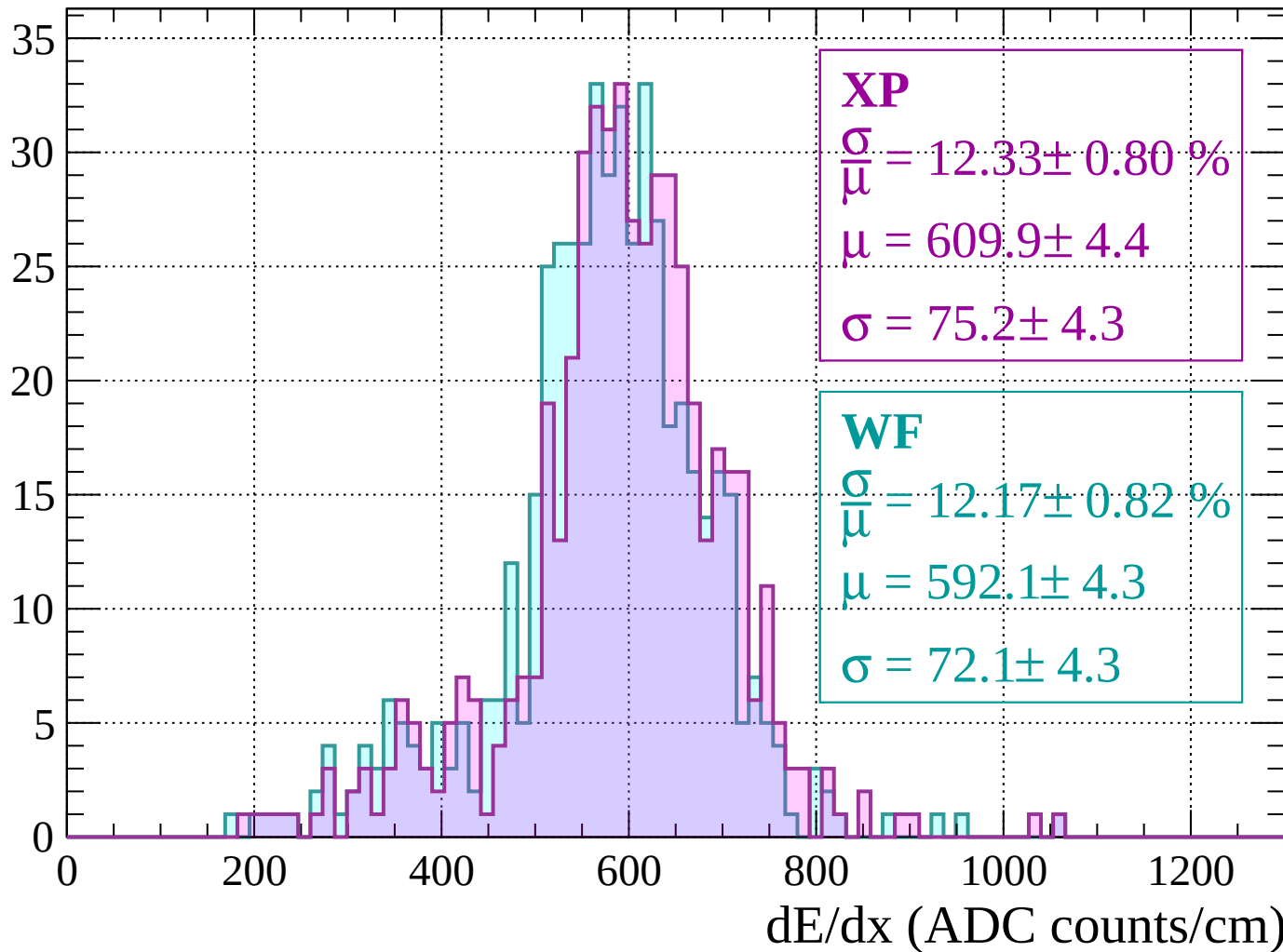
$$\mu = 488.4 \pm 2.5$$

$$\sigma = 53.7 \pm 2.3$$



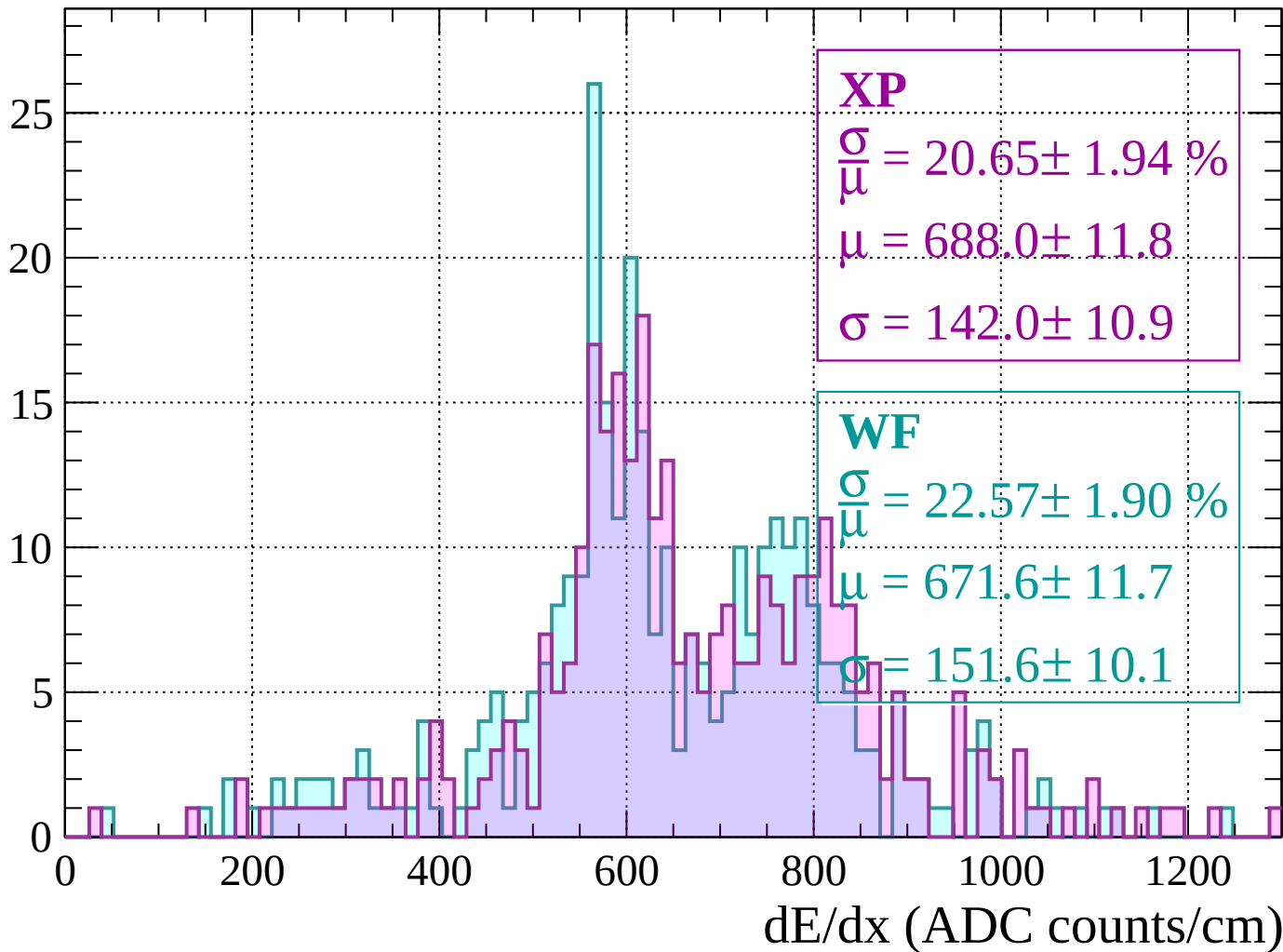
Energy loss | $-120 < p < -80$

Count



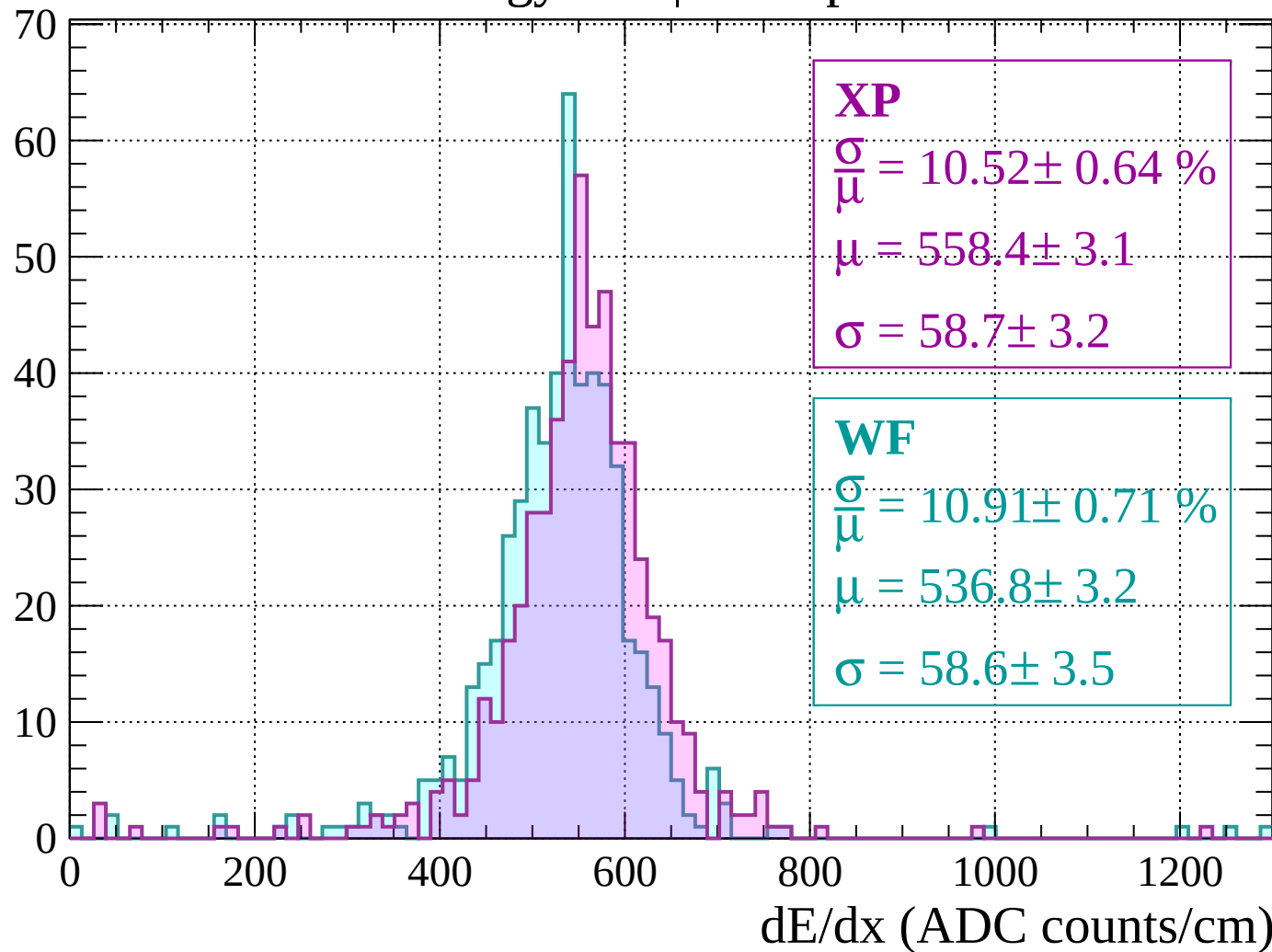
Energy loss | $-80 < p < -40$

Count



Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

Count

XP

$$\frac{\sigma}{\mu} = 12.67 \pm 0.83 \%$$

$$\mu = 496.2 \pm 3.8$$

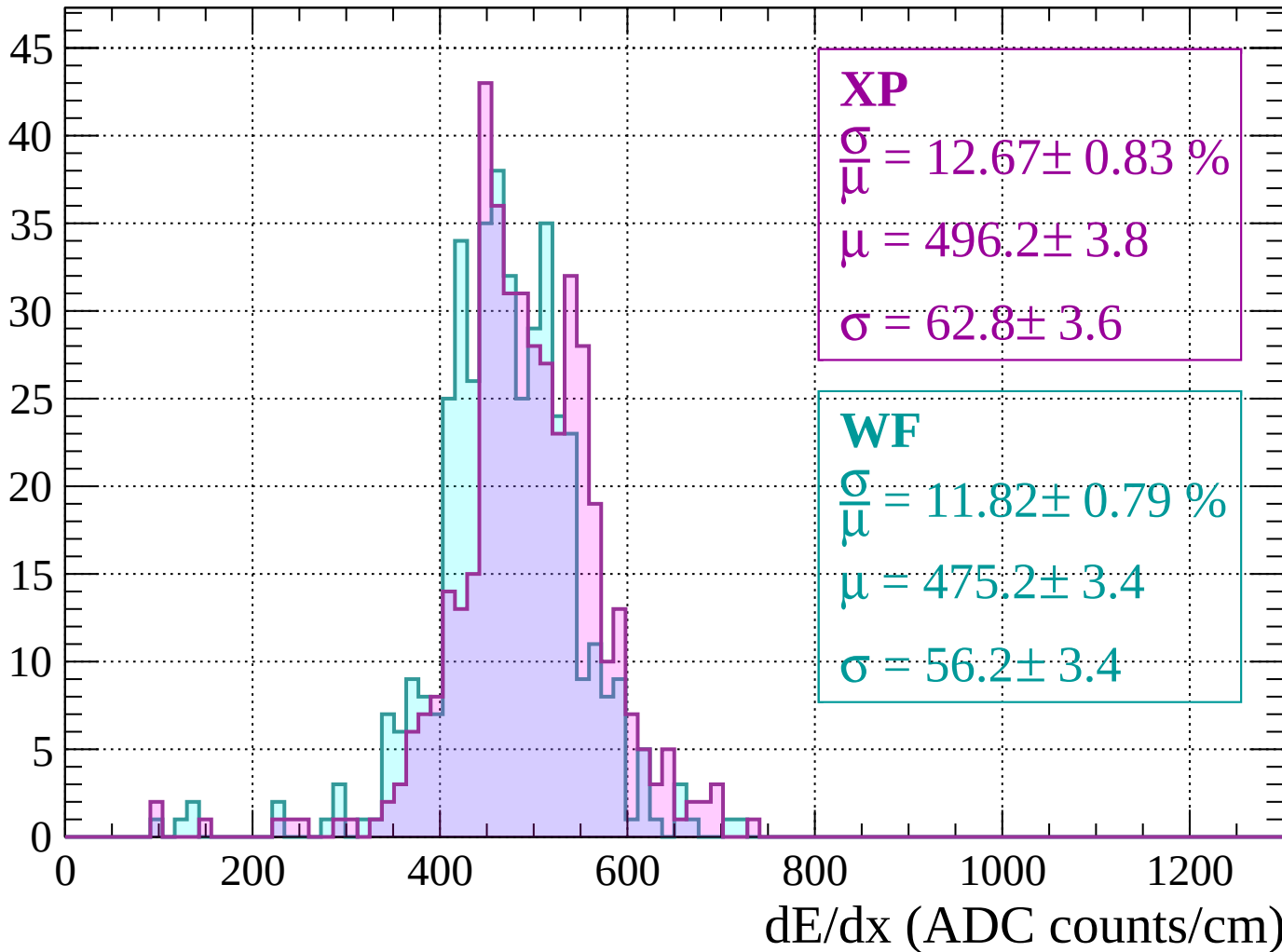
$$\sigma = 62.8 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 11.82 \pm 0.79 \%$$

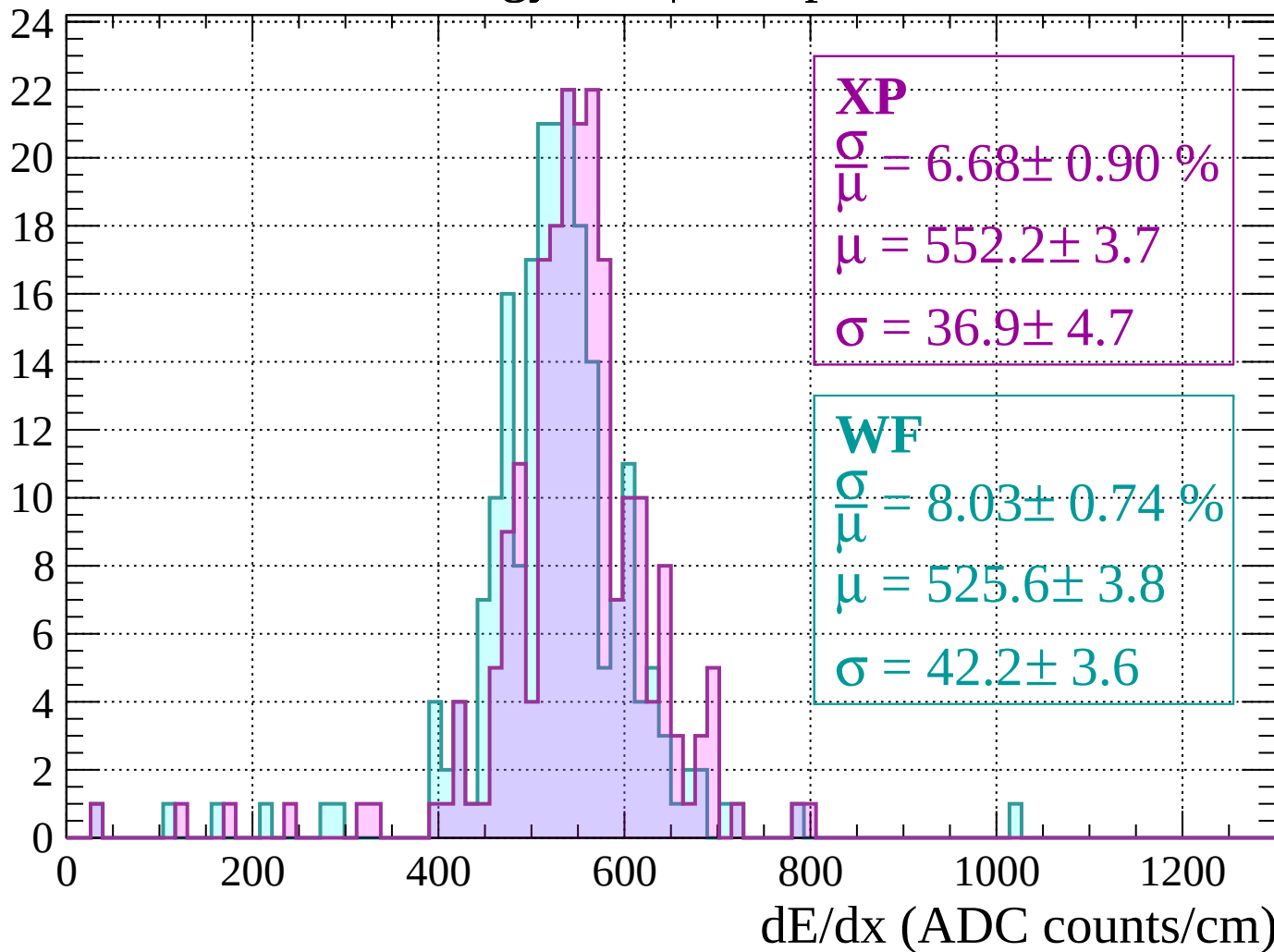
$$\mu = 475.2 \pm 3.4$$

$$\sigma = 56.2 \pm 3.4$$



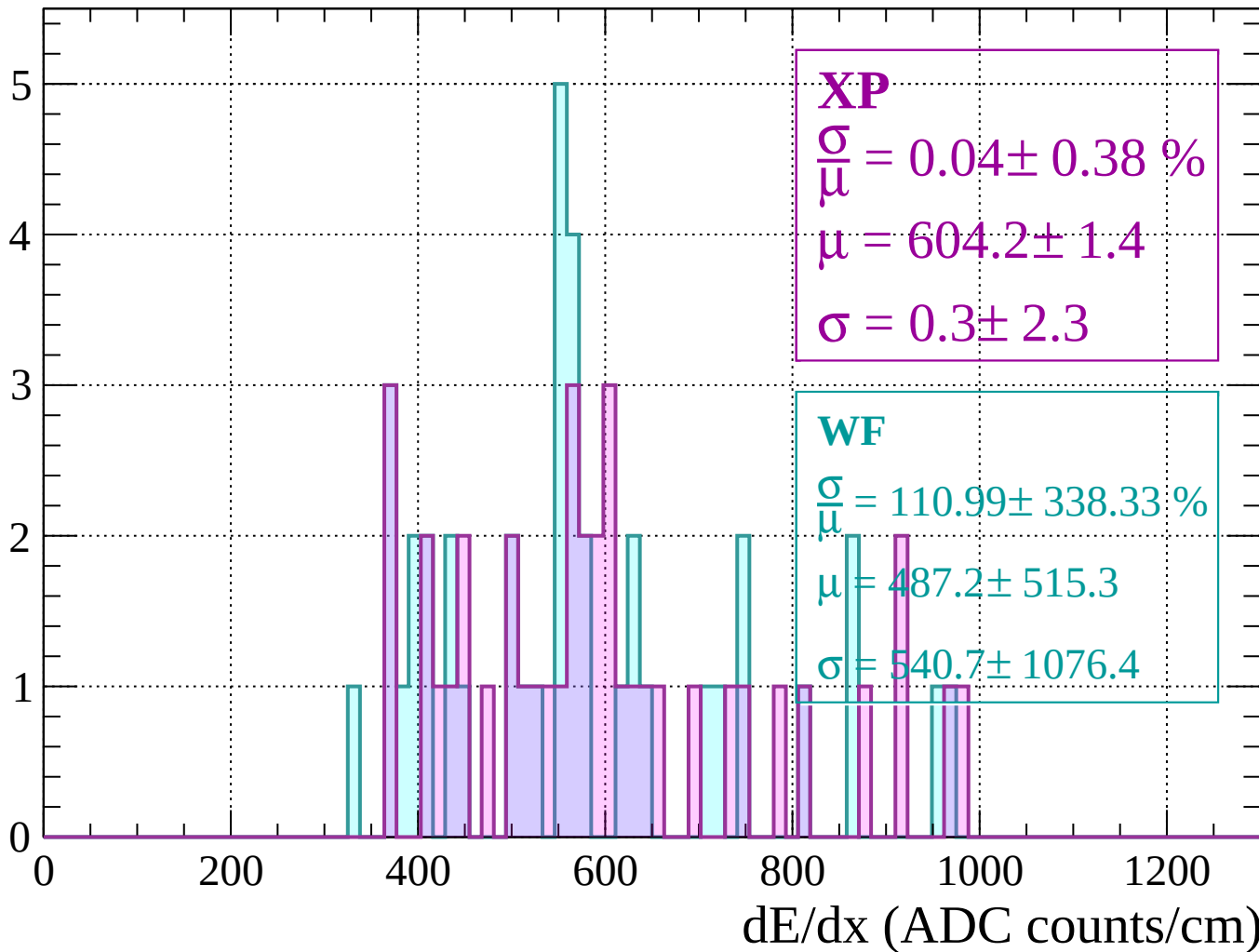
Energy loss | $40 < p < 80$

Count



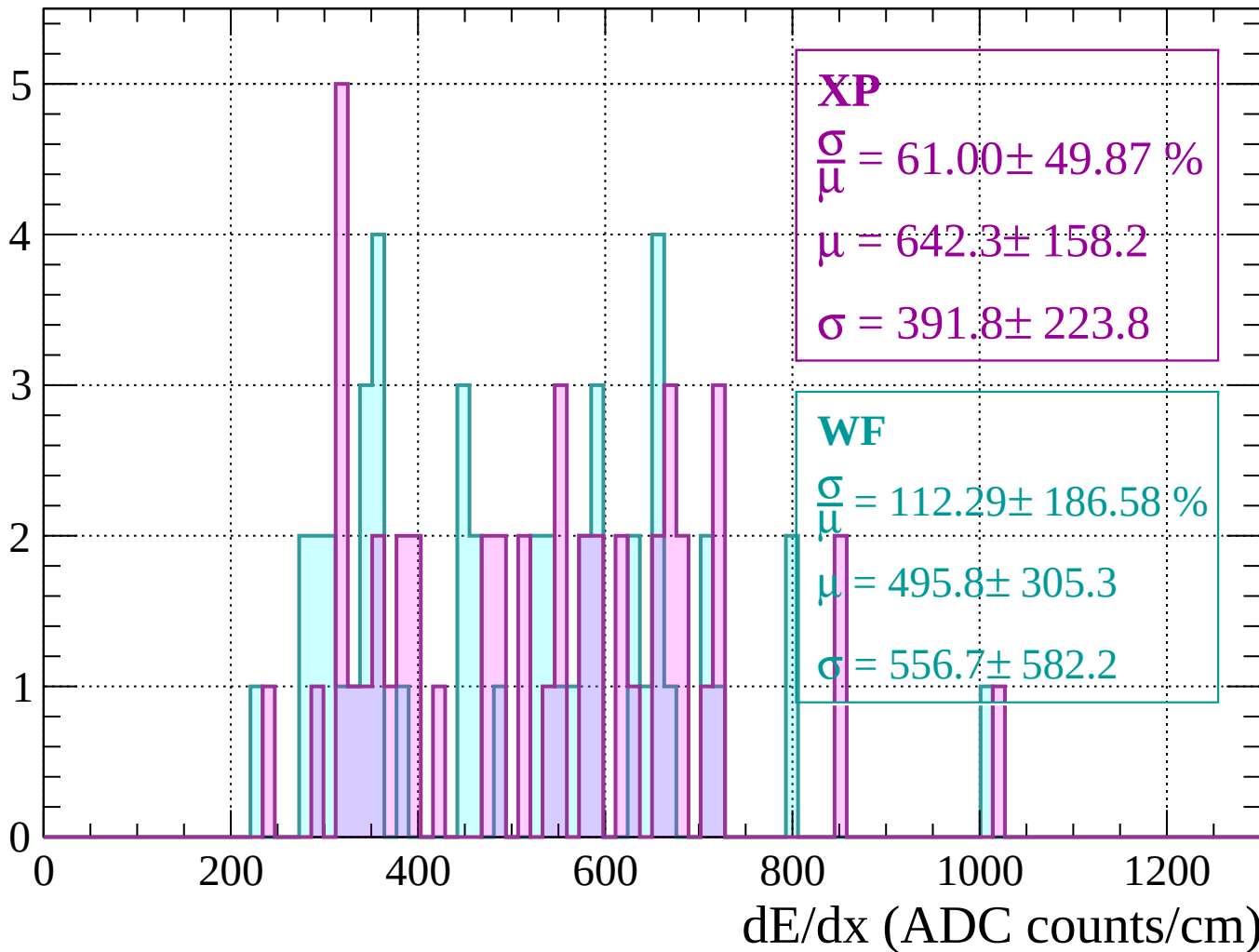
Energy loss | $80 < p < 120$

Count



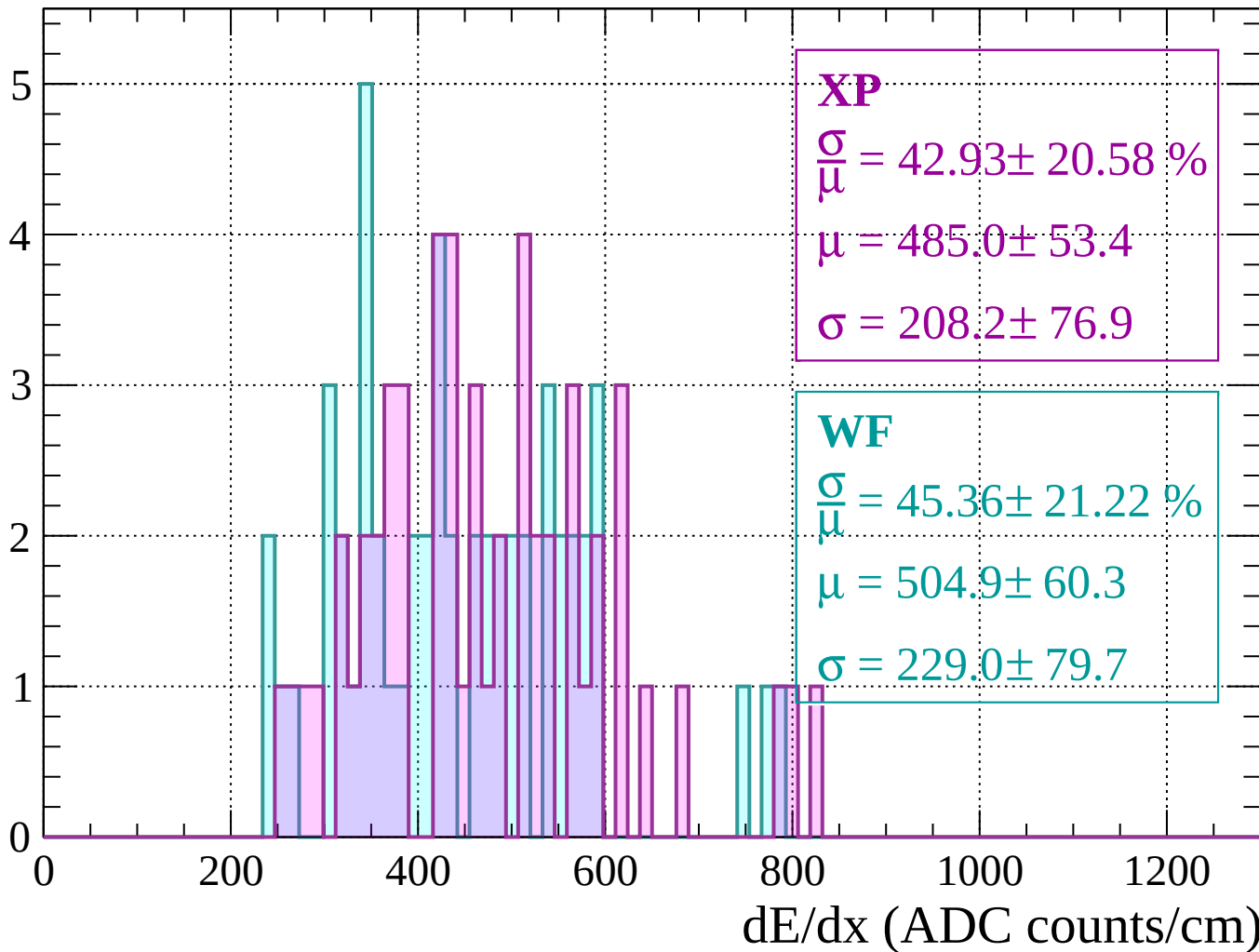
Energy loss | $120 < p < 160$

Count



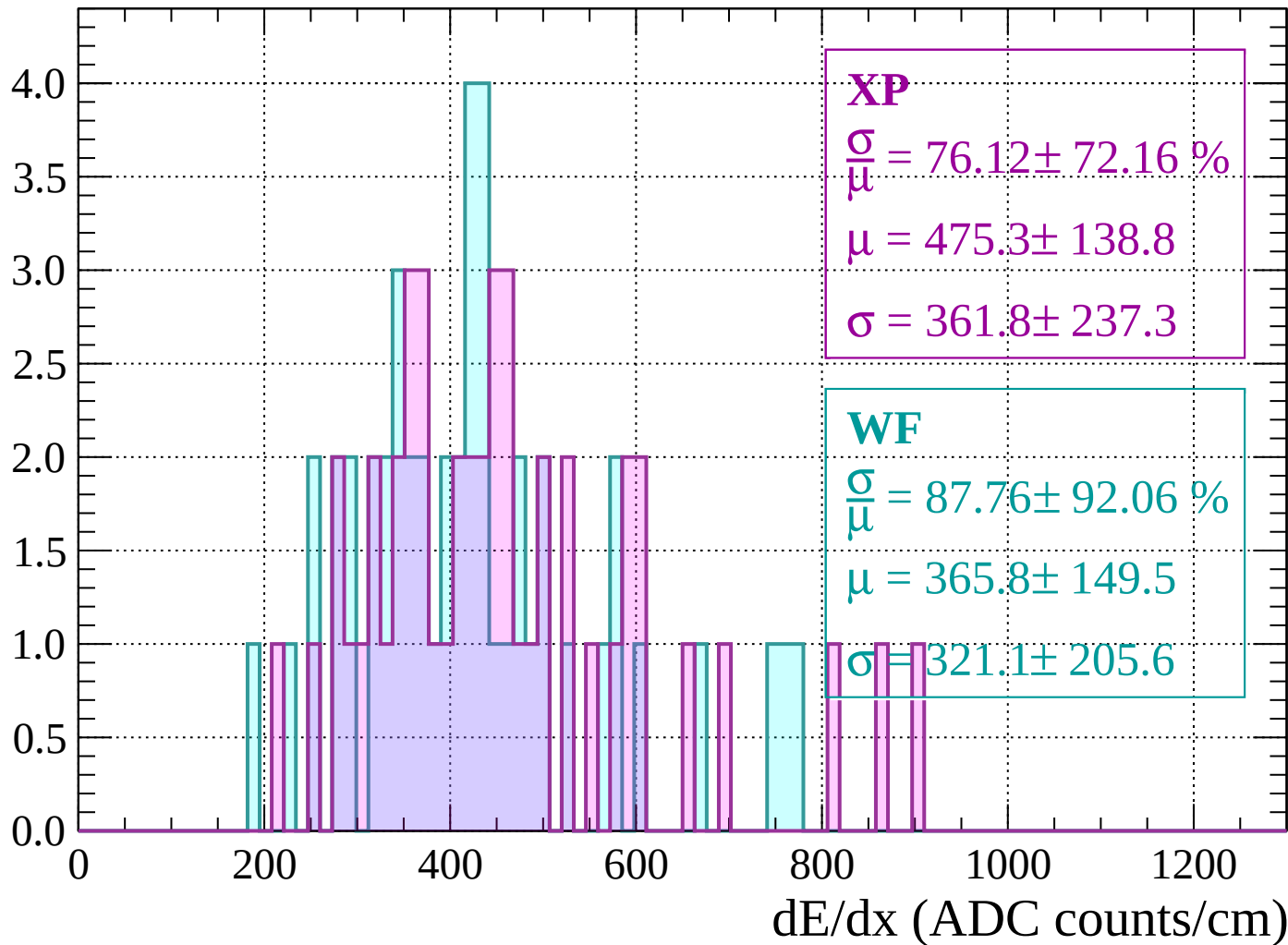
Energy loss | $160 < p < 200$

Count



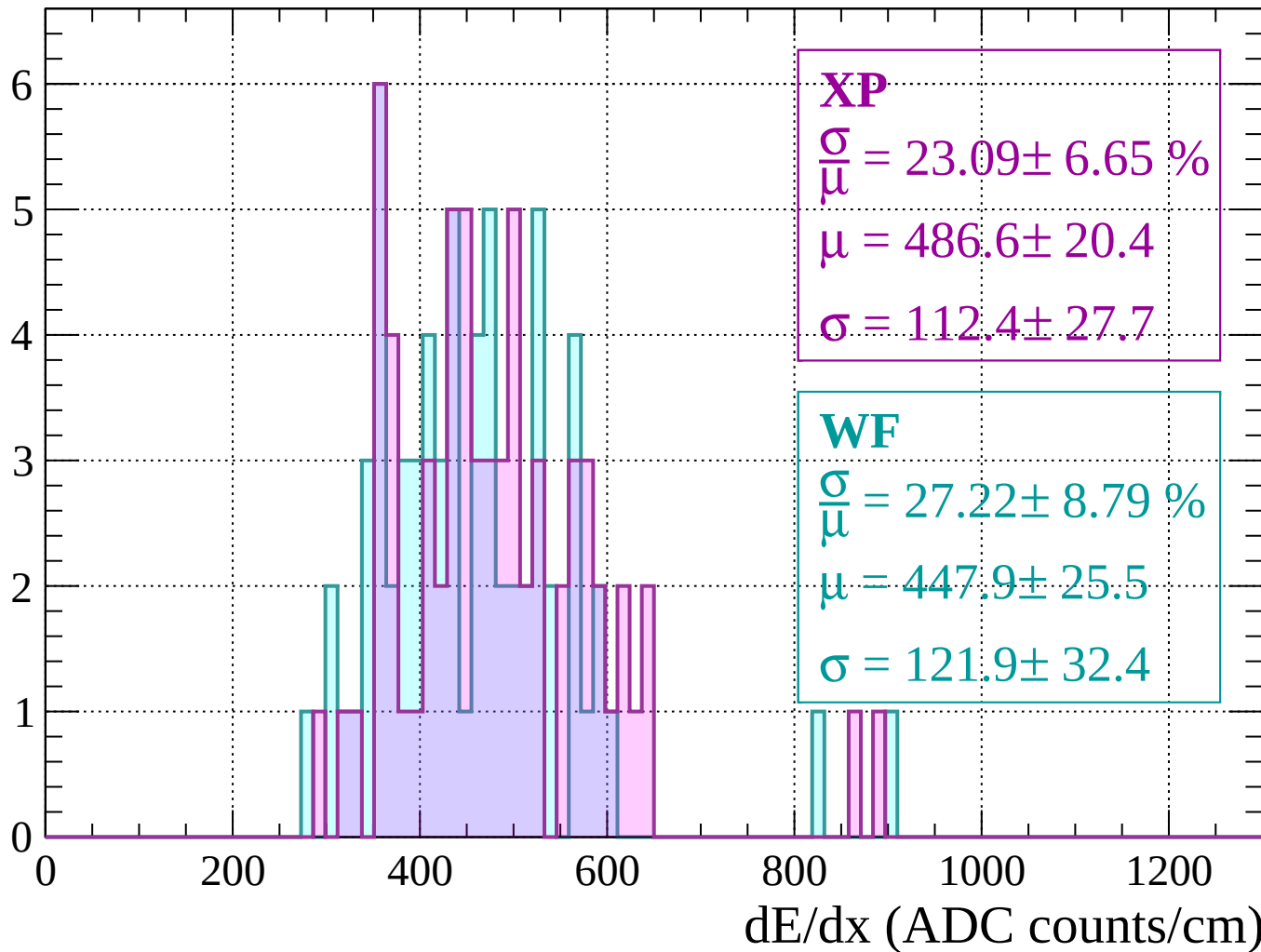
Energy loss | $200 < p < 240$

Count



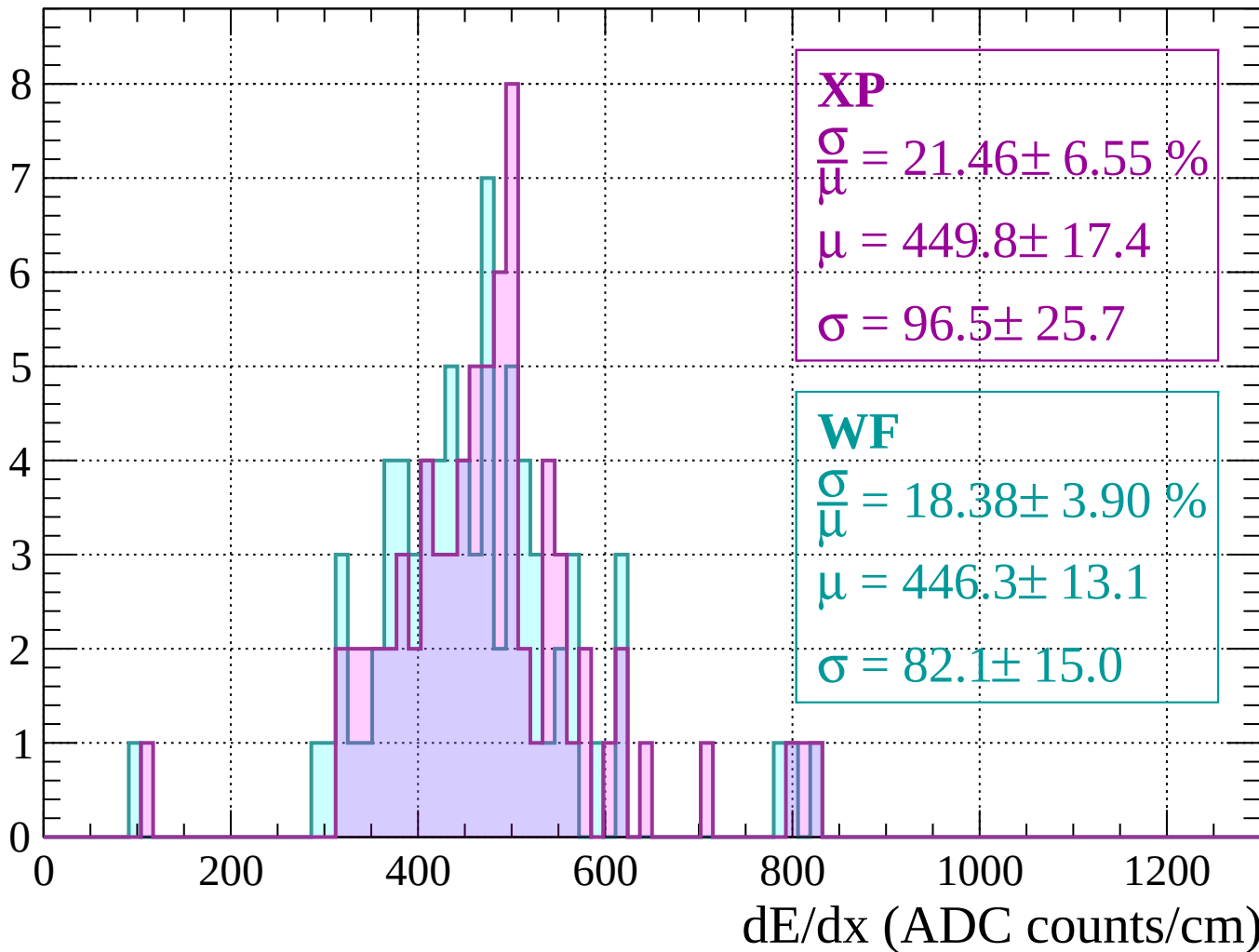
Energy loss | $240 < p < 280$

Count



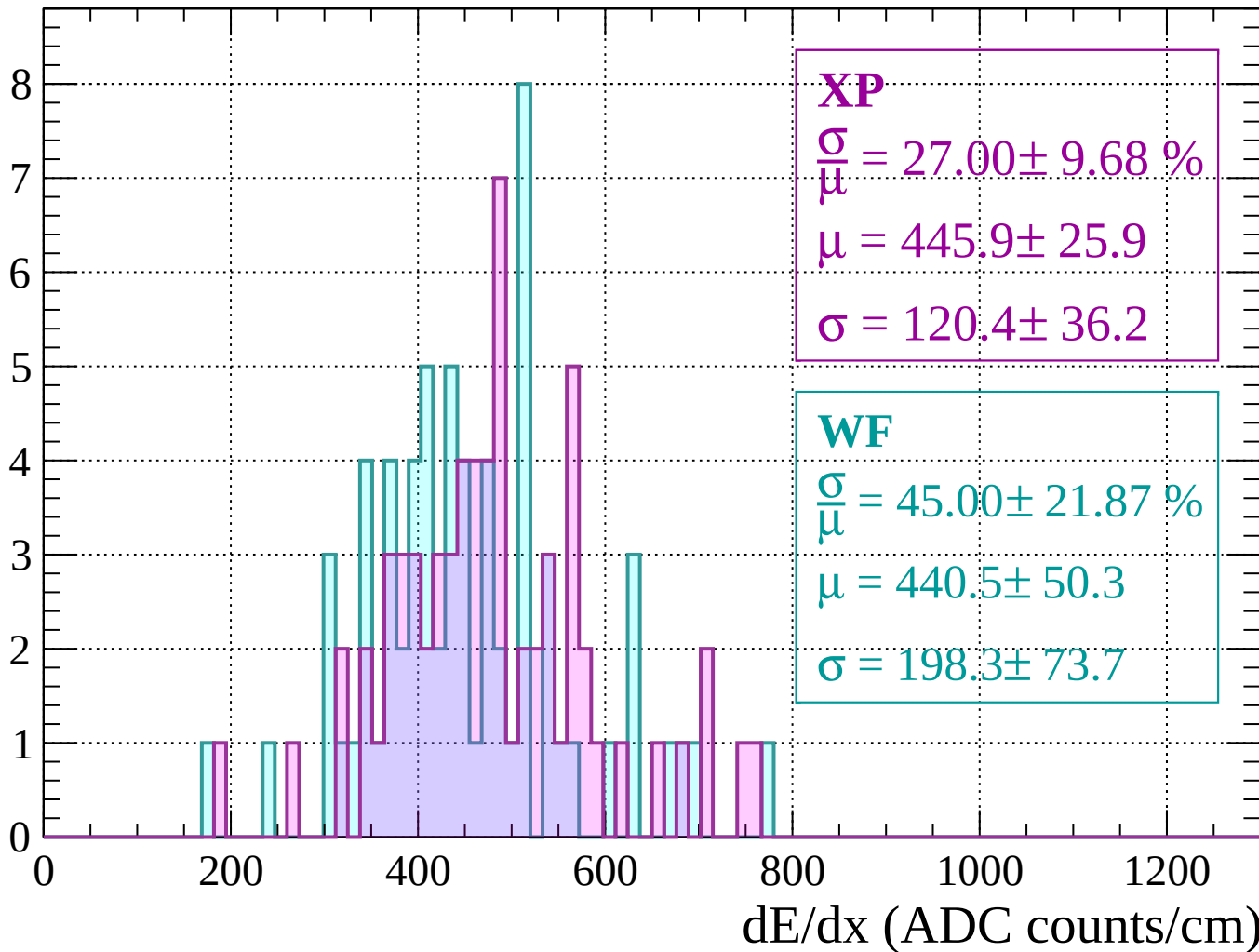
Energy loss | $280 < p < 320$

Count



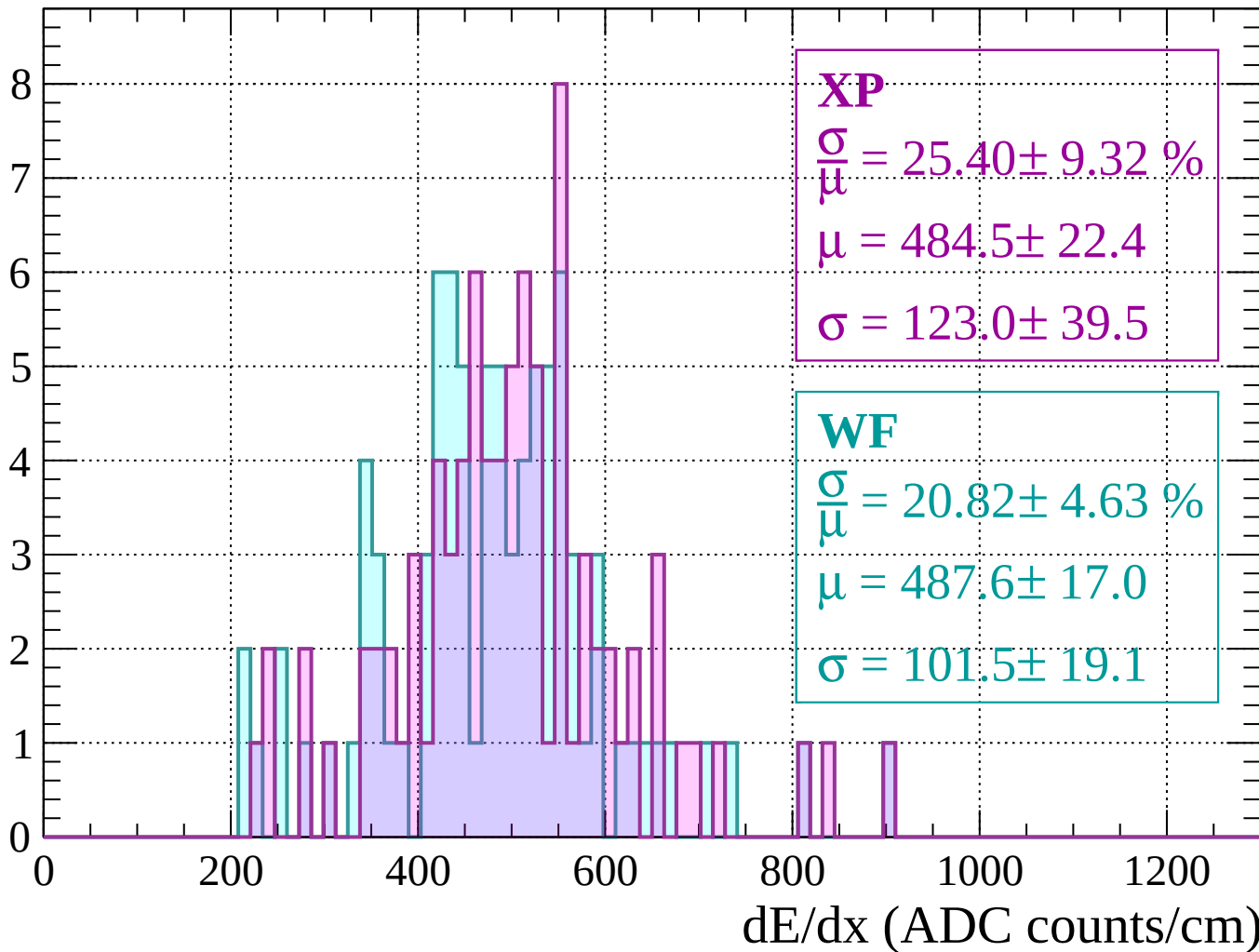
Energy loss | $320 < p < 360$

Count



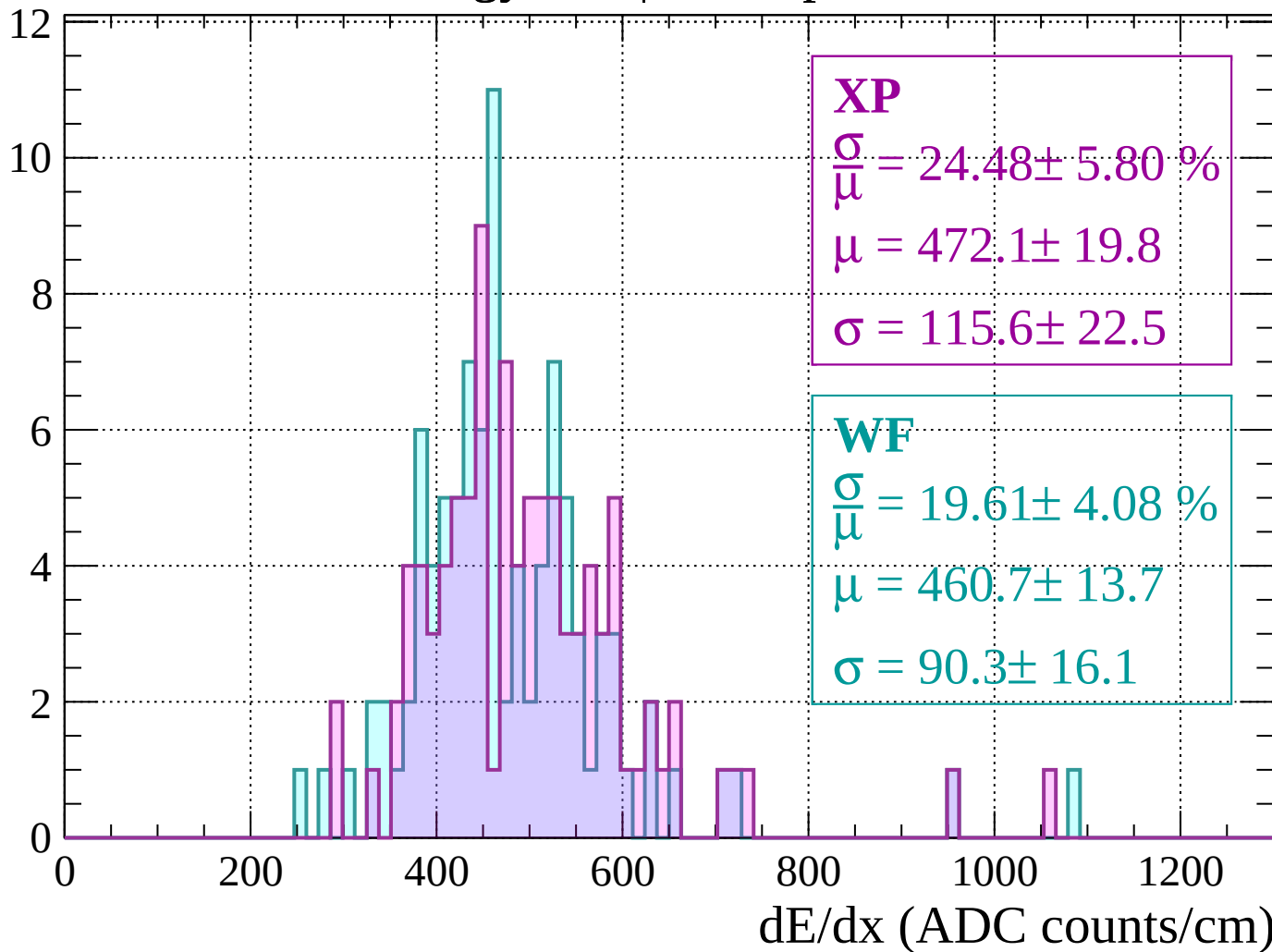
Energy loss | $360 < p < 400$

Count



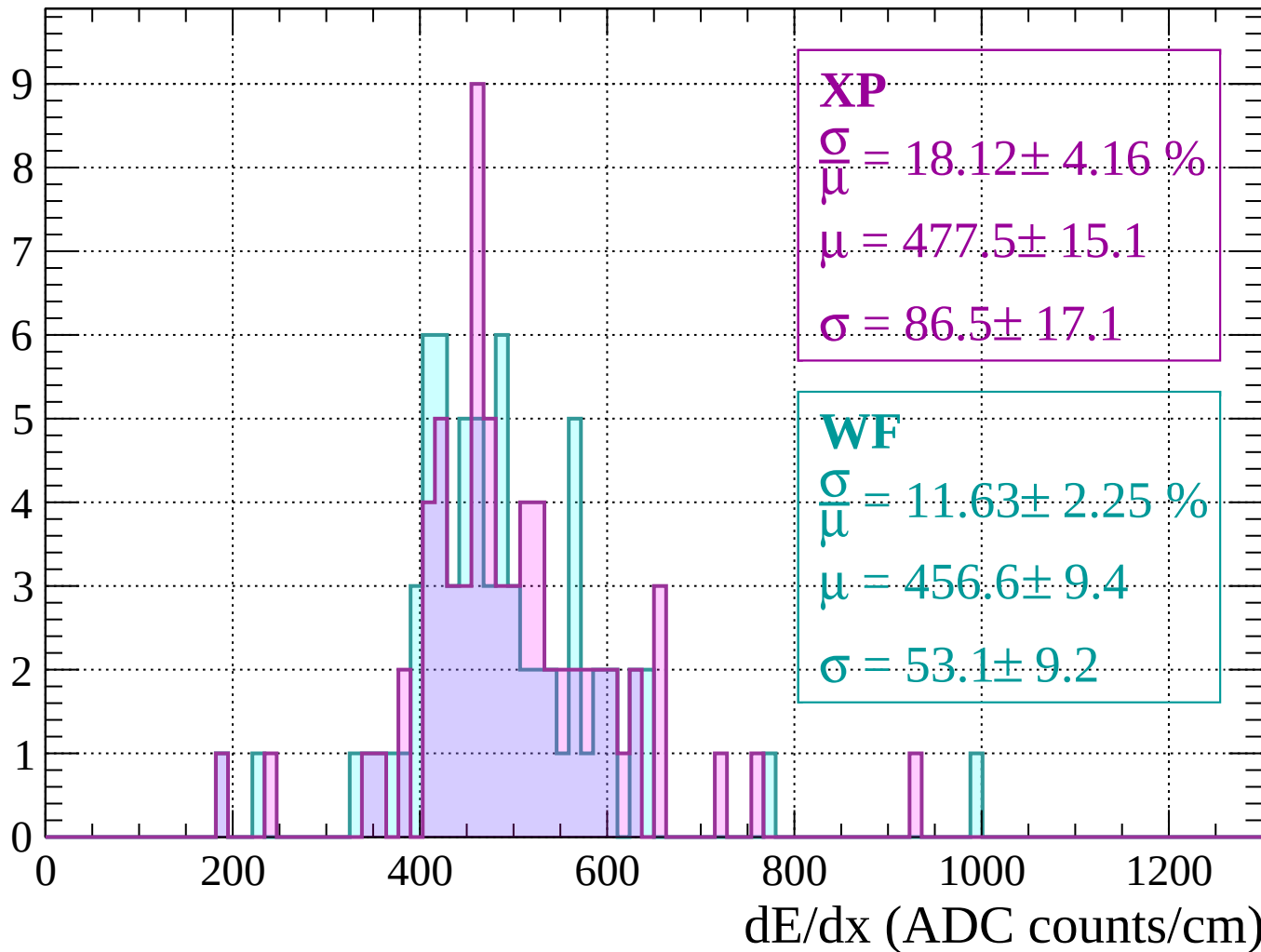
Energy loss | $400 < p < 440$

Count



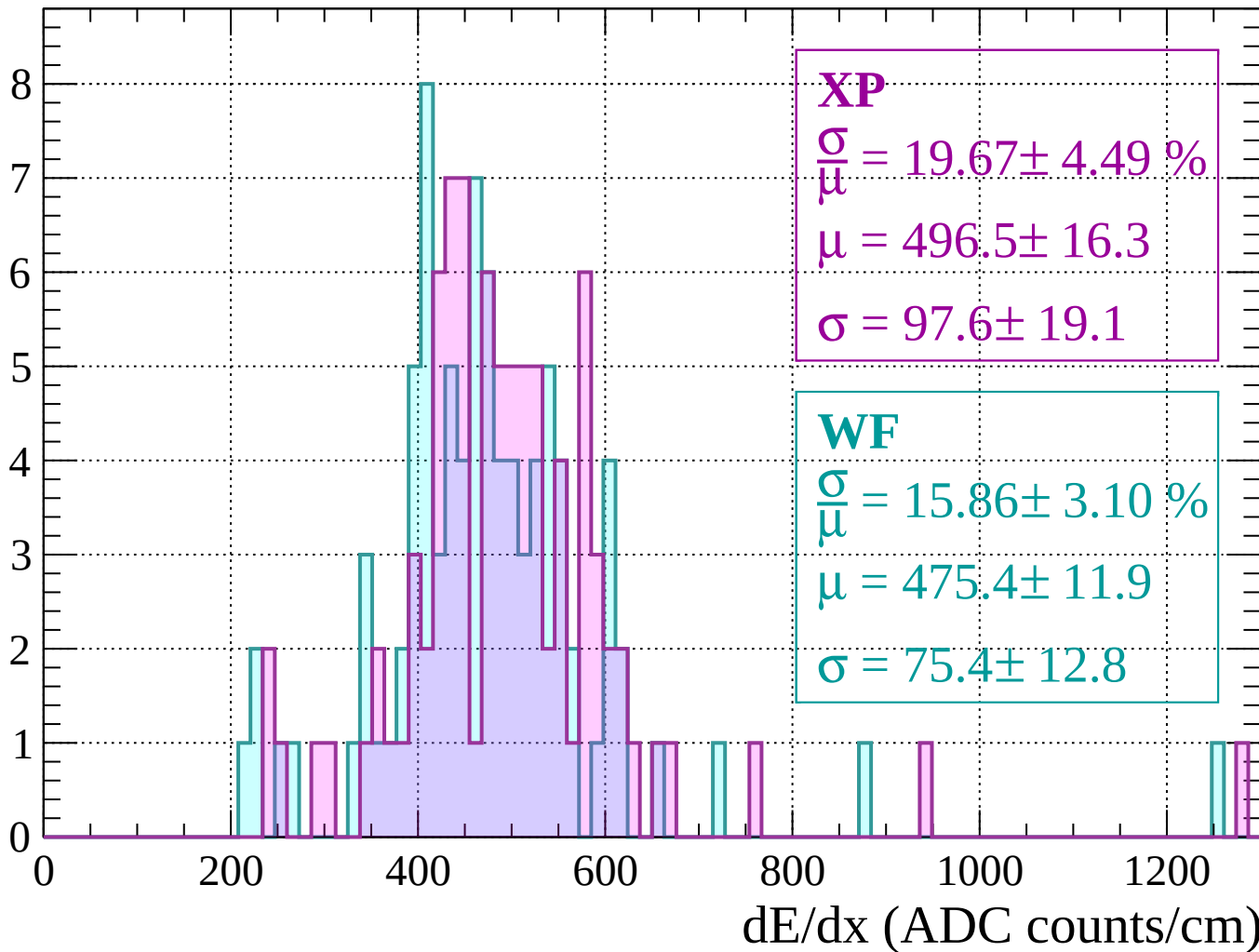
Energy loss | $440 < p < 480$

Count



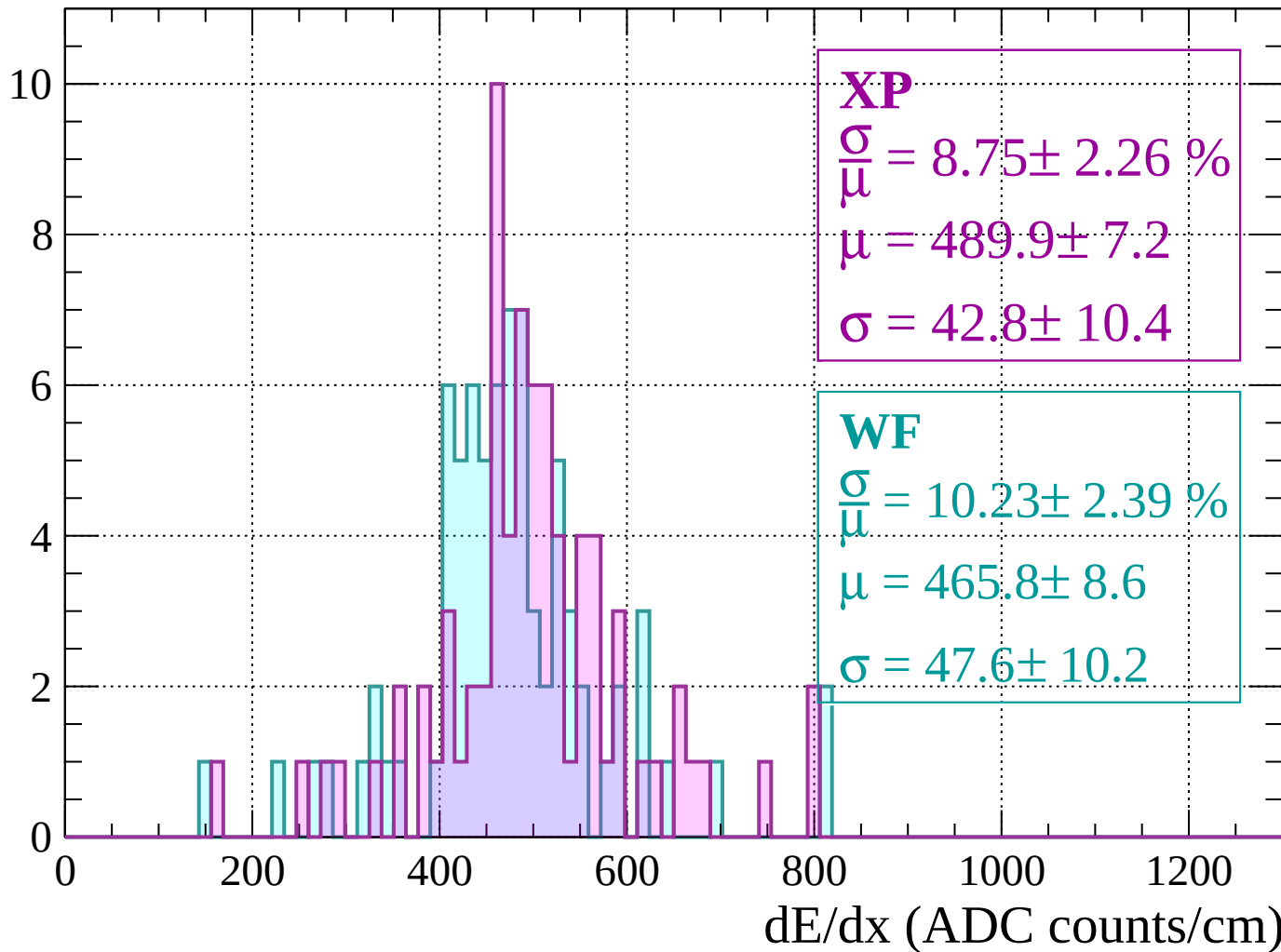
Energy loss | $480 < p < 520$

Count



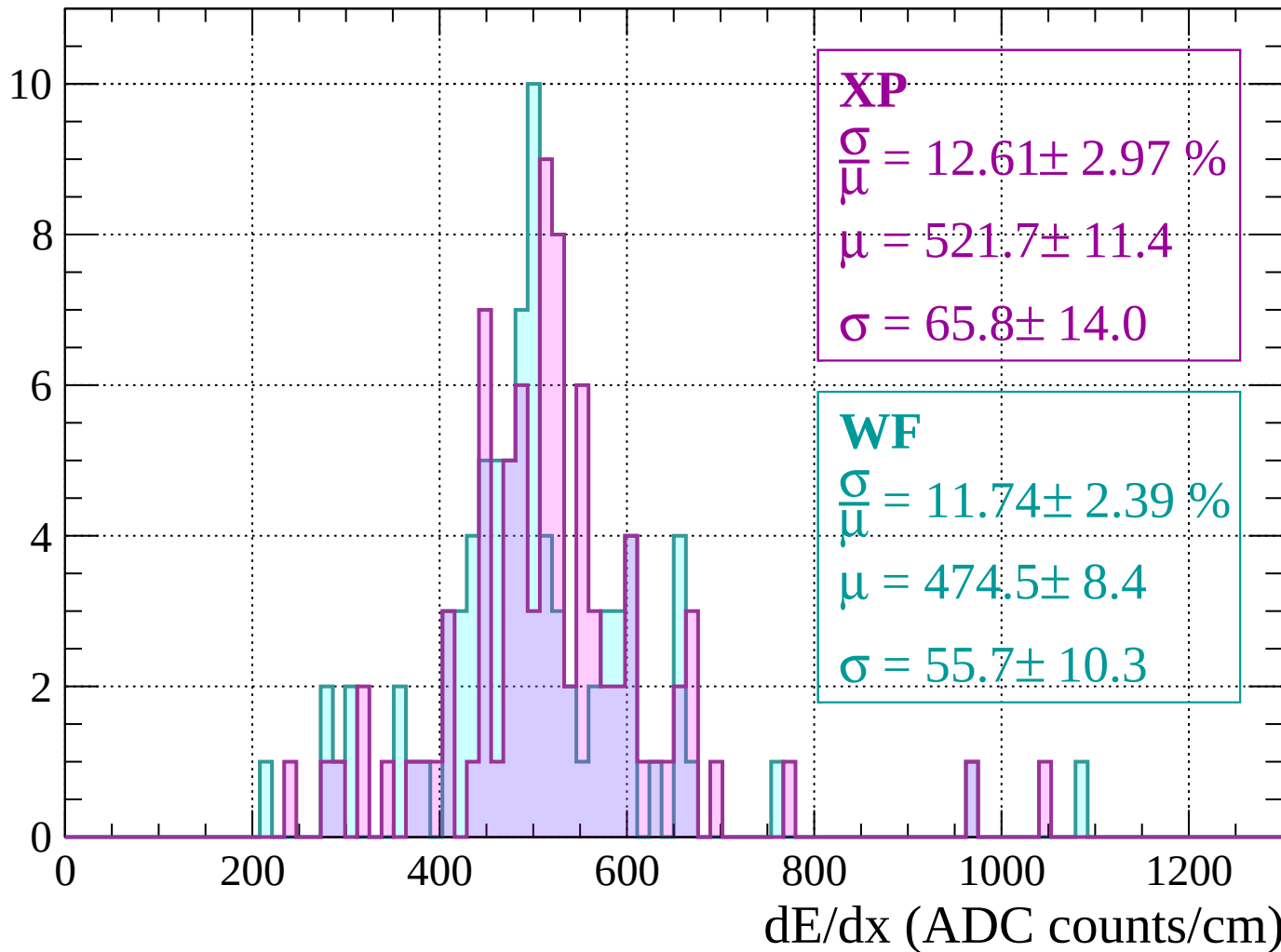
Energy loss | $520 < p < 560$

Count



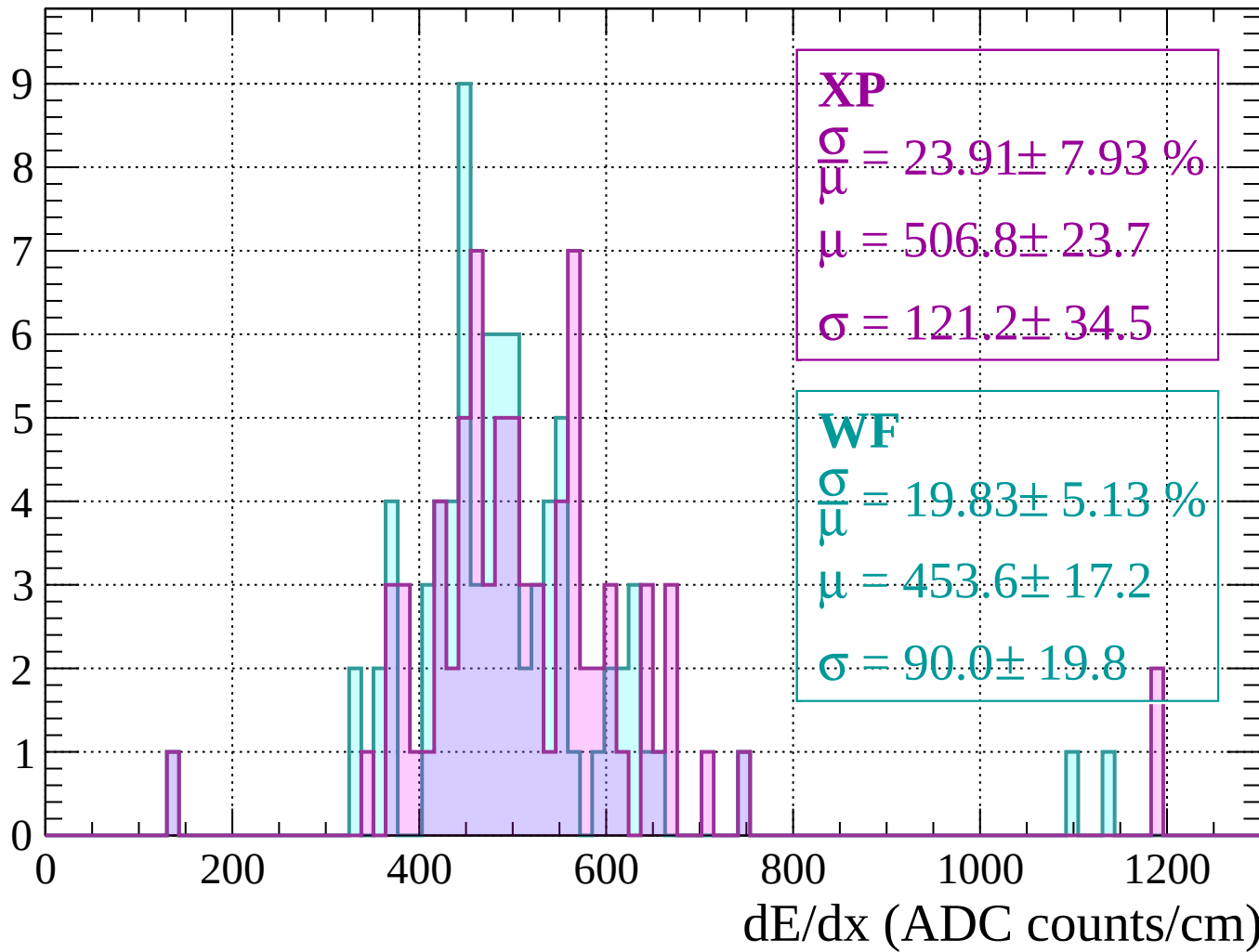
Energy loss | $560 < p < 600$

Count



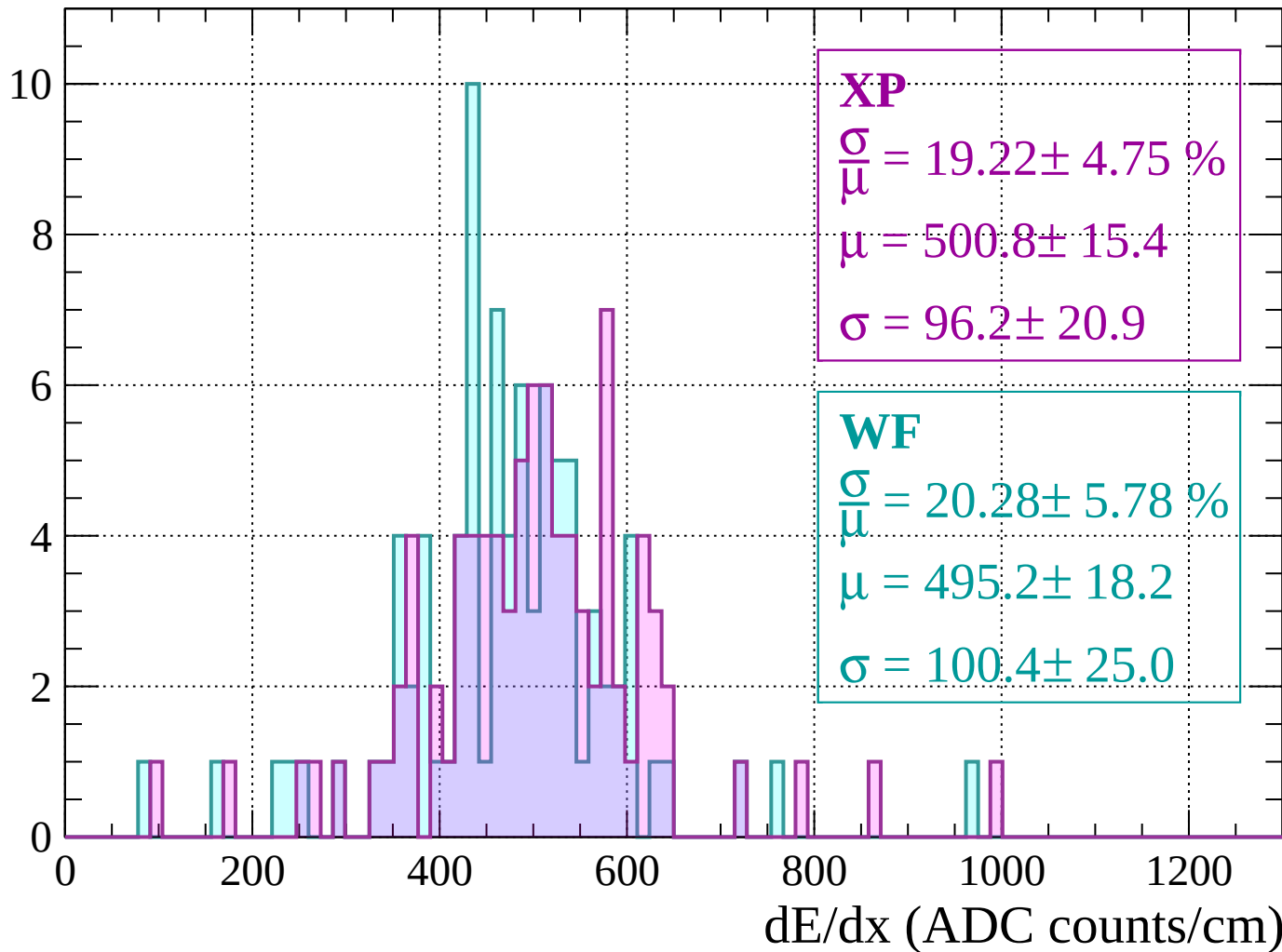
Energy loss | $600 < p < 640$

Count



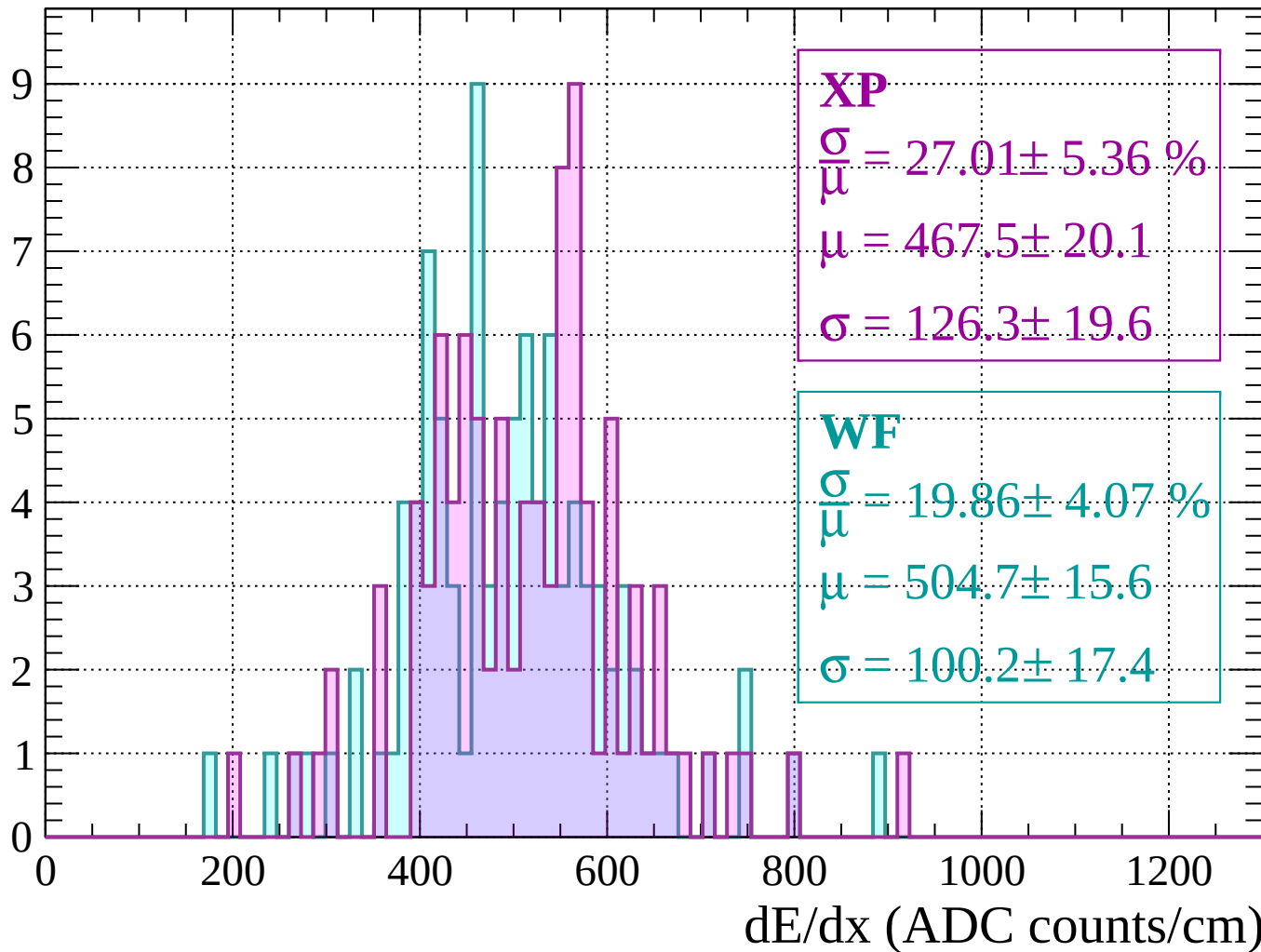
Energy loss | $640 < p < 680$

Count



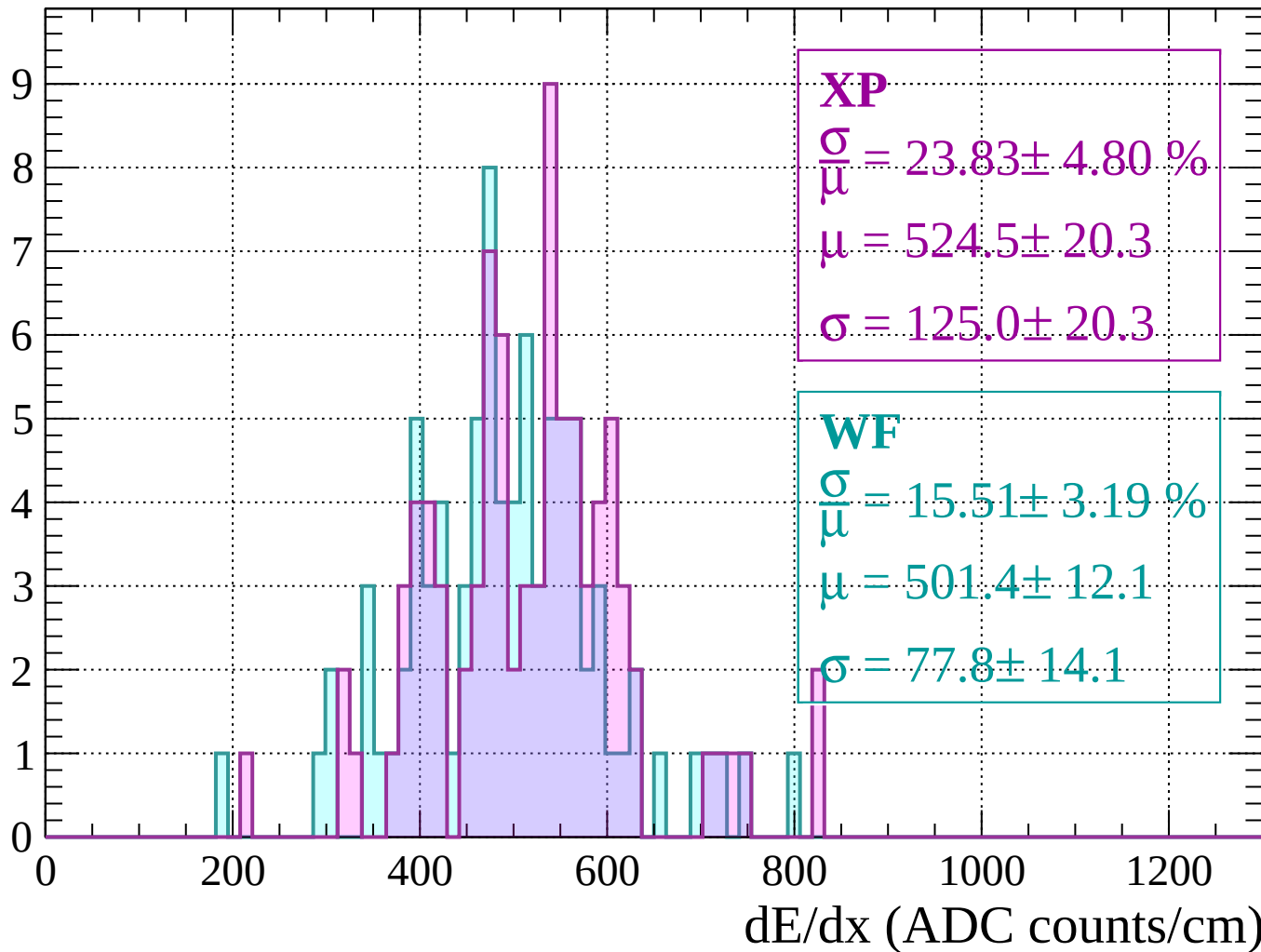
Energy loss | $680 < p < 720$

Count



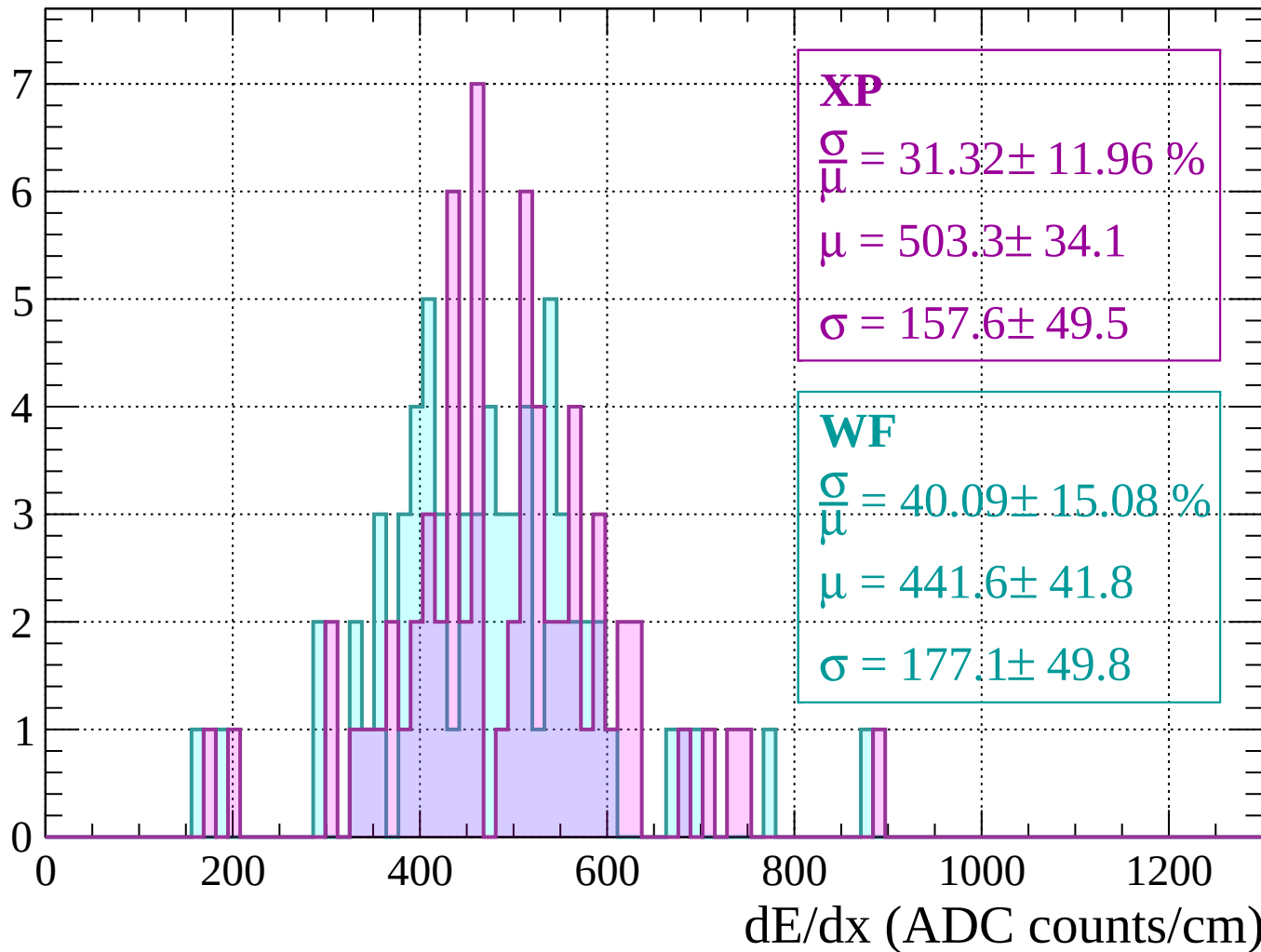
Energy loss | $720 < p < 760$

Count



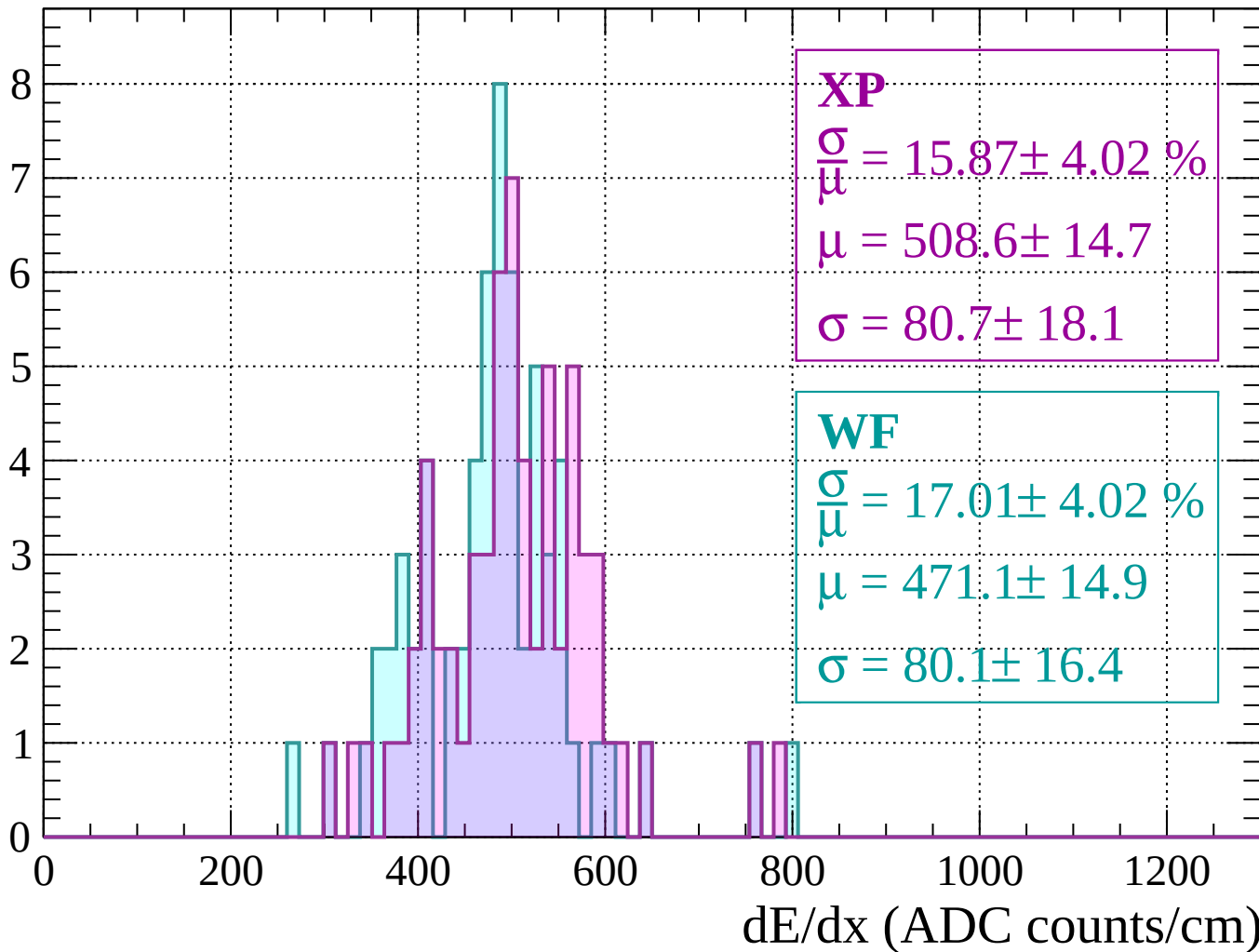
Energy loss | $760 < p < 800$

Count



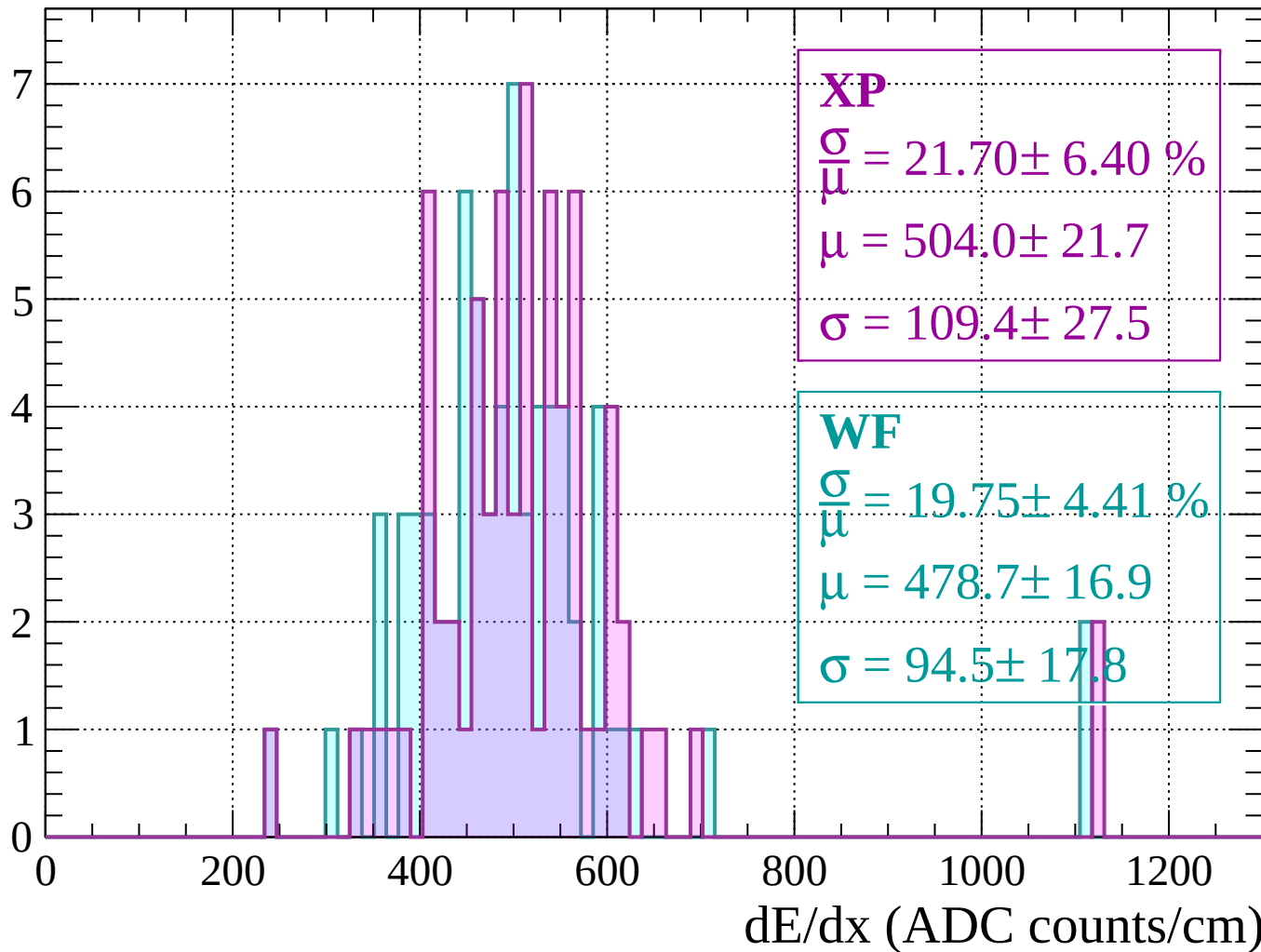
Energy loss | $800 < p < 840$

Count



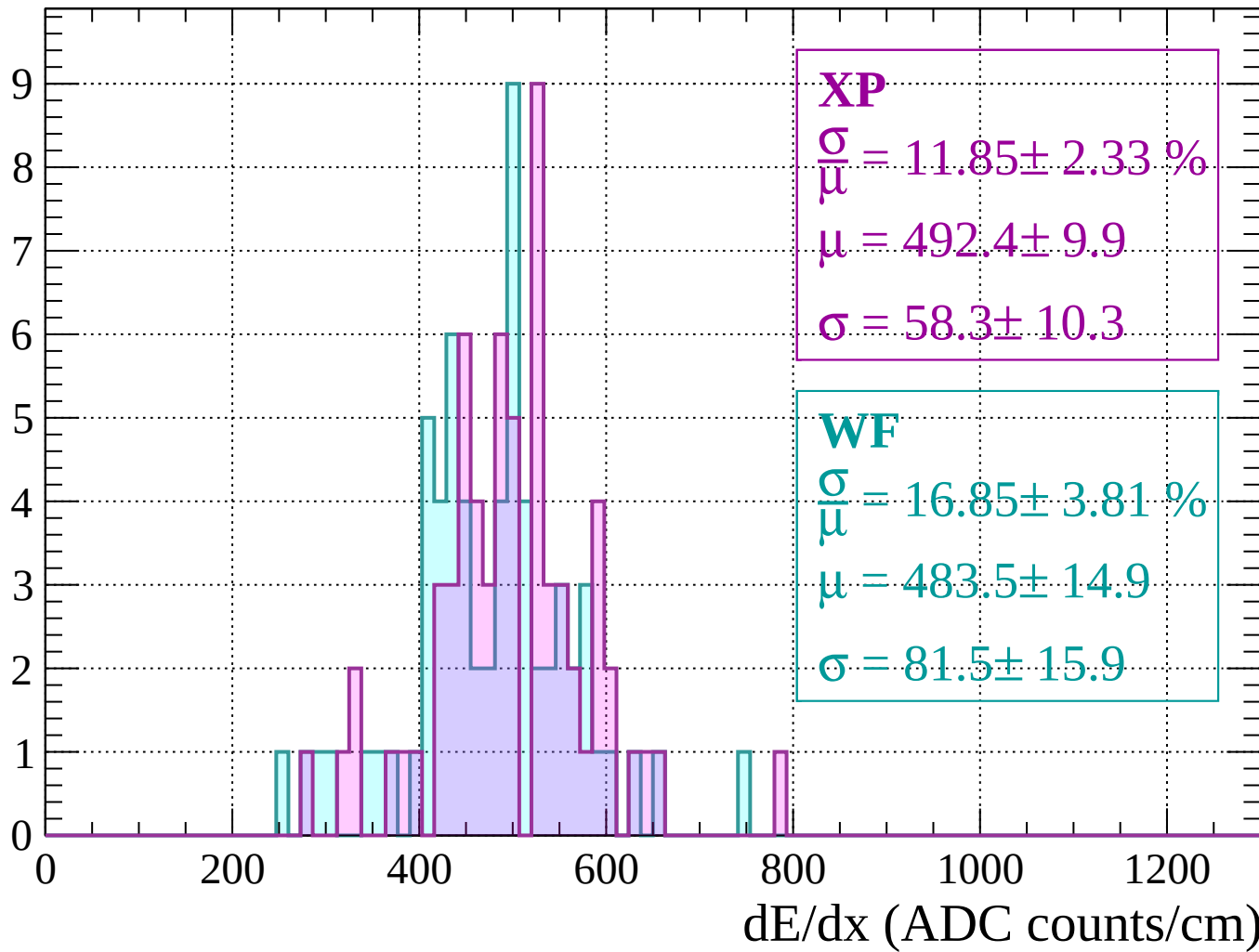
Energy loss | $840 < p < 880$

Count



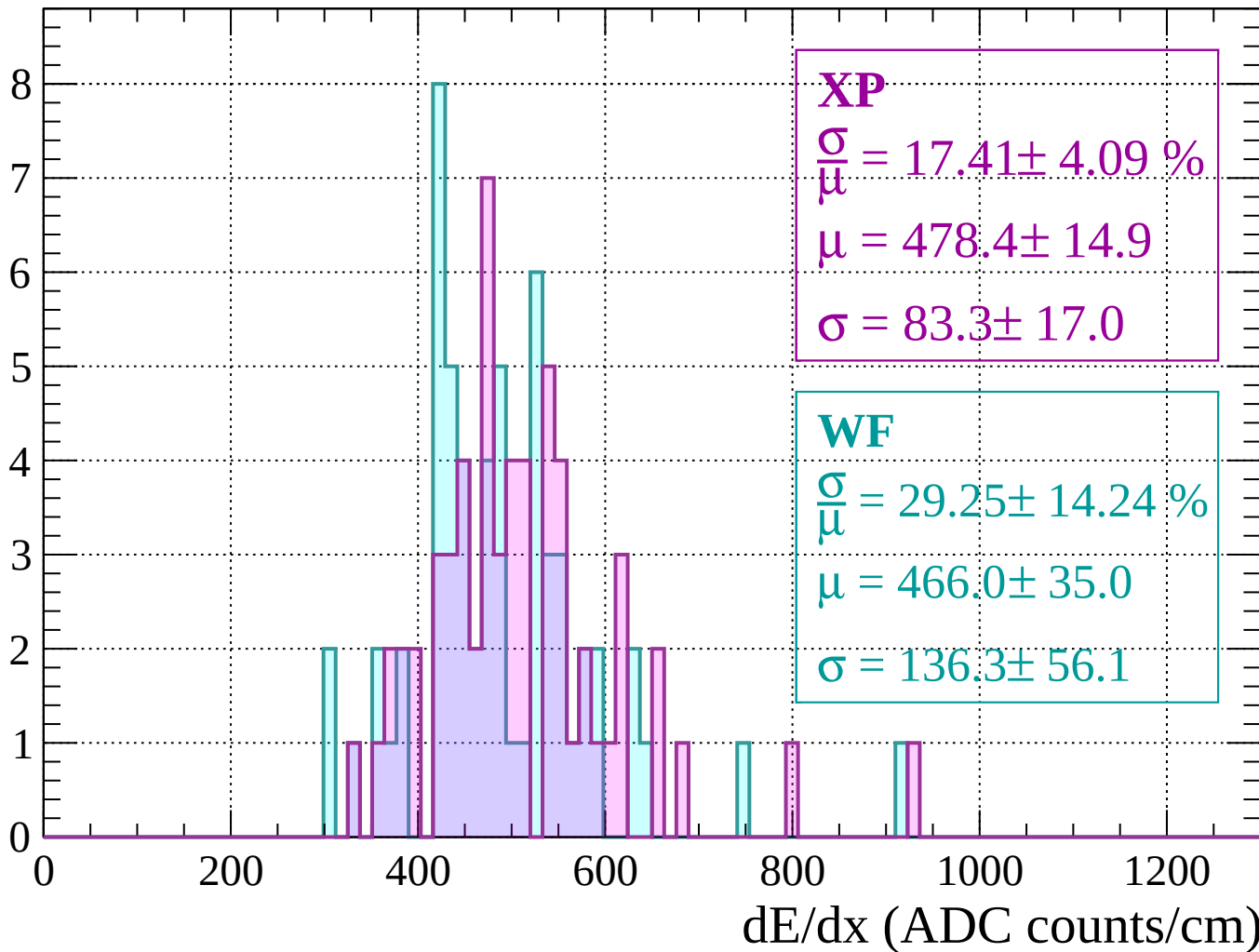
Energy loss | $880 < p < 920$

Count



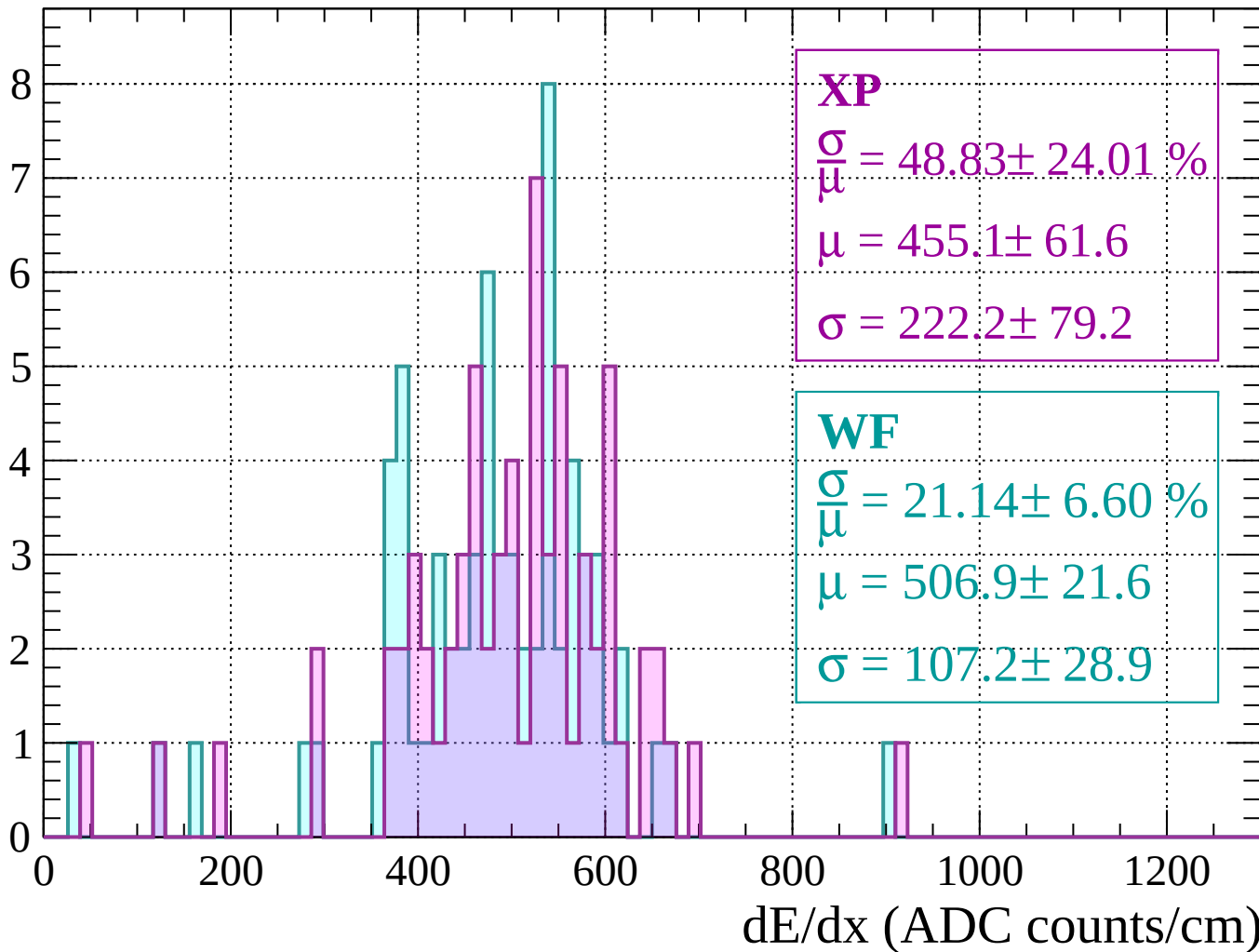
Energy loss | $920 < p < 960$

Count



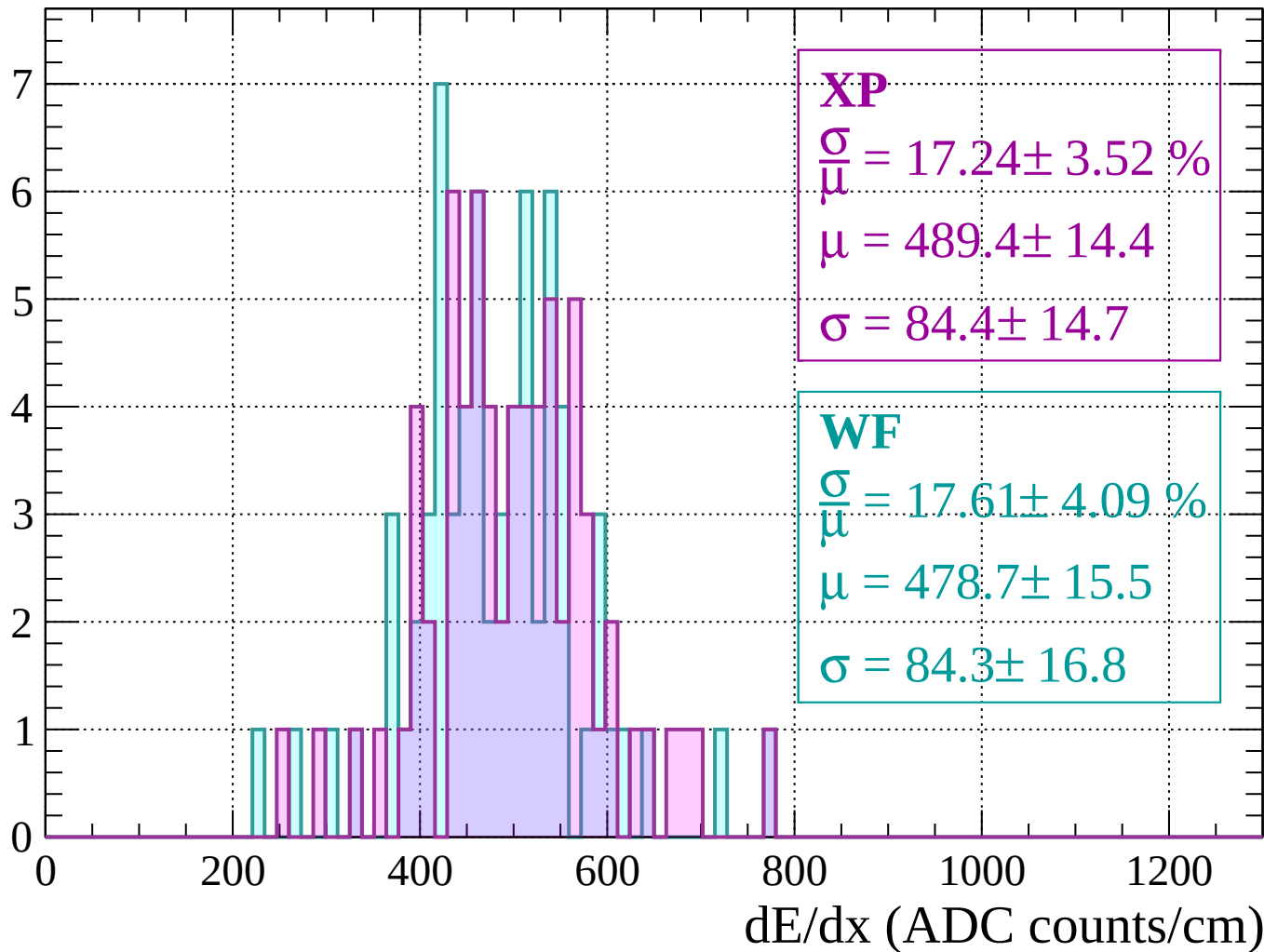
Energy loss | $960 < p < 1000$

Count



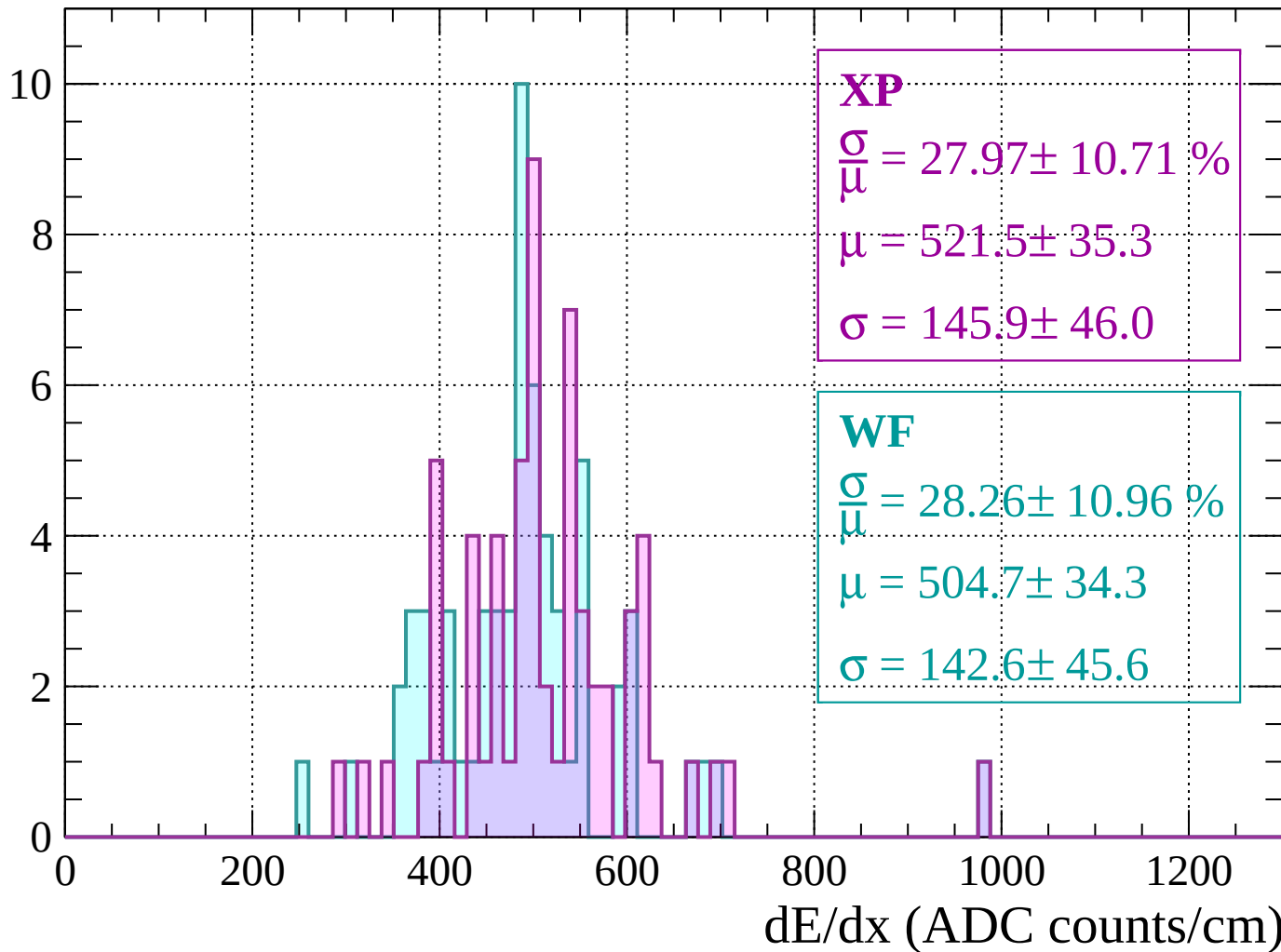
Energy loss | $1000 < p < 1040$

Count



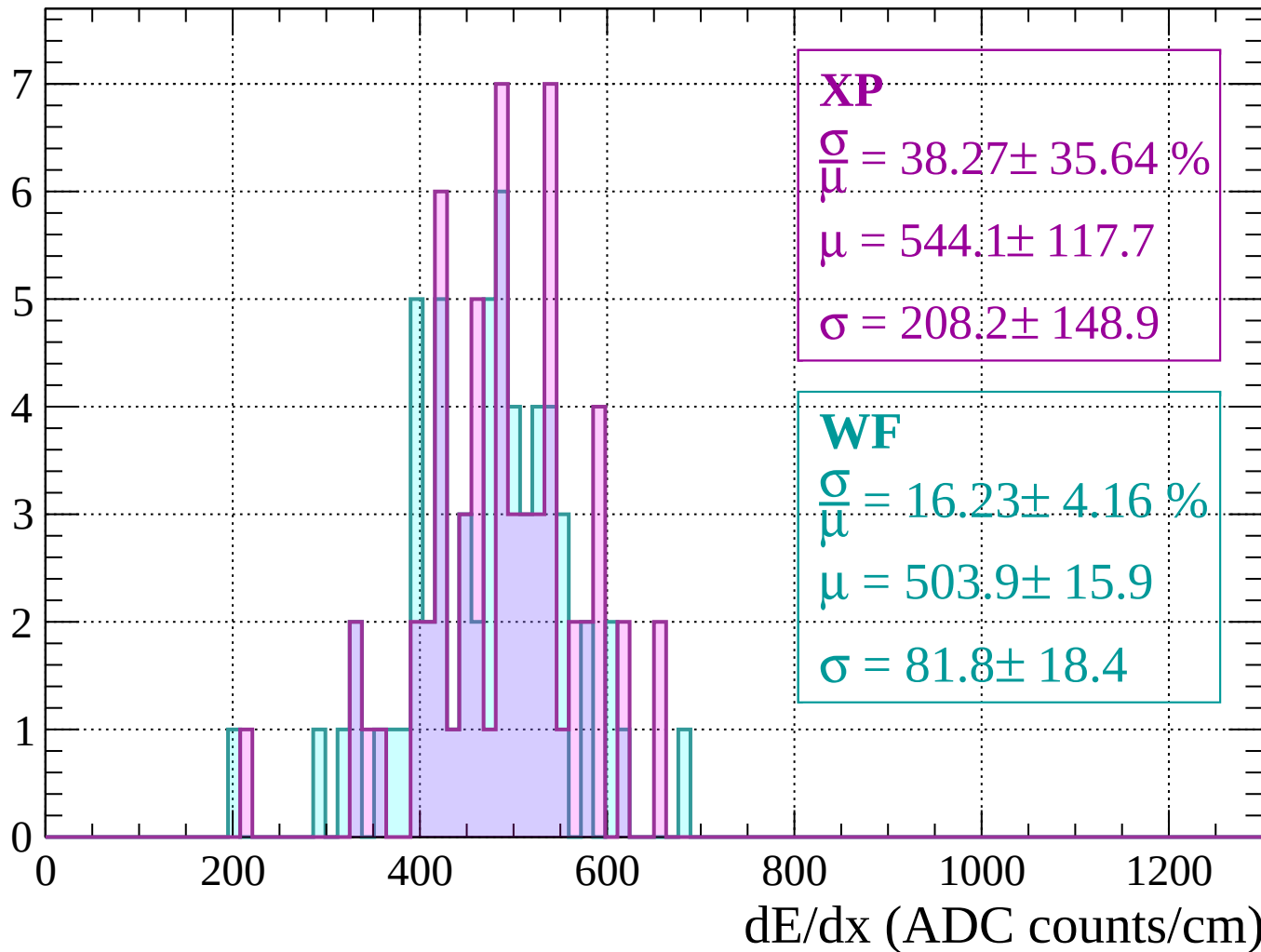
Energy loss | $1040 < p < 1080$

Count



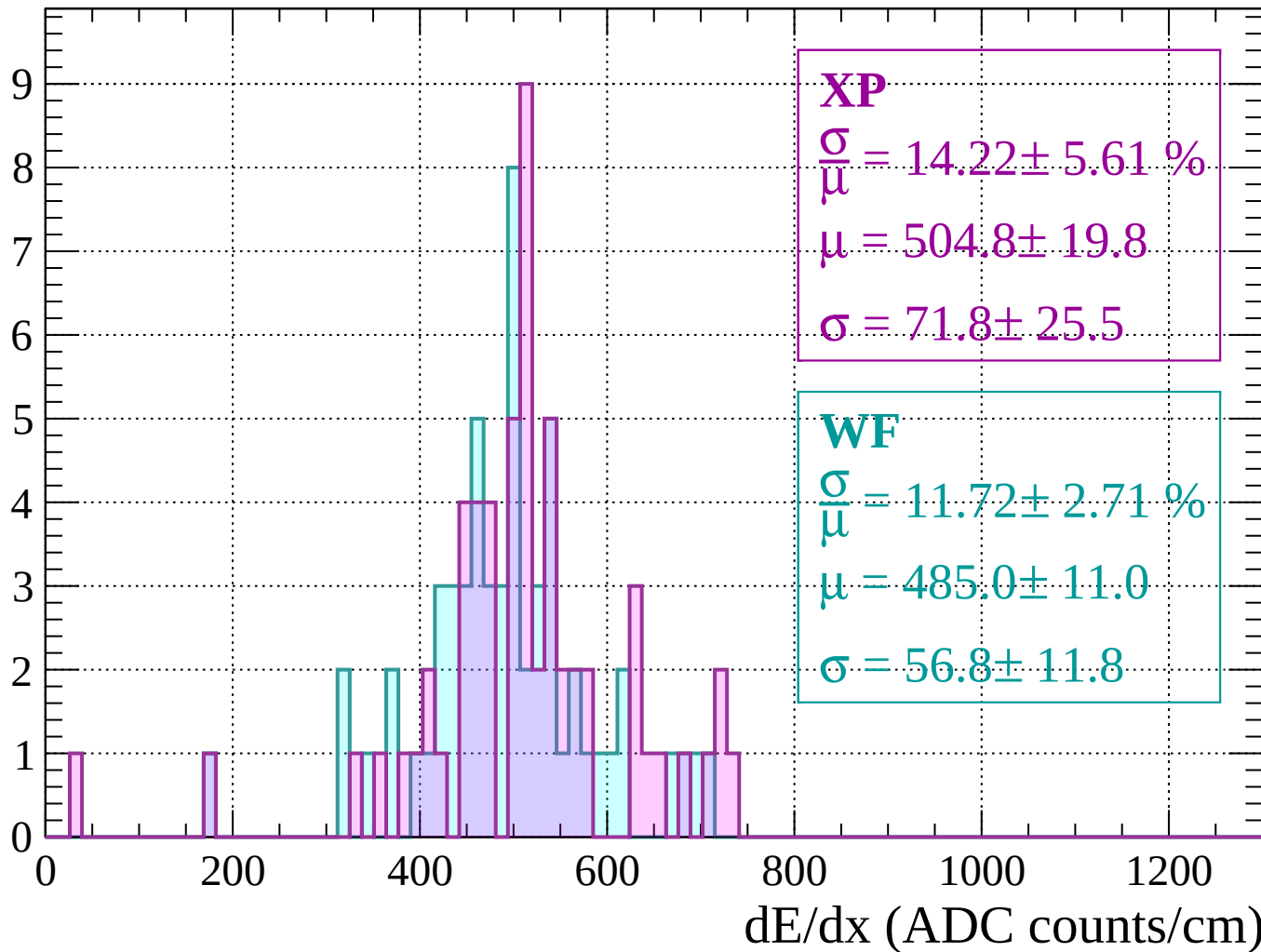
Energy loss | $1080 < p < 1120$

Count



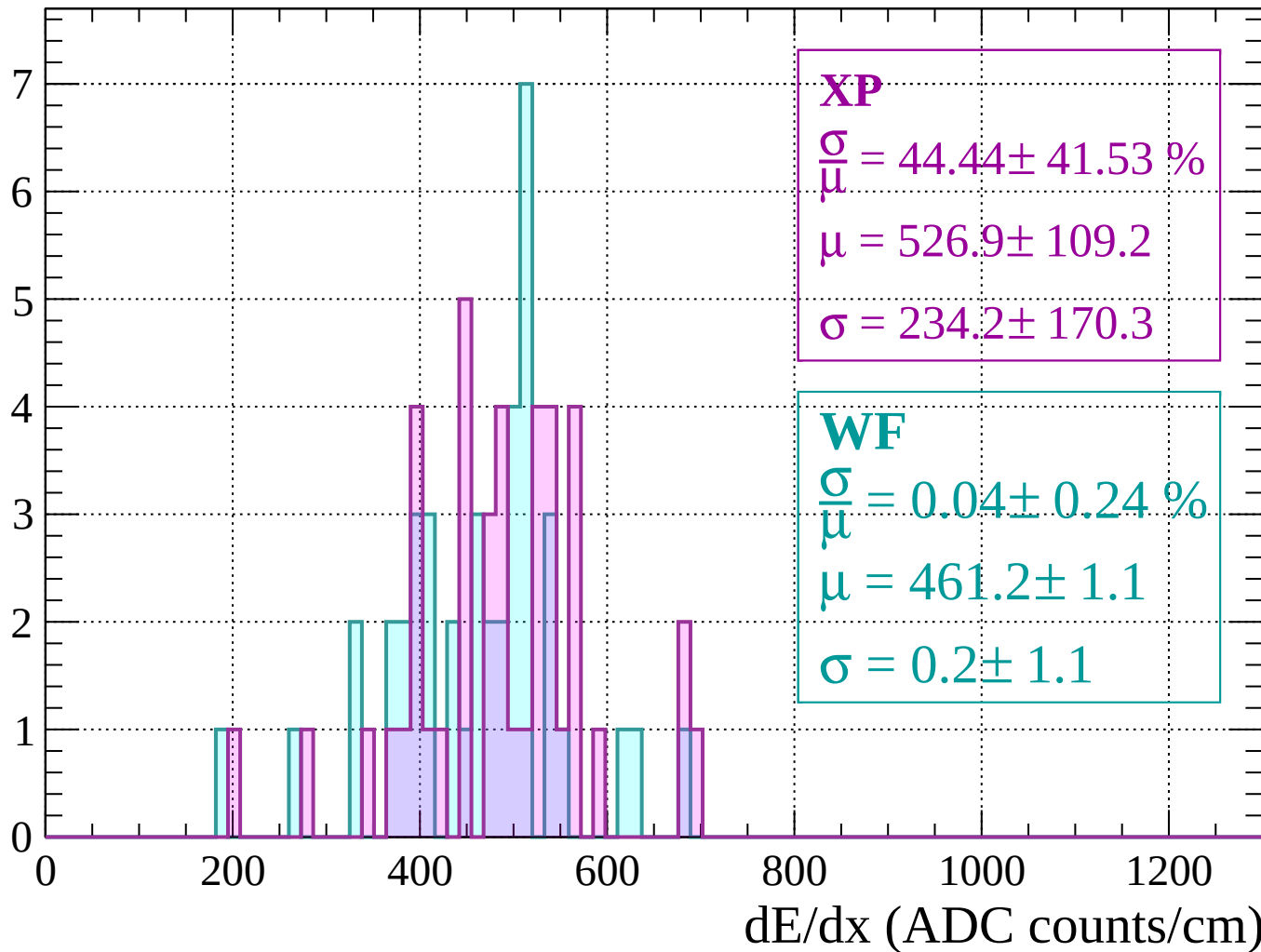
Energy loss | $1120 < p < 1160$

Count



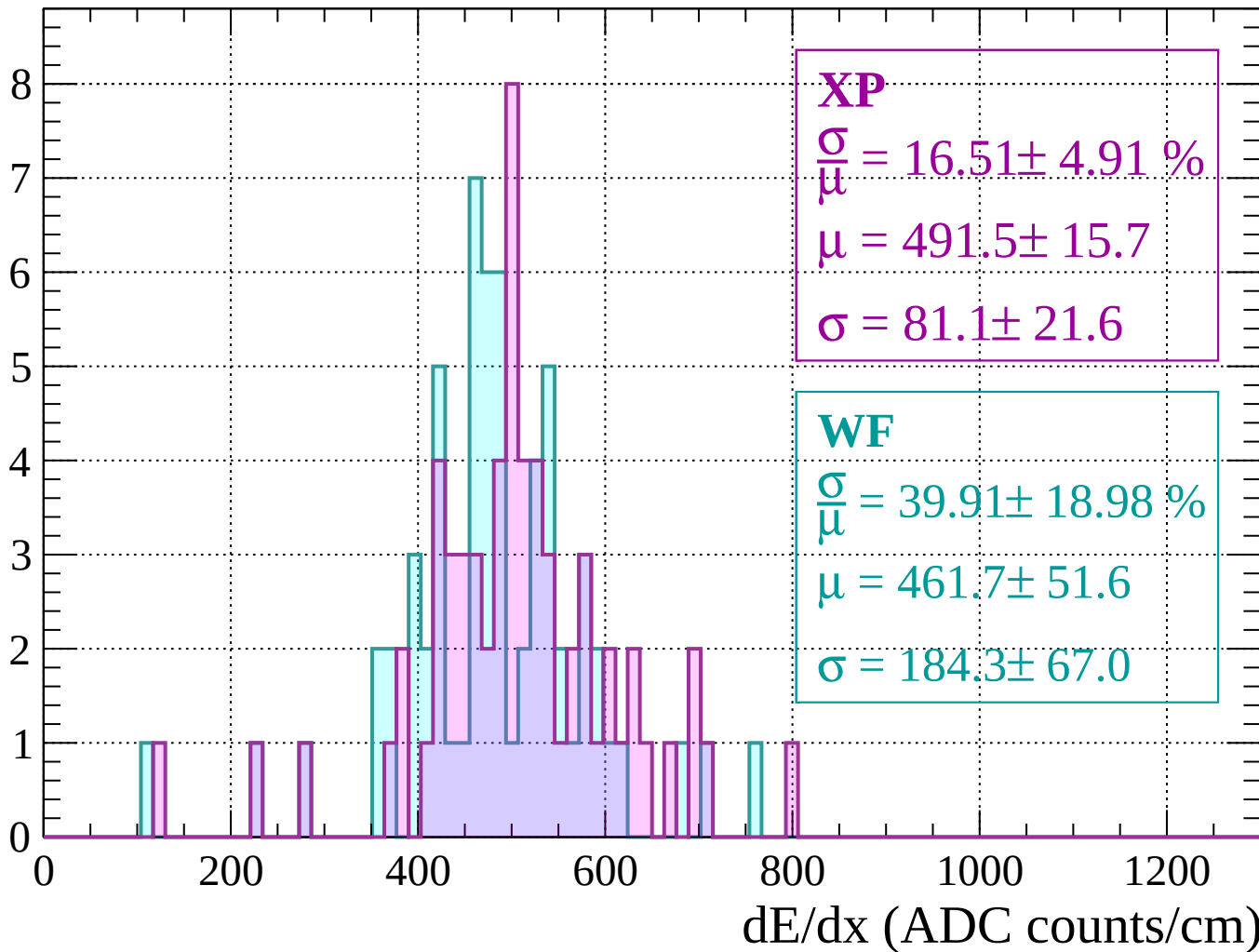
Energy loss | $1160 < p < 1200$

Count



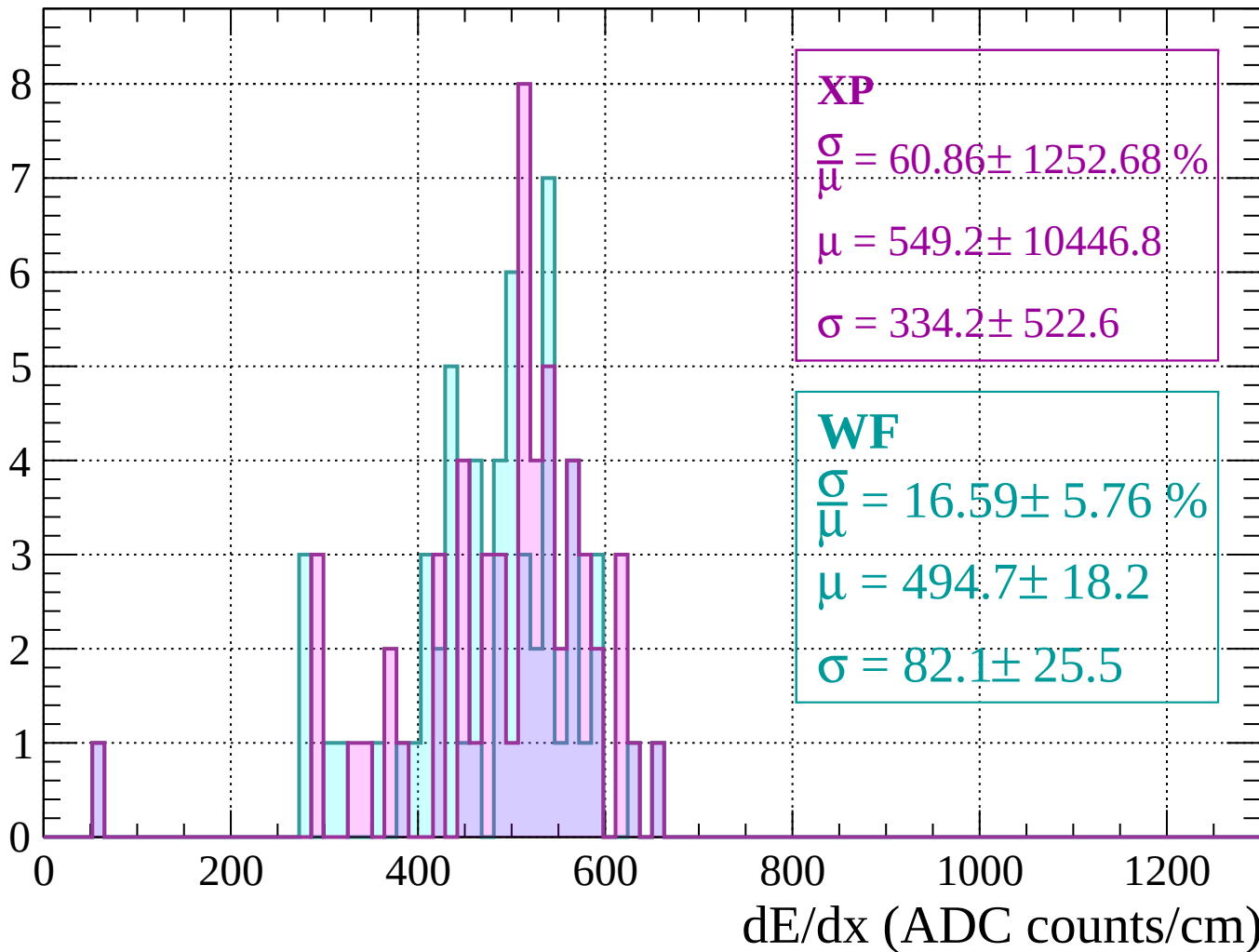
Energy loss | $1200 < p < 1240$

Count



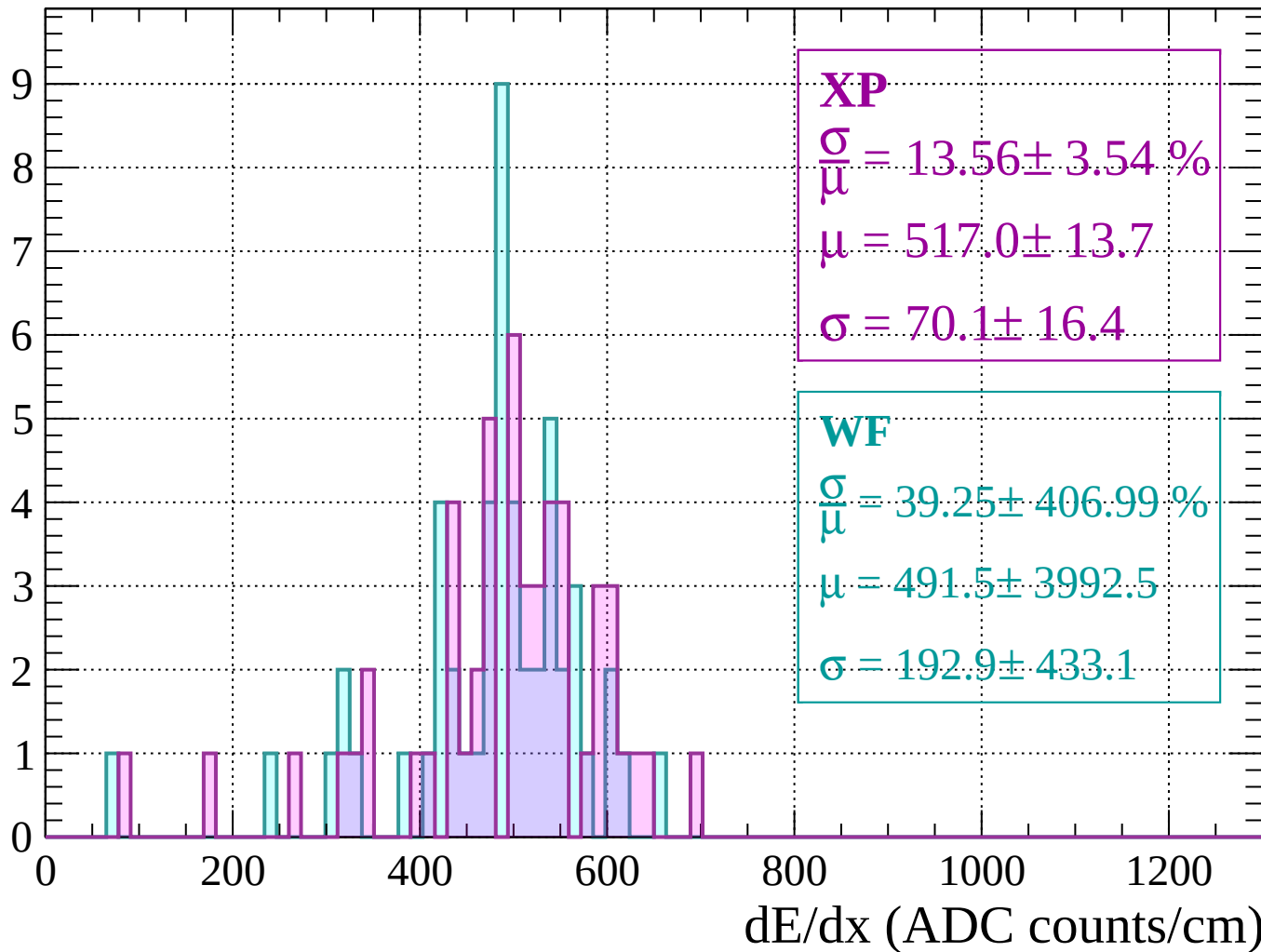
Energy loss | $1240 < p < 1280$

Count



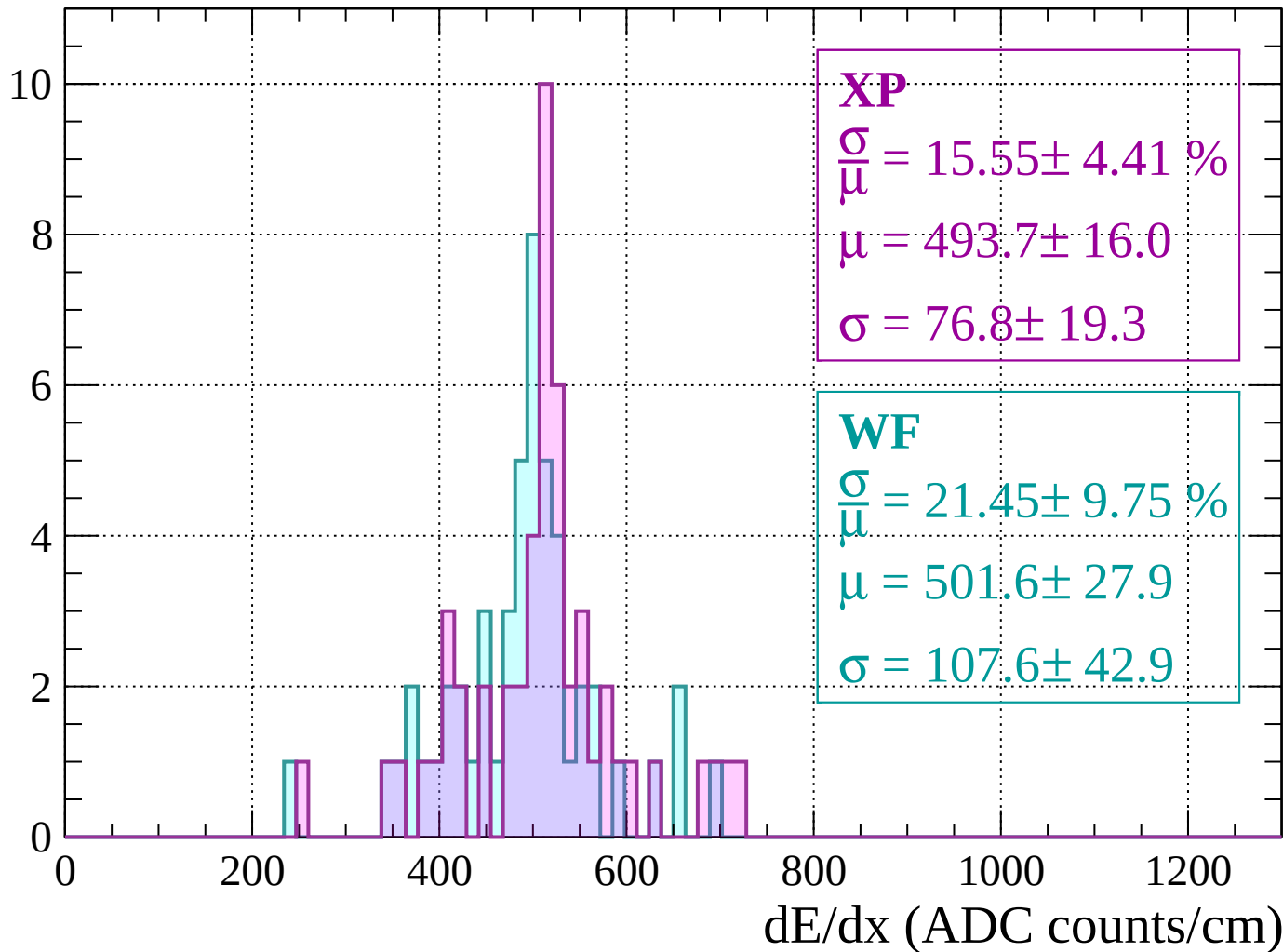
Energy loss | $1280 < p < 1320$

Count



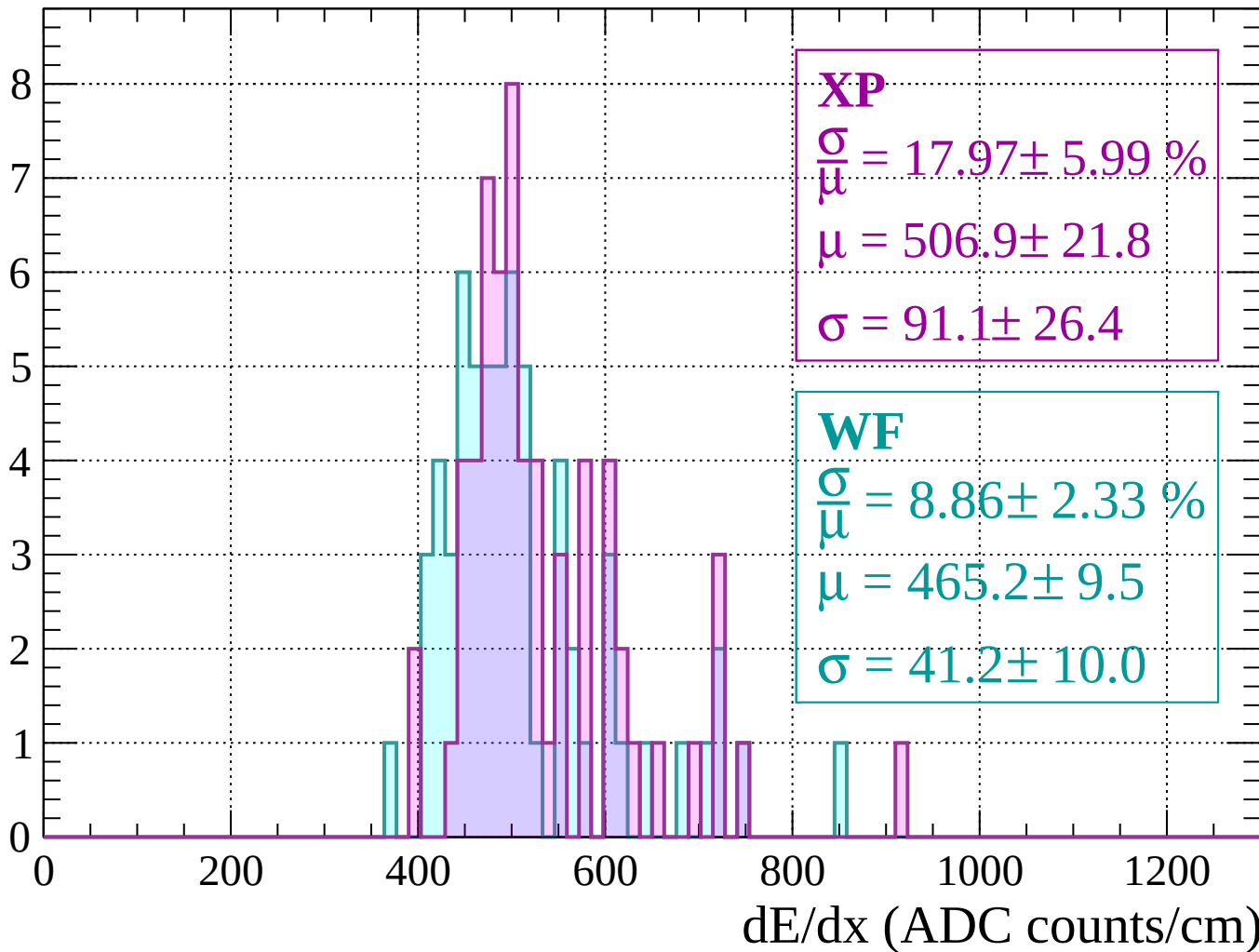
Energy loss | $1320 < p < 1360$

Count



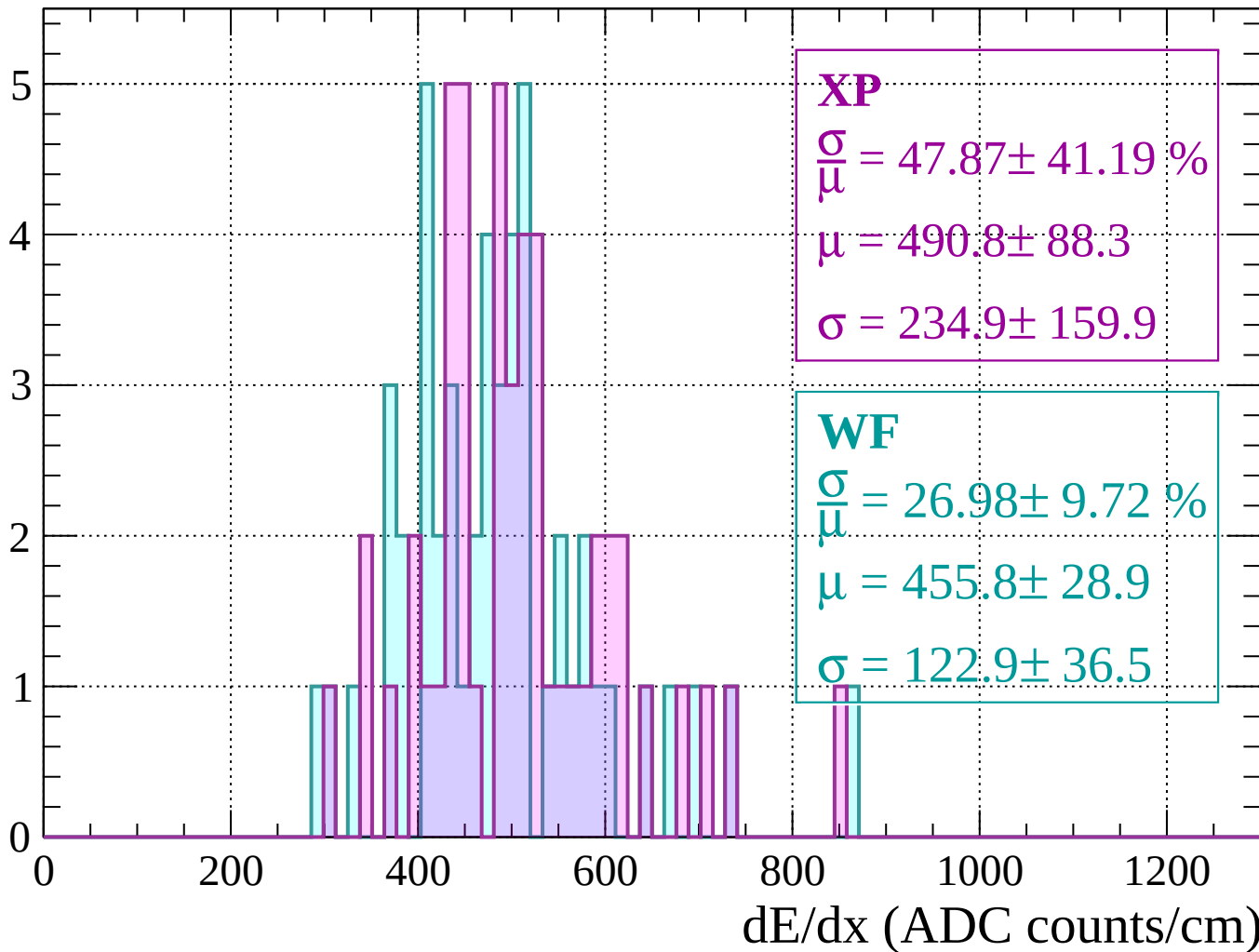
Energy loss | $1360 < p < 1400$

Count



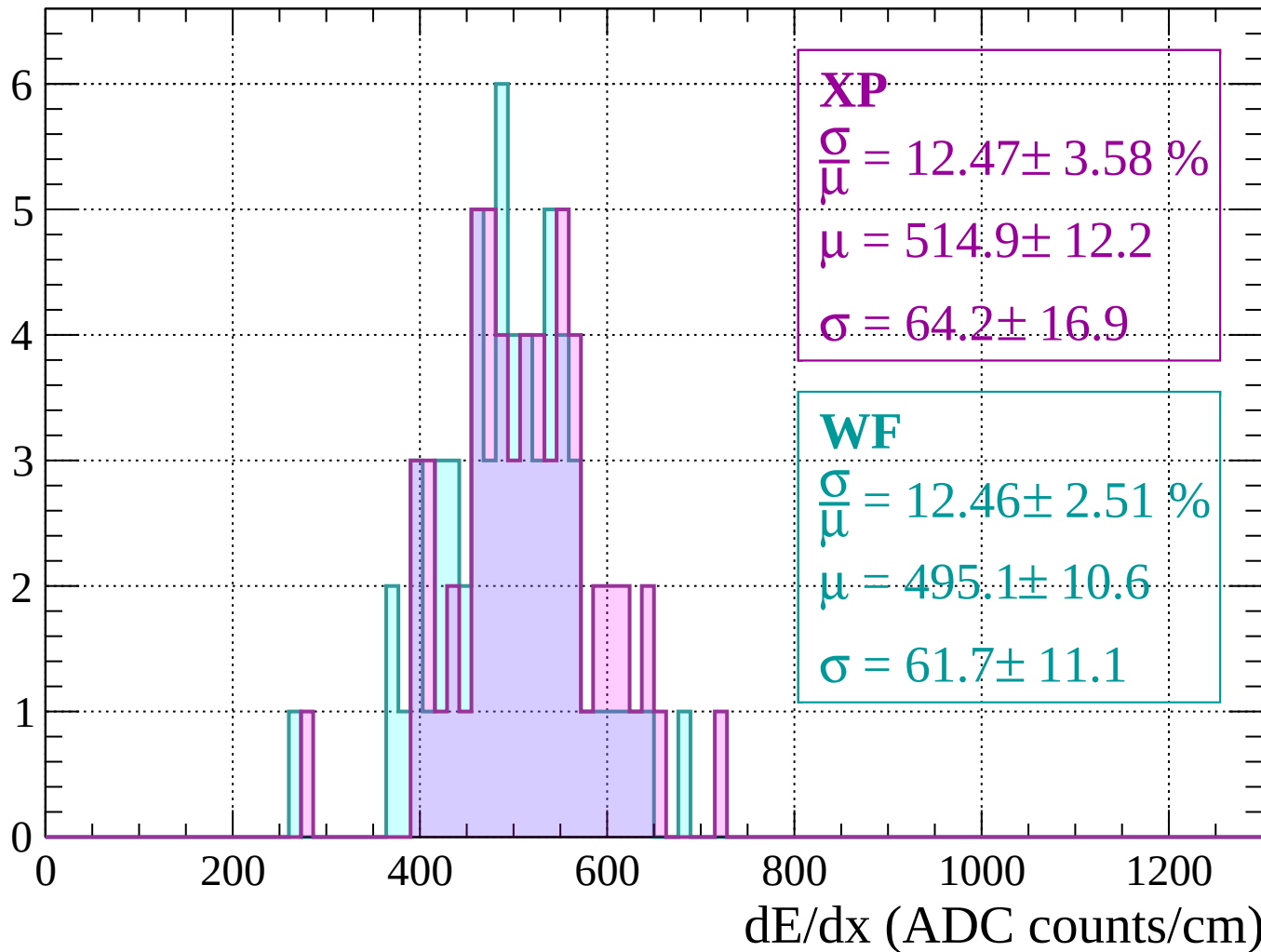
Energy loss | $1400 < p < 1440$

Count



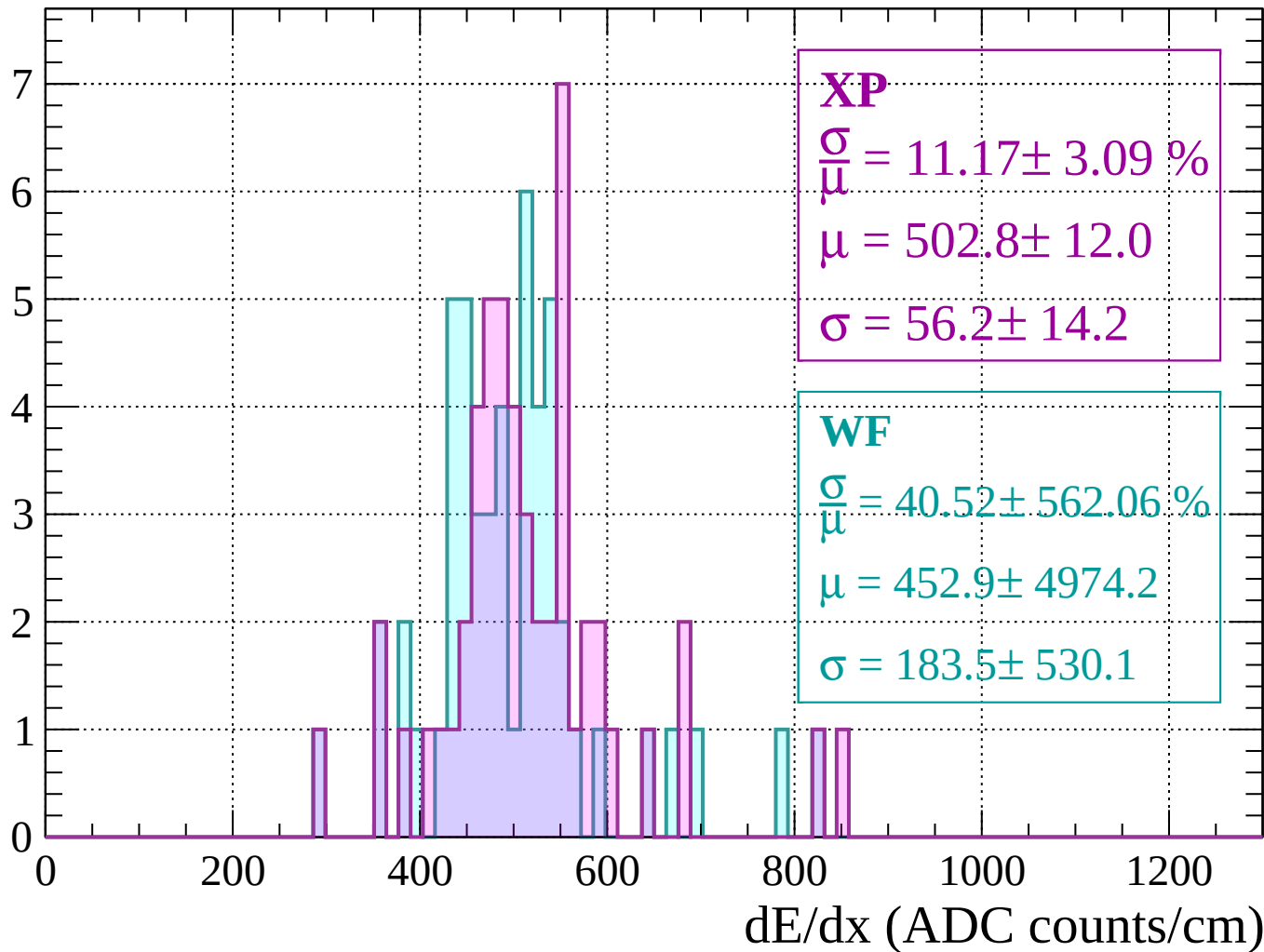
Energy loss | $1440 < p < 1480$

Count



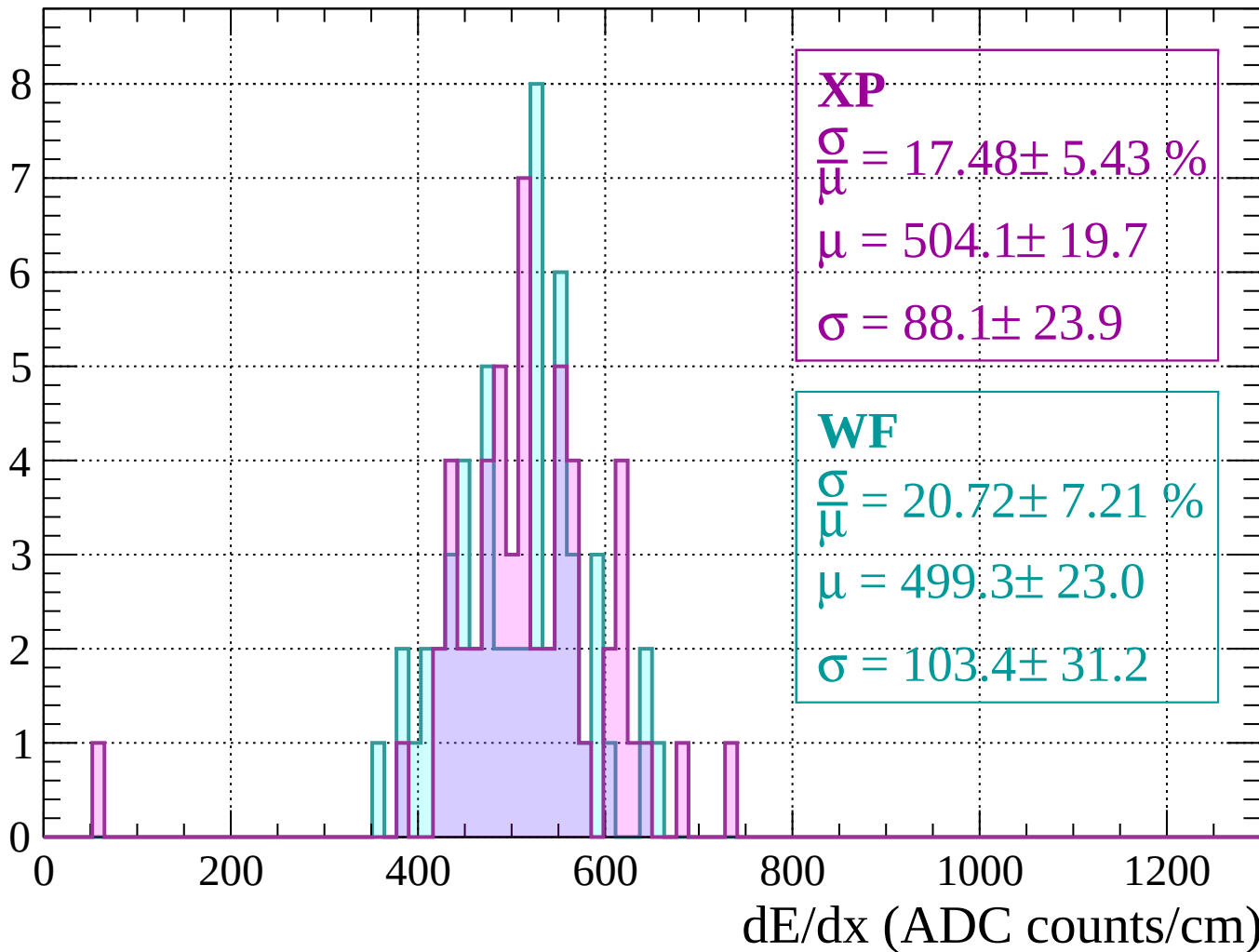
Energy loss | $1480 < p < 1520$

Count



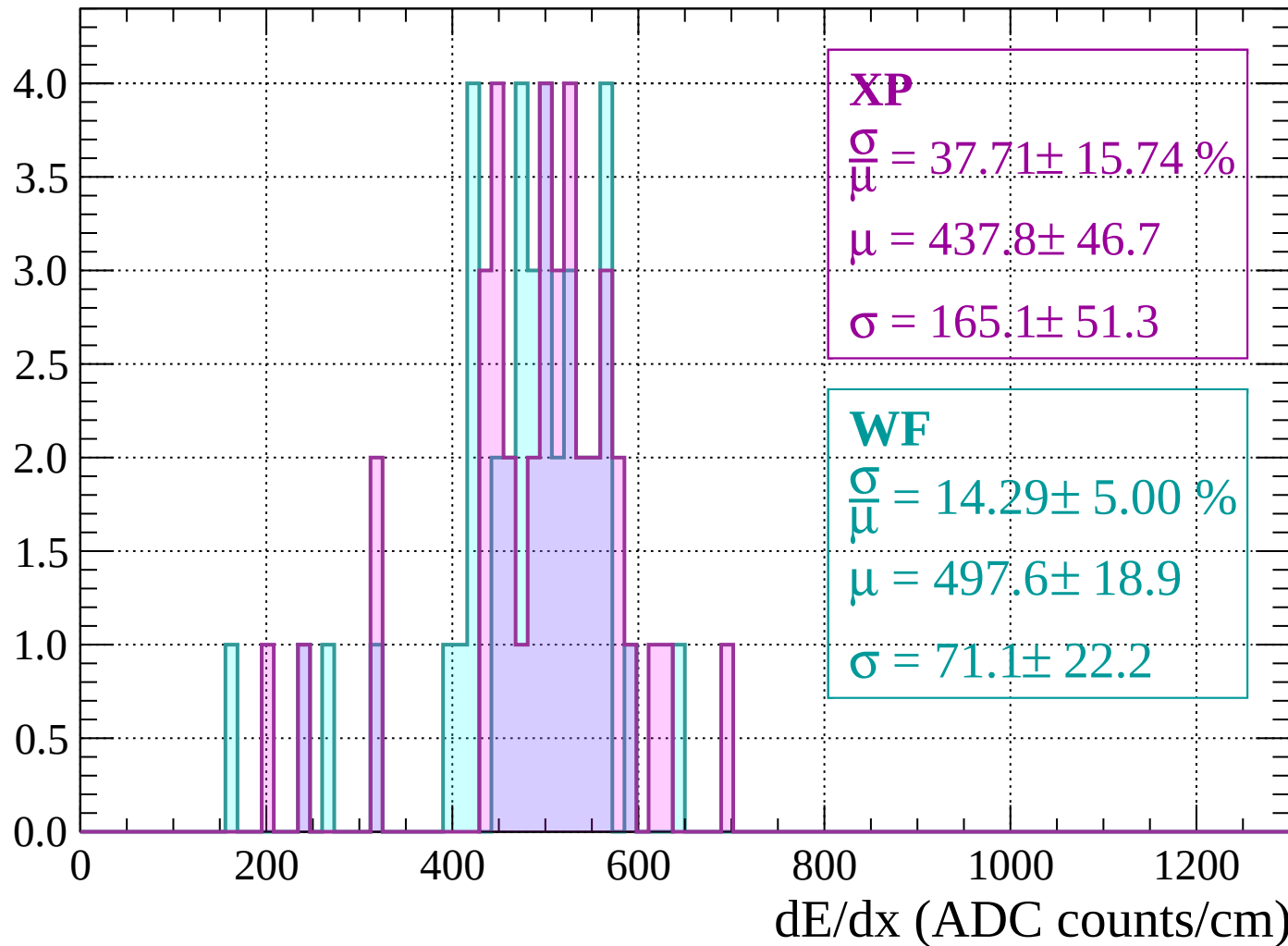
Energy loss | $1520 < p < 1560$

Count



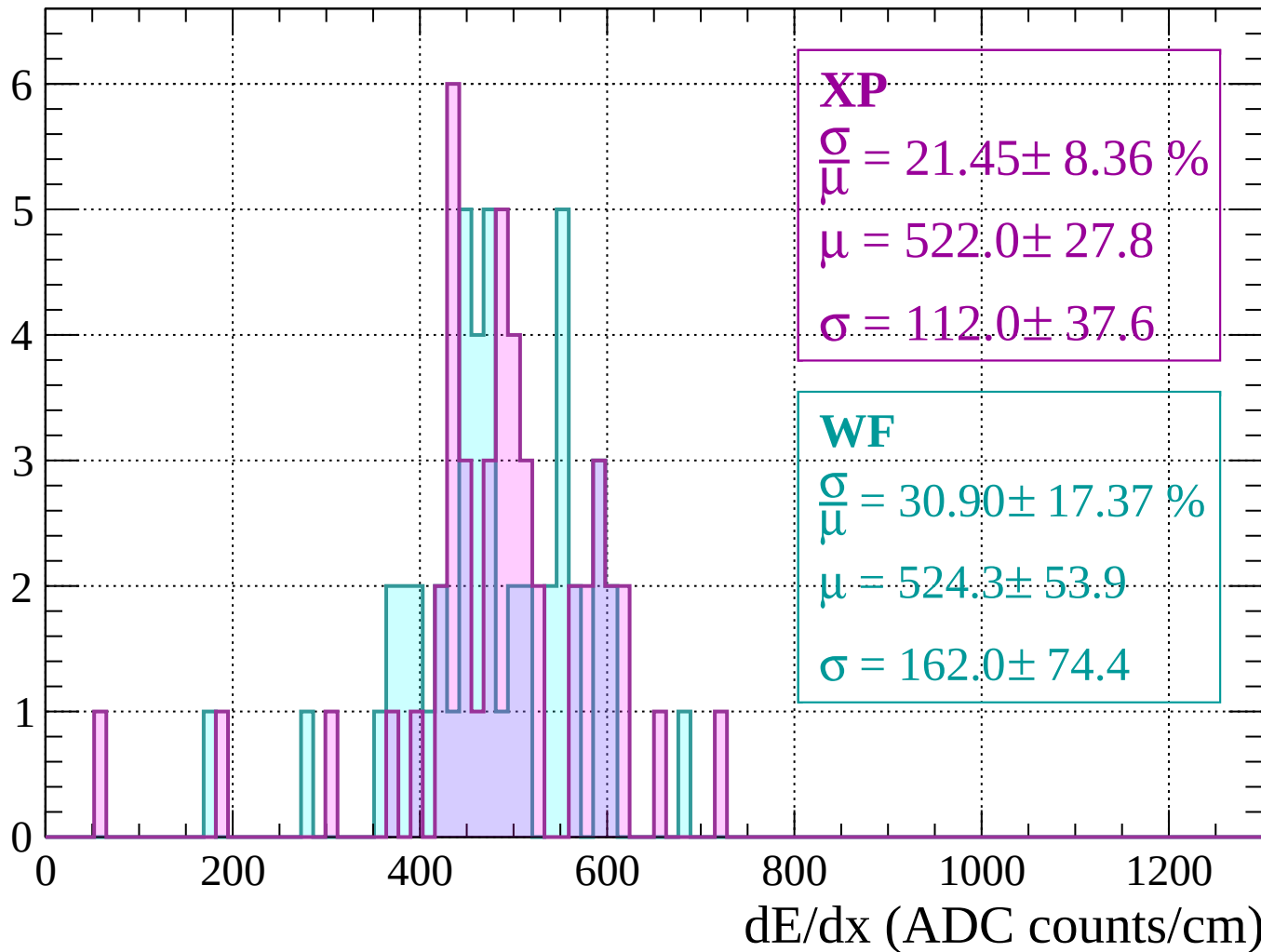
Energy loss | $1560 < p < 1600$

Count



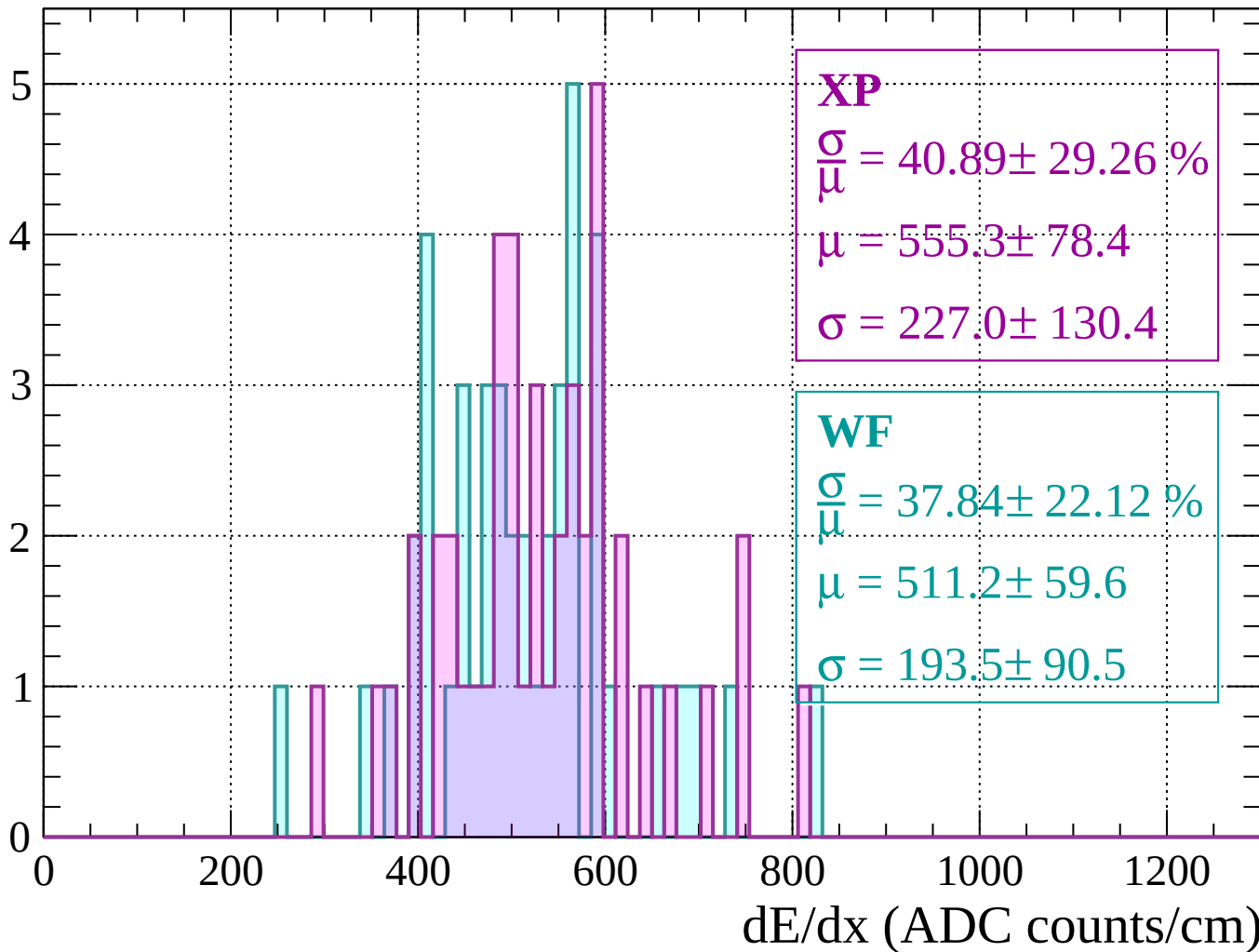
Energy loss | $1600 < p < 1640$

Count



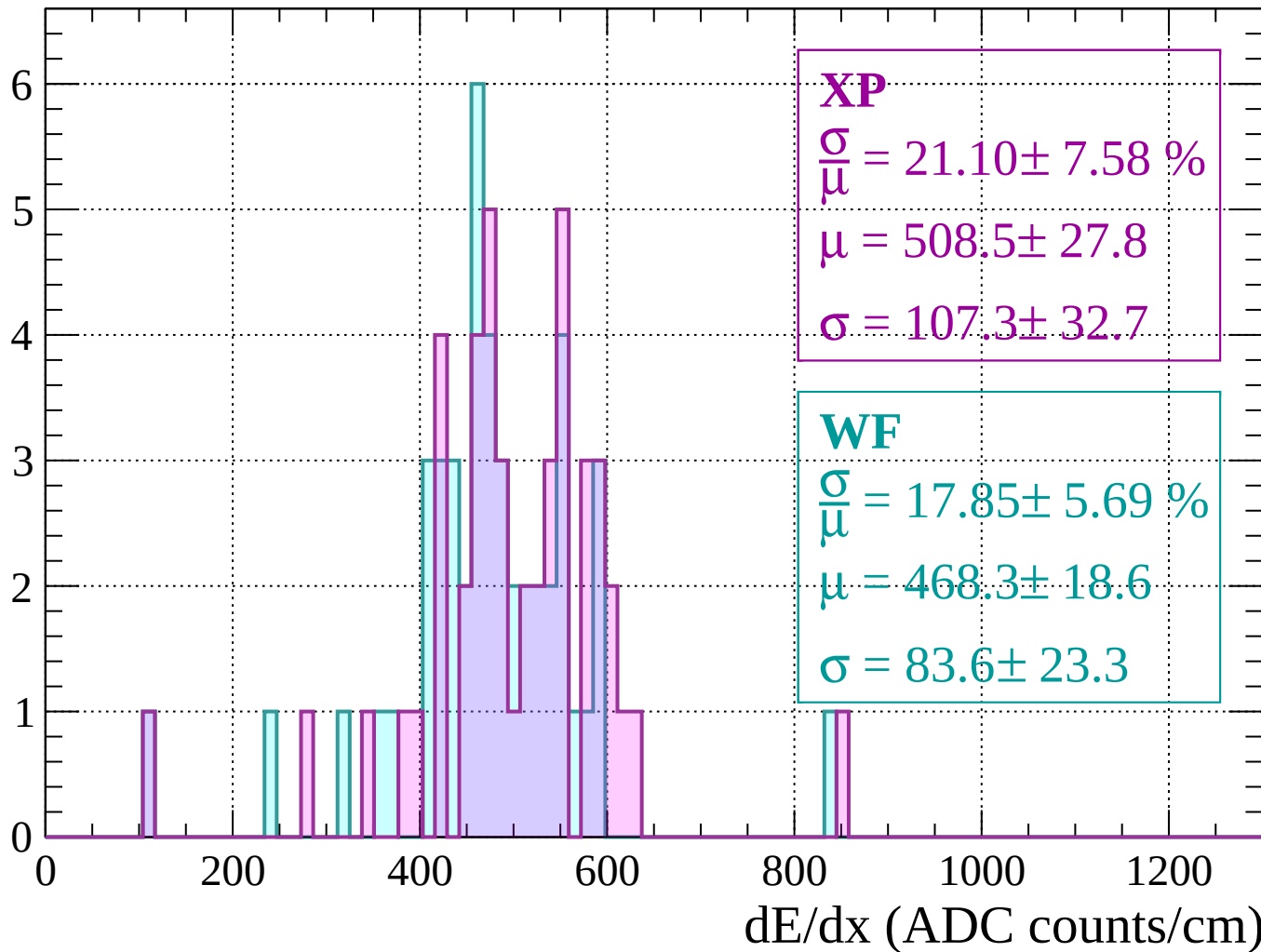
Energy loss | $1640 < p < 1680$

Count



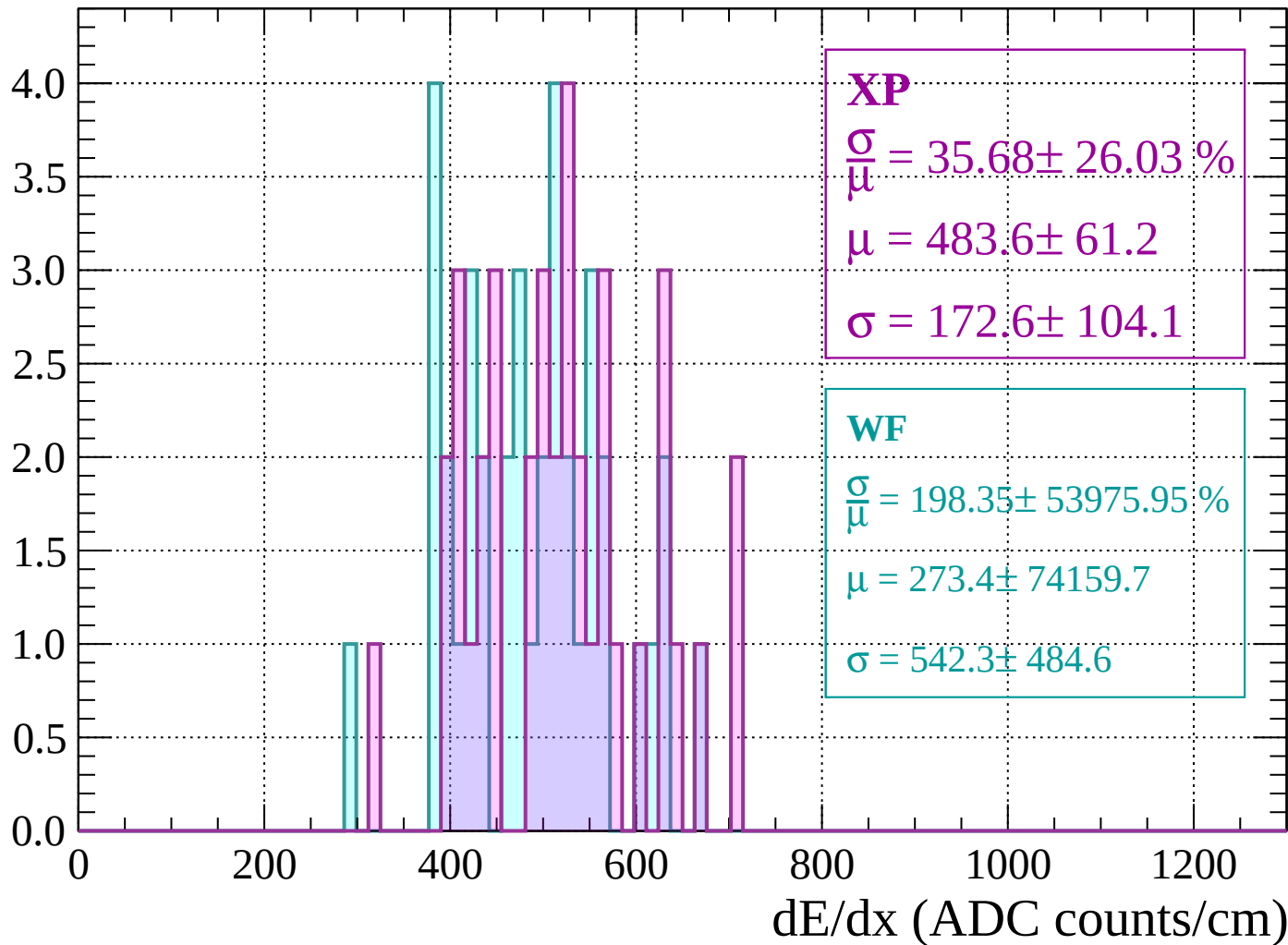
Energy loss | $1680 < p < 1720$

Count



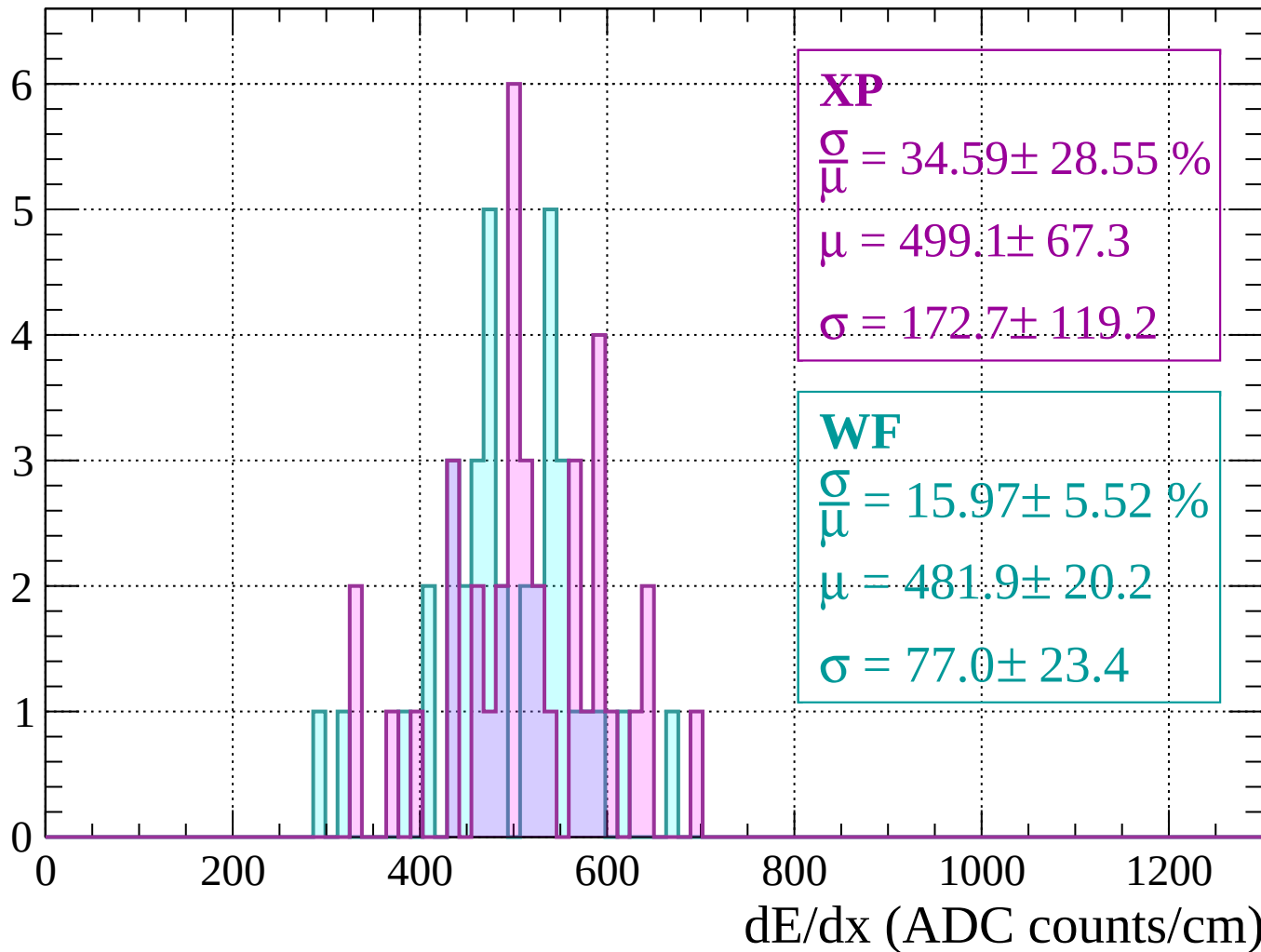
Energy loss | $1720 < p < 1760$

Count



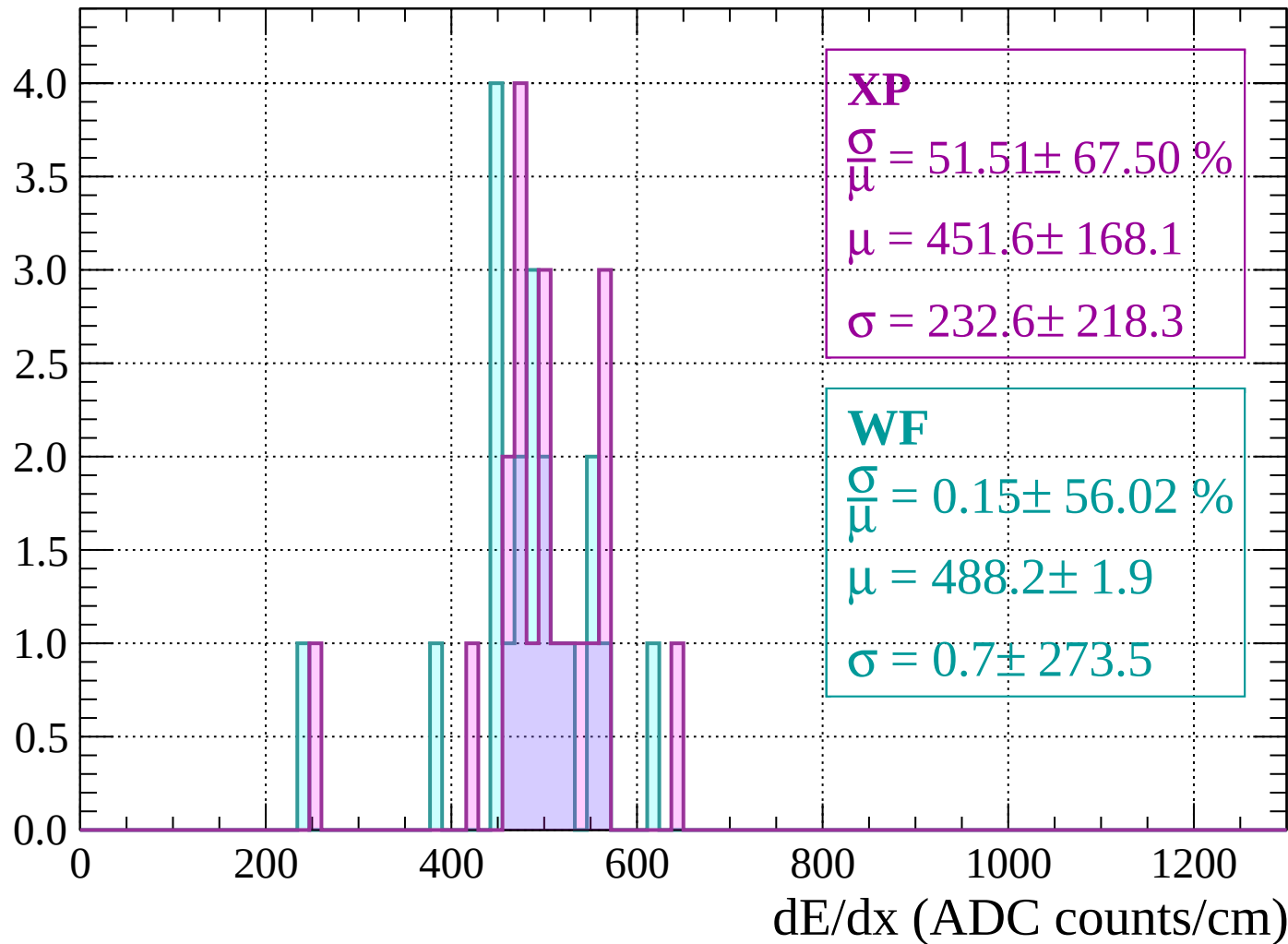
Energy loss | $1760 < p < 1800$

Count



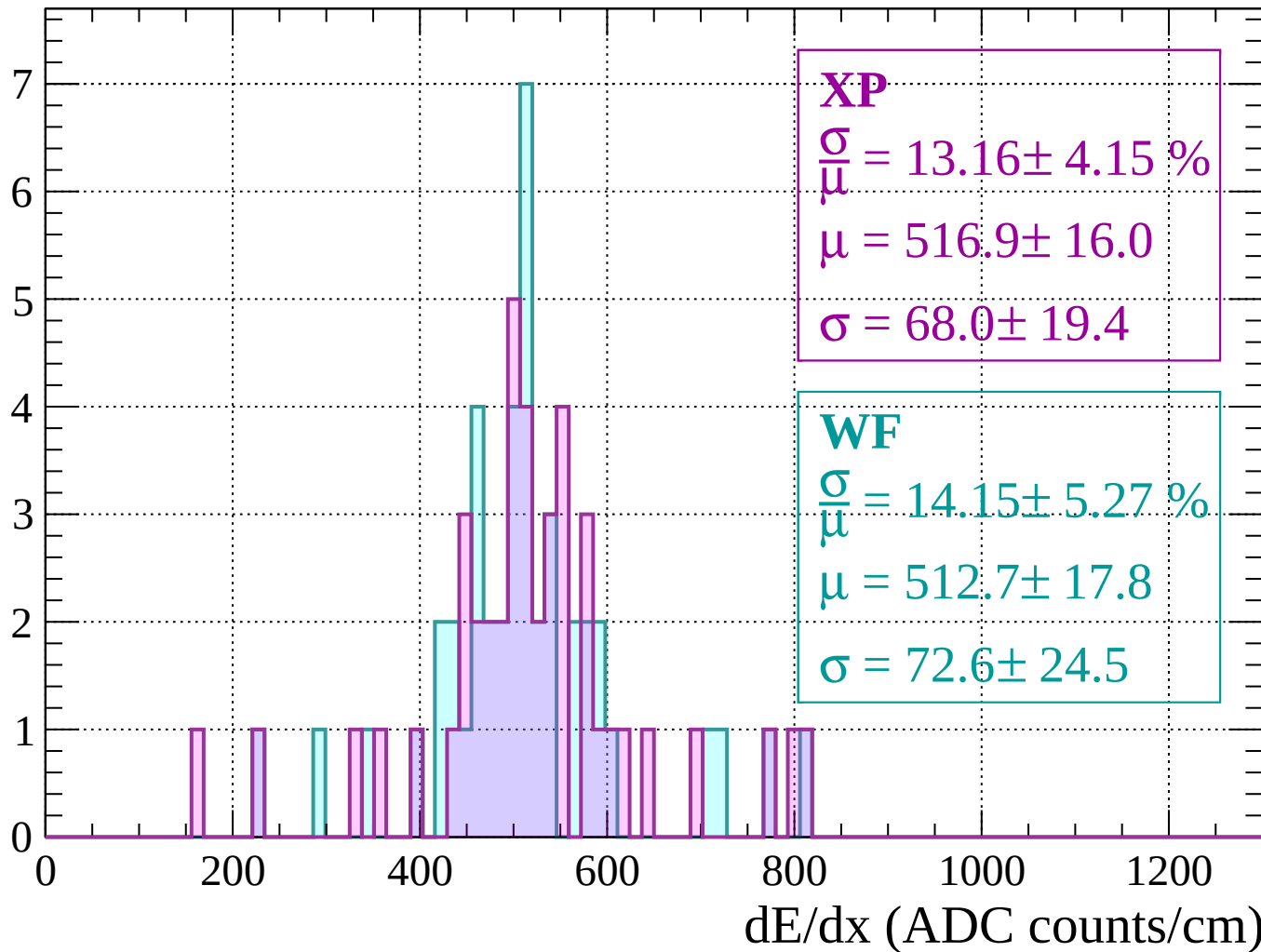
Energy loss | $1800 < p < 1840$

Count



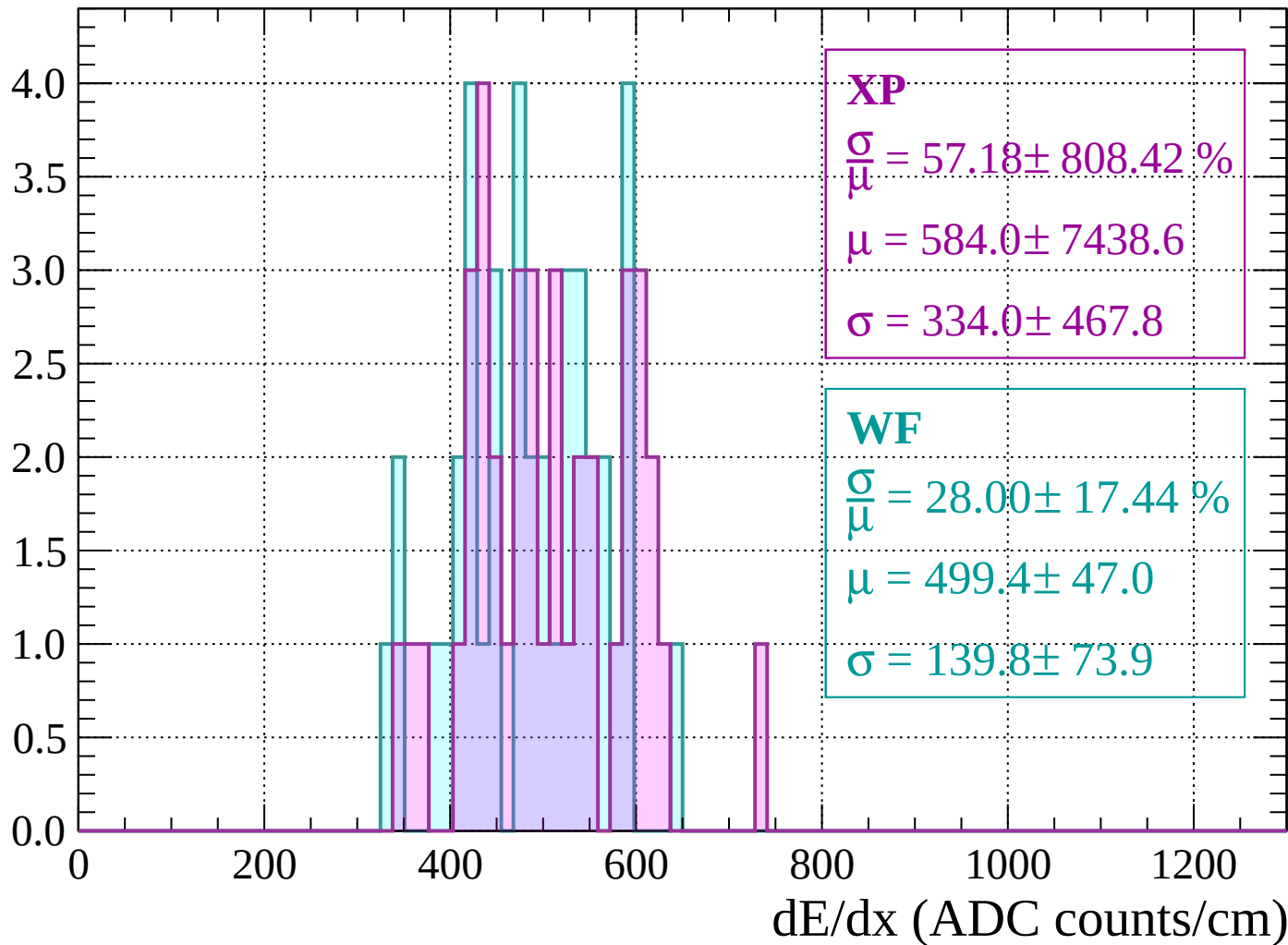
Energy loss | $1840 < p < 1880$

Count



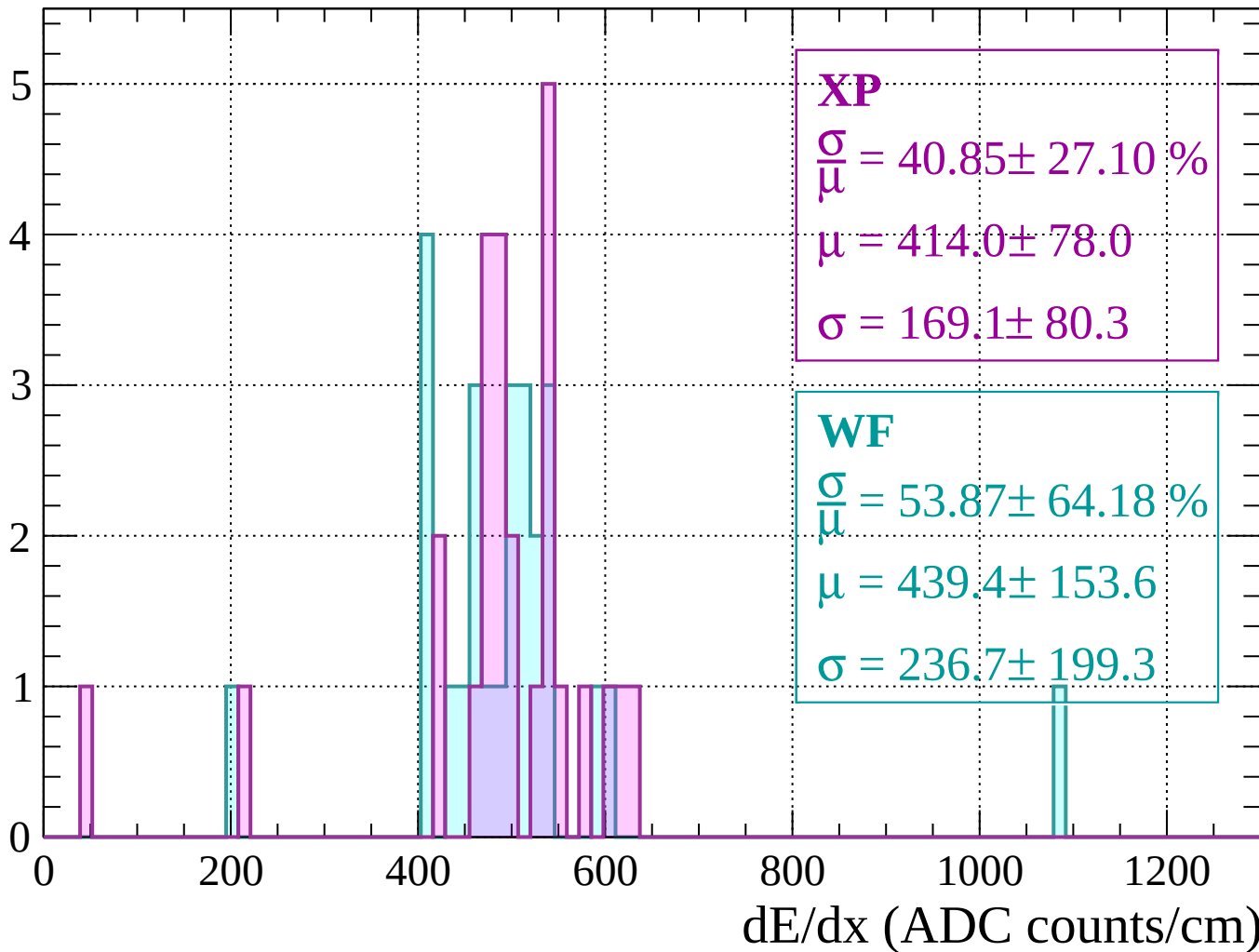
Energy loss | $1880 < p < 1920$

Count



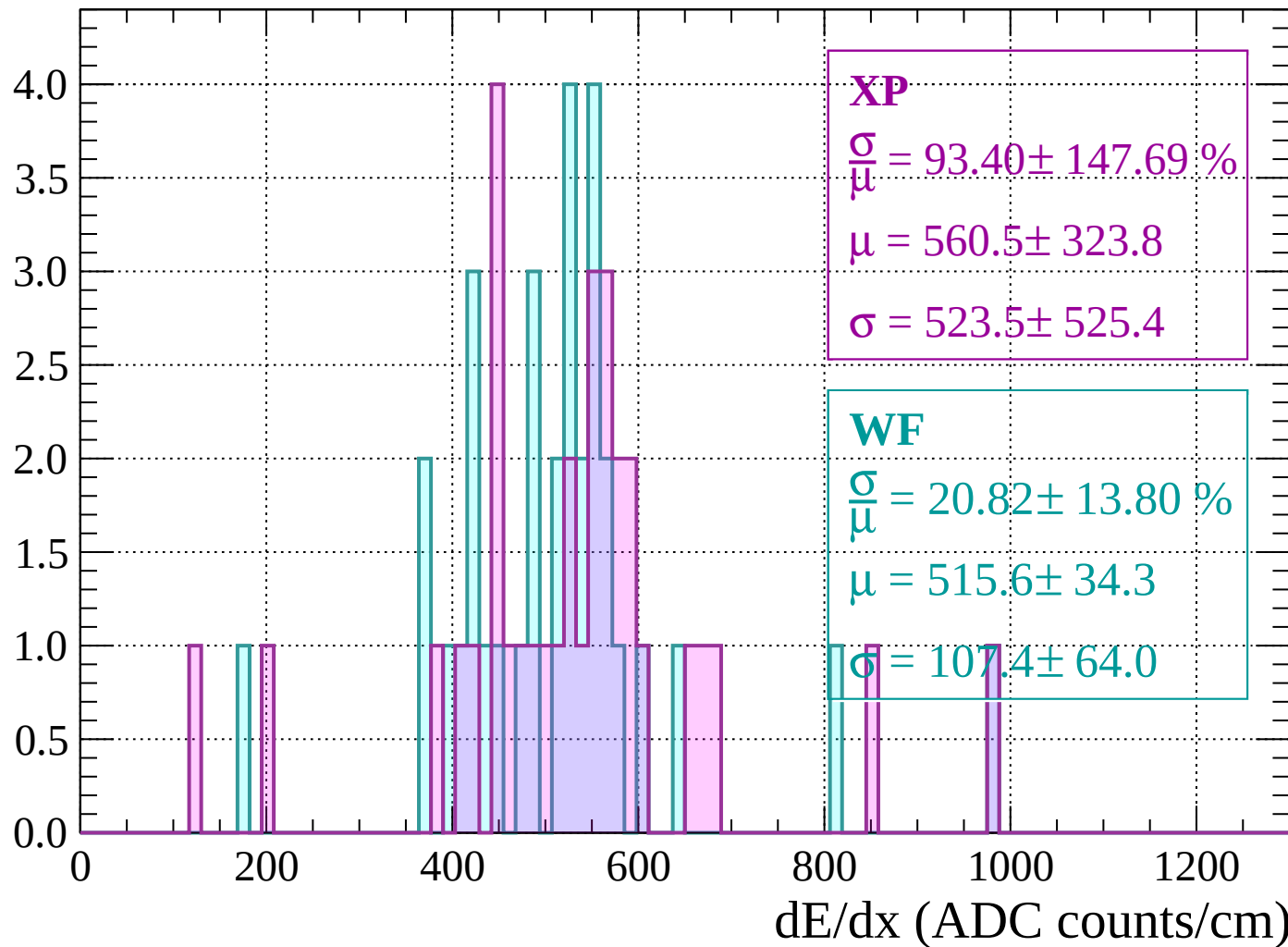
Energy loss | $1920 < p < 1960$

Count



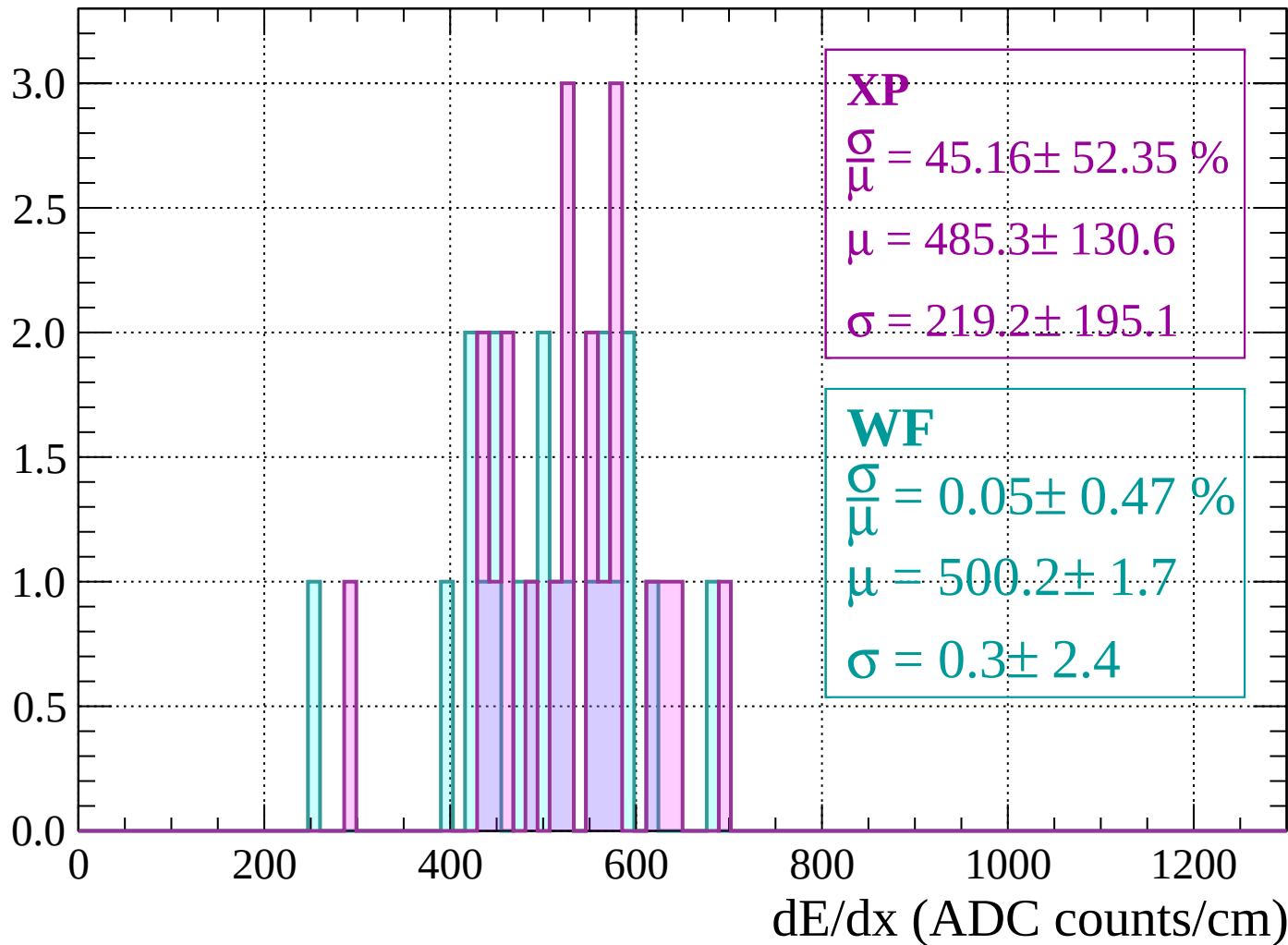
Energy loss | 1960 < p < 2000

Count



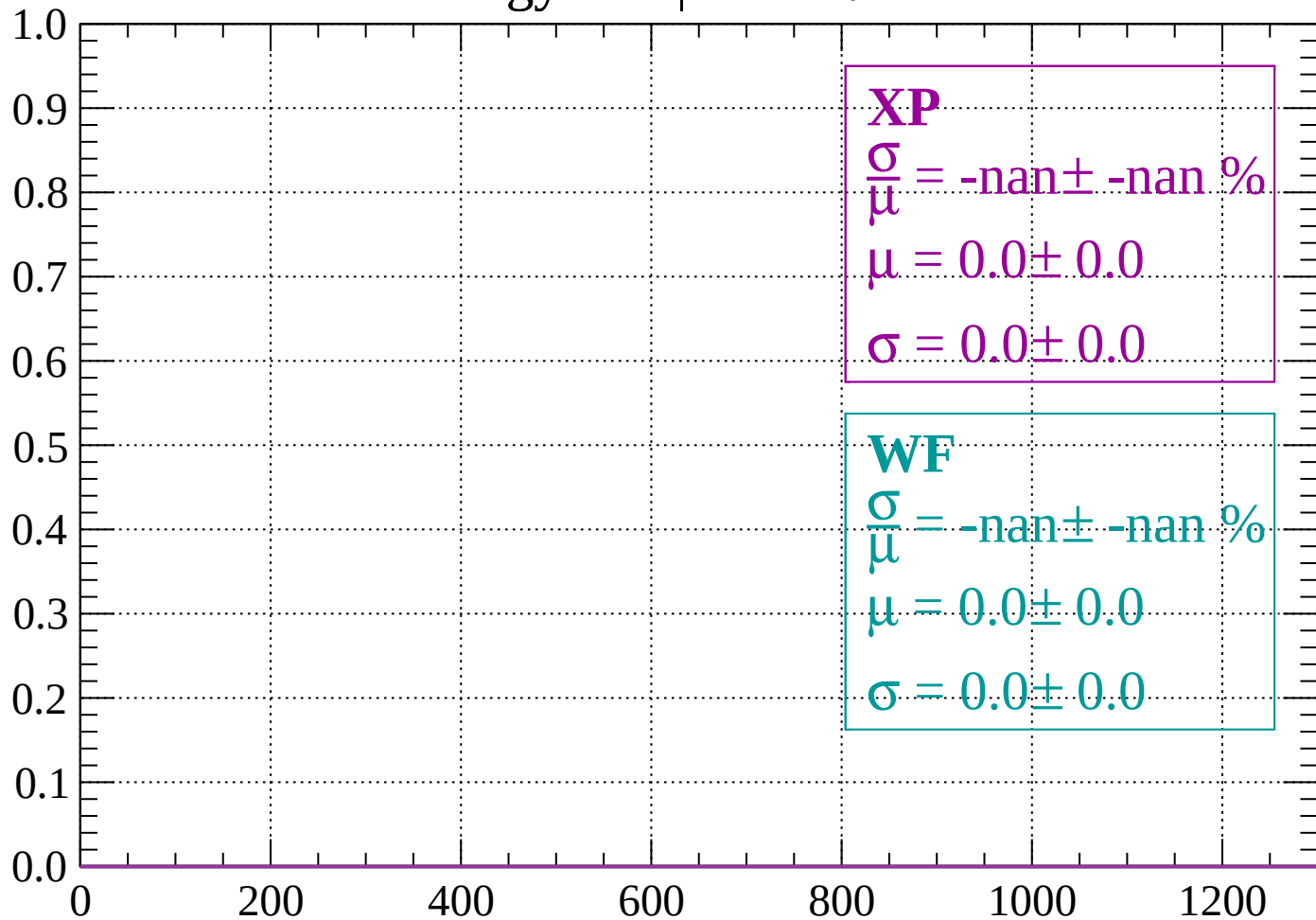
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

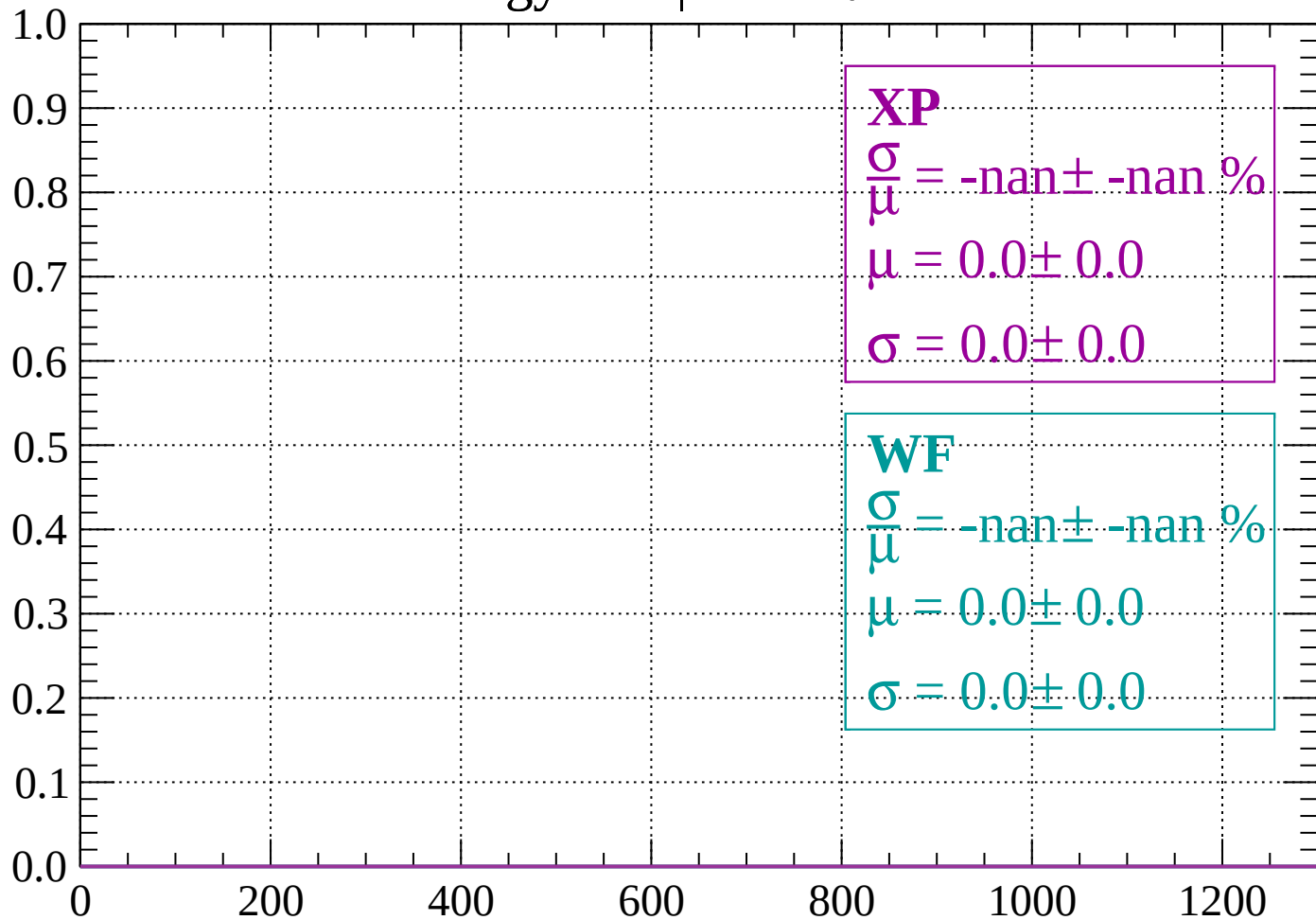
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

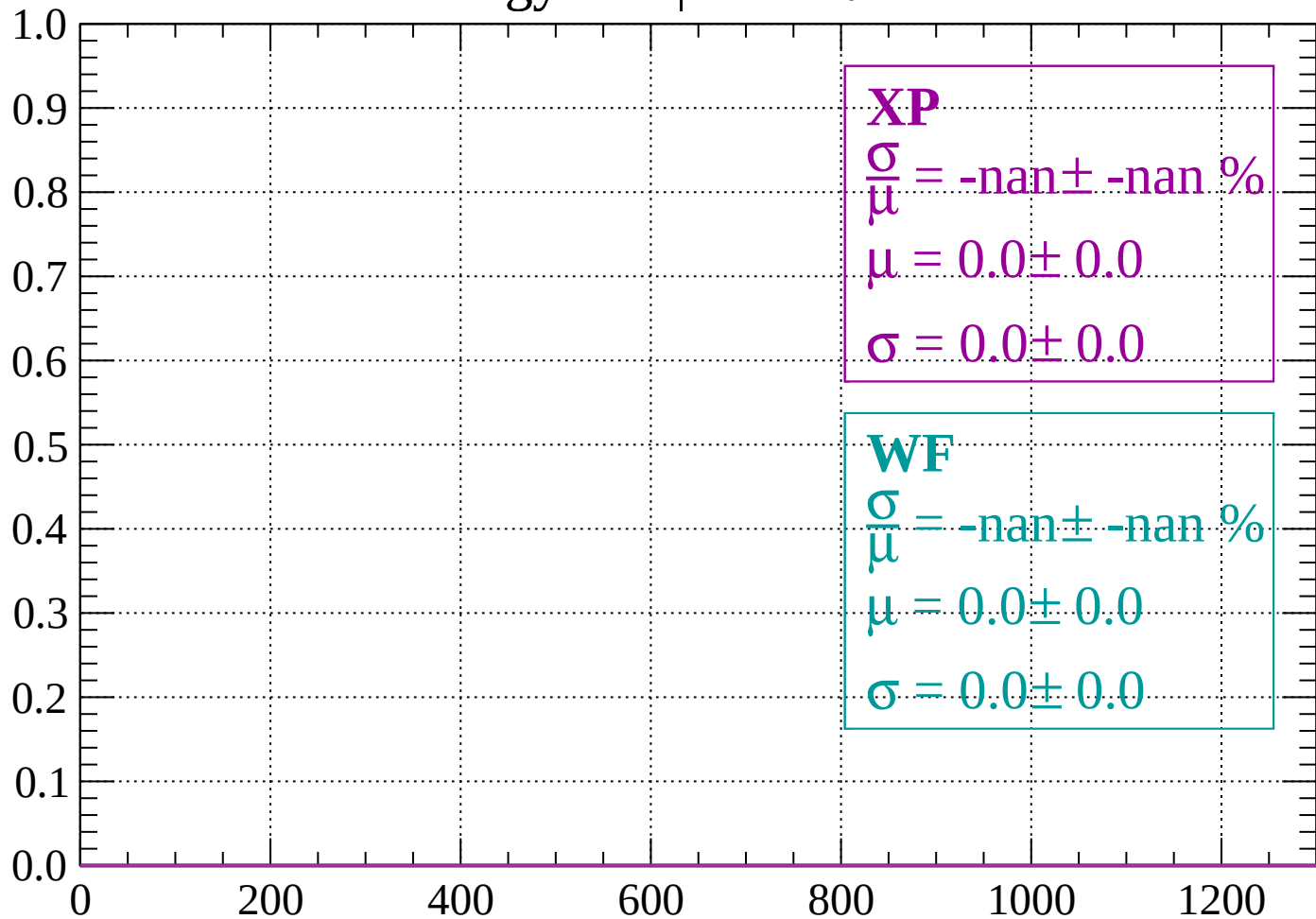
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

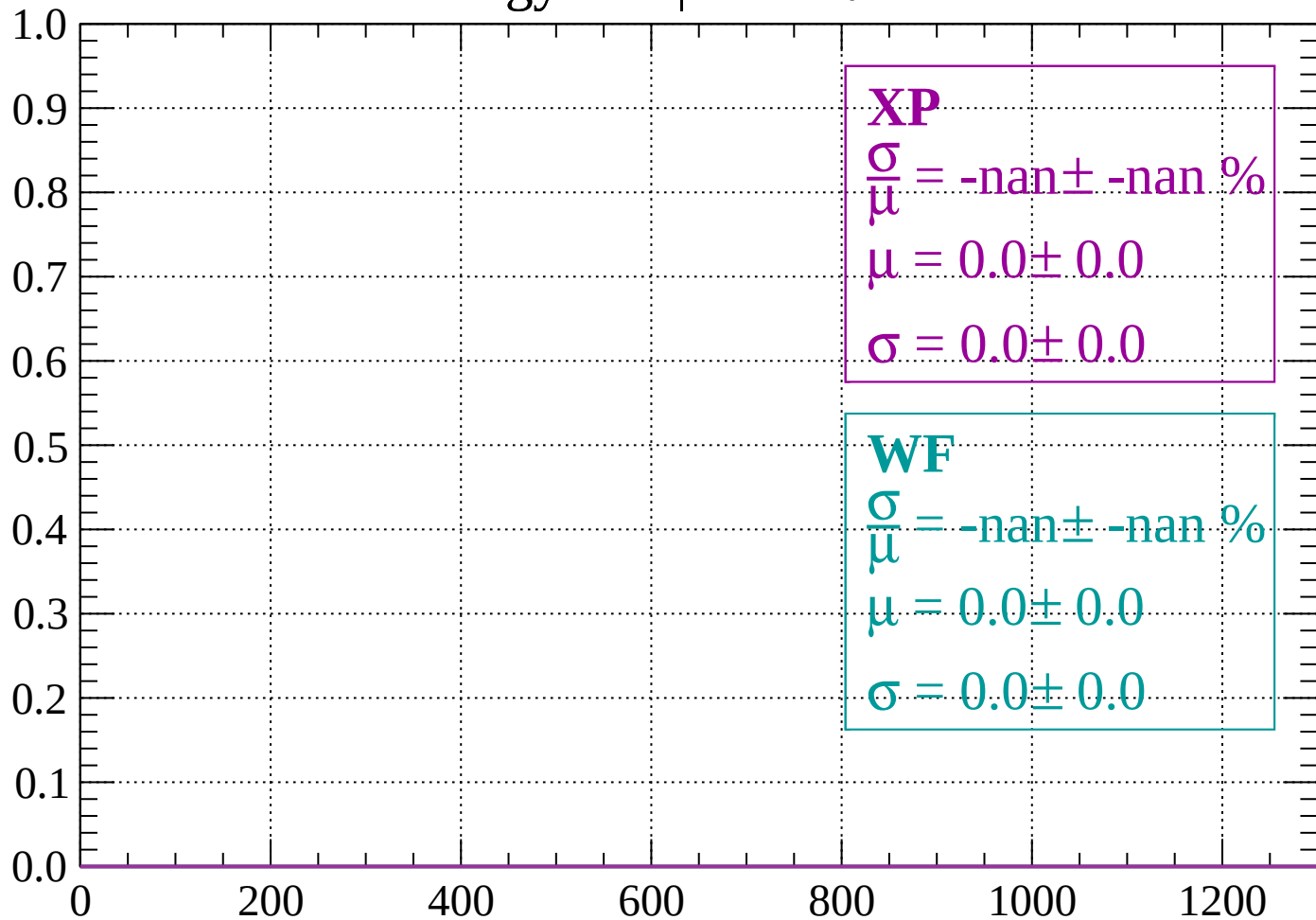
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

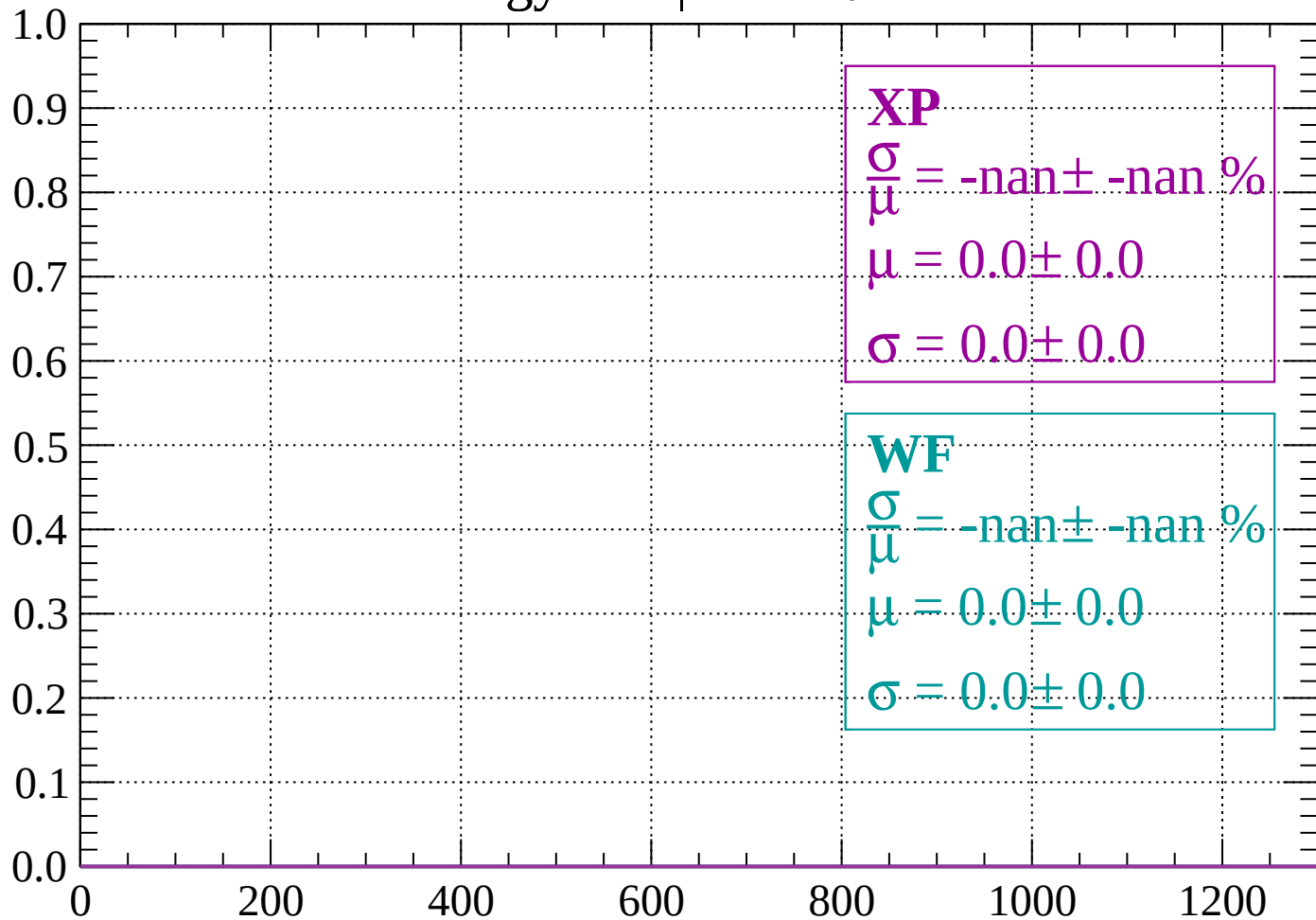
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

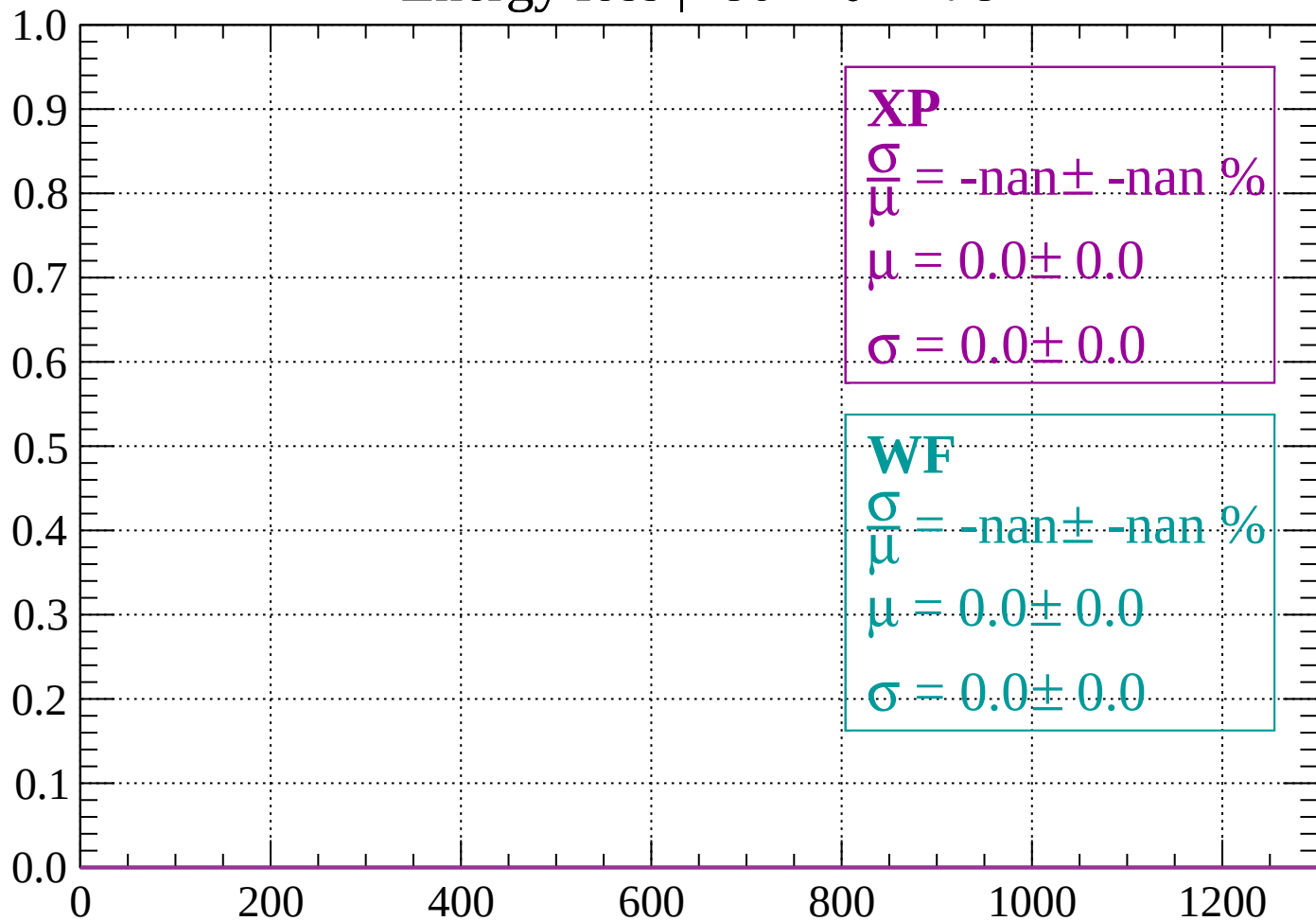
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

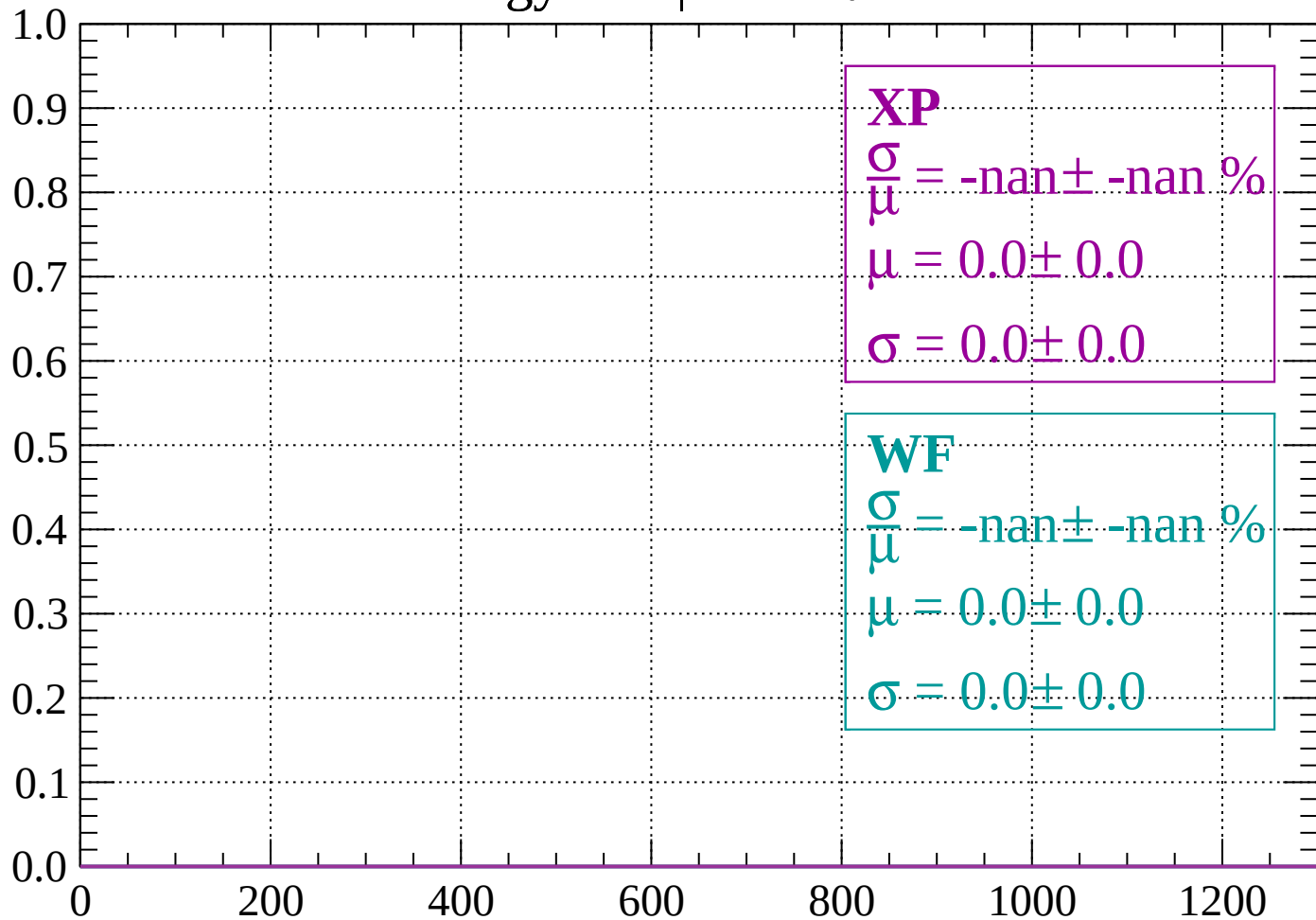
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

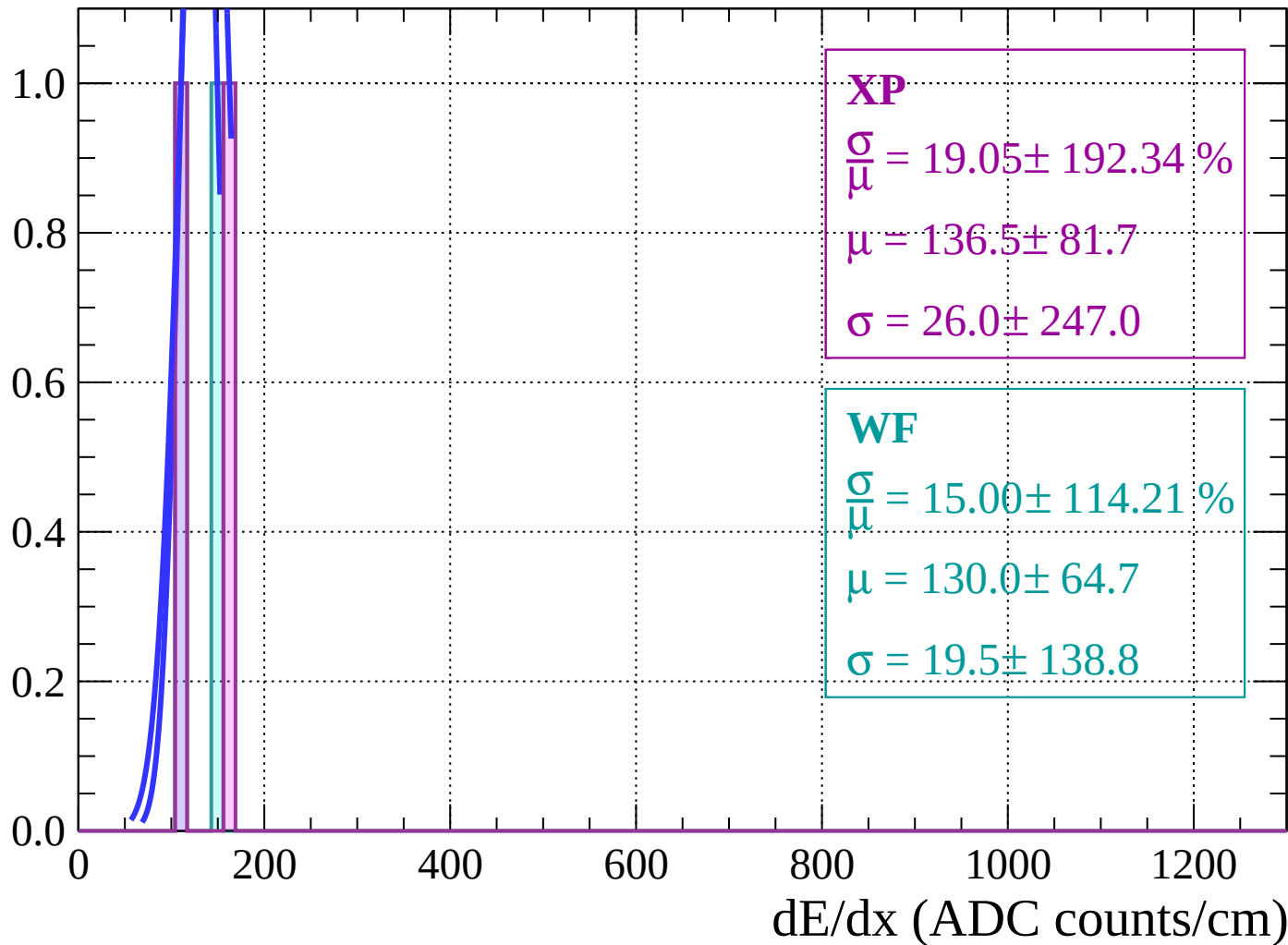
Count



dE/dx (ADC counts/cm)

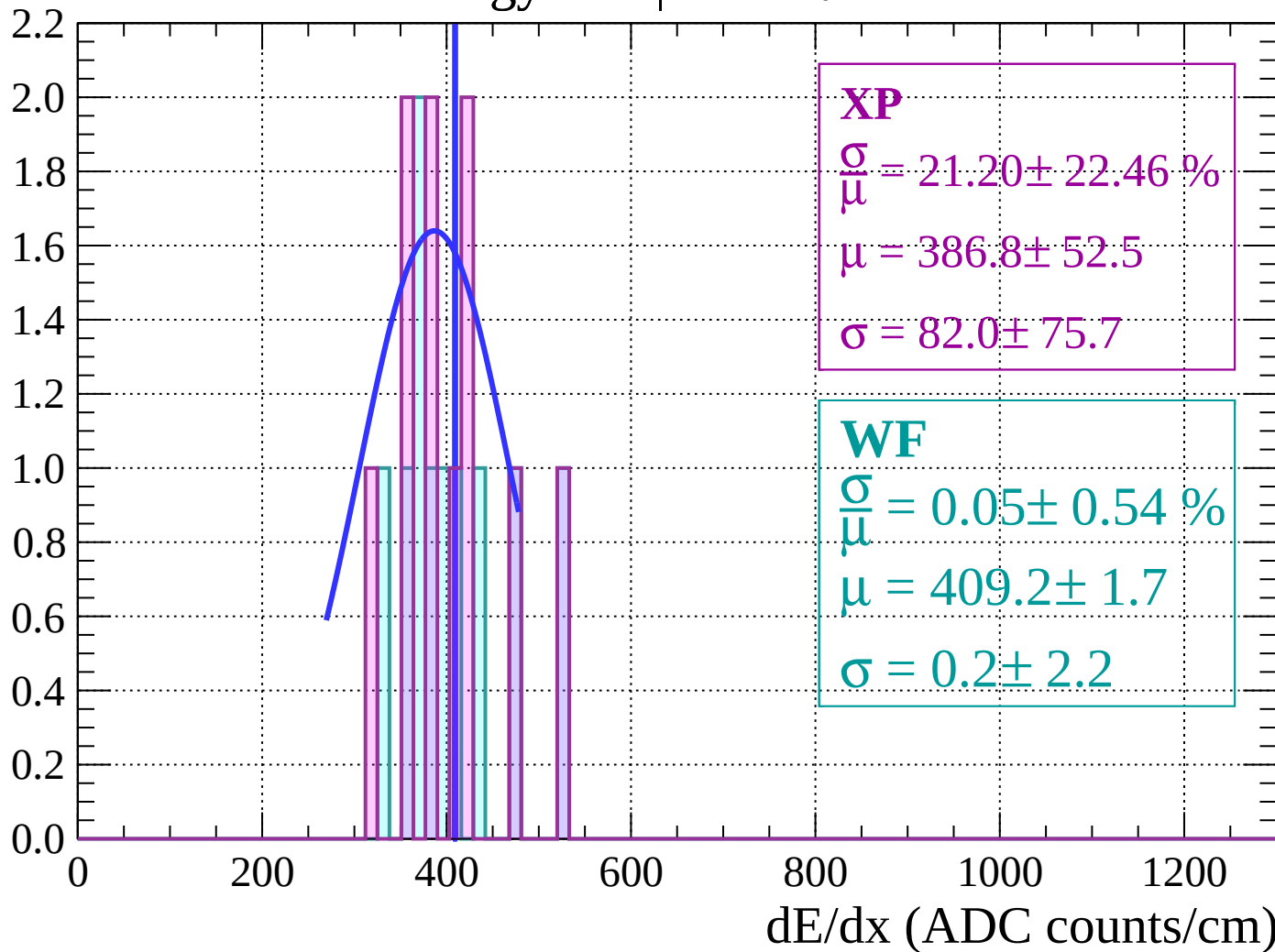
Energy loss | $-76 < \theta < -74$

Count



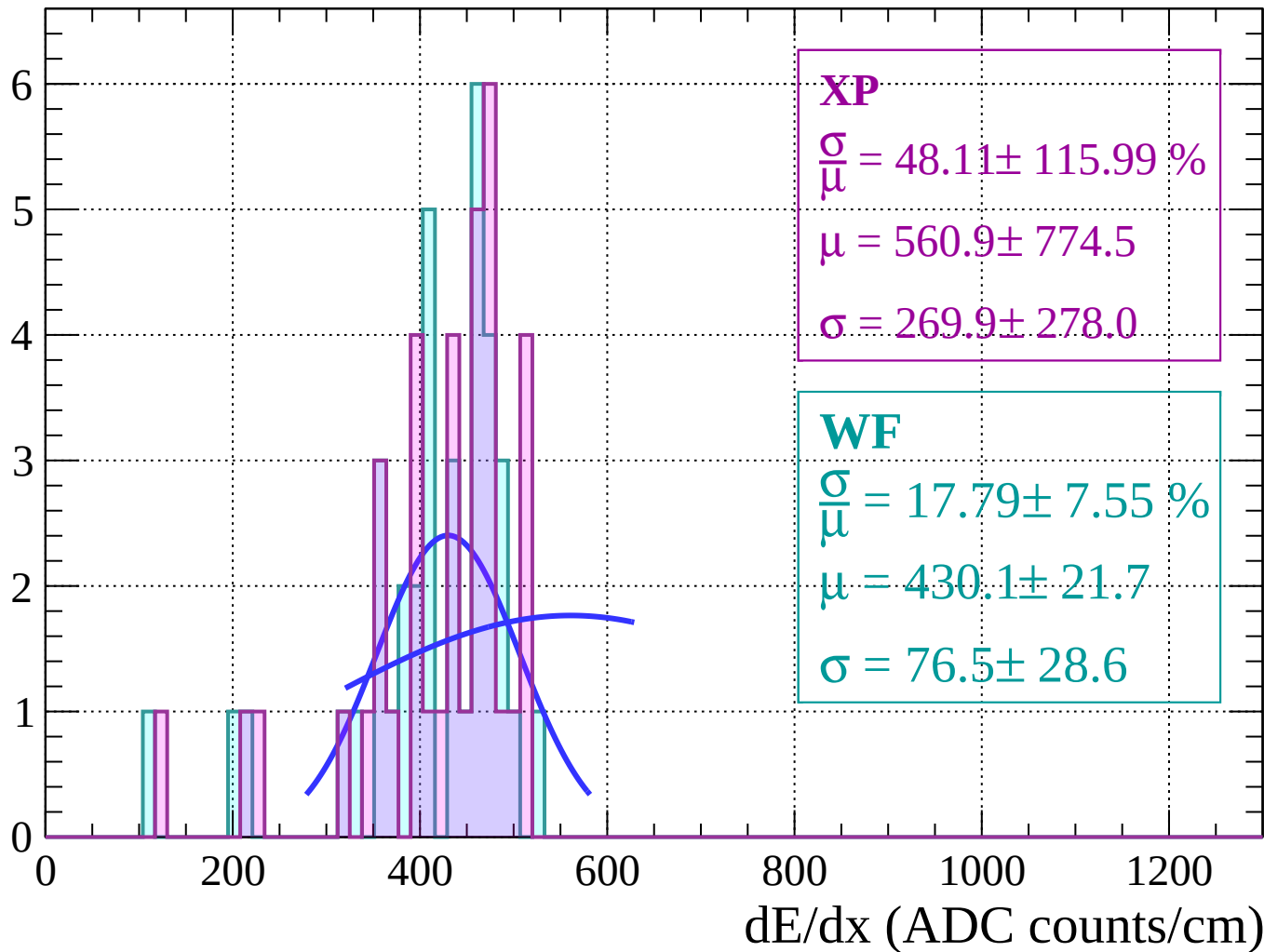
Energy loss | $-74 < \theta < -72$

Count



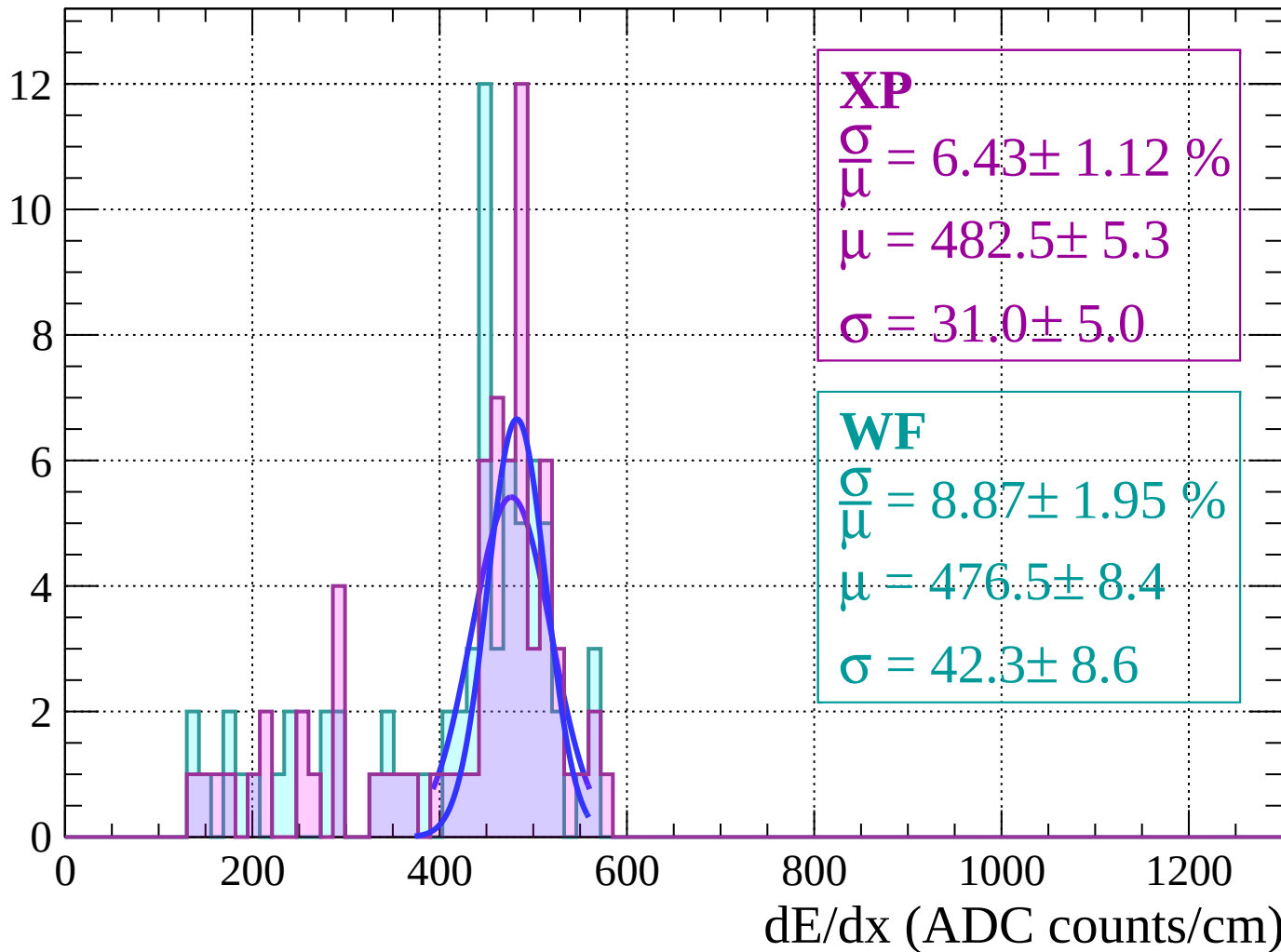
Energy loss | $-72 < \theta < -70$

Count



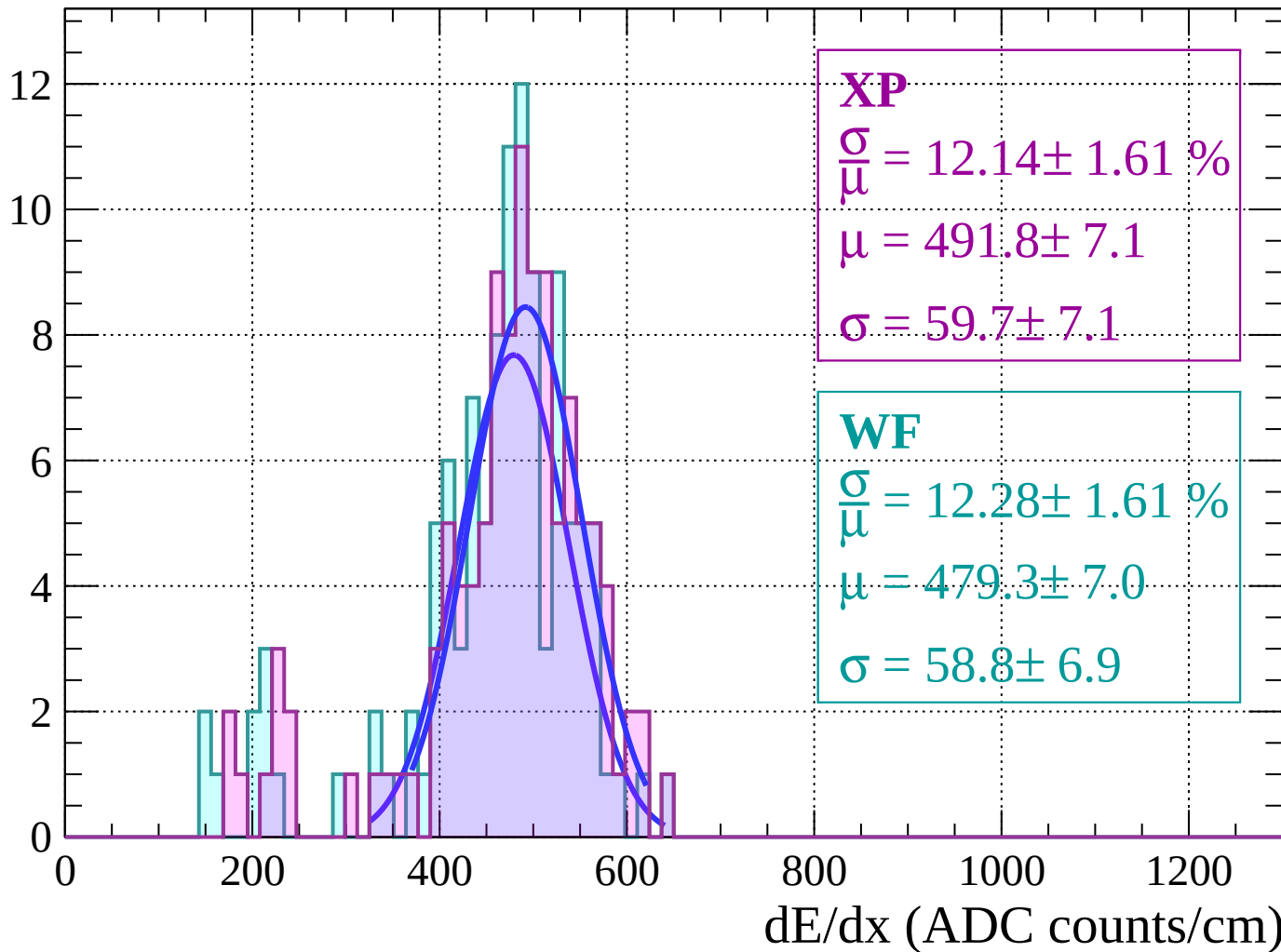
Energy loss | $-70 < \theta < -68$

Count



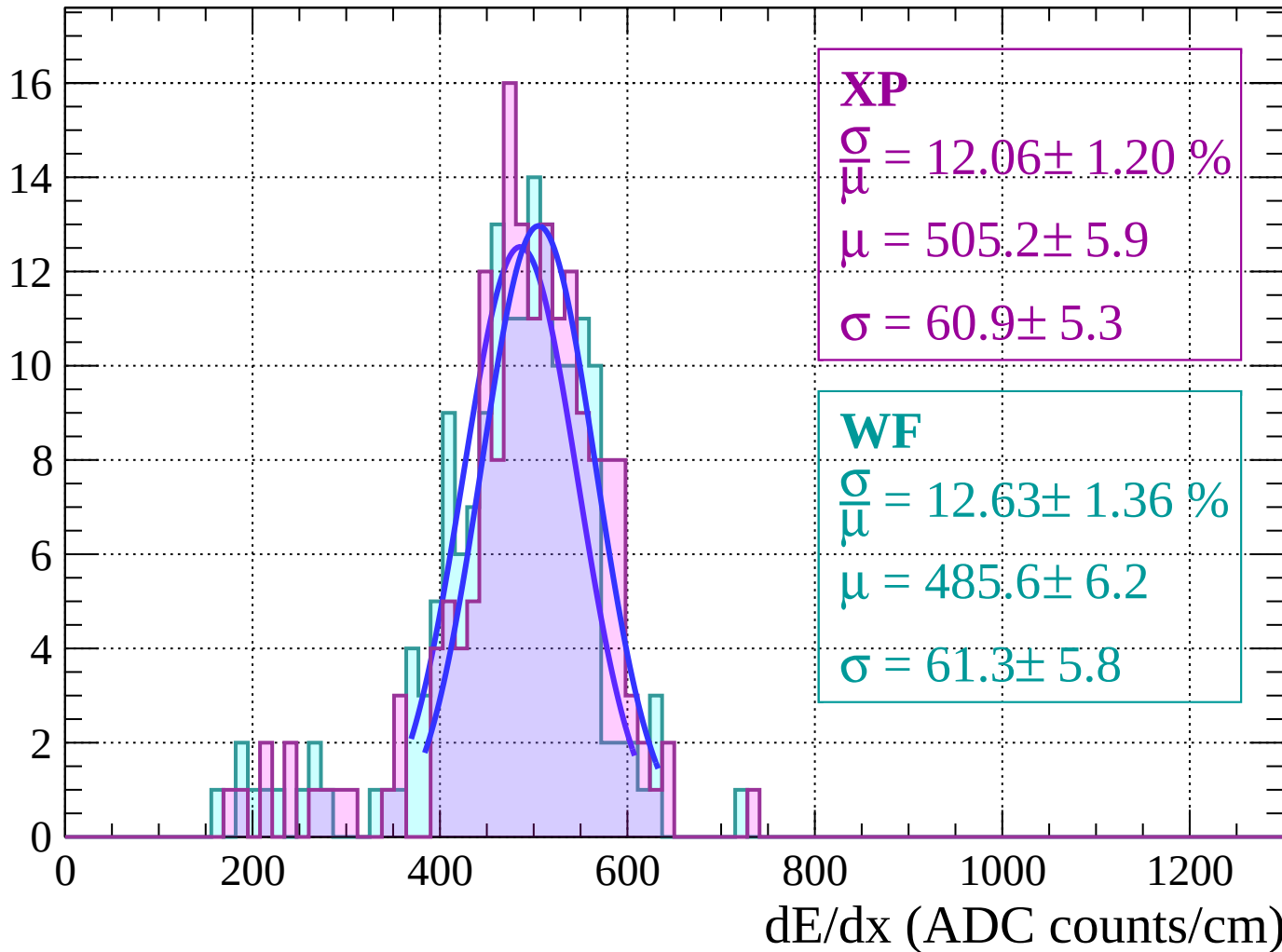
Energy loss $|-68 < \theta < -66$

Count



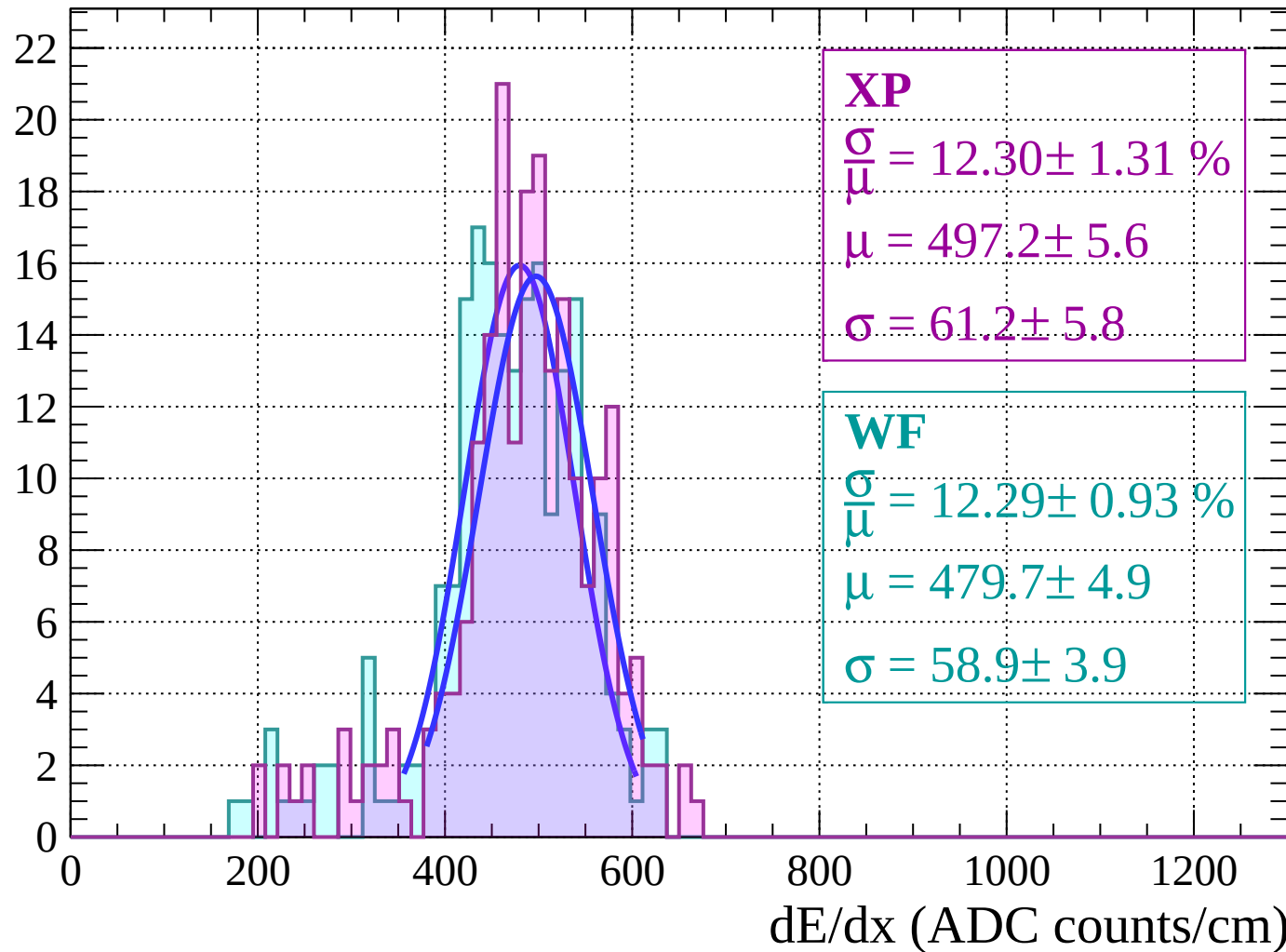
Energy loss $|-66 < \theta < -64$

Count



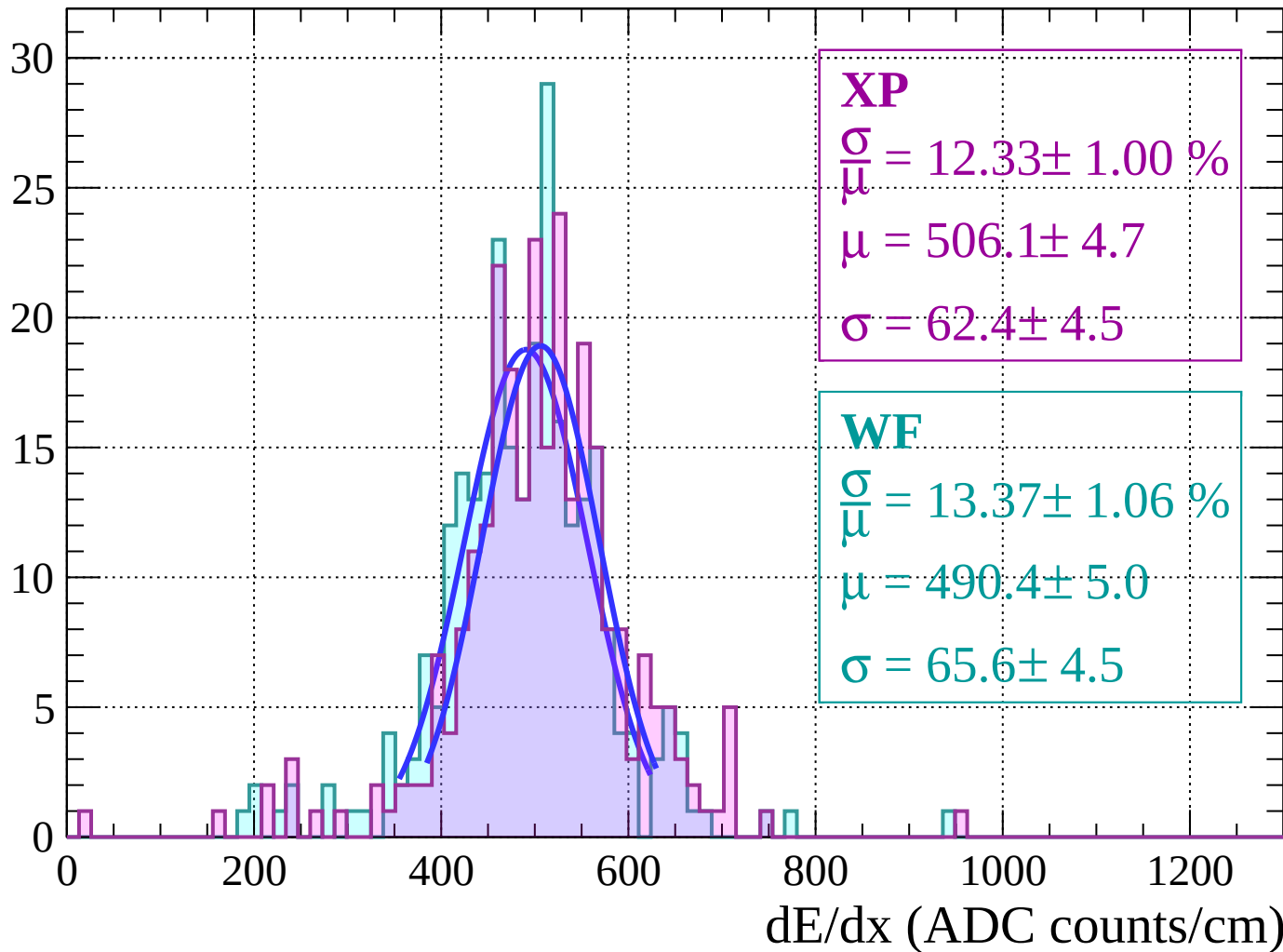
Energy loss | $-64 < \theta < -62$

Count



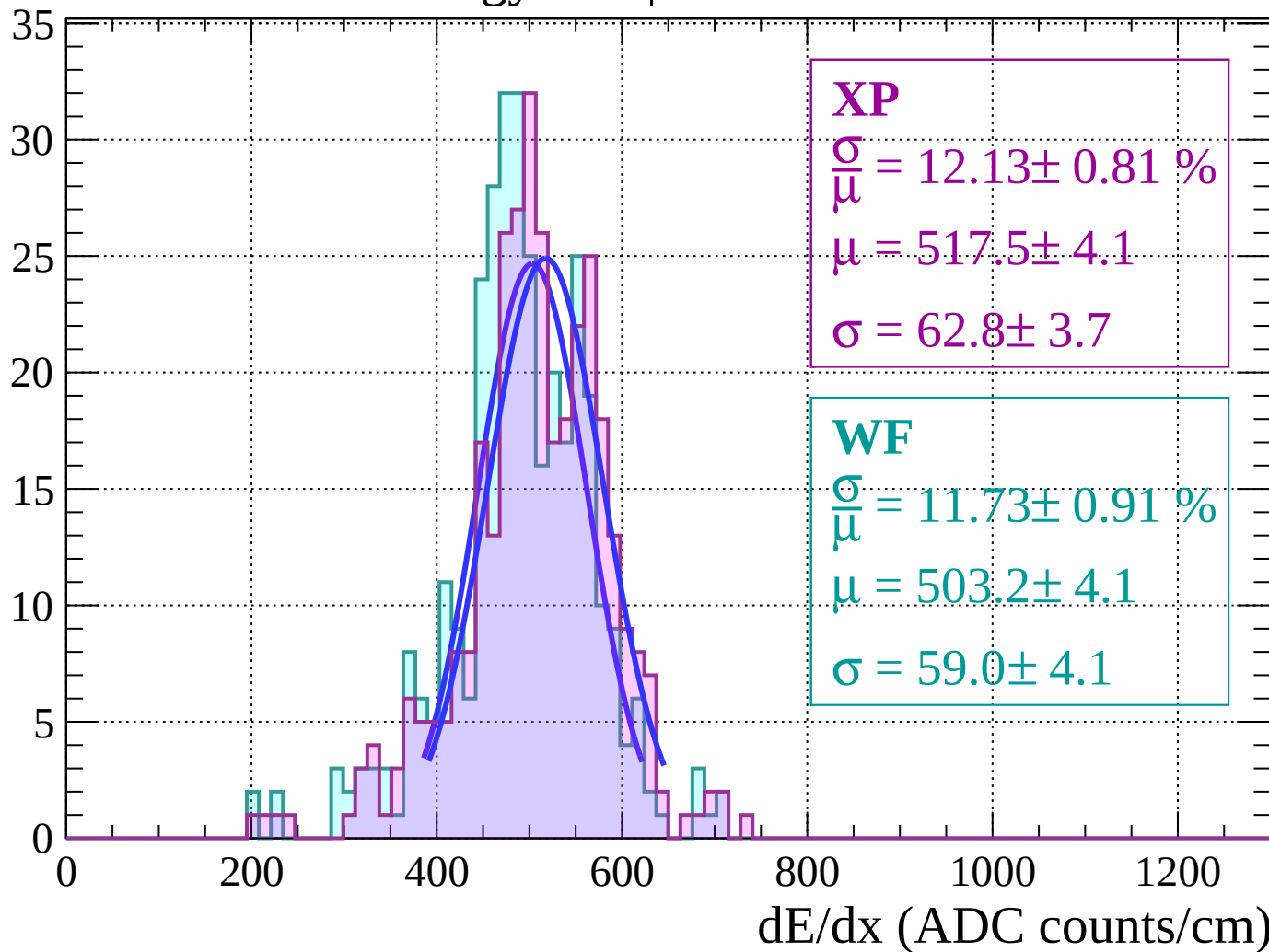
Energy loss | $-62 < \theta < -60$

Count



Energy loss | $-60 < \theta < -58$

Count



Energy loss $|-58 < \theta < -56$

Count

XP

$$\frac{\sigma}{\mu} = 13.44 \pm 0.76 \%$$

$$\mu = 512.6 \pm 4.2$$

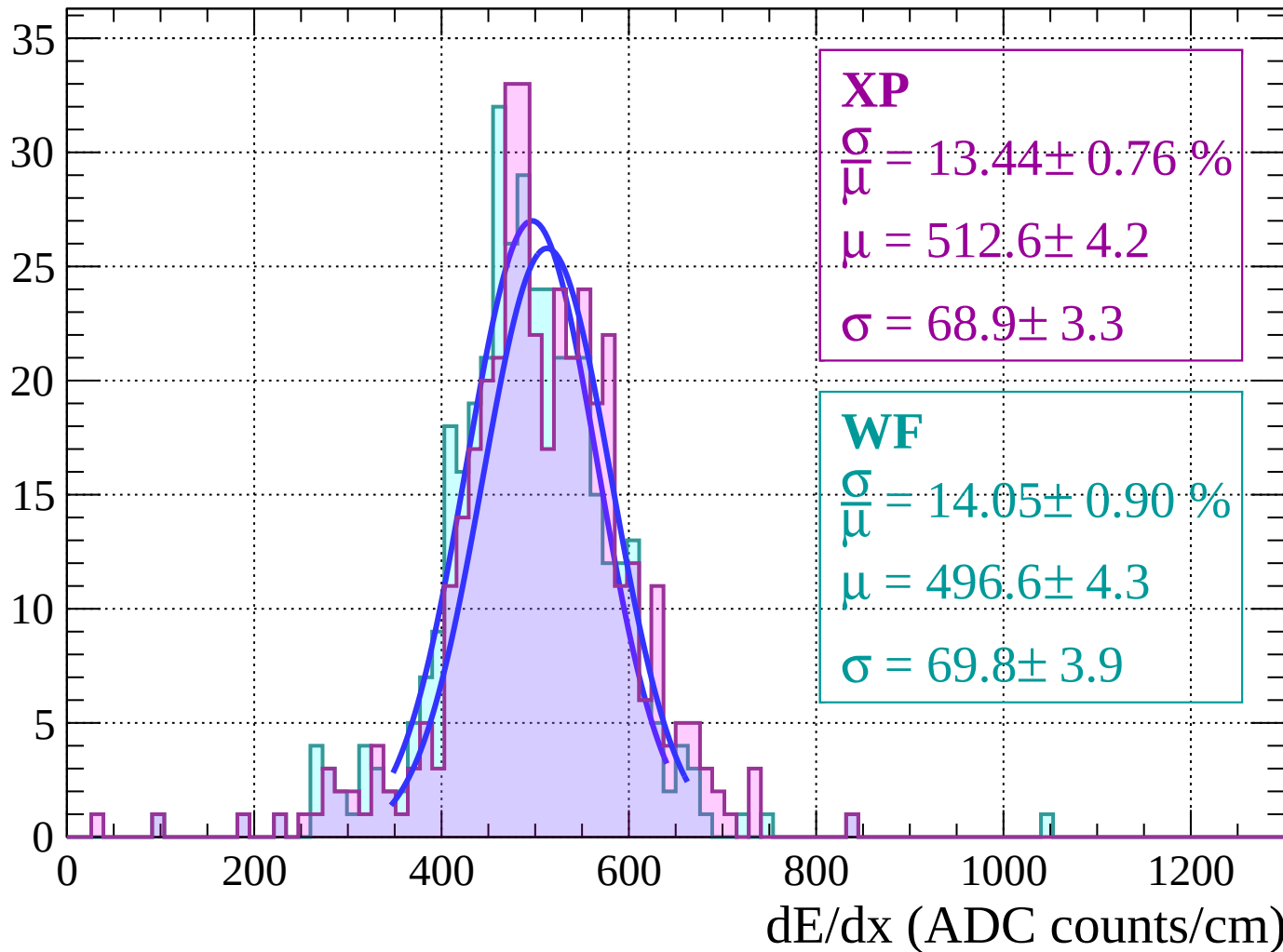
$$\sigma = 68.9 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 14.05 \pm 0.90 \%$$

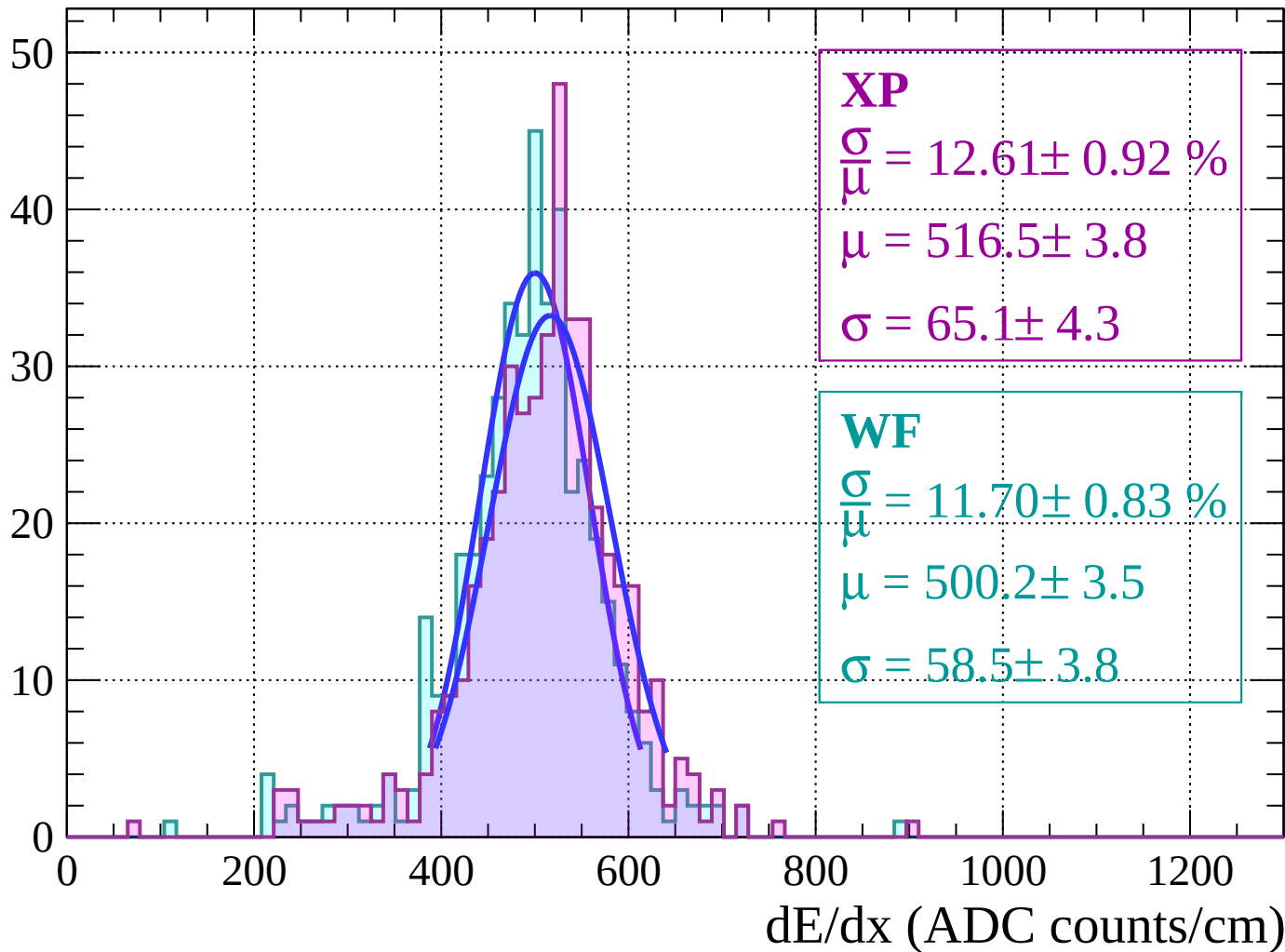
$$\mu = 496.6 \pm 4.3$$

$$\sigma = 69.8 \pm 3.9$$



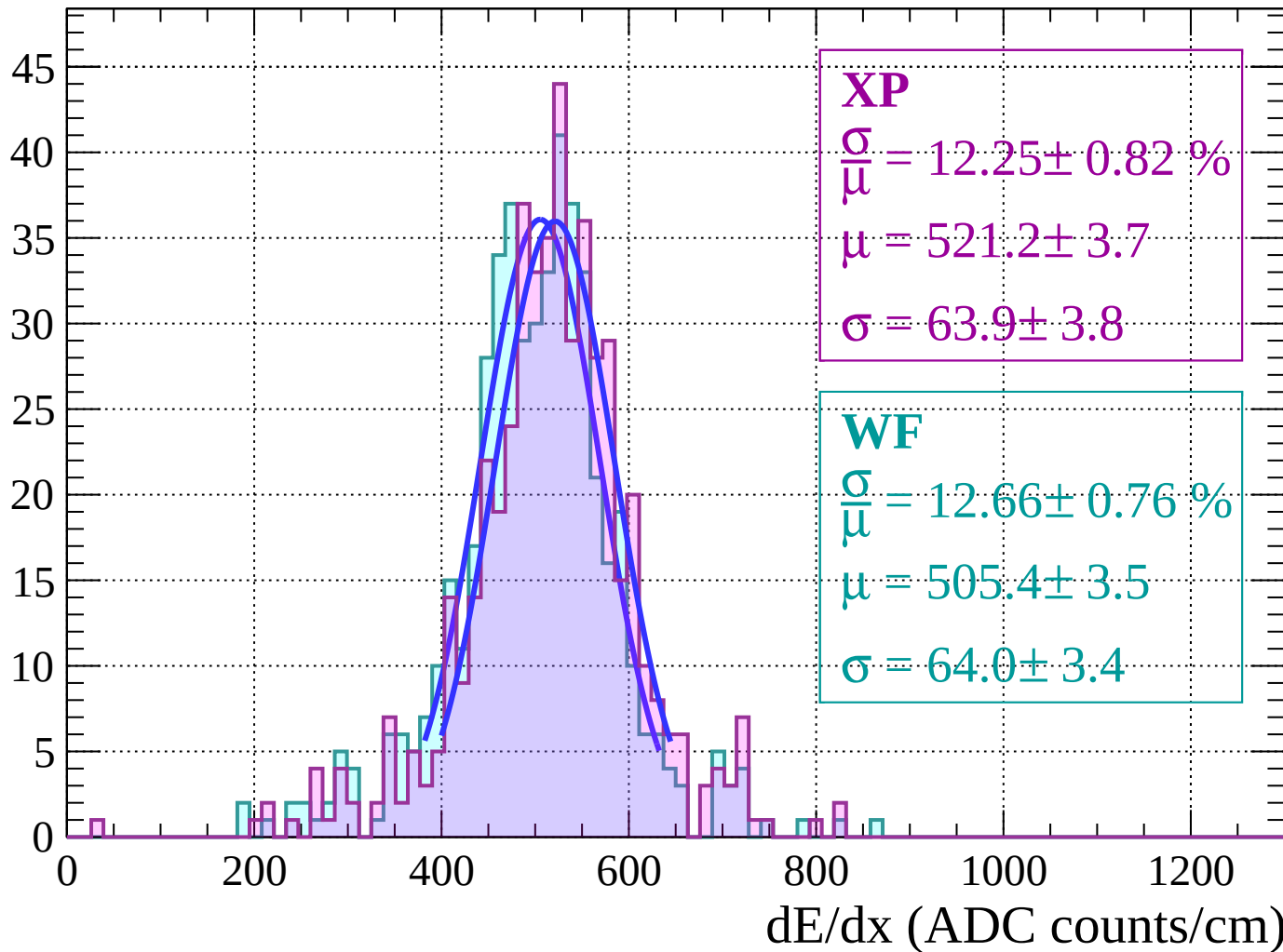
Energy loss $|-56 < \theta < -54$

Count



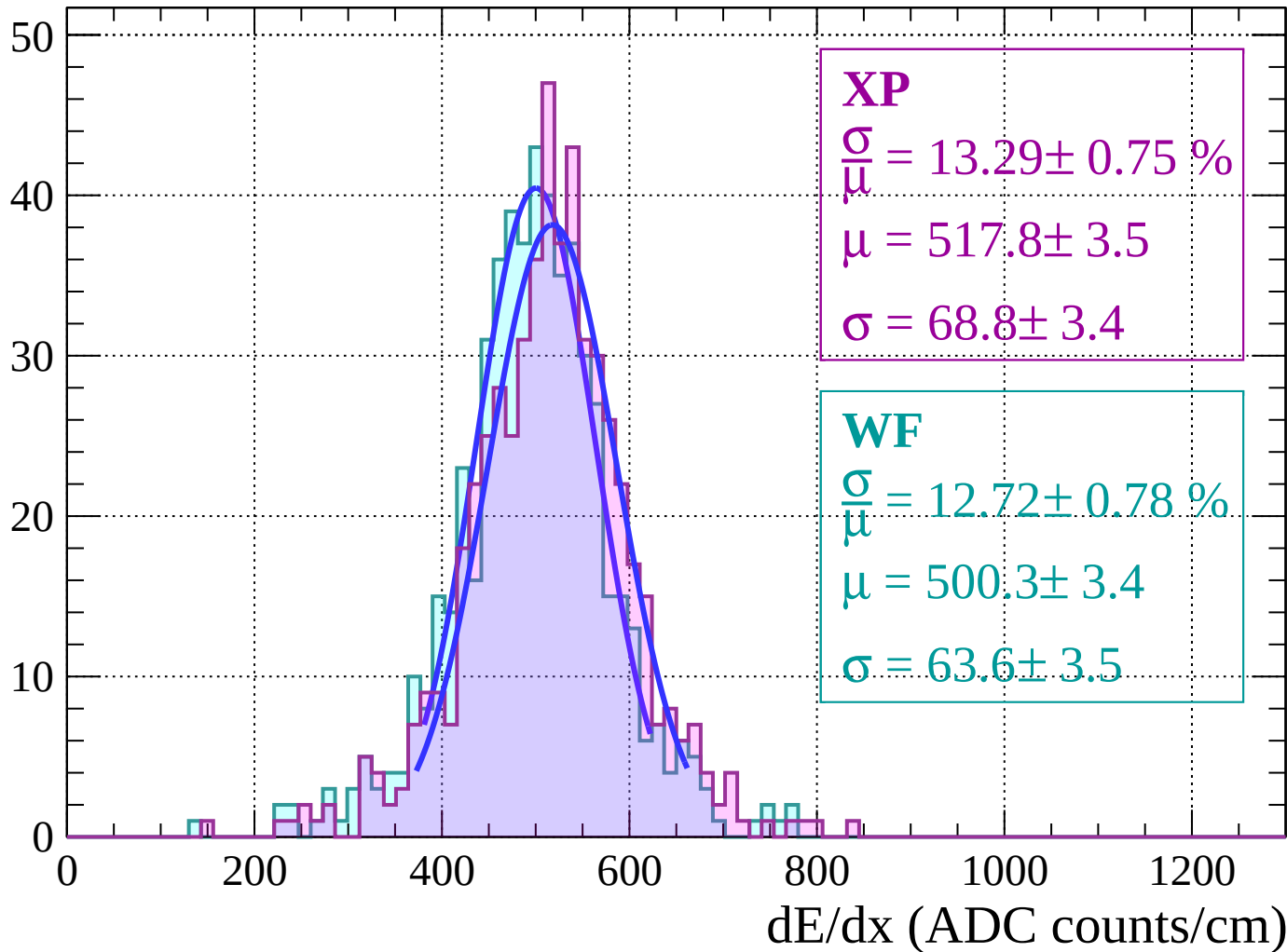
Energy loss | $-54 < \theta < -52$

Count



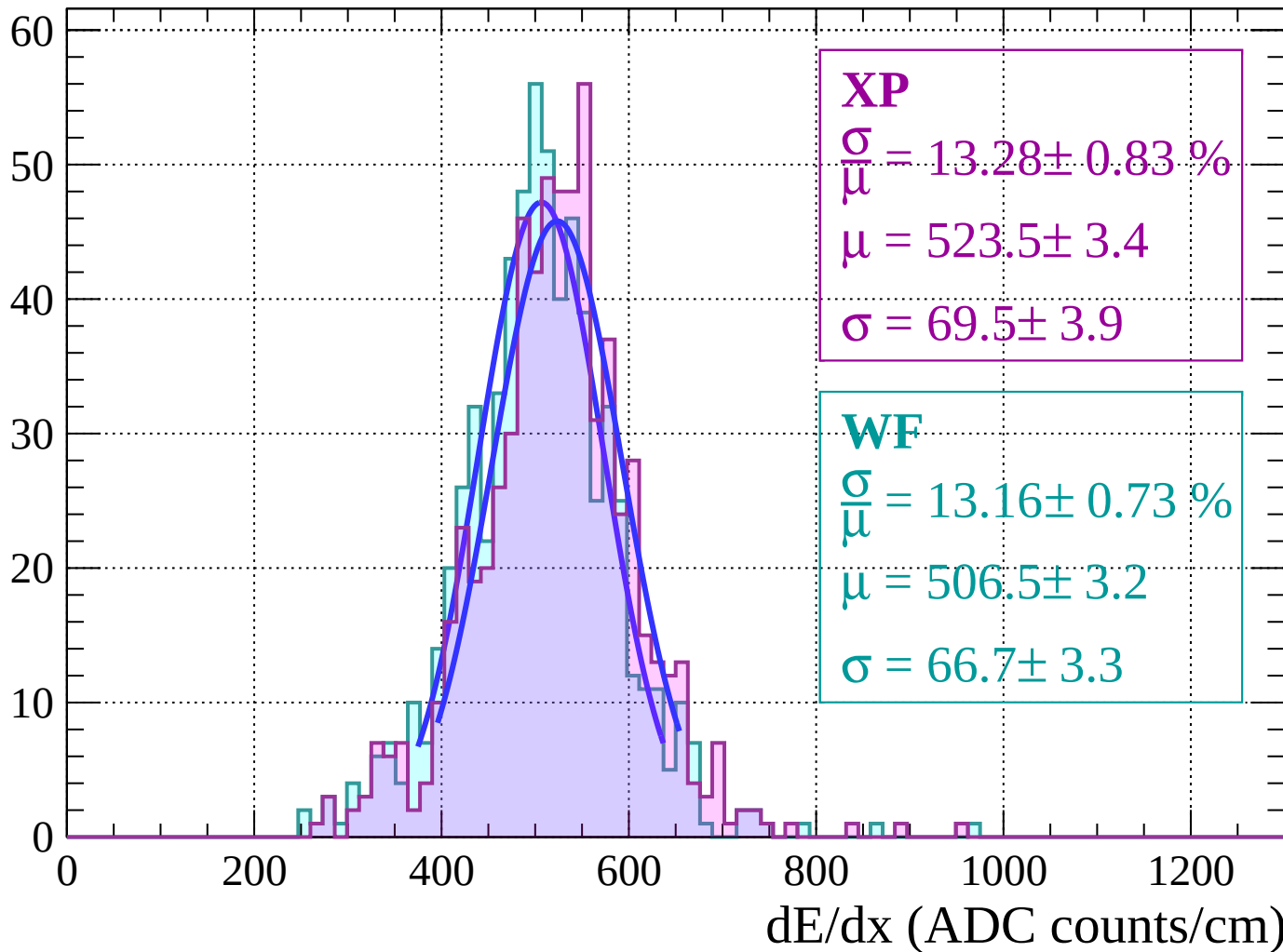
Energy loss $|-52 < \theta < -50$

Count



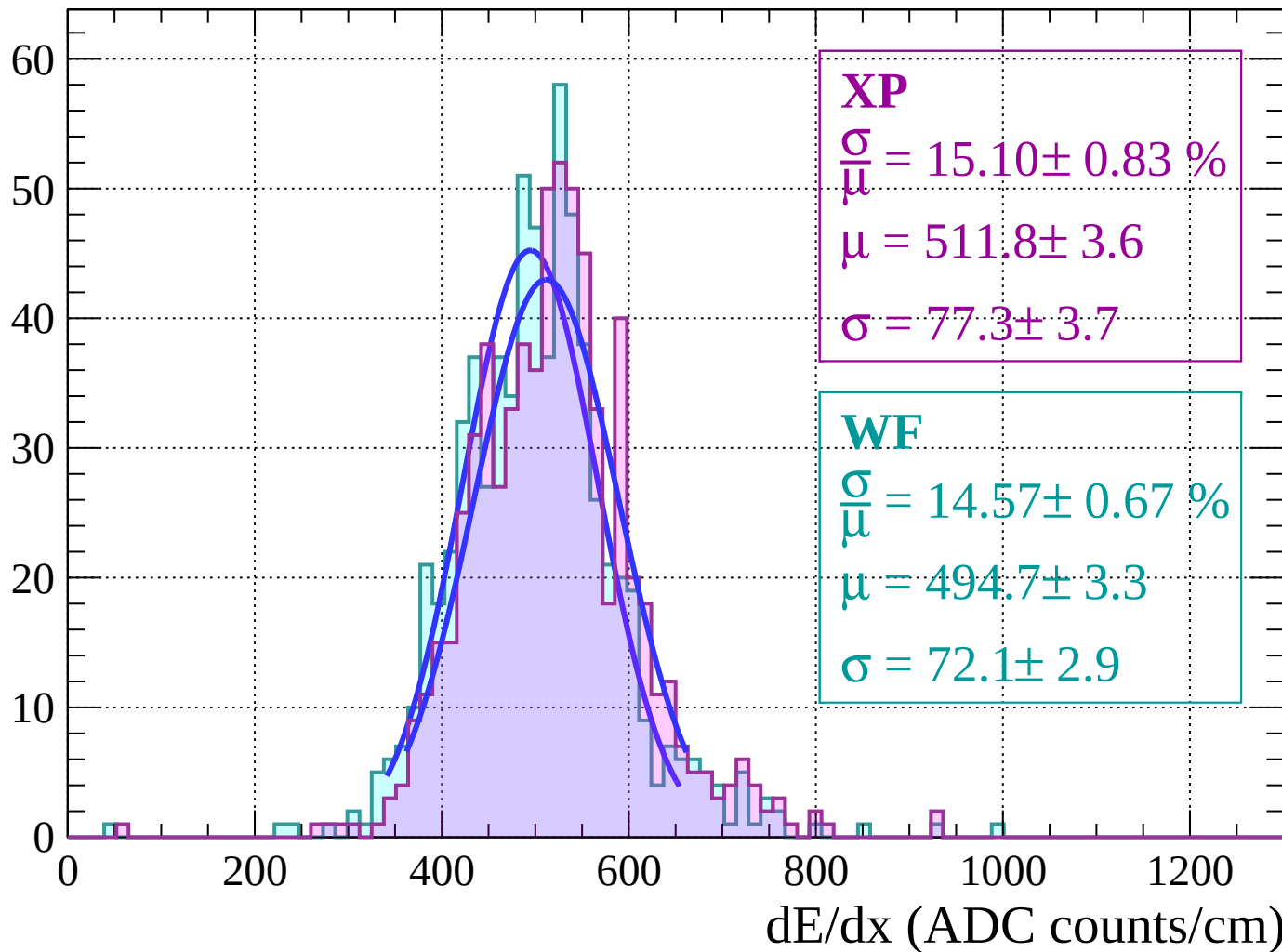
Energy loss | $-50 < \theta < -48$

Count



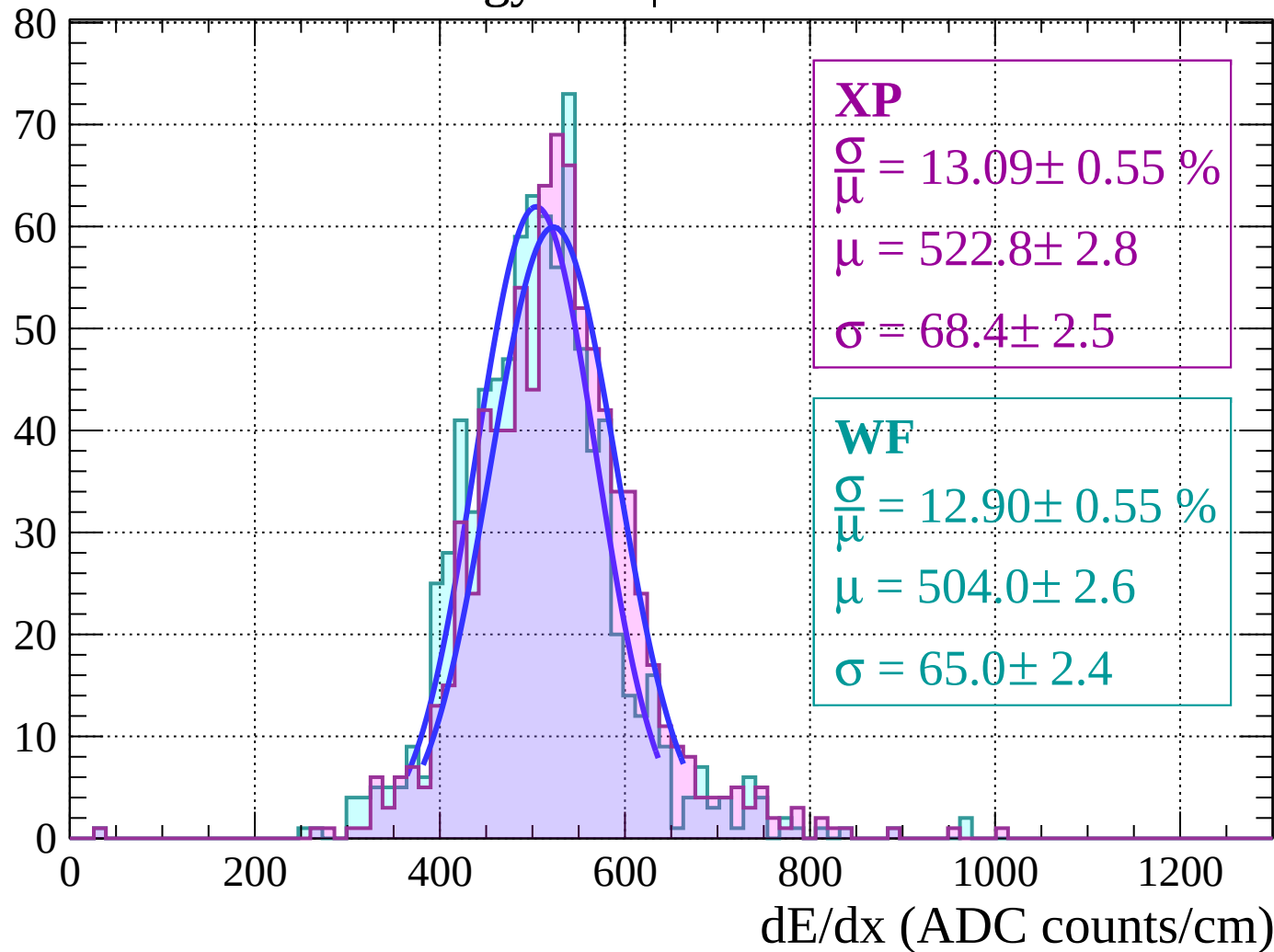
Energy loss | $-48 < \theta < -46$

Count



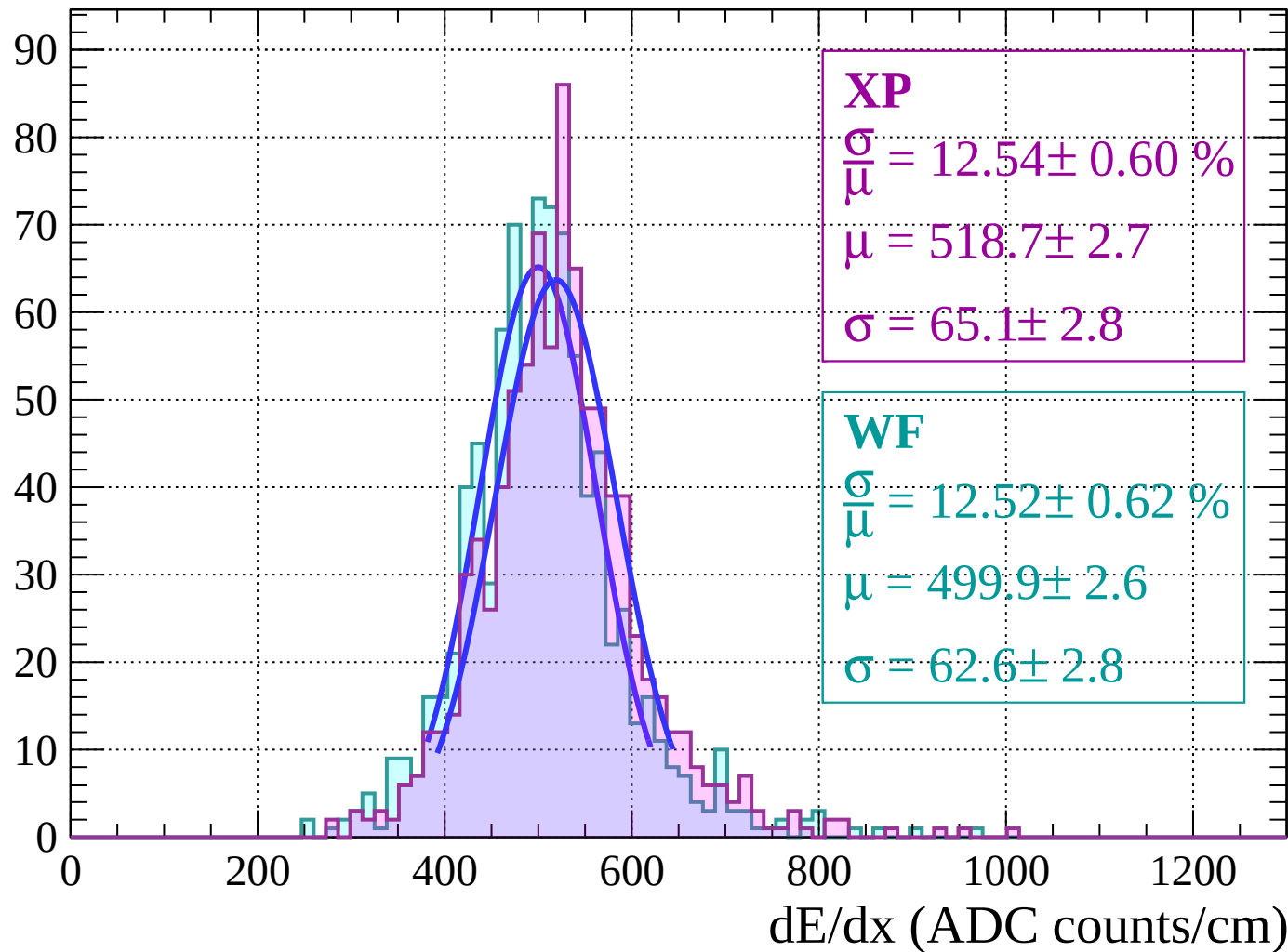
Energy loss $|-46 < \theta < -44$

Count



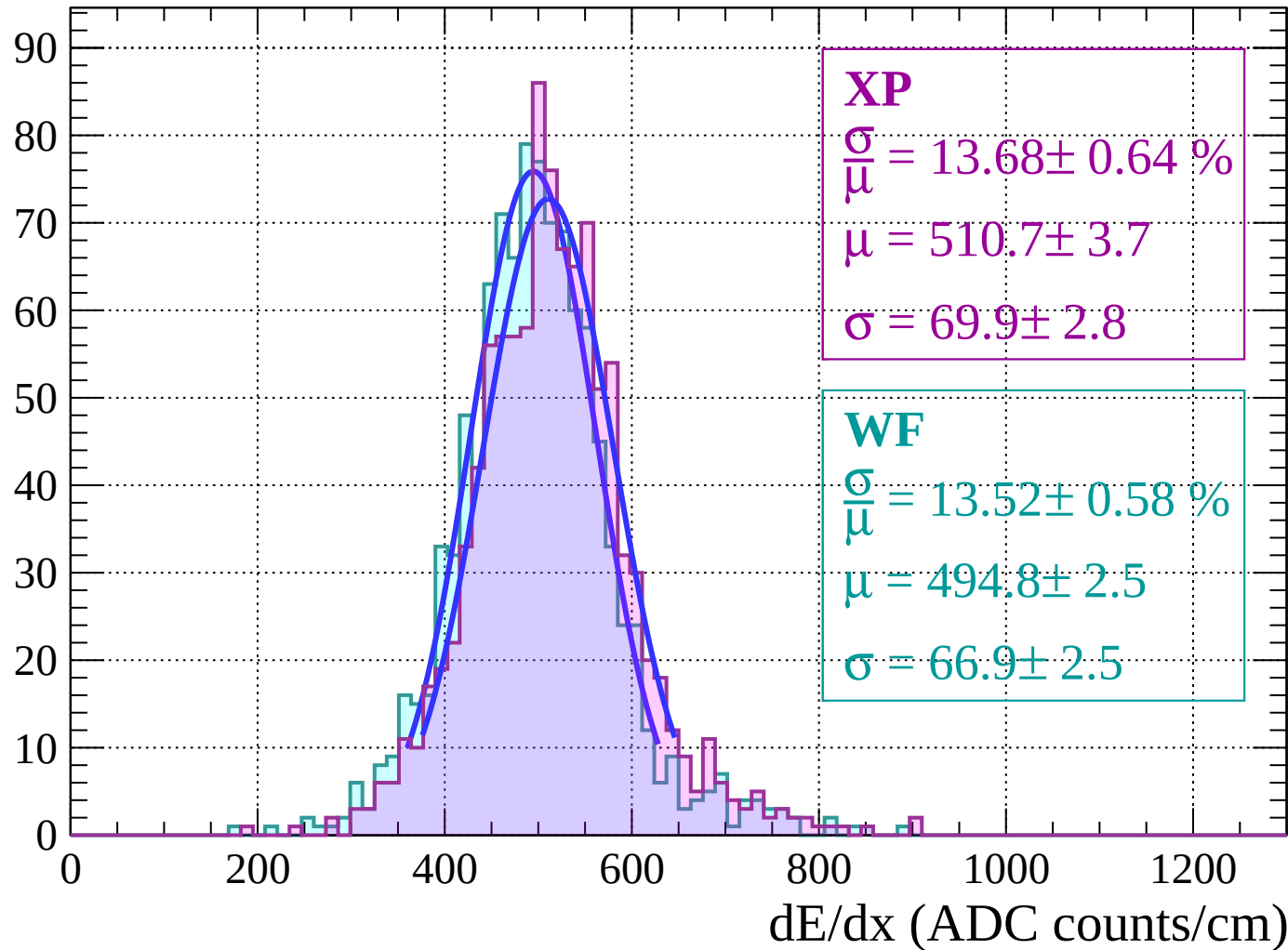
Energy loss $|-44 < \theta < -42$

Count



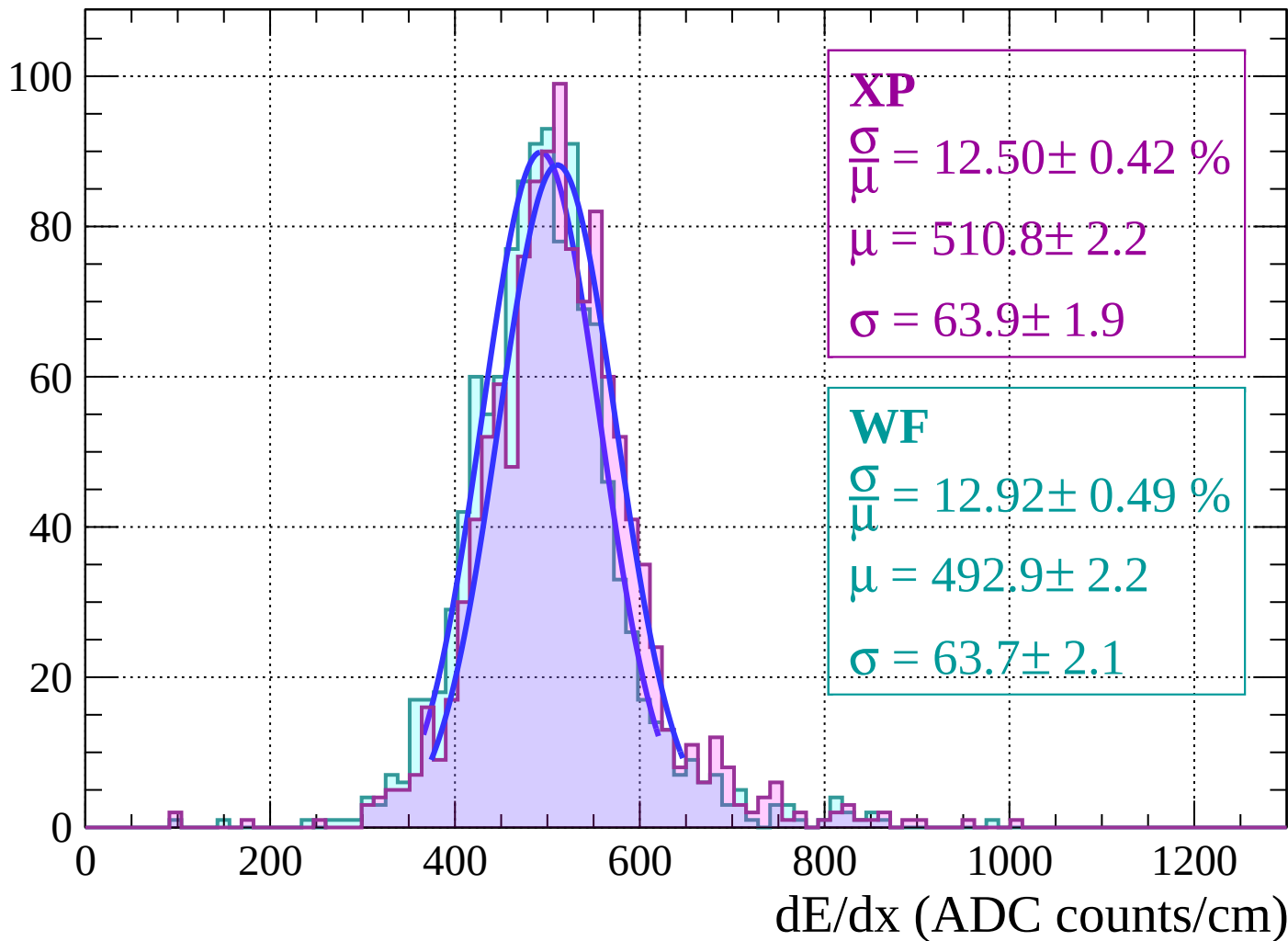
Energy loss | $-42 < \theta < -40$

Count



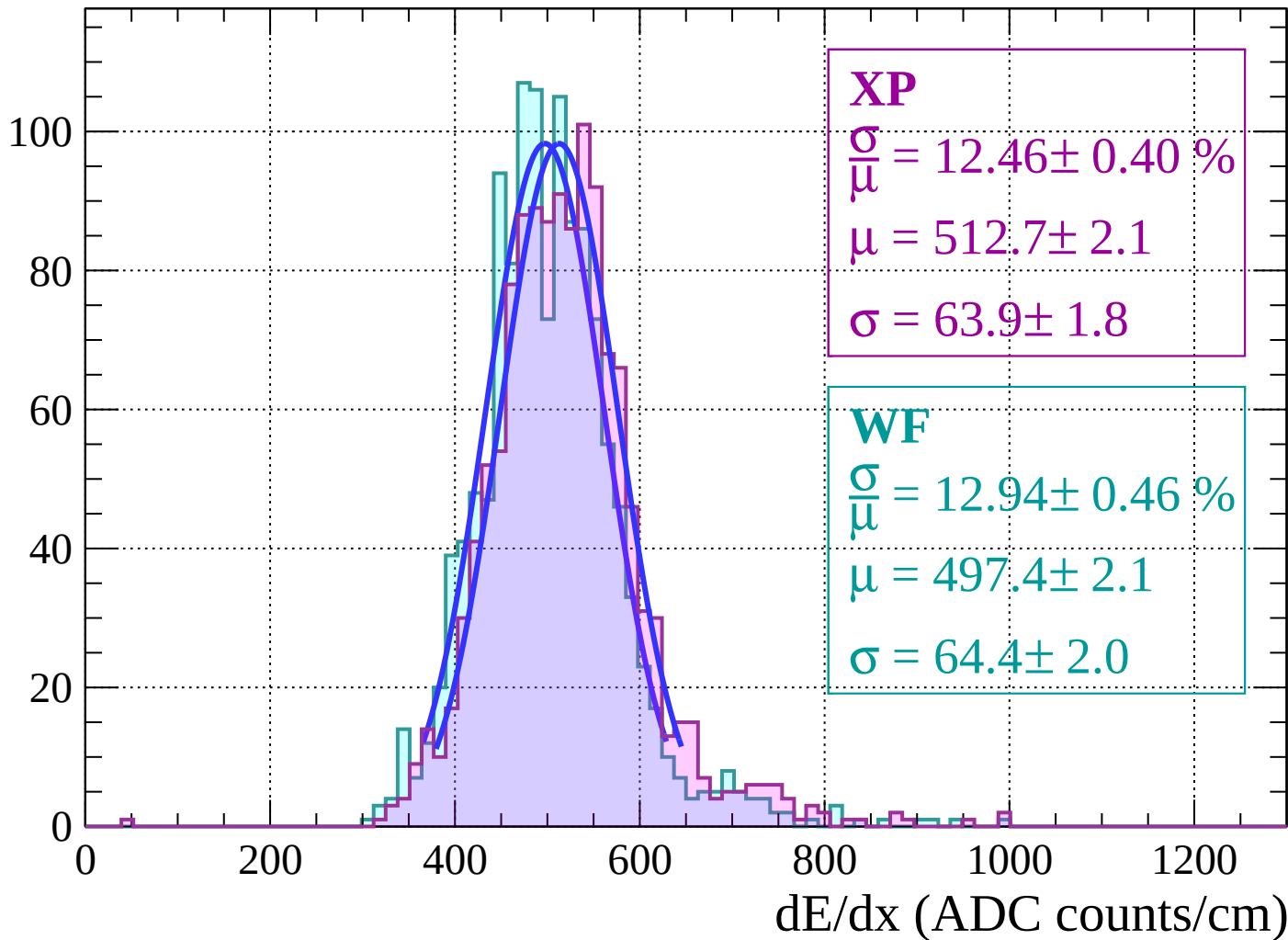
Energy loss | $-40 < \theta < -38$

Count



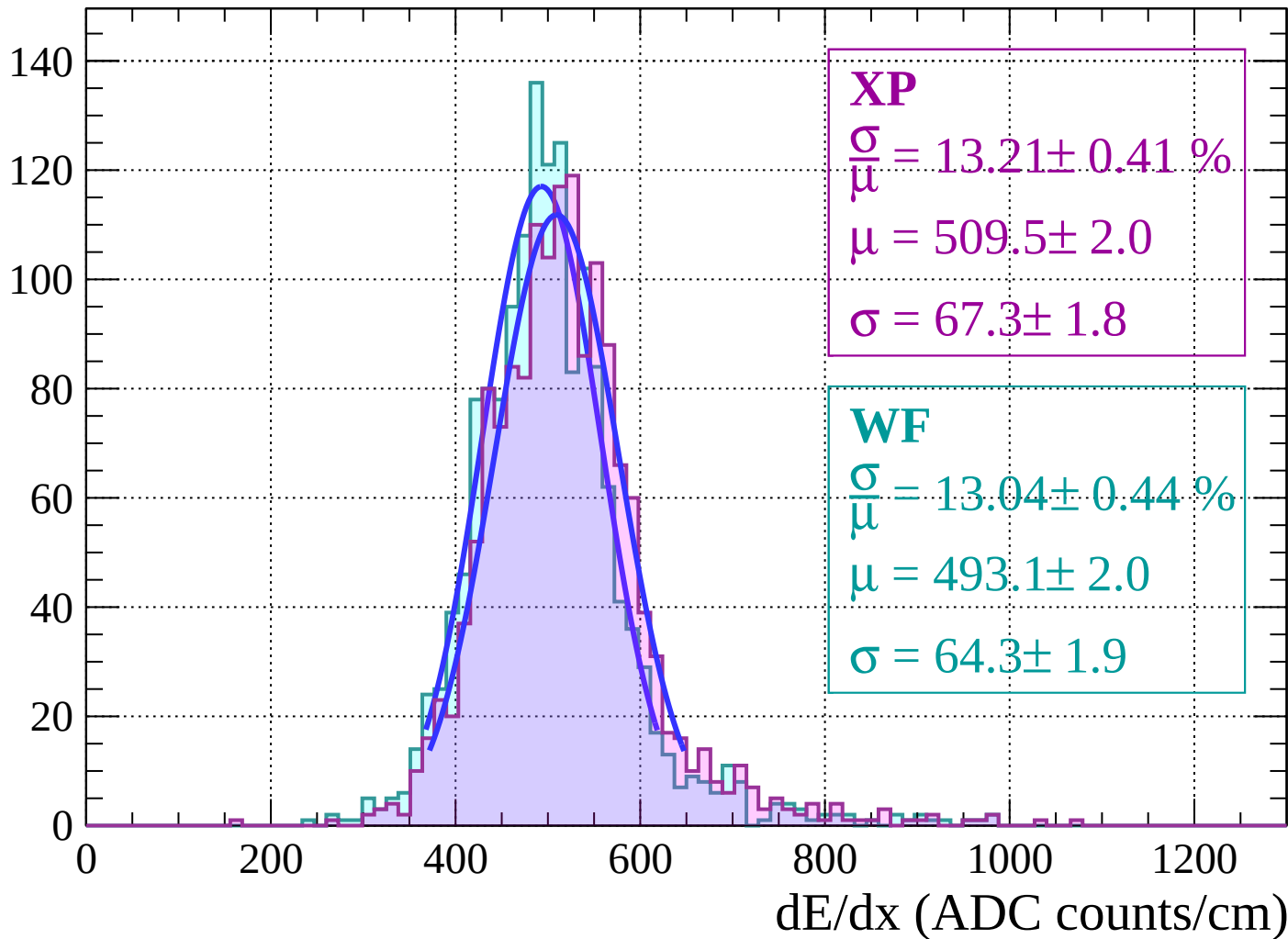
Energy loss $|-38 < \theta < -36$

Count



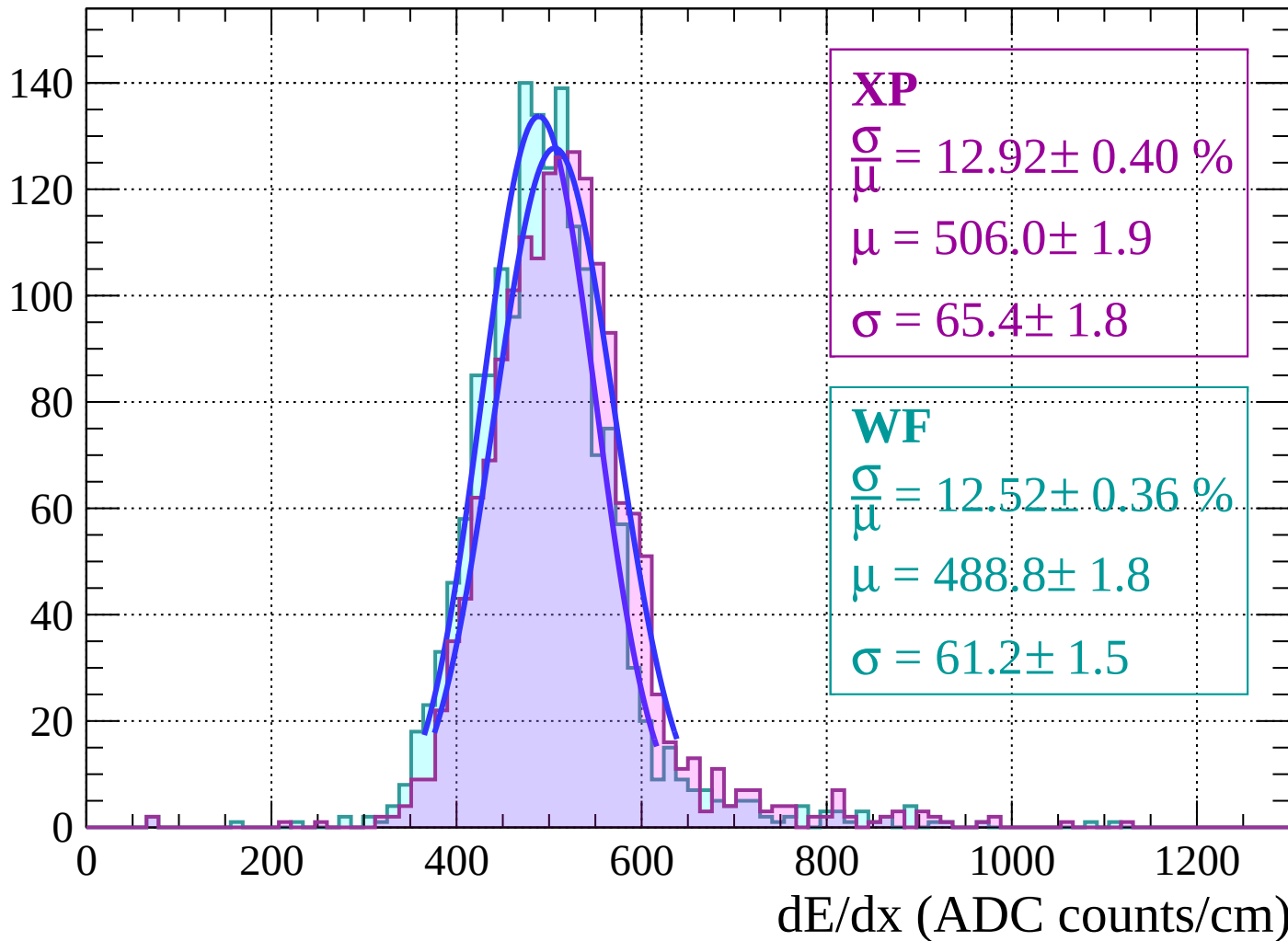
Energy loss $|-36 < \theta < -34$

Count



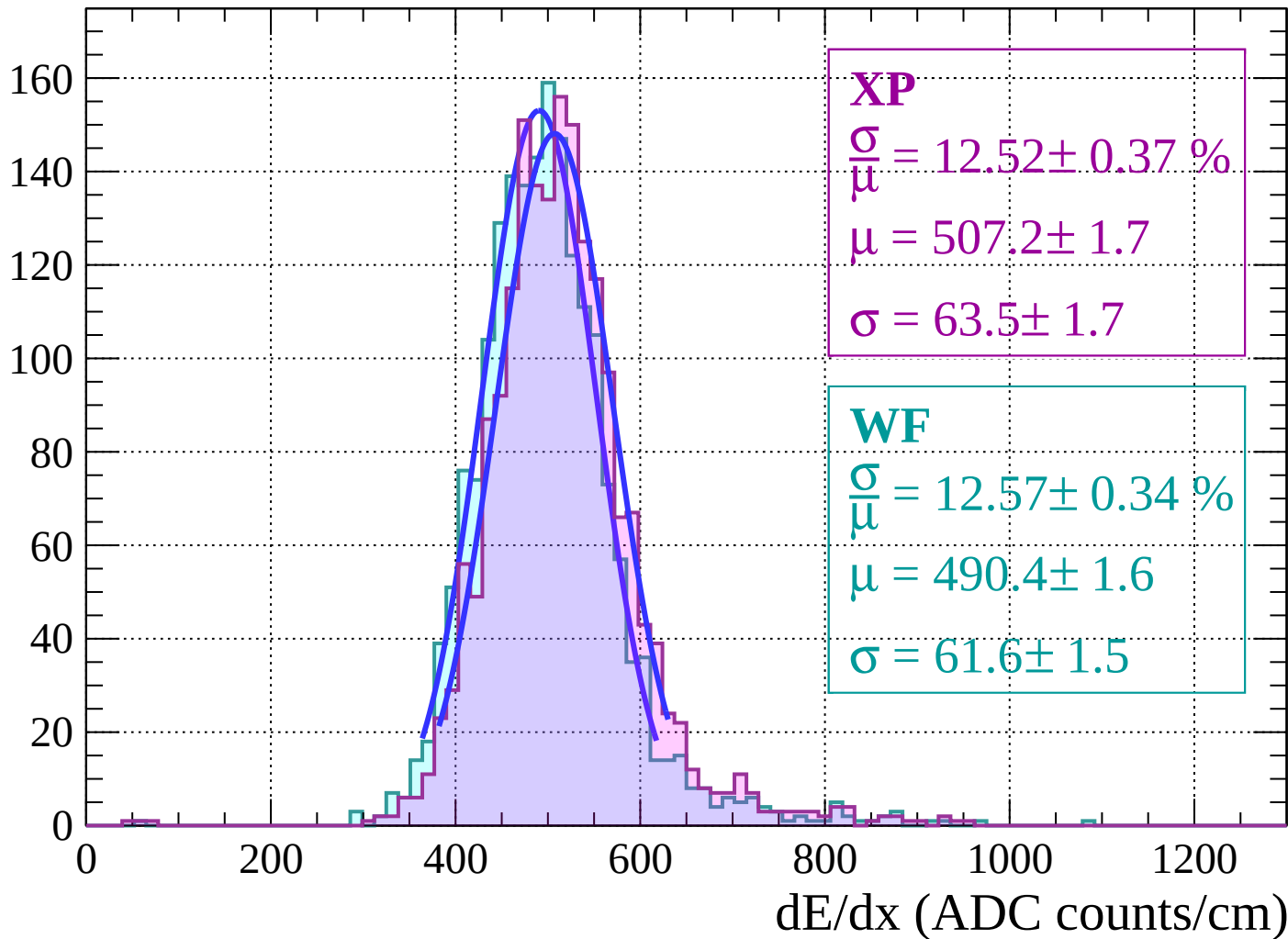
Energy loss $|-34 < \theta < -32$

Count



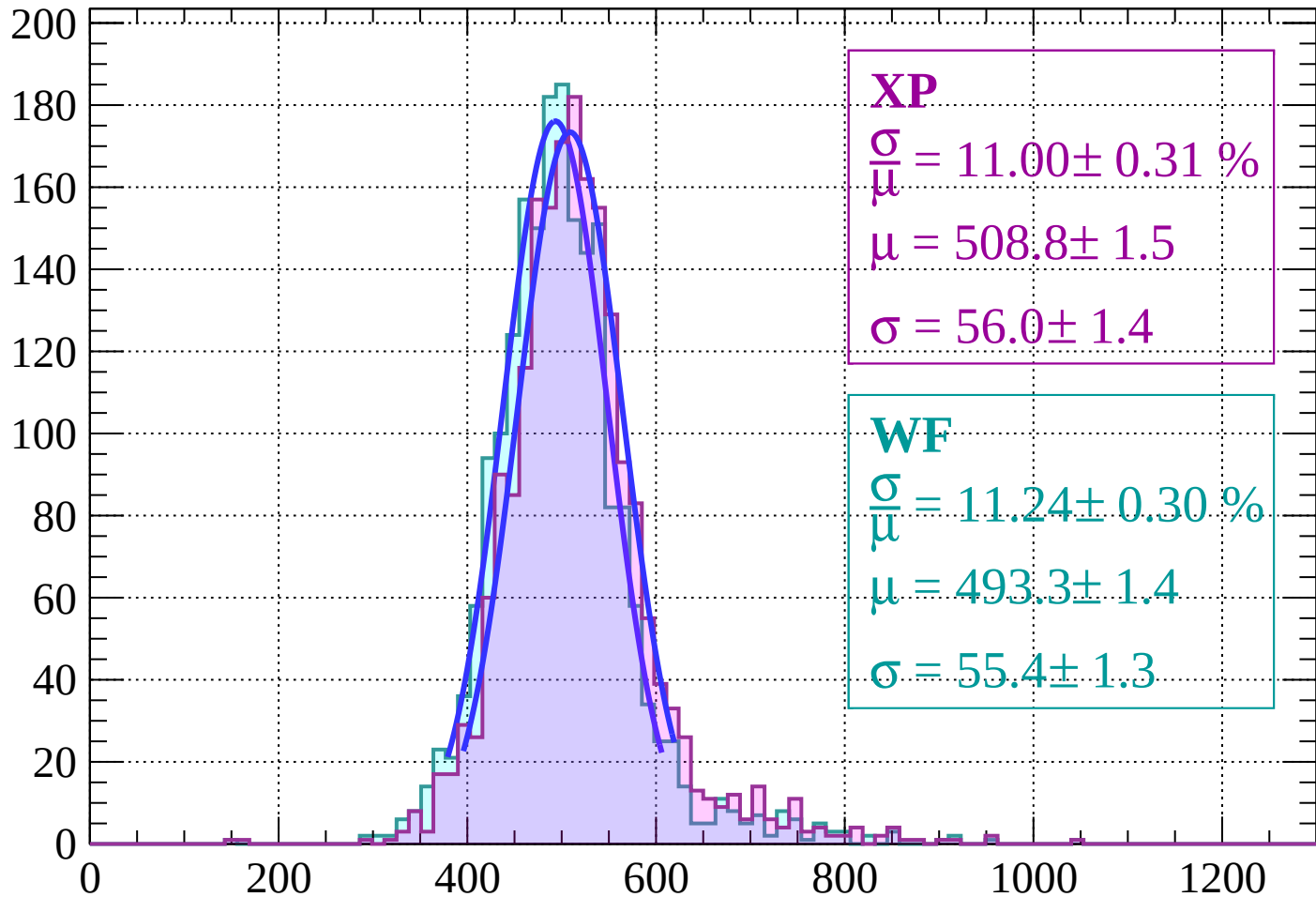
Energy loss $|-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count



XP

$$\frac{\sigma}{\mu} = 11.00 \pm 0.31 \%$$

$$\mu = 508.8 \pm 1.5$$

$$\sigma = 56.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.24 \pm 0.30 \%$$

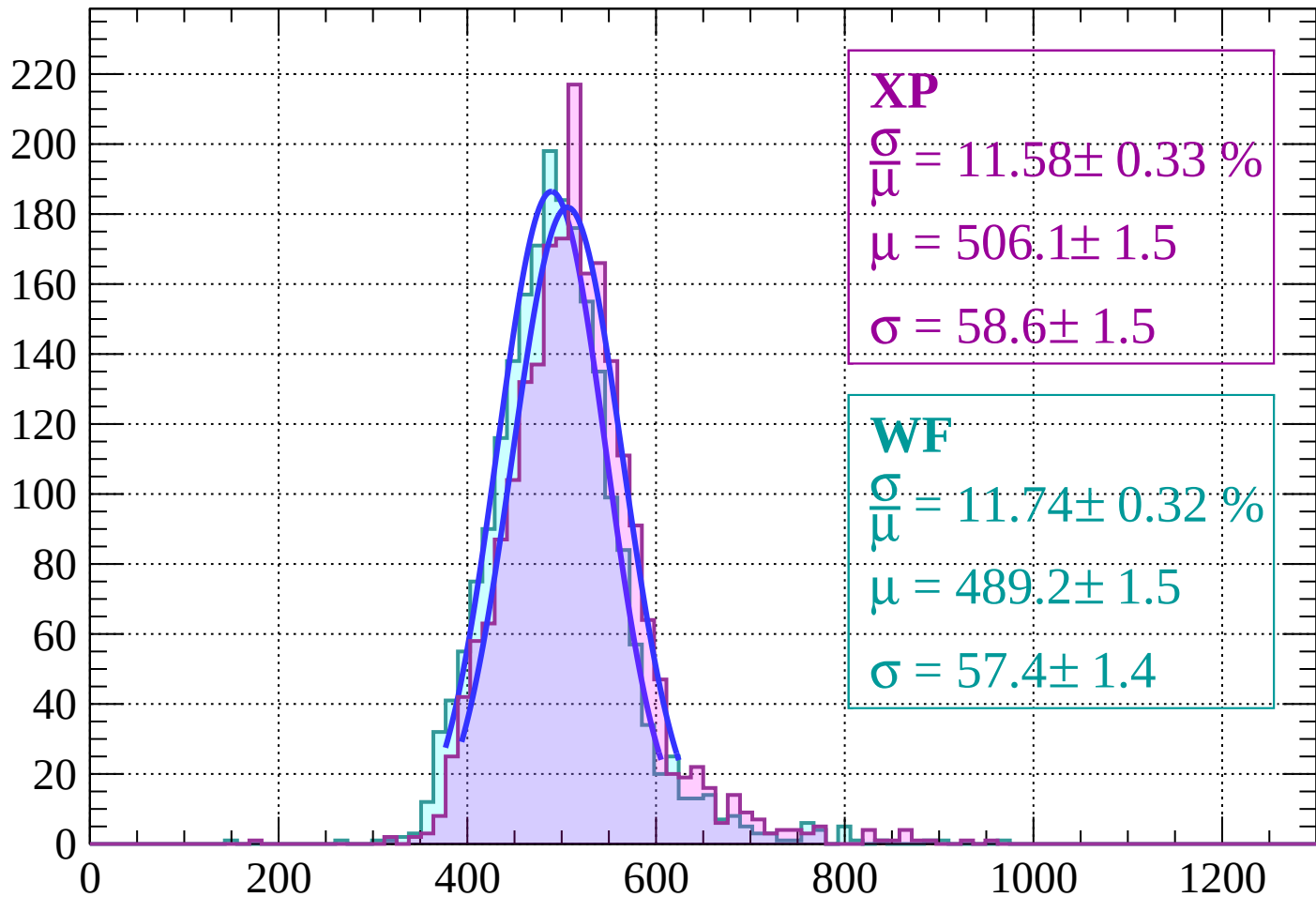
$$\mu = 493.3 \pm 1.4$$

$$\sigma = 55.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

Count



XP

$$\frac{\sigma}{\mu} = 11.58 \pm 0.33 \%$$

$$\mu = 506.1 \pm 1.5$$

$$\sigma = 58.6 \pm 1.5$$

WF

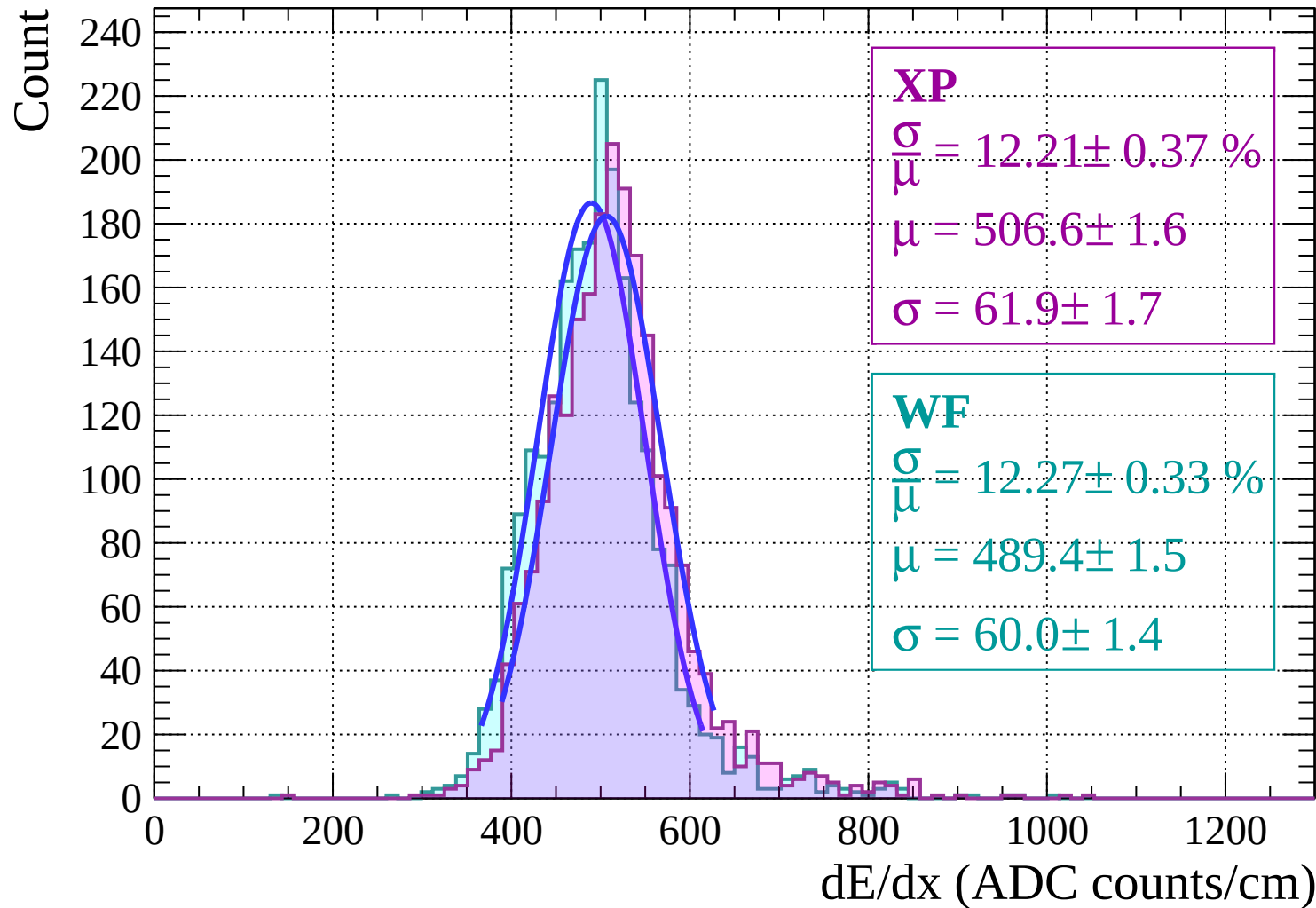
$$\frac{\sigma}{\mu} = 11.74 \pm 0.32 \%$$

$$\mu = 489.2 \pm 1.5$$

$$\sigma = 57.4 \pm 1.4$$

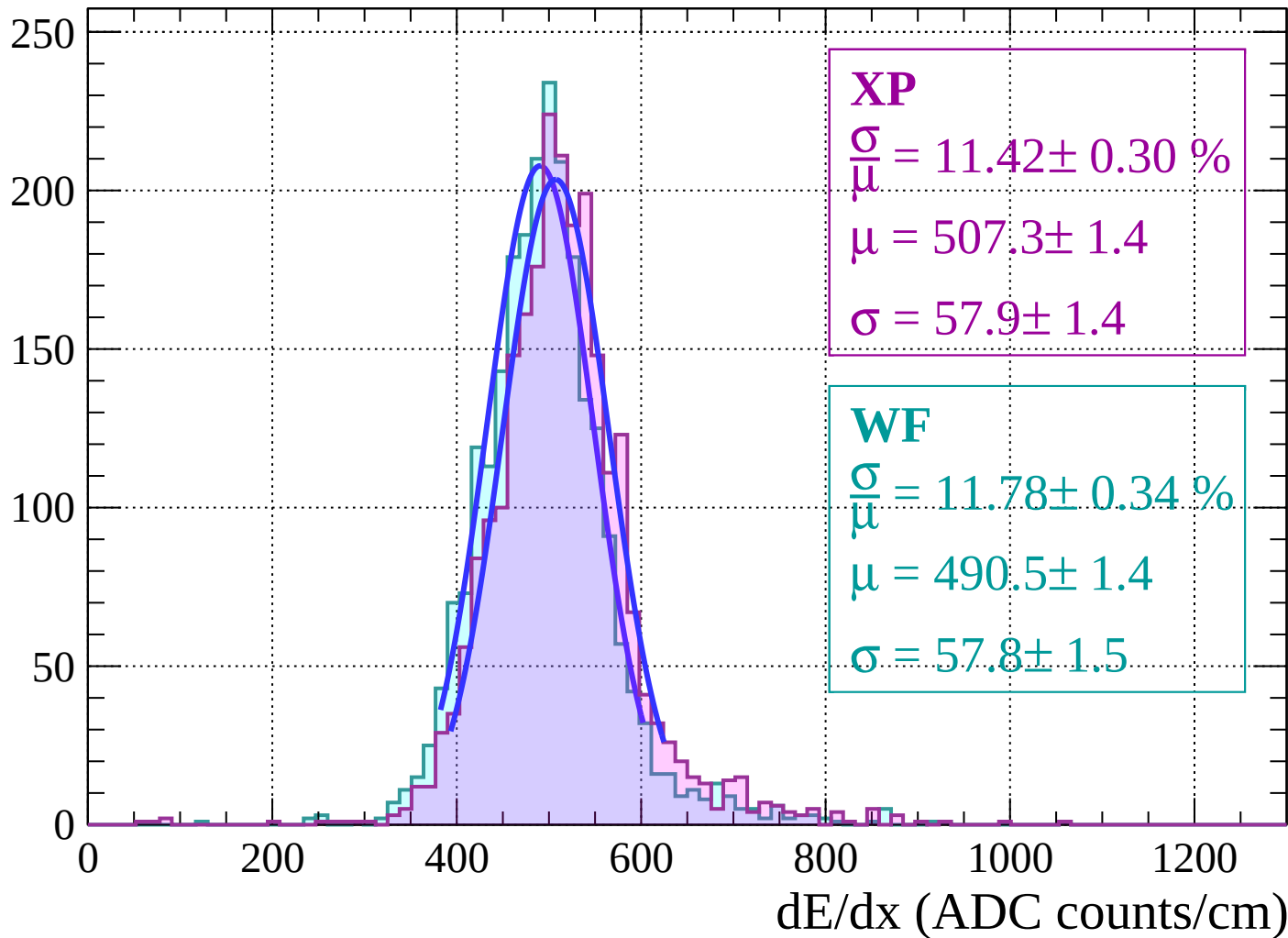
dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

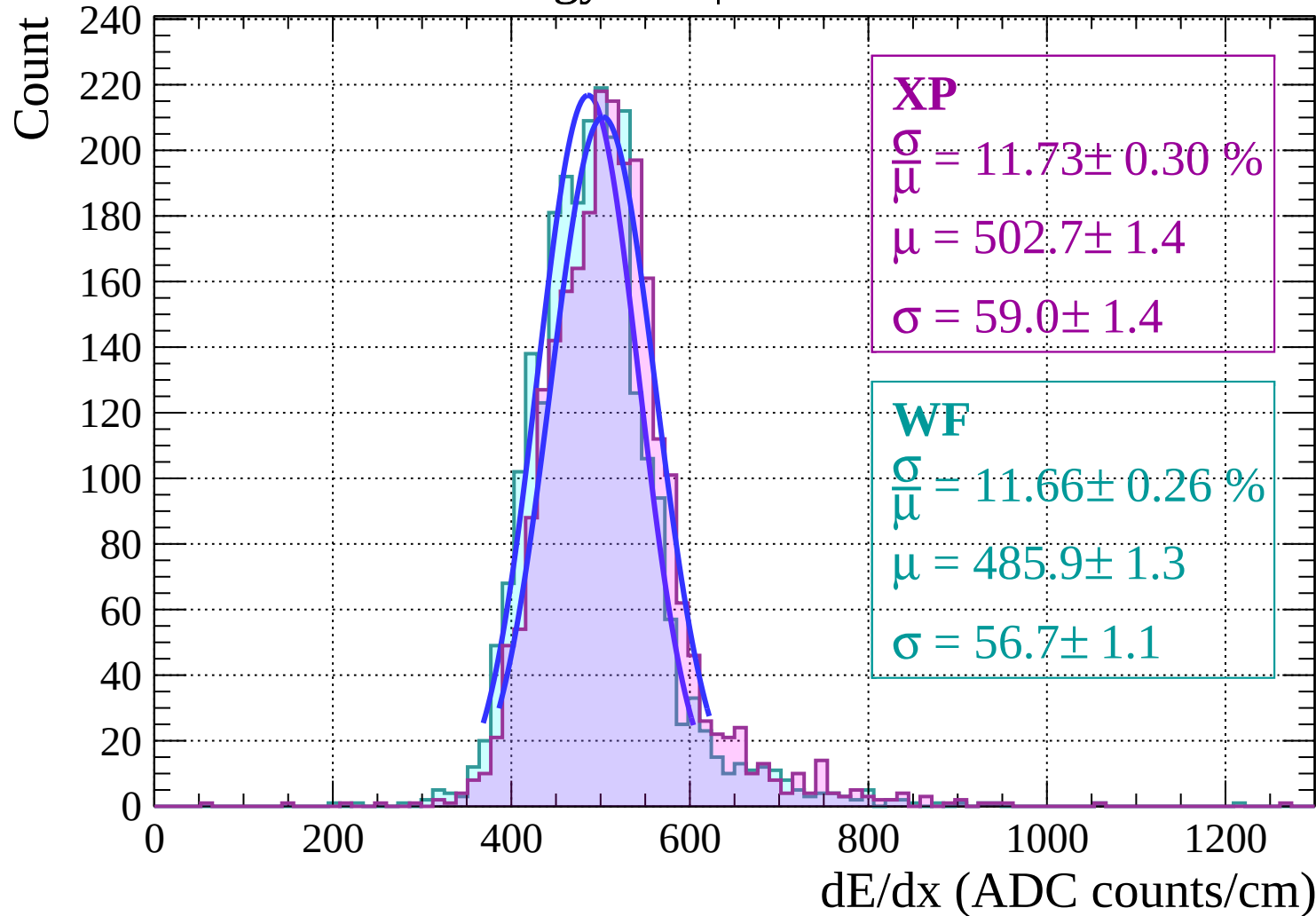


Energy loss $|-24 < \theta < -22$

Count

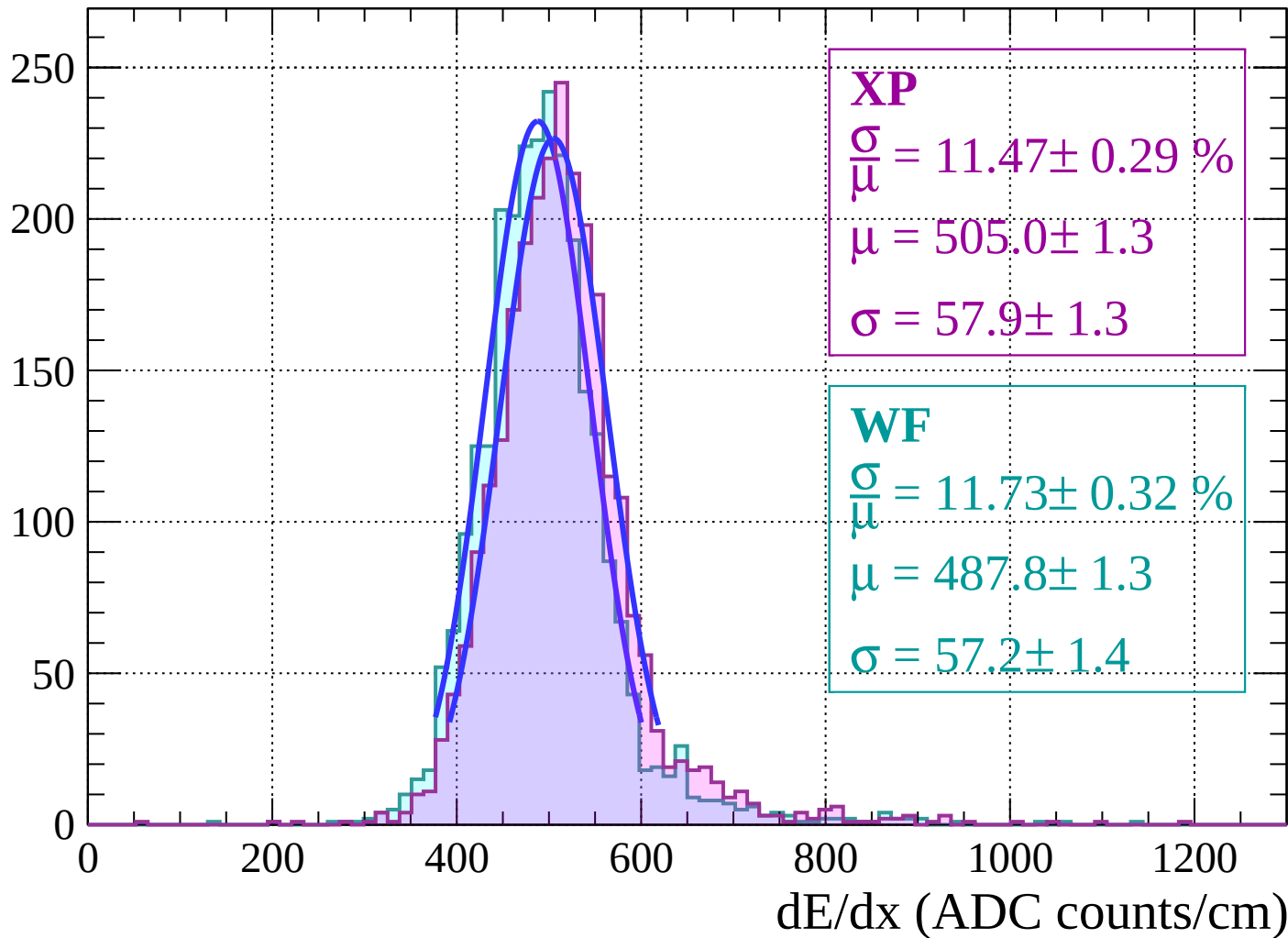


Energy loss $|-22 < \theta < -20$

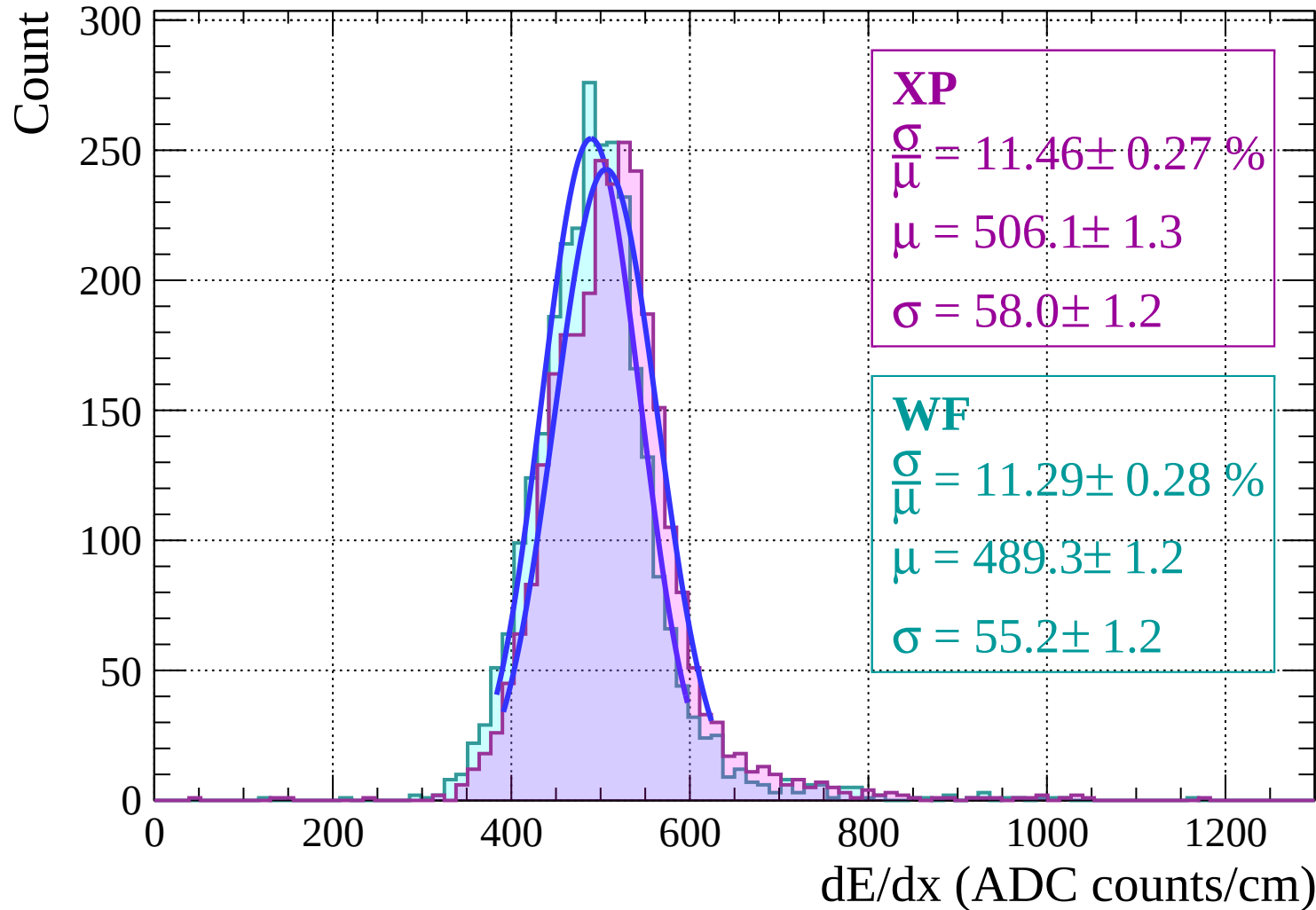


Energy loss | $-20 < \theta < -18$

Count

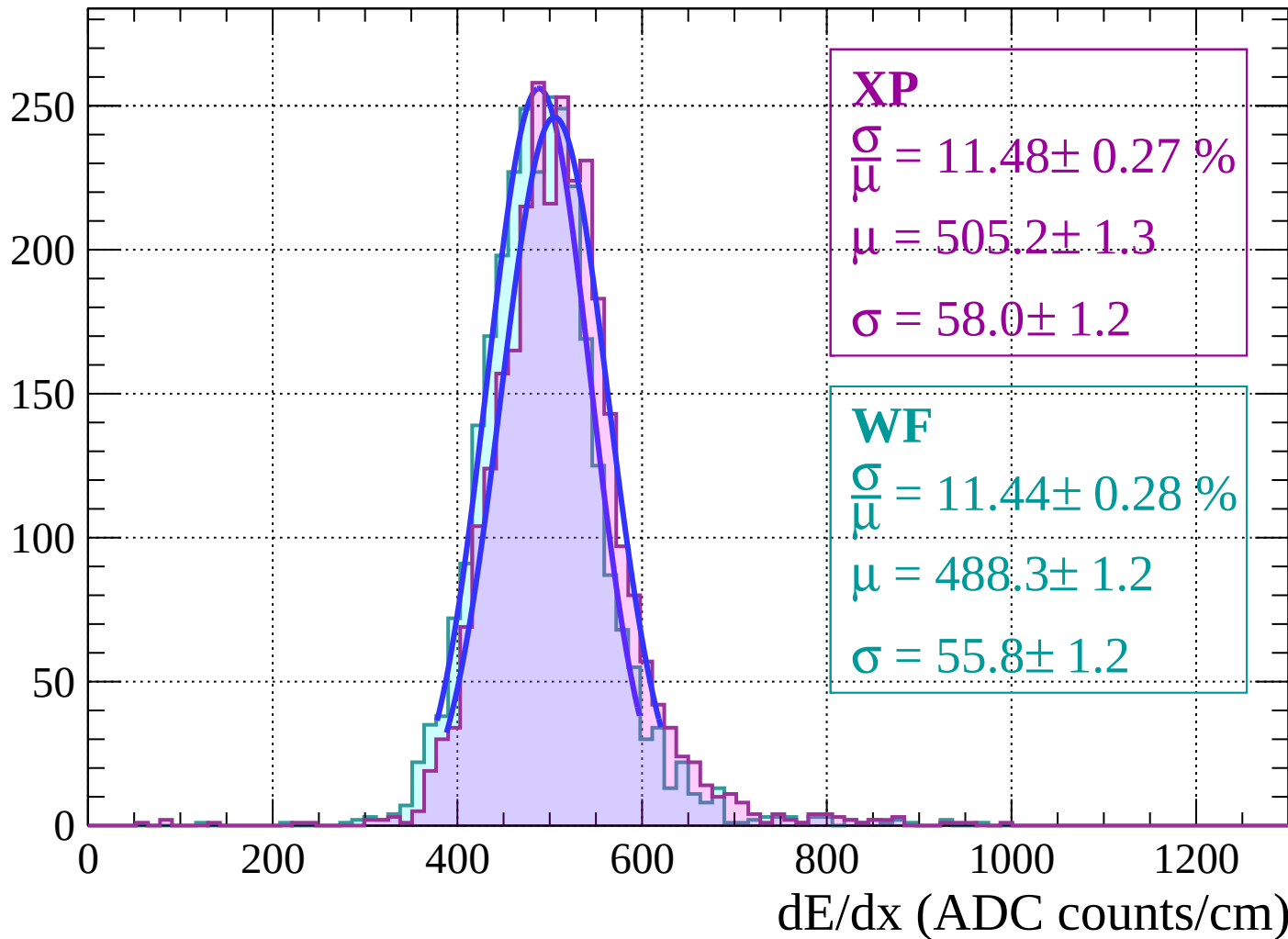


Energy loss | $-18 < \theta < -16$



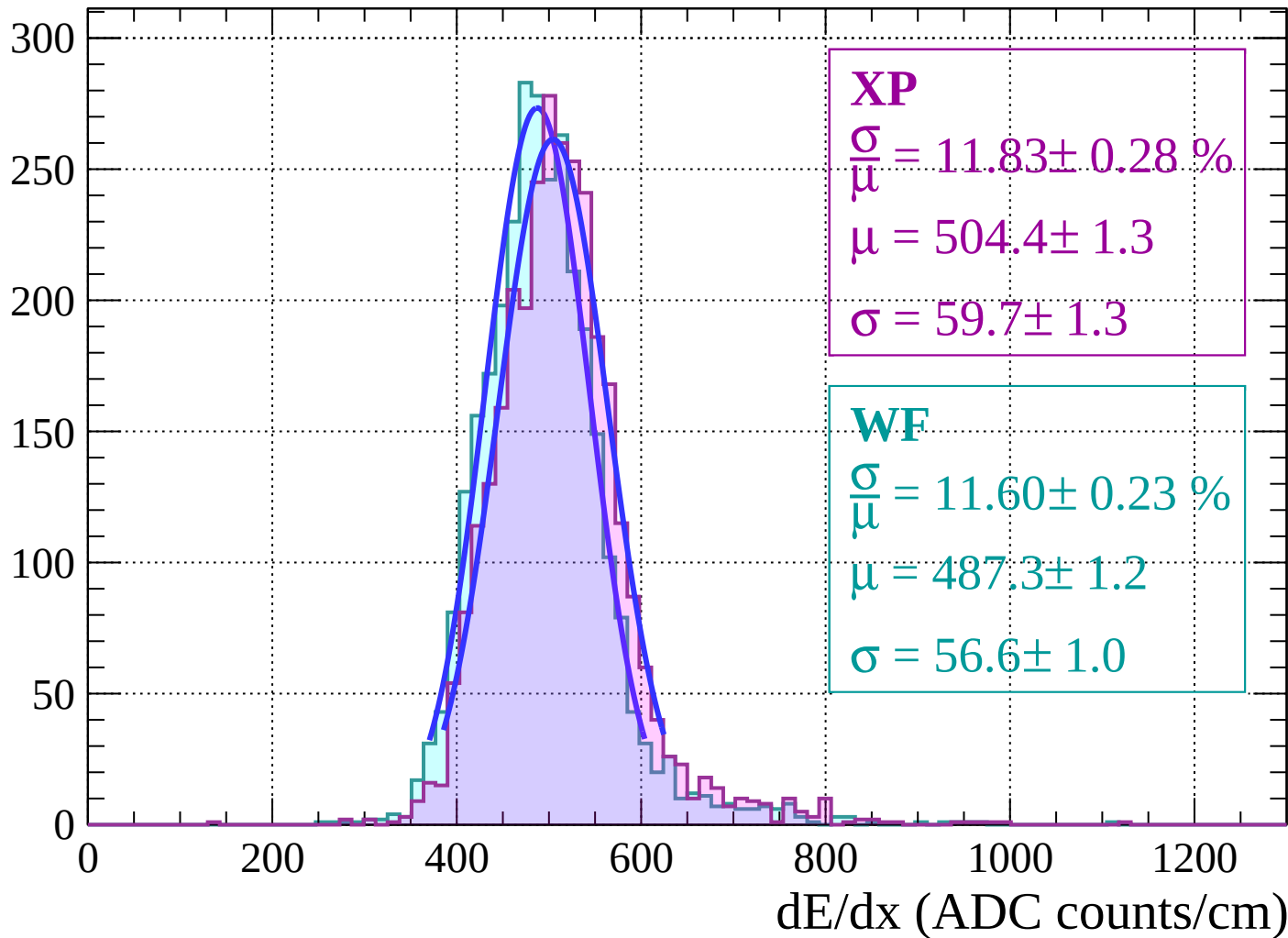
Energy loss $|-16 < \theta < -14$

Count



Energy loss | $-14 < \theta < -12$

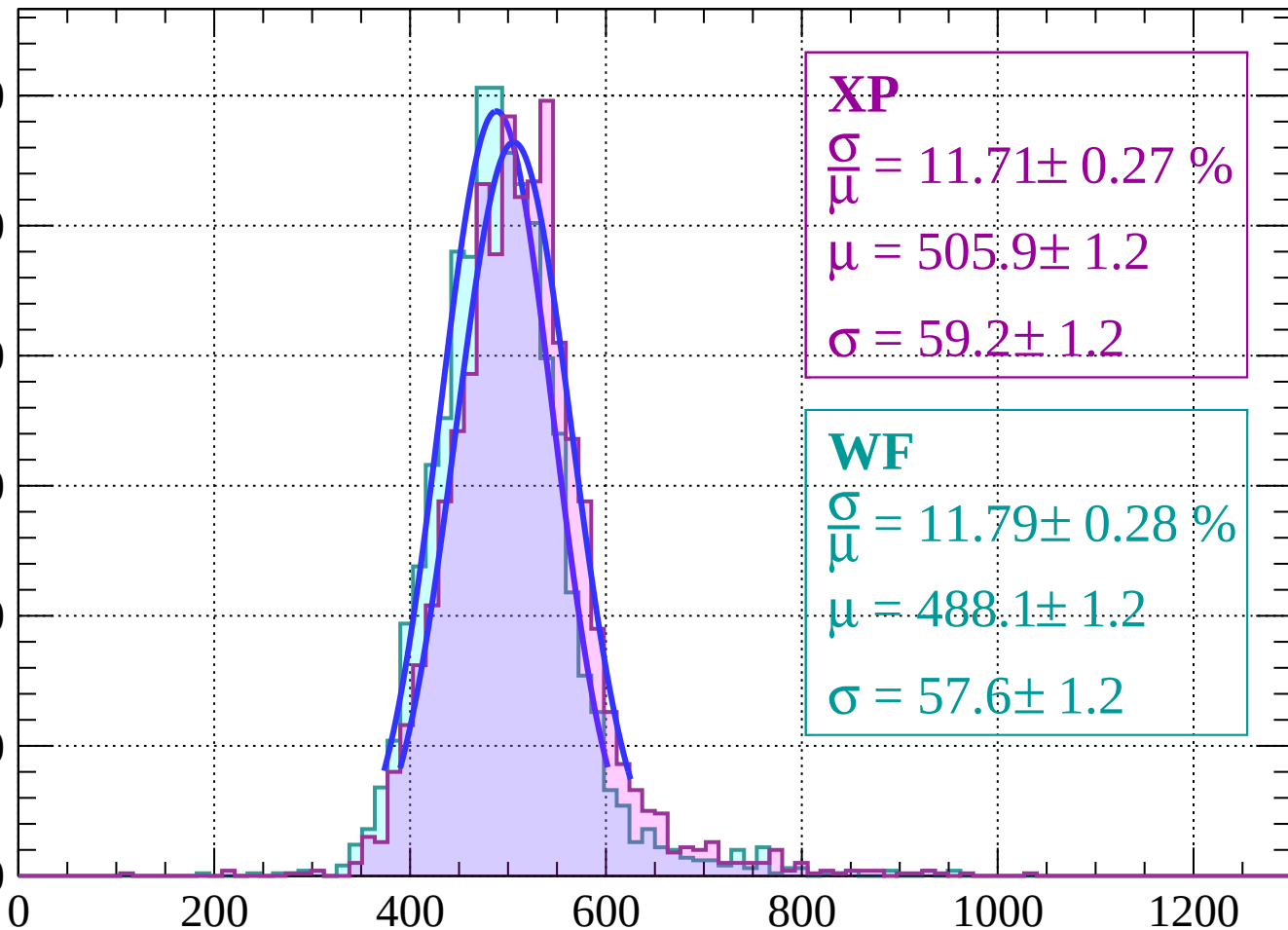
Count



Energy loss | $-12 < \theta < -10$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.71 \pm 0.27 \%$$

$$\mu = 505.9 \pm 1.2$$

$$\sigma = 59.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 11.79 \pm 0.28 \%$$

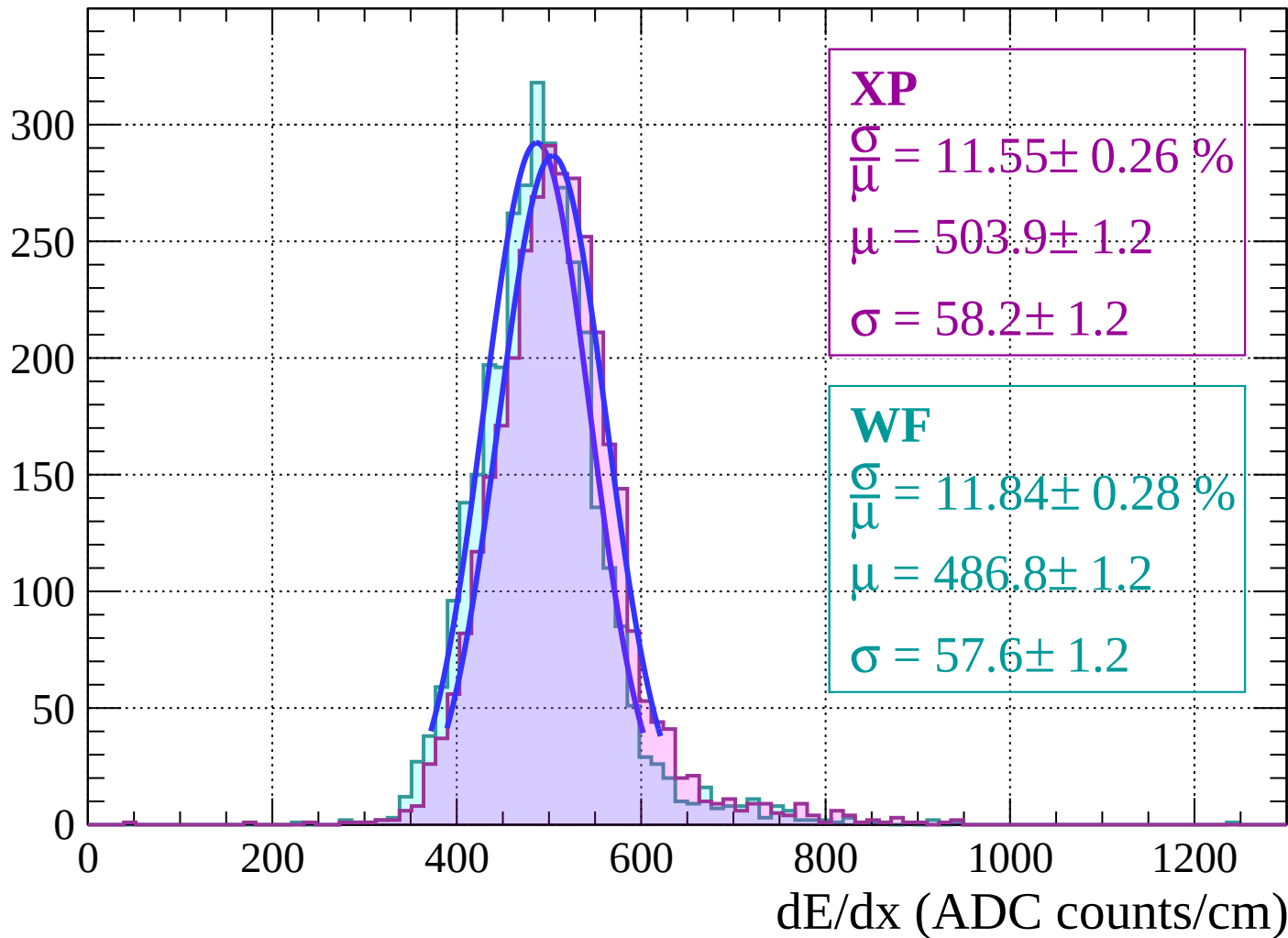
$$\mu = 488.1 \pm 1.2$$

$$\sigma = 57.6 \pm 1.2$$

dE/dx (ADC counts/cm)

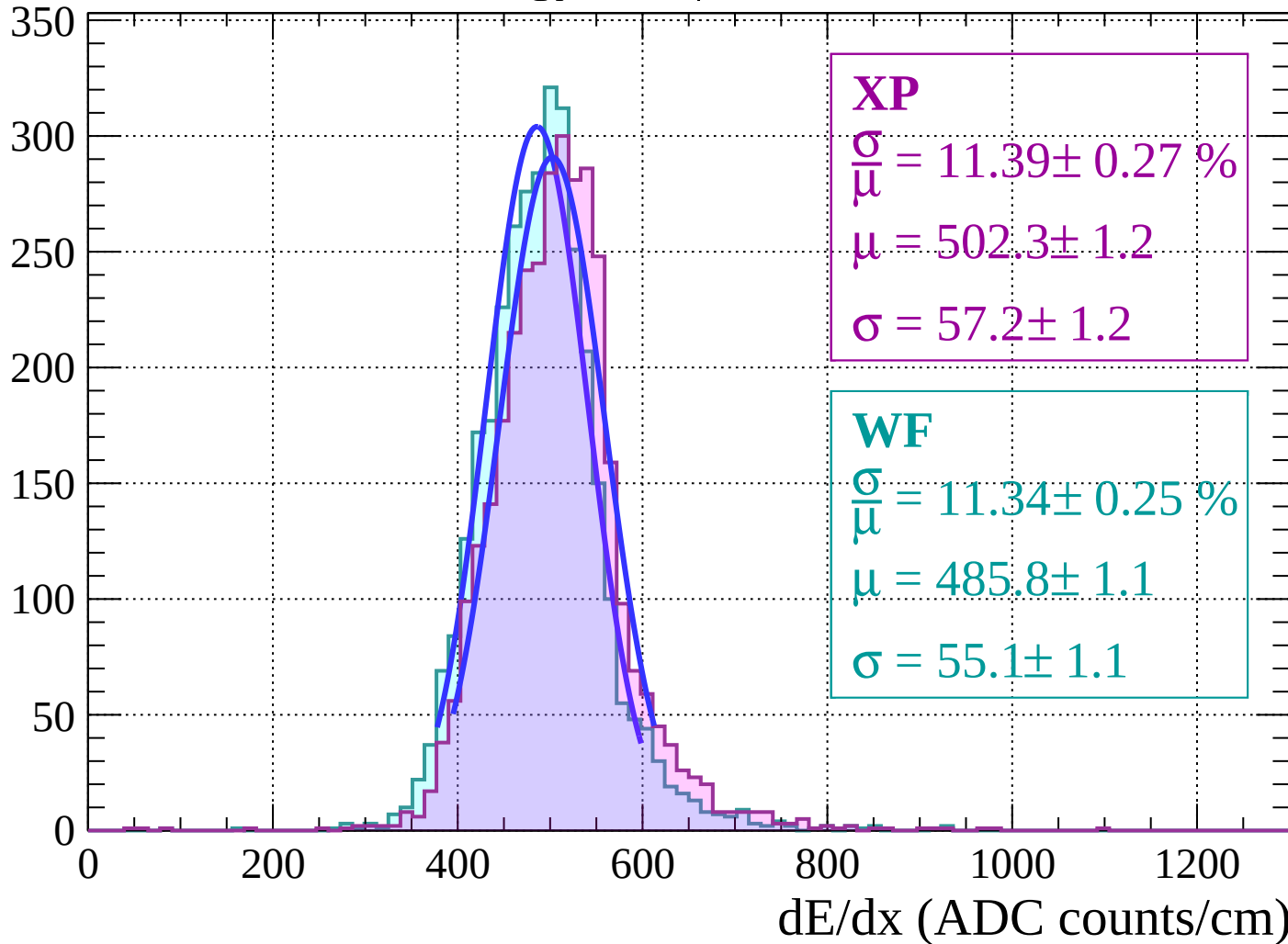
Energy loss | $-10 < \theta < -8$

Count



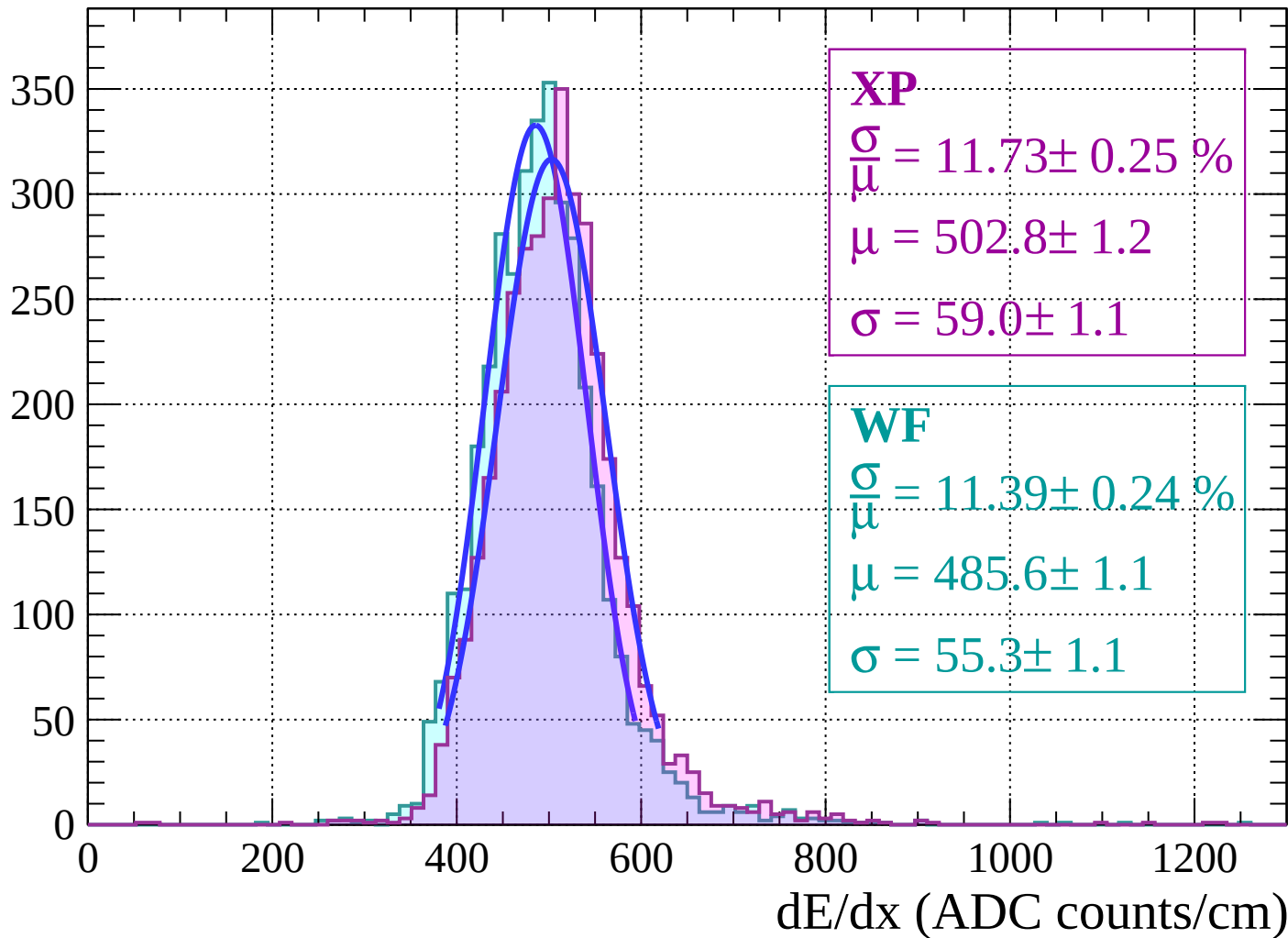
Energy loss | $-8 < \theta < -6$

Count

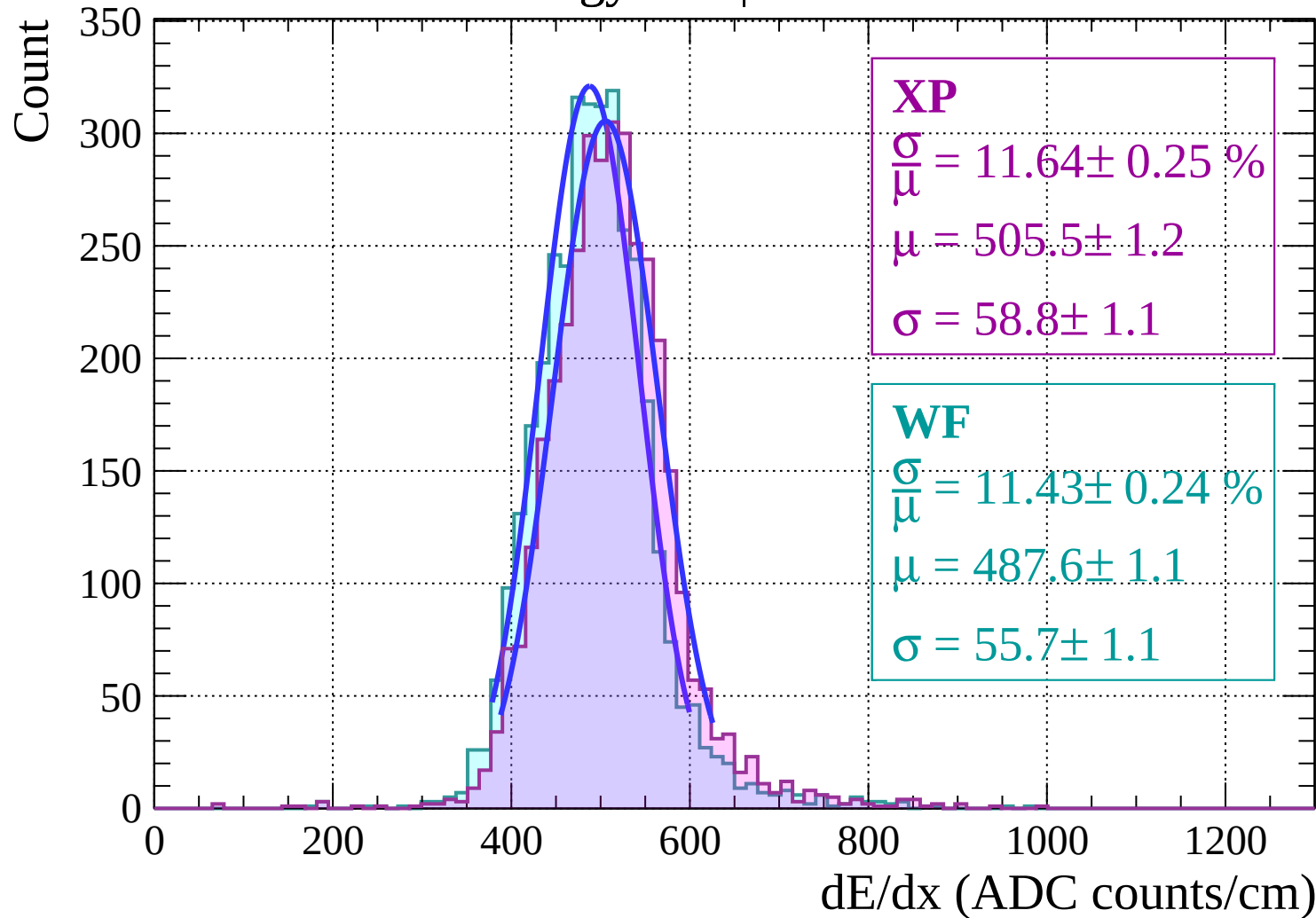


Energy loss $-6 < \theta < -4$

Count

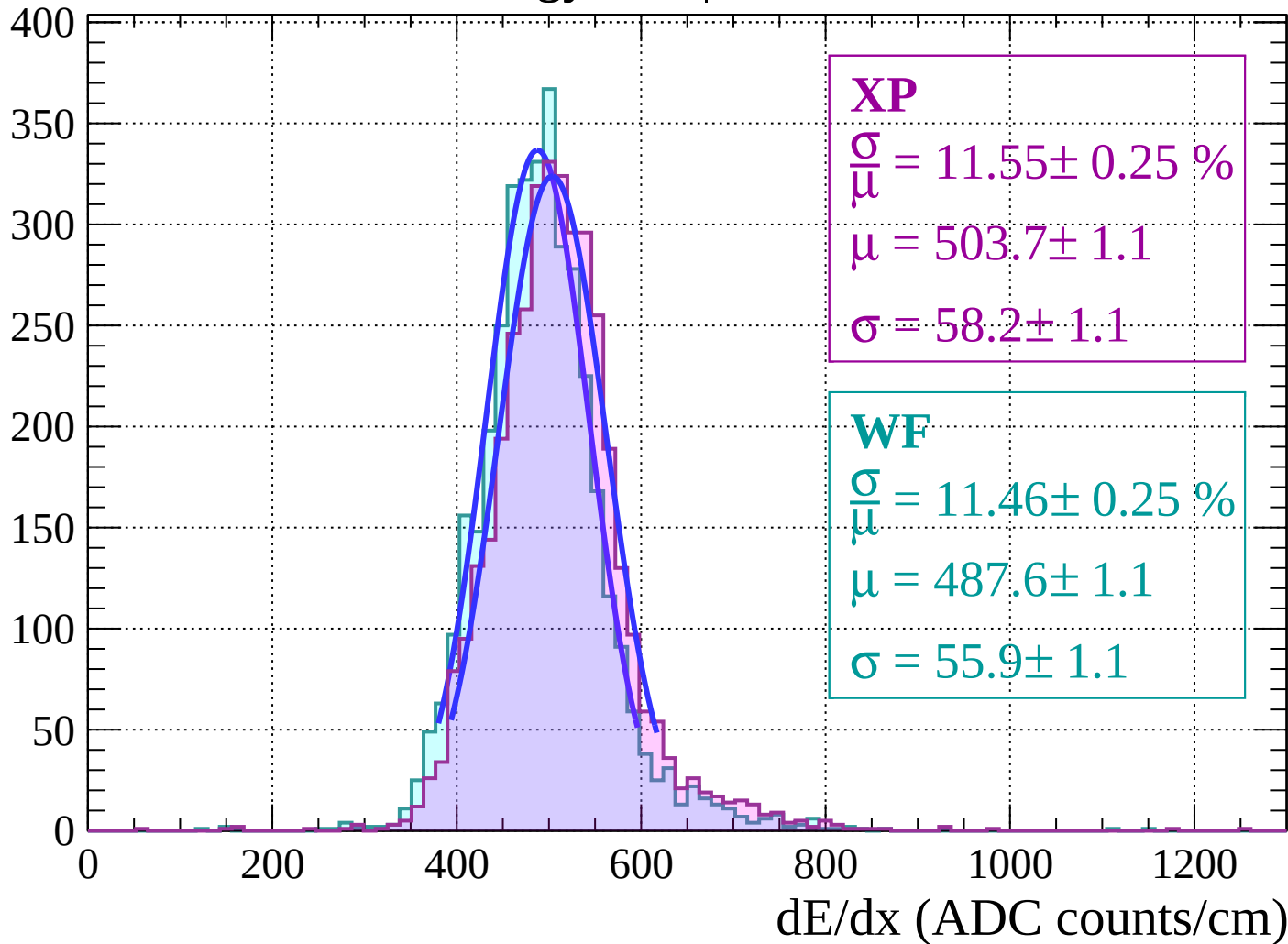


Energy loss $-4 < \theta < -2$

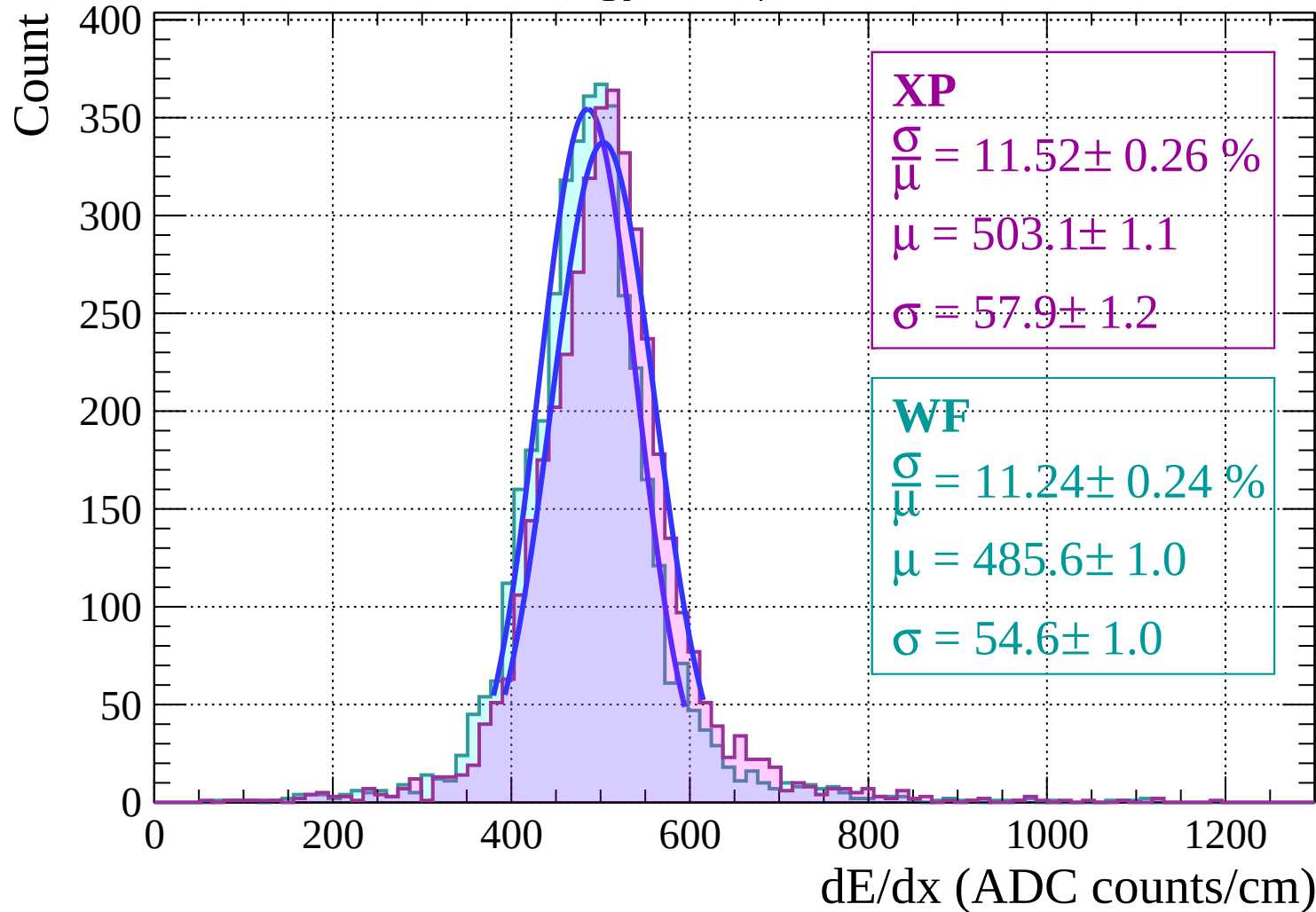


Energy loss | $-2 < \theta < 0$

Count



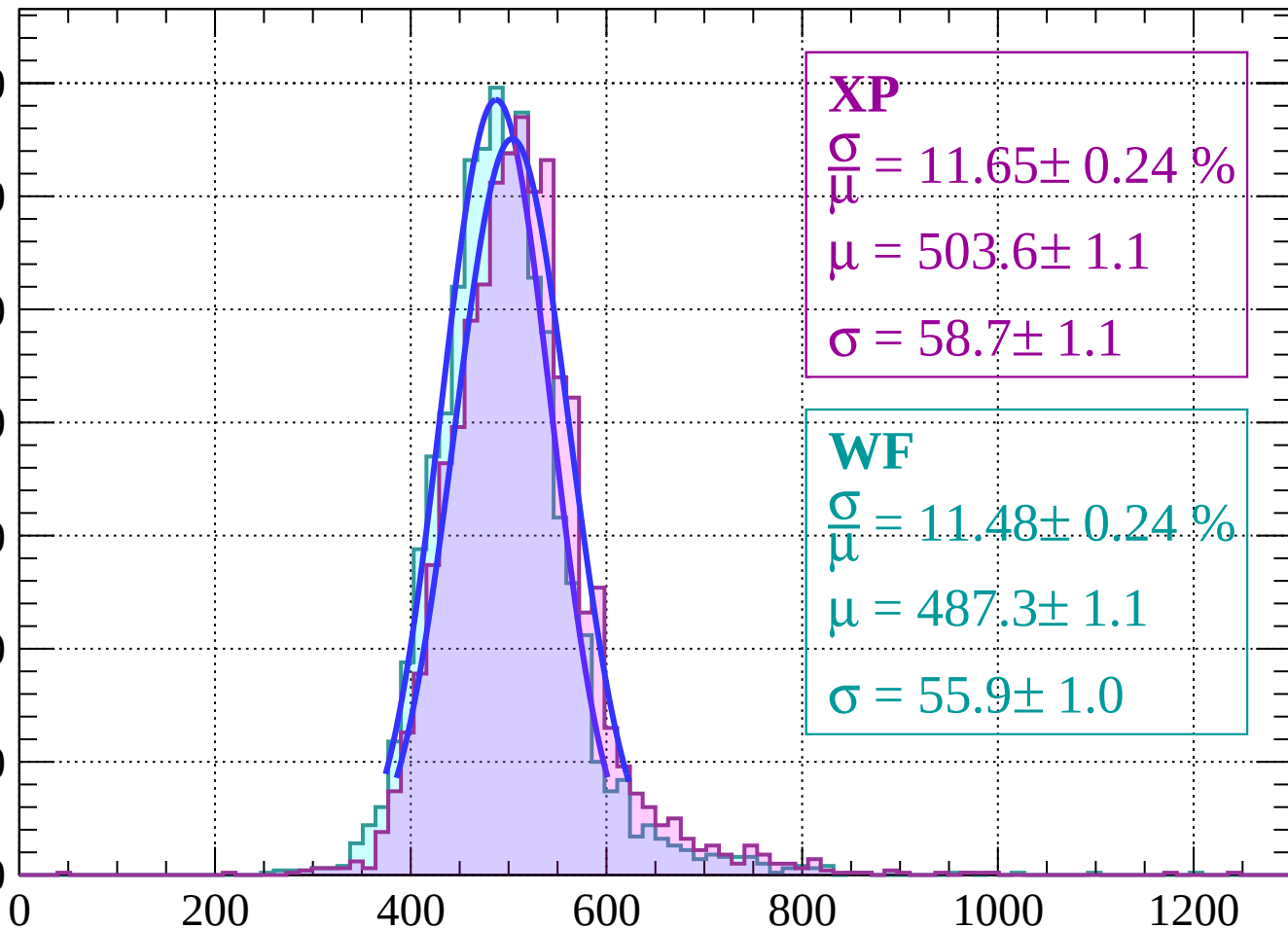
Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.65 \pm 0.24 \%$$

$$\mu = 503.6 \pm 1.1$$

$$\sigma = 58.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.48 \pm 0.24 \%$$

$$\mu = 487.3 \pm 1.1$$

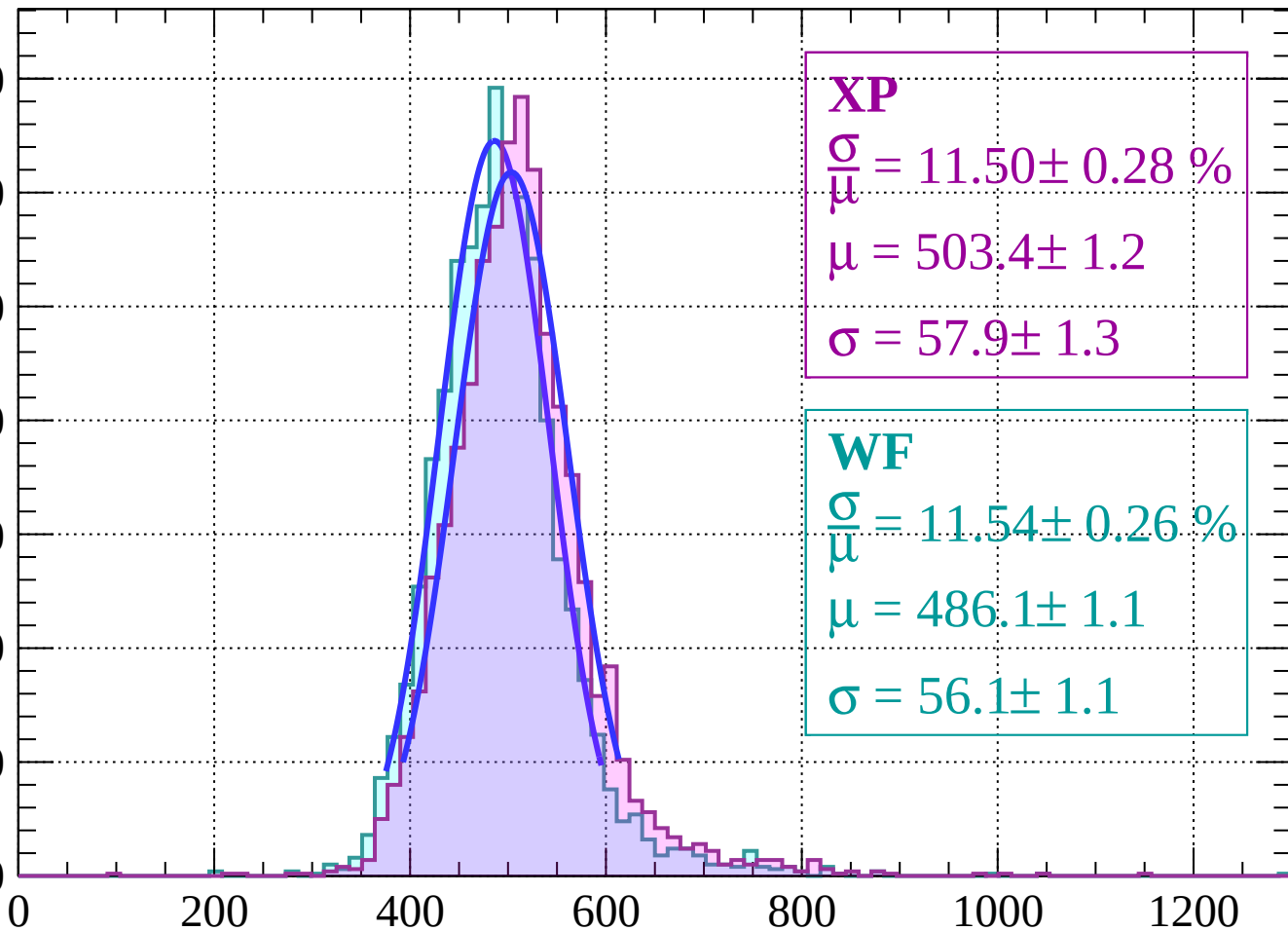
$$\sigma = 55.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$

Count

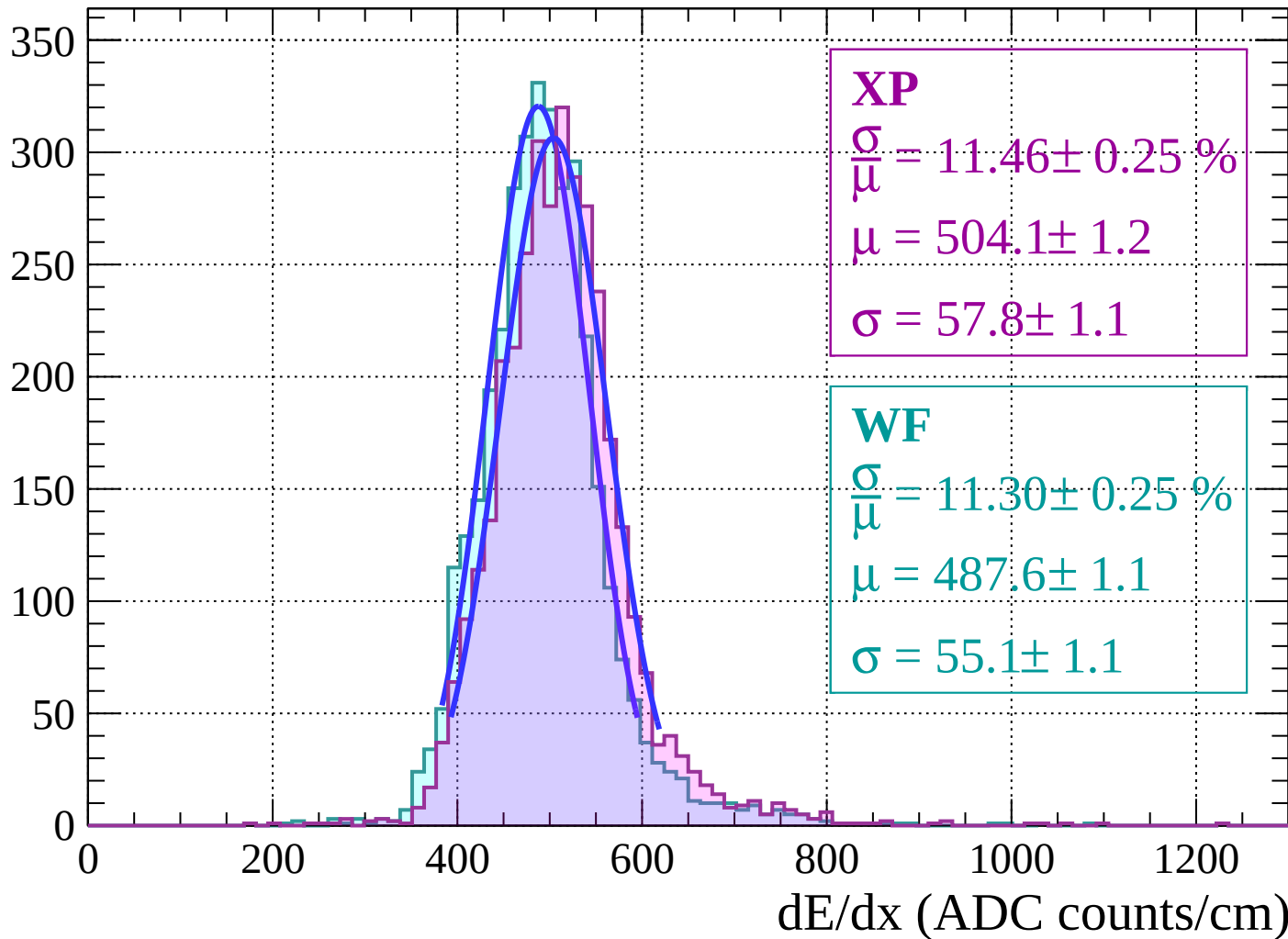
350
300
250
200
150
100
50
0



dE/dx (ADC counts/cm)

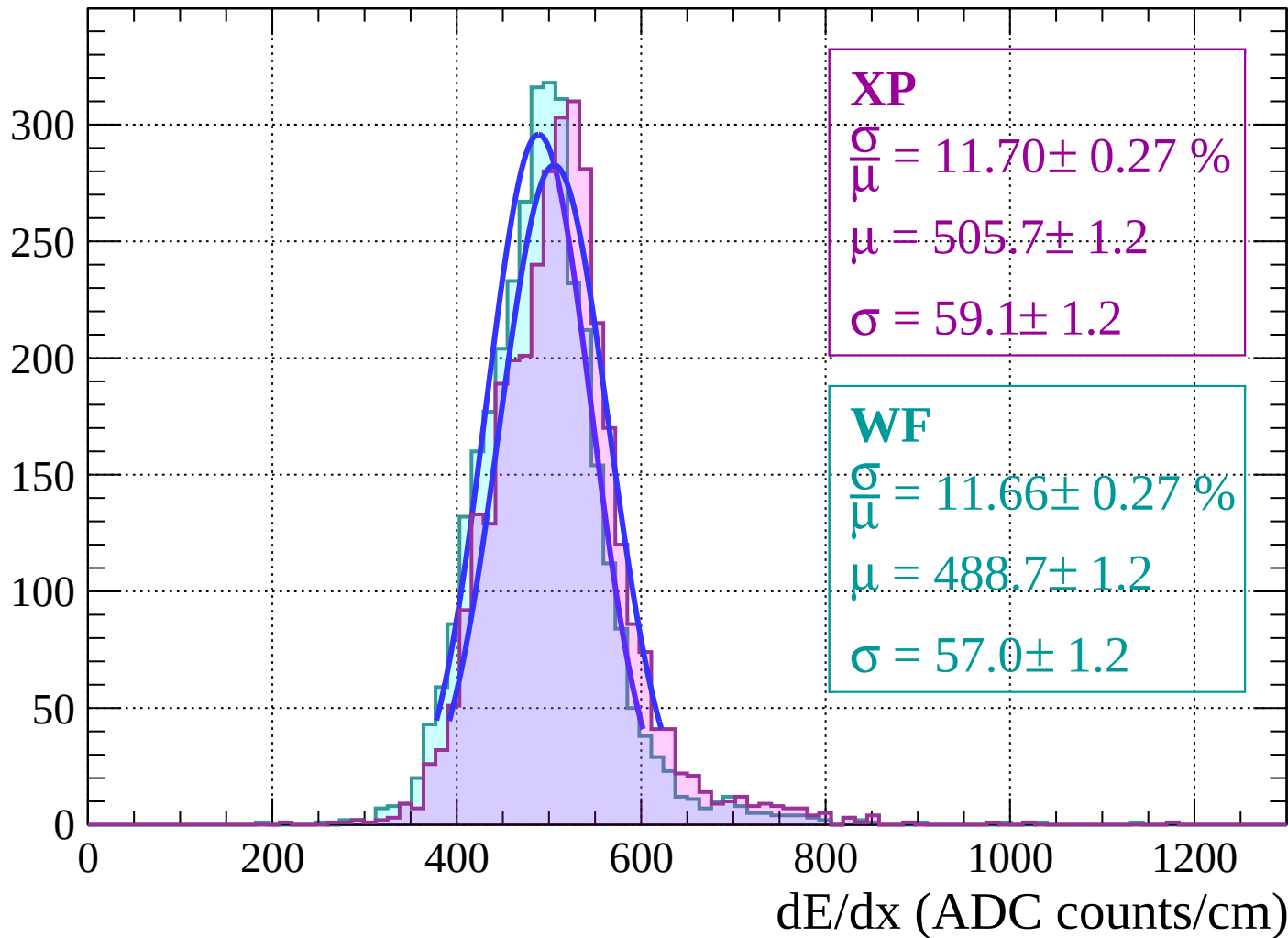
Energy loss | $6 < \theta < 8$

Count



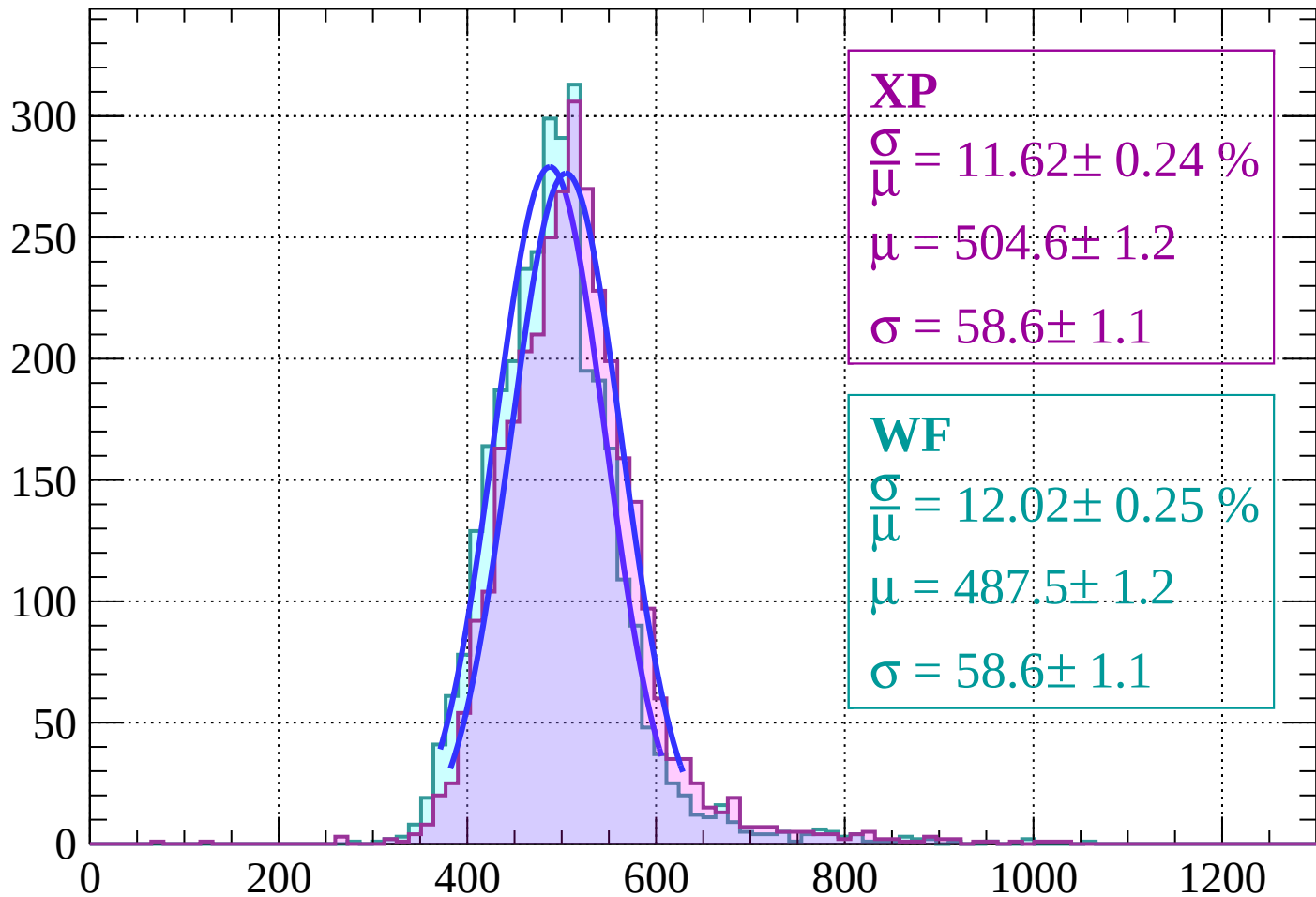
Energy loss | $8 < \theta < 10$

Count



Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 11.62 \pm 0.24 \%$$

$$\mu = 504.6 \pm 1.2$$

$$\sigma = 58.6 \pm 1.1$$

WF

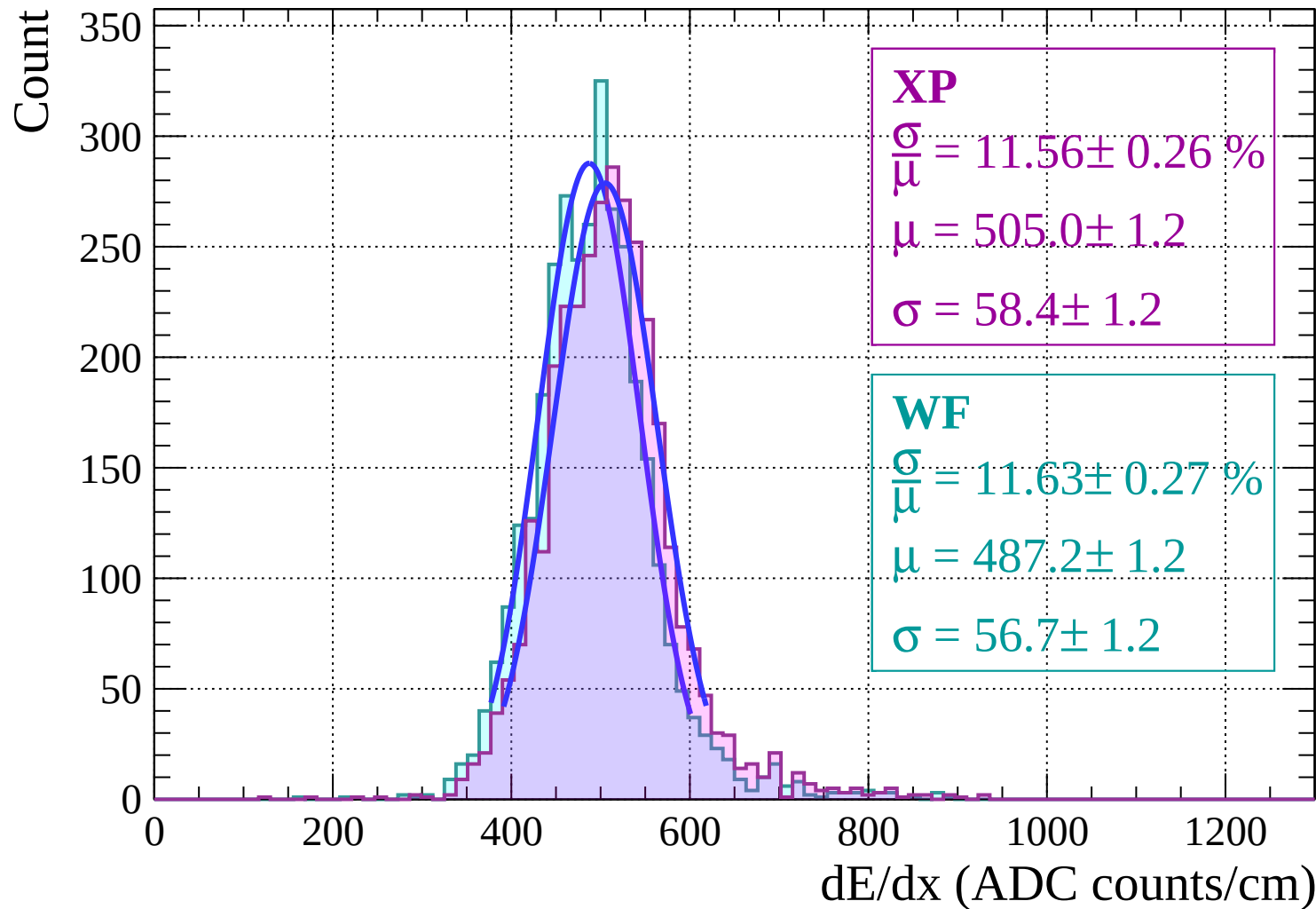
$$\frac{\sigma}{\mu} = 12.02 \pm 0.25 \%$$

$$\mu = 487.5 \pm 1.2$$

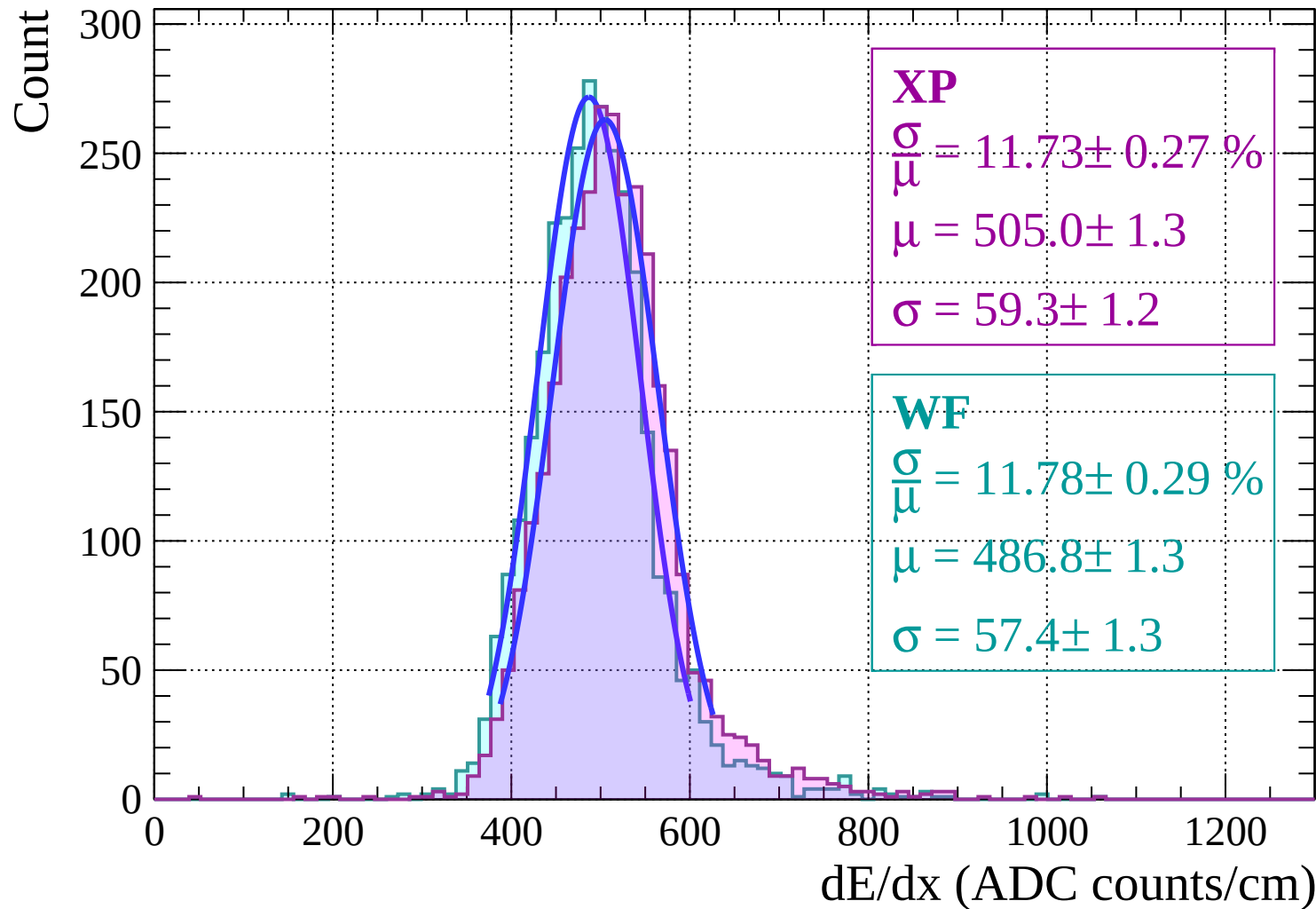
$$\sigma = 58.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

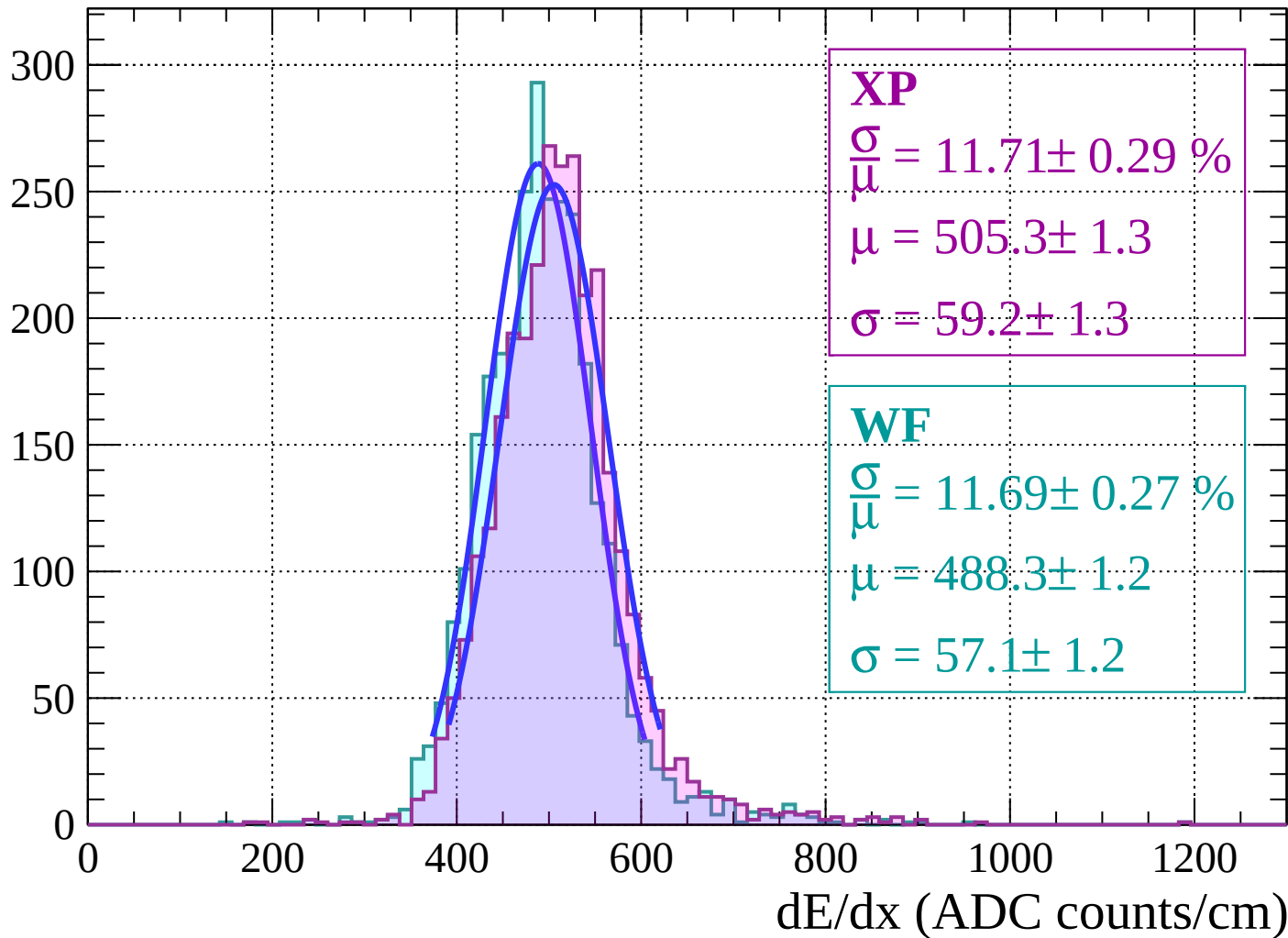


Energy loss | $14 < \theta < 16$



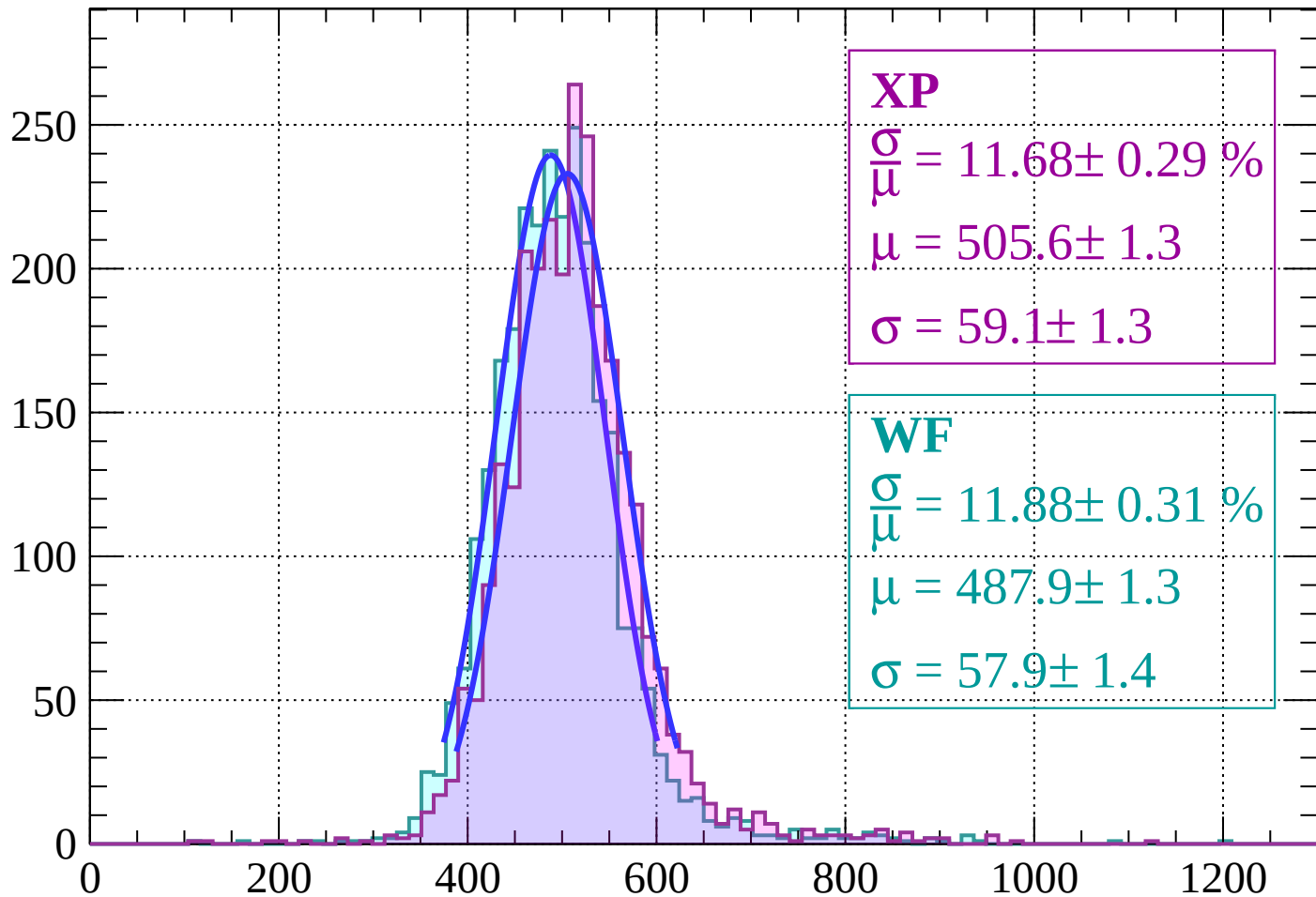
Energy loss | $16 < \theta < 18$

Count



Energy loss | $18 < \theta < 20$

Count



XP

$$\frac{\sigma}{\mu} = 11.68 \pm 0.29 \%$$

$$\mu = 505.6 \pm 1.3$$

$$\sigma = 59.1 \pm 1.3$$

WF

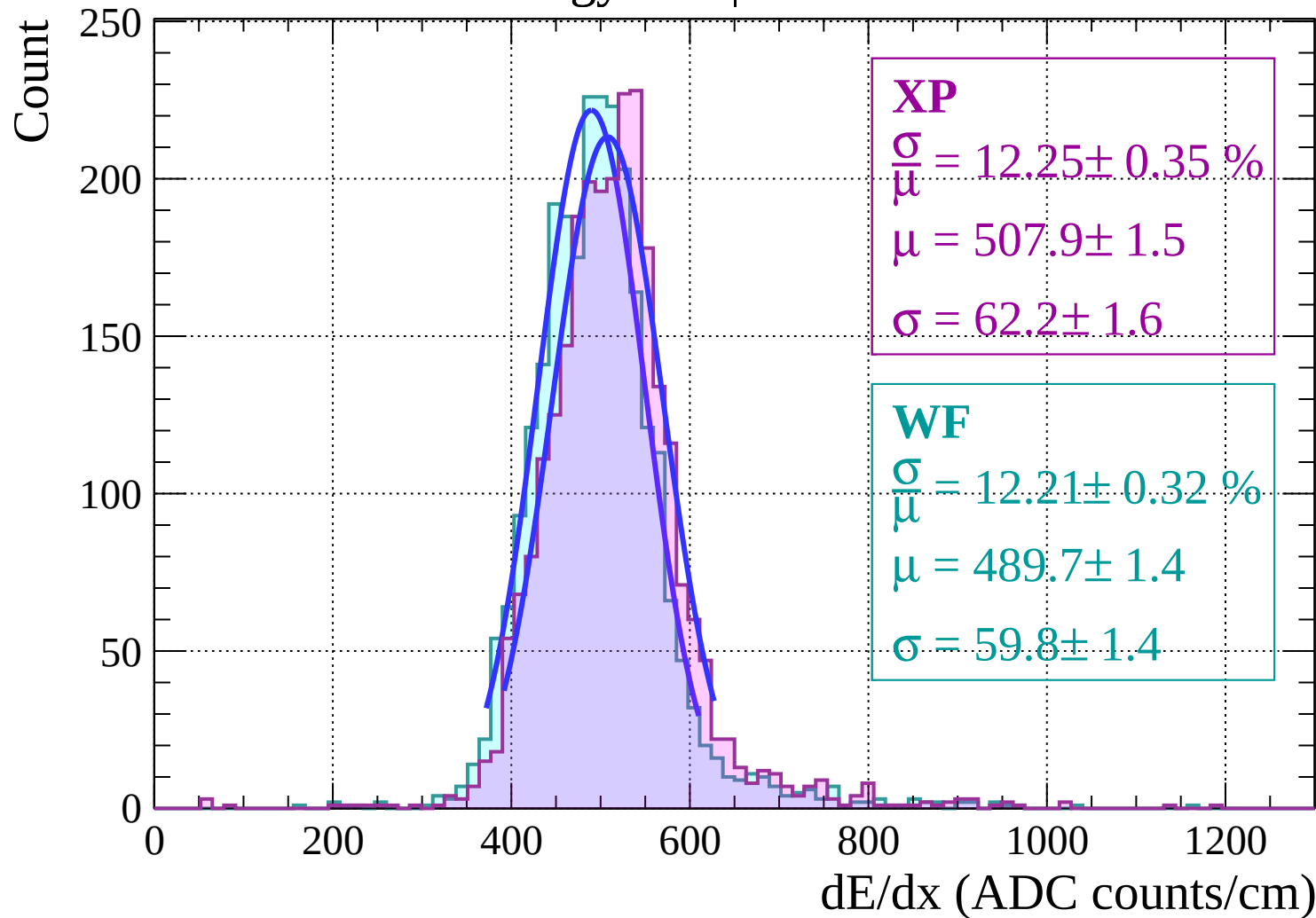
$$\frac{\sigma}{\mu} = 11.88 \pm 0.31 \%$$

$$\mu = 487.9 \pm 1.3$$

$$\sigma = 57.9 \pm 1.4$$

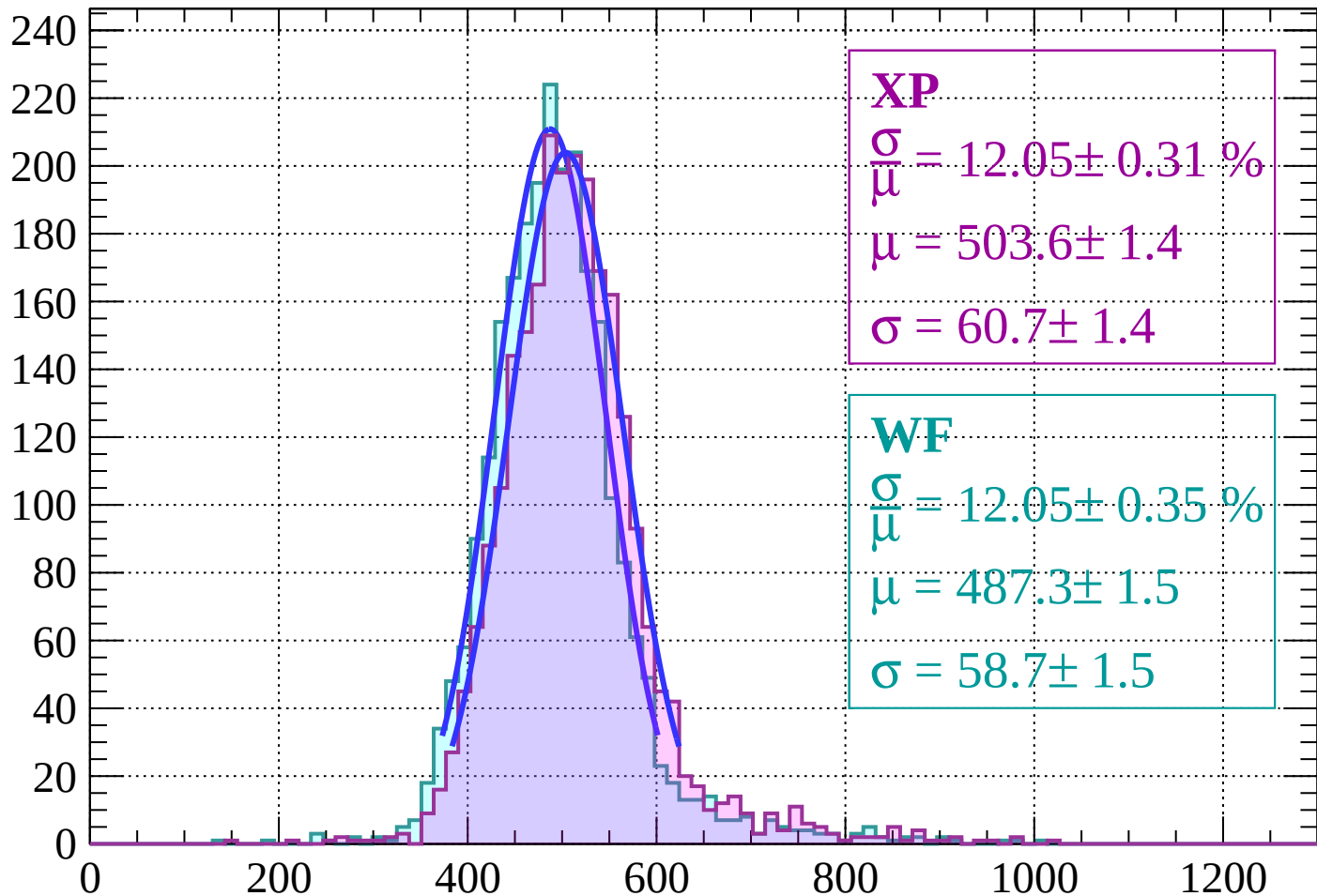
dE/dx (ADC counts/cm)

Energy loss | $20 < \theta < 22$



Energy loss | $22 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 12.05 \pm 0.31 \%$$

$$\mu = 503.6 \pm 1.4$$

$$\sigma = 60.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 12.05 \pm 0.35 \%$$

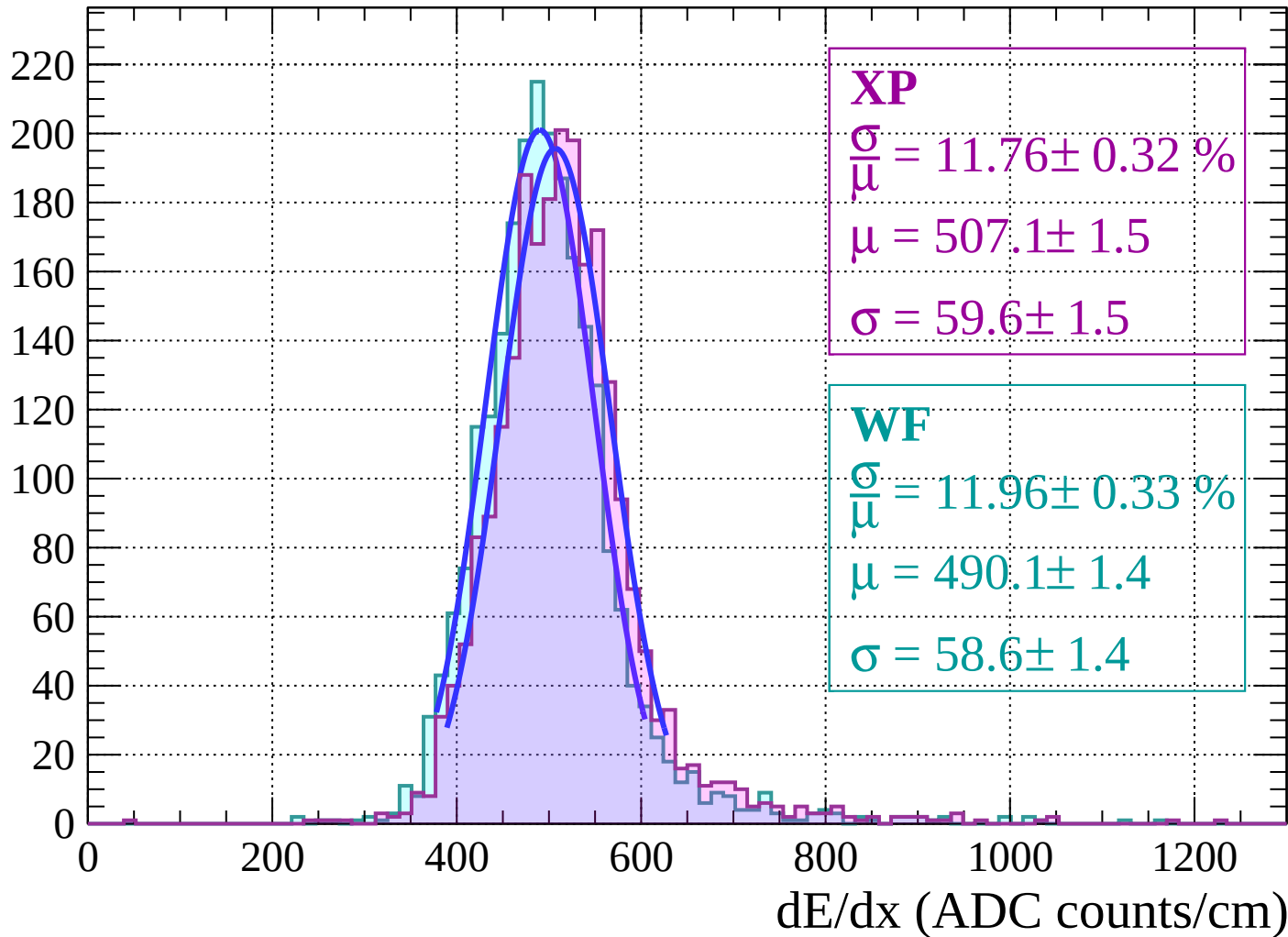
$$\mu = 487.3 \pm 1.5$$

$$\sigma = 58.7 \pm 1.5$$

dE/dx (ADC counts/cm)

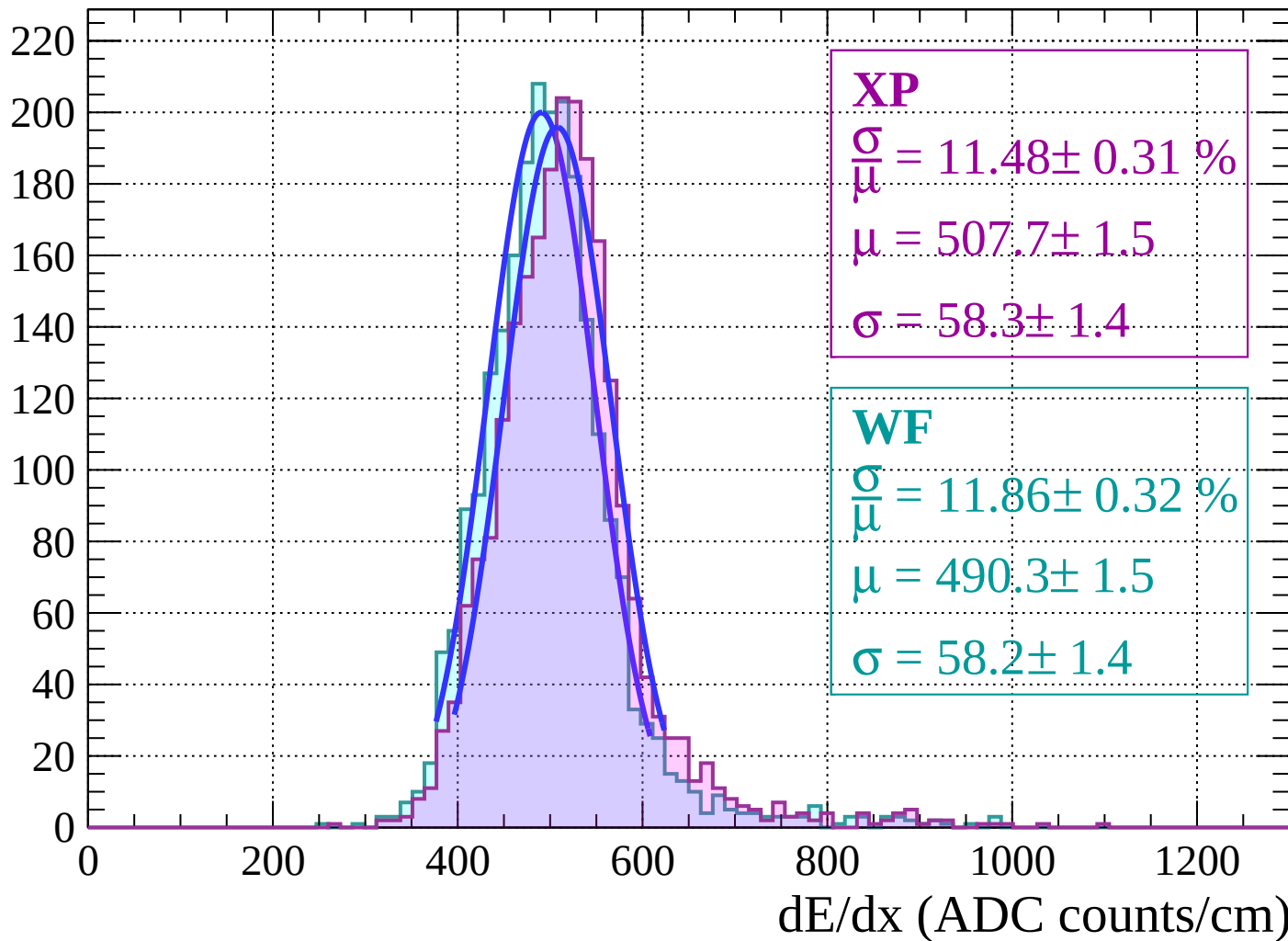
Energy loss | $24 < \theta < 26$

Count

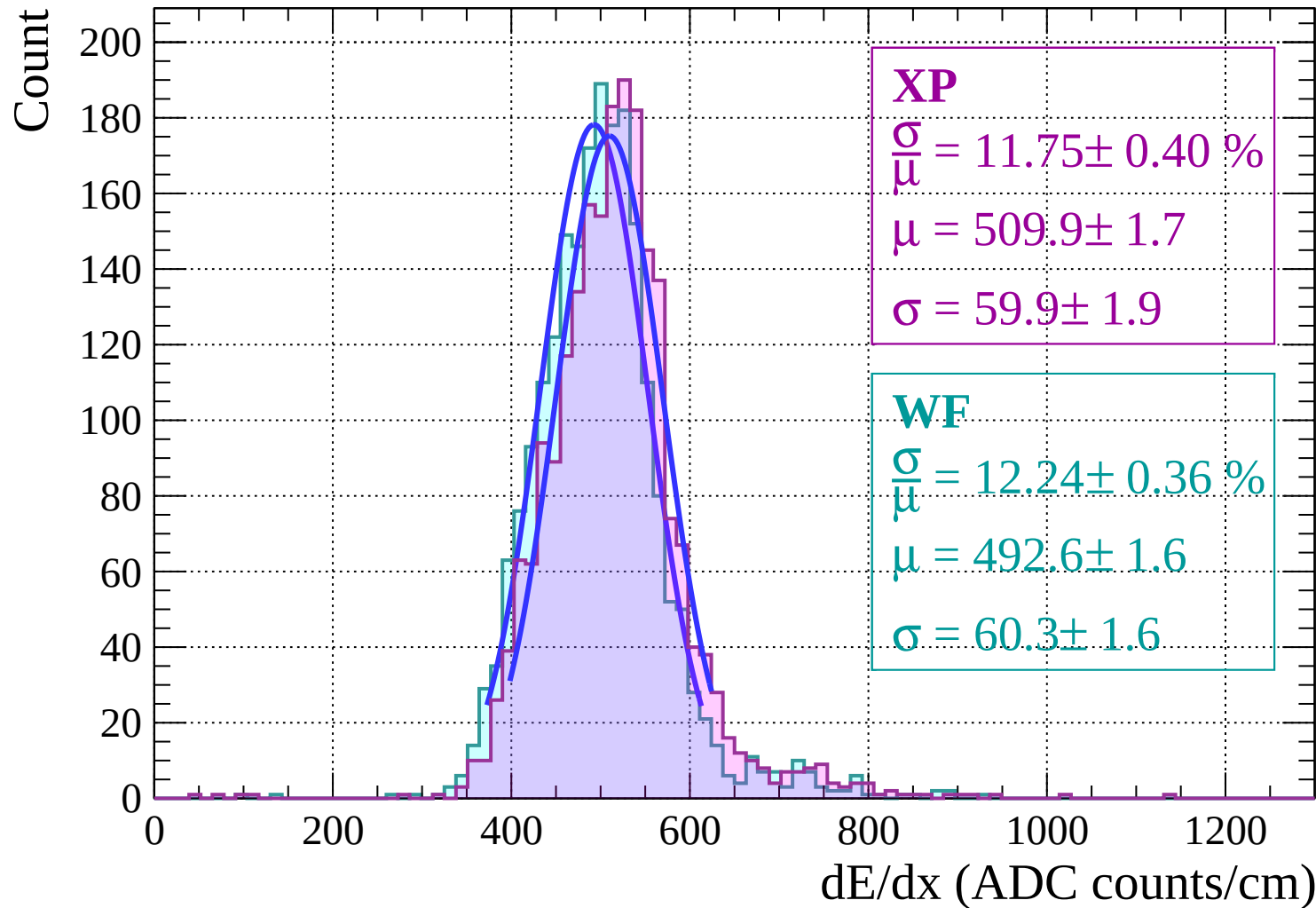


Energy loss | $26 < \theta < 28$

Count



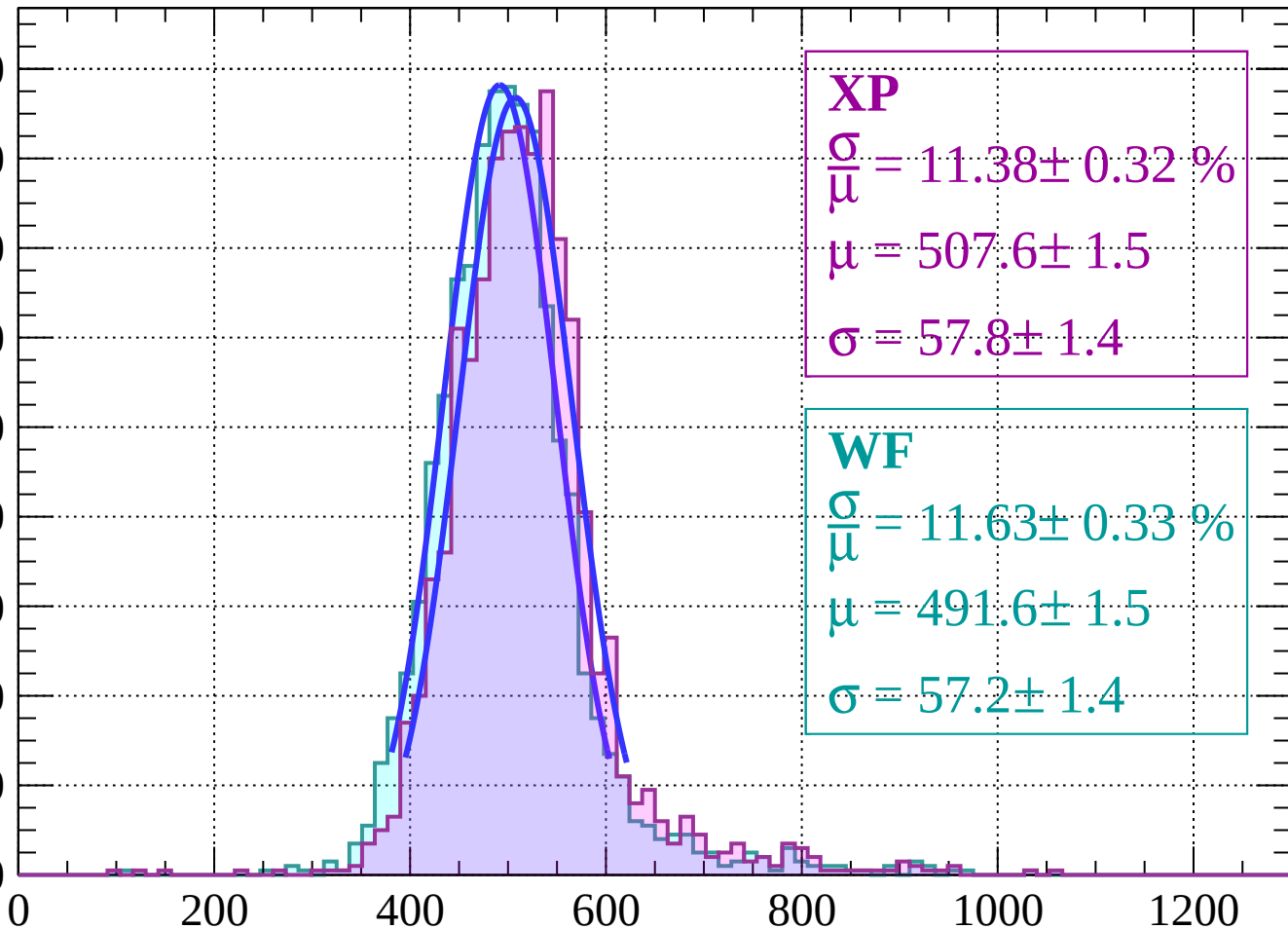
Energy loss | $28 < \theta < 30$



Energy loss | $30 < \theta < 32$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\bar{\mu}} = 11.38 \pm 0.32 \%$$

$$\mu = 507.6 \pm 1.5$$

$$\sigma = 57.8 \pm 1.4$$

WF

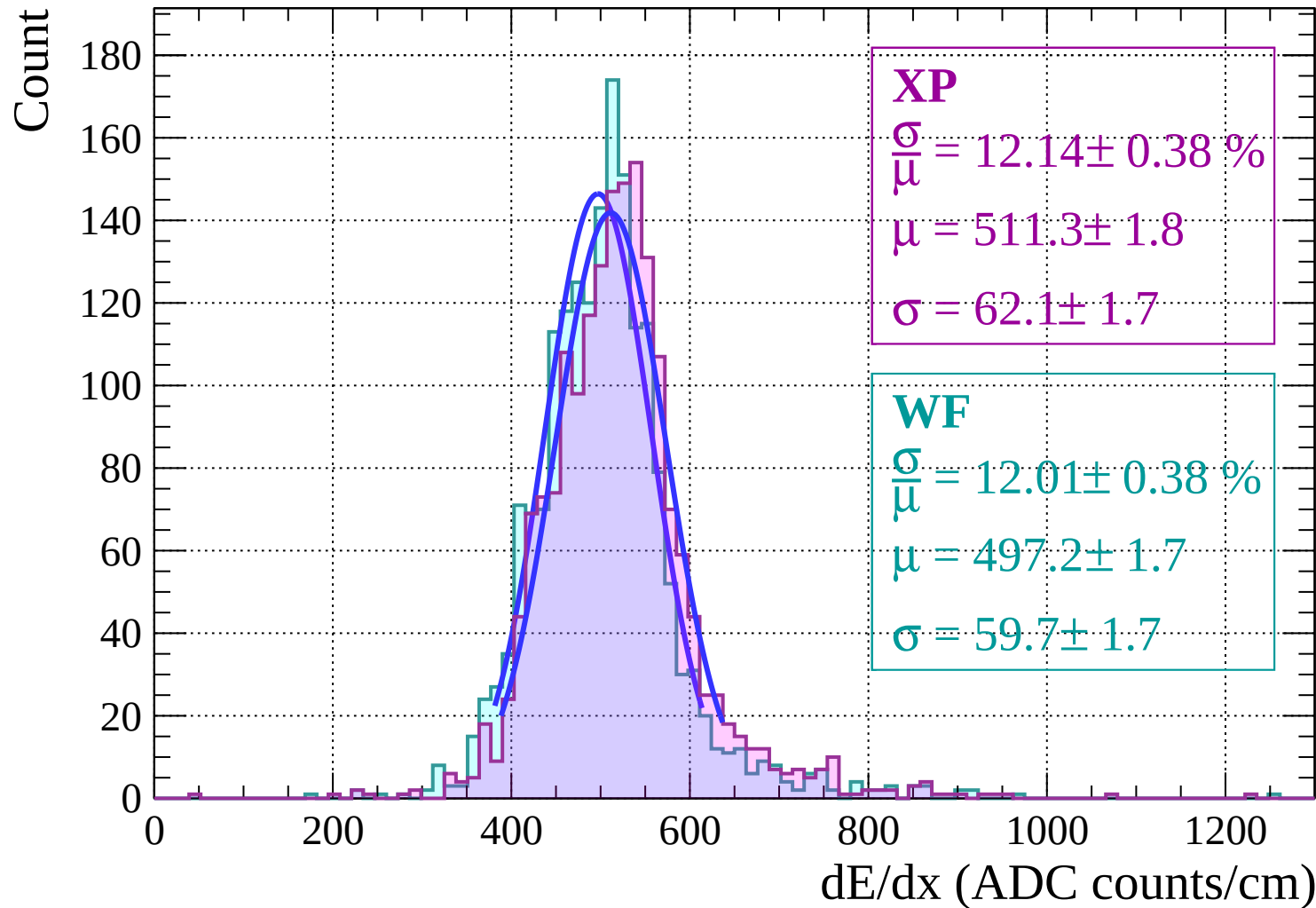
$$\frac{\sigma}{\bar{\mu}} = 11.63 \pm 0.33 \%$$

$$\mu = 491.6 \pm 1.5$$

$$\sigma = 57.2 \pm 1.4$$

dE/dx (ADC counts/cm)

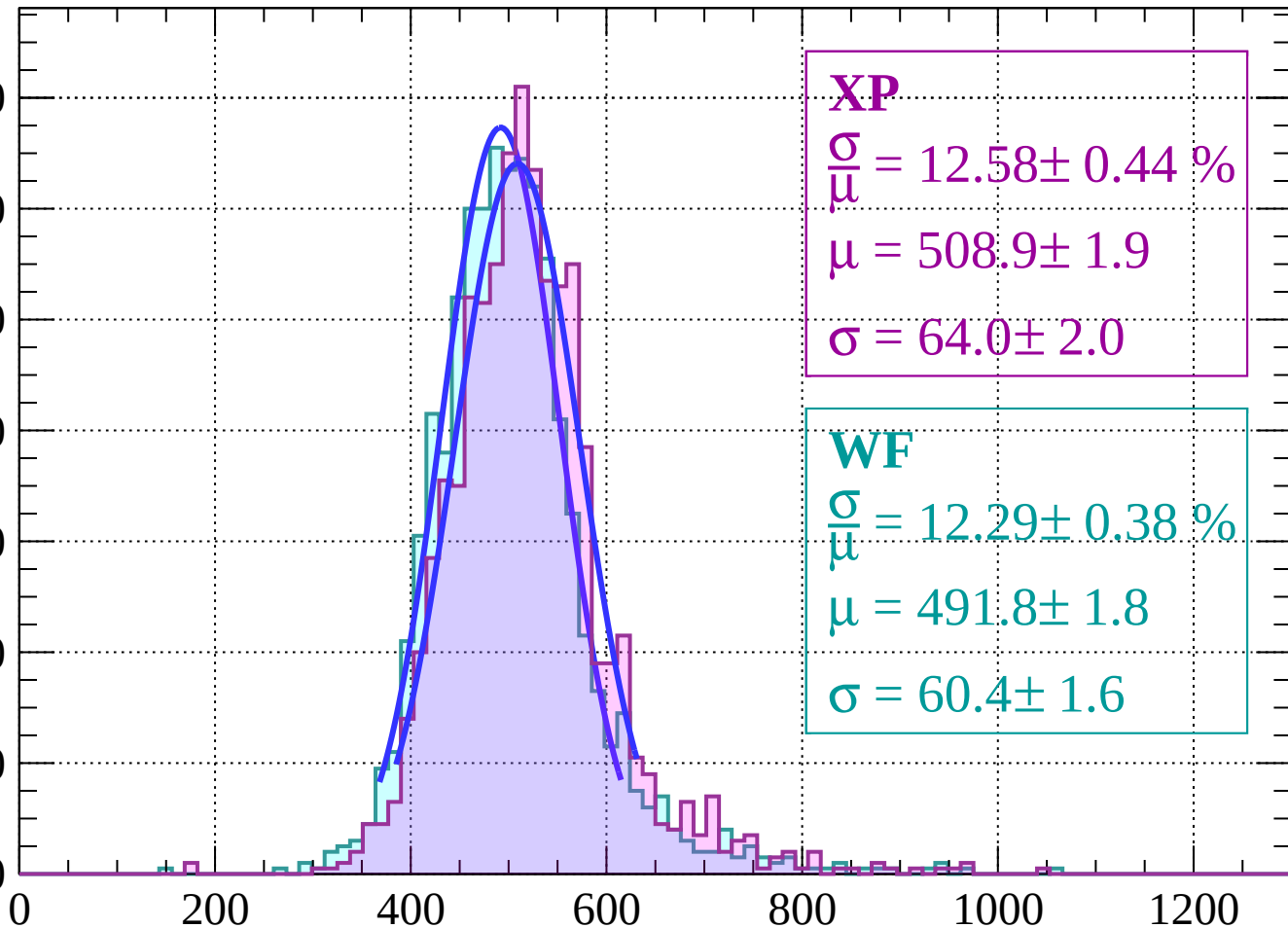
Energy loss | $32 < \theta < 34$



Energy loss | $34 < \theta < 36$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 12.58 \pm 0.44 \%$$

$$\mu = 508.9 \pm 1.9$$

$$\sigma = 64.0 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 12.29 \pm 0.38 \%$$

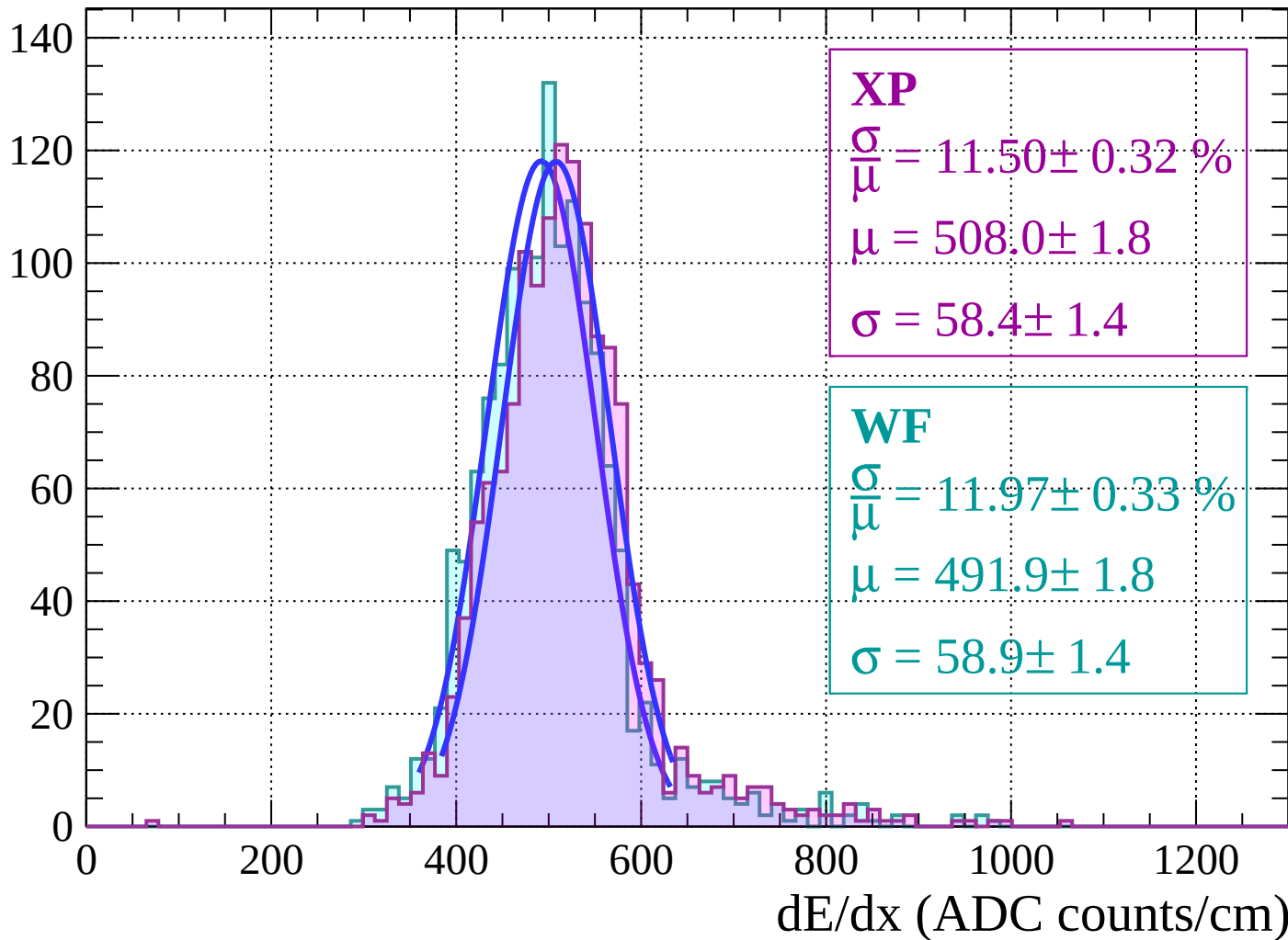
$$\mu = 491.8 \pm 1.8$$

$$\sigma = 60.4 \pm 1.6$$

dE/dx (ADC counts/cm)

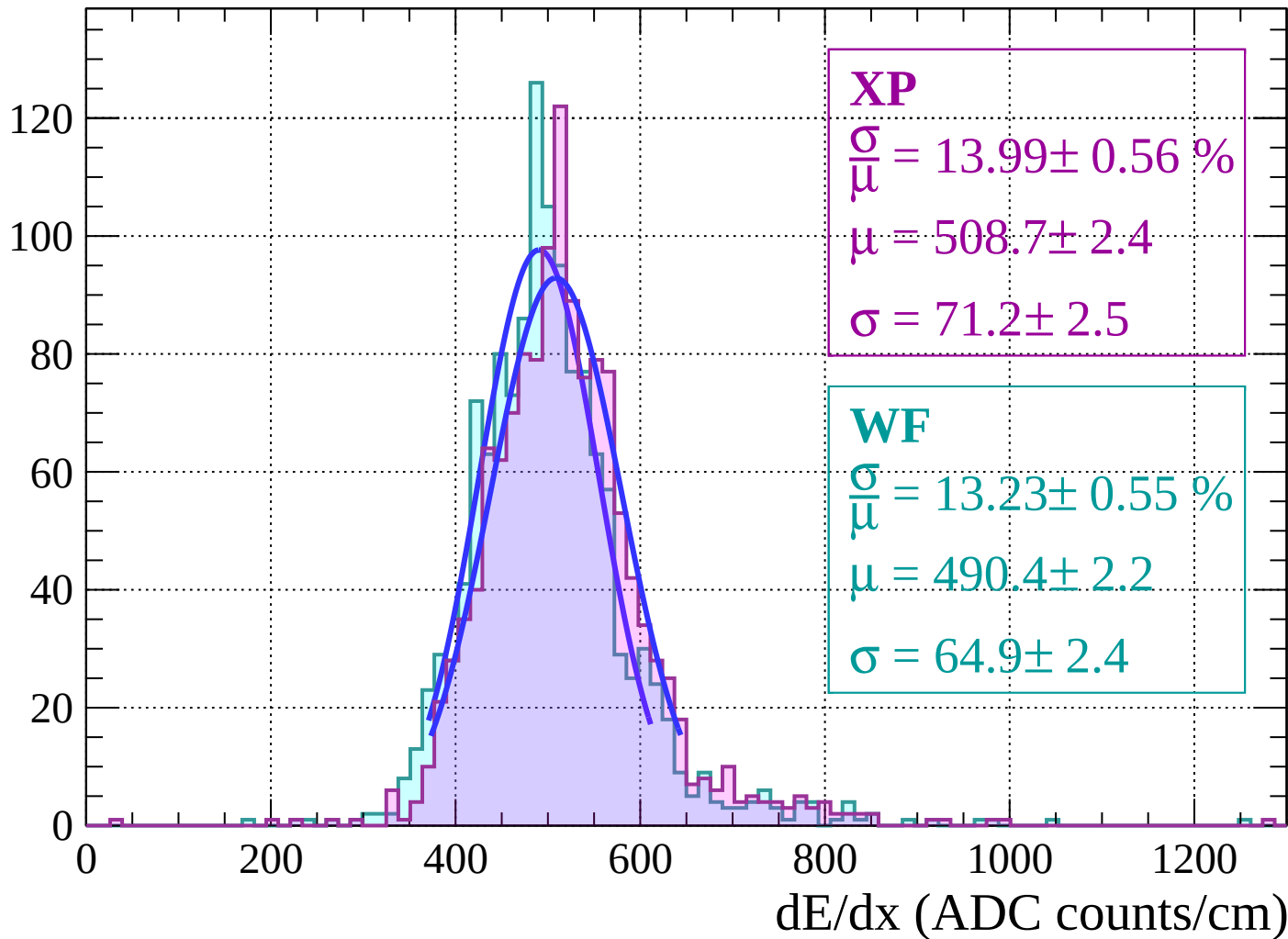
Energy loss | $36 < \theta < 38$

Count



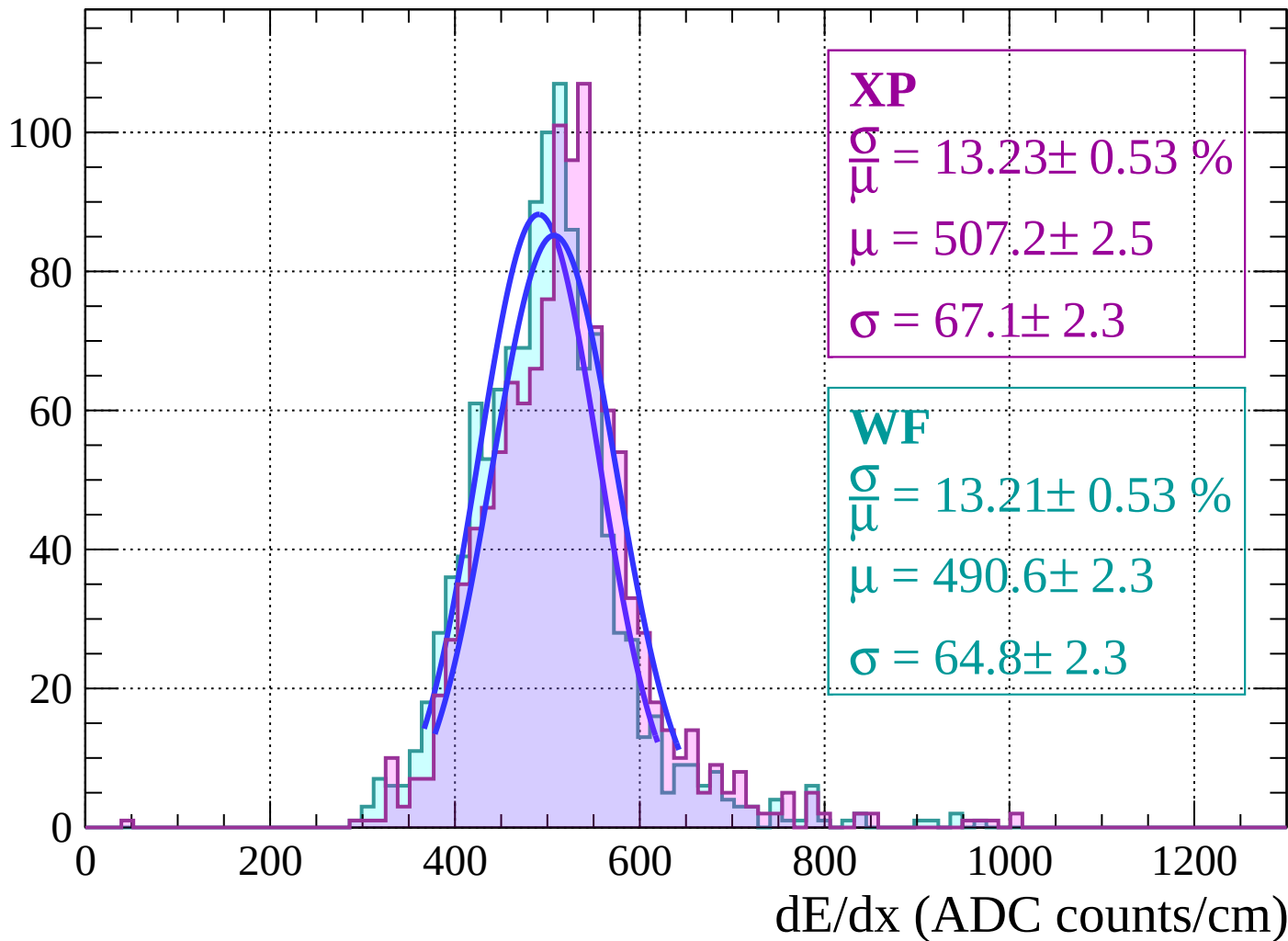
Energy loss | $38 < \theta < 40$

Count



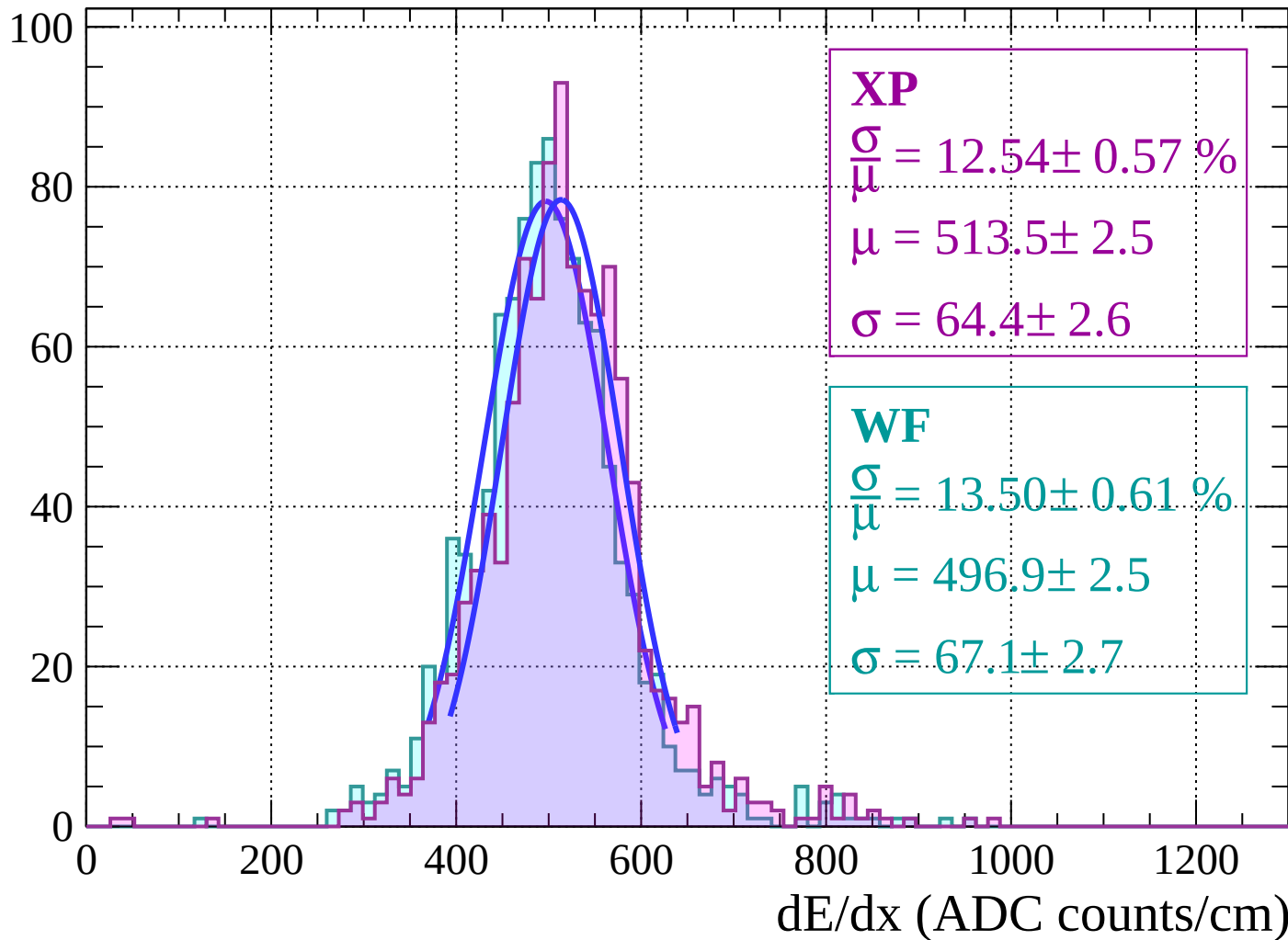
Energy loss | $40 < \theta < 42$

Count



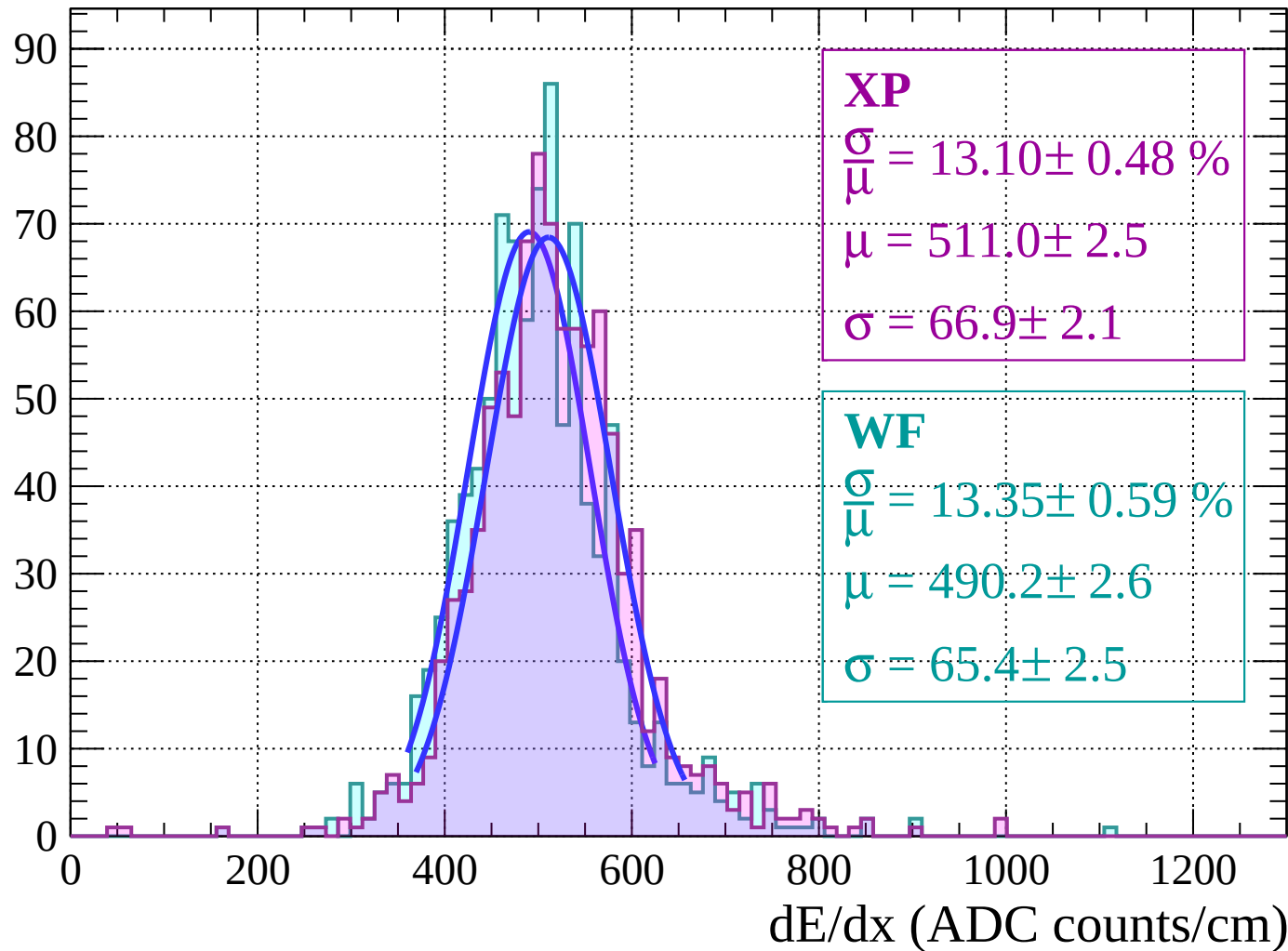
Energy loss | $42 < \theta < 44$

Count



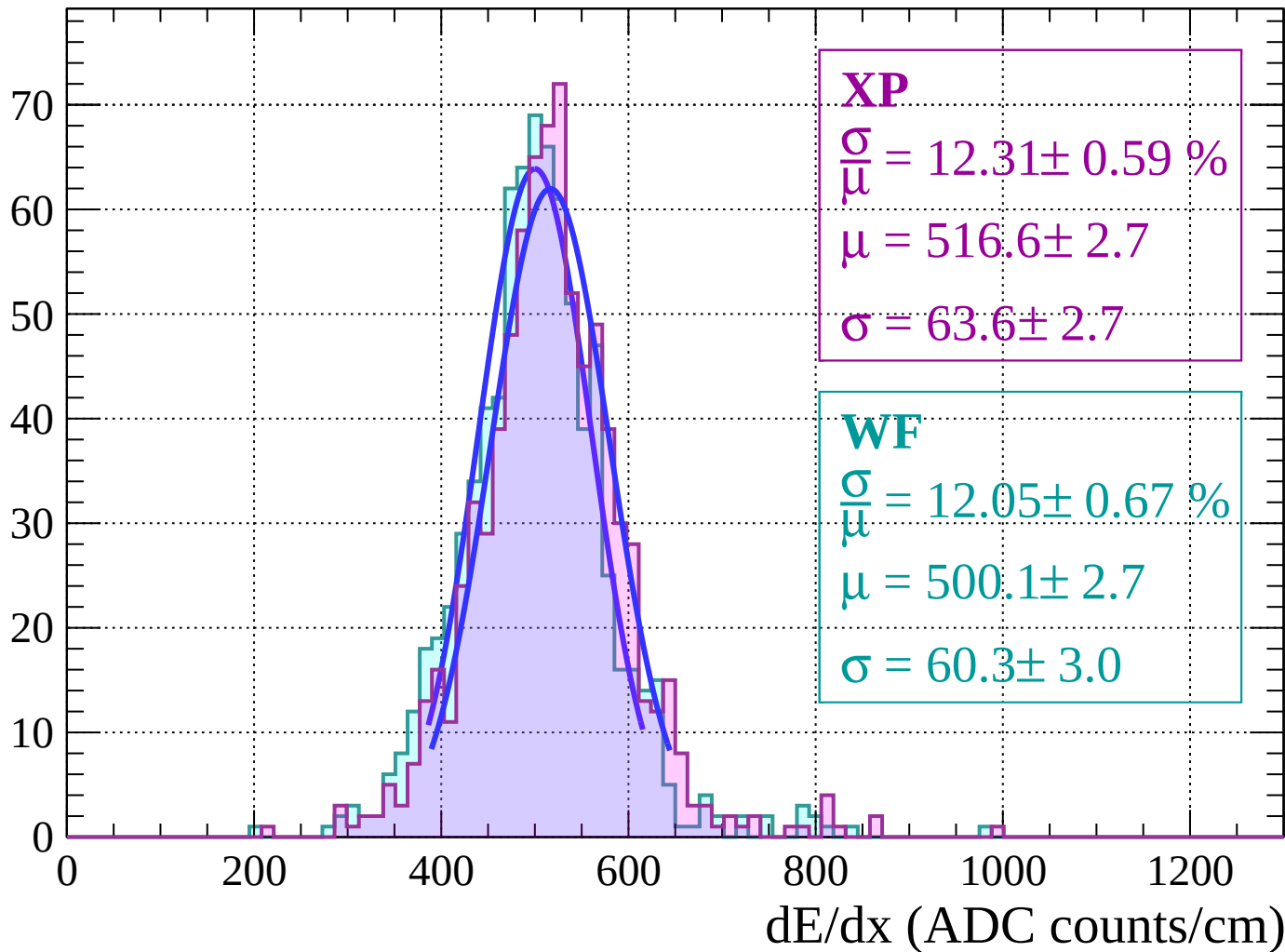
Energy loss | $44 < \theta < 46$

Count



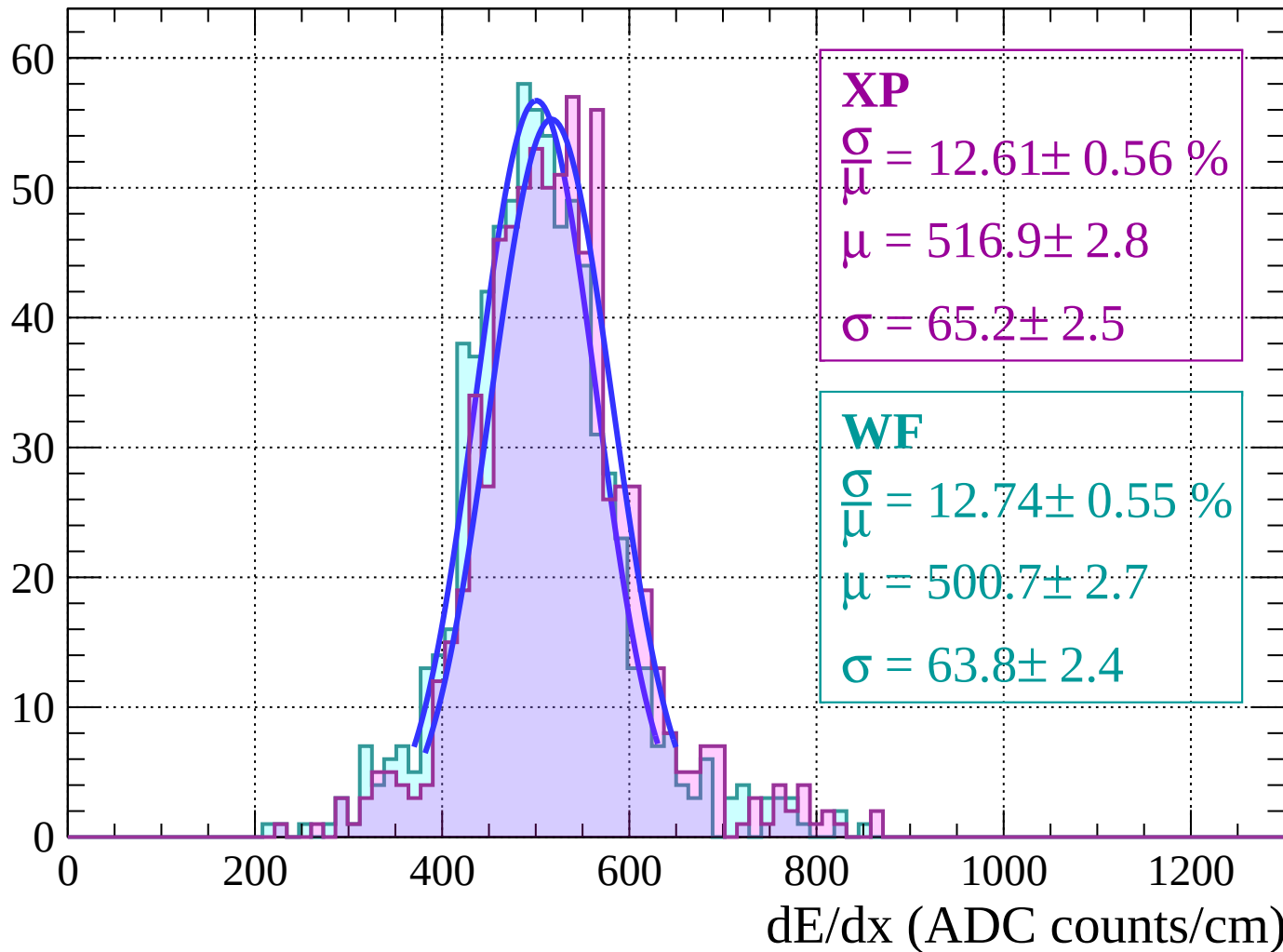
Energy loss | $46 < \theta < 48$

Count



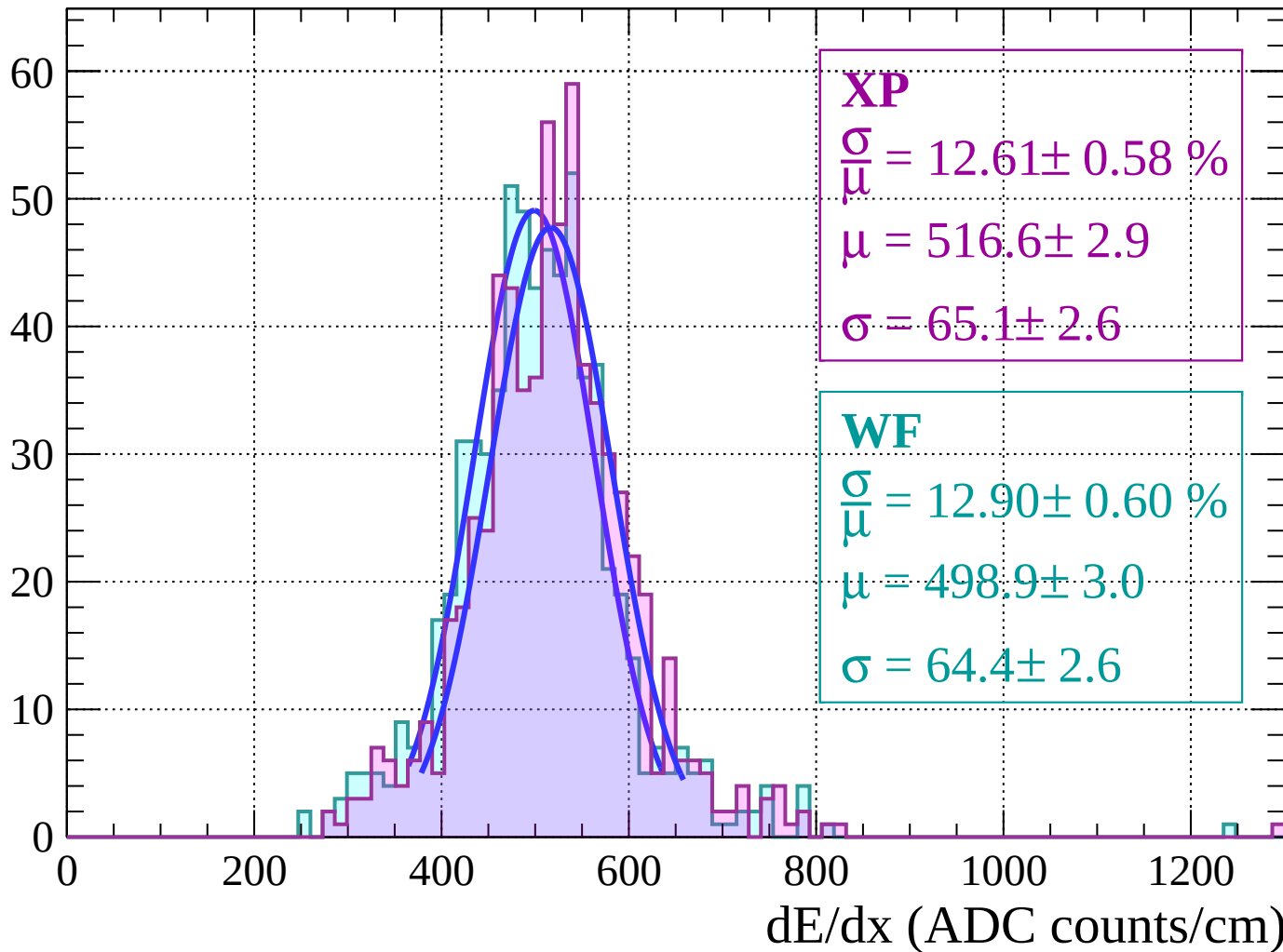
Energy loss | $48 < \theta < 50$

Count



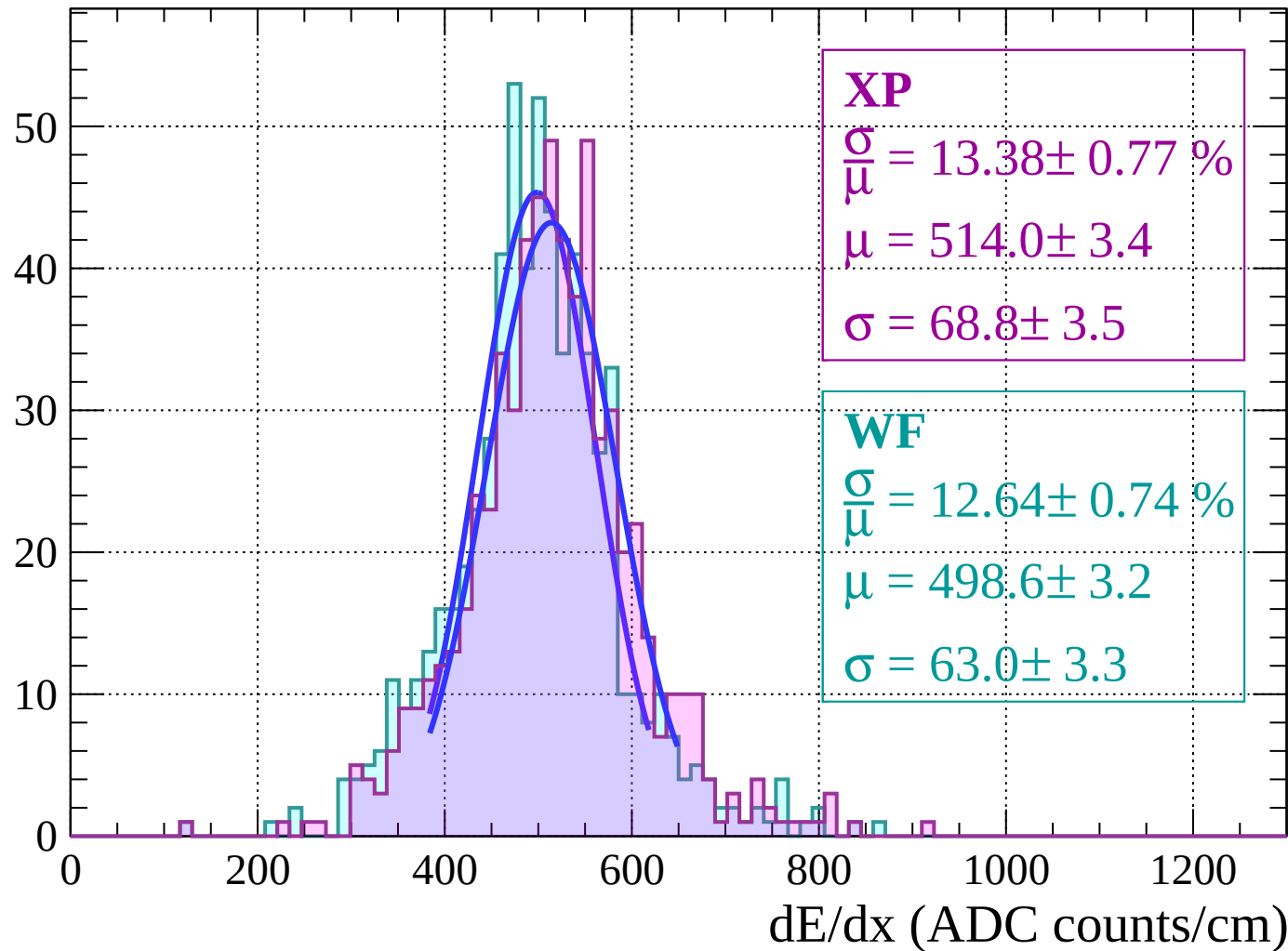
Energy loss | $50 < \theta < 52$

Count



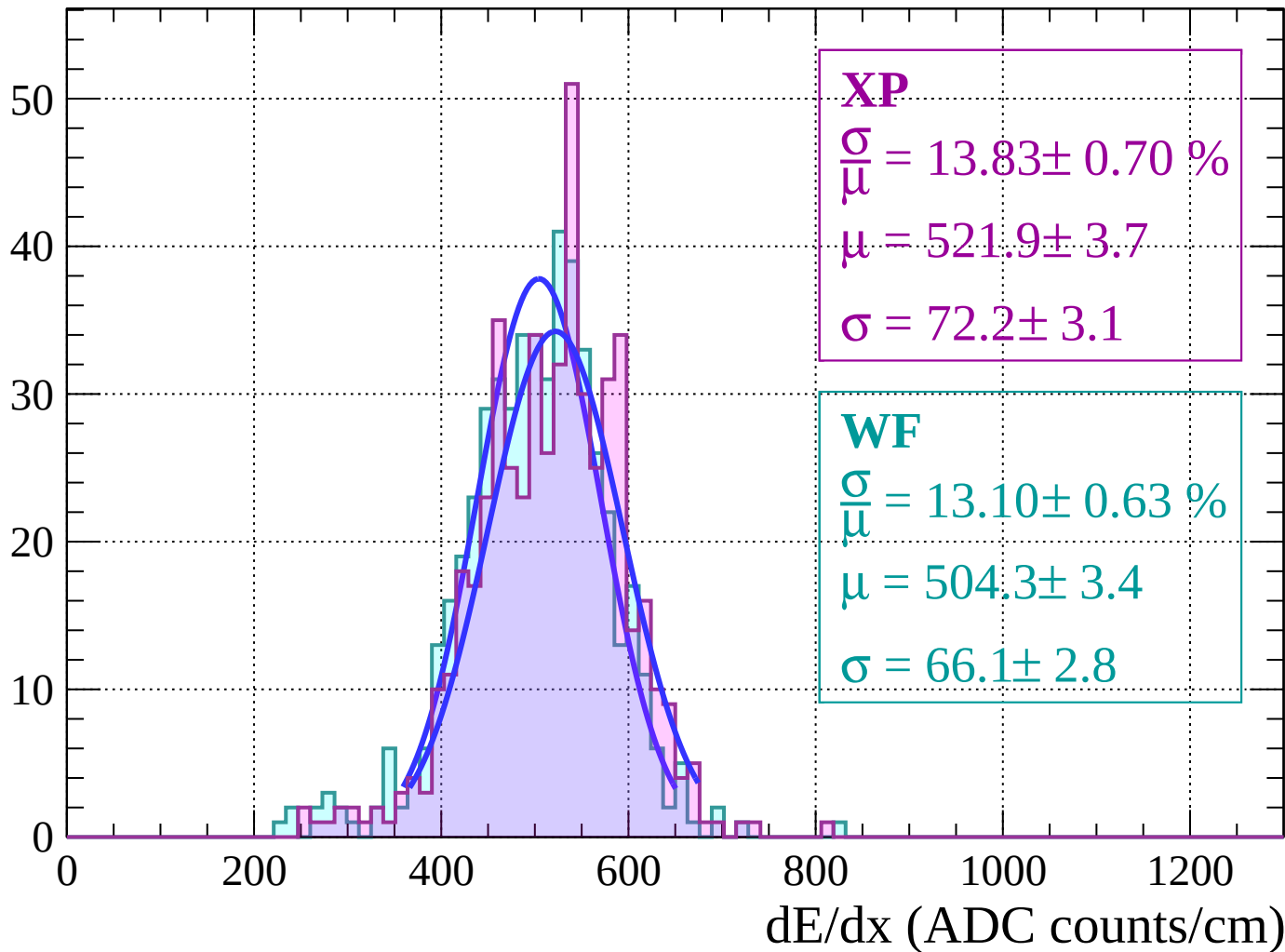
Energy loss | $52 < \theta < 54$

Count



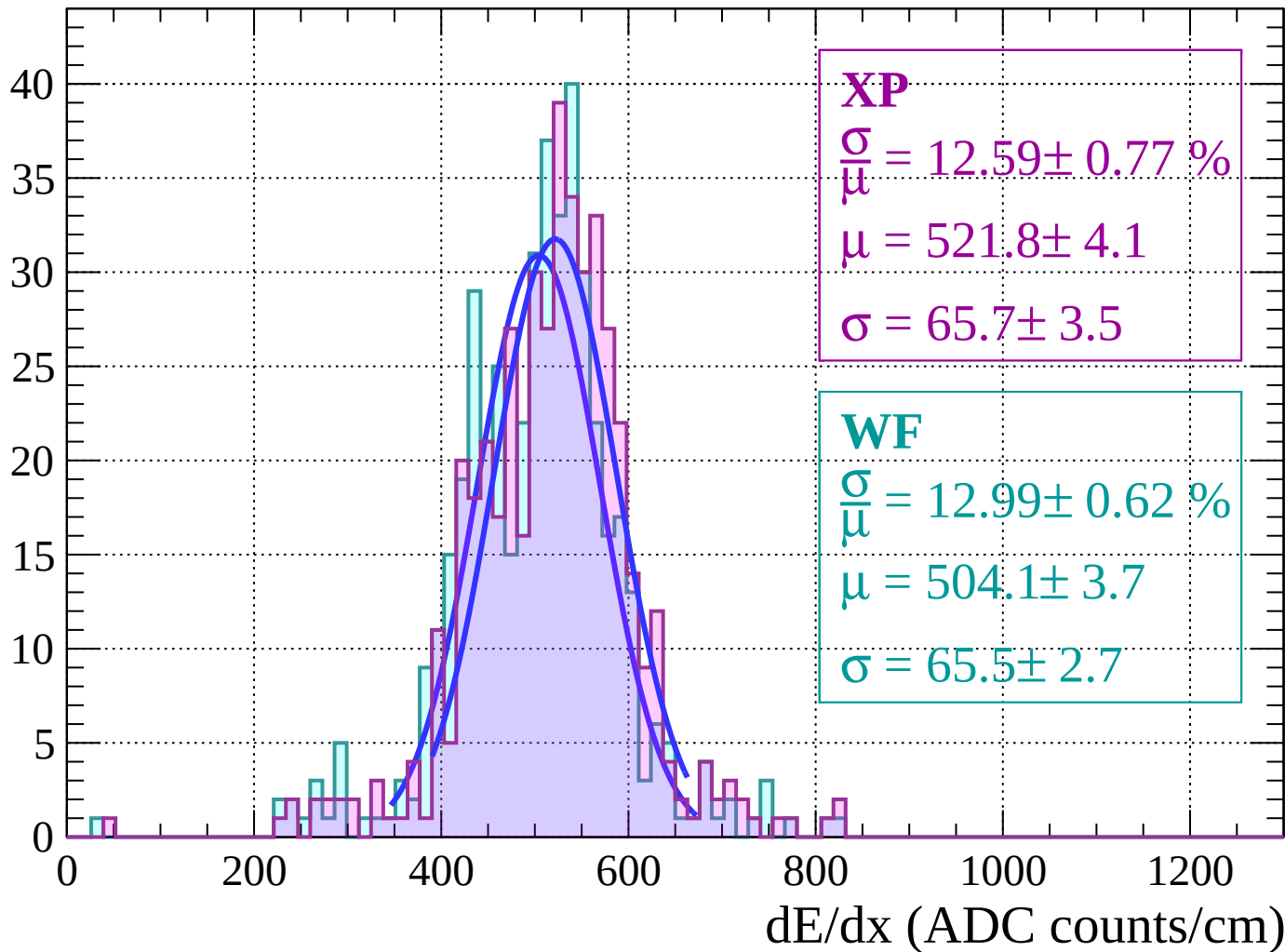
Energy loss | $54 < \theta < 56$

Count



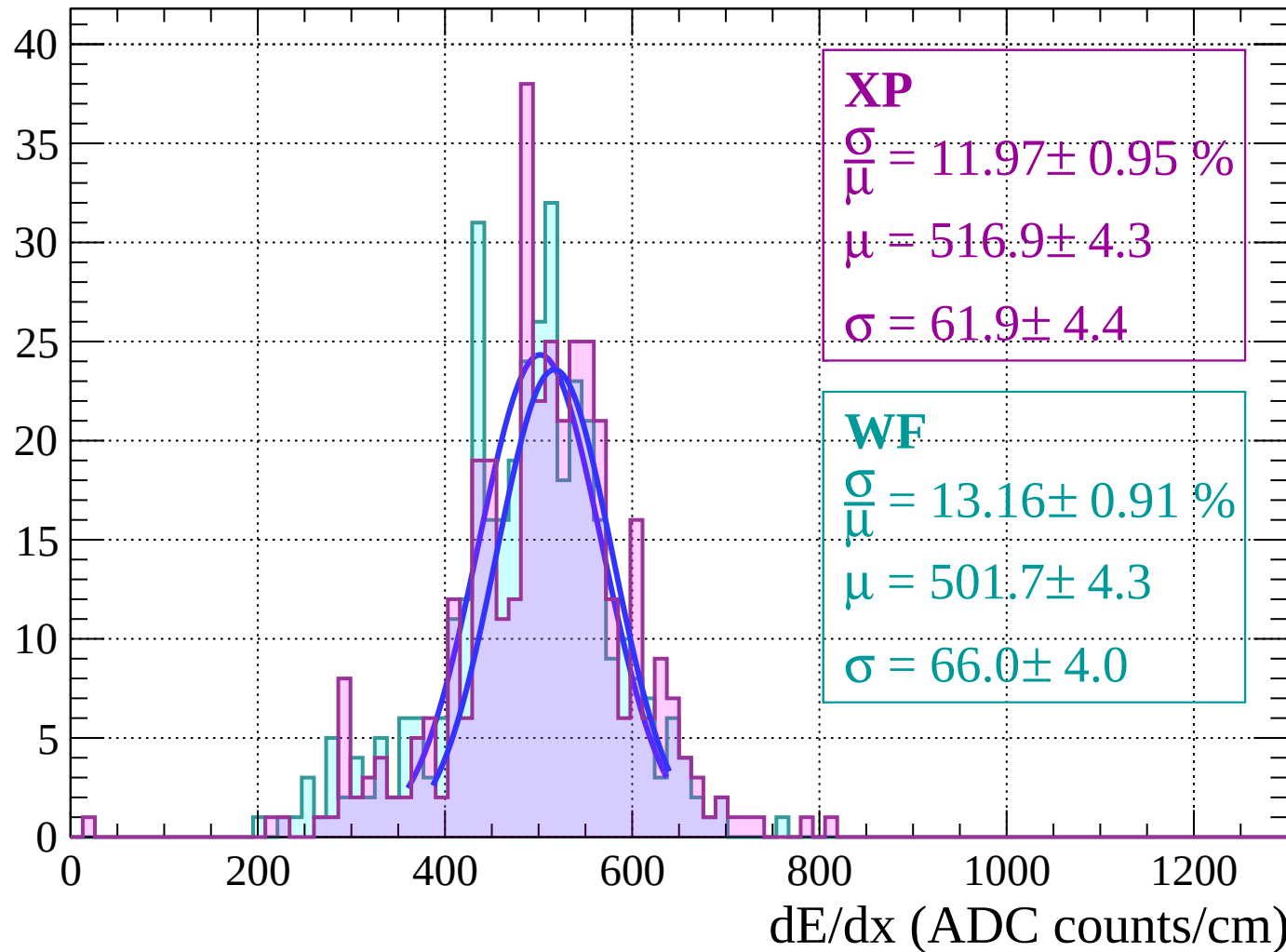
Energy loss | $56 < \theta < 58$

Count



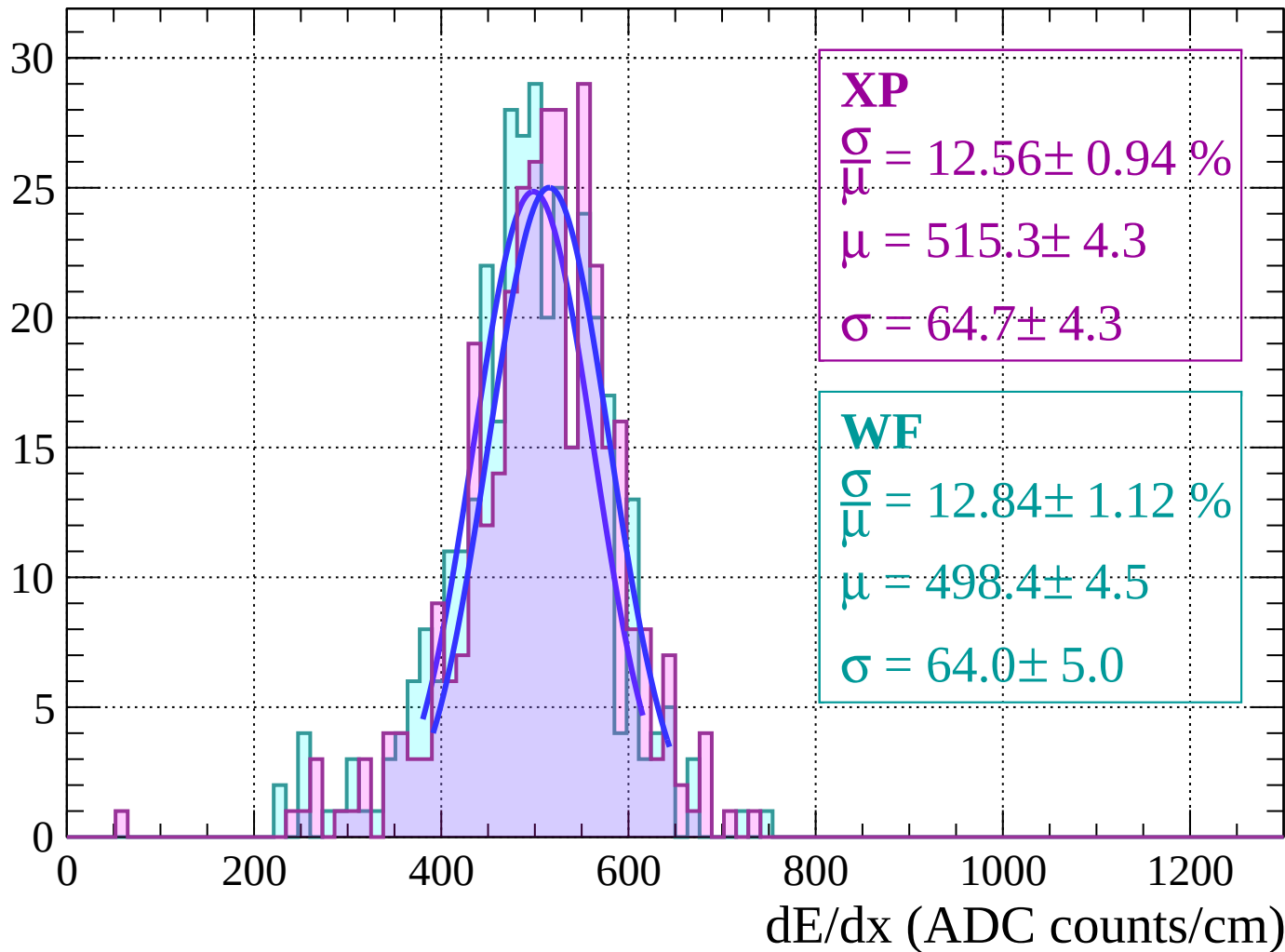
Energy loss | $58 < \theta < 60$

Count



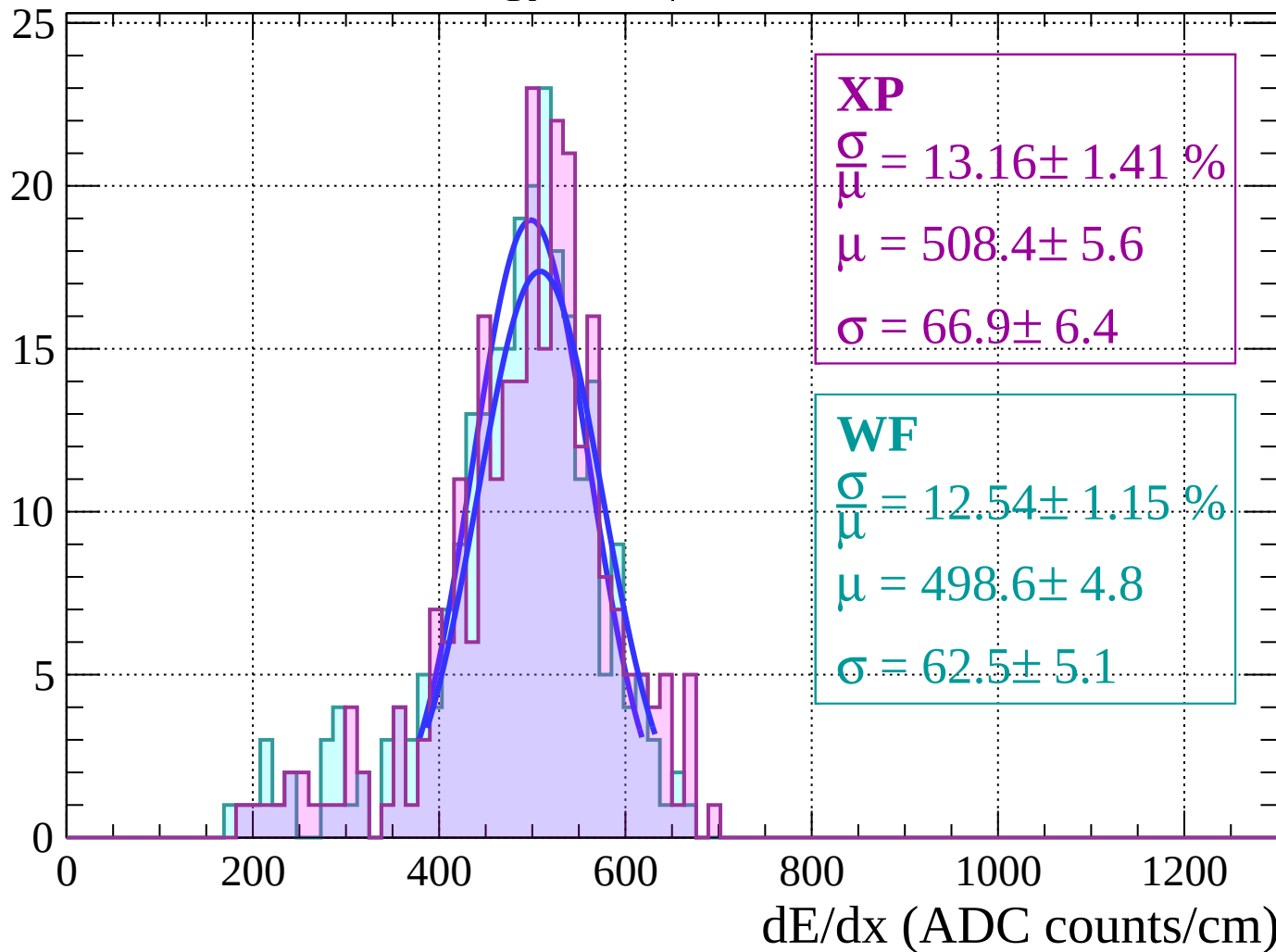
Energy loss | $60 < \theta < 62$

Count



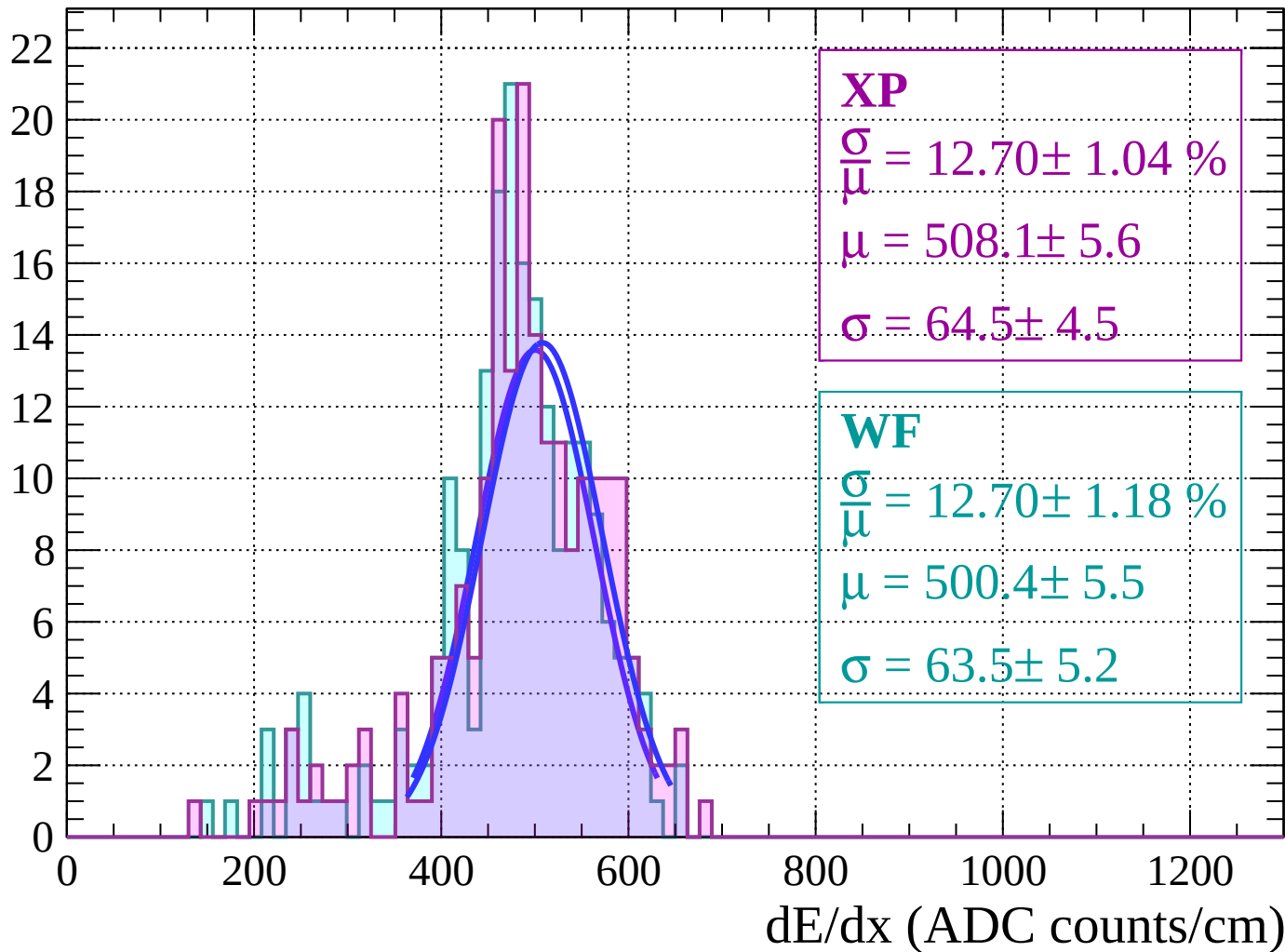
Energy loss | $62 < \theta < 64$

Count



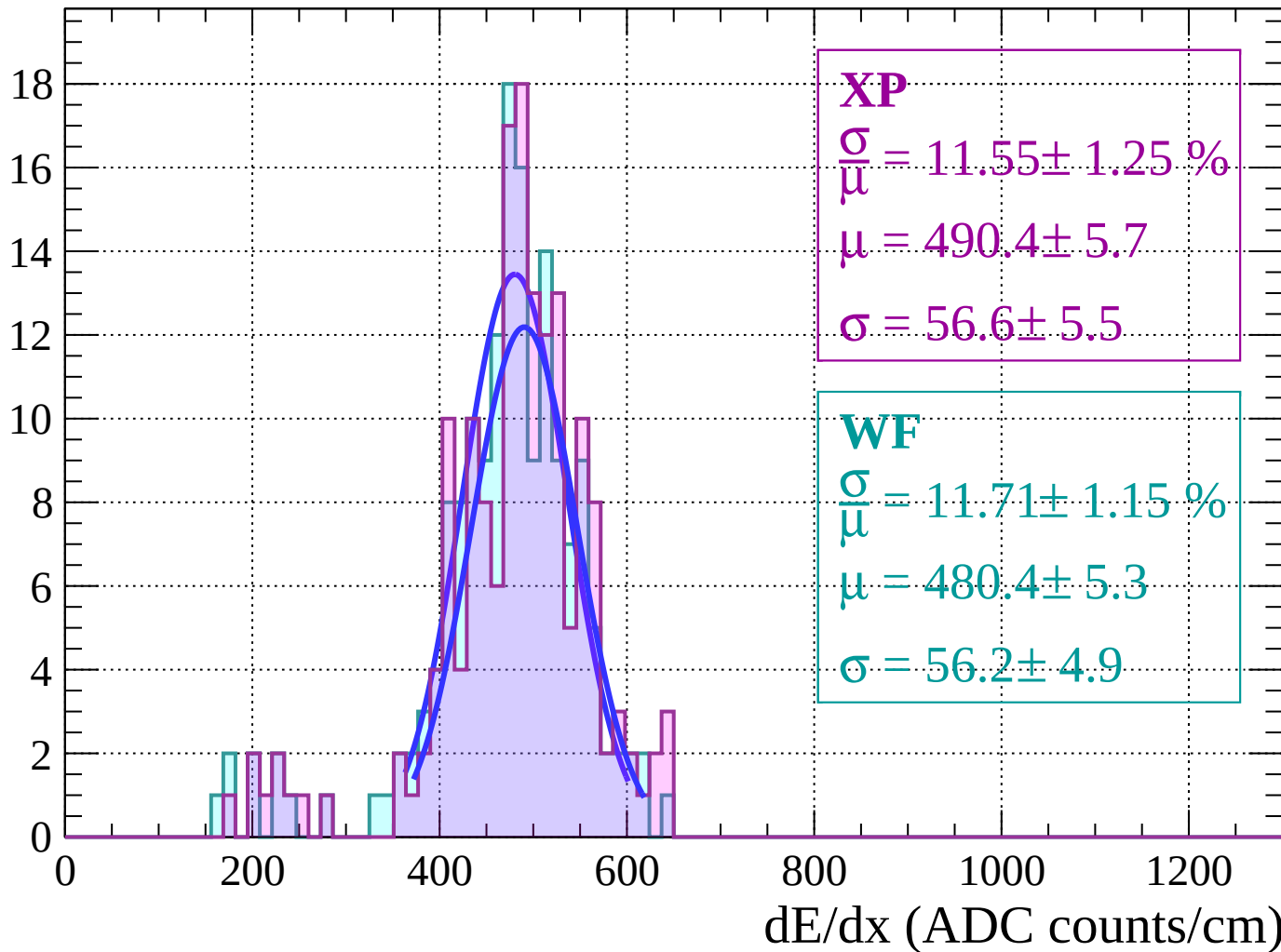
Energy loss | $64 < \theta < 66$

Count



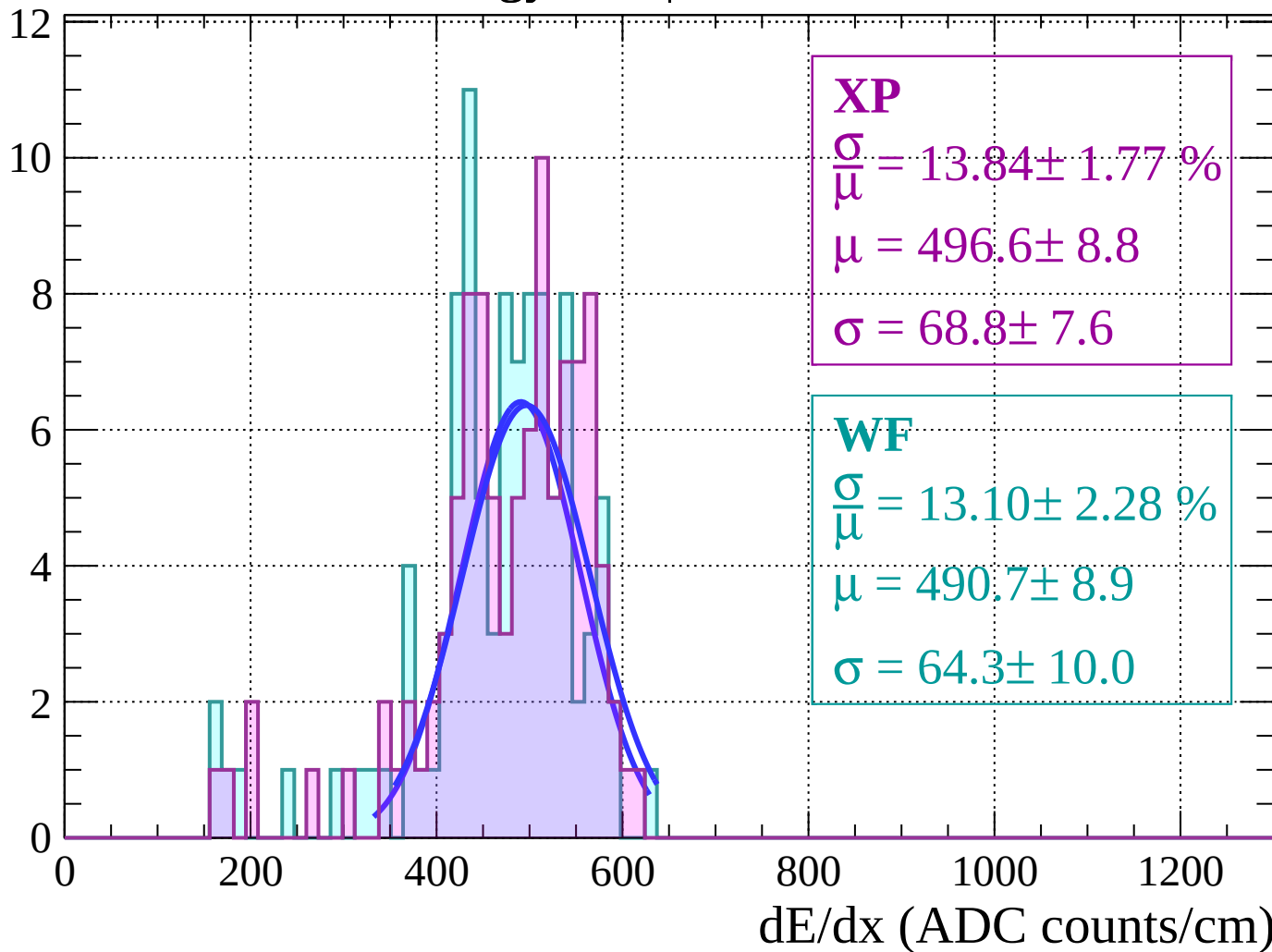
Energy loss | $66 < \theta < 68$

Count



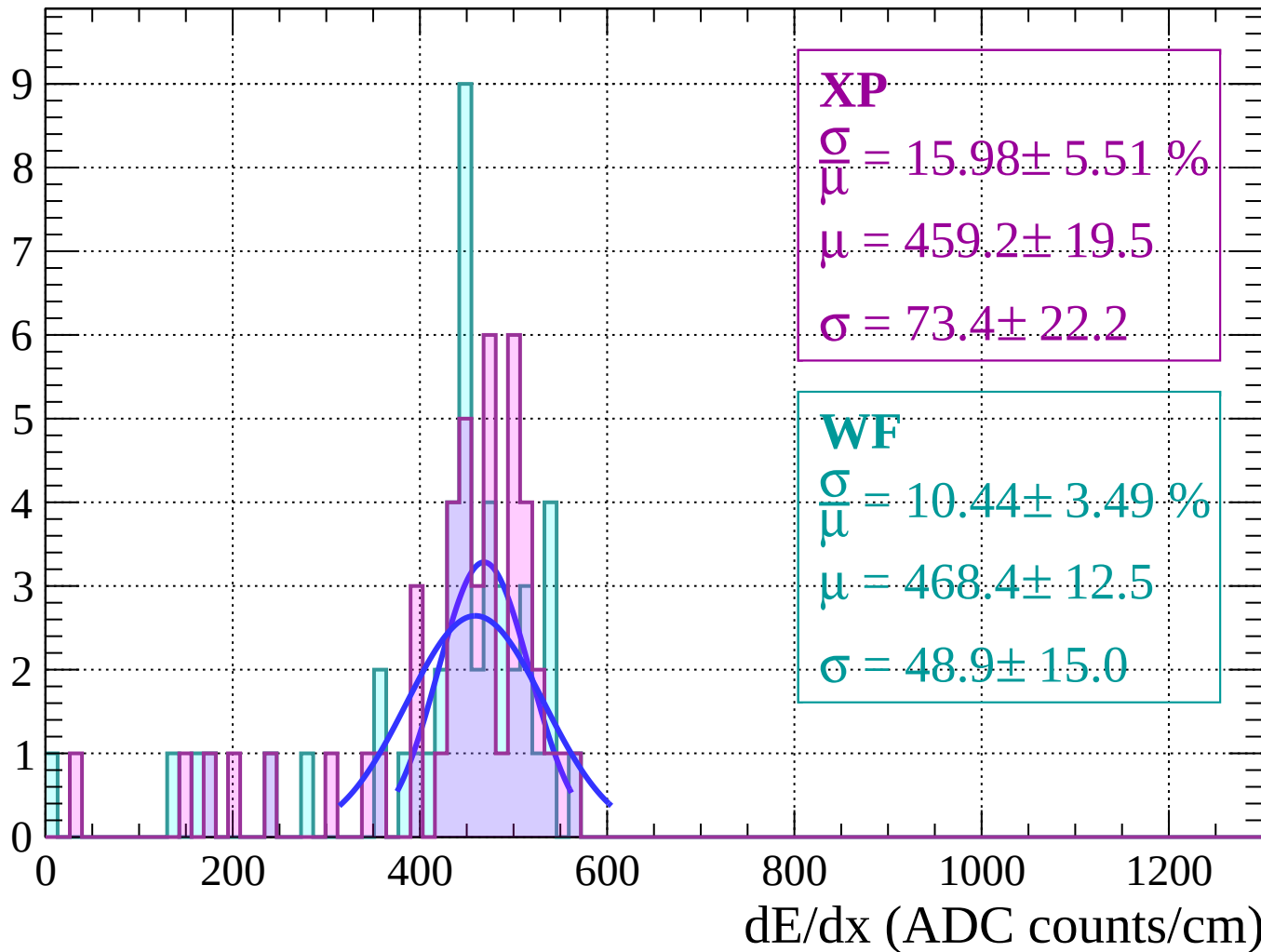
Energy loss | $68 < \theta < 70$

Count



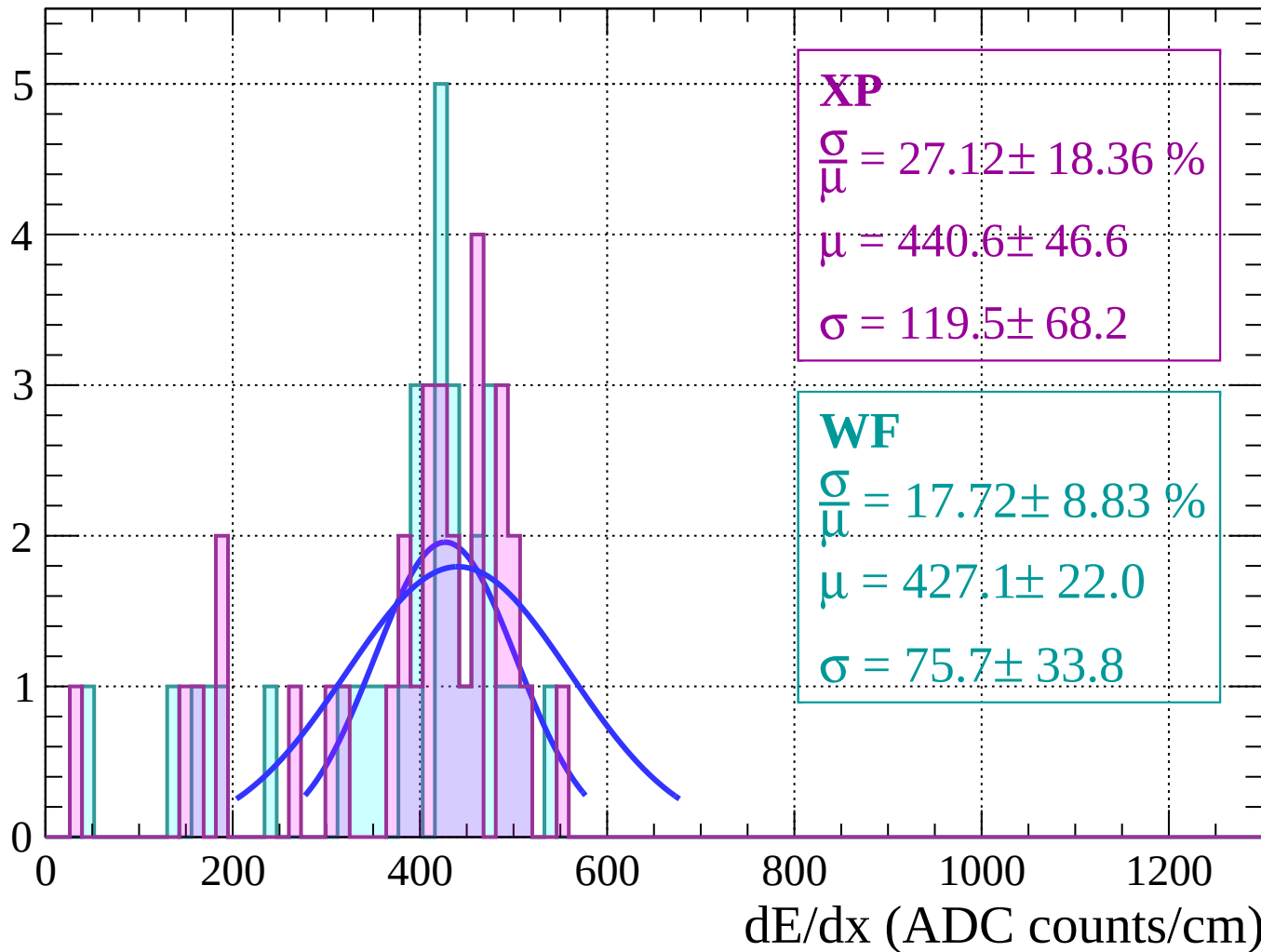
Energy loss | $70 < \theta < 72$

Count



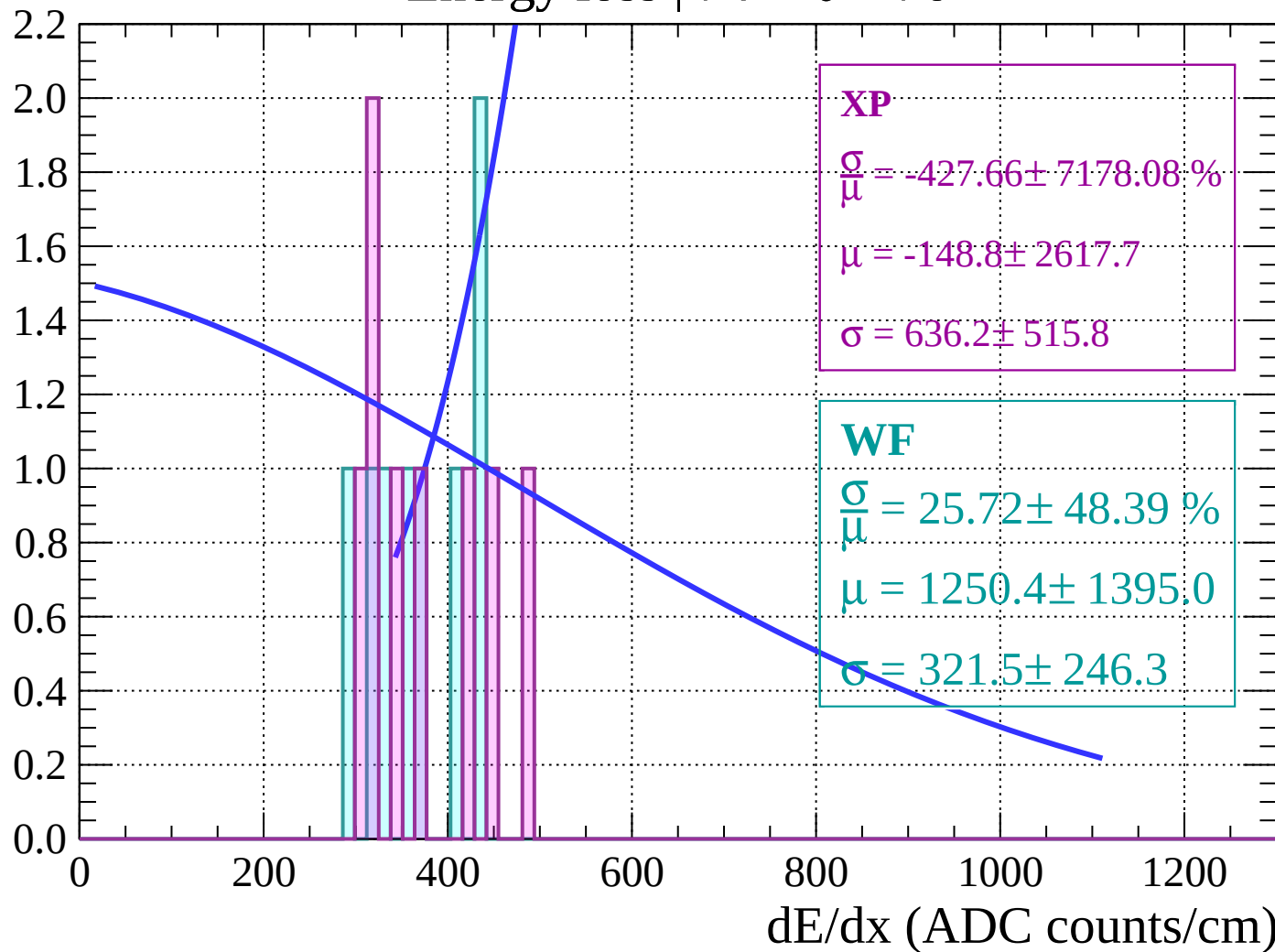
Energy loss | $72 < \theta < 74$

Count



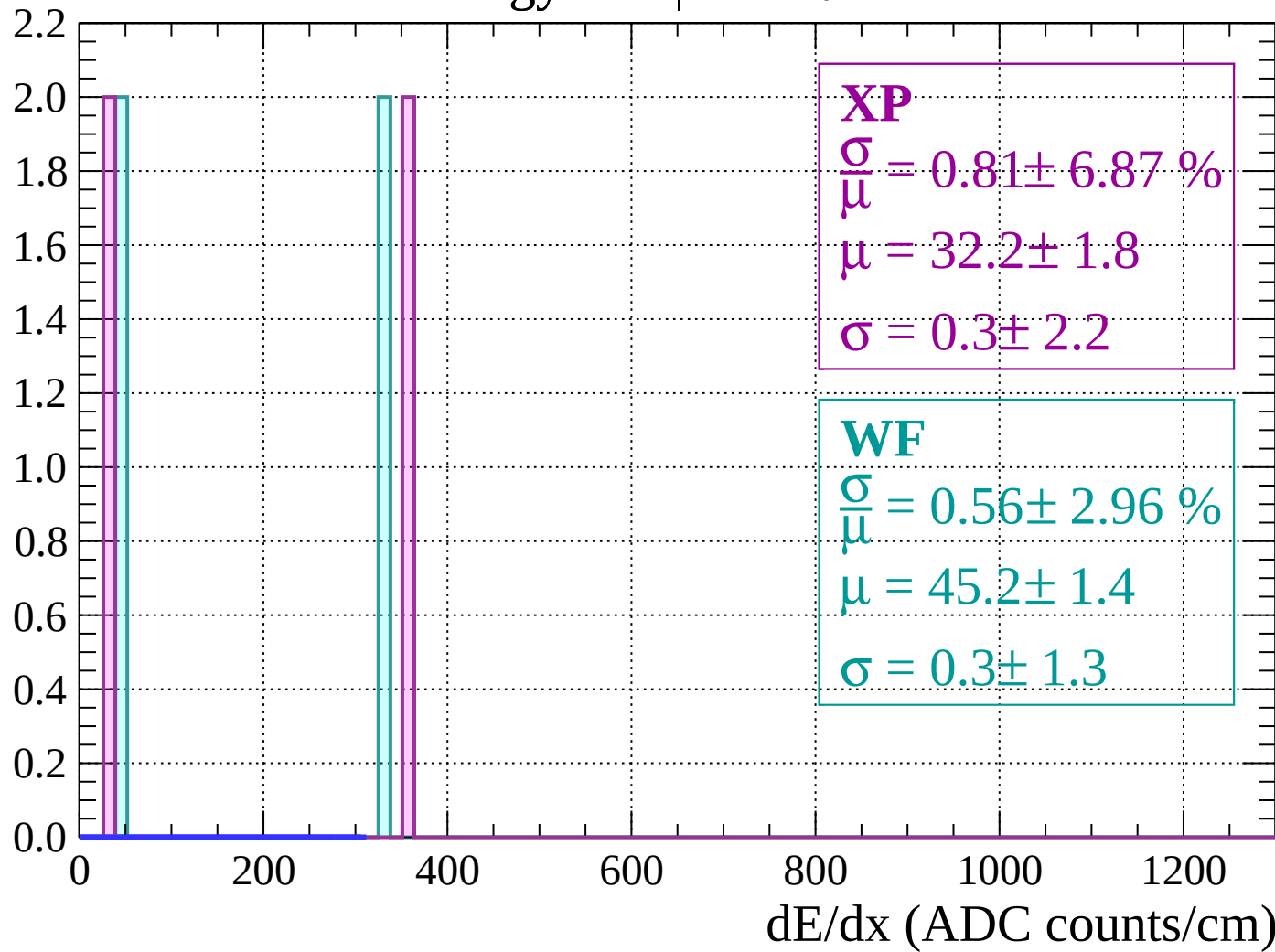
Energy loss | $74 < \theta < 76$

Count



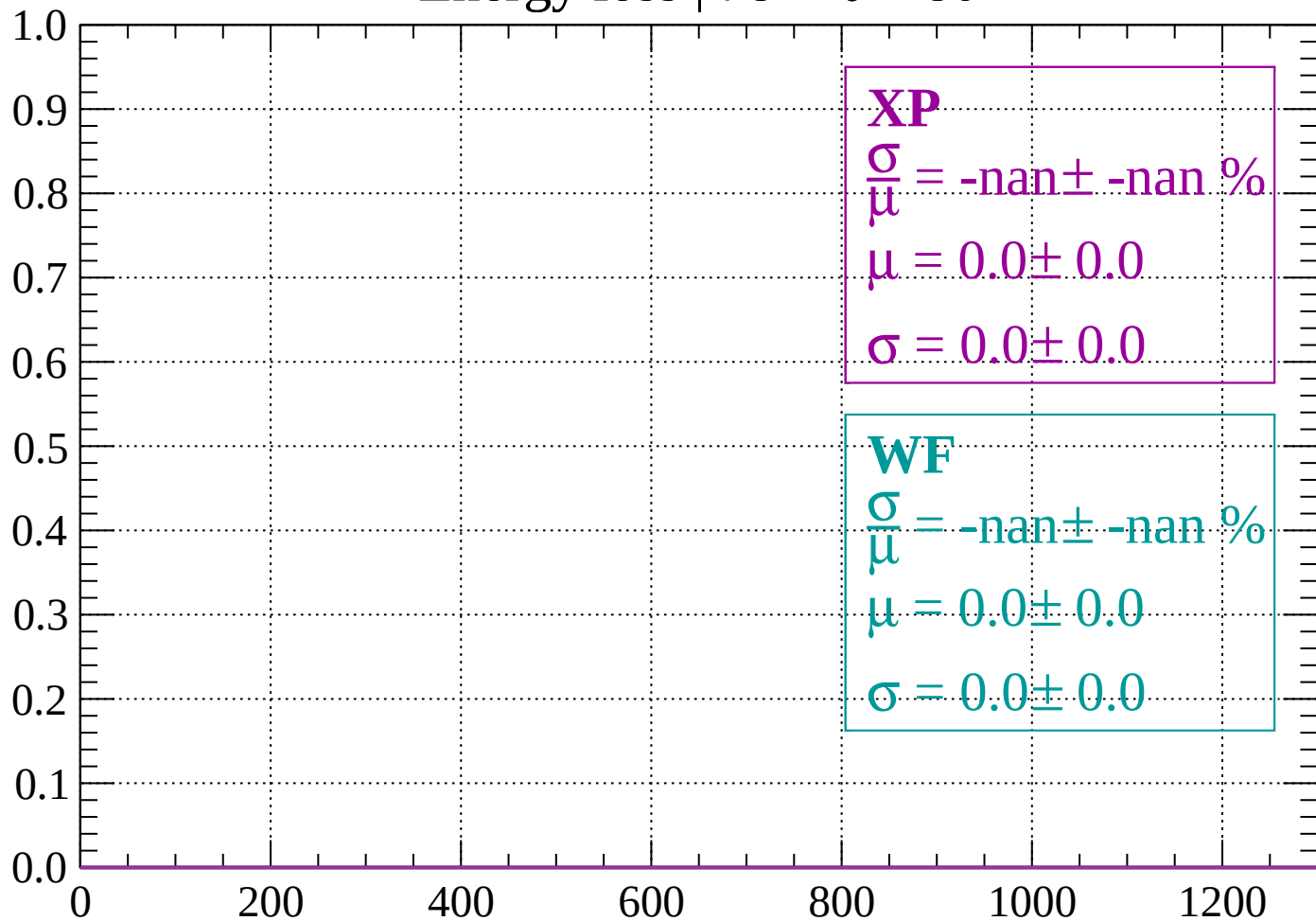
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

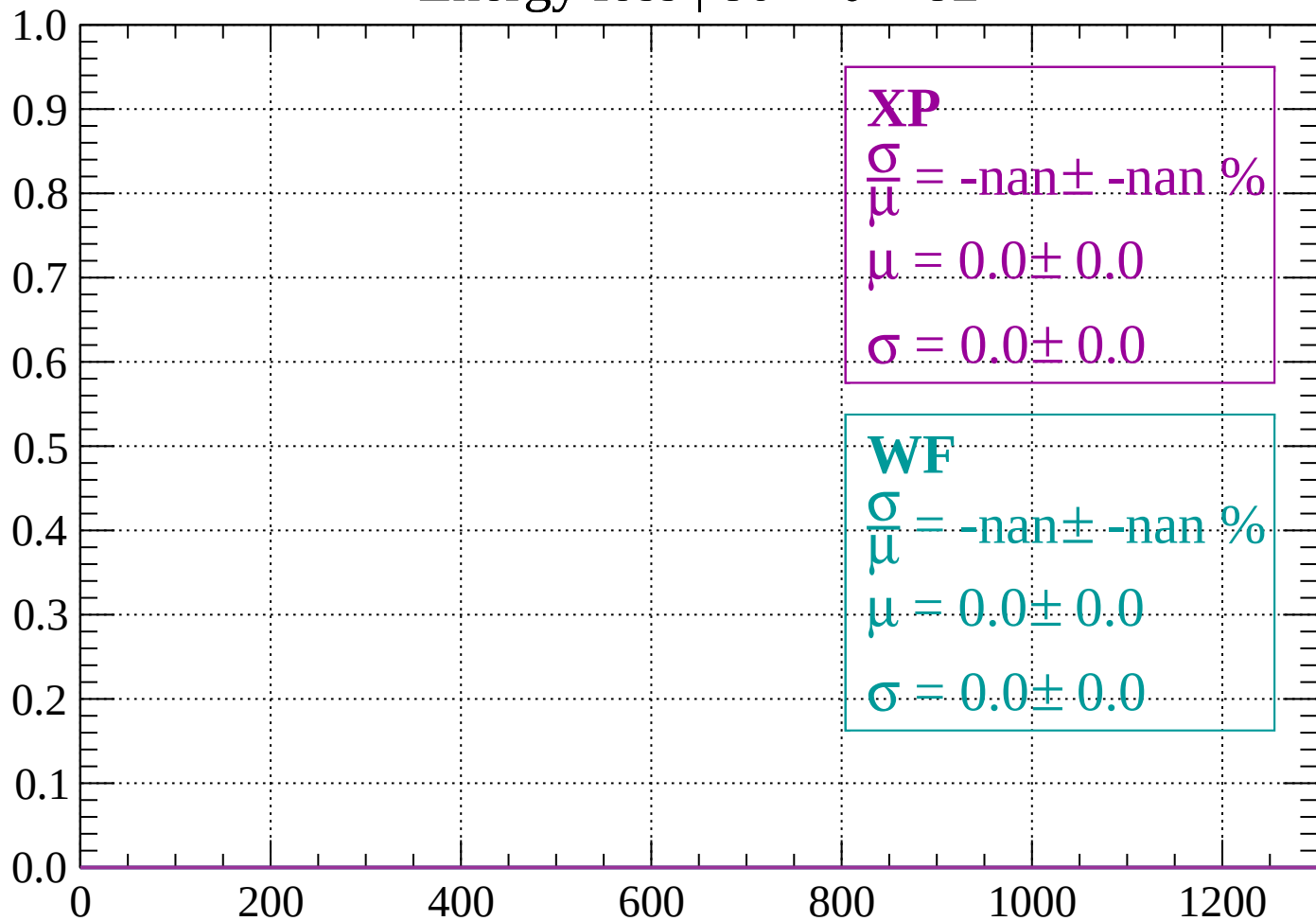
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

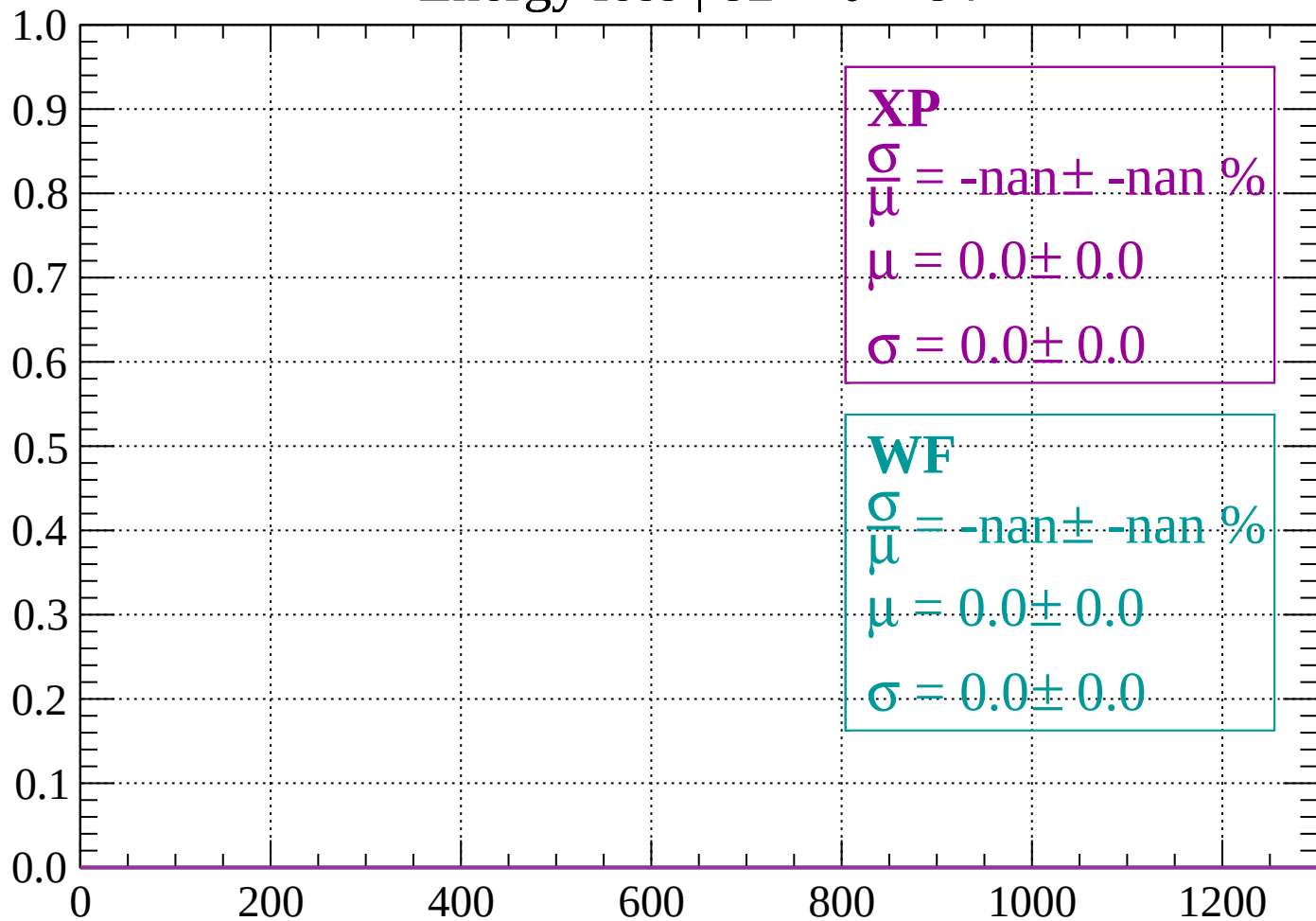
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

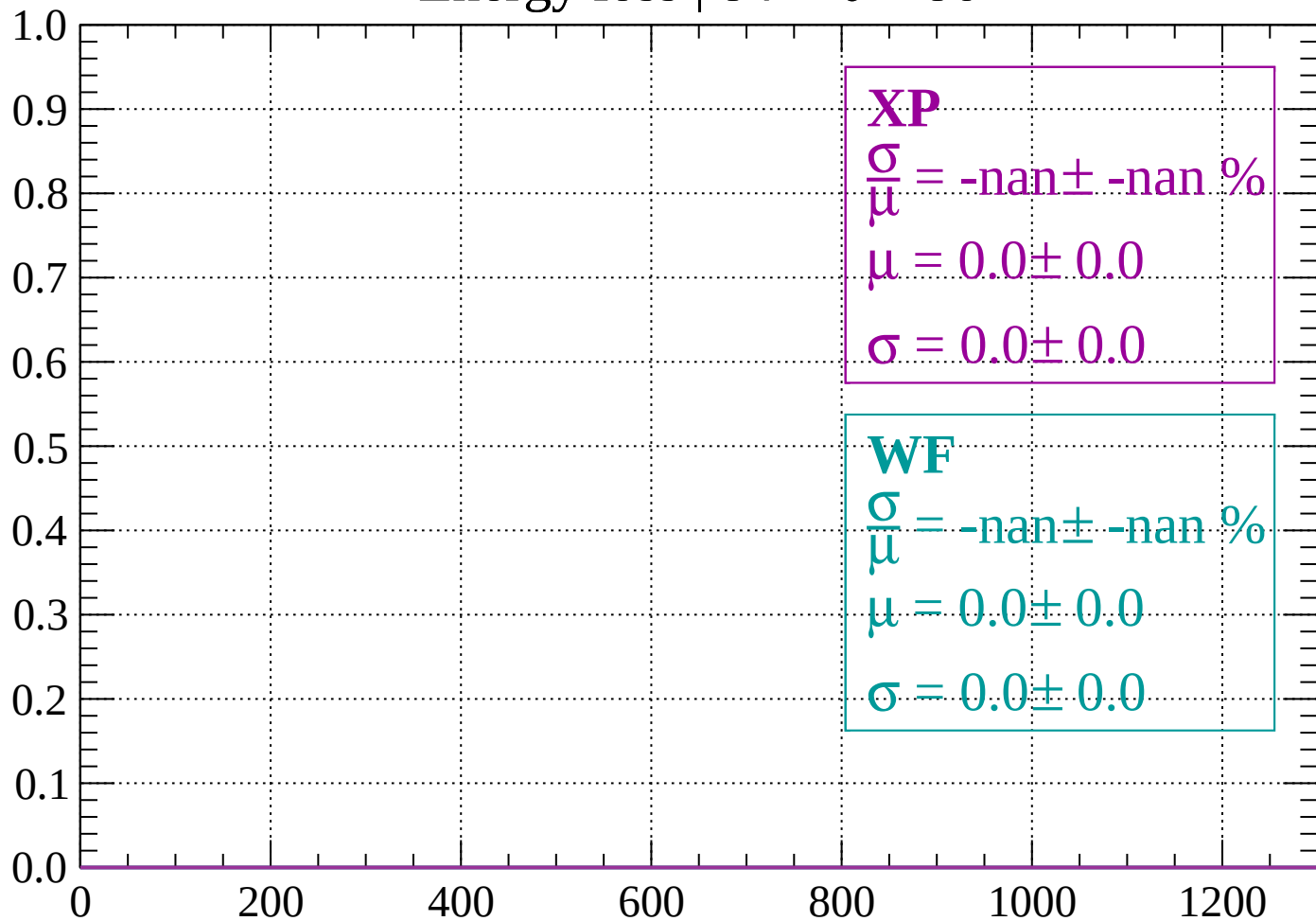
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

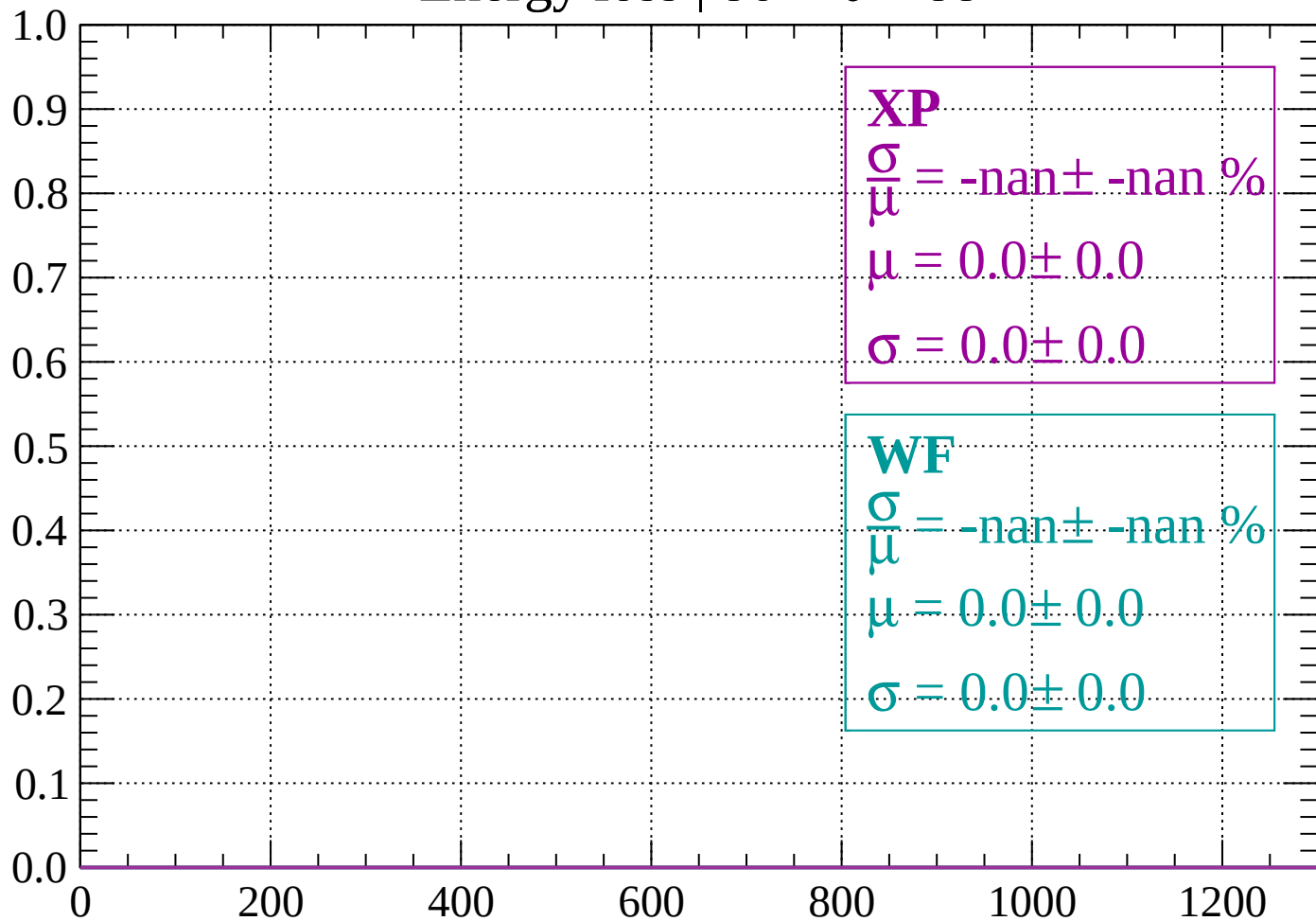
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

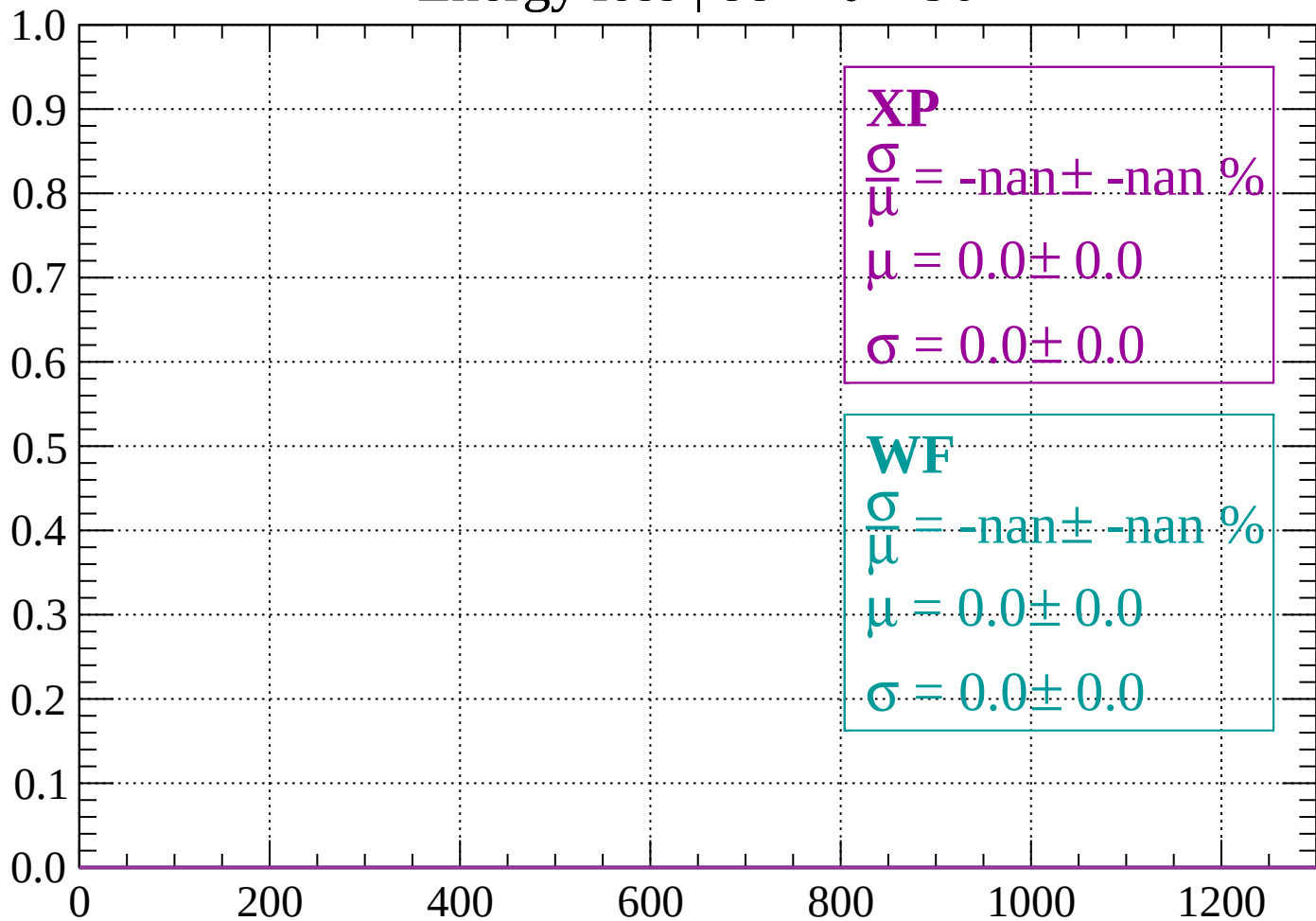
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

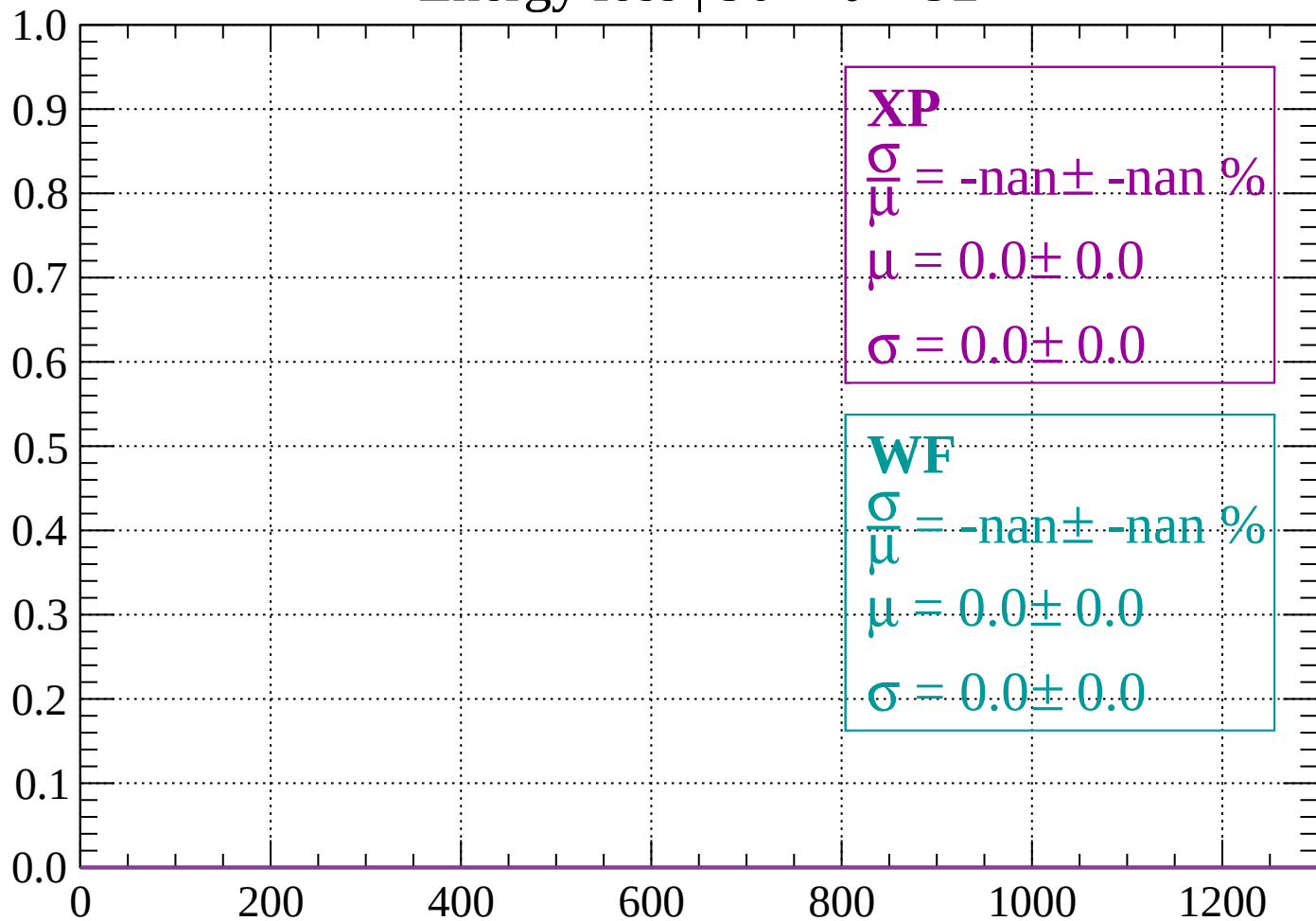
Count



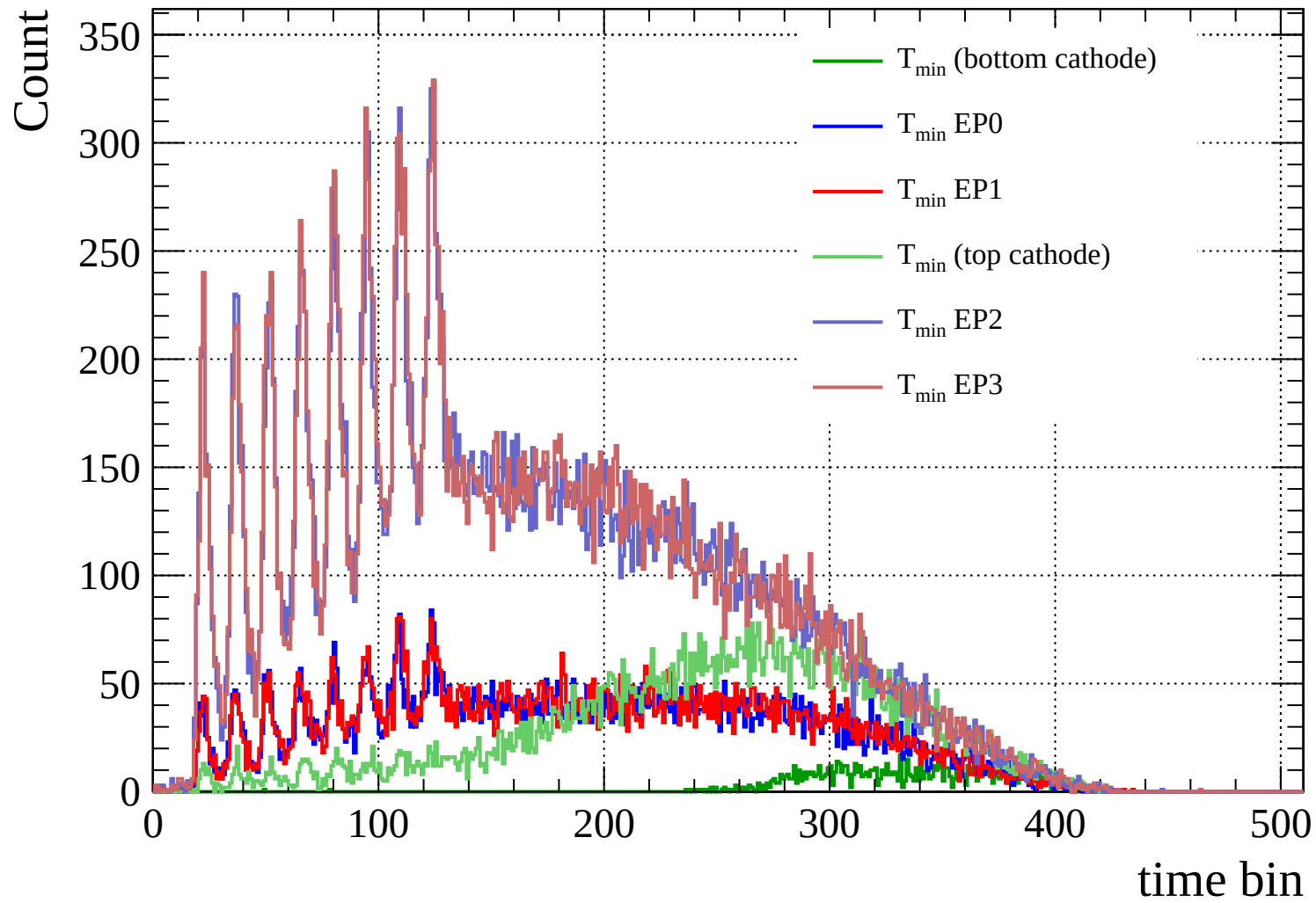
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

